

# 从 C++ 到 AI 计算：第三册实践与代码卷

本地量化推理引擎、完整源码、KV Cache、后端优化与验收

CPU Performance Study

本卷是《从 C++ 到 AI 计算：第三册》的配套实践与代码卷，围绕 `linux_cpu_inference` 贯穿项目展开：模型资产、tokenizer、张量布局、int8 kernel、reference decoder、KV cache、trace、7B 账本、服务调度、CPU/GPU 后端边界和工程验收都要落到可构建代码、完整源码、测试和报告。

# 前言

第三册原理卷把 AI 推理系统拆成模型资产、typed graph、张量布局、量化、KV cache、后端、服务运行时和验收体系。本实践与代码卷只做一件事：把这些概念压到同一个可运行项目里，并把完整源码阅读路径并入同一目录，让读者知道每个概念对应哪段代码、哪条命令、哪种输入、哪份报告和哪条失败路径。

贯穿项目是 [books/cpu-volume-3/labs/linux\\_cpu\\_inference](https://github.com/QuintusLiu/books/cpu-volume-3/labs/linux_cpu_inference)。它不是玩具目录，而是第三册已经持续维护的本地 CPU 量化推理切片：包含教学模型格式、tokenizer、safetensors manifest、gate、int8 linear、packed/AVX2 路径、tiny reference decoder、GPT-2 reference path、KV cache trace、7B compute ledger、serving SLO probe、自研 GPT-2 smoke、真实 small 开源模型 smoke、CLI、benchmark 和 CTest。本卷不维护第二份实现；源码章节通过 [lstinputlisting](#) 直接读取同一份工程文件，所有练习、讲解和完整代码都回到同一份源码。

本卷的学习顺序是从小证据到大系统：先让一个 tiny MLP 算对，再解释张量地址流和 int8 累加；再让 reference decoder、sampler 和 KV cache trace 固定语义；然后接模型导入、shape verifier、memory planner 和 IR lowering；最后用 CPU/GPU 后端边界、工业专项和回归门禁收束。代码走读不再被放到最后当附录，而是插入到相同主题部分中：讲完项目入口就读构建文件和调用链，讲完量化就读公共合同、tiny MLP 和 int8 kernel，讲到真实模型时读 tokenizer、safetensors 和 GPT-2，收束验收时读工具、benchmark、smoke 和测试。每章都要求读者交付代码运行结果或报告字段，而不是只抄概念定义。

## 核心思想

第三册实践的目标不是“写一个看起来像大模型的 demo”，而是建立一条能被复查的推理证据链：资产能校验，shape 能验证，kernel 能对照，trace 能定位，性能能降级，服务指标能解释，回归门禁能长期维护。

# 常见缩写和术语先导

| 缩写                     | 全称                               | 本卷中的含义                                                                                           |
|------------------------|----------------------------------|--------------------------------------------------------------------------------------------------|
| AI                     | Artificial Intelligence          | 这里特指本地推理工程，不讨论泛泛应用口号。                                                                            |
| LLM                    | Large Language Model             | decoder-only 文本生成模型是本卷主线。                                                                        |
| KV Cache               | Key/Value Cache                  | attention 历史 K/V 状态，属于请求生命周期。                                                                    |
| GEMM                   | General Matrix Multiply          | 矩阵乘，是预填充和训练常见核心算子。                                                                               |
| GEMV                   | General Matrix Vector Multiply   | 矩阵向量乘，是单 token decode 的核心形态。                                                                     |
| SIMD                   | Single Instruction Multiple Data | 一条指令处理多个 lane 的 CPU 执行方式。                                                                        |
| AVX2                   | Advanced Vector Extensions 2     | x86-64 上的 256-bit SIMD 指令集之一。                                                                    |
| FMA                    | Fused Multiply-Add               | 乘加融合指令，常出现在浮点 kernel 中。                                                                          |
| GGUF                   | GGML Universal File              | llama.cpp 生态常见模型文件，保存权重、量化格式、tokenizer 和模型元数据。                                                   |
| GGML                   | Georgi Gerganov Machine Learning | llama.cpp 生态底层张量、量化 block 和 CPU kernel 体系；本卷按它的 block 布局读取 <a href="#">Q4_K/Q5_0/Q6_K/Q8_0</a> 。 |
| Q4_K/Q5_0<br>Q6_K/Q8_0 | Quantized GGML formats           | GGML 量化权重格式名。数字表示主要位宽，后缀表示 block 布局族；它们不是 C++ 类型名，进入代码前必须先经过格式 verifier。                         |
| RMSNorm                | Root Mean Square Normalization   | Transformer 中按 hidden 维做均方根归一化的层。                                                                |
| RoPE                   | Rotary Position Embedding        | 旋转位置编码，把位置信息注入 Q/K 向量。                                                                           |
| GQA                    | Grouped-Query Attention          | 多个 query head 共享一组 KV head 的 attention 形状。                                                       |
| SwiGLU                 | Swish-Gated Linear Unit          | LLaMA-like MLP 常用的门控激活结构。                                                                        |
| SLO                    | Service Level Objective          | 服务目标，例如 TTFT、TPOT、p95/p99。                                                                       |
| TTFT                   | Time To First Token              | 从请求进入到首 token 输出的时间。                                                                             |
| TPOT                   | Time Per Output Token            | decode 阶段每个输出 token 的平均或尾部时间。                                                                    |
| IR                     | Intermediate Representation      | 编译器内部表示，用于 pass、lowering 和计划生成。                                                                  |
| RAG                    | Retrieval-Augmented Generation   | 检索增强生成，需要把外部文档和 prompt 编排起来。                                                                     |
| MoE                    | Mixture of Experts               | 多专家模型，会改变路由、负载和内存账本。                                                                             |
| LoRA                   | Low-Rank Adaptation              | 低秩适配权重，常用于微调和多租户加载。                                                                              |

这些缩写不要死记。每个缩写在正文里都要落到一个代码对象或报告字段：KV cache 对应 trace 里的 slot 和 position，SLO 对应 serving probe 的 CSV，IR 对应 executable plan，SIMD/AVX2 对应 kernel 反汇编和 benchmark，GGUF/GGML 对应 manifest、tensor bytes、direct/fallback tensor count 和 caw A/B 报告。

## 本卷结构

本卷按第三册原书的四个大块组织，但目录不再采用“前面做实践、后面贴源码”的割裂方式。每个大块都把实践任务、代码走读和验收证据放在同一条主线里：[linux\\_cpu\\_inference](#)。

第一部分从模型资产到量化 kernel。你会运行 tiny MLP，先读构建入口、项目边界和调用链总图，再检查 tokenizer 和 safetensors manifest，手算张量地址，比较 scalar、packed 和 AVX2 路径，并在公共合同、tiny MLP 与 int8 kernel 源码里解释 scale、accumulator 和 requantize 的边界。

第二部分进入 Transformer、KV cache、sampler 和服务运行时。你会跑 reference decoder trace，解释每个 checkpoint 的 shape、stride、dtype 和 layout，构造 KV cache 状态表，分析 admission、queue wait、TTFT、TPOT、取消和拒绝。

第三部分处理模型导入、真实模型加载闭环、memory planner 和 IR lowering。你会把 manifest、shape verifier、tensor role、state edge、workspace、liveness 和 executable plan 写成可检查报告，并在同一部分中读 reference decoder、tokenizer、safetensors 和 GPT-2 reference path 的完整实现，显式运行 LCQI 自研 GPT-2 safetensors smoke 和一个 small 级别开源 GGUF 模型 smoke，而不是只停留在 tiny fixture。

第四部分收束到后端优化和验收。你会比较 CPU kernel 证据、GPU 后端边界、工业专项能力和回归门禁，并直接读工具、smoke、benchmark、ledger、serving probe 和测试源码。最终交付不是“跑通一次”，而是可复查的工程档案：命令、输入、输出、trace、benchmark、不可验证边界和回归规则。

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第零部分

# 推理链路、张量账本与量化

## 第 1 章实践：端到端推理链路和项目总入口

本章对应原书中的第三册“端到端链路：从模型文件到下一个 token”。不要把它当作概念复述，本章要把模型目录、manifest、tokenizer、typed graph、backend plan、request session、KV cache、oracle 和 profiler 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `tiny_mlp_i8.txt`，核心命令或测试入口是 `lcqi_cli`。

Listing 1.1 第 1 章实践：端到端推理链路和项目总入口的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  <- DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_cli / tiny_mlp_i8.txt
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：只跑一个 tiny MLP 推理请求。读者要先手写预期结果，再运行项目观察输出。动作是：把 `readfiles->tokenize->run_inference->logits->predicted_class` 串起来。如果输出里没有 `predicted_class` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：模型文件存在不等于推理链路可信；只有 CLI 输出、测试断言、输入合同和失败路径同时成立，才算完成入口闭环这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

### 四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：最终大作业的总目录：每章报告都放入 `reports/volume3-practice/`，第一章提交项目地图、命令记录和一次成功/失败输入对照。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

# 代码走读：构建入口、项目边界与调用链总图

## 源码阅读覆盖范围

LCQI 的源码边界由三个层次组成。第一层是仓库构建入口，它决定 CMake 怎样把第三册的实验代码纳入构建和测试。第二层是 `labs/linux_cpu_inference`，它包含真正的库、CLI、benchmark、trace、ledger、serving probe 和测试目标。第三层是工具脚本，它们下载或复用外部模型资产，生成 smoke 和 benchmark 报告。

### 源码阅读任务

先读本章的 CMake 文件，再读 README。读者要能回答三个问题：一是 `lcqi` 静态库包含哪些实现文件；二是哪些可执行文件只依赖自研代码，哪些依赖可选外部库；三是真实 GPT-2 和 SmolLM2 smoke 为什么不进入默认 CTest。

## 根构建入口

第三册根目录的 CMake 文件只做项目级选择：是否构建第三册 labs，以及如何把 `labs/linux_cpu_inference` 接入。它应该保持薄层，避免把具体目标定义分散到多个地方。

**Listing 1.2** cpu-volume-3/README.md

```

1 # 从 C++ 到 AI 计算：第三册
2
3 第三册用于承接 AI、AI 编译器、算子开发和本地大模型推理引擎。第二册只讨论硬件原
  ↳ 理、多核、高性能计算、并发、锁、无锁数据结构、并行算法和分布式计算，不再
  ↳ 直接进入 AI。
4
5 本册贯穿项目是一台能推理开源 7B 左右 decoder-only 大模型的本地引擎，并把 AI 编
  ↳ 译器作为引擎内部主线：模型图导入、typed tensor IR、算子融合、内存规划、
  ↳ CPU/GPU lowering、kernel selection 和 runtime dispatch 都要能落回编译原理
  ↳ 与底层机器账本。工程路线从 CPU 上可解释的 reference 和量化切片开始，逐步
  ↳ 扩展到 tokenizer、模型加载、Transformer 层、KV cache、CPU SIMD/多线程后
  ↳ 端、GPU 后端、正确性报告和性能对标。
6
7 正式书稿工程位于：
8
9 ```text
10 source/latex/
11 ```
12
13 构建命令：
14
15 ```bash
16 make -C source/latex check
17 make -C source/latex pdf
18 make -C source/latex epub
19 ```
20
21 生成文件：
22
23 ```text

```

```

24 source/latex/main.pdf
25 source/latex/main.epub
26 ```
27
28 当前保留的第一阶段切片：
29
30 ```text
31 labs/linux_cpu_inference/
32 ```
33
34 该目录目前包含最小 MLP/int8 权重教学切片、tiny reference decoder、GPT-2 ↵
    ↵ reference path、int8 scalar baseline、packed layout、AVX2 SIMD 路径和 ↵
    ↵ shape sweep benchmark，用来验证张量、线性层、量化权重、tokenizer、↵
    ↵ safetensors、GPT-2 forward、KV cache、oracle 和 benchmark 的基本边界。当↵
    ↵ 前 x86-64 AVX2 结果见：
35
36 ```text
37 results/lcqi-x86-avx2-benchmark-2026-06-25.csv
38 results/lcqi-x86-avx2-decode4096-2026-06-25.csv
39 ```
40
41 自研 GPT-2 smoke 入口如下：
42
43 ```bash
44 make cpu3-gpt2-smoke
45 make cpu3-gpt2-benchmark-compare
46 ```
47
48 第一条命令下载 `openai-community/gpt2` 的标准 safetensors 资产到用户缓存目录，↵
    ↵ 用 LCQI C++ reference path 生成 1 个 token，并把报告写入：
49
50 ```text
51 results/lcqi-gpt2-smoke.txt
52 ```
53
54 第二条命令用同一个 GPT-2 F32 模型对比 LCQI full-prefix 和 cached-KV 生成路径，↵
    ↵ 并在本机存在 llama.cpp/SmolLM2 缓存时附带成熟引擎参考，报告写入：
55
56 ```text
57 results/lcqi-gpt2-benchmark-compare.txt
58 ```
59
60 它仍然不是生产级 AI Infra。生产级门禁见：
61
62 ```text
63 reports/ai-infra-maturity-audit.md
64 ```
65
66 第三册后续必须把 lab 推进到“小型 CPU 推理 runtime + 手写高性能算子 + 严格 ↵
    ↵ benchmark 报告 + 真实系统对标”，否则只能证明系统学习，不能证明生产级能↵
    ↵ 力。

```

Listing 1.3 cpu-volume-3/CMakeLists.txt

```

1 cmake_minimum_required(VERSION 3.31)
2 project(CPUVolume3 LANGUAGES CXX)
3
4 set(CMAKE_CXX_STANDARD 20)
5 set(CMAKE_CXX_STANDARD_REQUIRED ON)
6 set(CMAKE_CXX_EXTENSIONS OFF)
7 set(CMAKE_EXPORT_COMPILE_COMMANDS ON)
8
9 option(CPU_VOLUME_3_BUILD_LABS "Build CPU Volume 3 labs" ON)
10
11 if(CPU_VOLUME_3_BUILD_LABS)
12     enable_testing()
13     add_subdirectory(labs/linux_cpu_inference)
14 endif()

```

实践与代码卷也提供一个薄 CMake 入口，它把同一份 LCQI 源码作为可测试对象接入。这样本卷不会复制实现，只复用第三册主工程。

Listing 1.4 cpu-volume-3-practice/CMakeLists.txt

```

1 cmake_minimum_required(VERSION 3.31)
2 project(CPUVolume3Practice LANGUAGES CXX)
3
4 set(CMAKE_CXX_STANDARD 20)
5 set(CMAKE_CXX_STANDARD_REQUIRED ON)
6 set(CMAKE_CXX_EXTENSIONS OFF)
7 set(CMAKE_EXPORT_COMPILE_COMMANDS ON)
8
9 enable_testing()
10 add_subdirectory(..cpu-volume-3/labs/linux_cpu_inference linux_cpu_inference)

```

本卷自身的 Makefile 负责构建 PDF/EPUB，并通过 `test` 目标回到第三册真实 LCQI 工程做 CMake/CTest 验证。实践与代码卷如果只排版、不测试，就不能宣称“完整源码可运行”。

Listing 1.5 cpu-volume-3-practice/Makefile

```

1 LATEX_DIR := source/latex
2
3 .PHONY: all check pdf epub test clean
4
5 all: pdf epub
6
7 check:
8     $(MAKE) -C $(LATEX_DIR) check
9
10 pdf:
11     $(MAKE) -C $(LATEX_DIR) pdf
12
13 epub:
14     $(MAKE) -C $(LATEX_DIR) epub
15

```

```

16 test:
17     cmake -S . -B build/lcqi-debug -DCMAKE_BUILD_TYPE=Debug
18     cmake --build build/lcqi-debug
19     ctest --test-dir build/lcqi-debug --output-on-failure
20
21 clean:
22     $(MAKE) -C $(LATEX_DIR) clean

```

Listing 1.6 cpu-volume-3-practice/source/latex/Makefile

```

1 LATEXMK ?= latexmk
2 ENGINE ?= -xelatex
3 MAIN ?= main.tex
4 EPUB_BUILDER ?= ../../../../cpu-volume-1/source/latex/scripts/build_epub.py
5
6 .PHONY: all pdf epub clean check
7
8 all: pdf epub
9
10 pdf:
11     $(LATEXMK) $(ENGINE) -interaction=nonstopmode -halt-on-error $(MAIN)
12
13 epub:
14     @BOOK_TITLE="从 C++ 到 AI 计算：第三册实践与代码卷" \
15     BOOK_IMPORT_TITLE="CPU Volume 3 Practice Code" \
16     BOOK_ID_SEED="cpu-volume-3-practice-code-weread-20260629" \
17     EPUB_ASCII_TITLES=1 \
18     BOOK_SUBTITLE="本地量化推理引擎、完整源码、KV Cache、后端优化与验收" \
19     BOOK_AUTHOR="CPU Performance Study" \
20     python3 $(EPUB_BUILDER) --book-dir . --output main.epub --no-cover-page --↵
    ↵ forbid-images --linearize-tables --wechat-compatible --no-embed-fonts
21
22 clean:
23     $(LATEXMK) -C
24
25 check:
26     @python3 scripts/check_inputs.py
27     @command -v xelatex >/dev/null 2>&1 || { echo "missing xelatex"; exit 1; }
28     @command -v latexmk >/dev/null 2>&1 || { echo "missing latexmk"; exit 1; }

```

## LCQI 目标定义

`linux_cpu_inference/CMakeLists.txt` 是代码走读中的第一份核心文件。这里能看到 `lcqi` 库、`lcqi_cli`、`lcqi_bench`、`lcqi_trace`、`lcqi_gpt2`、`lcqi_ledger`、`lcqi_serving_probe`、可选 `lcqi_onednn_bench` 和 `lcqi_tests` 的完整关系。

Listing 1.7 linux\_cpu\_inference/CMakeLists.txt

```

1 find_package(Threads REQUIRED)
2
3 add_library(lcqi STATIC
4     src/f32_kernels.cpp

```

```

5     src/ggml_matvec.cpp
6     src/ggml_tensors.cpp
7     src/gguf.cpp
8     src/gpt2_reference.cpp
9     src/inference.cpp
10    src/int8_kernels.cpp
11    src/llama_gguf.cpp
12    src/model.cpp
13    src/q4_k.cpp
14    src/reference_decoder.cpp
15    src/safetensors.cpp
16    src/tokenizer.cpp
17 )
18
19 if(CMAKE_SYSTEM_PROCESSOR MATCHES "x86_64|AMD64|i[3-6]86")
20     if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
21         target_sources(lcqi PRIVATE src/f32_kernels_avx2.cpp)
22         target_sources(lcqi PRIVATE src/ggml_matvec_avx2.cpp)
23         target_sources(lcqi PRIVATE src/int8_kernels_avx2.cpp)
24         target_sources(lcqi PRIVATE src/q4_k_avx2.cpp)
25         set_source_files_properties(src/f32_kernels_avx2.cpp PROPERTIES ↵
↵ COMPILE_OPTIONS "-mavx2;-mfma")
26         set_source_files_properties(src/ggml_matvec_avx2.cpp PROPERTIES ↵
↵ COMPILE_OPTIONS "-mavx2")
27         set_source_files_properties(src/int8_kernels_avx2.cpp PROPERTIES ↵
↵ COMPILE_OPTIONS "-mavx2;-mfma")
28         set_source_files_properties(src/q4_k_avx2.cpp PROPERTIES ↵
↵ COMPILE_OPTIONS "-mavx2")
29         target_compile_definitions(lcqi PRIVATE LCQI_ENABLE_AVX2=1)
30     endif()
31 endif()
32
33 if(NOT CMAKE_SYSTEM_NAME STREQUAL "Linux")
34     message(WARNING "LCQI is written for the book's local Linux CPU target; ↵
↵ current system is ${CMAKE_SYSTEM_NAME}.")
35 endif()
36
37 target_include_directories(lcqi
38     PUBLIC
39     ${CMAKE_CURRENT_SOURCE_DIR}/include
40 )
41
42 target_compile_features(lcqi PUBLIC cxx_std_20)
43 target_link_libraries(lcqi PUBLIC Threads::Threads)
44
45 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
46     target_compile_options(lcqi PRIVATE -Wall -Wextra -Wpedantic)
47 endif()
48
49 add_executable(lcqi_cli
50     src/main.cpp
51 )

```

```
52
53 target_link_libraries(lcqi_cli PRIVATE lcqi)
54
55 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
56     target_compile_options(lcqi_cli PRIVATE -Wall -Wextra -Wpedantic)
57 endif()
58
59 add_executable(lcqi_bench
60     src/bench.cpp
61 )
62
63 target_link_libraries(lcqi_bench PRIVATE lcqi)
64
65 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
66     target_compile_options(lcqi_bench PRIVATE -Wall -Wextra -Wpedantic)
67 endif()
68
69 add_executable(lcqi_gguf_q4_bench
70     src/gguf_q4_bench.cpp
71 )
72
73 target_link_libraries(lcqi_gguf_q4_bench PRIVATE lcqi)
74
75 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
76     target_compile_options(lcqi_gguf_q4_bench PRIVATE -Wall -Wextra -Wpedantic)
77 endif()
78
79 add_executable(lcqi_trace
80     src/trace.cpp
81 )
82
83 target_link_libraries(lcqi_trace PRIVATE lcqi)
84
85 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
86     target_compile_options(lcqi_trace PRIVATE -Wall -Wextra -Wpedantic)
87 endif()
88
89 add_executable(lcqi_gpt2
90     src/gpt2_cli.cpp
91 )
92
93 target_link_libraries(lcqi_gpt2 PRIVATE lcqi)
94
95 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
96     target_compile_options(lcqi_gpt2 PRIVATE -Wall -Wextra -Wpedantic)
97 endif()
98
99 add_executable(lcqi_llama_gguf
100     src/llama_cli.cpp
101 )
102
103 target_link_libraries(lcqi_llama_gguf PRIVATE lcqi)
```



```

104
105 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
106     target_compile_options(lcqi_llama_gguf PRIVATE -Wall -Wextra -Wpedantic)
107 endif()
108
109 add_executable(lcqi_ledger
110     src/ledger.cpp
111 )
112
113 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
114     target_compile_options(lcqi_ledger PRIVATE -Wall -Wextra -Wpedantic)
115 endif()
116
117 add_executable(lcqi_serving_probe
118     src/serving_probe.cpp
119 )
120
121 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
122     target_compile_options(lcqi_serving_probe PRIVATE -Wall -Wextra -Wpedantic)
123 endif()
124
125 find_package(dnnl QUIET)
126 if(dnnl_FOUND)
127     add_executable(lcqi_onednn_bench
128         src/onednn_bench.cpp
129     )
130     target_link_libraries(lcqi_onednn_bench PRIVATE DNNL::dnnl)
131     target_compile_features(lcqi_onednn_bench PRIVATE cxx_std_20)
132     if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
133         target_compile_options(lcqi_onednn_bench PRIVATE -Wall -Wextra -↵
134             ↵ Wpedantic)
135     endif()
136 endif()
137
138 add_executable(lcqi_tests
139     tests/lcqi_tests.cpp
140 )
141
142 target_link_libraries(lcqi_tests PRIVATE lcqi)
143
144 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
145     target_compile_options(lcqi_tests PRIVATE -Wall -Wextra -Wpedantic)
146 endif()
147
148 add_test(NAME lcqi_tests COMMAND ${CMAKE_TARGET_FILE:lcqi_tests})
149 add_test(NAME lcqi_gpt2_tiny COMMAND ${CMAKE_TARGET_FILE:lcqi_gpt2})
150 set_tests_properties(lcqi_gpt2_tiny PROPERTIES
151     PASS_REGULAR_EXPRESSION "generated_ids 1 2 3 3 3"
152 )
153 add_test(NAME lcqi_trace COMMAND ${CMAKE_TARGET_FILE:lcqi_trace})
154 set_tests_properties(lcqi_trace PROPERTIES
155     PASS_REGULAR_EXPRESSION "tokenizer,0,-1,5,1,i32,contiguous,10,4,0;1;2;3;4"

```

```

155 )
156 add_test(NAME lcqi_ledger COMMAND ${TARGET_FILE:lcqi_ledger})
157 set_tests_properties(lcqi_ledger PROPERTIES
158     PASS_REGULAR_EXPRESSION "bandwidth_limit,int4_at_100_gbps,23.602,tokens/s"
159 )
160 add_test(NAME lcqi_serving_probe COMMAND ${TARGET_FILE:lcqi_serving_probe})
161 set_tests_properties(lcqi_serving_probe PROPERTIES
162     PASS_REGULAR_EXPRESSION "request,req_too_long,0,memory_budget_exceeded"
163 )

```

## 项目说明与模型资产

README 是复现实验的入口。它写清楚当前能力、后续边界、构建命令、tiny 推理命令、benchmark 字段、reference trace 字段、GPT-2 smoke、7B ledger、serving SLO probe 和 SmolLM2 small smoke。读源码前先读 README，可以避免把 tiny fixture、GPT-2 reference path 和外部 llama.cpp smoke 混成同一个能力等级。

**Listing 1.8** linux\_cpu\_inference/README.md

```

1 # Linux CPU Quantized Inference
2
3 这是第三册贯穿项目的第一阶段切片：Linux 本地 CPU 量化线性层和最小推理流程。第三
  ↳ 册的贯穿目标不是这个 MLP 本身，而是一台能推理开源 7B 左右 decoder-only 大
  ↳ 模型的本地引擎，并逐步覆盖 tokenizer、模型加载、Transformer 层、KV cache
  ↳ 、CPU/GPU 后端、正确性报告和性能对标。
4
5 目标平台是 x86-64 Linux 实验环境。完整性能报告应记录 CPU 型号、内核版本、编译器
  ↳ 版本、是否能使用 PMU、NUMA 和线程亲和能力。若运行环境缺少 perf、NUMA、线
  ↳ 程亲和性或硬件性能计数器权限，报告必须把相关结论降级为“未验证”，不能把源
  ↳ 码路径存在当作实测性能证据。后续 GPU 后端会作为独立 backend 接入，不改变
  ↳ 当前切片的语义合同。
6
7 当前版本支持：
8
9 - 教学文本模型格式 `LCQI_MODEL_V1`
10 - 最小 tokenizer 合同格式 `LCQI_TOKENIZER_V1`
11 - safetensors header manifest 解析：metadata、tensor name、dtype、shape、data
  ↳ offsets、offset 边界和 dtype/shape 字节数校验
12 - 两层 MLP 教学切片
13 - int8 权重加 per-layer scale
14 - float 输入和激活
15 - 量化线性层、ReLU、分类 argmax
16 - int8 linear scalar baseline、packed layout、AVX2 路径和 shape sweep benchmark
17 - GPT-2 F32 dot kernel: cached generation 热路径使用可替换 dot 边界，x86-64/GNU
  ↳ /Clang 构建接入 AVX2/FMA 实现，其他平台保留标量 fallback
18 - tiny reference decoder: RMSNorm、float linear、RoPE、GQA KV cache、decode
  ↳ attention、SwiGLU、lm head
19 - GPT-2 reference path: byte-level BPE tokenizer、`config.json`、F32/F16/BF16
  ↳ safetensors 张量读取、HuggingFace GPT-2 name mapping、LayerNorm、绝对位置
  ↳ 嵌入、packed QKV、causal attention、GELU、tied lm head、full-prefix
  ↳ greedy generation、KV cache greedy generation 和分阶段 benchmark
20 - reference trace CSV: 从 prompt/tokenizer 合同到 logits/sampler/token，逐

```

```

    ↪ checkpoint 输出 shape、stride、dtype、layout、checksum、max_abs、values ↪
    ↪ 摘要
21 - 7B ledger CSV: 输出典型 7B shape 的 FLOPs、KV 容量、权重/KV 字节和 bandwidth-↪
    ↪ only tokens/s 上限
22 - serving SLO probe CSV: 输出 admission、batch state、KV 预算、queue wait、TTFT↪
    ↪ 、TPOT、取消回收和拒绝率
23 - 自研 GPT-2 冒烟: 加载 `openai-community/gpt2` 的 `config.json`、`vocab.json`↪
    ↪ 、`merges.txt`、`model.safetensors`、在 LCQI C++ reference path 上生成 1 ↪
    ↪ 个 token, 并把文件 hash、prompt ids、generated ids 和文本输出写入报告
24 - 自研 GPT-2 benchmark 对比: 同一个 GPT-2 F32 safetensors 模型分别跑 full-↪
    ↪ prefix 和 cached-KV, 并在可用时附带 llama.cpp/SmolLM2 Q4_K_M 作为成熟引擎↪
    ↪ 参考
25 - 自研 GPT-2 优化 A/B: 从指定 baseline commit 建临时 worktree, 当前实现和 ↪
    ↪ baseline 都通过 `books/cpu-volume-3-practice` 的 Release CMake 入口构建, ↪
    ↪ 再用同一个 GPT-2 F32 模型多轮测量 full-prefix 与 cached-KV
26 - 真实 small 开源模型冒烟: 通过 llama.cpp 加载 SmolLM2-135M-Instruct 的 GGUF 量↪
    ↪ 化文件, 在本地 CPU 上生成一段 assistant 文本, 并把模型 hash、命令、输出和↪
    ↪ 性能行写入报告
27 - 自研 SmolLM2/GGUF reference decode: 读取同一个 GGUF 内的 tokenizer tokens/↪
    ↪ merges 和 `F32/Q8_0/Q5_0/Q6_K/Q4_K` 混合权重, 加载 30 层 SmolLM2, 执行 ↪
    ↪ RMSNorm、normal RoPE、GQA KV cache、SwiGLU 和 tied lm head; 默认把 shape ↪
    ↪ 兼容的 `Q4_K` linear tensor 留在 GGUF block bytes 中直接执行, 其余格式走 ↪
    ↪ f32 fallback; `LCQI_LLAMA_GGML_DIRECT=1` 可打开 `Q5_0/Q6_K/Q8_0` 实验直算↪
    ↪ 路径用于分析
28 - CLI 推理和简单 benchmark
29 - CTest 正确性测试
30
31 后续扩展方向:
32
33 - 真实 7B 级模型 metadata、tokenizer 和量化权重导入
34 - SentencePiece tokenizer、safetensors 分片加载和更多模型方言 name mapping
35 - GPT-2 reference path 的 Transformers/PyTorch golden logits 对齐报告
36 - KV cache 分页、内存规划和可选量化
37 - CPU packed weight、SIMD、NUMA 和线程亲和优化
38 - GPU backend adapter 和 device kernel
39 - 与现有框架的 prefill/decode/内存/误差对标报告
40
41 构建:
42
43 ```bash
44 cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -↪
    ↪ DCMAKE_BUILD_TYPE=Debug
45 cmake --build books/cpu-volume-3/build/lcqi-debug
46 ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
47 ```
48
49 运行:
50
51 ```bash
52 books/cpu-volume-3/build/lcqi-debug/labs/linux_cpu_inference/lcqi_cli \
53 books/cpu-volume-3/labs/linux_cpu_inference/models/tiny_mlp_i8.txt \
54 1.0 2.0 -1.0 0.5 1000

```

```

55  ```
56
57 kernel benchmark:
58
59 ```bash
60 books/cpu-volume-3/build/lcqi-debug/labs/linux_cpu_inference/lcqi_bench 200
61 ```
62
63 输出 CSV 字段:
64
65 ```text
66 target_arch,input_size,output_size,output_block,repeat,backend,available,↵
67   ↵ average_us,max_abs_diff,checksum
68 ```
69
70 当前 benchmark 按 backend 逐行输出: `scalar`、`packed_scalar`、`avx2`。x86-64 ↵
71   ↵ 构建会编译 AVX2 路径; 不支持 AVX2/FMA 的机器会把该 backend 标为 `↵
72   ↵ available=0`, 避免把源码存在误报成当前环境实测性能。
73
74 这还不是生产级高性能 kernel。当前 benchmark 建立 scalar baseline、packed layout↵
75   ↵ 、AVX2 基线正确性和 shape sweep 口径。要达到完整 AI Infra 证据, 还必须继↵
76   ↵ 续补反汇编分析、AVX-512、perf counter、oneDNN/OpenBLAS/llama.cpp/ggml 对↵
77   ↵ 照、sanitizer、fuzz 和 CI。
78
79 safetensors manifest gate:
80
81 `lcqi_tests` 会运行时生成一个最小 safetensors fixture, 验证 header 长度、`↵
82   ↵ __metadata__`、tensor name、dtype、shape、data offsets、payload 边界和 ↵
83   ↵ dtype/shape 推导出的字节数。它还会生成 offset 超出 payload、dtype/shape ↵
84   ↵ 字节数不匹配的坏文件, 确认加载器会拒绝。这个 gate 只覆盖模型资产目录表, ↵
85   ↵ 不等于已经支持完整 LLaMA/Qwen 权重映射、分片合并或 mmap 执行; 后续真实模↵
86   ↵ 型加载必须在这个 manifest 上继续接 name mapping、shape verifier、quant ↵
87   ↵ descriptor 和 first token trace。
88
89 reference decoder trace:
90
91 ```bash
92 books/cpu-volume-3/build/lcqi-debug/labs/linux_cpu_inference/lcqi_trace
93 ```
94
95 输出 CSV 字段:
96
97 ```text
98 name,token_position,layer_id,shape,stride,dtype,layout,checksum,max_abs,values
99 ```
100
101 这条 trace 固定 prompt fixture `tok1 tok2 tok3`, tokenizer 输出完整 `tokens↵
102   ↵ =[0,1,2,3,4]`, 其中 `0/4` 是 BOS/EOS; decoder trace 会剥离 BOS/EOS 后进入↵
103   ↵ tiny decoder。这样报告能同时固定 tokenizer 合同和 decoder 输入合同, 避免↵
104   ↵ 把二者混成一个不可解释的 token 数组。随后 trace 把 embedding、RMSNorm、Q/↵
105   ↵ K/V、RoPE、KV cache slot、attention、SwiGLU、layer output、final norm、↵
106   ↵ logits、argmax sampler 和 generated token 串成可回归检查的硬链路。`↵

```

```

    ↪ lcqi_tests` 会检查 tokenizer contract hash、关键 checkpoint 的 shape、↪
    ↪ stride、dtype、layout 和 logits golden, CTest 也会直接运行 `lcqi_trace`。
90
91 当前 tokenizer fixture 的关键输出:
92
93 ```text
94 tokenizer,0,-1,5,1,i32,contiguous,10,4,0;1;2;3;4
95 tokenizer_contract,0,-1,scalar,scalar,u64,fnv1a,13452902845388333734,0,tiny-↪
    ↪ chat-template-v1
96 ```
97
98 GPT-2 自研 reference smoke:
99
100 ```bash
101 books/cpu-volume-3/build/lcqi-debug/labs/linux_cpu_inference/lcqi_gpt2
102 make cpu3-gpt2-smoke
103 make cpu3-gpt2-benchmark-compare
104 make cpu3-gpt2-hotspot-profile
105 python3 books/cpu-volume-3/tools/run_gpt2_optimization_ab.py --rounds 3
106 python3 books/cpu-volume-3/tools/run_gpt2_hotspot_profile.py --token-counts ↪
    ↪ 4,16 --rounds 3
107 ```
108
109 第一条命令不需要下载模型，直接运行仓库内 tiny GPT-2 fixture，输出固定 prompt ↪
    ↪ ids、logits、predicted token 和 greedy generated ids。`lcqi_tests` 会覆盖↪
    ↪ tiny GPT-2 forward/generation、byte-level BPE 编码解码、F32/F16/BF16 ↪
    ↪ safetensors 读取、HuggingFace 风格 tiny GPT-2 checkpoint 加载和坏 shape ↪
    ↪ 拒绝。
110
111 第二条命令显式依赖网络，只下载 `openai-community/gpt2` 的 `config.json`、`vocab↪
    ↪ .json`、`merges.txt`、`model.safetensors` 到 `~/cache/lcqi-gpt2-smoke/↪
    ↪ openai-community--gpt2`，模型文件不进入 Git。脚本会校验四个文件的 bytes ↪
    ↪ 和 SHA256，构建 `lcqi_gpt2`，运行：
112
113 ```text
114 Hello, my name is
115 ```
116
117 当前验证报告写到：
118
119 ```text
120 books/cpu-volume-3/results/lcqi-gpt2-smoke.txt
121 ```
122
123 最近一次本机报告的关键行是：
124
125 ```text
126 prompt_ids 15496 11 616 1438 318
127 generated_ids 15496 11 616 1438 318 1757
128 generated_text Hello, my name is John
129 validation=PASS
130 ```

```

```

131
132 这说明 LCQI 自研 C++ reference path 已经能跑标准 GPT-2 safetensors 的 tokenizer↵
    ↵ 、权重导入、forward 和 greedy generation。默认真实模型路径现在使用 `↵
    ↵ cached_kv`，也可以用 `--engine full` 复现旧的 full-prefix 重算路径：
133
134 ```bash
135 books/cpu-volume-3/build/lcqi-release/labs/linux_cpu_inference/lcqi_gpt2 \
136 --benchmark --engine cached --threads 0 \
137 ~/.cache/lcqi-gpt2-smoke/openai-community--gpt2 \
138 "Hello, my name is" 4
139 ```
140
141 `--threads 0` 表示自动选择 cached decoder worker 数；`--threads 1` 强制串行；↵
    ↵ `--threads N` 固定 worker 数。当前 auto 上限是 16，避免在短 decode 上把线↵
    ↵ 程调度开销放大到超过收益。
142
143 `make cpu3-gpt2-benchmark-compare` 会生成：
144
145 ```text
146 books/cpu-volume-3/results/lcqi-gpt2-benchmark-compare.txt
147 ```
148
149 `run_gpt2_optimization_ab.py` 会生成：
150
151 ```text
152 books/cpu-volume-3/results/lcqi-gpt2-optimization-ab.txt
153 ```
154
155 这个 A/B 报告专门回答“当前 LCQI 代码相对指定 baseline 改快了吗”。脚本默认 ↵
    ↵ baseline 是 `4feb3f`，会临时创建 baseline worktree，分别构建 baseline 和↵
    ↵ 当前工作区的 Release `lcqi_gpt2`，再多轮运行同一条 prompt。上一轮工作区复↵
    ↵ 用优化的 2 轮 smoke 摘要保存在 `books/cpu-volume-3/reports/lcqi-gpt2-↵
    ↵ optimization-ab-summary.md`；本轮线程优化的正式 raw 报告保存在 `books/cpu↵
    ↵ -volume-3/reports/lcqi-gpt2-threaded-manual-ab-caw.txt`，汇总保存在 `↵
    ↵ books/cpu-volume-3/reports/lcqi-gpt2-threaded-optimization-summary.md`。↵
    ↵ 以 `73f40c7` 作为 baseline，在 `caw` 上 5 轮中位数显示：4 token cached-KV↵
    ↵ 生成阶段从 `395.397 ms` 降到 `76.4141 ms`，16 token 从 `989.060 ms` 降到↵
    ↵ `185.479 ms`，生成吞吐分别提升到 `52.3464 token/s` 和 `86.2633 token/s`↵
    ↵ 。随后的 F32 dot kernel 优化以 `cb4d89f` 为 baseline，raw 报告保存在 `↵
    ↵ books/cpu-volume-3/reports/lcqi-gpt2-f32-dot-ab-caw.txt`，汇总保存在 `↵
    ↵ books/cpu-volume-3/reports/lcqi-gpt2-f32-dot-optimization-summary.md`。在↵
    ↵ 同一台 `caw`、同一 GPT-2 F32 模型、同样 `--threads 0` 自动 16 workers ↵
    ↵ 下，4 token 生成阶段中位数从 `74.4247 ms` 到 `62.3628 ms`，16 token 从 ↵
    ↵ `181.257 ms` 到 `157.230 ms`；decode token/s 分别提升到 `130.073` 和 ↵
    ↵ `127.405`。再往后，row-range kernel 和去 `std::function` 调度开销的 raw ↵
    ↵ 报告保存在 `books/cpu-volume-3/reports/lcqi-gpt2-f32-rowrange-ab-caw.txt`↵
    ↵ ，汇总保存在 `books/cpu-volume-3/reports/lcqi-gpt2-f32-rowrange-↵
    ↵ optimization-summary.md`。这一轮相对 `ab72ccb` 的端到端提升很小，16 token↵
    ↵ 生成中位数约 `153.354 ms` 到 `152.574 ms`；但直接比较 row-range 与 4-row↵
    ↵ AVX2 row-range 时，4 token 从 `62.0243 ms` 到 `60.3965 ms`，16 token 从 ↵
    ↵ `154.087 ms` 到 `151.222 ms`。本轮 prefill-skip 优化的 raw 报告保存在 `↵
    ↵ books/cpu-volume-3/reports/lcqi-gpt2-prefill-skip-ab-caw.txt`，汇总保存在↵

```



↪ `books/cpu-volume-3/reports/lcqi-gpt2-prefill-skip-optimization-summary.md`。它跳过 prompt 前缀 token 上不会被使用的 `lm\_head` 预测：4 token 生成  
 ↪ 中位数从 `60.5154 ms` 到 `52.1949 ms`，16 token 从 `153.574 ms` 到  
 ↪ `147.424 ms`，生成文本和 token id 全部一致。端到端 GPT-2 数字受调度、温度  
 ↪ 和共享机器状态影响很大，因此正式结论必须看多轮中位数、min/max 和生成文本  
 ↪ 一致性，不能只挑一次最快结果。

156

157 `run\_gpt2\_hotspot\_profile.py` 专门回答“现在热点到底在哪”。它通过 `--profile`  
 ↪ hotspots` 打开 cached decode 内部计时，字段包括 `hotspot\_lm\_head\_pct`、`  
 ↪ hotspot\_mlp\_fc\_pct`、`hotspot\_mlp\_projection\_pct`、`  
 ↪ hotspot\_qkv\_projection\_pct` 和 `hotspot\_attention\_pct`。`caw` 上的 3 轮  
 ↪ Release profile 摘要保存在 `books/cpu-volume-3/reports/lcqi-gpt2-hotspot-  
 ↪ profile-summary.md`，原始报告保存在 `books/cpu-volume-3/reports/lcqi-gpt2-  
 ↪ -hotspot-profile-caw.txt`。关键结论是：4 token 与 16 token 两个口径下，`  
 ↪ lm\_head` 中位数约 `30.2%`，MLP 两个线性投影合计约 `46%`，QKV 投影约 `17%`  
 ↪ ，cached attention 本体只有 `0.06%` 到 `0.11%`。所以当前短上下文 GPT-2 的  
 ↪ 慢点不是 KV attention，而是标量 F32 matvec 和 `50257 x 768` vocab 扫描。

158

159 当前优化点集中在 cached-KV generation：`Gpt2CachedGreedyDecoder` 把模型级 shape  
 ↪ validation、KV cache、临时 workspace 和 worker pool 的生命周期提升到整段  
 ↪ 生成；`Gpt2ForwardWorkspace` 复用 hidden、normed、Q/K/V、packed QKV、  
 ↪ attention、MLP、scores 和 logits 缓冲区；生成路径只需要 greedy token 时走  
 ↪ logits-free argmax，避免把完整 logits vector 作为 API 结果反复分配；  
 ↪ prompt prefill 前缀通过 `advance\_without\_prediction()` 只更新 KV cache，  
 ↪ 不跑 final norm 和 `lm\_head`，因为这些预测结果不会被用作输出；QKV、  
 ↪ attention output projection、MLP 两个 projection 和 tied `lm\_head` 都在  
 ↪ cached session 内按输出行并行；每个线程 chunk 再通过 `  
 ↪ linear\_f32\_rows\_unchecked` 或 `max\_dot\_f32\_rows\_unchecked` 进入 F32 row-  
 ↪ range kernel，x86-64 可用时使用 AVX2/FMA，其他平台走标量 fallback。KV  
 ↪ cache 的 checked public API 仍保留，内部 cached attention 在一次边界校验  
 ↪ 后使用私有 unchecked span 扫描热路径。它不改变 GPT-2 数学语义，只改变会话  
 ↪ 生命周期、任务分片、无用工作剔除和热循环调度。

160

161 `make cpu3-gpt2-benchmark-compare` 仍用于和外部成熟引擎放在同一报告里看工程差  
 ↪ 距。同机 llama.cpp/SmolLM2 Q4\_K\_M 是成熟引擎参照，不是同模型质量排名：  
 ↪ LCQI 跑的是 GPT-2 F32 reference path，llama.cpp 跑的是 GGUF 量化 small  
 ↪ instruct 模型。它揭示的是工程差距：LCQI 已经消除了“每步重算前缀”的主要算  
 ↪ 法错误，补上 cached F32 row parallelism，并把热路径接到 AVX2/FMA row-  
 ↪ range kernel；但还缺 packed/quantized weight、GGUF 量化 block、mmap/lazy  
 ↪ load、采样器、分页 KV cache、NUMA/亲和和成熟调度。

162

163 它仍然不是生产级推理引擎：当前 GPT-2 路径还没有把 logits 和 Transformers/  
 ↪ PyTorch 逐数值对齐成外部 golden 报告，也没有 GGUF 自研导入、分页 KV cache  
 ↪ 、量化权重执行和高性能 SIMD/packed kernel。

164

165 7B ledger:

166

167 ```bash

168 books/cpu-volume-3/build/lcqi-debug/labs/linux\_cpu\_inference/lcqi\_ledger \  
 169 > books/cpu-volume-3/results/lcqi-sevenb-ledger-2026-06-24.csv  
 170 ```

171

```

172 输出 CSV 字段：
173
174 ```text
175 section,item,value,unit,notes
176 ```
177
178 这份账本固定 `L=32,H=4096,n_q=32,n_kv=8,head_dim=128,context=4096`，报告 decode↵
    ↵ FLOPs、KV 容量、fp16/int8/int4 每 token 字节和不同有效带宽下的 tokens/s ↵
    ↵ 上限。CTest 会检查 `int4_at_100_gbps=23.602 tokens/s` 这一行，防止账本公↵
    ↵ 式漂移。
179
180 serving SLO probe:
181
182 ```bash
183 books/cpu-volume-3/build/lcqi-debug/labs/linux_cpu_inference/lcqi_serving_probe↵
    ↵ \
184 > books/cpu-volume-3/results/lcqi-serving-slo-probe-2026-06-24.csv
185 ```
186
187 输出 CSV 字段：
188
189 ```text
190 row_type,request_id,accepted,rejection_reason,prompt_tokens,reserved_tokens,↵
    ↵ kv_reserved_mib,queue_wait_ms,prefill_ms,ttft_ms,decode_step_p95_ms,↵
    ↵ tpot_ms,completion_tokens,finish_reason,cancel_reclaim_ms,metric_name,↵
    ↵ metric_value
191 ```
192
193 probe 固定四个请求：短请求、RAG 请求、取消请求和超长请求。它演示 admission 先按↵
    ↵ `prompt_tokens + max_new_tokens` 预留 KV，取消请求释放预算，超长请求返回↵
    ↵ `memory_budget_exceeded`，并输出 `ttft_p95_ms`、`queue_wait_p95_ms`、↵
    ↵ `rejection_rate` 等服务化指标。
194
195 真实 small 开源模型 smoke:
196
197 ```bash
198 make cpu3-smolllm2-smoke
199 ```
200
201 这条命令显式依赖网络和外部构建，因此不放进默认 CTest。脚本会把 llama.cpp 和模型↵
    ↵ 文件缓存到 `~/.cache/lcqi-smolllm2-small-smoke`，模型不进入 Git。固定模型↵
    ↵ 是 `HuggingFaceTB/SmolLM2-135M-Instruct` 的 GGUF 量化版本，来源仓库为 `↵
    ↵ bartowski/SmolLM2-135M-Instruct-GGUF`，文件为 `SmolLM2-135M-Instruct-↵
    ↵ Q4_K_M.gguf`。脚本会校验文件大小 `105454432` 字节和 SHA256:
202
203 ```text
204 2e8040ceae7815abe0dcb3540b9995eaa1fa0d2ca9e797d0a635ae4433c68c2d
205 ```
206
207 脚本构建固定 commit 的 `llama-simple-chat`，用 `-ngl 0` 强制 CPU 路径，输入短 ↵
    ↵ prompt，检查 assistant response 非空，并生成：
208

```



```

209 ```text
210 books/cpu-volume-3/results/lcqi-smolllm2-small-model-smoke.txt
211 ```
212
213 这份报告只能证明 llama.cpp 成熟外部引擎上的真实 GGUF small 模型已经完成加载、↵
    ↵ tokenizer/chat template、prefill、decode 和文本输出。它不能证明 LCQI 自研↵
    ↵ C++ 引擎已经达到 llama.cpp 的性能，也不能替代 tiny fixture 的逐层 golden↵
    ↵ trace。
214
215 LCQI 自研 SmolLM2/GGUF reference decode:
216
217 ```bash
218 books/cpu-volume-3/build/lcqi-release/labs/linux_cpu_inference/lcqi_llama_gguf ↵
    ↵ \
219 ~/.cache/lcqi-smolllm2-small-smoke/models/SmolLM2-135M-Instruct-Q4_K_M.gguf \
220 --prompt "Hello" --max-new 2 --benchmark --decode-text
221 ```
222
223 caw 上的最新报告写到:
224
225 ```text
226 books/cpu-volume-3/results/lcqi-smolllm2-gguf-reference-decode-caw.txt
227 ```
228
229 关键行是:
230
231 ```text
232 weight_execution gguf_mixed_q4_k_direct
233 generated_text ... <|im_start|>assistant
234 Hello!
235 benchmark_prompt_tokens 31
236 benchmark_generated_tokens 2
237 benchmark_decode_tokens_per_second 64.8586
238 benchmark_quantized_weight_bytes 103668480
239 benchmark_f32_weight_bytes 481436928
240 benchmark_direct_quantized_weight_bytes 7962624
241 benchmark_fallback_dequantized_weight_bytes 481436928
242 benchmark_q4_k_direct_tensors 16
243 benchmark_ggml_direct_tensors 0
244 benchmark_f32_fallback_tensors 256
245 benchmark_hotspot_w_down_ms 63.5371
246 benchmark_hotspot_q4_k_direct_calls 512
247 benchmark_hotspot_ggml_direct_calls 0
248 benchmark_hotspot_f32_fallback_calls 6208
249 ```
250
251 这条路径证明 LCQI 自研代码已经能读取同一个 SmolLM2 GGUF、解析 tokenizer arrays↵
    ↵ 、处理混合量化权重并端到端生成文本。默认执行路径不再只是“加载时全部解成 ↵
    ↵ f32”: shape 兼容的 `Q4_K` linear tensor 会保留 GGUF block bytes，并在真实↵
    ↵ decoder 里通过 `matvec_q4_k_q8` 直接执行；`Q5_0/Q6_K/Q8_0/F32` tensor 仍↵
    ↵ 然 materialize 成 f32 fallback。当前只有约 `7.68%` 的加载权重字节进入默认↵
    ↵ direct Q4_K 路径，prefill 仍逐 token 执行；与 llama.cpp 的差距主要来自更↵

```

↪ 完整的低比特权重覆盖、prompt batching、packed kernel、线程调度、prefetch  
 ↪ 、KV cache 和整图执行。

252

253 实验路径 `LCQI\_LLAMA\_GGML\_DIRECT=1` 会尝试让 shape 兼容的 `Q5\_0/Q6\_K/Q8\_0` ↪  
 ↪ linear tensor 也保留 GGUF block bytes，并通过 `matvec\_ggml\_quantized\_q8\_0` ↪  
 ↪ 执行。caw 同输入报告显示它把 direct 量化字节覆盖率从 `0.076809` 提高到 ↪  
 ↪ `0.708479`，但 prefill 中位数从默认 `370.847 ms` 退化到 `1211.710 ms`，↪  
 ↪ decode step 从 `15.4182 ms` 退化到 `42.4420 ms`。原因是这些 GGML direct ↪  
 ↪ kernel 仍按 block/row 标量组织，热点 `benchmark\_hotspot\_ggml\_direct\_ms` ↪  
 ↪ 达到 `1207.13 ms`，慢于当前 f32 AVX2 fallback。因此它保留为实验路径，不默↪  
 ↪ 认启用。

254

255 LCQI 与 llama.cpp 同输入对比：

256

257 ```bash

258 python3 books/cpu-volume-3/tools/run\_smollm2\_same\_input\_compare.py \

259 --cache-dir ~/.cache/lcqi-smollm2-same-input-compare \

260 --model-path ~/.cache/lcqi-smollm2-small-smoke/models/SmollM2-135M-Instruct-↪  
 ↪ Q4\_K\_M.gguf \

261 --build-dir books/cpu-volume-3/build/lcqi-release \

262 --rounds 5 --max-new 2

263 ```

264

265 这个脚本会构建固定 commit 的 llama.cpp，运行 `llama-tokenize`、`llama-simple` ↪  
 ↪ 和 `llama-bench`，并确认 LCQI 和 llama.cpp 看到的 prompt token ids 完全↪  
 ↪ 致。caw 上的同输入报告保存在：

266

267 ```text

268 books/cpu-volume-3/reports/lcqi-smollm2-threaded-q4-compare-caw.txt

269 ```

270

271 关键结论：同一个模型、同一个展开 chat prompt、同样 31 个 token id、同样 `max-↪  
 ↪ new=2` 下，当前默认 LCQI 使用 `8` 个 worker，对大行数 f32 fallback 和 ↪  
 ↪ Q4\_K direct row-range 都做输出行切分。默认 Q4\_K direct prefill 中位数从串↪  
 ↪ 行 `363.759 ms` 降到 `161.436 ms`，decode step 从串行 `15.1652 ms` 降到 ↪  
 ↪ `7.26678 ms`；f32 fallback hotspot 从 `338.173 ms` 降到 `146.731 ms`。相↪  
 ↪ 对同输入 llama.cpp，LCQI prefill 仍慢 `7.381619x`，decode step 仍慢 ↪  
 ↪ `2.471694x`。同一二进制里关闭 `LCQI\_LLAMA\_Q4\_DIRECT=0` 后，LCQI prefill ↪  
 ↪ 为 `175.634 ms`，decode step 为 `7.23524 ms`，`w\_down` 热点为 `36.7329 ms`↪  
 ↪ ；默认 Q4\_K direct 后分别为 `161.436 ms`、`7.26678 ms`、`22.3253 ms`。所↪  
 ↪ 以这轮默认优化的 A/B 证据是：线程池带来 prefill `2.253271x`、decode step ↪  
 ↪ `2.086922x`、f32 hotspot `2.304714x`；Q4\_K direct 在并行后带来 prefill ↪  
 ↪ `1.087948x`、`w\_down` `1.645349x`，decode 中位数 `0.995660x` 基本持平，并↪  
 ↪ 减少 `56,623,104` 字节 f32 materialized 权重。实验 GGML direct 仍只保留为↪  
 ↪ opt-in：之前的 caw 负结果已经证明，单纯提高 `Q5\_0/Q6\_K/Q8\_0` 覆盖率而没↪  
 ↪ 有 packed multi-row/SIMD/threaded kernel，会让端到端倒退。

tiny MLP 模型文件是最短闭环的资产。它决定输入维度、hidden 维度、输出维度、权重量化值、scale 和 bias；测试中的 logits golden 依赖这份文件。

Listing 1.9 models/tiny\_mlp\_i8.txt

1 LCQI\_MODEL\_V1

```

2 input_size 4
3 hidden_size 3
4 output_size 2
5 hidden_scale 0.1
6 hidden_weights
7 5 -3 2 1
8 -4 6 1 -2
9 3 2 -5 4
10 hidden_bias
11 0.1 -0.2 0.0
12 output_scale 0.05
13 output_weights
14 4 -6 2
15 -3 5 8
16 output_bias
17 -0.5 0.4

```

tiny tokenizer fixture 固定 BOS、EOS、UNK、词表和模板 hash。tokenizer 合同如果漂移，后面的 decoder trace 和测试都会被污染。

**Listing 1.10** models/tiny\_tokenizer.txt

```

1 LCQI_TOKENIZER_V1
2 bos_id 0
3 eos_id 4
4 unk_id -1
5 chat_template_hash tiny-chat-template-v1
6 vocab_size 5
7 vocab
8 0 <bos>
9 1 tok1
10 2 tok2
11 3 tok3
12 4 <eos>

```

## 为什么要先讲调用链

实践与代码卷如果只把 **\*.cpp** 和 **\*.hpp** 全部贴出来，读者仍然不知道从哪里下手。LCQI 的代码同时包含 tiny MLP、int8 kernel、tokenizer、safetensors、reference decoder、GPT-2、benchmark、trace、ledger、serving probe 和外部模型脚本；这些文件不是平铺关系，而是几条调用链。先讲调用链，再给完整源码，读者才能把每个文件放到正确位置。

### 核心思想

读 LCQI 的顺序不是“哪个文件长先读哪个”，而是“入口、合同、资产、执行、证据”。入口说明目标怎样构建，合同说明类型怎样连接，资产说明模型从哪里来，执行说明 token 或张量怎样流动，证据说明测试和报告守住了哪些边界。

## 一、全工程入口：哪些文件决定能不能复现

| 文件 | 负责什么 | 读源码时要确认什么 |
|----|------|-----------|
|----|------|-----------|

|            |                                                                              |                                                                                                   |
|------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 第三册README  | 第三册 AI 计算工程路线、当前 LCQI 能力、GPT-2 smoke 和结果报告位置。                                | 能力声明必须能在 <code>labs/linux_cpu_inference</code> 、 <code>tools</code> 和 <code>results</code> 中找到证据。 |
| 第三册CMake   | 第三册根 CMake，负责把 <code>linux_cpu_inference</code> 接入。                          | 根 CMake 应保持薄层；真正 target 定义必须在 lab 内部，避免入口文件承担实现细节。                                                |
| 本卷CMake    | 本卷复用同一份 LCQI lab。                                                            | 本卷不复制源码，只把同一工程接入自己的测试构建，防止两份代码漂移。                                                                 |
| 本卷Makefile | 本卷的书籍构建入口。                                                                   | 它构建 PDF/EPUB，也提供 <code>test</code> 目标去构建第三册 LCQI；实践与代码卷不能只排版不验证。                                  |
| LCQICMake  | <code>lcqi</code> 库、CLI、benchmark、trace、GPT-2、ledger、serving probe、CTest 目标。 | 读这里能看到哪些文件进入库，哪些是可执行程序，哪些依赖可选外部库。                                                                 |
| LCQIREADME | 本地构建、tiny 推理、benchmark、trace、GPT-2、SmolLM2 smoke 的命令。                        | 先按 README 跑命令，再读源码；否则不知道某个函数是否真的能从命令行触达。                                                          |

上表里的短标签对应下面这些真实路径：

- `books/cpu-volume-3/README.md`
- `books/cpu-volume-3/CMakeLists.txt`
- `books/cpu-volume-3-practice/CMakeLists.txt`
- `books/cpu-volume-3-practice/Makefile`
- `books/cpu-volume-3/labs/linux_cpu_inference/CMakeLists.txt`
- `books/cpu-volume-3/labs/linux_cpu_inference/README.md`

这些入口文件回答“代码在哪里”和“怎样运行”。读者进入本卷代码走读章时，先不要直接跳到 `gpt2_reference.cpp`。先确认 `lcqi` 静态库包含 `model.cpp`、`inference.cpp`、`int8_kernels.cpp`、`f32_kernels.cpp`、`gguf.cpp`、`ggml_tensors.cpp`、`q4_k.cpp`、`llama_gguf.cpp`、`reference_decoder.cpp`、`tokenizer.cpp`、`safetensors.cpp` 和 `gpt2_reference.cpp`；再确认 `lcqi_cli`、`lcqi_bench`、`lcqi_gguf_q4_bench`、`lcqi_llama_gguf`、`lcqi_trace`、`lcqi_gpt2`、`lcqi_ledger`、`lcqi_serving_probe` 和 `lcqi_tests` 分别从哪条路径进入。

## 二、Tiny MLP 调用链：最短闭环怎样跑通

Tiny MLP 是最小推理闭环，用来证明“模型资产、shape 检查、int8 linear、激活函数、logits、argmax、CLI 输出”可以完整连起来。它不是大模型能力本身，但它是后面所有复杂路径的可手算底座。

| 文件                                  | 关键类型或函数                                                                                   | 这段代码的意思                                                   |
|-------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <code>models/tiny_mlp_i8.txt</code> | 输入维度、hidden 维度、输出维度、int8 权重、scale、bias。                                                   | 这是模型资产。测试中的 <code>golden logits</code> 来自这里，资产改动必须带着测试改动。 |
| <code>include/lcqi/model.hpp</code> | <code>QuantizedLinearLayer</code> 、 <code>TinyMlpModel</code> 、 <code>load_model</code> 。 | 定义 tiny MLP 如何在内存中表示；只负责模型结构和加载入口，不做推理。                   |
| <code>src/model.cpp</code>          | <code>read_i8_vector</code> 、 <code>validate_layer</code> 、 <code>load_model</code> 。     | 解析文本模型，拒绝坏 key、坏维度、权重数量不匹配和非法 scale，把错误挡在执行前。             |

|                                         |                                                                                                                  |                                                                                                |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <code>include/lcqi/inference.hpp</code> | <code>InferenceResult</code> 、 <code>linear_i8</code> 、 <code>relu_inplace</code> 、 <code>run_inference</code> 。 | 定义推理输出要包含 <code>hidden</code> 、 <code>logits</code> 和 <code>predicted class</code> ，方便测试定位错误层。 |
| <code>src/inference.cpp</code>          | 输入 <code>shape</code> 检查、两层 <code>int8 linear</code> 、 <code>ReLU</code> 、 <code>argmax</code> 。                 | 最短执行路径：模型加输入变成 <code>logits</code> 。这里的代码应保持清楚，不能混入 <code>CLI</code> 或文件解析。                    |
| <code>src/main.cpp</code>               | 解析命令行输入并打印结果。                                                                                                    | <code>CLI</code> 是薄层；它只接模型路径和输入，不复制 <code>run_inference</code> 逻辑。                             |

这条链路按下面顺序读：`main()` 取得模型路径和输入向量，调用 `load_model()` 把 `tiny_mlp_i8.txt` 变成 `TinyMlpModel`，再调用 `run_inference()`。在 `run_inference()` 内部，第一层 `linear_i8()` 把 `float` 输入和 `int8` 权重相乘，乘以 `scale` 后加 `bias`；`relu_inplace()` 把负 `hidden` 截断；第二层 `linear_i8()` 得到 `logits`；最后 `argmax()` 给出 `predicted class`。

读这个路径时要抓住边界条件。输入长度不等于 `input_size` 要立即拒绝；权重数量不是 `rows*columns` 要在加载时拒绝；`logits` 为空不能取 `argmax`。这些错误如果拖到 `SIMD kernel` 或 `benchmark` 里才暴露，定位成本会高很多。

### 三、Int8 Kernel 调用链：reference、packed、AVX2 和 benchmark 怎样对照

| 文件                                                                             | 关键代码                                                                                                                                              | 这段代码的意思                                                                                                                                                                                         |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>include/lcqi/int8_kernels.hpp</code>                                     | <code>PackedLinearI8</code> 、 <code>KernelBenchmarkCase</code> 、 <code>benchmark_linear_i8_case</code> 。                                          | 定义 <code>kernel</code> 对标口径： <code>shape</code> 、 <code>repeat</code> 、 <code>backend</code> 、平均耗时、 <code>checksum</code> 、 <code>max diff</code> 。                                             |
| <code>src/int8_kernels.cpp</code>                                              | <code>pack_linear_i8</code> 、 <code>linear_i8_packed</code> 、 <code>linear_i8_packed_with_workspace</code> 、 <code>deterministic fixture</code> 。 | <code>scalar</code> 和 <code>packed</code> 路径是正确性基线， <code>workspace</code> 路径减少重复分配， <code>fixture</code> 让 <code>benchmark</code> 和测试可复现。                                                      |
| <code>src/int8_kernels_avx2.cpp</code>                                         | <code>linear_i8_packed_avx2_available</code> 、 <code>linear_i8_packed_avx2</code> 。                                                               | 平台优化路径。必须先判断机器能力，再执行 <code>AVX2/FMA</code> ；不可用时报告 <code>unavailable</code> ，而不是假装测过。                                                                                                           |
| <code>include/lcqi/f32_kernels.hpp</code> 与 <code>src/f32_kernels.cpp</code>   | <code>dot_f32</code> 、 <code>linear_f32_rows_unchecked</code> 、 <code>max_dot_f32_rows_unchecked</code> 。                                         | GPT-2 F32 hot path 的 <code>kernel</code> 边界。单行 <code>dot</code> 是基础， <code>row-range linear</code> 一次写一段输出行， <code>row-range max</code> 给 <code>logits-free greedy lm_head</code> 用。            |
| <code>src/f32_kernels_avx2.cpp</code>                                          | <code>dot_f32_avx2_available</code> 、 <code>dot_f32_avx2_unchecked</code> 、4-row <code>row-range kernel</code> 。                                  | x86 平台优化路径。它在一段输出行内部复用 <code>input load</code> ，同时不改变 GPT-2 层结构、线程分片或 <code>greedy tie-breaking</code> 。                                                                                        |
| <code>include/lcqi/gguf.hpp</code> 与 <code>src/gguf.cpp</code>                 | <code>GgufManifest</code> 、 <code>GgufTensorInfo</code> 、 <code>load_gguf_manifest</code> 、 <code>read_gguf_tensor_bytes</code> 。                 | 读取 GGUF header、metadata、tensor 表、alignment 和真实 tensor payload；它只负责格式边界，不做推理数学。                                                                                                                  |
| <code>include/lcqi/ggml_tensors.hpp</code> 与 <code>src/ggml_tensors.cpp</code> | <code>dequantize_ggml_tensor</code> 、 <code>ggml_f16_to_f32</code> 。                                                                              | 把 GGUF 中的 <code>F32</code> 、 <code>Q8_0</code> 、 <code>Q5_0</code> 、 <code>Q6_K</code> 和 <code>Q4_K</code> 张量按 GGML block 布局解释出来。它是格式 oracle，不决定某个 tensor 最终走 <code>f32 fallback</code> 还是量化直算。 |



|                                                                                                                      |   |                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------|
| <code>include/lcqi/q4_k.hpp</code><br><code>src/q4_k.cpp</code>                                                      | 与 | <code>dequantize_q4_k</code> 、 <code>quantize_q8_k_input</code> 、 <code>matvec_q4_k_q8</code> 。                                       |
| <code>include/lcqi/ggml_matvec.hpp</code> 、 <code>src/ggml_matvec.cpp</code> 与 <code>src/ggml_matvec_avx2.cpp</code> |   | <code>matvec_ggml_quantized_f32</code> 、 <code>matvec_ggml_quantized_q8_0</code> 、 <code>quantize_q8_0_input</code> 。                 |
| <code>include/lcqi/llama_gguf.hpp</code> 与 <code>src/llama_gguf.cpp</code>                                           | 与 | <code>load_llama_gguf_reference_model</code> 、 <code>load_gpt2_tokenizer_from_gguf</code> 、 <code>llama_gguf_generate_greedy</code> 。 |
| <code>src/llama_cli.cpp</code>                                                                                       |   | <code>lcqi_llama_gguf</code> 。                                                                                                        |
| <code>src/q4_k_avx2.cpp</code>                                                                                       |   | <code>q4_k_q8_avx2_available</code> 、 <code>matvec_q4_k_q8_avx2_unchecked</code> 。                                                    |
| <code>src/gguf_q4_bench.cpp</code>                                                                                   |   | <code>lcqi_gguf_q4_bench</code> 。                                                                                                     |
| <code>src/bench.cpp</code>                                                                                           |   | shape 列表、repeat 参数、backend 输出。                                                                                                        |

实现 llama.cpp 同格式的 **Q4\_K** block 解码和 **Q4\_KxQ8\_K** 单算子路径；单测用手写 block 固定 nibble 顺序、scale/min 解包和误差边界。

**Q5\_0/Q6\_K/Q8\_0** direct matvec 的实验路径。它能显著提高量化字节覆盖率，但 caw 证明当前 block/row 组织端到端更慢，所以只用 **LCQI\_LLAMA\_GGML\_DIRECT=1** 显式打开。

自研 SmolLM2/GGUF decoder path。它从同一个 GGUF 读取 tokenizer tokens/merges、混合量化权重、RMSNorm、normal RoPE、GQA KV cache、SwiGLU 和 tied lm head；默认执行 **Q4\_K** direct 加 f32 fallback，实验开关可接入 **Q5\_0/Q6\_K/Q8\_0** direct。

命令行加载真实 SmolLM2 GGUF，支持 token id prompt 和文本 prompt，输出生成 token、文本、load/prefill/decode 时间、KV 字节、量化权重字节和解码后 f32 字节。

x86 AVX2 路径。它用 **maddubs** 把 32 个 unsigned nibble 与 32 个 signed int8 激活先归约成整数和，再套用 **Q4\_K** 的 scale/min；报告必须同时给 scalar 与 auto 后端，防止把机器波动写成优化收益。

从真实 SmolLM2 GGUF 选择 **Q4\_K** 矩阵，比较 f32 解码基线、**Q4\_Kxf32** 和 **Q4\_KxQ8\_K**，输出速度、误差、checksum 和字节账本。

benchmark 程序只负责测 kernel 入口。性能结论必须和 **max\_abs\_diff**、CPU、编译器、构建类型一起报告。

kernel 文件要按“可信基线到优化路径”的顺序读。先看 deterministic layer 和 input 怎么生成，保证每次运行输入相同；再看 **linear\_i8\_packed()** 怎样把 int8 权重、scale、bias 和 float 输入算成输出；再看 workspace 路径怎样复用临时缓冲；最后看 AVX2 路径怎样在支持平台上替换内层循环。Q4\_K 路径也遵守同一顺序：先读 scalar 解码，再读 AVX2 **maddubs** 子块归约，最后看 benchmark 怎样强制 scalar 和 auto 后端并列输出。测试用 packed 与 scalar 的 **max\_abs\_diff** 对照，benchmark 再比较耗时。

#### 四、Tokenizer、Safetensors 与 Reference Decoder：真实模型前的三道门

| 文件                                     | 关键类型或函数                 | 这段代码的意思                                     |
|----------------------------------------|-------------------------|---------------------------------------------|
| <code>models/tiny_tokenizer.txt</code> | BOS、EOS、UNK、词表、模板 hash。 | 固定 tokenizer 合同，避免 prompt 到 token id 的规则漂移。 |

|                                                                                          |                                                                                                                                 |                                                                        |
|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <code>include/lcqi/tokenizer.hpp</code> 与 <code>src/tokenizer.cpp</code>                 | <code>TokenizerModel</code> 、 <code>load_tokenizer</code> 、 <code>encode_prompt</code> 、 <code>tokenizer_contract_hash</code> 。 | 把 prompt 切成稳定 token id；未知 token 是否允许，必须由 UNK 合同决定。                     |
| <code>include/lcqi/safetensors.hpp</code> 与 <code>src/safetensors.cpp</code>             | <code>SafeTensorManifest</code> 、 <code>SafeTensorFile</code> 、manifest parser、dtype 转换。                                        | 解析模型字节之前先验证 header、dtype、shape、offset 和 payload 边界，防止坏资产进入数学层。         |
| <code>include/lcqi/reference_decoder.hpp</code> 与 <code>src/reference_decoder.cpp</code> | embedding、RMSNorm、RoPE、KV cache、attention、SwiGLU、trace checkpoint。                                                              | Tiny decoder-only reference path。它用 checkpoint 解释每一步张量状态，不只返回最终 token。 |
| <code>src/trace.cpp</code>                                                               | trace CSV 输出。                                                                                                                   | 把 reference decoder 的 checkpoint 打印成可复查表格，定位错误发生在哪个阶段。                 |

这三道门分别解决三个问题。Tokenizer 解决“人写的 prompt 如何变成 token id”；Safetensors 解决“磁盘里的模型字节如何变成带 shape 的张量”；Reference Decoder 解决“token 和权重如何经过 decoder-only 层生成 logits”。任何一层含糊，后面的 GPT-2 路径都会变成看似能跑、实则无法定位的黑盒。

## 五、GPT-2 调用链：从 HuggingFace 资产到生成 token

GPT-2 路径是本卷代码走读里最长的一条链，因为它要处理真实模型格式。读 `gpt2_reference.cpp` 时按模块读，不要从第一行一路滚到最后。

| 文件                                               | 关键代码                                                                                                | 这段代码的意思                                                                                                                    |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <code>include/lcqi/gpt2_reference.hpp</code>     | 配置、模型、KV cache、cached decoder、tokenizer 与 forward/generate API。                                     | 公开 GPT-2 reference path 的合同：配置、权重、KV cache、tokenizer、full-prefix、cached generation，以及 prompt prefill 中只推进 cache、不做无用预测的入口。 |
| <code>src/gpt2_reference.cpp</code>              | JSON parser、byte-level BPE、safetensors loader、LayerNorm、QKV、causal attention、GELU、lm head、KV cache。 | 正确性优先实现。它把 HuggingFace GPT-2 权重名、Conv1D 转置、tokenizer 和 forward 规则全部显式写出来。                                                  |
| <code>src/gpt2_cli.cpp</code>                    | tiny mode、真实模型目录模式、generation、benchmark 输出。                                                         | 命令行入口。无参数跑 tiny fixture；传模型目录时加载真实 GPT-2 资产；benchmark 对比 full-prefix 和 cached-KV。                                          |
| <code>tools/run_gpt2_smoke.py</code>             | 下载标准 GPT-2 资产，调用 <code>lcqi_gpt2</code> 。                                                           | 证明自研 safetensors、BPE、forward、KV cache 和 greedy generation 能跑真实开源模型。                                                        |
| <code>tools/run_gpt2_benchmark_compare.py</code> | 运行 LCQI full-prefix、cached-KV，并可附带成熟引擎参考。                                                           | 用同一模型口径比较算法差距和 kernel 差距，不把不同模型的速度混成一个结论。                                                                                  |

GPT-2 代码按五段读。第一段是配置和权重加载：`load_gpt2_config()` 读 `config.json`，`load_gpt2_reference_model()` 从 safetensors 中取 embedding、每层 LayerNorm、QKV、MLP 和 final norm。HuggingFace GPT-2 的 Conv1D 权重形状和普通线性层方向不同，所以代码显式调用转置函数，不能跳过。

第二段是 tokenizer: `load_gpt2_tokenizer()` 读 `vocab.json` 和 `merges.txt`, `gpt2_encode()` 执行 byte-level BPE, `gpt2_decode()` 把 id 还原为文本。这里的重点是字节到 Unicode 映射、BPE merge rank 和 unknown token 边界。

第三段是 full-prefix forward: `run_gpt2_forward()` 每生成一个新 token 都用完整 token 前缀重新跑所有层。它慢, 但容易验证。流程是 token embedding 加 position embedding, 逐层执行 LayerNorm、packed QKV、causal attention、残差、MLP、残差, 最后 final LayerNorm 和 tied lm head 得到 logits。

第四段是 cached-KV forward: `Gpt2KvCache` 保存每层每个 position 的 key/value, `run_gpt2_forward_cached()` 每次只处理最新 token, 并让 attention 读取过去 cache。这里要特别看 `validate_written()`: 读取未写入 cache slot 必须报错, 否则生成错误会很难定位。

第五段是 greedy generation: `gpt2_generate_greedy()` 用 full-prefix 路径, `gpt2_generate_greedy_cached()` 用 KV cache 路径。两个函数都要检查 prompt 非空、`max_new_tokens` 非负、不会超过 `max_positions`, 并在生成 EOS 时停止。

## 六、报告、服务探针、外部对标和测试

| 文件                                            | 关键代码                                                                                          | 这段代码的意思                                                  |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <code>src/ledger.cpp</code>                   | 7B 参数量、KV cache、bandwidth-only 上限。                                                            | 输出的是工程账本, 不是实测速度; 它帮助读者估算权重和 KV 内存压力。                    |
| <code>src/serving_probe.cpp</code>            | 短请求、RAG 请求、取消请求、超长请求, TTFT/TPOT 字段。                                                           | 把服务层边界变成字段: 哪些请求接受, 哪些拒绝, 取消后资源是否回收。                     |
| <code>src/onednn_bench.cpp</code>             | oneDNN matmul benchmark。                                                                      | 可选外部库对标入口。没有 oneDNN 环境时不能声称已验证。                          |
| <code>tools/run_smollm2_small_smoke.py</code> | 下载 SmolLM2-135M-Instruct GGUF, 构建固定 commit 的 llama.cpp smoke。                                 | 这是成熟外部引擎上的真实 small 模型 smoke, 用来建立外部参照, 不代表 LCQI 已具备同等能力。 |
| <code>tools/run_lcqi_benchmark.sh</code>      | 收集 LCQI 本地 benchmark。                                                                         | 在固定机器上重复运行同一组 shape, 形成可复查 CSV 或文本报告。                    |
| <code>tests/lcqi_tests.cpp</code>             | tiny MLP、tokenizer、safetensors、GPT-2 tiny、HF-style checkpoint、bad shape、decoder、trace、kernel。 | 这是本卷最重要的代码证据。新增功能如果没有进入这里, 就不能写成已经守住的能力。                 |

读测试文件时, 不要只看最后是否返回 0。它里面的 fixture 本身就是源码讲解的一部分: 测试会手写 safetensors header, 构造 F16/BF16/F32 payload, 生成 tiny GPT-2 权重, 验证 bad shape 会被拒绝, 比较 packed int8 kernel 和 scalar 路径。这些测试告诉读者“代码真正承诺了什么”。如果一个能力只出现在 README 或 CLI 输出里, 却没有进入测试或 smoke 报告, 就只能算实验入口, 不能算稳定能力。

### 源码阅读任务

读完本章后, 再去读相邻主题中的完整源码。每看到一个文件, 都把它归到五类之一: 入口、合同、资产解析、执行路径、证据。归不进去的文件, 就说明工程边界需要重新解释或重构。



## 第 2 章实践：张量布局、地址流与第一个 MatVec

本章对应原书中的第三册“张量布局、地址流与第一个矩阵向量内核”。不要把它当作概念复述，本章要把 TensorView、dtype、shape、stride、row-major、地址公式、bias、ReLU、checksum 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `linear_i8`，核心命令或测试入口是 `lcqi_tests`。

**Listing 2.1** 第 2 章实践：张量布局、地址流与第一个 MatVec 的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  <- DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_tests / linear_i8
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：手算一个 2 行 4 列权重矩阵的地址偏移。读者要先手写预期结果，再运行项目观察输出。动作是：把 `weight[out,k]` 展开成 `base+(out*K+k)*sizeof(i8)`，再对应到 `linear_i8` 的循环。如果输出里没有 `logit` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：张量不是裸数组；每个 kernel 入口都要能解释 dtype、shape、stride、scale 和输出缓冲这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

### 四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交一页地址流账本：输入向量、hidden 权重、输出 logits 各自的 shape、连续性、读写次数和错误边界。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

## 第3章实践：GEMM、packing 与 SIMD 证据

本章对应原书中的第三册“从矩阵向量乘走向 GEMM、packing 与 SIMD”。不要把它当作概念复述，本章要把 packed weight、scalar baseline、AVX2 backend、output block、shape sweep、反汇编和降级放到同一个可运行项目里检查。贯穿代码仍然是 [books/cpu-volume-3/labs/linux\\_cpu\\_inference](#)；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `output_block`，核心命令或测试入口是 `lcqi_bench`。

Listing 3.1 第3章实践：GEMM、packing 与 SIMD 证据的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  <- DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_bench / output_block
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：同一组输入同时跑 `scalar`、`packed_scalar` 和 `avx2`。读者要先手写预期结果，再运行项目观察输出。动作是：比较 CSV 中 `backend`、`available`、`average_us`、`max_abs_diff` 和 `checksum`。如果输出里没有 `max_abs_diff` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：SIMD 路径只有在结果和 baseline 对齐后才有资格谈速度；`available=0` 也是证据，不是失败这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 [books/cpu-volume-3/labs/linux\\_cpu\\_inference/README.md](#)、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

### 四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交一个 shape sweep 表，至少解释一个 SIMD 有收益、一个收益不明显、一个当前环境不可验证的场景。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

## 第 4 章实践：int8 量化、累加器与重新量化

本章对应原书中的第三册“量化系统总览”和“int8 量化、累加器与重新量化”。不要把它当作概念复述，本章要把 int8 weight、float activation、scale、accumulator、requantize、oracle tolerance 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `TinyMlpModel`，核心命令或测试入口是 `lcqi_tests`。

Listing 4.1 第 4 章实践：int8 量化、累加器与重新量化的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_tests / TinyMlpModel
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：构造一行 int8 权重 `[2, -1, 3, 0]` 和输入 `[1, 2, -1, 0.5]`。读者要先手写预期结果，再运行项目观察输出。动作是：先手算 int32/float accumulator，再和 `linear_i8` 输出比较。如果输出里没有 `scale` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：量化不是把 float 强转成 int8；必须记录 scale 作用位置、累加宽度、误差阈值和 golden 输出这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

### 四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交量化合同卡：每个 tensor 的 dtype、scale 粒度、accumulator 类型、误差门限和拒绝条件。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

## 代码走读：公共合同、Tiny MLP 与 Int8 Kernel

### 为什么先读头文件

头文件是系统对外承诺的最小合同。实现可以重写，kernel 可以替换，benchmark 可以补充平台路径，但头文件里暴露的数据结构、函数入口和返回字段决定了测试、CLI、工具脚本和章节讲解怎样调用这套工程。

#### 核心思想

读 LCQI 时不要从最长的 `gpt2_reference.cpp` 开始。先读公共头文件，明确“模型如何表示”“推理返回什么”“trace 记录什么”“KV cache 怎样寻址”“tokenizer 的坏输入怎样处理”，再进入实现。这样读到长循环和 shape 检查时，知道它们在维护哪份合同。

### Tiny MLP 与推理入口

`model.hpp` 定义 tiny MLP 的模型对象、权重、bias、scale 和加载入口。它的责任是把文本模型资产解释成 C++ 对象，不负责执行推理。

Listing 4.2 `include/lcqi/model.hpp`

```

1 #pragma once
2
3 #include <stdint>
4 #include <filesystem>
5 #include <string>
6 #include <vector>
7
8 namespace lcqi {
9
10 struct QuantizedLinearLayer {
11     std::int32_t input_size = 0;
12     std::int32_t output_size = 0;
13     float scale = 1.0F;
14     std::vector<std::int8_t> weights;
15     std::vector<float> bias;
16 };
17
18 struct TinyMlpModel {
19     std::int32_t input_size = 0;
20     std::int32_t hidden_size = 0;
21     std::int32_t output_size = 0;
22     QuantizedLinearLayer hidden;
23     QuantizedLinearLayer output;
24 };
25
26 TinyMlpModel load_model(const std::filesystem::path& path);
27
28 } // namespace lcqi

```

`inference.hpp` 定义推理结果和 `run_inference` 入口。返回值包含 hidden、logits 和 pre-

dicted class，测试可以用这些字段定位是加载错、kernel 错，还是 argmax 错。

**Listing 4.3** include/lcqi/inference.hpp

```

1 #pragma once
2
3 #include <cstdint>
4 #include <span>
5 #include <vector>
6
7 #include <lcqi/model.hpp>
8
9 namespace lcqi {
10
11 struct InferenceResult {
12     std::vector<float> logits;
13     std::int32_t predicted_class = 0;
14 };
15
16 void linear_i8(const QuantizedLinearLayer& layer,
17               std::span<const float> input,
18               std::span<float> output);
19
20 void relu_inplace(std::span<float> values);
21
22 InferenceResult run_inference(const TinyMlpModel& model,
23                               std::span<const float> input);
24
25 } // namespace lcqi

```

## Int8 Kernel、Tokenizer 与 Safetensors

int8 kernel 头文件定义 packed linear、deterministic fixture、scalar/workspace/AVX2 路径和误差比较。注意 AVX2 是运行时能力，不是源码存在就等于当前机器已验证。

**Listing 4.4** include/lcqi/int8\_kernels.hpp

```

1 #pragma once
2
3 #include <cstdint>
4 #include <span>
5 #include <vector>
6
7 #include <lcqi/model.hpp>
8
9 namespace lcqi {
10
11 struct PackedLinearI8 {
12     std::int32_t input_size = 0;
13     std::int32_t output_size = 0;
14     std::int32_t output_block_size = 0;
15     float scale = 1.0F;
16     std::vector<std::int8_t> packed_weights;

```

```

17     std::vector<float> bias;
18 };
19
20 struct KernelBenchmarkCase {
21     std::int32_t input_size = 0;
22     std::int32_t output_size = 0;
23     std::int32_t output_block_size = 0;
24     std::int32_t repeat = 0;
25 };
26
27 struct KernelBackendBenchmark {
28     const char* name = "";
29     bool available = false;
30     double average_us = 0.0;
31     float max_abs_diff = 0.0F;
32     float checksum = 0.0F;
33 };
34
35 struct KernelBenchmarkResult {
36     KernelBenchmarkCase benchmark_case;
37     std::vector<KernelBackendBenchmark> backends;
38 };
39
40 PackedLinearI8 pack_linear_i8(const QuantizedLinearLayer& layer,
41                               std::int32_t output_block_size);
42
43 void linear_i8_packed(const PackedLinearI8& layer,
44                      std::span<const float> input,
45                      std::span<float> output);
46
47 void linear_i8_packed_with_workspace(const PackedLinearI8& layer,
48                                     std::span<const float> input,
49                                     std::span<float> output,
50                                     std::span<float> accumulator_workspace);
51
52 bool linear_i8_packed_avx2_available() noexcept;
53
54 void linear_i8_packed_avx2(const PackedLinearI8& layer,
55                            std::span<const float> input,
56                            std::span<float> output);
57
58 QuantizedLinearLayer make_deterministic_i8_layer(std::int32_t input_size,
59                                                  std::int32_t output_size,
60                                                  float scale);
61
62 std::vector<float> make_deterministic_input(std::int32_t input_size);
63
64 float max_abs_diff(std::span<const float> lhs, std::span<const float> rhs);
65
66 KernelBenchmarkResult benchmark_linear_i8_case(const KernelBenchmarkCase& ←
67         ↪ benchmark_case);
68

```







```

41         std::size_t row_begin,
42         std::size_t row_end,
43         float* output) noexcept;
44
45 void linear_f32_rows_avx2_unchecked(const float* weights,
46                                     const float* input,
47                                     const float* bias,
48                                     std::size_t input_size,
49                                     std::size_t row_begin,
50                                     std::size_t row_end,
51                                     float* output) noexcept;
52
53 [[nodiscard]] F32RowMax max_dot_f32_rows_scalar_unchecked(const float* weights,
54                                                            const float* input,
55                                                            std::size_t ↵
56                    ↵ input_size,
57                                                            std::size_t row_begin↵
58                    ↵ ,
59                                                            std::size_t row_end) ↵
60                    ↵ noexcept;
61
62 [[nodiscard]] F32RowMax max_dot_f32_rows_unchecked(const float* weights,
63                                                      const float* input,
64                                                      std::size_t input_size,
65                                                      std::size_t row_begin,
66                                                      std::size_t row_end) ↵
67                                                      ↵ noexcept;
68
69 [[nodiscard]] F32RowMax max_dot_f32_rows_avx2_unchecked(const float* weights,
70                                                          const float* input,
71                                                          std::size_t input_size,
72                                                          std::size_t row_begin,
73                                                          std::size_t row_end) ↵
74                                                          ↵ noexcept;
75
76 } // namespace lcqi

```

tokenizer 头文件固定 BOS/EOS/UNK、词表和 template hash。prompt 进入模型前，必须先变成稳定 token id；未知 token 行为也必须被写进合同。

**Listing 4.6** include/lcqi/tokenizer.hpp

```

1 #pragma once
2
3 #include <cstdint>
4 #include <filesystem>
5 #include <string>
6 #include <string_view>
7 #include <vector>
8
9 namespace lcqi {
10
11 struct TokenEntry {

```

```

12     std::string text;
13     std::int32_t id = 0;
14 };
15
16 struct TokenizerModel {
17     std::int32_t bos_id = -1;
18     std::int32_t eos_id = -1;
19     std::int32_t unk_id = -1;
20     std::string chat_template_hash;
21     std::vector<TokenEntry> vocab;
22 };
23
24 [[nodiscard]] TokenizerModel load_tokenizer(const std::filesystem::path& path);
25
26 [[nodiscard]] std::vector<std::int32_t> encode_prompt(
27     const TokenizerModel& tokenizer,
28     std::string_view prompt);
29
30 [[nodiscard]] std::uint64_t tokenizer_contract_hash(const TokenizerModel& ↵
    ↵ tokenizer);
31
32 } // namespace lcqi

```

safetensors 头文件固定 dtype、shape、offset 和 payload 边界。它负责模型字节怎样被解释，不应该和数学推理逻辑混在一起。

Listing 4.7 include/lcqi/safetensors.hpp

```

1 #pragma once
2
3 #include <cstdint>
4 #include <filesystem>
5 #include <span>
6 #include <string>
7 #include <string_view>
8 #include <utility>
9 #include <vector>
10
11 namespace lcqi {
12
13 struct SafeTensorEntry {
14     std::string name;
15     std::string dtype;
16     std::vector<std::int64_t> shape;
17     std::uint64_t data_begin = 0;
18     std::uint64_t data_end = 0;
19
20     [[nodiscard]] std::uint64_t byte_size() const;
21 };
22
23 struct SafeTensorManifest {
24     std::uint64_t header_size = 0;
25     std::uint64_t data_start_offset = 0;

```

```

26     std::vector<std::pair<std::string, std::string>> metadata;
27     std::vector<SafeTensorEntry> tensors;
28
29     [[nodiscard]] const SafeTensorEntry* find_tensor(std::string_view name) ↵
    ↵ const;
30 };
31
32 struct SafeTensorFile {
33     std::filesystem::path path;
34     SafeTensorManifest manifest;
35     std::vector<std::uint8_t> bytes;
36 };
37
38 [[nodiscard]] SafeTensorManifest load_safetensors_manifest(
39     const std::filesystem::path& path);
40
41 [[nodiscard]] SafeTensorFile load_safetensors_file(const std::filesystem::path& ↵
    ↵ path);
42
43 [[nodiscard]] std::vector<float> read_safetensor_f32_tensor(
44     const SafeTensorFile& file,
45     std::string_view name);
46
47 [[nodiscard]] std::span<const std::uint8_t> read_safetensor_tensor_bytes(
48     const SafeTensorFile& file,
49     std::string_view name);
50
51 } // namespace lcqi

```

## GGUF 与 Q4\_K 合同

GGUF 头文件固定 llama.cpp 生态模型文件的读取边界：版本、metadata、alignment、tensor 名、shape、ggml type 和 tensor payload offset。它也保留 tokenizer 需要的字符串数组 metadata，例如 `tokenizer.ggml.tokens` 和 `tokenizer.ggml.merges`。它不负责模型图执行，只把“真实 GGUF 文件里有哪些 tensor、每个 tensor 在哪里、哪些 metadata 能支撑 tokenizer”变成可检查对象。

Listing 4.8 include/lcqi/gguf.hpp

```

1 #pragma once
2
3 #include <cstdint>
4 #include <filesystem>
5 #include <span>
6 #include <string>
7 #include <string_view>
8 #include <vector>
9
10 namespace lcqi {
11
12     enum class GgmlType : std::int32_t {
13         f32 = 0,

```

```

14     f16 = 1,
15     q4_0 = 2,
16     q4_1 = 3,
17     q5_0 = 6,
18     q5_1 = 7,
19     q8_0 = 8,
20     q8_1 = 9,
21     q2_k = 10,
22     q3_k = 11,
23     q4_k = 12,
24     q5_k = 13,
25     q6_k = 14,
26     q8_k = 15,
27     i8 = 24,
28     i16 = 25,
29     i32 = 26,
30     i64 = 27,
31     f64 = 28,
32     bf16 = 30,
33     q1_0 = 41,
34 };
35
36 struct GgmlTypeLayout {
37     std::int64_t block_size = 1;
38     std::uint64_t type_size = 0;
39     const char* name = "";
40     bool quantized = false;
41 };
42
43 struct GgufMetadataEntry {
44     std::string key;
45     std::string type;
46     std::string value_preview;
47     std::vector<std::string> string_values;
48 };
49
50 struct GgufTensorInfo {
51     std::string name;
52     std::vector<std::int64_t> shape;
53     GgmlType type = GgmlType::f32;
54     std::uint64_t relative_offset = 0;
55     std::uint64_t absolute_offset = 0;
56     std::uint64_t byte_size = 0;
57
58     [[nodiscard]] std::uint64_t element_count() const;
59 };
60
61 struct GgufManifest {
62     std::uint32_t version = 0;
63     std::uint64_t tensor_count = 0;
64     std::uint64_t metadata_count = 0;
65     std::uint64_t alignment = 32;

```

```

66     std::uint64_t data_offset = 0;
67     std::vector<GgufMetadataEntry> metadata;
68     std::vector<GgufTensorInfo> tensors;
69
70     [[nodiscard]] const GgufTensorInfo* find_tensor(std::string_view name) ↵
    ↵ const;
71     [[nodiscard]] const GgufMetadataEntry* find_metadata(std::string_view key) ↵
    ↵ const;
72 };
73
74 [[nodiscard]] GgmlTypeLayout ggml_type_layout(GgmlType type);
75
76 [[nodiscard]] const char* ggml_type_name(GgmlType type);
77
78 [[nodiscard]] std::uint64_t ggml_tensor_byte_size(GgmlType type,
79                                                    std::span<const std::int64_t>↵
    ↵ shape);
80
81 [[nodiscard]] GgufManifest load_gguf_manifest(const std::filesystem::path& path↵
    ↵ );
82
83 [[nodiscard]] std::vector<std::uint8_t> read_gguf_tensor_bytes(
84     const std::filesystem::path& path,
85     const GgufTensorInfo& tensor);
86
87 } // namespace lcqi

```

GGML tensor 头文件是 GGUF 与模型数学之间的格式层。SmolLM2 的 Q4\_K\_M 文件内部并不是所有矩阵都是 Q4\_K：远端 caw 扫描显示它包含 F32、Q8\_0、Q5\_0、Q6\_K 和 Q4\_K。因此完整模型加载不能只写一个 Q4\_K 解码器，必须把这些格式统一变成后续执行层能消费的张量。

**Listing 4.9** include/lcqi/ggml\_tensors.hpp

```

1  #pragma once
2
3  #include <lcqi/gguf.hpp>
4
5  #include <cstdint>
6  #include <span>
7  #include <vector>
8
9  namespace lcqi {
10
11  constexpr std::int64_t LCQI_Q8_0_BLOCK_VALUES = 32;
12  constexpr std::int64_t LCQI_Q8_0_BLOCK_BYTES = 34;
13  constexpr std::int64_t LCQI_Q5_0_BLOCK_VALUES = 32;
14  constexpr std::int64_t LCQI_Q5_0_BLOCK_BYTES = 22;
15  constexpr std::int64_t LCQI_Q6_K_BLOCK_VALUES = 256;
16  constexpr std::int64_t LCQI_Q6_K_BLOCK_BYTES = 210;
17
18  [[nodiscard]] float ggml_f16_to_f32(std::uint16_t bits);
19
20  [[nodiscard]] std::vector<float> dequantize_ggml_tensor(

```

```

21     GgmlType type,
22     std::span<const std::uint8_t> bytes,
23     std::int64_t element_count);
24
25 void dequantize_ggml_tensor_to(GgmlType type,
26                               std::span<const std::uint8_t> bytes,
27                               std::span<float> output);
28
29 } // namespace lcqi

```

Q4\_K 头文件固定本轮量化切片的数学合同。一个 Q4\_K block 解码出 256 个权重, Q8\_K input 是每 256 个激活临时量化一次。这里暴露的 `matvec_q4_k_q8` 是单算子入口, 不是完整 LLaMA-like decoder; 完整 decoder 还要接 RMSNorm、RoPE、attention、SwiGLU、KV cache 和 sampler。

**Listing 4.10** `include/lcqi/q4_k.hpp`

```

1 #pragma once
2
3 #include <array>
4 #include <cstdint>
5 #include <span>
6 #include <vector>
7
8 namespace lcqi {
9
10 constexpr std::int64_t LCQI_QK_K_BLOCK_VALUES = 256;
11 constexpr std::int64_t LCQI_Q4_K_BLOCK_BYTES = 144;
12 constexpr std::int64_t LCQI_Q8_K_BLOCK_BYTES = 292;
13 constexpr std::int64_t LCQI_Q4_K_SUBBLOCK_VALUES = 32;
14 constexpr std::int64_t LCQI_Q4_K_SUBBLOCKS = 8;
15 constexpr std::int64_t LCQI_Q8_K_BSUM_GROUPS = 16;
16
17 struct Q8KBlock {
18     float d = 0.0F;
19     std::array<std::int8_t, LCQI_QK_K_BLOCK_VALUES> qs{};
20     std::array<std::int16_t, LCQI_Q8_K_BSUM_GROUPS> bsums{};
21 };
22
23 [[nodiscard]] std::vector<float> dequantize_q4_k(std::span<const std::uint8_t> ↵
24     ↵ bytes,
25     std::int64_t element_count);
26
27 void dequantize_q4_k_to(std::span<const std::uint8_t> bytes,
28     std::span<float> output);
29
30 [[nodiscard]] std::vector<Q8KBlock> quantize_q8_k_input(std::span<const float> ↵
31     ↵ input);
32
33 void quantize_q8_k_input_to(std::span<const float> input,
34     std::span<Q8KBlock> output);
35
36 [[nodiscard]] bool q4_k_q8_avx2_available() noexcept;
37

```

```

36 [[nodiscard]] const char* q4_k_q8_active_backend() noexcept;
37
38 float dot_q4_k_f32(std::span<const std::uint8_t> q4_blocks,
39                  std::span<const float> input);
40
41 float dot_q4_k_q8(std::span<const std::uint8_t> q4_blocks,
42                  std::span<const Q8KBlock> input);
43
44 void matvec_q4_k_f32(std::span<const std::uint8_t> q4_rows,
45                    std::int64_t row_count,
46                    std::int64_t column_count,
47                    std::span<const float> input,
48                    std::span<float> output);
49
50 void matvec_q4_k_q8(std::span<const std::uint8_t> q4_rows,
51                   std::int64_t row_count,
52                   std::int64_t column_count,
53                   std::span<const Q8KBlock> input,
54                   std::span<float> output);
55
56 void matvec_q4_k_q8_rows_unchecked(std::span<const std::uint8_t> q4_rows,
57                                    std::int64_t row_count,
58                                    std::int64_t column_count,
59                                    std::span<const Q8KBlock> input,
60                                    std::span<float> output,
61                                    std::int64_t row_begin,
62                                    std::int64_t row_end) noexcept;
63
64 } // namespace lcqi

```

GGML matvec 头文件固定 `Q5_0/Q6_K/Q8_0` direct 实验路径的合同。`matvec_ggml_quantized_f32` 是 exact oracle: 它不先 materialize 整个矩阵, 而是直接从 GGUF block bytes 和 f32 input 做点积, 用来对照解码后的 f32 权重。`quantize_q8_0_input` 把 activation 每 32 个元素临时量化一次, `matvec_ggml_quantized_q8_0` 再执行低比特权重乘 `Q8_0` activation。这个头文件只承诺单算子数学和边界检查, 不承诺端到端更快; 是否能进入默认路径必须看 caw 同输入 A/B。

Listing 4.11 include/lcqi/ggml\_matvec.hpp

```

1 #pragma once
2
3 #include <lcqi/gguf.hpp>
4
5 #include <array>
6 #include <cstdint>
7 #include <span>
8 #include <vector>
9
10 namespace lcqi {
11
12 constexpr std::int64_t LCQI_Q8_0_INPUT_BLOCK_VALUES = 32;
13

```



```

14 struct Q8_0InputBlock {
15     float d = 0.0F;
16     std::array<std::int8_t, LCQI_Q8_0_INPUT_BLOCK_VALUES> qs{};
17     std::int16_t sum = 0;
18 };
19
20 [[nodiscard]] bool ggml_type_has_f32_direct_matvec(GgmlType type) noexcept;
21
22 [[nodiscard]] bool ggml_type_has_q8_0_direct_matvec(GgmlType type) noexcept;
23
24 [[nodiscard]] std::vector<Q8_0InputBlock> quantize_q8_0_input(std::span<const float> ←
    ↪ float> input);
25
26 void quantize_q8_0_input_to(std::span<const float> input,
27                             std::span<Q8_0InputBlock> output);
28
29 void matvec_ggml_quantized_f32(GgmlType type,
30                                 std::span<const std::uint8_t> rows,
31                                 std::int64_t row_count,
32                                 std::int64_t column_count,
33                                 std::span<const float> input,
34                                 std::span<float> output);
35
36 void matvec_ggml_quantized_q8_0(GgmlType type,
37                                 std::span<const std::uint8_t> rows,
38                                 std::int64_t row_count,
39                                 std::int64_t column_count,
40                                 std::span<const Q8_0InputBlock> input,
41                                 std::span<float> output);
42
43 } // namespace lcqi

```

LLaMA GGUF 头文件把真实 GGUF 文件接到 decoder 数学层。当前实现有三种权重执行状态：关闭 direct 时是 `f32_dequantized_reference`；默认是 `gguf_mixed_q4_k_direct`，只让 shape 兼容的 `Q4_K` linear tensor 保留原始 block bytes；显式设置 `LCQI_LLAMA_GGML_DIRECT=1` 时才进入 `gguf_mixed_quantized_direct_experimental`，尝试把 `Q5_0/Q6_K/Q8_0` 也接入 direct。这个边界很重要，因为它防止把“能跑真实模型”误写成“已经拥有 llama.cpp 同等级的低比特执行引擎”。

**Listing 4.12** include/lcqi/llama\_gguf.hpp

```

1 #pragma once
2
3 #include <lcqi/gpt2_reference.hpp>
4 #include <lcqi/gguf.hpp>
5 #include <lcqi/reference_decoder.hpp>
6
7 #include <cstdint>
8 #include <cstdint>
9 #include <filesystem>
10 #include <span>
11 #include <string>
12 #include <vector>

```

```

13
14 namespace lcqi {
15
16 enum class LlamaGgufLinearStorage : std::uint8_t {
17     f32,
18     q4_k,
19     ggml_quantized,
20 };
21
22 struct LlamaGgufLinearWeights {
23     std::int32_t input_size = 0;
24     std::int32_t output_size = 0;
25     GgmlType source_type = GgmlType::f32;
26     LlamaGgufLinearStorage storage = LlamaGgufLinearStorage::f32;
27     std::vector<float> f32_weights;
28     std::vector<std::uint8_t> quantized_bytes;
29 };
30
31 struct LlamaGgufDecoderLayerWeights {
32     std::vector<float> rms_attention_weight;
33     LlamaGgufLinearWeights wq;
34     LlamaGgufLinearWeights wk;
35     LlamaGgufLinearWeights wv;
36     LlamaGgufLinearWeights wo;
37     std::vector<float> rms_mlp_weight;
38     LlamaGgufLinearWeights w_gate;
39     LlamaGgufLinearWeights w_up;
40     LlamaGgufLinearWeights w_down;
41 };
42
43 struct LlamaGgufDecoderModel {
44     DecoderConfig config;
45     std::vector<float> token_embedding;
46     std::vector<LlamaGgufDecoderLayerWeights> layers;
47     std::vector<float> final_norm_weight;
48     LlamaGgufLinearWeights lm_head;
49 };
50
51 struct LlamaGgufModel {
52     LlamaGgufDecoderModel decoder;
53     std::string architecture;
54     std::string name;
55     std::int32_t bos_token_id = -1;
56     std::int32_t eos_token_id = -1;
57     std::int32_t unknown_token_id = -1;
58     bool lm_head_tied_to_embedding = true;
59 };
60
61 struct LlamaGgufLoadReport {
62     double manifest_ms = 0.0;
63     double weights_ms = 0.0;
64     std::uint64_t quantized_weight_bytes = 0;

```

```

65     std::uint64_t f32_weight_bytes = 0;
66     std::uint64_t direct_quantized_weight_bytes = 0;
67     std::uint64_t fallback_dequantized_weight_bytes = 0;
68     std::int64_t tensors_loaded = 0;
69     std::int64_t q4_k_direct_tensors = 0;
70     std::int64_t ggml_direct_tensors = 0;
71     std::int64_t f32_fallback_tensors = 0;
72 };
73
74 struct LlamaGgufLoadedModel {
75     LlamaGgufModel model;
76     LlamaGgufLoadReport report;
77 };
78
79 struct LlamaGgufHotspotReport {
80     double rms_norm_ms = 0.0;
81     double attention_ms = 0.0;
82     double rope_ms = 0.0;
83     double wq_ms = 0.0;
84     double wk_ms = 0.0;
85     double wv_ms = 0.0;
86     double wo_ms = 0.0;
87     double w_gate_ms = 0.0;
88     double w_up_ms = 0.0;
89     double w_down_ms = 0.0;
90     double lm_head_ms = 0.0;
91     double q4_k_direct_ms = 0.0;
92     double ggml_direct_ms = 0.0;
93     double f32_fallback_ms = 0.0;
94     std::uint64_t q4_k_direct_calls = 0;
95     std::uint64_t ggml_direct_calls = 0;
96     std::uint64_t f32_fallback_calls = 0;
97 };
98
99 struct LlamaGgufExecutionOptions {
100     std::int32_t worker_count = 0;
101     std::int32_t parallel_min_rows = 512;
102 };
103
104 struct LlamaGgufGenerationResult {
105     std::vector<std::int32_t> prompt_ids;
106     std::vector<std::int32_t> generated_ids;
107     LlamaGgufHotspotReport hotspots;
108     double load_ms = 0.0;
109     double prefill_ms = 0.0;
110     double decode_ms = 0.0;
111     double total_ms = 0.0;
112     std::size_t kv_cache_bytes = 0;
113     std::size_t prefill_steps = 0;
114     std::size_t decode_steps = 0;
115     std::int32_t worker_count = 1;
116     std::int32_t predicted_first_token = -1;

```

```

117 };
118
119 [[nodiscard]] LlamaGgufLoadedModel load_llama_gguf_reference_model(
120     const std::filesystem::path& gguf_path);
121
122 [[nodiscard]] Gpt2Tokenizer load_gpt2_tokenizer_from_gguf(
123     const std::filesystem::path& gguf_path);
124
125 [[nodiscard]] std::vector<std::int32_t> llama_gguf_chat_prompt_ids(
126     const Gpt2Tokenizer& tokenizer,
127     std::string_view user_prompt,
128     std::int32_t bos_token_id);
129
130 [[nodiscard]] LlamaGgufGenerationResult llama_gguf_generate_greedy(
131     const LlamaGgufModel& model,
132     std::span<const std::int32_t> prompt_ids,
133     std::int32_t max_new_tokens);
134
135 [[nodiscard]] LlamaGgufGenerationResult llama_gguf_generate_greedy(
136     const LlamaGgufModel& model,
137     std::span<const std::int32_t> prompt_ids,
138     std::int32_t max_new_tokens,
139     const LlamaGgufExecutionOptions& options);
140
141 } // namespace lcqi

```

## Reference Decoder 与 GPT-2 合同

reference decoder 头文件定义张量、层配置、trace checkpoint、采样结果和 decode 函数。它是定位推理错误的主要合同，不只返回最终 token。

**Listing 4.13** include/lcqi/reference\_decoder.hpp

```

1  #pragma once
2
3  #include <stdint>
4  #include <span>
5  #include <string>
6  #include <vector>
7
8  namespace lcqi {
9
10     struct LinearWeightsF32 {
11         std::int32_t input_size = 0;
12         std::int32_t output_size = 0;
13         std::vector<float> weights;
14         std::vector<float> bias;
15     };
16
17     struct DecoderConfig {
18         std::int32_t hidden_size = 0;
19         std::int32_t query_heads = 0;

```

```

20     std::int32_t kv_heads = 0;
21     std::int32_t head_dim = 0;
22     std::int32_t intermediate_size = 0;
23     std::int32_t vocab_size = 0;
24     std::int32_t max_context = 0;
25     float rms_epsilon = 1.0e-5F;
26     float rope_base = 10000.0F;
27 };
28
29 struct DecoderLayerWeightsF32 {
30     std::vector<float> rms_attention_weight;
31     LinearWeightsF32 wq;
32     LinearWeightsF32 wk;
33     LinearWeightsF32 wv;
34     LinearWeightsF32 wo;
35     std::vector<float> rms_mlp_weight;
36     LinearWeightsF32 w_gate;
37     LinearWeightsF32 w_up;
38     LinearWeightsF32 w_down;
39 };
40
41 struct ReferenceDecoderModel {
42     DecoderConfig config;
43     std::vector<float> token_embedding;
44     std::vector<DecoderLayerWeightsF32> layers;
45     std::vector<float> final_norm_weight;
46     LinearWeightsF32 lm_head;
47 };
48
49 struct ReferenceDecodeResult {
50     std::vector<float> logits;
51     std::int32_t predicted_token = 0;
52 };
53
54 struct ReferenceTensorCheckpoint {
55     std::string name;
56     std::int32_t token_position = 0;
57     std::int32_t layer_id = 0;
58     std::vector<std::int32_t> shape;
59     std::vector<std::int32_t> stride;
60     std::string dtype;
61     std::string layout;
62     float checksum = 0.0F;
63     float max_abs = 0.0F;
64     std::vector<float> values;
65 };
66
67 struct ReferenceDecodeTraceResult {
68     ReferenceDecodeResult result;
69     std::vector<ReferenceTensorCheckpoint> checkpoints;
70 };
71

```

```

72 class ReferenceKVCache {
73 public:
74     ReferenceKVCache(const DecoderConfig& config, std::int32_t layer_count);
75
76     void append(std::int32_t layer_id,
77                 std::int32_t model_position,
78                 std::span<const float> key,
79                 std::span<const float> value);
80
81     [[nodiscard]] std::span<const float> key(std::int32_t layer_id,
82                                              std::int32_t model_position,
83                                              std::int32_t kv_head) const;
84
85     [[nodiscard]] std::span<const float> value(std::int32_t layer_id,
86                                              std::int32_t model_position,
87                                              std::int32_t kv_head) const;
88
89     [[nodiscard]] std::int32_t layer_count() const noexcept;
90     [[nodiscard]] std::int32_t filled_tokens() const noexcept;
91
92 private:
93     [[nodiscard]] std::size_t base_offset(std::int32_t layer_id,
94                                           std::int32_t model_position,
95                                           std::int32_t kv_head) const;
96
97     void validate_address(std::int32_t layer_id,
98                          std::int32_t model_position,
99                          std::int32_t kv_head) const;
100
101     DecoderConfig config_;
102     std::int32_t layer_count_ = 0;
103     std::int32_t filled_tokens_ = 0;
104     std::vector<float> keys_;
105     std::vector<float> values_;
106     std::vector<std::uint8_t> written_;
107 };
108
109 void rms_norm(std::span<const float> input,
110              std::span<const float> weight,
111              float epsilon,
112              std::span<float> output);
113
114 void linear_f32(const LinearWeightsF32& layer,
115                std::span<const float> input,
116                std::span<float> output);
117
118 void apply_rope(std::span<float> heads,
119                std::int32_t head_count,
120                std::int32_t head_dim,
121                std::int32_t model_position,
122                float rope_base);
123
124 void attention_decode(const DecoderConfig& config,

```

```

124         const ReferenceKVCache& cache,
125         std::int32_t layer_id,
126         std::int32_t model_position,
127         std::span<const float> query,
128         std::span<float> output);
129
130 ReferenceDecodeResult run_reference_decode(const ReferenceDecoderModel& model,
131                                         std::span<const std::int32_t> ↵
132                                         ↵ token_ids);
133
134 ReferenceDecodeTraceResult run_reference_decode_with_trace(
135     const ReferenceDecoderModel& model,
136     std::span<const std::int32_t> token_ids);
137
138 ReferenceDecoderModel make_tiny_reference_decoder_model();
139 } // namespace lcqi

```

GPT-2 reference 头文件定义 GPT-2 配置、byte-level BPE tokenizer、HuggingFace safetensors 加载、forward、cached forward 和 greedy generation。它单独存在，是因为 GPT-2 使用 LayerNorm、绝对位置嵌入、packed QKV、GELU 和 tied lm head，不能直接塞进 LLaMA-like reference decoder。读 `Gpt2CachedGreedyDecoder` 时要区分三个入口：`step()` 推进一个 token 并返回 greedy 预测；`step_with_logits()` 保留完整 logits 给测试和对照；`advance_without_prediction()` 只推进 embedding、Transformer layers 和 KV cache，用在 prompt prefill 的前缀 token 上。第三个入口不是“少算一点但可能改变结果”，而是删掉不会被任何 caller 使用的预测结果。

Listing 4.14 include/lcqi/gpt2\_reference.hpp

```

1 #pragma once
2
3 #include <cstdint>
4 #include <filesystem>
5 #include <memory>
6 #include <span>
7 #include <string>
8 #include <string_view>
9 #include <unordered_map>
10 #include <vector>
11
12 namespace lcqi {
13
14 struct Gpt2Config {
15     std::int32_t hidden_size = 0;
16     std::int32_t head_count = 0;
17     std::int32_t layer_count = 0;
18     std::int32_t max_positions = 0;
19     std::int32_t vocab_size = 0;
20     std::int32_t intermediate_size = 0;
21     std::int32_t bos_token_id = -1;
22     std::int32_t eos_token_id = -1;
23     float layer_norm_epsilon = 1.0e-5F;
24     std::string activation_function = "gelu_new";
25 };

```



```

26
27 struct Gpt2LinearF32 {
28     std::int32_t input_size = 0;
29     std::int32_t output_size = 0;
30     std::vector<float> weights;
31     std::vector<float> bias;
32 };
33
34 struct Gpt2LayerWeightsF32 {
35     std::vector<float> ln_1_weight;
36     std::vector<float> ln_1_bias;
37     Gpt2LinearF32 c_attn;
38     Gpt2LinearF32 c_proj;
39     std::vector<float> ln_2_weight;
40     std::vector<float> ln_2_bias;
41     Gpt2LinearF32 c_fc;
42     Gpt2LinearF32 mlp_c_proj;
43 };
44
45 struct Gpt2ReferenceModel {
46     Gpt2Config config;
47     std::vector<float> token_embedding;
48     std::vector<float> position_embedding;
49     std::vector<Gpt2LayerWeightsF32> layers;
50     std::vector<float> final_ln_weight;
51     std::vector<float> final_ln_bias;
52     std::vector<float> lm_head_weight;
53     bool tie_lm_head_to_embedding = true;
54 };
55
56 struct Gpt2ForwardResult {
57     std::vector<float> logits;
58     std::int32_t predicted_token = 0;
59 };
60
61 struct Gpt2HotspotProfile {
62     std::int64_t decoder_steps = 0;
63     std::int64_t layer_steps = 0;
64     double total_step_ms = 0.0;
65     double embedding_ms = 0.0;
66     double layer_norm_ms = 0.0;
67     double final_norm_ms = 0.0;
68     double qkv_projection_ms = 0.0;
69     double kv_append_ms = 0.0;
70     double attention_ms = 0.0;
71     double attention_projection_ms = 0.0;
72     double residual_add_ms = 0.0;
73     double mlp_fc_ms = 0.0;
74     double gelu_ms = 0.0;
75     double mlp_projection_ms = 0.0;
76     double lm_head_ms = 0.0;
77     double logits_result_ms = 0.0;

```

```

78 };
79
80 struct Gpt2ExecutionOptions {
81     std::int32_t worker_count = 0;
82     std::int32_t parallel_min_rows = 512;
83 };
84
85 class Gpt2KvCache;
86
87 namespace detail {
88 class Gpt2ParallelWorkerPool;
89
90 void attend_cached_position(const Gpt2KvCache& cache,
91                             std::int32_t layer_id,
92                             std::int32_t model_position,
93                             std::span<const float> query,
94                             std::span<float> scores,
95                             std::span<float> output);
96 } // namespace detail
97
98 class Gpt2ForwardWorkspace {
99 public:
100     explicit Gpt2ForwardWorkspace(const Gpt2Config& config);
101
102     void reset_for_config(const Gpt2Config& config);
103
104     [[nodiscard]] std::span<float> hidden() noexcept;
105     [[nodiscard]] std::span<float> normed() noexcept;
106     [[nodiscard]] std::span<float> query() noexcept;
107     [[nodiscard]] std::span<float> key() noexcept;
108     [[nodiscard]] std::span<float> value() noexcept;
109     [[nodiscard]] std::span<float> attention() noexcept;
110     [[nodiscard]] std::span<float> projected() noexcept;
111     [[nodiscard]] std::span<float> qkv_packed() noexcept;
112     [[nodiscard]] std::span<float> mlp_fc() noexcept;
113     [[nodiscard]] std::span<float> mlp_out() noexcept;
114     [[nodiscard]] std::span<float> logits() noexcept;
115     [[nodiscard]] std::span<float> scores_prefix(std::int32_t count);
116
117 private:
118     Gpt2Config config_;
119     std::vector<float> hidden_;
120     std::vector<float> normed_;
121     std::vector<float> query_;
122     std::vector<float> key_;
123     std::vector<float> value_;
124     std::vector<float> attention_;
125     std::vector<float> projected_;
126     std::vector<float> qkv_packed_;
127     std::vector<float> mlp_fc_;
128     std::vector<float> mlp_out_;
129     std::vector<float> logits_;

```

```

130     std::vector<float> scores_;
131 };
132
133 class Gpt2KvCache {
134 public:
135     explicit Gpt2KvCache(const Gpt2Config& config);
136
137     void append(std::int32_t layer_id,
138                std::int32_t model_position,
139                std::span<const float> key,
140                std::span<const float> value);
141
142     [[nodiscard]] std::span<const float> key(std::int32_t layer_id,
143                                              std::int32_t model_position,
144                                              std::int32_t head) const;
145
146     [[nodiscard]] std::span<const float> value(std::int32_t layer_id,
147                                                std::int32_t model_position,
148                                                std::int32_t head) const;
149
150     [[nodiscard]] std::int32_t filled_tokens() const noexcept;
151     [[nodiscard]] std::size_t byte_size() const noexcept;
152
153 private:
154     friend void detail::attend_cached_position(const Gpt2KvCache& cache,
155                                              std::int32_t layer_id,
156                                              std::int32_t model_position,
157                                              std::span<const float> query,
158                                              std::span<float> scores,
159                                              std::span<float> output);
160
161     [[nodiscard]] std::size_t base_offset(std::int32_t layer_id,
162                                           std::int32_t model_position,
163                                           std::int32_t head) const;
164     [[nodiscard]] std::size_t base_offset_unchecked(std::int32_t layer_id,
165                                                    std::int32_t model_position↵
166     ↵ ,
167                                                    std::int32_t head) const ↵
168     ↵ noexcept;
169     [[nodiscard]] std::span<const float> key_unchecked(std::int32_t layer_id,
170                                                       std::int32_t ↵
171     ↵ model_position,
172                                                       std::int32_t head) const↵
173     ↵ noexcept;
174     [[nodiscard]] std::span<const float> value_unchecked(std::int32_t layer_id,
175                                                         std::int32_t ↵
176     ↵ model_position,
177                                                         std::int32_t head) ↵
178     ↵ const noexcept;
179     void validate_address(std::int32_t layer_id,
180                          std::int32_t model_position,
181                          std::int32_t head) const;

```

```

176 void validate_written(std::int32_t layer_id,
177                      std::int32_t model_position) const;
178
179 Gpt2Config config_;
180 std::int32_t filled_tokens_ = 0;
181 std::vector<float> keys_;
182 std::vector<float> values_;
183 std::vector<std::uint8_t> written_;
184 };
185
186 class Gpt2CachedGreedyDecoder {
187 public:
188     explicit Gpt2CachedGreedyDecoder(const Gpt2ReferenceModel& model);
189     Gpt2CachedGreedyDecoder(const Gpt2ReferenceModel& model,
190                             Gpt2HotspotProfile* hotspot_profile);
191     Gpt2CachedGreedyDecoder(const Gpt2ReferenceModel& model,
192                             Gpt2HotspotProfile* hotspot_profile,
193                             const Gpt2ExecutionOptions& execution_options);
194     ~Gpt2CachedGreedyDecoder();
195
196     Gpt2CachedGreedyDecoder(const Gpt2CachedGreedyDecoder&) = delete;
197     Gpt2CachedGreedyDecoder& operator=(const Gpt2CachedGreedyDecoder&) = delete;↵
198     ↵ ;
199     Gpt2CachedGreedyDecoder(Gpt2CachedGreedyDecoder&&) noexcept;
200     Gpt2CachedGreedyDecoder& operator=(Gpt2CachedGreedyDecoder&&) noexcept;
201
202     void advance_without_prediction(std::int32_t token_id);
203     [[nodiscard]] std::int32_t step(std::int32_t token_id);
204     [[nodiscard]] Gpt2ForwardResult step_with_logits(std::int32_t token_id);
205     [[nodiscard]] const Gpt2KvCache& cache() const noexcept;
206     [[nodiscard]] std::int32_t filled_tokens() const noexcept;
207     [[nodiscard]] std::size_t kv_cache_bytes() const noexcept;
208     [[nodiscard]] std::int32_t worker_count() const noexcept;
209
210 private:
211     const Gpt2ReferenceModel* model_;
212     Gpt2HotspotProfile* hotspot_profile_;
213     Gpt2ExecutionOptions execution_options_;
214     Gpt2KvCache cache_;
215     Gpt2ForwardWorkspace workspace_;
216     std::unique_ptr<detail::Gpt2ParallelWorkerPool> worker_pool_;
217 };
218
219 struct Gpt2Tokenizer {
220     std::int32_t bos_token_id = -1;
221     std::int32_t eos_token_id = -1;
222     std::unordered_map<std::string, std::int32_t> vocab;
223     std::vector<std::string> id_to_token;
224     std::unordered_map<std::string, std::int32_t> bpe_ranks;
225 };
226
227 [[nodiscard]] Gpt2Tokenizer load_gpt2_tokenizer(

```

```

227     const std::filesystem::path& vocab_json,
228     const std::filesystem::path& merges_txt,
229     std::int32_t bos_token_id,
230     std::int32_t eos_token_id);
231
232 [[nodiscard]] std::vector<std::int32_t> gpt2_encode(
233     const Gpt2Tokenizer& tokenizer,
234     std::string_view text);
235
236 [[nodiscard]] std::string gpt2_decode(
237     const Gpt2Tokenizer& tokenizer,
238     std::span<const std::int32_t> token_ids);
239
240 [[nodiscard]] Gpt2Config load_gpt2_config(const std::filesystem::path& ↵
    ↵ config_json);
241
242 [[nodiscard]] Gpt2ReferenceModel load_gpt2_reference_model(
243     const std::filesystem::path& config_json,
244     const std::filesystem::path& safetensors_path);
245
246 [[nodiscard]] Gpt2ReferenceModel load_gpt2_from_directory(
247     const std::filesystem::path& model_directory);
248
249 [[nodiscard]] Gpt2ForwardResult run_gpt2_forward(
250     const Gpt2ReferenceModel& model,
251     std::span<const std::int32_t> token_ids);
252
253 [[nodiscard]] Gpt2ForwardResult run_gpt2_forward_cached(
254     const Gpt2ReferenceModel& model,
255     Gpt2KvCache& cache,
256     std::int32_t token_id);
257
258 [[nodiscard]] std::vector<std::int32_t> gpt2_generate_greedy(
259     const Gpt2ReferenceModel& model,
260     std::span<const std::int32_t> prompt_token_ids,
261     std::int32_t max_new_tokens);
262
263 [[nodiscard]] std::vector<std::int32_t> gpt2_generate_greedy_cached(
264     const Gpt2ReferenceModel& model,
265     std::span<const std::int32_t> prompt_token_ids,
266     std::int32_t max_new_tokens);
267
268 [[nodiscard]] Gpt2ReferenceModel make_tiny_gpt2_reference_model();
269
270 } // namespace lcqi

```

## 最短闭环：从模型文件到 logits

tiny MLP 路径不是为了证明 LCQI 已经是完整大模型引擎，而是为了建立最短的、可手算的、可测试的推理闭环。读者要把模型加载、输入检查、int8 linear、ReLU、第二层 logits、argmax 和 CLI 输出连起来看。

## 模型加载与基础推理

`model.cpp` 负责解析 tiny 文本模型、权重、scale、bias 和 shape。坏格式、坏维度和坏数值都应该在这里被拒绝，不能拖到 kernel 内部才崩溃。

Listing 4.15 `src/model.cpp`

```

1  #include <lcqi/model.hpp>
2
3  #include <fstream>
4  #include <limits>
5  #include <stdexcept>
6  #include <string>
7
8  namespace lcqi {
9  namespace {
10
11  constexpr std::int32_t LCQI_INT8_MIN_VALUE = -128;
12  constexpr std::int32_t LCQI_INT8_MAX_VALUE = 127;
13
14  std::string read_key(std::istream& input) {
15      std::string key;
16      if (!(input >> key)) {
17          throw std::runtime_error("unexpected end of model file");
18      }
19      return key;
20  }
21
22  void expect_key(std::istream& input, const std::string& expected) {
23      const std::string key = read_key(input);
24      if (key != expected) {
25          throw std::runtime_error("expected key '" + expected + "', got '" + key +
26                                  " + "'");
27      }
28
29  std::int32_t read_i32(std::istream& input, const std::string& key) {
30      expect_key(input, key);
31      std::int32_t value = 0;
32      if (!(input >> value)) {
33          throw std::runtime_error("invalid int32 value for key '" + key + "'");
34      }
35      return value;
36  }
37
38  float read_float(std::istream& input, const std::string& key) {
39      expect_key(input, key);
40      float value = 0.0F;
41      if (!(input >> value)) {
42          throw std::runtime_error("invalid float value for key '" + key + "'");
43      }
44      return value;
45  }

```

```

46
47 std::vector<std::int8_t> read_i8_vector(std::istream& input,
48                                         const std::string& key,
49                                         std::int32_t count) {
50     expect_key(input, key);
51     if (count < 0) {
52         throw std::runtime_error("negative vector size for key '" + key + "'");
53     }
54     std::vector<std::int8_t> values(static_cast<std::size_t>(count));
55     for (std::int32_t i = 0; i < count; ++i) {
56         std::int32_t raw = 0;
57         if (!(input >> raw)) {
58             throw std::runtime_error("invalid int8 payload for key '" + key + "'");
59         }
60         if (raw < LCQI_INT8_MIN_VALUE || raw > LCQI_INT8_MAX_VALUE) {
61             throw std::runtime_error("int8 payload out of range for key '" + key + "'");
62         }
63         values[static_cast<std::size_t>(i)] = static_cast<std::int8_t>(raw);
64     }
65     return values;
66 }
67
68 std::vector<float> read_float_vector(std::istream& input,
69                                      const std::string& key,
70                                      std::int32_t count) {
71     expect_key(input, key);
72     if (count < 0) {
73         throw std::runtime_error("negative vector size for key '" + key + "'");
74     }
75     std::vector<float> values(static_cast<std::size_t>(count));
76     for (std::int32_t i = 0; i < count; ++i) {
77         if (!(input >> values[static_cast<std::size_t>(i)])) {
78             throw std::runtime_error("invalid float payload for key '" + key + "'");
79         }
80     }
81     return values;
82 }
83
84 void validate_layer(const QuantizedLinearLayer& layer, const std::string& name) {
85     if (layer.input_size <= 0 || layer.output_size <= 0) {
86         throw std::runtime_error(name + " layer has invalid shape");
87     }
88     const std::size_t expected_weights =
89         static_cast<std::size_t>(layer.input_size) *
90         static_cast<std::size_t>(layer.output_size);
91     if (layer.weights.size() != expected_weights) {
92         throw std::runtime_error(name + " layer weight size mismatch");
93     }

```



```

94     if (layer.bias.size() != static_cast<std::size_t>(layer.output_size)) {
95         throw std::runtime_error(name + " layer bias size mismatch");
96     }
97 }
98
99 std::int32_t checked_element_count(std::int32_t rows,
100                                   std::int32_t columns,
101                                   const std::string& name) {
102     if (rows <= 0 || columns <= 0) {
103         throw std::runtime_error(name + " layer has invalid shape");
104     }
105     const std::int64_t count =
106         static_cast<std::int64_t>(rows) * static_cast<std::int64_t>(columns);
107     if (count > static_cast<std::int64_t>(std::numeric_limits<std::int32_t>::max()
↵
108         throw std::runtime_error(name + " layer element count is too large");
109     }
110     return static_cast<std::int32_t>(count);
111 }
112
113 } // namespace
114
115 TinyMlpModel load_model(const std::filesystem::path& path) {
116     std::ifstream input(path);
117     if (!input) {
118         throw std::runtime_error("cannot open model file: " + path.string());
119     }
120
121     expect_key(input, "LCQI_MODEL_V1");
122
123     TinyMlpModel model;
124     model.input_size = read_i32(input, "input_size");
125     model.hidden_size = read_i32(input, "hidden_size");
126     model.output_size = read_i32(input, "output_size");
127
128     const std::int32_t hidden_weight_count =
129         checked_element_count(model.input_size, model.hidden_size, "hidden");
130     const std::int32_t output_weight_count =
131         checked_element_count(model.hidden_size, model.output_size, "output");
132
133     model.hidden.input_size = model.input_size;
134     model.hidden.output_size = model.hidden_size;
135     model.hidden.scale = read_float(input, "hidden_scale");
136     model.hidden.weights = read_i8_vector(
137         input,
138         "hidden_weights",
139         hidden_weight_count);
140     model.hidden.bias = read_float_vector(input, "hidden_bias", model.
↵
141         hidden_size);
142
143     model.output.input_size = model.hidden_size;
144     model.output.output_size = model.output_size;

```

```

144     model.output.scale = read_float(input, "output_scale");
145     model.output.weights = read_i8_vector(
146         input,
147         "output_weights",
148         output_weight_count);
149     model.output.bias = read_float_vector(input, "output_bias", model.↵
↵ output_size);
150
151     validate_layer(model.hidden, "hidden");
152     validate_layer(model.output, "output");
153     return model;
154 }
155
156 } // namespace lcqi

```

`inference.cpp` 执行 int8 linear、ReLU、第二层 logits 和 predicted class。它是系统中最短的“模型资产到输出”路径。

**Listing 4.16** `src/inference.cpp`

```

1  #include <lcqi/inference.hpp>
2
3  #include <algorithm>
4  #include <cmath>
5  #include <stdexcept>
6
7  namespace lcqi {
8  namespace {
9
10 void validate_input_size(std::span<const float> input, std::int32_t expected) {
11     if (input.size() != static_cast<std::size_t>(expected)) {
12         throw std::runtime_error("input size does not match model shape");
13     }
14 }
15
16 std::int32_t argmax(std::span<const float> values) {
17     if (values.empty()) {
18         throw std::runtime_error("cannot take argmax of empty logits");
19     }
20     std::int32_t best_index = 0;
21     float best_value = values[0];
22     for (std::int32_t i = 1; i < static_cast<std::int32_t>(values.size()); ++i)↵
↵ {
23         const float candidate = values[static_cast<std::size_t>(i)];
24         if (candidate > best_value) {
25             best_value = candidate;
26             best_index = i;
27         }
28     }
29     return best_index;
30 }
31
32 } // namespace

```

```

33
34 void linear_i8(const QuantizedLinearLayer& layer,
35               std::span<const float> input,
36               std::span<float> output) {
37     if (input.size() != static_cast<std::size_t>(layer.input_size)) {
38         throw std::runtime_error("linear_i8 input size mismatch");
39     }
40     if (output.size() != static_cast<std::size_t>(layer.output_size)) {
41         throw std::runtime_error("linear_i8 output size mismatch");
42     }
43
44     for (std::int32_t out = 0; out < layer.output_size; ++out) {
45         const std::size_t row_base =
46             static_cast<std::size_t>(out) * static_cast<std::size_t>(layer.↵
↵ input_size);
47         float acc = layer.bias[static_cast<std::size_t>(out)];
48         for (std::int32_t in = 0; in < layer.input_size; ++in) {
49             const std::int8_t weight =
50                 layer.weights[row_base + static_cast<std::size_t>(in)];
51             acc += static_cast<float>(weight) * layer.scale *
52                 input[static_cast<std::size_t>(in)];
53         }
54         output[static_cast<std::size_t>(out)] = acc;
55     }
56 }
57
58 void relu_inplace(std::span<float> values) {
59     for (float& value : values) {
60         value = std::max(0.0F, value);
61     }
62 }
63
64 InferenceResult run_inference(const TinyMlpModel& model,
65                               std::span<const float> input) {
66     validate_input_size(input, model.input_size);
67
68     std::vector<float> hidden(static_cast<std::size_t>(model.hidden_size), 0.0F↵
↵ );
69     std::vector<float> logits(static_cast<std::size_t>(model.output_size), 0.0F↵
↵ );
70
71     linear_i8(model.hidden, input, hidden);
72     relu_inplace(hidden);
73     linear_i8(model.output, hidden, logits);
74
75     InferenceResult result;
76     result.predicted_class = argmax(logits);
77     result.logits = std::move(logits);
78     return result;
79 }
80
81 } // namespace lcqi

```

`main.cpp` 把模型路径和输入接到 `run_inference`，并打印结果。CLI 应保持薄层，不复制模型解析和推理逻辑。

Listing 4.17 `src/main.cpp`

```

1  #include <lcqi/inference.hpp>
2  #include <lcqi/model.hpp>
3
4  #include <chrono>
5  #include <cstdint>
6  #include <cstdlib>
7  #include <filesystem>
8  #include <iostream>
9  #include <span>
10 #include <stdexcept>
11 #include <string>
12 #include <vector>
13
14 namespace {
15
16 constexpr std::int32_t LCQI_DEFAULT_REPEAT = 1000;
17 constexpr int LCQI_EXPECTED_ARGC_MIN = 2;
18 constexpr double LCQI_SECONDS_TO_MICROSECONDS = 1000000.0;
19
20 std::vector<float> parse_input(int argc, char** argv, std::int32_t ↵
    ↵ expected_size) {
21     if (argc < LCQI_EXPECTED_ARGC_MIN + expected_size) {
22         throw std::runtime_error("usage: lcqi_cli <model.txt> <input floats...>↵
    ↵ [repeat]");
23     }
24     std::vector<float> input(static_cast<std::size_t>(expected_size));
25     for (std::int32_t i = 0; i < expected_size; ++i) {
26         input[static_cast<std::size_t>(i)] =
27             std::stof(argv[LCQI_EXPECTED_ARGC_MIN + i]);
28     }
29     return input;
30 }
31
32 std::int32_t parse_repeat(int argc, char** argv, std::int32_t input_size) {
33     const int repeat_index = LCQI_EXPECTED_ARGC_MIN + input_size;
34     if (argc <= repeat_index) {
35         return LCQI_DEFAULT_REPEAT;
36     }
37     const std::int32_t repeat = static_cast<std::int32_t>(std::stoi(argv[↵
    ↵ repeat_index]));
38     if (repeat <= 0) {
39         throw std::runtime_error("repeat must be positive");
40     }
41     return repeat;
42 }
43
44 } // namespace
45

```

```

46 int main(int argc, char** argv) {
47     try {
48         if (argc < LCQI_EXPECTED_ARGC_MIN) {
49             throw std::runtime_error("usage: lcqi_cli <model.txt> <input floats>
↪ ...> [repeat]");
50         }
51
52         const std::filesystem::path model_path = argv[1];
53         const lcqi::TinyMlpModel model = lcqi::load_model(model_path);
54         const std::vector<float> input = parse_input(argc, argv, model.
↪ input_size);
55         const std::int32_t repeat = parse_repeat(argc, argv, model.input_size);
56
57         const lcqi::InferenceResult result = lcqi::run_inference(model, input);
58         std::cout << "predicted_class " << result.predicted_class << "\n";
59         std::cout << "logits";
60         for (const float value : result.logits) {
61             std::cout << ' ' << value;
62         }
63         std::cout << "\n";
64
65         const auto begin = std::chrono::steady_clock::now();
66         std::int32_t checksum = 0;
67         for (std::int32_t i = 0; i < repeat; ++i) {
68             checksum += lcqi::run_inference(model, input).predicted_class;
69         }
70         const auto end = std::chrono::steady_clock::now();
71         const std::chrono::duration<double> elapsed = end - begin;
72         const double average_us =
73             elapsed.count() * LCQI_SECONDS_TO_MICROSECONDS / static_cast<double>
↪ >(repeat);
74         std::cout << "repeat " << repeat << "\n";
75         std::cout << "average_us " << average_us << "\n";
76         std::cout << "checksum " << checksum << "\n";
77         return EXIT_SUCCESS;
78     } catch (const std::exception& error) {
79         std::cerr << "error: " << error.what() << "\n";
80         return EXIT_FAILURE;
81     }
82 }

```

## Int8 Kernel 基础实现

`int8_kernels.cpp` 包含 scalar、packed、workspace、deterministic fixture 和误差比较。它是 AVX2 路径的 reference 对照，也是 benchmark 的正确性底座。

**Listing 4.18** `src/int8_kernels.cpp`

```

1 #include <lcqi/int8_kernels.hpp>
2
3 #include <lcqi/inference.hpp>
4

```

```

5  #include <algorithm>
6  #include <chrono>
7  #include <cmath>
8  #include <stdexcept>
9  #include <string>
10
11 namespace lcqi {
12 namespace {
13
14 constexpr std::int32_t LCQI_I8_PATTERN_MODULUS = 23;
15 constexpr std::int32_t LCQI_I8_PATTERN_CENTER = 11;
16 constexpr std::int32_t LCQI_INPUT_PATTERN_MODULUS = 17;
17 constexpr std::int32_t LCQI_INPUT_PATTERN_CENTER = 8;
18 constexpr float LCQI_INPUT_PATTERN_SCALE = 0.125F;
19 constexpr double LCQI_SECONDS_TO_MICROSECONDS = 1000000.0;
20 constexpr std::int32_t LCQI_SIMD_OUTPUT_BLOCK = 8;
21 constexpr std::size_t LCQI_BENCHMARK_BACKEND_COUNT = 3;
22 constexpr const char* LCQI_BACKEND_SCALAR = "scalar";
23 constexpr const char* LCQI_BACKEND_PACKED_SCALAR = "packed_scalar";
24 constexpr const char* LCQI_BACKEND_AVX2 = "avx2";
25
26 void require_positive(std::int32_t value, const char* name) {
27     if (value <= 0) {
28         throw std::runtime_error(std::string(name) + " must be positive");
29     }
30 }
31
32 std::size_t checked_size(std::int32_t value, const char* name) {
33     if (value < 0) {
34         throw std::runtime_error(std::string(name) + " cannot be negative");
35     }
36     return static_cast<std::size_t>(value);
37 }
38
39 void validate_quantized_layer(const QuantizedLinearLayer& layer) {
40     require_positive(layer.input_size, "input_size");
41     require_positive(layer.output_size, "output_size");
42     const std::size_t expected_weights =
43         checked_size(layer.input_size, "input_size") *
44         checked_size(layer.output_size, "output_size");
45     if (layer.weights.size() != expected_weights) {
46         throw std::runtime_error("quantized layer weight size mismatch");
47     }
48     if (layer.bias.size() != checked_size(layer.output_size, "output_size")) {
49         throw std::runtime_error("quantized layer bias size mismatch");
50     }
51 }
52
53 std::int32_t round_up_to_block(std::int32_t value, std::int32_t block_size) {
54     require_positive(value, "value");
55     require_positive(block_size, "block_size");
56     return ((value + block_size - 1) / block_size) * block_size;

```

```

57 }
58
59 float checksum(std::span<const float> values) {
60     float sum = 0.0F;
61     for (const float value : values) {
62         sum += value;
63     }
64     return sum;
65 }
66
67 void add_unavailable_backend(KernelBenchmarkResult& result, const char* name) {
68     result.backends.push_back(KernelBackendBenchmark{
69         .name = name,
70         .available = false,
71         .average_us = 0.0,
72         .max_abs_diff = 0.0F,
73         .checksum = 0.0F,
74     });
75 }
76
77 template <typename Callable>
78 double measure_average_us(std::int32_t repeat, Callable&& callable) {
79     require_positive(repeat, "repeat");
80     const auto begin = std::chrono::steady_clock::now();
81     for (std::int32_t index = 0; index < repeat; ++index) {
82         callable();
83     }
84     const auto end = std::chrono::steady_clock::now();
85     const std::chrono::duration<double> elapsed = end - begin;
86     return elapsed.count() * LCQI_SECONDS_TO_MICROSECONDS / static_cast<double>
    ↵ >(repeat);
87 }
88
89 } // namespace
90
91 PackedLinearI8 pack_linear_i8(const QuantizedLinearLayer& layer,
92                             std::int32_t output_block_size) {
93     validate_quantized_layer(layer);
94     require_positive(output_block_size, "output_block_size");
95
96     PackedLinearI8 packed;
97     packed.input_size = layer.input_size;
98     packed.output_size = layer.output_size;
99     packed.output_block_size = output_block_size;
100     packed.scale = layer.scale;
101     packed.bias = layer.bias;
102
103     const std::int32_t padded_outputs =
104         round_up_to_block(layer.output_size, output_block_size);
105     packed.packed_weights.assign(
106         checked_size(padded_outputs, "padded_outputs") *
107         checked_size(layer.input_size, "input_size"),

```



```

108     0);
109
110     std::size_t packed_index = 0;
111     for (std::int32_t output_block = 0; output_block < padded_outputs;
112          output_block += output_block_size) {
113         for (std::int32_t input_index = 0; input_index < layer.input_size; ++
↵ input_index) {
114             for (std::int32_t offset = 0; offset < output_block_size; ++offset)↵
↵ {
115                 const std::int32_t output_index = output_block + offset;
116                 if (output_index < layer.output_size) {
117                     const std::size_t source =
118                         checked_size(output_index, "output_index") *
119                         checked_size(layer.input_size, "input_size") +
120                         checked_size(input_index, "input_index");
121                     packed.packed_weights[packed_index] = layer.weights[source↵
↵ ];
122                 }
123                 ++packed_index;
124             }
125         }
126     }
127     return packed;
128 }
129
130 void linear_i8_packed(const PackedLinearI8& layer,
131                      std::span<const float> input,
132                      std::span<float> output) {
133     std::vector<float> accumulators(
134         checked_size(layer.output_block_size, "output_block_size"),
135         0.0F);
136     linear_i8_packed_with_workspace(layer, input, output, accumulators);
137 }
138
139 void linear_i8_packed_with_workspace(const PackedLinearI8& layer,
140                                     std::span<const float> input,
141                                     std::span<float> output,
142                                     std::span<float> accumulator_workspace) {
143     require_positive(layer.input_size, "packed input_size");
144     require_positive(layer.output_size, "packed output_size");
145     require_positive(layer.output_block_size, "packed output_block_size");
146     if (input.size() != checked_size(layer.input_size, "input_size")) {
147         throw std::runtime_error("linear_i8_packed input size mismatch");
148     }
149     if (output.size() != checked_size(layer.output_size, "output_size")) {
150         throw std::runtime_error("linear_i8_packed output size mismatch");
151     }
152     if (layer.bias.size() != checked_size(layer.output_size, "output_size")) {
153         throw std::runtime_error("linear_i8_packed bias size mismatch");
154     }
155
156     const std::int32_t padded_outputs =

```

```

157     round_up_to_block(layer.output_size, layer.output_block_size);
158     const std::size_t expected_packed_size =
159         checked_size(padded_outputs, "padded_outputs") *
160         checked_size(layer.input_size, "input_size");
161     if (layer.packed_weights.size() != expected_packed_size) {
162         throw std::runtime_error("linear_i8_packed packed weight size mismatch"
163     ↪ );
164     }
165     if (accumulator_workspace.size() != checked_size(layer.output_block_size, "
166     ↪ output_block_size")) {
167         throw std::runtime_error("linear_i8_packed accumulator workspace size
168     ↪ mismatch");
169     }
170
171     std::size_t packed_index = 0;
172     for (std::int32_t output_block = 0; output_block < padded_outputs;
173     ↪ output_block += layer.output_block_size) {
174         std::fill(accumulator_workspace.begin(), accumulator_workspace.end(),
175     ↪ 0.0F);
176         for (std::int32_t offset = 0; offset < layer.output_block_size; ++
177     ↪ offset) {
178             const std::int32_t output_index = output_block + offset;
179             if (output_index < layer.output_size) {
180                 accumulator_workspace[checked_size(offset, "offset")] =
181                 layer.bias[checked_size(output_index, "output_index")];
182             }
183         }
184
185         for (std::int32_t input_index = 0; input_index < layer.input_size; ++
186     ↪ input_index) {
187             const float input_value = input[checked_size(input_index, "
188     ↪ input_index")];
189             for (std::int32_t offset = 0; offset < layer.output_block_size; ++
190     ↪ offset) {
191                 const std::int32_t output_index = output_block + offset;
192                 const std::int8_t weight = layer.packed_weights[packed_index
193     ↪ ++];
194                 if (output_index < layer.output_size) {
195                     accumulator_workspace[checked_size(offset, "offset")] +=
196                     static_cast<float>(weight) * layer.scale * input_value;
197                 }
198             }
199         }
200
201         for (std::int32_t offset = 0; offset < layer.output_block_size; ++
202     ↪ offset) {
203             const std::int32_t output_index = output_block + offset;
204             if (output_index < layer.output_size) {
205                 output[checked_size(output_index, "output_index")] =
206                 accumulator_workspace[checked_size(offset, "offset")];
207             }
208         }
209     }

```

```

199     }
200 }
201
202 QuantizedLinearLayer make_deterministic_i8_layer(std::int32_t input_size,
203                                                  std::int32_t output_size,
204                                                  float scale) {
205     require_positive(input_size, "input_size");
206     require_positive(output_size, "output_size");
207     QuantizedLinearLayer layer;
208     layer.input_size = input_size;
209     layer.output_size = output_size;
210     layer.scale = scale;
211     const std::size_t weight_count =
212         checked_size(input_size, "input_size") * checked_size(output_size, "↵
↵ output_size");
213     layer.weights.resize(weight_count);
214     layer.bias.resize(checked_size(output_size, "output_size"));
215     for (std::int32_t output_index = 0; output_index < output_size; ++↵
↵ output_index) {
216         layer.bias[checked_size(output_index, "output_index")] =
217             static_cast<float>((output_index % LCQI_INPUT_PATTERN_MODULUS) -
218                               LCQI_INPUT_PATTERN_CENTER) *
219             LCQI_INPUT_PATTERN_SCALE;
220         for (std::int32_t input_index = 0; input_index < input_size; ++↵
↵ input_index) {
221             const std::int32_t raw =
222                 ((output_index * input_size + input_index) % ↵
↵ LCQI_I8_PATTERN_MODULUS) -
223                 LCQI_I8_PATTERN_CENTER;
224             layer.weights[checked_size(output_index, "output_index") *
225                           checked_size(input_size, "input_size") +
226                           checked_size(input_index, "input_index")] =
227                 static_cast<std::int8_t>(raw);
228         }
229     }
230     return layer;
231 }
232
233 std::vector<float> make_deterministic_input(std::int32_t input_size) {
234     require_positive(input_size, "input_size");
235     std::vector<float> input(checked_size(input_size, "input_size"));
236     for (std::int32_t input_index = 0; input_index < input_size; ++input_index)↵
↵ {
237         input[checked_size(input_index, "input_index")] =
238             static_cast<float>((input_index % LCQI_INPUT_PATTERN_MODULUS) -
239                               LCQI_INPUT_PATTERN_CENTER) *
240             LCQI_INPUT_PATTERN_SCALE;
241     }
242     return input;
243 }
244
245 float max_abs_diff(std::span<const float> lhs, std::span<const float> rhs) {

```

```

246     if (lhs.size() != rhs.size()) {
247         throw std::runtime_error("max_abs_diff size mismatch");
248     }
249     float diff = 0.0F;
250     for (std::size_t index = 0; index < lhs.size(); ++index) {
251         diff = std::max(diff, std::fabs(lhs[index] - rhs[index]));
252     }
253     return diff;
254 }
255
256 KernelBenchmarkResult benchmark_linear_i8_case(const KernelBenchmarkCase& ←
    ↪ benchmark_case) {
257     require_positive(benchmark_case.input_size, "benchmark input_size");
258     require_positive(benchmark_case.output_size, "benchmark output_size");
259     require_positive(benchmark_case.output_block_size, "benchmark ←
    ↪ output_block_size");
260     require_positive(benchmark_case.repeat, "benchmark repeat");
261
262     const QuantizedLinearLayer layer =
263         make_deterministic_i8_layer(benchmark_case.input_size,
264                                     benchmark_case.output_size,
265                                     0.015625F);
266     const PackedLinearI8 packed = pack_linear_i8(layer, benchmark_case.↪
    ↪ output_block_size);
267     const PackedLinearI8 packed_simd = pack_linear_i8(layer, ↪
    ↪ LCQI_SIMD_OUTPUT_BLOCK);
268     const std::vector<float> input = make_deterministic_input(benchmark_case.↪
    ↪ input_size);
269     std::vector<float> scalar_output(checked_size(benchmark_case.output_size, "↪
    ↪ output_size"),
270                                     0.0F);
271     std::vector<float> packed_output(scalar_output.size(), 0.0F);
272     std::vector<float> simd_output(scalar_output.size(), 0.0F);
273     std::vector<float> packed_workspace(
274         checked_size(benchmark_case.output_block_size, "output_block_size"),
275         0.0F);
276
277     linear_i8(layer, input, scalar_output);
278     linear_i8_packed_with_workspace(packed, input, packed_output, ↪
    ↪ packed_workspace);
279
280     KernelBenchmarkResult result;
281     result.benchmark_case = benchmark_case;
282     result.backends.reserve(LCQI_BENCHMARK_BACKEND_COUNT);
283     result.backends.push_back(KernelBackendBenchmark{
284         .name = LCQI_BACKEND_SCALAR,
285         .available = true,
286         .average_us = measure_average_us(benchmark_case.repeat,
287                                         [&]() { linear_i8(layer, input, ↪
    ↪ scalar_output); }),
288         .max_abs_diff = 0.0F,
289         .checksum = checksum(scalar_output),

```

```

290     });
291     result.backends.push_back(KernelBackendBenchmark{
292         .name = LCQI_BACKEND_PACKED_SCALAR,
293         .available = true,
294         .average_us = measure_average_us(
295             benchmark_case.repeat,
296             [&]() { linear_i8_packed_with_workspace(packed,
297                                                         input,
298                                                         packed_output,
299                                                         packed_workspace); }),
300         .max_abs_diff = max_abs_diff(scalar_output, packed_output),
301         .checksum = checksum(packed_output),
302     });
303     if (linear_i8_packed_avx2_available()) {
304         linear_i8_packed_avx2(packed_simd, input, simd_output);
305         result.backends.push_back(KernelBackendBenchmark{
306             .name = LCQI_BACKEND_AVX2,
307             .available = true,
308             .average_us = measure_average_us(
309                 benchmark_case.repeat,
310                 [&]() { linear_i8_packed_avx2(packed_simd, input, simd_output); }↵
311         ↵ },
312             .max_abs_diff = max_abs_diff(scalar_output, simd_output),
313             .checksum = checksum(simd_output),
314         });
315     } else {
316         add_unavailable_backend(result, LCQI_BACKEND_AVX2);
317     }
318     return result;
319 }
320 } // namespace lcqi
321
322 #if !defined(LCQI_ENABLE_AVX2)
323 namespace lcqi {
324
325 bool linear_i8_packed_avx2_available() noexcept {
326     return false;
327 }
328
329 void linear_i8_packed_avx2(const PackedLinearI8&, std::span<const float>, std::span↵
330 ↵ span<float>) {
331     throw std::runtime_error("AVX2 packed kernel is not enabled in this build")↵
332     ↵ ;
333 }
334 } // namespace lcqi
335 #endif

```

`int8_kernels_avx2.cpp` 是平台优化路径。读者要先看可用性判断，再看向量化实现；如果机器不支持 AVX2/FMA，报告必须写 `unavailable`。

Listing 4.19 src/int8\_kernels\_avx2.cpp

```

1  #include <lcqi/int8_kernels.hpp>
2
3  #if defined(LCQI_ENABLE_AVX2)
4
5  #include <immintrin.h>
6
7  #include <algorithm>
8  #include <cstdint>
9  #include <cstring>
10 #include <stdexcept>
11 #include <string>
12
13 namespace lcqi {
14 namespace {
15
16 constexpr std::int32_t LCQI_AVX2_OUTPUT_BLOCK = 8;
17
18 void require_positive(std::int32_t value, const char* name) {
19     if (value <= 0) {
20         throw std::runtime_error(std::string(name) + " must be positive");
21     }
22 }
23
24 std::size_t checked_size(std::int32_t value, const char* name) {
25     if (value < 0) {
26         throw std::runtime_error(std::string(name) + " cannot be negative");
27     }
28     return static_cast<std::size_t>(value);
29 }
30
31 std::int32_t round_up_to_block(std::int32_t value, std::int32_t block_size) {
32     require_positive(value, "value");
33     require_positive(block_size, "block_size");
34     return ((value + block_size - 1) / block_size) * block_size;
35 }
36
37 void validate_avx2_contract(const PackedLinearI8& layer,
38                             std::span<const float> input,
39                             std::span<float> output) {
40     require_positive(layer.input_size, "packed input_size");
41     require_positive(layer.output_size, "packed output_size");
42     if (layer.output_block_size != LCQI_AVX2_OUTPUT_BLOCK) {
43         throw std::runtime_error("AVX2 packed kernel requires output_block_size ←
44     ↪ == 8");
45     }
46     if (input.size() != checked_size(layer.input_size, "input_size")) {
47         throw std::runtime_error("AVX2 packed kernel input size mismatch");
48     }
49     if (output.size() != checked_size(layer.output_size, "output_size")) {
50         throw std::runtime_error("AVX2 packed kernel output size mismatch");
51     }
52 }

```

```

51     if (layer.bias.size() != checked_size(layer.output_size, "output_size")) {
52         throw std::runtime_error("AVX2 packed kernel bias size mismatch");
53     }
54     const std::int32_t padded_outputs =
55         round_up_to_block(layer.output_size, layer.output_block_size);
56     const std::size_t expected_packed_size =
57         checked_size(padded_outputs, "padded_outputs") *
58         checked_size(layer.input_size, "input_size");
59     if (layer.packed_weights.size() != expected_packed_size) {
60         throw std::runtime_error("AVX2 packed kernel packed weight size ↵
↵ mismatch");
61     }
62 }
63
64 } // namespace
65
66 bool linear_i8_packed_avx2_available() noexcept {
67     #if defined(__GNUC__) || defined(__clang__)
68         __builtin_cpu_init();
69         return __builtin_cpu_supports("avx2") && __builtin_cpu_supports("fma");
70     #else
71         return true;
72     #endif
73 }
74
75 void linear_i8_packed_avx2(const PackedLinearI8& layer,
76                             std::span<const float> input,
77                             std::span<float> output) {
78     if (!linear_i8_packed_avx2_available()) {
79         throw std::runtime_error("AVX2/FMA is not available on this CPU");
80     }
81     validate_avx2_contract(layer, input, output);
82
83     const std::int32_t padded_outputs =
84         round_up_to_block(layer.output_size, layer.output_block_size);
85     std::size_t packed_index = 0;
86     alignas(32) float block_output[LCQI_AVX2_OUTPUT_BLOCK] = {};
87
88     for (std::int32_t output_block = 0; output_block < padded_outputs;
89         output_block += LCQI_AVX2_OUTPUT_BLOCK) {
90         _mm256 accumulator = _mm256_setzero_ps();
91
92         for (std::int32_t offset = 0; offset < LCQI_AVX2_OUTPUT_BLOCK; ++offset ↵
↵ ) {
93             const std::int32_t output_index = output_block + offset;
94             block_output[checked_size(offset, "offset")] =
95                 output_index < layer.output_size
96                 ? layer.bias[checked_size(output_index, "output_index")]
97                 : 0.0F;
98         }
99         accumulator = _mm256_load_ps(block_output);
100

```



```

101     for (std::int32_t input_index = 0; input_index < layer.input_size; ++input_index) {
102         std::uint64_t packed_bytes = 0;
103         std::memcpy(&packed_bytes,
104                     layer.packed_weights.data() + packed_index,
105                     sizeof(packed_bytes));
106         const __m128i packed_i8 =
107             _mm_cvtsi64_si128(static_cast<long long>(packed_bytes));
108         packed_index += LCQI_AVX2_OUTPUT_BLOCK;
109         const __m256i weights_i32 = _mm256_cvtepi8_epi32(packed_i8);
110         const __m256 weights_f32 = _mm256_cvtepi32_ps(weights_i32);
111         const __m256 input_scaled =
112             _mm256_set1_ps(input[checked_size(input_index, "input_index")])
113         * layer.scale);
114         accumulator = _mm256_fmadd_ps(weights_f32, input_scaled,
115         accumulator);
116     }
117     _mm256_store_ps(block_output, accumulator);
118     for (std::int32_t offset = 0; offset < LCQI_AVX2_OUTPUT_BLOCK; ++offset) {
119         const std::int32_t output_index = output_block + offset;
120         if (output_index < layer.output_size) {
121             output[checked_size(output_index, "output_index")] =
122                 block_output[checked_size(offset, "offset")];
123         }
124     }
125 }
126
127 } // namespace lcqi
128
129 #endif

```

第零部分

# Transformer、KV Cache 与服务运行时

## 第 5 章实践：Transformer 切片与 KV Cache 状态

本章对应原书中的第三册“AI 推理原理、KV cache 与计算账本”和“KV cache 实现细节”。不要把它当作概念复述，本章要把 RMSNorm、Q/K/V、RoPE、GQA、KV slot、position、state edge 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `kv_cache`，核心命令或测试入口是 `lcqi_trace`。

**Listing 5.1** 第 5 章实践：Transformer 切片与 KV Cache 状态的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_trace / kv_cache
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：固定 prompt `tok1tok2tok3`，观察第一个 decode step。读者要先手写预期结果，再运行项目观察输出。动作是：把 trace 中 tokenizer、q/k/v、`kv_cache_slot`、attention 和 logits 串成状态表。如果输出里没有 `token_position` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：KV cache 是请求状态，不是临时 activation；position、layer、head、layout 和生命周期必须进入 trace 这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

### 四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交 KV cache 状态表：每层每个 token 写入哪里、何时读取、何时禁止复用。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

## 第6章实践：Reference Decoder、logits 与采样

本章对应原书中的第三册“Reference decoder 实现”和“推理算法：logits、softmax、采样”。不要把它当作概念复述，本章要把 reference decoder、logits golden、argmax sampler、tokenizer contract hash、BOS/EOS 放到同一个可运行项目里检查。贯穿代码仍然是 [books/cpu-volume-3/labs/linux\\_cpu\\_inference](#)；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 **argmax**，核心命令或测试入口是 **lcqi\_trace**。

**Listing 6.1** 第6章实践：Reference Decoder、logits 与采样的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_trace / argmax
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：让 trace 固定输出 tokenizer contract 和 logits golden。读者要先手写预期结果，再运行项目观察输出。动作是：比较 generated token、logits 数组和 tokenizer hash。如果输出里没有 **tokenizer\_contract** 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：采样前必须先证明 logits 可复查；sampler 的随机性、温度和 top-k 只能建立在 reference 通过之后这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 [books/cpu-volume-3/labs/linux\\_cpu\\_inference/README.md](#)、相关 **include/lcqi** 头文件和一个 **src** 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

### 四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交采样报告：argmax、top-k、temperature 三种策略各自需要哪些额外字段才能复现。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

## 第 7 章实践：服务调度、SLO 与资源预算

本章对应原书中的第三册“业务服务实战：请求、调度、队列、取消与资源预算”。不要把它当作概念复述，本章要把 admission、KV budget、queue wait、TTFT、TPOT、cancel、reject、p95 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `ttft_ms`，核心命令或测试入口是 `lcqi_serving_probe`。

Listing 7.1 第 7 章实践：服务调度、SLO 与资源预算的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_serving_probe / ttft_ms
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：运行 serving probe 中短请求、RAG 请求、取消请求和超长请求。读者要先手写预期结果，再运行项目观察输出。动作是：解释 `memory_budget_exceeded`、cancel reclaim、`queue_wait_p95_ms` 和 `rejection_rate`。如果输出里没有 `rejection_rate` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：推理服务不是只看 tokens/s；是否接收请求、如何预留 KV、取消后是否回收，决定系统是否可运营这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

### 四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交 SLO 表：每类请求的 prompt/new token、预算、是否接受、TTFT、TPOT、拒绝原因和未验证边界。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

## 第 8 章实践：RAG、Agent 与状态图边界

本章对应原书中的第三册“上层 AI 应用编排：RAG、Agent、工具和状态图”。不要把它当作概念复述，本章要把 retrieval、tool call、prompt assembly、state graph、budget、trace id 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `req_rag`，核心命令或测试入口是 `lcqi_serving_probe`。

Listing 8.1 第 8 章实践：RAG、Agent 与状态图边界的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_serving_probe / req_rag
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：把 RAG 请求拆成 retrieve、rerank、prompt build、prefill、decode 五段。读者要先手写预期结果，再运行项目观察输出。动作是：为每段写输入输出和失败后能否重试。如果输出里没有 `prompt_tokens` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：上层编排不能把外部工具结果和模型状态混成字符串；每个边界都要有身份、预算和可重放记录这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

### 四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交一个 RAG 状态图：节点、边、超时、重试、缓存键、prompt 版本和审计字段。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

第零部分

# 模型导入、AI 编译器与内存计划



## 第9章实践：模型导入、manifest 与 Shape Verifier

本章对应原书中的第三册“模型导入、manifest 与 shape verifier”。不要把它当作概念复述，本章要把 safetensors header、metadata、tensor name、dtype、shape、`data_offsets`、payload boundary 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `load_safetensors_manifest`，核心命令或测试入口是 `lcqi_tests`。

**Listing 9.1** 第9章实践：模型导入、manifest 与 Shape Verifier 的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_tests / load_safetensors_manifest
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：加载测试临时 safetensors fixture。读者要先手写预期结果，再运行项目观察输出。动作是：故意构造 offset 越界和 dtype/shape 字节数不匹配。如果输出里没有 `data_offsets` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：模型导入的第一职责是早拒绝：名字、shape、dtype、offset 和 payload 都必须在 kernel 运行前被验证这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

### 四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交 manifest gate 报告：合法 fixture 和两个坏 fixture 的字段、错误原因和拒绝位置。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

## 第 10 章实践：真实模型加载闭环与首 Token Trace

本章对应原书中的第三册“真实模型加载闭环：Tokenizer、权重格式与首 token trace”。不要把它当作概念复述，本章要把 tokenizer、weight mapping、chat template、first token trace、checksum 放到同一个可运行项目里检查。贯穿代码仍然是 [books/cpu-volume-3/labs/linux\\_cpu\\_inference](#)；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

本章新增一个明确边界：tiny fixture 负责手算和逐层 oracle；LCQI GPT-2 reference path 负责证明自研代码已经能加载标准 GPT-2 safetensors、执行 byte-level BPE、跑 GPT-2 forward 并 greedy 生成；SmolLM2 small smoke 负责证明外部 llama.cpp 路径能加载 GGUF small 模型。三者不能互相冒充。tiny MLP 算对，不等于 GPT-2 完成；GPT-2 reference 能出一个 token，不等于高性能 KV cache runtime 完成；SmolLM2 能输出文本，也不等于 LCQI 自研 GGUF 后端完成。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里有三组核心对象。第一组是 [tiny\\_tokenizer.txt](#)、reference decoder 和 [lcqi\\_trace](#)，用于手算 token 合同和逐 checkpoint 追踪。第二组是 [gpt2\\_reference.hpp/.cpp](#)、[lcqi\\_gpt2](#) 和 [tools/run\\_gpt2\\_smoke.py](#)，用于自研 GPT-2 safetensors 路径。第三组是 [tools/run\\_smolllm2\\_small\\_smoke.py](#)，用于下载并校验 SmolLM2-135M-Instruct 的 GGUF 量化文件，再通过 llama.cpp 在 CPU 上实际生成 assistant 文本。

Listing 10.1 第 10 章实践：真实模型加载闭环与首 Token Trace 的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_trace / tiny_tokenizer.txt
books/cpu-volume-3/build/lcqi-debug/labs/linux_cpu_inference/lcqi_gpt2
make cpu3-gpt2-smoke
make cpu3-smolllm2-smoke
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：从 tokenizer fixture 开始，不跳到真实 7B。读者要先手写预期结果，再运行项目观察输出。动作是：解释 BOS/EOS、UNK、template hash 和 decoder 输入 tokens 的边界。如果输出里没有 `tokens=[0,1,2,3,4]` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：真实模型加载不是下载权重；必须先让 tokenizer 合同、权重 role、shape verifier 和首 token trace 形成闭环。这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、自研 GPT-2 reference smoke：先把标准 safetensors 跑通

GPT-2 不是 LLaMA。它使用 LayerNorm，不是 RMSNorm；使用 learned absolute position embedding，不是 RoPE；attention 里的 Q/K/V 在 HuggingFace 权重中是 packed Conv1D 权重；MLP 使用 GELU，不是 SwiGLU；`lm_head` 通常和 token embedding 共享权重。把 GPT-2 硬塞进原来的 LLaMA-like tiny decoder，会让概念混乱，所以 LCQI 新增了独立的 `gpt2_reference` 模块。

仓库内最小验收分两层。第一层不下载模型，运行 tiny GPT-2 fixture：

**Listing 10.2** 仓库内 tiny GPT-2 数学闭环

```
books/cpu-volume-3/build/lcqi-debug/labs/linux_cpu_inference/lcqi_gpt2
```

它固定 prompt ids 为 123，检查 logits、predicted token 和 greedy generated ids。对应测试还会临时写出一个 HuggingFace 风格 `model.safetensors`，再用 loader 读回来，覆盖 tensor name mapping、shape verifier、Conv1D 转置和 tied lm head。坏 shape 必须在 loader 阶段被拒绝。

第二层下载真实 GPT-2 资产：

**Listing 10.3** LCQI 自研 GPT-2 smoke

```
make cpu3-gpt2-smoke
```

脚本下载 `openai-community/gpt2` 的 `config.json`、`vocab.json`、`merges.txt` 和 `model.safetensors` 到用户缓存目录，不提交模型文件。最近一次报告写到：

**Listing 10.4** GPT-2 smoke 报告关键字段

```
books/cpu-volume-3/results/lcqi-gpt2-smoke.txt
prompt_ids 15496 11 616 1438 318
generated_ids 15496 11 616 1438 318 1757
generated_text Hello, my name is John
validation=PASS
```

这说明自研 LCQI C++ reference path 已经能跑标准 GPT-2 safetensors 的 tokenizer、权重导入、forward 和 greedy generation。边界也要写清：当前实现为了可读和可验证，会重跑完整前缀，没有 KV cache、没有 packed weight、没有多线程，也没有把 logits 与 PyTorch/Transformers 的逐数值 golden 报告纳入仓库。因此它是正确性闭环，不是性能结论。

#### 四、真实 small 模型 smoke：不用 tiny，先让开源模型实际跑起来

本章的真实模型 smoke 固定为 `SmolLM2-135M-Instruct`，规模是 135M 参数，不是 tiny 随机夹具。脚本使用 GGUF 文件 `SmolLM2-135M-Instruct-Q4_K_M.gguf`，来源仓库是 `bartowski/SmolLM2-135M-Instruct-GGUF`，原模型组织是 `HuggingFaceTB`，license 字段为 `apache-2.0`。脚本不会把模型文件提交到仓库，而是缓存到用户目录；验收时检查文件大小 105454432 字节和 SHA256：

**Listing 10.5** SmolLM2 small GGUF 文件身份

```
model_source_id=HuggingFaceTB/SmolLM2-135M-Instruct
model_gguf_repo_id=bartowski/SmolLM2-135M-Instruct-GGUF
model_file=SmolLM2-135M-Instruct-Q4_K_M.gguf
model_expected_bytes=105454432
model_expected_sha256=2↵
↵ e8040ceae7815abe0dcb3540b9995eaa1fa0d2ca9e797d0a635ae4433c68c2d
```

运行命令是：

**Listing 10.6** 真实 small 模型 CPU 冒烟

```
make cpu3-smolllm2-smoke
```

这条命令会做四步。第一，下载或复用 GGUF 文件，并校验 SHA256，防止“文件名对了但内容不对”。第二，拉取固定 commit 的 `llama.cpp` 并构建 `llama-simple-chat`，避免依赖本机临时安装。第三，用 `-ngl0` 强制 CPU 路径，输入一个短 prompt，确认程序能走过 tokenizer、chat template、prefill、decode 和 sampler。第四，把报告写到：

Listing 10.7 真实模型 smoke 报告路径

```
books/cpu-volume-3/results/lcqi-smolllm2-small-model-smoke.txt
```

报告里必须至少出现这些字段：`model_expected_sha256`、`llama_cpp_commit`、`command`、`exit_code=0`、`validation=PASS` 和 `assistant_response`。如果只有下载日志，没有 assistant response，不能算通过；如果 assistant response 内容答错了，也不能改写成“质量通过”。本章只验证真实模型资产可加载、可 tokenize、可 decode、可生成非空文本。

## 五、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：运行 GPT-2 smoke。报告不许只贴“生成了 John”，必须解释 `config.json`、`vocab.json`、`merges.txt`、`model.safetensors` 各自对应哪段代码，为什么 GPT-2 需要 byte-level BPE，为什么 HuggingFace Conv1D 权重要转置，为什么 `lm_head` 可以 tied 到 embedding。

任务四：运行真实 small 模型 smoke。报告不许只贴“命令执行成功”，必须解释脚本为什么校验模型 hash、为什么固定 `llama.cpp` commit、为什么使用 `-ngl0`，以及为什么 assistant response 非空只是 smoke，不是质量评测。

任务五：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。GPT-2 反例可以是：tensor name 缺失、shape 和 config 不一致、vocab 中缺少 BPE merge 后 token、`max_new_tokens` 超过 context。真实模型 smoke 的反例可以是：模型 SHA256 不匹配、网络不可用、`cmake` 缺失、assistant response 为空、命令超时。每个反例都要写清期望处理方式。

## 六、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。新增 GPT-2 smoke 后，最低验收还包括：真实 GPT-2 文件身份有 hash，prompt ids 可解释，generated ids 可解释，文本反解码可解释，并明确声明这不是高性能推理结论。新增 SmolLM2 smoke 后，最低验收还包括：模型身份有 hash，模型规模不是 tiny，CPU 路径实际运行，报告里有非空 assistant response，并明确声明这不是自研 LCQI 完整 GGUF 支持。

递进大作业：提交 GPT-2 reference path 的下一阶段迁移清单：KV cache、prefill/decode 分离、packed weight、线程并行、Transformers golden logits、更多模型方言 name mapping、分片 safetensors 和性能报告。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

## 代码走读: Reference Decoder、Tokenizer、Safetensors 与 GPT-2

### 从 tiny MLP 走向 decoder-only 模型

Reference decoder 和 GPT-2 reference path 是 LCQI 从教学闭环走向真实模型导入的关键层。这里的代码更长，是因为真实模型不只有矩阵乘法：它还需要 tokenizer、位置、归一化、QKV 拆分、attention mask、KV cache、激活函数、lm head、权重命名映射、dtype 转换和坏资产拒绝。

### Tiny Reference Decoder

`reference_decoder.cpp` 实现 embedding、RMSNorm、Q/K/V、RoPE、KV cache、attention、SwiGLU、final norm、logits 和 sampler，并在关键位置生成 trace checkpoint。读者应按 checkpoint 顺序读，而不是从最终 logits 倒推。

Listing 10.8 src/reference\_decoder.cpp

```

1 #include <lcqi/reference_decoder.hpp>
2
3 #include <algorithm>
4 #include <array>
5 #include <cmath>
6 #include <limits>
7 #include <stdexcept>
8 #include <string>
9
10 namespace lcqi {
11 namespace {
12
13 constexpr float LCQI_SOFTMAX_NEGATIVE_INFINITY = -std::numeric_limits<float>::infinity();
14 constexpr std::int32_t LCQI_ROPE_PAIR_WIDTH = 2;
15 constexpr const char* LCQI_TRACE_DTYPE_F32 = "f32";
16 constexpr const char* LCQI_TRACE_LAYOUT_CONTIGUOUS = "contiguous";
17 constexpr const char* LCQI_TRACE_LAYOUT_ROW_MAJOR = "row_major";
18 constexpr const char* LCQI_TRACE_LAYOUT_KV_CONTIGUOUS = "kv_contiguous";
19
20 void require_positive(std::int32_t value, const char* name) {
21     if (value <= 0) {
22         throw std::runtime_error(std::string(name) + " must be positive");
23     }
24 }
25
26 void validate_config(const DecoderConfig& config) {
27     require_positive(config.hidden_size, "hidden_size");
28     require_positive(config.query_heads, "query_heads");
29     require_positive(config.kv_heads, "kv_heads");
30     require_positive(config.head_dim, "head_dim");
31     require_positive(config.intermediate_size, "intermediate_size");
32     require_positive(config.vocab_size, "vocab_size");

```



```

33     require_positive(config.max_context, "max_context");
34     if (config.query_heads % config.kv_heads != 0) {
35         throw std::runtime_error("query_heads must be divisible by kv_heads");
36     }
37     if (config.hidden_size != config.query_heads * config.head_dim) {
38         throw std::runtime_error("hidden_size must equal query_heads * head_dim↵
↵ ");
39     }
40     if (config.head_dim % LCQI_ROPE_PAIR_WIDTH != 0) {
41         throw std::runtime_error("head_dim must be even for RoPE");
42     }
43     if (config.rope_base <= 1.0F) {
44         throw std::runtime_error("rope_base must be greater than 1");
45     }
46 }
47
48 std::size_t checked_size(std::int32_t value, const char* name) {
49     if (value < 0) {
50         throw std::runtime_error(std::string(name) + " cannot be negative");
51     }
52     return static_cast<std::size_t>(value);
53 }
54
55 std::int32_t kv_width(const DecoderConfig& config) {
56     return config.kv_heads * config.head_dim;
57 }
58
59 void validate_linear(const LinearWeightsF32& layer, const char* name) {
60     require_positive(layer.input_size, "linear input_size");
61     require_positive(layer.output_size, "linear output_size");
62     const std::size_t expected_weights =
63         checked_size(layer.input_size, name) * checked_size(layer.output_size, ↵
↵ name);
64     if (layer.weights.size() != expected_weights) {
65         throw std::runtime_error(std::string(name) + " weight size mismatch");
66     }
67     if (!layer.bias.empty() &&
68         layer.bias.size() != checked_size(layer.output_size, name)) {
69         throw std::runtime_error(std::string(name) + " bias size mismatch");
70     }
71 }
72
73 void validate_span_size(std::size_t actual, std::int32_t expected, const char* ↵
↵ name) {
74     if (actual != checked_size(expected, name)) {
75         throw std::runtime_error(std::string(name) + " size mismatch");
76     }
77 }
78
79 std::int32_t argmax(std::span<const float> values) {
80     if (values.empty()) {
81         throw std::runtime_error("cannot take argmax of empty values");

```

```

82     }
83     std::int32_t best_index = 0;
84     float best_value = values[0];
85     for (std::int32_t index = 1; index < static_cast<std::int32_t>(values.size()
↵    )); ++index) {
86         const float candidate = values[static_cast<std::size_t>(index)];
87         if (candidate > best_value) {
88             best_value = candidate;
89             best_index = index;
90         }
91     }
92     return best_index;
93 }
94
95 float silu(float value) {
96     return value / (1.0F + std::exp(-value));
97 }
98
99 std::span<const float> row_span(std::span<const float> values,
100                                std::int32_t row,
101                                std::int32_t row_width) {
102     const std::size_t begin = checked_size(row, "row") * checked_size(row_width,
↵    , "row_width");
103     return values.subspan(begin, checked_size(row_width, "row_width"));
104 }
105
106 void add_inplace(std::span<float> target, std::span<const float> source) {
107     if (target.size() != source.size()) {
108         throw std::runtime_error("residual add size mismatch");
109     }
110     for (std::size_t index = 0; index < target.size(); ++index) {
111         target[index] += source[index];
112     }
113 }
114
115 void swiglu_mlp(const DecoderLayerWeightsF32& layer,
116                std::span<const float> input,
117                std::span<float> output) {
118     std::vector<float> gate(checked_size(layer.w_gate.output_size, "gate"), 0.0
↵    F);
119     std::vector<float> up(checked_size(layer.w_up.output_size, "up"), 0.0F);
120     std::vector<float> hidden(gate.size(), 0.0F);
121     linear_f32(layer.w_gate, input, gate);
122     linear_f32(layer.w_up, input, up);
123     if (gate.size() != up.size()) {
124         throw std::runtime_error("SwiGLU gate/up size mismatch");
125     }
126     for (std::size_t index = 0; index < hidden.size(); ++index) {
127         hidden[index] = silu(gate[index]) * up[index];
128     }
129     linear_f32(layer.w_down, hidden, output);
130 }

```



```

131
132 float checksum(std::span<const float> values) {
133     float sum = 0.0F;
134     for (const float value : values) {
135         sum += value;
136     }
137     return sum;
138 }
139
140 float max_abs(std::span<const float> values) {
141     float result = 0.0F;
142     for (const float value : values) {
143         result = std::max(result, std::fabs(value));
144     }
145     return result;
146 }
147
148 std::vector<std::int32_t> contiguous_stride(std::span<const std::int32_t> shape ←
    ↪ ) {
149     std::vector<std::int32_t> stride(shape.size(), 1);
150     std::int32_t running = 1;
151     for (std::int32_t index = static_cast<std::int32_t>(shape.size()) - 1; ←
    ↪ index >= 0; --index) {
152         stride[checked_size(index, "stride index")] = running;
153         running *= shape[checked_size(index, "shape index")];
154     }
155     return stride;
156 }
157
158 void add_checkpoint(std::vector<ReferenceTensorCheckpoint>* checkpoints,
159                     const char* name,
160                     std::int32_t token_position,
161                     std::int32_t layer_id,
162                     std::span<const std::int32_t> shape,
163                     const char* layout,
164                     std::span<const float> values) {
165     if (checkpoints == nullptr) {
166         return;
167     }
168     ReferenceTensorCheckpoint checkpoint;
169     checkpoint.name = name;
170     checkpoint.token_position = token_position;
171     checkpoint.layer_id = layer_id;
172     checkpoint.shape.assign(shape.begin(), shape.end());
173     checkpoint.stride = contiguous_stride(shape);
174     checkpoint.dtype = LCQI_TRACE_DTYPE_F32;
175     checkpoint.layout = layout;
176     checkpoint.checksum = checksum(values);
177     checkpoint.max_abs = max_abs(values);
178     checkpoint.values.assign(values.begin(), values.end());
179     checkpoints->push_back(std::move(checkpoint));
180 }

```

```

181
182 void swiglu_mlp_with_trace(const DecoderLayerWeightsF32& layer,
183                             std::span<const float> input,
184                             std::span<float> output,
185                             std::int32_t token_position,
186                             std::int32_t layer_id,
187                             std::vector<ReferenceTensorCheckpoint>* checkpoints)↵
    ↵ {
188     std::vector<float> gate(check_size(layer.w_gate.output_size, "gate"), 0.0↵
    ↵ F);
189     std::vector<float> up(check_size(layer.w_up.output_size, "up"), 0.0F);
190     std::vector<float> hidden(gate.size(), 0.0F);
191     linear_f32(layer.w_gate, input, gate);
192     linear_f32(layer.w_up, input, up);
193     if (gate.size() != up.size()) {
194         throw std::runtime_error("SwiGLU gate/up size mismatch");
195     }
196     const std::array<std::int32_t, 1> intermediate_shape{layer.w_gate.↵
    ↵ output_size};
197     add_checkpoint(checkpoints,
198                     "mlp_gate",
199                     token_position,
200                     layer_id,
201                     intermediate_shape,
202                     LCQI_TRACE_LAYOUT_CONTIGUOUS,
203                     gate);
204     add_checkpoint(checkpoints,
205                     "mlp_up",
206                     token_position,
207                     layer_id,
208                     intermediate_shape,
209                     LCQI_TRACE_LAYOUT_CONTIGUOUS,
210                     up);
211     for (std::size_t index = 0; index < hidden.size(); ++index) {
212         hidden[index] = silu(gate[index]) * up[index];
213     }
214     add_checkpoint(checkpoints,
215                     "mlp_hidden",
216                     token_position,
217                     layer_id,
218                     intermediate_shape,
219                     LCQI_TRACE_LAYOUT_CONTIGUOUS,
220                     hidden);
221     linear_f32(layer.w_down, hidden, output);
222 }
223
224 void forward_layer(const DecoderConfig& config,
225                   const DecoderLayerWeightsF32& layer,
226                   std::int32_t layer_id,
227                   std::int32_t model_position,
228                   ReferenceKVCache& cache,
229                   std::span<float> hidden,

```

```

230         std::vector<ReferenceTensorCheckpoint>* checkpoints = ↵
↵ nullptr) {
231     validate_span_size(hidden.size(), config.hidden_size, "hidden");
232
233     std::vector<float> normed(check_size(config.hidden_size, "hidden_size"), ↵
↵ 0.0F);
234     std::vector<float> query(check_size(config.hidden_size, "hidden_size"), ↵
↵ 0.0F);
235     std::vector<float> key(check_size(kv_width(config), "kv_width"), 0.0F);
236     std::vector<float> value(check_size(kv_width(config), "kv_width"), 0.0F);
237     std::vector<float> attention(check_size(config.hidden_size, "hidden_size"↵
↵ ), 0.0F);
238     std::vector<float> attention_projected(check_size(config.hidden_size, "↵
↵ hidden_size"), 0.0F);
239     std::vector<float> mlp(check_size(config.hidden_size, "hidden_size"), 0.0↵
↵ F);
240
241     const std::array<std::int32_t, 1> hidden_shape{config.hidden_size};
242     const std::array<std::int32_t, 2> query_shape{config.query_heads, config.↵
↵ head_dim};
243     const std::array<std::int32_t, 2> kv_shape{config.kv_heads, config.head_dim↵
↵ };
244     const std::array<std::int32_t, 4> kv_slot_shape{
245         1, 1, config.kv_heads, config.head_dim};
246
247     rms_norm(hidden, layer.rms_attention_weight, config.rms_epsilon, normed);
248     add_checkpoint(checkpoints,
249         "attn_norm",
250         model_position,
251         layer_id,
252         hidden_shape,
253         LCQI_TRACE_LAYOUT_CONTIGUOUS,
254         normed);
255     linear_f32(layer.wq, normed, query);
256     linear_f32(layer.wk, normed, key);
257     linear_f32(layer.wv, normed, value);
258     add_checkpoint(checkpoints,
259         "q_before_rope",
260         model_position,
261         layer_id,
262         query_shape,
263         LCQI_TRACE_LAYOUT_CONTIGUOUS,
264         query);
265     add_checkpoint(checkpoints,
266         "k_before_rope",
267         model_position,
268         layer_id,
269         kv_shape,
270         LCQI_TRACE_LAYOUT_CONTIGUOUS,
271         key);
272     add_checkpoint(checkpoints,
273         "v",

```

```

274         model_position,
275         layer_id,
276         kv_shape,
277         LCQI_TRACE_LAYOUT_CONTIGUOUS,
278         value);
279     apply_rope(query, config.query_heads, config.head_dim, model_position, ↵
↵ config.rope_base);
280     apply_rope(key, config.kv_heads, config.head_dim, model_position, config.↵
↵ rope_base);
281     add_checkpoint(checkpoints,
282         "q_after_rope",
283         model_position,
284         layer_id,
285         query_shape,
286         LCQI_TRACE_LAYOUT_CONTIGUOUS,
287         query);
288     add_checkpoint(checkpoints,
289         "k_after_rope",
290         model_position,
291         layer_id,
292         kv_shape,
293         LCQI_TRACE_LAYOUT_CONTIGUOUS,
294         key);
295
296     cache.append(layer_id, model_position, key, value);
297     add_checkpoint(checkpoints,
298         "kv_cache_key_slot",
299         model_position,
300         layer_id,
301         kv_slot_shape,
302         LCQI_TRACE_LAYOUT_KV_CONTIGUOUS,
303         key);
304     attention_decode(config, cache, layer_id, model_position, query, attention)↵
↵ ;
305     add_checkpoint(checkpoints,
306         "attention_out",
307         model_position,
308         layer_id,
309         hidden_shape,
310         LCQI_TRACE_LAYOUT_CONTIGUOUS,
311         attention);
312     linear_f32(layer.wo, attention, attention_projected);
313     add_inplace(hidden, attention_projected);
314     add_checkpoint(checkpoints,
315         "post_attention_residual",
316         model_position,
317         layer_id,
318         hidden_shape,
319         LCQI_TRACE_LAYOUT_CONTIGUOUS,
320         hidden);
321
322     rms_norm(hidden, layer.rms_mlp_weight, config.rms_epsilon, normed);

```

```

323     add_checkpoint(checkpoints,
324                     "mlp_norm",
325                     model_position,
326                     layer_id,
327                     hidden_shape,
328                     LCQI_TRACE_LAYOUT_CONTIGUOUS,
329                     normed);
330     if (checkpoints == nullptr) {
331         swiglu_mlp(layer, normed, mlp);
332     } else {
333         swiglu_mlp_with_trace(layer, normed, mlp, model_position, layer_id, ←
↪ checkpoints);
334     }
335     add_inplace(hidden, mlp);
336     add_checkpoint(checkpoints,
337                     "layer_out",
338                     model_position,
339                     layer_id,
340                     hidden_shape,
341                     LCQI_TRACE_LAYOUT_CONTIGUOUS,
342                     hidden);
343 }
344
345 LinearWeightsF32 make_linear(std::int32_t input_size,
346                               std::int32_t output_size,
347                               std::vector<float> weights,
348                               std::vector<float> bias = {}) {
349     LinearWeightsF32 layer;
350     layer.input_size = input_size;
351     layer.output_size = output_size;
352     layer.weights = std::move(weights);
353     layer.bias = std::move(bias);
354     validate_linear(layer, "tiny linear");
355     return layer;
356 }
357
358 std::vector<float> zeros(std::int32_t count) {
359     return std::vector<float>(checked_size(count, "zero count"), 0.0F);
360 }
361
362 } // namespace
363
364 ReferenceDecodeTraceResult run_reference_decode_with_trace(
365     const ReferenceDecoderModel& model,
366     std::span<const std::int32_t> token_ids) {
367     validate_config(model.config);
368     if (token_ids.empty()) {
369         throw std::runtime_error("reference decode needs at least one token");
370     }
371     if (token_ids.size() > checked_size(model.config.max_context, "max_context" ←
↪ )) {
372         throw std::runtime_error("token_ids exceed max_context");

```

```

373     }
374     if (model.token_embedding.size() !=
375         checked_size(model.config.vocab_size, "vocab_size") *
376         checked_size(model.config.hidden_size, "hidden_size")) {
377         throw std::runtime_error("token embedding size mismatch");
378     }
379     if (model.final_norm_weight.size() != checked_size(model.config.hidden_size,
380 ↪ , "hidden_size")) {
381         throw std::runtime_error("final norm weight size mismatch");
382     }
383     validate_linear(model.lm_head, "lm_head");
384
385     ReferenceDecodeTraceResult trace;
386     ReferenceKVCache cache(model.config, static_cast<std::int32_t>(model.layers.
387 ↪ .size()));
388     std::vector<float> hidden(checked_size(model.config.hidden_size, "
389 ↪ hidden_size"), 0.0F);
390     const std::array<std::int32_t, 1> hidden_shape{model.config.hidden_size};
391     for (std::int32_t position = 0; position < static_cast<std::int32_t>(
392 ↪ token_ids.size());
393         ++position) {
394         const std::int32_t token_id = token_ids[checked_size(position, "
395 ↪ position")];
396         if (token_id < 0 || token_id >= model.config.vocab_size) {
397             throw std::runtime_error("token id out of vocabulary");
398         }
399         const std::span<const float> embedding =
400             row_span(model.token_embedding, token_id, model.config.hidden_size)
401 ↪ ;
402         std::copy(embedding.begin(), embedding.end(), hidden.begin());
403         add_checkpoint(&trace.checkpoints,
404             "embedding",
405             position,
406             -1,
407             hidden_shape,
408             LCQI_TRACE_LAYOUT_ROW_MAJOR,
409             hidden);
410         for (std::int32_t layer_id = 0; layer_id < static_cast<std::int32_t>(
411 ↪ model.layers.size());
412             ++layer_id) {
413             forward_layer(model.config,
414                 model.layers[checked_size(layer_id, "layer_id")],
415                 layer_id,
416                 position,
417                 cache,
418                 hidden,
419                 &trace.checkpoints);
420         }
421     }
422
423     std::vector<float> normed(checked_size(model.config.hidden_size, "
424 ↪ hidden_size"), 0.0F);

```

```

417     std::vector<float> logits(checkered_size(model.config.vocab_size, "vocab_size" ←
    ↪ "), 0.0F);
418     const std::array<std::int32_t, 1> logits_shape{model.config.vocab_size};
419     const std::int32_t last_position = static_cast<std::int32_t>(token_ids.size ←
    ↪ ()) - 1;
420     rms_norm(hidden, model.final_norm_weight, model.config.rms_epsilon, normed) ←
    ↪ ;
421     add_checkpoint(&trace.checkpoints,
422                   "final_norm",
423                   last_position,
424                   -1,
425                   hidden_shape,
426                   LCQI_TRACE_LAYOUT_CONTIGUOUS,
427                   normed);
428     linear_f32(model.lm_head, normed, logits);
429     add_checkpoint(&trace.checkpoints,
430                   "logits",
431                   last_position,
432                   -1,
433                   logits_shape,
434                   LCQI_TRACE_LAYOUT_CONTIGUOUS,
435                   logits);
436
437     trace.result.predicted_token = argmax(logits);
438     trace.result.logits = std::move(logits);
439     return trace;
440 }
441
442 ReferenceKVCache::ReferenceKVCache(const DecoderConfig& config, std::int32_t ←
    ↪ layer_count)
443 : config_(config),
444   layer_count_(layer_count) {
445     validate_config(this->config_);
446     require_positive(this->layer_count_, "layer_count");
447     const std::size_t element_count =
448         checked_size(this->layer_count_, "layer_count") *
449         checked_size(this->config_.max_context, "max_context") *
450         checked_size(this->config_.kv_heads, "kv_heads") *
451         checked_size(this->config_.head_dim, "head_dim");
452     this->keys_.assign(element_count, 0.0F);
453     this->values_.assign(element_count, 0.0F);
454     this->written_.assign(
455         checked_size(this->layer_count_, "layer_count") *
456         checked_size(this->config_.max_context, "max_context"),
457         0);
458 }
459
460 void ReferenceKVCache::append(std::int32_t layer_id,
461                               std::int32_t model_position,
462                               std::span<const float> key,
463                               std::span<const float> value) {
464     validate_span_size(key.size(), kv_width(this->config_), "KV key");

```



```

465     validate_span_size(value.size(), kv_width(this->config_), "KV value");
466     this->validate_address(layer_id, model_position, 0);
467     for (std::int32_t kv_head = 0; kv_head < this->config_.kv_heads; ++kv_head) {
468         const std::size_t target = this->base_offset(layer_id, model_position,
469         kv_head);
469         const std::size_t source =
470             checked_size(kv_head, "kv_head") * checked_size(this->config_.
471         head_dim, "head_dim");
471         std::copy_n(key.begin() + static_cast<std::ptrdiff_t>(source),
472             this->config_.head_dim,
473             this->keys_.begin() + static_cast<std::ptrdiff_t>(target));
474         std::copy_n(value.begin() + static_cast<std::ptrdiff_t>(source),
475             this->config_.head_dim,
476             this->values_.begin() + static_cast<std::ptrdiff_t>(target));
477     };
478     const std::size_t written_index =
479         checked_size(layer_id, "layer_id") * checked_size(this->config_.
480         max_context, "max_context") +
481         checked_size(model_position, "model_position");
481     this->written_[written_index] = 1;
482     this->filled_tokens_ = std::max(this->filled_tokens_, model_position + 1);
483 }
484
485 std::span<const float> ReferenceKVCache::key(std::int32_t layer_id,
486                                             std::int32_t model_position,
487                                             std::int32_t kv_head) const {
488     this->validate_address(layer_id, model_position, kv_head);
489     const std::size_t offset = this->base_offset(layer_id, model_position,
490     kv_head);
491     return std::span<const float>(this->keys_.data() + offset,
492     checked_size(this->config_.head_dim, "
493     head_dim"));
494 }
495
496 std::span<const float> ReferenceKVCache::value(std::int32_t layer_id,
497                                             std::int32_t model_position,
498                                             std::int32_t kv_head) const {
499     this->validate_address(layer_id, model_position, kv_head);
500     const std::size_t offset = this->base_offset(layer_id, model_position,
501     kv_head);
502     return std::span<const float>(this->values_.data() + offset,
503     checked_size(this->config_.head_dim, "
504     head_dim"));
505 }
506
507 std::int32_t ReferenceKVCache::layer_count() const noexcept {
508     return this->layer_count_;
509 }
510
511 std::int32_t ReferenceKVCache::filled_tokens() const noexcept {

```

```

508     return this->filled_tokens_;
509 }
510
511 std::size_t ReferenceKVCache::base_offset(std::int32_t layer_id,
512                                           std::int32_t model_position,
513                                           std::int32_t kv_head) const {
514     return (((checked_size(layer_id, "layer_id") *
515                    checked_size(this->config.max_context, "max_context")) +
516            checked_size(model_position, "model_position")) *
517            checked_size(this->config.kv_heads, "kv_heads") +
518            checked_size(kv_head, "kv_head")) *
519            checked_size(this->config.head_dim, "head_dim");
520 }
521
522 void ReferenceKVCache::validate_address(std::int32_t layer_id,
523                                         std::int32_t model_position,
524                                         std::int32_t kv_head) const {
525     if (layer_id < 0 || layer_id >= this->layer_count_) {
526         throw std::runtime_error("KV layer_id out of range");
527     }
528     if (model_position < 0 || model_position >= this->config.max_context) {
529         throw std::runtime_error("KV model_position out of range");
530     }
531     if (kv_head < 0 || kv_head >= this->config.kv_heads) {
532         throw std::runtime_error("KV head out of range");
533     }
534 }
535
536 void rms_norm(std::span<const float> input,
537              std::span<const float> weight,
538              float epsilon,
539              std::span<float> output) {
540     if (input.size() != weight.size() || input.size() != output.size()) {
541         throw std::runtime_error("RMSNorm size mismatch");
542     }
543     if (input.empty()) {
544         throw std::runtime_error("RMSNorm input is empty");
545     }
546     float sum_square = 0.0F;
547     for (const float value : input) {
548         sum_square += value * value;
549     }
550     const float mean_square = sum_square / static_cast<float>(input.size());
551     const float scale = 1.0F / std::sqrt(mean_square + epsilon);
552     for (std::size_t index = 0; index < input.size(); ++index) {
553         output[index] = input[index] * scale * weight[index];
554     }
555 }
556
557 void linear_f32(const LinearWeightsF32& layer,
558               std::span<const float> input,
559               std::span<float> output) {

```

```

560     validate_linear(layer, "linear_f32");
561     validate_span_size(input.size(), layer.input_size, "linear input");
562     validate_span_size(output.size(), layer.output_size, "linear output");
563     for (std::int32_t out = 0; out < layer.output_size; ++out) {
564         const std::size_t row_base =
565             checked_size(out, "out") * checked_size(layer.input_size, "↵
↵ input_size");
566         float accumulator = layer.bias.empty() ? 0.0F : layer.bias[checked_size↵
↵ (out, "out")];
567         for (std::int32_t in = 0; in < layer.input_size; ++in) {
568             accumulator += layer.weights[row_base + checked_size(in, "in")] *
569                 input[checked_size(in, "in")];
570         }
571         output[checked_size(out, "out")] = accumulator;
572     }
573 }
574
575 void apply_rope(std::span<float> heads,
576               std::int32_t head_count,
577               std::int32_t head_dim,
578               std::int32_t model_position,
579               float rope_base) {
580     require_positive(head_count, "head_count");
581     require_positive(head_dim, "head_dim");
582     validate_span_size(heads.size(), head_count * head_dim, "RoPE heads");
583     if (head_dim % LCQI_ROPE_PAIR_WIDTH != 0) {
584         throw std::runtime_error("RoPE head_dim must be even");
585     }
586     if (model_position < 0) {
587         throw std::runtime_error("RoPE model_position cannot be negative");
588     }
589     if (rope_base <= 1.0F) {
590         throw std::runtime_error("RoPE base must be greater than 1");
591     }
592     for (std::int32_t head = 0; head < head_count; ++head) {
593         for (std::int32_t offset = 0; offset < head_dim; offset += ↵
↵ LCQI_ROPE_PAIR_WIDTH) {
594             const float exponent = static_cast<float>(offset) / static_cast<↵
↵ float>(head_dim);
595             const float theta = 1.0F / std::pow(rope_base, exponent);
596             const float angle = static_cast<float>(model_position) * theta;
597             const float cosine = std::cos(angle);
598             const float sine = std::sin(angle);
599             const std::size_t first =
600                 checked_size(head, "head") * checked_size(head_dim, "head_dim")↵
↵ +
601                 checked_size(offset, "offset");
602             const float x0 = heads[first];
603             const float x1 = heads[first + 1];
604             heads[first] = x0 * cosine - x1 * sine;
605             heads[first + 1] = x0 * sine + x1 * cosine;
606         }

```

```

607     }
608 }
609
610 void attention_decode(const DecoderConfig& config,
611                     const ReferenceKVCache& cache,
612                     std::int32_t layer_id,
613                     std::int32_t model_position,
614                     std::span<const float> query,
615                     std::span<float> output) {
616     validate_config(config);
617     validate_span_size(query.size(), config.hidden_size, "attention query");
618     validate_span_size(output.size(), config.hidden_size, "attention output");
619     if (model_position < 0 || model_position >= cache.filled_tokens()) {
620         throw std::runtime_error("attention model_position has not been written↵
↵ ");
621     }
622     std::fill(output.begin(), output.end(), 0.0F);
623
624     const std::int32_t group_size = config.query_heads / config.kv_heads;
625     const float score_scale = 1.0F / std::sqrt(static_cast<float>(config.↵
↵ head_dim));
626     std::vector<float> scores(check_size(model_position + 1, "score count"),
627                             LCQI_SOFTMAX_NEGATIVE_INFINITY);
628     std::vector<float> probabilities(scores.size(), 0.0F);
629
630     for (std::int32_t query_head = 0; query_head < config.query_heads; ++↵
↵ query_head) {
631         const std::int32_t kv_head = query_head / group_size;
632         const std::size_t query_base =
633             checked_size(query_head, "query_head") * checked_size(config.↵
↵ head_dim, "head_dim");
634         float max_score = LCQI_SOFTMAX_NEGATIVE_INFINITY;
635         for (std::int32_t position = 0; position <= model_position; ++position)↵
↵ {
636             const std::span<const float> key = cache.key(layer_id, position, ↵
↵ kv_head);
637             float dot = 0.0F;
638             for (std::int32_t dim = 0; dim < config.head_dim; ++dim) {
639                 dot += query[query_base + checked_size(dim, "dim")] *
640                     key[checked_size(dim, "dim")];
641             }
642             const float score = dot * score_scale;
643             scores[checked_size(position, "position")] = score;
644             max_score = std::max(max_score, score);
645         }
646
647         float probability_sum = 0.0F;
648         for (std::int32_t position = 0; position <= model_position; ++position)↵
↵ {
649             const std::size_t index = checked_size(position, "position");
650             const float unnormalized = std::exp(scores[index] - max_score);
651             probabilities[index] = unnormalized;

```

```

652         probability_sum += unnormalized;
653     }
654     if (probability_sum <= 0.0F) {
655         throw std::runtime_error("attention probability sum is invalid");
656     }
657     for (std::int32_t position = 0; position <= model_position; ++position) ←
    ↪ {
658         const float probability =
659             probabilities[checked_size(position, "position")] / ←
    ↪ probability_sum;
660         const std::span<const float> value = cache.value(layer_id, position ←
    ↪ , kv_head);
661         for (std::int32_t dim = 0; dim < config.head_dim; ++dim) {
662             output[query_base + checked_size(dim, "dim")] +=
663                 probability * value[checked_size(dim, "dim")];
664         }
665     }
666 }
667 }
668
669 ReferenceDecodeResult run_reference_decode(const ReferenceDecoderModel& model,
670     std::span<const std::int32_t> ←
    ↪ token_ids) {
671     validate_config(model.config);
672     if (token_ids.empty()) {
673         throw std::runtime_error("reference decode needs at least one token");
674     }
675     if (token_ids.size() > checked_size(model.config.max_context, "max_context" ←
    ↪ )) {
676         throw std::runtime_error("token_ids exceed max_context");
677     }
678     if (model.token_embedding.size() !=
679         checked_size(model.config.vocab_size, "vocab_size") *
680         checked_size(model.config.hidden_size, "hidden_size")) {
681         throw std::runtime_error("token embedding size mismatch");
682     }
683     if (model.final_norm_weight.size() != checked_size(model.config.hidden_size ←
    ↪ , "hidden_size")) {
684         throw std::runtime_error("final norm weight size mismatch");
685     }
686     validate_linear(model.lm_head, "lm_head");
687
688     ReferenceKVCache cache(model.config, static_cast<std::int32_t>(model.layers ←
    ↪ .size()));
689     std::vector<float> hidden(checked_size(model.config.hidden_size, " ←
    ↪ hidden_size"), 0.0F);
690     for (std::int32_t position = 0; position < static_cast<std::int32_t>( ←
    ↪ token_ids.size());
691         ++position) {
692         const std::int32_t token_id = token_ids[checked_size(position, " ←
    ↪ position")];
693         if (token_id < 0 || token_id >= model.config.vocab_size) {

```

```

694         throw std::runtime_error("token id out of vocabulary");
695     }
696     const std::span<const float> embedding =
697         row_span(model.token_embedding, token_id, model.config.hidden_size)↵
↵ ;
698     std::copy(embedding.begin(), embedding.end(), hidden.begin());
699     for (std::int32_t layer_id = 0; layer_id < static_cast<std::int32_t>(↵
↵ model.layers.size());
700         ++layer_id) {
701         forward_layer(model.config,
702                       model.layers[checked_size(layer_id, "layer_id")],
703                       layer_id,
704                       position,
705                       cache,
706                       hidden);
707     }
708 }
709
710 std::vector<float> normed(checked_size(model.config.hidden_size, "↵
↵ hidden_size"), 0.0F);
711 std::vector<float> logits(checked_size(model.config.vocab_size, "vocab_size↵
↵ "), 0.0F);
712 rms_norm(hidden, model.final_norm_weight, model.config.rms_epsilon, normed)↵
↵ ;
713 linear_f32(model.lm_head, normed, logits);
714
715 ReferenceDecodeResult result;
716 result.predicted_token = argmax(logits);
717 result.logits = std::move(logits);
718 return result;
719 }
720
721 ReferenceDecoderModel make_tiny_reference_decoder_model() {
722     ReferenceDecoderModel model;
723     model.config.hidden_size = 4;
724     model.config.query_heads = 2;
725     model.config.kv_heads = 1;
726     model.config.head_dim = 2;
727     model.config.intermediate_size = 6;
728     model.config.vocab_size = 5;
729     model.config.max_context = 8;
730     model.config.rms_epsilon = 1.0e-5F;
731     model.config.rope_base = 10000.0F;
732
733     model.token_embedding = {
734         0.20F, -0.10F, 0.05F, 0.30F,
735         -0.40F, 0.10F, 0.25F, -0.20F,
736         0.15F, 0.35F, -0.30F, 0.05F,
737         0.50F, -0.25F, 0.10F, -0.15F,
738         -0.05F, 0.20F, 0.40F, -0.35F,
739     };
740

```

```

741 DecoderLayerWeightsF32 layer;
742 layer.rms_attention_weight = {1.0F, 0.9F, 1.1F, 1.0F};
743 layer.wq = make_linear(4,
744                        4,
745                        {
746                            0.20F, -0.10F, 0.05F, 0.30F,
747                            -0.15F, 0.25F, 0.10F, -0.05F,
748                            0.05F, 0.10F, -0.20F, 0.15F,
749                            0.30F, -0.05F, 0.20F, 0.10F,
750                        });
751 layer.wk = make_linear(4,
752                        2,
753                        {
754                            0.10F, 0.20F, -0.10F, 0.05F,
755                            -0.05F, 0.15F, 0.25F, -0.20F,
756                        });
757 layer.wv = make_linear(4,
758                        2,
759                        {
760                            0.25F, -0.05F, 0.10F, 0.15F,
761                            -0.10F, 0.30F, 0.05F, 0.20F,
762                        });
763 layer.wo = make_linear(4,
764                        4,
765                        {
766                            0.20F, 0.10F, -0.05F, 0.15F,
767                            -0.10F, 0.25F, 0.20F, -0.05F,
768                            0.05F, -0.20F, 0.30F, 0.10F,
769                            0.15F, 0.05F, -0.10F, 0.25F,
770                        });
771 layer.rms_mlp_weight = {0.95F, 1.05F, 1.0F, 0.9F};
772 layer.w_gate = make_linear(4,
773                            6,
774                            {
775                                0.20F, -0.10F, 0.05F, 0.10F,
776                                -0.05F, 0.15F, 0.20F, -0.10F,
777                                0.10F, 0.05F, -0.15F, 0.25F,
778                                -0.20F, 0.10F, 0.15F, 0.05F,
779                                0.05F, -0.25F, 0.10F, 0.20F,
780                                0.15F, 0.05F, 0.05F, -0.15F,
781                            });
782 layer.w_up = make_linear(4,
783                          6,
784                          {
785                              0.10F, 0.20F, -0.05F, 0.05F,
786                              -0.15F, 0.05F, 0.25F, 0.10F,
787                              0.20F, -0.10F, 0.05F, 0.15F,
788                              0.05F, 0.10F, -0.20F, 0.25F,
789                              -0.05F, 0.15F, 0.10F, -0.10F,
790                              0.25F, -0.05F, 0.05F, 0.10F,
791                          });
792 layer.w_down = make_linear(6,

```



```

793         4,
794         {
795             0.10F, -0.05F, 0.15F, 0.20F, -0.10F, 0.05F,
796             -0.20F, 0.10F, 0.05F, -0.05F, 0.15F, 0.20F,
797             0.05F, 0.15F, -0.10F, 0.10F, 0.20F, -0.05F,
798             0.20F, 0.05F, 0.10F, -0.15F, -0.05F, 0.10F,
799         });
800     model.layers.push_back(std::move(layer));
801     model.final_norm_weight = {1.0F, 0.95F, 1.05F, 1.0F};
802     model.lm_head = make_linear(4,
803                                 5,
804                                 {
805                                     0.20F, -0.10F, 0.05F, 0.15F,
806                                     -0.05F, 0.25F, 0.10F, -0.20F,
807                                     0.15F, 0.05F, -0.25F, 0.10F,
808                                     -0.10F, 0.20F, 0.15F, 0.05F,
809                                     0.05F, -0.15F, 0.20F, 0.25F,
810                                 },
811                                 zeros(5));
812     return model;
813 }
814
815 } // namespace lcqi

```

## Tokenizer 与 Safetensors 实现

`tokenizer.cpp` 把 prompt 变成 token id, 并固定 unknown token 行为。读者应检查 BOS/EOS 是否稳定插入, UNK 是否按合同处理。

Listing 10.9 src/tokenizer.cpp

```

1  #include <lcqi/tokenizer.hpp>
2
3  #include <cstdint>
4  #include <fstream>
5  #include <stdexcept>
6  #include <string>
7  #include <string_view>
8
9  namespace lcqi {
10 namespace {
11
12 constexpr std::uint64_t LCQI_FNV_OFFSET_BASIS = 1469598103934665603ULL;
13 constexpr std::uint64_t LCQI_FNV_PRIME = 1099511628211ULL;
14 constexpr std::int32_t LCQI_INVALID_TOKEN_ID = -1;
15
16 std::string read_key(std::istream& input) {
17     std::string key;
18     if (!(input >> key)) {
19         throw std::runtime_error("unexpected end of tokenizer file");
20     }
21     return key;

```

```

22 }
23
24 void expect_key(std::istream& input, const std::string& expected) {
25     const std::string key = read_key(input);
26     if (key != expected) {
27         throw std::runtime_error("expected tokenizer key '" + expected + "', ←
↪ got '" + key + "'");
28     }
29 }
30
31 std::int32_t read_i32(std::istream& input, const std::string& key) {
32     expect_key(input, key);
33     std::int32_t value = 0;
34     if (!(input >> value)) {
35         throw std::runtime_error("invalid tokenizer int32 value for key '" + ←
↪ key + "'");
36     }
37     return value;
38 }
39
40 std::string read_string(std::istream& input, const std::string& key) {
41     expect_key(input, key);
42     std::string value;
43     if (!(input >> value)) {
44         throw std::runtime_error("invalid tokenizer string value for key '" + ←
↪ key + "'");
45     }
46     return value;
47 }
48
49 std::int32_t lookup_token_id(const TokenizerModel& tokenizer, std::string_view ←
↪ text) {
50     for (const TokenEntry& entry : tokenizer.vocab) {
51         if (entry.text == text) {
52             return entry.id;
53         }
54     }
55     return LCQI_INVALID_TOKEN_ID;
56 }
57
58 void validate_tokenizer(const TokenizerModel& tokenizer) {
59     if (tokenizer.vocab.empty()) {
60         throw std::runtime_error("tokenizer vocab is empty");
61     }
62     for (std::size_t i = 0; i < tokenizer.vocab.size(); ++i) {
63         const TokenEntry& left = tokenizer.vocab[i];
64         if (left.id < 0) {
65             throw std::runtime_error("tokenizer token id must be non-negative")↪
↪ ;
66         }
67         for (std::size_t j = i + 1; j < tokenizer.vocab.size(); ++j) {
68             const TokenEntry& right = tokenizer.vocab[j];

```

```

69         if (left.id == right.id) {
70             throw std::runtime_error("duplicate tokenizer token id");
71         }
72         if (left.text == right.text) {
73             throw std::runtime_error("duplicate tokenizer token text");
74         }
75     }
76 }
77 if (tokenizer.bos_id != LCQI_INVALID_TOKEN_ID &&
78     lookup_token_id(tokenizer, "<bos>") != tokenizer.bos_id) {
79     throw std::runtime_error("tokenizer BOS id does not match vocab");
80 }
81 if (tokenizer.eos_id != LCQI_INVALID_TOKEN_ID &&
82     lookup_token_id(tokenizer, "<eos>") != tokenizer.eos_id) {
83     throw std::runtime_error("tokenizer EOS id does not match vocab");
84 }
85 }
86
87 std::uint64_t hash_byte(std::uint64_t hash, unsigned char value) {
88     hash ^= static_cast<std::uint64_t>(value);
89     hash *= LCQI_FNV_PRIME;
90     return hash;
91 }
92
93 std::uint64_t hash_string(std::uint64_t hash, std::string_view text) {
94     for (const unsigned char value : text) {
95         hash = hash_byte(hash, value);
96     }
97     return hash;
98 }
99
100 std::uint64_t hash_i32(std::uint64_t hash, std::int32_t value) {
101     const std::uint32_t bits = static_cast<std::uint32_t>(value);
102     for (std::int32_t shift = 0; shift < 32; shift += 8) {
103         hash = hash_byte(hash, static_cast<unsigned char>((bits >> shift) & 0x
104             ↪ xFFU));
105     }
106     return hash;
107 }
108 } // namespace
109
110 TokenizerModel load_tokenizer(const std::filesystem::path& path) {
111     std::ifstream input(path);
112     if (!input) {
113         throw std::runtime_error("cannot open tokenizer file: " + path.string()
114             ↪ );
115     }
116
117     expect_key(input, "LCQI_TOKENIZER_V1");
118
119     TokenizerModel tokenizer;

```

```

119     tokenizer.bos_id = read_i32(input, "bos_id");
120     tokenizer.eos_id = read_i32(input, "eos_id");
121     tokenizer.unk_id = read_i32(input, "unk_id");
122     tokenizer.chat_template_hash = read_string(input, "chat_template_hash");
123     const std::int32_t vocab_size = read_i32(input, "vocab_size");
124     if (vocab_size <= 0) {
125         throw std::runtime_error("tokenizer vocab_size must be positive");
126     }
127
128     tokenizer.vocab.reserve(static_cast<std::size_t>(vocab_size));
129     expect_key(input, "vocab");
130     for (std::int32_t i = 0; i < vocab_size; ++i) {
131         TokenEntry entry;
132         if (!(input >> entry.id >> entry.text)) {
133             throw std::runtime_error("invalid tokenizer vocab entry");
134         }
135         tokenizer.vocab.push_back(entry);
136     }
137
138     validate_tokenizer(tokenizer);
139     return tokenizer;
140 }
141
142 std::vector<std::int32_t> encode_prompt(const TokenizerModel& tokenizer,
143                                         std::string_view prompt) {
144     std::vector<std::int32_t> ids;
145     if (tokenizer.bos_id != LCQI_INVALID_TOKEN_ID) {
146         ids.push_back(tokenizer.bos_id);
147     }
148
149     std::size_t begin = 0;
150     while (begin < prompt.size()) {
151         while (begin < prompt.size() && prompt[begin] == ' ') {
152             ++begin;
153         }
154         if (begin >= prompt.size()) {
155             break;
156         }
157         std::size_t end = begin;
158         while (end < prompt.size() && prompt[end] != ' ') {
159             ++end;
160         }
161         const std::string_view token = prompt.substr(begin, end - begin);
162         const std::int32_t token_id = lookup_token_id(tokenizer, token);
163         if (token_id == LCQI_INVALID_TOKEN_ID) {
164             if (tokenizer.unk_id == LCQI_INVALID_TOKEN_ID) {
165                 throw std::runtime_error("tokenizer encountered unknown token");
166             }
167             ids.push_back(tokenizer.unk_id);
168         } else {
169             ids.push_back(token_id);

```

```

170     }
171     begin = end;
172 }
173
174 if (tokenizer.eos_id != LCQI_INVALID_TOKEN_ID) {
175     ids.push_back(tokenizer.eos_id);
176 }
177 return ids;
178 }
179
180 std::uint64_t tokenizer_contract_hash(const TokenizerModel& tokenizer) {
181     std::uint64_t hash = LCQI_FNV_OFFSET_BASIS;
182     hash = hash_string(hash, tokenizer.chat_template_hash);
183     hash = hash_i32(hash, tokenizer.bos_id);
184     hash = hash_i32(hash, tokenizer.eos_id);
185     hash = hash_i32(hash, tokenizer.unk_id);
186     for (const TokenEntry& entry : tokenizer.vocab) {
187         hash = hash_string(hash, entry.text);
188         hash = hash_i32(hash, entry.id);
189     }
190     return hash;
191 }
192
193 } // namespace lcqi

```

**safetensors.cpp** 解析 header、metadata、dtype、shape 和 data offsets。坏 dtype、坏 shape、坏 offset 和 payload 太短都应该被拒绝。

**Listing 10.10** src/safetensors.cpp

```

1 #include <lcqi/safetensors.hpp>
2
3 #include <algorithm>
4 #include <cctype>
5 #include <cstdint>
6 #include <cmath>
7 #include <cstring>
8 #include <fstream>
9 #include <limits>
10 #include <stdexcept>
11 #include <string>
12 #include <string_view>
13 #include <vector>
14
15 namespace lcqi {
16 namespace {
17
18 constexpr std::size_t LCQI_SAFETENSORS_PREFIX_BYTES = 8;
19 constexpr std::uint64_t LCQI_SAFETENSORS_MAX_HEADER_BYTES = 16U * 1024U * 1024U↵
    ↵ ;
20 constexpr std::size_t LCQI_SAFETENSORS_OFFSET_COUNT = 2;
21 constexpr std::uint64_t LCQI_BYTE_BITS = 8;
22 constexpr std::uint64_t LCQI_DECIMAL_BASE = 10;

```

```

23 constexpr std::uint64_t LCQI_BYTE_MASK = 0xFFU;
24 constexpr unsigned char LCQI_JSON_CONTROL_CHAR_LIMIT = 0x20U;
25 constexpr std::uint64_t LCQI_DTYPE_BYTES_F64 = 8;
26 constexpr std::uint64_t LCQI_DTYPE_BYTES_F32 = 4;
27 constexpr std::uint64_t LCQI_DTYPE_BYTES_F16 = 2;
28 constexpr std::uint64_t LCQI_DTYPE_BYTES_I64 = 8;
29 constexpr std::uint64_t LCQI_DTYPE_BYTES_I32 = 4;
30 constexpr std::uint64_t LCQI_DTYPE_BYTES_I16 = 2;
31 constexpr std::uint64_t LCQI_DTYPE_BYTES_I8 = 1;
32 constexpr std::uint64_t LCQI_DTYPE_BYTES_U8 = 1;
33 constexpr std::uint32_t LCQI_F32_EXPONENT_MASK = 0x7F800000U;
34 constexpr std::uint32_t LCQI_F16_SIGN_MASK = 0x8000U;
35 constexpr std::uint32_t LCQI_F16_EXPONENT_MASK = 0x7C00U;
36 constexpr std::uint32_t LCQI_F16_MANTISSA_MASK = 0x03FFU;
37 constexpr std::uint32_t LCQI_F16_EXPONENT_BIAS = 15U;
38 constexpr std::uint32_t LCQI_F32_EXPONENT_BIAS = 127U;
39 constexpr std::uint32_t LCQI_F32_MANTISSA_BITS = 23U;
40 constexpr std::uint32_t LCQI_F16_MANTISSA_BITS = 10U;
41 constexpr std::uint32_t LCQI_F16_TO_F32_SIGN_SHIFT = 16U;
42
43 class HeaderCursor {
44 public:
45     explicit HeaderCursor(std::string_view text) : text_(text) {}
46
47     void expect(char expected) {
48         this->skip_ws();
49         if (this->eof() || this->text_[this->position_] != expected) {
50             throw std::runtime_error("invalid safetensors header syntax");
51         }
52         ++this->position_;
53     }
54
55     [[nodiscard]] bool consume(char expected) {
56         this->skip_ws();
57         if (!this->eof() && this->text_[this->position_] == expected) {
58             ++this->position_;
59             return true;
60         }
61         return false;
62     }
63
64     [[nodiscard]] std::string parse_string() {
65         this->skip_ws();
66         this->expect('\"');
67         std::string result;
68         while (!this->eof()) {
69             const char ch = this->text_[this->position_++];
70             if (ch == '\"') {
71                 return result;
72             }
73             if (static_cast<unsigned char>(ch) < LCQI_JSON_CONTROL_CHAR_LIMIT) ←
↪ {

```

```

74         throw std::runtime_error("control character in safetensors ↵
↵ string");
75     }
76     if (ch == '\\') {
77         result.push_back(this->parse_escape());
78     } else {
79         result.push_back(ch);
80     }
81 }
82 throw std::runtime_error("unterminated safetensors string");
83 }
84
85 [[nodiscard]] std::uint64_t parse_u64() {
86     this->skip_ws();
87     if (this->eof() ||
88 ↵ !std::isdigit(static_cast<unsigned char>(this->text_[this->↵
↵ position_]))) {
89         throw std::runtime_error("expected unsigned integer in safetensors ↵
↵ header");
90     }
91     std::uint64_t value = 0;
92     while (!this->eof() &&
93 ↵ std::isdigit(static_cast<unsigned char>(this->text_[this->↵
↵ position_]))) {
94         const std::uint64_t digit =
95             static_cast<std::uint64_t>(this->text_[this->position_] - '0');
96         if (value >
97 ↵ (std::numeric_limits<std::uint64_t>::max() - digit) / ↵
↵ LCQI_DECIMAL_BASE) {
98             throw std::runtime_error("safetensors integer overflow");
99         }
100         value = value * LCQI_DECIMAL_BASE + digit;
101         ++this->position_;
102     }
103     return value;
104 }
105
106 [[nodiscard]] std::vector<std::int64_t> parse_i64_array() {
107     std::vector<std::int64_t> values;
108     this->expect('[');
109     if (this->consume(' ')) {
110         return values;
111     }
112     while (true) {
113         const std::uint64_t raw = this->parse_u64();
114         if (raw > static_cast<std::uint64_t>(std::numeric_limits<std::↵
↵ int64_t>::max())) {
115             throw std::runtime_error("safetensors shape value is too large"↵
↵ );
116         }
117         values.push_back(static_cast<std::int64_t>(raw));
118         if (this->consume(' ')) {

```



```

119         return values;
120     }
121     this->expect(',');
122 }
123 }
124
125 [[nodiscard]] std::vector<std::uint64_t> parse_u64_array() {
126     std::vector<std::uint64_t> values;
127     this->expect('[');
128     if (this->consume(' ')) {
129         return values;
130     }
131     while (true) {
132         values.push_back(this->parse_u64());
133         if (this->consume(' ')) {
134             return values;
135         }
136         this->expect(',');
137     }
138 }
139
140 [[nodiscard]] bool eof_after_ws() {
141     this->skip_ws();
142     return this->eof();
143 }
144
145 private:
146     std::string_view text_;
147     std::size_t position_ = 0;
148
149     [[nodiscard]] bool eof() const {
150         return this->position_ >= this->text_.size();
151     }
152
153     void skip_ws() {
154         while (!this->eof() &&
155             std::isspace(static_cast<unsigned char>(this->text_[this->position_])) {
156             ++this->position_;
157         }
158     }
159
160     [[nodiscard]] char parse_escape() {
161         if (this->eof()) {
162             throw std::runtime_error("unterminated safetensors escape");
163         }
164         const char escaped = this->text_[this->position_++];
165         switch (escaped) {
166             case '"':
167             case '\\':
168             case '/':
169                 return escaped;

```

```

170         case 'b':
171             return '\\b';
172         case 'f':
173             return '\\f';
174         case 'n':
175             return '\\n';
176         case 'r':
177             return '\\r';
178         case 't':
179             return '\\t';
180         default:
181             throw std::runtime_error("unsupported safetensors string escape↵
↵ ");
182     }
183 }
184 };
185
186 std::uint64_t read_le_u64(std::span<const std::uint8_t> bytes) {
187     if (bytes.size() < LCQI_SAFETENSORS_PREFIX_BYTES) {
188         throw std::runtime_error("safetensors file is too small");
189     }
190     std::uint64_t value = 0;
191     for (std::size_t i = 0; i < LCQI_SAFETENSORS_PREFIX_BYTES; ++i) {
192         value |= static_cast<std::uint64_t>(bytes[i]) << (LCQI_BYTE_BITS * i);
193     }
194     return value;
195 }
196
197 std::string read_file_bytes(const std::filesystem::path& path) {
198     std::ifstream input(path, std::ios::binary);
199     if (!input) {
200         throw std::runtime_error("cannot open safetensors file: " + path.string↵
↵ ());
201     }
202     input.seekg(0, std::ios::end);
203     const std::streamoff size = input.tellg();
204     if (size < 0) {
205         throw std::runtime_error("cannot determine safetensors file size");
206     }
207     input.seekg(0, std::ios::beg);
208     std::string bytes(static_cast<std::size_t>(size), '\\0');
209     if (!bytes.empty() && !input.read(bytes.data(), static_cast<std::streamsize>↵
↵ >(bytes.size())) {
210         throw std::runtime_error("cannot read safetensors file");
211     }
212     return bytes;
213 }
214
215 std::vector<std::uint8_t> read_file_u8(const std::filesystem::path& path) {
216     std::ifstream input(path, std::ios::binary);
217     if (!input) {
218         throw std::runtime_error("cannot open safetensors file: " + path.string↵

```

```

    ↪ ();
219 }
220 input.seekg(0, std::ios::end);
221 const std::streamoff size = input.tellg();
222 if (size < 0) {
223     throw std::runtime_error("cannot determine safetensors file size");
224 }
225 input.seekg(0, std::ios::beg);
226 std::vector<std::uint8_t> bytes(static_cast<std::size_t>(size), 0);
227 if (!bytes.empty() &&
228     !input.read(reinterpret_cast<char*>(bytes.data()),
229                 static_cast<std::streamsize>(bytes.size()))) {
230     throw std::runtime_error("cannot read safetensors file");
231 }
232 return bytes;
233 }
234
235 void parse_metadata_value(HeaderCursor& cursor,
236                           SafeTensorManifest& manifest,
237                           const std::string& key) {
238     if (key == "__metadata__") {
239         cursor.expect('{');
240         if (cursor.consume('}')) {
241             return;
242         }
243         while (true) {
244             const std::string metadata_key = cursor.parse_string();
245             cursor.expect(':');
246             const std::string metadata_value = cursor.parse_string();
247             manifest.metadata.emplace_back(metadata_key, metadata_value);
248             if (cursor.consume('}')) {
249                 return;
250             }
251             cursor.expect(',');
252         }
253     }
254     throw std::runtime_error("unexpected safetensors metadata value");
255 }
256
257 SafeTensorEntry parse_tensor_entry(HeaderCursor& cursor, const std::string& ↪
    ↪ name) {
258     SafeTensorEntry entry;
259     entry.name = name;
260     bool saw_dtype = false;
261     bool saw_shape = false;
262     bool saw_offsets = false;
263
264     cursor.expect('{');
265     if (cursor.consume('}')) {
266         throw std::runtime_error("empty safetensors tensor entry");
267     }
268     while (true) {

```

```

269     const std::string field = cursor.parse_string();
270     cursor.expect(':');
271     if (field == "dtype") {
272         entry.dtype = cursor.parse_string();
273         saw_dtype = true;
274     } else if (field == "shape") {
275         entry.shape = cursor.parse_i64_array();
276         saw_shape = true;
277     } else if (field == "data_offsets") {
278         const std::vector<std::uint64_t> offsets = cursor.parse_u64_array();
279         ↵ ;
280         if (offsets.size() != LCQI_SAFETENSORS_OFFSET_COUNT) {
281             throw std::runtime_error("safetensors data_offsets must contain ↵
282             ↵ two values");
283         }
284         entry.data_begin = offsets[0];
285         entry.data_end = offsets[1];
286         saw_offsets = true;
287     } else {
288         throw std::runtime_error("unsupported safetensors tensor field: " + ↵
289         ↵ field);
290     }
291     if (cursor.consume('}')) {
292         break;
293     }
294     cursor.expect(',');
295 }
296
297 if (!saw_dtype || !saw_shape || !saw_offsets) {
298     throw std::runtime_error("safetensors tensor entry is missing required ↵
299     ↵ fields");
300 }
301 if (entry.data_end < entry.data_begin) {
302     throw std::runtime_error("safetensors tensor offset range is invalid");
303 }
304 return entry;
305 }
306
307 void parse_header(std::string_view header, SafeTensorManifest& manifest) {
308     HeaderCursor cursor(header);
309     cursor.expect('{');
310     if (cursor.consume('}')) {
311         throw std::runtime_error("safetensors header is empty");
312     }
313
314     while (true) {
315         const std::string key = cursor.parse_string();
316         cursor.expect(':');
317         if (key == "__metadata__") {
318             parse_metadata_value(cursor, manifest, key);
319         } else {
320             manifest.tensors.push_back(parse_tensor_entry(cursor, key));

```

```

317     }
318     if (cursor.consume('}')) {
319         break;
320     }
321     cursor.expect(',');
322 }
323
324 if (!cursor.eof_after_ws()) {
325     throw std::runtime_error("trailing data after safetensors header");
326 }
327 }
328
329 std::uint64_t expected_tensor_bytes(const SafeTensorEntry& entry);
330
331 void validate_manifest(const SafeTensorManifest& manifest,
332                       std::uint64_t file_size) {
333     const std::uint64_t payload_size = file_size - manifest.data_start_offset;
334     for (std::size_t i = 0; i < manifest.tensors.size(); ++i) {
335         const SafeTensorEntry& left = manifest.tensors[i];
336         if (left.name.empty() || left.dtype.empty()) {
337             throw std::runtime_error("safetensors tensor has empty name or ↵
↵ dtype");
338         }
339         if (left.data_end > payload_size) {
340             throw std::runtime_error("safetensors tensor data_offsets exceed ↵
↵ payload size");
341         }
342         const std::uint64_t expected_bytes = expected_tensor_bytes(left);
343         if (left.byte_size() != expected_bytes) {
344             throw std::runtime_error("safetensors tensor byte size does not ↵
↵ match dtype and shape");
345         }
346         for (const std::int64_t dim : left.shape) {
347             if (dim < 0) {
348                 throw std::runtime_error("safetensors shape dimension is ↵
↵ negative");
349             }
350         }
351         for (std::size_t j = i + 1; j < manifest.tensors.size(); ++j) {
352             const SafeTensorEntry& right = manifest.tensors[j];
353             if (left.name == right.name) {
354                 throw std::runtime_error("duplicate safetensors tensor name");
355             }
356             const bool overlaps =
357                 left.data_begin < right.data_end && right.data_begin < left.↵
↵ data_end;
358             if (overlaps) {
359                 throw std::runtime_error("overlapping safetensors tensor data ↵
↵ range");
360             }
361         }
362     }

```

```

363 }
364
365 std::uint64_t dtype_byte_size(std::string_view dtype) {
366     if (dtype == "F64") {
367         return LCQI_DTYPE_BYTES_F64;
368     }
369     if (dtype == "F32") {
370         return LCQI_DTYPE_BYTES_F32;
371     }
372     if (dtype == "F16" || dtype == "BF16") {
373         return LCQI_DTYPE_BYTES_F16;
374     }
375     if (dtype == "I64" || dtype == "U64") {
376         return LCQI_DTYPE_BYTES_I64;
377     }
378     if (dtype == "I32" || dtype == "U32") {
379         return LCQI_DTYPE_BYTES_I32;
380     }
381     if (dtype == "I16" || dtype == "U16") {
382         return LCQI_DTYPE_BYTES_I16;
383     }
384     if (dtype == "I8") {
385         return LCQI_DTYPE_BYTES_I8;
386     }
387     if (dtype == "U8" || dtype == "B00L") {
388         return LCQI_DTYPE_BYTES_U8;
389     }
390     throw std::runtime_error("unsupported safetensors dtype: " + std::string(
391         dtype));
392 }
393
394 std::uint64_t expected_tensor_bytes(const SafeTensorEntry& entry) {
395     std::uint64_t element_count = 1;
396     for (const std::int64_t dim : entry.shape) {
397         if (dim < 0) {
398             throw std::runtime_error("safetensors shape dimension is negative");
399         }
400         const std::uint64_t extent = static_cast<std::uint64_t>(dim);
401         if (extent != 0 &&
402             element_count > std::numeric_limits<std::uint64_t>::max() / extent) {
403             throw std::runtime_error("safetensors shape element count overflow");
404         }
405         element_count *= extent;
406     }
407     const std::uint64_t dtype_bytes = dtype_byte_size(entry.dtype);
408     if (dtype_bytes != 0 &&
409         element_count > std::numeric_limits<std::uint64_t>::max() / dtype_bytes) {
410         throw std::runtime_error("safetensors tensor byte size overflow");
411     }
412 }

```

```

410     }
411     return element_count * dtype_bytes;
412 }
413
414 SafeTensorManifest parse_manifest_from_bytes(std::span<const std::uint8_t> ↵
    ↵ bytes) {
415     const std::uint64_t header_size = read_le_u64(bytes);
416     if (header_size == 0 || header_size > LCQI_SAFETENSORS_MAX_HEADER_BYTES) {
417         throw std::runtime_error("safetensors header size is invalid");
418     }
419     const std::uint64_t data_start_offset =
420         static_cast<std::uint64_t>(LCQI_SAFETENSORS_PREFIX_BYTES) + header_size↵
    ↵ ;
421     if (data_start_offset > bytes.size()) {
422         throw std::runtime_error("safetensors header extends beyond file size")↵
    ↵ ;
423     }
424
425     SafeTensorManifest manifest;
426     manifest.header_size = header_size;
427     manifest.data_start_offset = data_start_offset;
428     const char* header_begin =
429         reinterpret_cast<const char*>(bytes.data() + ↵
    ↵ LCQI_SAFETENSORS_PREFIX_BYTES);
430     const std::string_view header(header_begin, static_cast<std::size_t>(↵
    ↵ header_size));
431     parse_header(header, manifest);
432     validate_manifest(manifest, static_cast<std::uint64_t>(bytes.size()));
433     return manifest;
434 }
435
436 std::uint16_t read_le_u16(std::span<const std::uint8_t> bytes, std::size_t ↵
    ↵ offset) {
437     return static_cast<std::uint16_t>(
438         static_cast<std::uint16_t>(bytes[offset]) |
439         (static_cast<std::uint16_t>(bytes[offset + 1]) << LCQI_BYTE_BITS));
440 }
441
442 std::uint32_t read_le_u32(std::span<const std::uint8_t> bytes, std::size_t ↵
    ↵ offset) {
443     std::uint32_t value = 0;
444     for (std::size_t i = 0; i < LCQI_DTYPE_BYTES_F32; ++i) {
445         value |= static_cast<std::uint32_t>(bytes[offset + i]) << (↵
    ↵ LCQI_BYTE_BITS * i);
446     }
447     return value;
448 }
449
450 float bits_to_float(std::uint32_t bits) {
451     float value = 0.0F;
452     std::memcpy(&value, &bits, sizeof(value));
453     return value;

```



```

454 }
455
456 float f16_to_f32(std::uint16_t bits) {
457     const std::uint32_t sign = (static_cast<std::uint32_t>(bits) & ↵
    ↵ LCQI_F16_SIGN_MASK)
458         << LCQI_F16_TO_F32_SIGN_SHIFT;
459     const std::uint32_t exponent = static_cast<std::uint32_t>(bits) & ↵
    ↵ LCQI_F16_EXPONENT_MASK;
460     const std::uint32_t mantissa = static_cast<std::uint32_t>(bits) & ↵
    ↵ LCQI_F16_MANTISSA_MASK;
461
462     if (exponent == 0U) {
463         if (mantissa == 0U) {
464             return bits_to_float(sign);
465         }
466         float value = static_cast<float>(mantissa) /
467             static_cast<float>(1U << LCQI_F16_MANTISSA_BITS);
468         value = std::ldexp(value, 1 - static_cast<int>(LCQI_F16_EXPONENT_BIAS))↵
    ↵ ;
469         return (sign == 0U) ? value : -value;
470     }
471
472     if (exponent == LCQI_F16_EXPONENT_MASK) {
473         const std::uint32_t f32_mantissa =
474             mantissa << (LCQI_F32_MANTISSA_BITS - LCQI_F16_MANTISSA_BITS);
475         return bits_to_float(sign | LCQI_F32_EXPONENT_MASK | f32_mantissa);
476     }
477
478     const std::uint32_t f32_exponent =
479         ((exponent >> LCQI_F16_MANTISSA_BITS) - LCQI_F16_EXPONENT_BIAS +
480          LCQI_F32_EXPONENT_BIAS)
481         << LCQI_F32_MANTISSA_BITS;
482     const std::uint32_t f32_mantissa =
483         mantissa << (LCQI_F32_MANTISSA_BITS - LCQI_F16_MANTISSA_BITS);
484     return bits_to_float(sign | f32_exponent | f32_mantissa);
485 }
486
487 float bf16_to_f32(std::uint16_t bits) {
488     return bits_to_float(static_cast<std::uint32_t>(bits) << 16U);
489 }
490
491 std::span<const std::uint8_t> tensor_payload_span(const SafeTensorFile& file,
492                                                    const SafeTensorEntry& entry)↵
    ↵ {
493     const std::uint64_t begin = file.manifest.data_start_offset + entry.↵
    ↵ data_begin;
494     const std::uint64_t end = file.manifest.data_start_offset + entry.data_end;
495     if (begin > end || end > file.bytes.size()) {
496         throw std::runtime_error("safetensors tensor byte range is invalid");
497     }
498     return std::span<const std::uint8_t>(
499         file.bytes.data() + static_cast<std::size_t>(begin),

```

```

500     static_cast<std::size_t>(end - begin));
501 }
502
503 } // namespace
504
505 std::uint64_t SafeTensorEntry::byte_size() const {
506     return this->data_end - this->data_begin;
507 }
508
509 const SafeTensorEntry* SafeTensorManifest::find_tensor(std::string_view name) ←
    ↪ const {
510     const auto found = std::find_if(
511         this->tensors.begin(),
512         this->tensors.end(),
513         [name](const SafeTensorEntry& entry) {
514             return entry.name == name;
515         });
516     if (found == this->tensors.end()) {
517         return nullptr;
518     }
519     return &(*found);
520 }
521
522 SafeTensorManifest load_safetensors_manifest(const std::filesystem::path& path) ←
    ↪ {
523     const std::string bytes = read_file_bytes(path);
524     return parse_manifest_from_bytes(
525         std::span<const std::uint8_t>(
526             reinterpret_cast<const std::uint8_t*>(bytes.data()),
527             bytes.size()));
528 }
529
530 SafeTensorFile load_safetensors_file(const std::filesystem::path& path) {
531     SafeTensorFile file;
532     file.path = path;
533     file.bytes = read_file_u8(path);
534     file.manifest = parse_manifest_from_bytes(file.bytes);
535     return file;
536 }
537
538 std::span<const std::uint8_t> read_safetensor_tensor_bytes(
539     const SafeTensorFile& file,
540     std::string_view name) {
541     const SafeTensorEntry* entry = file.manifest.find_tensor(name);
542     if (entry == nullptr) {
543         throw std::runtime_error("missing safetensors tensor: " + std::string(←
    ↪ name));
544     }
545     return tensor_payload_span(file, *entry);
546 }
547
548 std::vector<float> read_safetensor_f32_tensor(const SafeTensorFile& file,

```

```

549         std::string_view name) {
550     const SafeTensorEntry* entry = file.manifest.find_tensor(name);
551     if (entry == nullptr) {
552         throw std::runtime_error("missing safetensors tensor: " + std::string(↵
↵ name));
553     }
554     const std::span<const std::uint8_t> bytes = tensor_payload_span(file, *↵
↵ entry);
555     std::vector<float> values;
556     if (entry->dtype == "F32") {
557         values.resize(bytes.size() / LCQI_DTYPE_BYTES_F32);
558         for (std::size_t index = 0; index < values.size(); ++index) {
559             values[index] = bits_to_float(
560                 read_le_u32(bytes, index * LCQI_DTYPE_BYTES_F32));
561         }
562         return values;
563     }
564     if (entry->dtype == "F16") {
565         values.resize(bytes.size() / LCQI_DTYPE_BYTES_F16);
566         for (std::size_t index = 0; index < values.size(); ++index) {
567             values[index] = f16_to_f32(
568                 read_le_u16(bytes, index * LCQI_DTYPE_BYTES_F16));
569         }
570         return values;
571     }
572     if (entry->dtype == "BF16") {
573         values.resize(bytes.size() / LCQI_DTYPE_BYTES_F16);
574         for (std::size_t index = 0; index < values.size(); ++index) {
575             values[index] = bf16_to_f32(
576                 read_le_u16(bytes, index * LCQI_DTYPE_BYTES_F16));
577         }
578         return values;
579     }
580     throw std::runtime_error("safetensors tensor is not float-compatible: " + ↵
↵ entry->name);
581 }
582
583 } // namespace lcqi

```

## GPT-2 Reference Path

`gpt2_reference.cpp` 包括 JSON 配置读取、byte-level BPE、safetensors F32/F16/BF16 张量读取、HuggingFace 权重名前缀解析、LayerNorm、packed QKV、causal attention、GELU、tied lm head、full-prefix greedy generation 和 cached-KV generation。它仍然是正确性优先的 reference path，但现在已经开始承接前三册的优化方法：先建立 baseline，再改 hot path，再用测试和 A/B benchmark 决定改动是否能进入书。

## 一次可审查的 cached-KV 优化

优化不能从“感觉慢”开始。LCQI 先用 `makecpu3-gpt2-benchmark-compare` 证明 full-prefix 每生成一个 token 都重算前缀，而 cached-KV 已经把前缀复用下来；随后再看 cached path 内部。旧实现每处理一个 token、每经过一层 Transformer，都会重新分配 `normed`、`query`、`key`、

`value`、`attention`、`projected`、`mlp_fc`、`mlp_out` 和 `attention scores`。同时，生成路径只需要下一个 greedy token，却仍然每步构造完整 `Gpt2ForwardResult.logits`。这些问题不是算法语义错误，但会在真实 GPT-2 的 decode loop 里形成额外分配、重复 shape validation、重复边界转换和不必要的结果写回。

本轮优化对应三册里的三条原则。第一册要求从热循环和二进制可观察行为出发，所以先量 `lm_head`、MLP、QKV、attention projection 和 attention 本体各占多少，再决定改哪里。第二册要求减少重复分配和内存扰动，所以新增 `Gpt2ForwardWorkspace`，把 hidden、QKV、MLP、scores 和 logits 缓冲区的生命周期提升到整段 cached generation。第三册要求按 decoder runtime 的状态组织代码，所以新增 `Gpt2CachedGreedyDecoder`，让 KV cache、workspace、worker pool 和模型验证跟随一次生成会话，而不是散落在每个 token 的临时对象里。

代码上有五处关键边界。第一，旧的 `run_gpt2_forward_cached` 仍然保留完整 logits API，测试和源码阅读可以继续用它和 full-prefix logits 做逐项对照。第二，CLI 的 `cached_kv` 生成路径改用 `Gpt2CachedGreedyDecoder::step`，只返回 greedy token；需要完整 logits 的测试使用 `step_with_logits`。第三，prompt prefill 的前缀 token 使用 `advance_without_prediction`，只更新 KV cache，不做 final norm 和 `lm_head`，因为这些中间预测会被后一个 prompt token 立即覆盖。第四，KV cache 对外仍保留 checked `key/value` API，内部 attention 在一次地址和 workspace 校验后使用私有 unchecked span 扫描热路径，避免把“不检查”的能力暴露给普通调用者。第五，线程池只属于 cached decoder 会话；`--threads0` 自动选择 worker 数，`--threads1` 强制串行，`--threadsN` 固定线程数，避免把全局可变线程服务塞进 reference model。

正确性证据在 `tests/lcqi_tests.cpp`。tiny GPT-2 同一个 prompt 同时覆盖 full-prefix logits、旧 cached logits、优化 decoder logits、强制两 worker 的并行 decoder logits 和 logits-free greedy token。它要求 predicted token、logits size、逐项 logit 误差、KV cache filled token 数和 greedy generated ids 全部一致。这样优化后的代码不是“跑得快但不知道还对不对”，而是先通过 tiny golden，再进入真实 GPT-2 benchmark。

性能证据不只看一次运行。新增 `tools/run_gpt2_optimization_ab.py` 会从指定 baseline commit 建临时 worktree，baseline 和当前工作区都通过 `books/cpu-volume-3-practice` 的 Release CMake 入口构建，再用同一份 GPT-2 F32 safetensors 模型跑多轮 full-prefix 与 cached-KV。后来远端 `caw` 不能直接复用本机 SSH remote 别名时，本轮采用等价的源码包 A/B：本机用 `gitarchive73f40c7` 导出 baseline，用当前工作区导出 current，二者在 `caw` 上分别 Release 构建并运行同一模型。raw 报告保存在 `books/cpu-volume-3/reports/lcqi-gpt2-threaded-manual-ab-caw.txt`，汇总保存在 `books/cpu-volume-3/reports/lcqi-gpt2-threaded-optimization-summary.md`。5 轮中位数显示，4 token cached-KV 生成阶段从 395.397ms 到 76.4141ms，16 token 从 989.060ms 到 185.479ms；生成吞吐分别到 52.3464token/s 和 86.2633token/s。

用户看到上一轮提升幅度后，正确反应不是继续猜，而是重新量热点。新增 `--profile-hotspots` 和 `tools/run_gpt2_hotspot_profile.py` 后，可以把 cached decode 的时间拆成 embedding、LayerNorm、QKV projection、KV append、attention、attention projection、MLP fc、GELU、MLP projection 和 lm head。远端 `caw` 使用 Intel Core Ultra 7 265KF、GCC 15.2.1、Release 构建、GPT-2 F32 safetensors，跑 `Hello, mynameis` 这个 prompt，优化前 3 轮中位数保存在 `books/cpu-volume-3/reports/lcqi-gpt2-hotspot-profile-summary.md`：生成 4 个 token 时，`lm_head` 约占 30.2423%，MLP 的 `c_fc` 和 `c_proj` 合计约 46.4228%，QKV projection 约 17.2127%，cached attention 本体只有 0.0558254%。这说明慢点不是 attention，而是大量互相独立的行点积。

这一轮优化据此把 `lm_head` greedy argmax 和 GPT-2 F32 `linear` 的输出行切给持久 worker pool。每一行仍然是同一个 dot product；不同线程只负责不同输出行。完整 logits 路径把 logits buffer 分片写入，logits-free greedy 路径每个分片先找本地 best token，再按分片顺序合并，用严格大于比较保留较小 token id 的 tie-breaking。这样并行改变调度，不改变数学定义。

还有一个细节说明为什么优化必须带着 profiler 走。第一次 auto 阈值只按“一百万次乘加”判断是否并行，结果 `768\times768` 的 attention output projection 仍走串行。中间 profile 立即显示 attention projection 从约 6% 反弹到约 25%，成为新瓶颈。最终阈值改成：输出行数至少 512，且输入列数至少 512 时也允许并行。最终 profile 保存在 `books/cpu-volume-3/reports/`



`lcqi-gpt2-threaded-hotspot-caw.txt`: 4 token 生成中位数 **75.9807ms**, 16 token **184.780ms**; `lm_head` 约 **20%**, MLP 两个投影合计约 **45%**, QKV 约 **19%**, attention projection 约 **12%**, attention 本体仍低于 **1%**。

用户继续问和 llama.cpp 的差距时, 下一步就不能再停留在“行并行”。LCQI 新增 `include/lcqi/f32_kernels.hpp`、`src/f32_kernels.cpp` 和 `src/f32_kernels_avx2.cpp`, 把 GPT-2 cached 热路径的单行点积抽成 `dot_f32_unchecked()`。checked `dot_f32()` 负责普通 span 长度检查, unchecked 入口给已经完成 shape validation 的 `linear`、logits 和 logits-free `lm_head` 使用; x86-64/GNU/Clang 构建会编译 AVX2/FMA 文件, 运行时再用 `dot_f32_avx2_available()` 判断是否能执行。这样 AVX intrinsics 不进入 GPT-2 层逻辑, 后续替换 AVX-512、int8 或 int4 block kernel 时, 边界仍然清楚。

`tests/lcqi_tests.cpp` 也随之增加 F32 dot kernel 对照: 同一组固定输入先跑 scalar, 再跑 dispatch; 如果当前 CPU 支持 AVX2/FMA, 再显式跑 AVX2 实现, 并要求误差落在 **1.0e-4** 内。测试还会传入长度不一致的 span, 确认 checked API 会拒绝。这样优化路径不是“编译过就算”, 而是有 scalar oracle、运行时能力判断和错误输入边界。

这轮 F32 dot 优化以 `cb4d89f` 为 baseline, 在 `caw` 上用同一 GPT-2 F32 模型、Release 构建和 `--threads0` 自动 16 workers 跑 5 轮。raw 报告保存在 `books/cpu-volume-3/reports/lcqi-gpt2-f32-dot-ab-caw.txt`, 汇总保存在 `books/cpu-volume-3/reports/lcqi-gpt2-f32-dot-optimization-summary.md`。4 token 生成中位数从 **74.4247ms** 到 **62.3628ms**, 约 **1.19x**; 16 token 从 **181.257ms** 到 **157.230ms**, 约 **1.15x**。decode 中位数分别从 **26.1887ms** 到 **23.0640ms**, 从 **132.201ms** 到 **117.734ms**。这个提升比线程化小, 因为它只缩短每一行内部 dot, 而没有减少行数、层数、vocab 扫描或内存流量。

继续顺着热点走, 下一步不是在 GPT-2 层里散落更多 intrinsics, 而是把 kernel 边界从“单行 dot”扩成“row range”。新增的 `linear_f32_rows_unchecked()` 一次计算一段输出行, `max_dot_f32_rows_unchecked()` 一次扫描一段 vocab 行并返回局部最大值; `Gpt2ParallelWorkerPool::parallel_for_rows()` 改成模板入口, 去掉每个热路径任务里的 `std::function` 构造和间接调用。AVX2 文件里还增加 4-row row-range kernel: 同一组 input 向量 load 后, 同时喂给 4 个输出行累加器。这个写法已经接近真实 matvec kernel 的形状: 不是每一行各自重新走完整控制流, 而是在一个小 tile 里复用输入。

这轮 row-range 优化的 raw 报告保存在 `books/cpu-volume-3/reports/lcqi-gpt2-f32-rowrange-ab-caw.txt`, 汇总保存在 `books/cpu-volume-3/reports/lcqi-gpt2-f32-rowrange-optimization-summary.md`。相对 `ab72ccb` 这个 F32 dot baseline, 7 轮中位数显示 4 token 端到端生成从 **60.6291ms** 到 **61.1045ms**, 属于噪声内波动; 16 token 从 **153.354ms** 到 **152.574ms**, 约 **1.01x**。但把 row-range 版本和 4-row AVX2 row-range 版本直接对比, 4 token 从 **62.0243ms** 到 **60.3965ms**, 16 token 从 **154.087ms** 到 **151.222ms**。所以这段代码可以保留, 但结论要诚实: 它证明 tile 化 row-range 有正收益, 却不是数量级变化。

最终 hotspot 仍然说明差距在哪里。row-range 后, 16 token profile 中 `lm_head` 约 **22.3075%**, MLP `c_fc` 约 **22.4147%**, MLP projection 约 **22.448%**, QKV projection 约 **18.6294%**, attention projection 约 **11.1687%**, cached attention 本体只有 **0.653274%**。也就是说, 小 kernel 和调度优化已经把 reference path 又往前推了一点, 但主要时间仍在 F32 row-major matvec。真实 llama.cpp/ggml 的大优势来自 GGUF、block quantization、packed layout、cache-friendly matmul/matvec、mmap 和成熟线程调度, 不是只把裸 F32 行循环写得更精巧。

这个结论必须保守阅读。真实端到端 GPT-2 benchmark 受调度、频率、温度和共享机器状态影响明显, 单次结果可能反向波动。因此报告里必须同时看 `median/min/max`、生成文本是否一致、baseline 和 current 是否同一构建入口。更大的差距仍然不在这轮 F32 row-range 里: LCQI 还没有 GGUF 量化导入、packed/quantized lm head、量化 block 权重、分页 KV cache、NUMA/亲和和成熟调度。也就是说, 本轮把 reference path 从“行点积有 SIMD 内核边界”推进到“输出行段有可替换 tile kernel”, 但还没有把它变成 llama.cpp 级推理引擎。

继续看热点时, 有一类优化不在 kernel 内部, 而在调用链上。cached generation 的 prompt prefill 要把 prompt 的每个 token 写入 KV cache; 但如果 prompt 长度是 **N**, 前 **N-1** 个 prompt token 的 next-token prediction 根本不会被输出, 也不会作为后续输入。只有最后一个 prompt token 的 prediction 会成为第一个新生成 token。旧代码每个 prompt token 都跑 final norm 和 `lm_head`, 等

于把前缀 token 的预测算出来后马上丢掉。

因此新增 `Gpt2CachedGreedyDecoder::advance_without_prediction()`。它复用同一条 cached layer path, 照样执行 embedding、每层 QKV、attention、MLP 和 KV append, 只是在层结束后直接返回, 不进入 final norm 和 `lm_head`。CLI 和 `gpt2_generate_greedy_cached()` 的 prefill 改成: 前缀 token 调 `advance_without_prediction()`, 最后一个 prompt token 调 `step()`。这不是近似优化, 因为被跳过的预测值在旧流程中没有任何 caller 读取。

远端 `caw` 上 7 轮 A/B 证明这条优化有效, raw 报告保存在 `books/cpu-volume-3/reports/lcqi-gpt2-prefill-skip-ab-caw.txt`, 汇总保存在 `books/cpu-volume-3/reports/lcqi-gpt2-prefill-skip-optimization-summary.md`。同一 GPT-2 F32 模型、同一 prompt、同一 `--threads0` 下, 4 token 生成中位数从 `60.5154ms` 到 `52.1949ms`, 约快 **13.75%**; 16 token 从 `153.574ms` 到 `147.424ms`, 约快 **4.00%**。收益在短生成里更明显, 因为 prompt prefill 占总时间比例更大; decode 段几乎不变, 因为真正生成阶段仍然必须预测下一个 token。所有轮次生成文本和 token id 都一致。

这一轮还做过一个没有进入主线的 packed F32 `lm_head` 实验: 把 `lm_head` 权重按 8 个输出一组重排, 让 AVX2 一次累计 8 个 token score。它能让 `lm_head_ms` 小幅下降, 但 `caw` 多轮 A/B 没有稳定端到端收益, 甚至会让部分 decode 口径略慢。这个结果本身也要写进工程判断: 不是每个看起来更接近 SIMD 的布局都值得进入教材主线; 没有稳定证据的优化只能留作实验记录, 不能伪装成成功案例。

继续追问“为什么提升还是不够大”时, 下一步不是再猜一个更复杂的内核, 而是把已经接受的 profile 再拆开看。row-range 之后, GPT-2 cached path 每个 token 会在 12 层里反复调用 QKV projection、attention projection、MLP `c_fc`、MLP projection; 再加上需要预测 token 时的 `lm_head`。这些矩阵向量乘每次本身不算巨型 GEMM, 却会频繁进入 `parallel_for_rows()`。也就是说, 热点不只有“每行点积慢”, 还有“每个小并行区唤醒 worker 的成本”。如果 worker pool 每次都拿 mutex、发 condition variable、再等完成信号, 那么短生成里会把调度成本支付几百次。

因此这一轮接受的优化没有改数学公式, 而是改 worker pool 的交接方式。旧版本用 `std::mutex`、`std::condition_variable` 和 `finished_` 条件变量发布任务; 新版本把任务元数据写入原子字段, 再用 `generation` 作为发布点。后台 worker 看到新的 generation 后按固定 worker index 取自己的行块, 做完后递减 `active_workers`; 主线程自己处理最后一个行块, 然后用 bounded spin/yield 等后台 worker 归零。代码里特意把这个池限制在 `Gpt2CachedGreedyDecoder` 会话内, 没有变成全局忙等线程服务; 当 `--threads1` 或模型规模不够并行时, 路径仍然直接串行执行。

这个设计有三个边界必须看懂。第一, 任务上下文仍然是栈上的 `ParallelTaskContext`, 所以所有任务元数据必须先写完, 再递增 `generation`; worker 对 `generation` 做 acquire load 后才允许读上下文。第二, 后台 worker 不再争抢全局 `next_worker_index`, 而是启动时固定自己的 worker index; 这样不会出现旧任务里空闲 worker 和下一轮任务元数据交叉的问题。第三, spin 不是无限空转: 短时间使用 x86 pause, 超过阈值后让出调度; 这适合当前 cached decode 的短小并行区, 但不应该被复制到长时间等待的 I/O 或服务端队列里。

`caw` 上 9 轮 A/B 证明这条优化是本轮真正站得住的提升。raw 报告保存在 `books/cpu-volume-3/reports/lcqi-gpt2-spinpool-ab-caw.txt`, 汇总保存在 `books/cpu-volume-3/reports/lcqi-gpt2-spinpool-optimization-summary.md`。相对旧 auto 16 worker baseline, 4 token 生成中位数从 `53.7887ms` 到 `37.5403ms`, 约快 **30.21%**; 16 token 从 `146.930ms` 到 `104.107ms`, 约快 **29.15%**; 64 token 从 `510.650ms` 到 `377.233ms`, 约快 **26.13%**。只和 auto8 condition-variable 版本比较, 也能看到 4/16/64 token 分别约 **25.67%**、**29.56%**、**20.44%** 的生成阶段下降; 固定 `--threads8` 下也有约 **18%** 到 **23%** 的下降。这说明收益不是只来自 worker 数从 16 改成 8, 而是高频并行区交接成本确实降下来了。

热点证据也吻合原因。auto 口径下, 4 token 的 QKV projection 中位数约快 **33.58%**, attention projection 约快 **63.30%**, MLP `c_fc` 约快 **30.36%**, MLP projection 约快 **29.08%**; 而 `lm_head` 只约快 **8.57%**。原因是 QKV 和 MLP 这些线性层在每层每 token 都调用小并行区, 调度成本占比高; `lm_head` 是一次更大的 vocab 扫描, 仍然主要受内存流量和点积内核影响。这个差异比一句“线程池更快”更重要, 因为它告诉读者优化到底打中了哪个成本。

本轮还试过 8-row AVX2 row-range 候选: 把现有一次处理 4 个输出行的 row-range kernel 扩

成一次处理 8 行。它没有进入主线，因为 `caw` 数据没有稳定收益：auto 口径下 4 token 从 **37.4937ms** 到 **37.7082ms**，16 token 从 **103.714ms** 到 **103.178ms**，64 token 从 **374.439ms** 到 **374.457ms**；固定 `--threads8` 下 64 token 还从 **371.807ms** 退到 **377.053ms**。这说明寄存器压力、load/store 形状和调度噪声抵消了多行复用 input 的设想，不能因为“8 行看起来比 4 行更充分”就合并。

至此可以重新回答和 llama.cpp 的差距。LCQI 现在已经有 cached KV、workspace 复用、prefill 跳过无用 prediction、F32 row-range AVX2、行并行和轻量 worker hand-off；这些能把 reference path 推进很多，但它仍然主要在 F32 row-major matvec 上花时间。llama.cpp/ggml 的优势不只是“也用了线程”，而是 GGUF 模型格式、block quantization、量化权重布局、成熟 matvec microkernel、mmap 加载、NUMA/亲和策略和大量 shape-specific 调参一起工作。LCQI 这轮优化的价值是把调度证据链讲清楚，并把失败候选挡在书外；下一阶段如果要继续接近 llama.cpp，应优先做 GGUF/量化权重路径和量化 matvec，而不是继续在裸 F32 row-major 上堆小修小补。

**Listing 10.11** src/f32\_kernels.cpp

```

1 #include <lcqi/f32_kernels.hpp>
2
3 #include <limits>
4 #include <stdexcept>
5
6 namespace lcqi {
7 namespace {
8
9 constexpr std::size_t LCQI_F32_AVX2_MIN_DOT_SIZE = 32;
10 constexpr std::size_t LCQI_F32_AVX2_MIN_ROW_COUNT = 4;
11
12 } // namespace
13
14 float dot_f32(std::span<const float> lhs, std::span<const float> rhs) {
15     if (lhs.size() != rhs.size()) {
16         throw std::runtime_error("dot_f32 size mismatch");
17     }
18     return dot_f32_unchecked(lhs.data(), rhs.data(), lhs.size());
19 }
20
21 float dot_f32_scalar_unchecked(const float* lhs,
22                                const float* rhs,
23                                std::size_t size) noexcept {
24     float sum = 0.0F;
25     for (std::size_t index = 0; index < size; ++index) {
26         sum += lhs[index] * rhs[index];
27     }
28     return sum;
29 }
30
31 float dot_f32_unchecked(const float* lhs,
32                          const float* rhs,
33                          std::size_t size) noexcept {
34     static const bool AVX2_AVAILABLE = dot_f32_avx2_available();
35     if (AVX2_AVAILABLE && size >= LCQI_F32_AVX2_MIN_DOT_SIZE) {
36         return dot_f32_avx2_unchecked(lhs, rhs, size);
37     }
38     return dot_f32_scalar_unchecked(lhs, rhs, size);
39 }

```

```

40
41 void linear_f32_rows_scalar_unchecked(const float* weights,
42                                     const float* input,
43                                     const float* bias,
44                                     std::size_t input_size,
45                                     std::size_t row_begin,
46                                     std::size_t row_end,
47                                     float* output) noexcept {
48     for (std::size_t row = row_begin; row < row_end; ++row) {
49         const float* weight_row = weights + row * input_size;
50         output[row] = dot_f32_scalar_unchecked(weight_row, input, input_size) +
51                     (bias == nullptr ? 0.0F : bias[row]);
52     }
53 }
54
55 void linear_f32_rows_unchecked(const float* weights,
56                               const float* input,
57                               const float* bias,
58                               std::size_t input_size,
59                               std::size_t row_begin,
60                               std::size_t row_end,
61                               float* output) noexcept {
62     static const bool AVX2_AVAILABLE = dot_f32_avx2_available();
63     if (AVX2_AVAILABLE && input_size >= LCQI_F32_AVX2_MIN_DOT_SIZE &&
64         row_end - row_begin >= LCQI_F32_AVX2_MIN_ROW_COUNT) {
65         linear_f32_rows_avx2_unchecked(weights,
66                                         input,
67                                         bias,
68                                         input_size,
69                                         row_begin,
70                                         row_end,
71                                         output);
72         return;
73     }
74     linear_f32_rows_scalar_unchecked(weights, input, bias, input_size, ↵
75     ↵ row_begin, row_end, output);
76 }
77 F32RowMax max_dot_f32_rows_scalar_unchecked(const float* weights,
78                                              const float* input,
79                                              std::size_t input_size,
80                                              std::size_t row_begin,
81                                              std::size_t row_end) noexcept {
82     F32RowMax best;
83     best.row = row_begin;
84     best.value = -std::numeric_limits<float>::infinity();
85     for (std::size_t row = row_begin; row < row_end; ++row) {
86         const float* weight_row = weights + row * input_size;
87         const float value = dot_f32_scalar_unchecked(weight_row, input, ↵
88         ↵ input_size);
89         if (row == row_begin || value > best.value) {
90             best.row = row;

```



```

90         best.value = value;
91     }
92 }
93 return best;
94 }
95
96 F32RowMax max_dot_f32_rows_unchecked(const float* weights,
97                                     const float* input,
98                                     std::size_t input_size,
99                                     std::size_t row_begin,
100                                     std::size_t row_end) noexcept {
101     static const bool AVX2_AVAILABLE = dot_f32_avx2_available();
102     if (AVX2_AVAILABLE && input_size >= LCQI_F32_AVX2_MIN_DOT_SIZE &&
103         row_end - row_begin >= LCQI_F32_AVX2_MIN_ROW_COUNT) {
104         return max_dot_f32_rows_avx2_unchecked(weights,
105                                                 input,
106                                                 input_size,
107                                                 row_begin,
108                                                 row_end);
109     }
110     return max_dot_f32_rows_scalar_unchecked(weights, input, input_size, ↵
111     ↵ row_begin, row_end);
112 }
113 } // namespace lcqi
114
115 #if !defined(LCQI_ENABLE_AVX2)
116 namespace lcqi {
117
118 bool dot_f32_avx2_available() noexcept {
119     return false;
120 }
121
122 float dot_f32_avx2_unchecked(const float* lhs,
123                             const float* rhs,
124                             std::size_t size) noexcept {
125     return dot_f32_scalar_unchecked(lhs, rhs, size);
126 }
127
128 void linear_f32_rows_avx2_unchecked(const float* weights,
129                                     const float* input,
130                                     const float* bias,
131                                     std::size_t input_size,
132                                     std::size_t row_begin,
133                                     std::size_t row_end,
134                                     float* output) noexcept {
135     linear_f32_rows_scalar_unchecked(weights,
136                                     input,
137                                     bias,
138                                     input_size,
139                                     row_begin,
140                                     row_end,

```

```

141         output);
142     }
143
144     F32RowMax max_dot_f32_rows_avx2_unchecked(const float* weights,
145                                               const float* input,
146                                               std::size_t input_size,
147                                               std::size_t row_begin,
148                                               std::size_t row_end) noexcept {
149         return max_dot_f32_rows_scalar_unchecked(weights, input, input_size, ↵
            ↵ row_begin, row_end);
150     }
151
152 } // namespace lcqi
153 #endif

```

Listing 10.12 src/f32\_kernels\_avx2.cpp

```

1 #include <lcqi/f32_kernels.hpp>
2
3 #if defined(LCQI_ENABLE_AVX2)
4
5 #include <immintrin.h>
6
7 #include <array>
8 #include <cstdint>
9 #include <limits>
10
11 namespace lcqi {
12     namespace {
13
14         constexpr std::size_t LCQI_F32_AVX2_LANE_COUNT = 8;
15         constexpr std::size_t LCQI_F32_AVX2_UNROLL_COUNT = 4;
16         constexpr std::size_t LCQI_F32_AVX2_UNROLLED_FLOATS =
17             LCQI_F32_AVX2_LANE_COUNT * LCQI_F32_AVX2_UNROLL_COUNT;
18
19         [[nodiscard]] float horizontal_sum(_m256 values) noexcept {
20             const __m128 low = _mm256_castps256_ps128(values);
21             const __m128 high = _mm256_extractf128_ps(values, 1);
22             const __m128 pair_sum = _mm_add_ps(low, high);
23             const __m128 first_fold = _mm_hadd_ps(pair_sum, pair_sum);
24             const __m128 second_fold = _mm_hadd_ps(first_fold, first_fold);
25             return _mm_cvtss_f32(second_fold);
26         }
27
28         void compute_four_row_dot(const float* row0,
29                                  const float* row1,
30                                  const float* row2,
31                                  const float* row3,
32                                  const float* input,
33                                  std::size_t input_size,
34                                  float* sums) noexcept {
35             std::size_t index = 0;

```

```

36     __m256 accumulator0 = _mm256_setzero_ps();
37     __m256 accumulator1 = _mm256_setzero_ps();
38     __m256 accumulator2 = _mm256_setzero_ps();
39     __m256 accumulator3 = _mm256_setzero_ps();
40
41     for (; index + LCQI_F32_AVX2_LANE_COUNT <= input_size;
42          index += LCQI_F32_AVX2_LANE_COUNT) {
43         const __m256 input_values = _mm256_loadu_ps(input + index);
44         accumulator0 =
45             _mm256_fmadd_ps(_mm256_loadu_ps(row0 + index), input_values, ↵
↵ accumulator0);
46         accumulator1 =
47             _mm256_fmadd_ps(_mm256_loadu_ps(row1 + index), input_values, ↵
↵ accumulator1);
48         accumulator2 =
49             _mm256_fmadd_ps(_mm256_loadu_ps(row2 + index), input_values, ↵
↵ accumulator2);
50         accumulator3 =
51             _mm256_fmadd_ps(_mm256_loadu_ps(row3 + index), input_values, ↵
↵ accumulator3);
52     }
53
54     sums[0] = horizontal_sum(accumulator0);
55     sums[1] = horizontal_sum(accumulator1);
56     sums[2] = horizontal_sum(accumulator2);
57     sums[3] = horizontal_sum(accumulator3);
58     for (; index < input_size; ++index) {
59         const float input_value = input[index];
60         sums[0] += row0[index] * input_value;
61         sums[1] += row1[index] * input_value;
62         sums[2] += row2[index] * input_value;
63         sums[3] += row3[index] * input_value;
64     }
65 }
66
67 } // namespace
68
69 bool dot_f32_avx2_available() noexcept {
70     #if defined(__GNUC__) || defined(__clang__)
71         __builtin_cpu_init();
72         return __builtin_cpu_supports("avx2") && __builtin_cpu_supports("fma");
73     #else
74         return true;
75     #endif
76 }
77
78 float dot_f32_avx2_unchecked(const float* lhs,
79                              const float* rhs,
80                              std::size_t size) noexcept {
81     std::size_t index = 0;
82     __m256 accumulator0 = _mm256_setzero_ps();
83     __m256 accumulator1 = _mm256_setzero_ps();

```

```

84     __m256 accumulator2 = _mm256_setzero_ps();
85     __m256 accumulator3 = _mm256_setzero_ps();
86
87     for (; index + LCQI_F32_AVX2_UNROLLED_FLOATS <= size;
88           index += LCQI_F32_AVX2_UNROLLED_FLOATS) {
89         const __m256 lhs0 = _mm256_loadu_ps(lhs + index);
90         const __m256 rhs0 = _mm256_loadu_ps(rhs + index);
91         const __m256 lhs1 = _mm256_loadu_ps(lhs + index + ↵
↵ LCQI_F32_AVX2_LANE_COUNT);
92         const __m256 rhs1 = _mm256_loadu_ps(rhs + index + ↵
↵ LCQI_F32_AVX2_LANE_COUNT);
93         const __m256 lhs2 = _mm256_loadu_ps(lhs + index + ↵
↵ LCQI_F32_AVX2_LANE_COUNT * 2);
94         const __m256 rhs2 = _mm256_loadu_ps(rhs + index + ↵
↵ LCQI_F32_AVX2_LANE_COUNT * 2);
95         const __m256 lhs3 = _mm256_loadu_ps(lhs + index + ↵
↵ LCQI_F32_AVX2_LANE_COUNT * 3);
96         const __m256 rhs3 = _mm256_loadu_ps(rhs + index + ↵
↵ LCQI_F32_AVX2_LANE_COUNT * 3);
97
98         accumulator0 = _mm256_fmadd_ps(lhs0, rhs0, accumulator0);
99         accumulator1 = _mm256_fmadd_ps(lhs1, rhs1, accumulator1);
100        accumulator2 = _mm256_fmadd_ps(lhs2, rhs2, accumulator2);
101        accumulator3 = _mm256_fmadd_ps(lhs3, rhs3, accumulator3);
102    }
103
104    __m256 accumulator = _mm256_add_ps(
105        _mm256_add_ps(accumulator0, accumulator1),
106        _mm256_add_ps(accumulator2, accumulator3));
107    for (; index + LCQI_F32_AVX2_LANE_COUNT <= size;
108          index += LCQI_F32_AVX2_LANE_COUNT) {
109        const __m256 lhs_values = _mm256_loadu_ps(lhs + index);
110        const __m256 rhs_values = _mm256_loadu_ps(rhs + index);
111        accumulator = _mm256_fmadd_ps(lhs_values, rhs_values, accumulator);
112    }
113
114    float sum = horizontal_sum(accumulator);
115    for (; index < size; ++index) {
116        sum += lhs[index] * rhs[index];
117    }
118    return sum;
119 }
120
121 void linear_f32_rows_avx2_unchecked(const float* weights,
122                                     const float* input,
123                                     const float* bias,
124                                     std::size_t input_size,
125                                     std::size_t row_begin,
126                                     std::size_t row_end,
127                                     float* output) noexcept {
128     std::size_t row = row_begin;
129     for (; row + LCQI_F32_AVX2_UNROLL_COUNT <= row_end;

```

```

130     row += LCQI_F32_AVX2_UNROLL_COUNT) {
131     const float* row0 = weights + row * input_size;
132     const float* row1 = row0 + input_size;
133     const float* row2 = row1 + input_size;
134     const float* row3 = row2 + input_size;
135     std::array<float, LCQI_F32_AVX2_UNROLL_COUNT> sums{};
136     compute_four_row_dot(row0, row1, row2, row3, input, input_size, sums.↵
↵ data());
137     for (std::size_t offset = 0; offset < LCQI_F32_AVX2_UNROLL_COUNT; ++↵
↵ offset) {
138         output[row + offset] =
139             sums[offset] + (bias == nullptr ? 0.0F : bias[row + offset]);
140     }
141 }
142 for (; row < row_end; ++row) {
143     const float* weight_row = weights + row * input_size;
144     output[row] = dot_f32_avx2_unchecked(weight_row, input, input_size) +
145         (bias == nullptr ? 0.0F : bias[row]);
146 }
147 }
148
149 F32RowMax max_dot_f32_rows_avx2_unchecked(const float* weights,
150                                           const float* input,
151                                           std::size_t input_size,
152                                           std::size_t row_begin,
153                                           std::size_t row_end) noexcept {
154     F32RowMax best;
155     best.row = row_begin;
156     best.value = -std::numeric_limits<float>::infinity();
157     std::size_t row = row_begin;
158     for (; row + LCQI_F32_AVX2_UNROLL_COUNT <= row_end;
159         row += LCQI_F32_AVX2_UNROLL_COUNT) {
160         const float* row0 = weights + row * input_size;
161         const float* row1 = row0 + input_size;
162         const float* row2 = row1 + input_size;
163         const float* row3 = row2 + input_size;
164         std::array<float, LCQI_F32_AVX2_UNROLL_COUNT> sums{};
165         compute_four_row_dot(row0, row1, row2, row3, input, input_size, sums.↵
↵ data());
166         for (std::size_t offset = 0; offset < LCQI_F32_AVX2_UNROLL_COUNT; ++↵
↵ offset) {
167             if (row + offset == row_begin || sums[offset] > best.value) {
168                 best.row = row + offset;
169                 best.value = sums[offset];
170             }
171         }
172     }
173     for (; row < row_end; ++row) {
174         const float* weight_row = weights + row * input_size;
175         const float value = dot_f32_avx2_unchecked(weight_row, input, ↵
↵ input_size);
176         if (row == row_begin || value > best.value) {

```

```

177         best.row = row;
178         best.value = value;
179     }
180 }
181 return best;
182 }
183
184 } // namespace lcqi
185
186 #endif

```

Listing 10.13 src/gpt2\_reference.cpp

```

1 #include <lcqi/gpt2_reference.hpp>
2
3 #include <lcqi/f32_kernels.hpp>
4 #include <lcqi/safetensors.hpp>
5
6 #include <algorithm>
7 #include <array>
8 #include <atomic>
9 #include <chrono>
10 #include <cmath>
11 #include <cstdint>
12 #include <fstream>
13 #include <limits>
14 #include <optional>
15 #include <sstream>
16 #include <stdexcept>
17 #include <string>
18 #include <string_view>
19 #include <thread>
20 #include <unordered_set>
21 #include <utility>
22
23 namespace lcqi {
24 namespace {
25
26 constexpr std::int32_t LCQI_GPT2_DEFAULT_BOS_EOS = 50256;
27 constexpr std::int32_t LCQI_GPT2_ATTENTION_PROJECTIONS = 3;
28 constexpr std::int32_t LCQI_GPT2_PAIR_TOKEN_COUNT = 2;
29 constexpr std::int32_t LCQI_GPT2_BYTE_COUNT = 256;
30 constexpr std::int32_t LCQI_GPT2_UNICODE_PRIVATE_BASE = 256;
31 constexpr std::int32_t LCQI_GPT2_ASCII_EXCLAMATION = 33;
32 constexpr std::int32_t LCQI_GPT2_ASCII_TILDE = 126;
33 constexpr std::int32_t LCQI_GPT2_LATIN_INVERTED_EXCLAMATION = 161;
34 constexpr std::int32_t LCQI_GPT2_LATIN_NOT_SIGN = 172;
35 constexpr std::int32_t LCQI_GPT2_LATIN_REGISTERED = 174;
36 constexpr std::int32_t LCQI_GPT2_LATIN_Y_DIAERESIS = 255;
37 constexpr std::int32_t LCQI_GPT2_UTF8_CONTINUATION_MASK = 0x3F;
38 constexpr std::int32_t LCQI_GPT2_UTF8_CONTINUATION_BITS = 0x80;
39 constexpr std::int32_t LCQI_GPT2_UTF8_TWO_BYTE_HEAD = 0xC0;

```

```

40 constexpr std::int32_t LCQI_GPT2_UTF8_THREE_BYTE_HEAD = 0xE0;
41 constexpr std::int32_t LCQI_GPT2_UTF8_FOUR_BYTE_HEAD = 0xF0;
42 constexpr std::int32_t LCQI_GPT2_UTF8_ONE_BYTE_LIMIT = 0x80;
43 constexpr std::int32_t LCQI_GPT2_UTF8_TWO_BYTE_LIMIT = 0x800;
44 constexpr std::int32_t LCQI_GPT2_UTF8_THREE_BYTE_LIMIT = 0x10000;
45 constexpr unsigned char LCQI_GPT2_ASCII_APOSTROPHE = 0x27U;
46 constexpr float LCQI_GPT2_GELU_SQRT_2_OVER_PI = 0.7978845608028654F;
47 constexpr float LCQI_GPT2_GELU_CUBIC_COEFF = 0.044715F;
48 constexpr float LCQI_GPT2_SOFTMAX_NEGATIVE_INFINITY =
49     -std::numeric_limits<float>::infinity();
50 constexpr std::int32_t LCQI_GPT2_MAX_AUTO_WORKERS = 8;
51 constexpr std::int32_t LCQI_GPT2_DEFAULT_PARALLEL_MIN_ROWS = 512;
52 constexpr std::int32_t LCQI_GPT2_PARALLEL_MIN_COLUMNS = 512;
53 constexpr std::int64_t LCQI_GPT2_PARALLEL_MIN_OPS = 1'000'000;
54 constexpr std::size_t LCQI_GPT2_PARALLEL_ROW_BLOCK_MULTIPLIER = 4;
55 constexpr std::size_t LCQI_GPT2_WORKER_SPIN_PAUSES_BEFORE_YIELD = 4096;
56
57 using Gpt2ProfileClock = std::chrono::steady_clock;
58
59 void pause_worker_spin(std::size_t& pause_count) noexcept {
60     #if defined(__GNUC__) && (defined(__x86_64__) || defined(__i386__))
61         __builtin_ia32_pause();
62     #endif
63     ++pause_count;
64     if (pause_count >= LCQI_GPT2_WORKER_SPIN_PAUSES_BEFORE_YIELD) {
65         pause_count = 0;
66         std::this_thread::yield();
67     }
68 }
69
70 } // namespace
71
72 namespace detail {
73
74 class Gpt2ParallelWorkerPool {
75 public:
76     explicit Gpt2ParallelWorkerPool(std::int32_t requested_workers)
77         : worker_count_(Gpt2ParallelWorkerPool::normalize_worker_count(↵
↵ requested_workers)) {
78         if (this->worker_count_ <= 1) {
79             return;
80         }
81         const std::size_t background_count =
82             static_cast<std::size_t>(this->worker_count_ - 1);
83         this->threads_.reserve(background_count);
84         for (std::size_t index = 0; index < background_count; ++index) {
85             this->threads_.emplace_back([this, index]() { this->worker_loop(↵
↵ index); });
86         }
87     }
88
89     ~Gpt2ParallelWorkerPool() {

```



```

90     this->stopping_.store(true, std::memory_order_release);
91     this->generation_.fetch_add(1, std::memory_order_release);
92     for (std::thread& thread : this->threads_) {
93         if (thread.joinable()) {
94             thread.join();
95         }
96     }
97 }
98
99 Gpt2ParallelWorkerPool(const Gpt2ParallelWorkerPool&) = delete;
100 Gpt2ParallelWorkerPool& operator=(const Gpt2ParallelWorkerPool&) = delete;
101
102 [[nodiscard]] std::int32_t worker_count() const noexcept {
103     return this->worker_count_;
104 }
105
106 template <typename Function>
107 void parallel_for_rows(std::size_t begin,
108                       std::size_t end,
109                       std::size_t grain,
110                       Function&& function) {
111     if (end <= begin) {
112         return;
113     }
114     const std::size_t row_count = end - begin;
115     if (this->worker_count_ <= 1 || row_count <= grain) {
116         function(begin, end);
117         return;
118     }
119
120     const std::size_t task_count =
121         std::min<std::size_t>(static_cast<std::size_t>(this->worker_count_)↵
↵ ,
122                               (row_count + grain - 1) / grain);
123     if (task_count <= 1) {
124         function(begin, end);
125         return;
126     }
127
128     auto task_context = ParallelTaskContext<Function>{function};
129     // Publish all task metadata before the generation bump; workers use ↵
↵ the
130     // generation acquire load as the hand-off point for this stack context↵
↵ .
131     this->task_begin_.store(begin, std::memory_order_release);
132     this->task_end_.store(end, std::memory_order_release);
133     this->task_count_.store(task_count, std::memory_order_release);
134     this->task_context_.store(&task_context, std::memory_order_release);
135     this->task_invoke_.store(&ParallelTaskContext<Function>::invoke,
136                             std::memory_order_release);
137     this->active_workers_.store(task_count - 1, std::memory_order_release);
138     this->generation_.fetch_add(1, std::memory_order_release);

```

```

139
140     const std::size_t main_worker = task_count - 1;
141     const auto [main_begin, main_end] = this->chunk_bounds(begin, end, ↵
↵ task_count, main_worker);
142     function(main_begin, main_end);
143
144     std::size_t pause_count = 0;
145     while (this->active_workers_.load(std::memory_order_acquire) != 0) {
146         pause_worker_spin(pause_count);
147     }
148     this->task_context_.store(nullptr, std::memory_order_release);
149     this->task_invoke_.store(nullptr, std::memory_order_release);
150 }
151
152 private:
153     template <typename Function>
154     struct ParallelTaskContext {
155         Function& function;
156
157         static void invoke(void* context, std::size_t begin, std::size_t end) {
158             auto* typed_context = static_cast<ParallelTaskContext*>(context);
159             typed_context->function(begin, end);
160         }
161     };
162
163     std::int32_t worker_count_ = 1;
164     std::vector<std::thread> threads_;
165     std::atomic<bool> stopping_ = false;
166     std::atomic<std::uint64_t> generation_ = 0;
167     std::atomic<std::size_t> task_begin_ = 0;
168     std::atomic<std::size_t> task_end_ = 0;
169     std::atomic<std::size_t> task_count_ = 0;
170     std::atomic<std::size_t> active_workers_ = 0;
171     std::atomic<void*> task_context_ = nullptr;
172     std::atomic<void (*) (void*, std::size_t, std::size_t)> task_invoke_ = ↵
↵ nullptr;
173
174     [[nodiscard]] static std::int32_t normalize_worker_count(std::int32_t ↵
↵ requested_workers) {
175         if (requested_workers > 0) {
176             return requested_workers;
177         }
178         const unsigned int hardware_workers = std::thread::hardware_concurrency↵
↵ ();
179         if (hardware_workers == 0) {
180             return 1;
181         }
182         const std::int32_t capped =
183             std::min<std::int32_t>(static_cast<std::int32_t>(hardware_workers),
184                                   LCQI_GPT2_MAX_AUTO_WORKERS);
185         return std::max(capped, 1);
186     }

```

```

187
188 [[nodiscard]] std::pair<std::size_t, std::size_t> chunk_bounds(
189     std::size_t begin,
190     std::size_t end,
191     std::size_t task_count,
192     std::size_t worker_index) const {
193     const std::size_t row_count = end - begin;
194     const std::size_t chunk_begin = begin + row_count * worker_index / ↵
↵ task_count;
195     const std::size_t chunk_end = begin + row_count * (worker_index + 1) / ↵
↵ task_count;
196     return {chunk_begin, chunk_end};
197 }
198
199 void worker_loop(std::size_t worker_index) {
200     std::uint64_t observed_generation = 0;
201     std::size_t pause_count = 0;
202     while (true) {
203         const std::uint64_t current_generation =
204             this->generation_.load(std::memory_order_acquire);
205         if (current_generation == observed_generation) {
206             if (this->stopping_.load(std::memory_order_acquire)) {
207                 return;
208             }
209             pause_worker_spin(pause_count);
210             continue;
211         }
212
213         observed_generation = current_generation;
214         pause_count = 0;
215         if (this->stopping_.load(std::memory_order_acquire)) {
216             return;
217         }
218
219         const std::size_t task_count = this->task_count_.load(std::↵
↵ memory_order_acquire);
220         if (this->generation_.load(std::memory_order_acquire) != ↵
↵ current_generation) {
221             continue;
222         }
223         const std::size_t background_task_count = task_count - 1;
224         if (worker_index >= background_task_count) {
225             continue;
226         }
227         void* task_context = this->task_context_.load(std::↵
↵ memory_order_acquire);
228         void (*task_invoke)(void*, std::size_t, std::size_t) =
229             this->task_invoke_.load(std::memory_order_acquire);
230         if (task_context == nullptr || task_invoke == nullptr) {
231             continue;
232         }
233         const std::size_t task_begin = this->task_begin_.load(std::↵

```

```

    ↪ memory_order_acquire);
234     const std::size_t task_end = this->task_end_.load(std::memory_order_acquire);
    ↪ memory_order_acquire);
235     if (this->generation_.load(std::memory_order_acquire) != current_generation) {
236         continue;
237     }
238     const auto [begin, end] =
239         this->chunk_bounds(task_begin,
240                             task_end,
241                             task_count,
242                             worker_index);
243     task_invoke(task_context, begin, end);
244
245     this->active_workers_.fetch_sub(1, std::memory_order_acq_rel);
246 }
247 }
248 };
249
250 } // namespace detail
251
252 namespace {
253
254 class JsonCursor {
255 public:
256     explicit JsonCursor(std::string_view text) : text_(text) {}
257
258     void expect(char expected) {
259         this->skip_ws();
260         if (this->eof() || this->text_[this->position_] != expected) {
261             throw std::runtime_error("invalid JSON syntax");
262         }
263         ++this->position_;
264     }
265
266     [[nodiscard]] bool consume(char expected) {
267         this->skip_ws();
268         if (!this->eof() && this->text_[this->position_] == expected) {
269             ++this->position_;
270             return true;
271         }
272         return false;
273     }
274
275     [[nodiscard]] std::string parse_string() {
276         this->skip_ws();
277         this->expect('"');
278         std::string result;
279         while (!this->eof()) {
280             const char ch = this->text_[this->position_++];
281             if (ch == '"') {
282                 return result;

```

```

283     }
284     if (ch == '\\') {
285         result += this->parse_escape();
286     } else {
287         result.push_back(ch);
288     }
289 }
290 throw std::runtime_error("unterminated JSON string");
291 }
292
293 [[nodiscard]] std::int64_t parse_i64() {
294     this->skip_ws();
295     bool negative = false;
296     if (this->consume('-')) {
297         negative = true;
298     }
299     if (this->eof() || this->text_[this->position_] < '0' ||
300         this->text_[this->position_] > '9') {
301         throw std::runtime_error("expected JSON integer");
302     }
303     std::int64_t value = 0;
304     while (!this->eof() && this->text_[this->position_] >= '0' &&
305         this->text_[this->position_] <= '9') {
306         const std::int64_t digit = this->text_[this->position_] - '0';
307         if (value >
308             (std::numeric_limits<std::int64_t>::max() - digit) / 10) {
309             throw std::runtime_error("JSON integer overflow");
310         }
311         value = value * 10 + digit;
312         ++this->position_;
313     }
314     return negative ? -value : value;
315 }
316
317 [[nodiscard]] double parse_number() {
318     this->skip_ws();
319     const std::size_t begin = this->position_;
320     if (!this->eof() && this->text_[this->position_] == '-') {
321         ++this->position_;
322     }
323     while (!this->eof() && this->text_[this->position_] >= '0' &&
324         this->text_[this->position_] <= '9') {
325         ++this->position_;
326     }
327     if (!this->eof() && this->text_[this->position_] == '.') {
328         ++this->position_;
329         while (!this->eof() && this->text_[this->position_] >= '0' &&
330             this->text_[this->position_] <= '9') {
331             ++this->position_;
332         }
333     }
334     if (!this->eof() &&

```

```

335         (this->text_[this->position_] == 'e' || this->text_[this->position_ ←
↪ ] == 'E')) {
336             ++this->position_;
337             if (!this->eof() &&
338                 (this->text_[this->position_] == '+' || this->text_[this-> ←
↪ position_] == '-')) {
339                 ++this->position_;
340             }
341             while (!this->eof() && this->text_[this->position_] >= '0' &&
342                 this->text_[this->position_] <= '9') {
343                 ++this->position_;
344             }
345         }
346         if (begin == this->position_) {
347             throw std::runtime_error("expected JSON number");
348         }
349         return std::stod(std::string(this->text_.substr(begin, this->position_ ←
↪ - begin)));
350     }
351
352     void skip_value() {
353         this->skip_ws();
354         if (this->eof()) {
355             throw std::runtime_error("unexpected end of JSON");
356         }
357         const char ch = this->text_[this->position_];
358         if (ch == '"') {
359             static_cast<void>(this->parse_string());
360             return;
361         }
362         if (ch == '{') {
363             this->expect('{');
364             if (this->consume(' ')) {
365                 return;
366             }
367             while (true) {
368                 static_cast<void>(this->parse_string());
369                 this->expect(':');
370                 this->skip_value();
371                 if (this->consume(' ')) {
372                     return;
373                 }
374                 this->expect(',');
375             }
376         }
377         if (ch == '[') {
378             this->expect('[');
379             if (this->consume(' ')) {
380                 return;
381             }
382             while (true) {
383                 this->skip_value();

```

```

384         if (this->consume(' ')) {
385             return;
386         }
387         this->expect(',');
388     }
389 }
390 if (ch == 't') {
391     this->expect_literal("true");
392     return;
393 }
394 if (ch == 'f') {
395     this->expect_literal("false");
396     return;
397 }
398 if (ch == 'n') {
399     this->expect_literal("null");
400     return;
401 }
402 static_cast<void>(this->parse_number());
403 }
404
405 [[nodiscard]] bool eof_after_ws() {
406     this->skip_ws();
407     return this->eof();
408 }
409
410 private:
411     std::string_view text_;
412     std::size_t position_ = 0;
413
414     [[nodiscard]] bool eof() const {
415         return this->position_ >= this->text_.size();
416     }
417
418     void skip_ws() {
419         while (!this->eof() &&
420             (this->text_[this->position_] == ' ' ||
421              this->text_[this->position_] == '\n' ||
422              this->text_[this->position_] == '\r' ||
423              this->text_[this->position_] == '\t')) {
424             ++this->position_;
425         }
426     }
427
428     void expect_literal(std::string_view literal) {
429         for (const char expected : literal) {
430             if (this->eof() || this->text_[this->position_] != expected) {
431                 throw std::runtime_error("invalid JSON literal");
432             }
433             ++this->position_;
434         }
435     }

```



```

436
437 [[nodiscard]] std::string parse_escape() {
438     if (this->eof()) {
439         throw std::runtime_error("unterminated JSON escape");
440     }
441     const char escaped = this->text_[this->position_++];
442     switch (escaped) {
443         case '"':
444         case '\\':
445         case '/':
446             return std::string(1, escaped);
447         case 'b':
448             return "\b";
449         case 'f':
450             return "\f";
451         case 'n':
452             return "\n";
453         case 'r':
454             return "\r";
455         case 't':
456             return "\t";
457         case 'u':
458             return this->parse_unicode_escape();
459         default:
460             throw std::runtime_error("unsupported JSON escape");
461     }
462 }
463
464 [[nodiscard]] std::string parse_unicode_escape() {
465     std::uint32_t value = 0;
466     for (std::int32_t i = 0; i < 4; ++i) {
467         if (this->eof()) {
468             throw std::runtime_error("unterminated JSON unicode escape");
469         }
470         const char ch = this->text_[this->position_++];
471         value <= 4U;
472         if (ch >= '0' && ch <= '9') {
473             value += static_cast<std::uint32_t>(ch - '0');
474         } else if (ch >= 'a' && ch <= 'f') {
475             value += static_cast<std::uint32_t>(10 + ch - 'a');
476         } else if (ch >= 'A' && ch <= 'F') {
477             value += static_cast<std::uint32_t>(10 + ch - 'A');
478         } else {
479             throw std::runtime_error("invalid JSON unicode escape");
480         }
481     }
482     std::string encoded;
483     if (value < LCQI_GPT2_UTF8_ONE_BYTE_LIMIT) {
484         encoded.push_back(static_cast<char>(value));
485     } else if (value < LCQI_GPT2_UTF8_TWO_BYTE_LIMIT) {
486         encoded.push_back(static_cast<char>(
487             LCQI_GPT2_UTF8_TWO_BYTE_HEAD | (value >> 6U)));

```

```

488         encoded.push_back(static_cast<char>(
489             LCQI_GPT2_UTF8_CONTINUATION_BITS |
490             (value & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
491     } else {
492         encoded.push_back(static_cast<char>(
493             LCQI_GPT2_UTF8_THREE_BYTE_HEAD | (value >> 12U)));
494         encoded.push_back(static_cast<char>(
495             LCQI_GPT2_UTF8_CONTINUATION_BITS |
496             ((value >> 6U) & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
497         encoded.push_back(static_cast<char>(
498             LCQI_GPT2_UTF8_CONTINUATION_BITS |
499             (value & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
500     }
501     return encoded;
502 }
503 };
504
505 std::string read_text_file(const std::filesystem::path& path) {
506     std::ifstream input(path);
507     if (!input) {
508         throw std::runtime_error("cannot open text file: " + path.string());
509     }
510     std::ostringstream buffer;
511     buffer << input.rdbuf();
512     return buffer.str();
513 }
514
515 std::size_t checked_size(std::int64_t value, const char* name) {
516     if (value < 0) {
517         throw std::runtime_error(std::string(name) + " cannot be negative");
518     }
519     return static_cast<std::size_t>(value);
520 }
521
522 std::string encode_codepoint(std::uint32_t value) {
523     std::string encoded;
524     if (value < LCQI_GPT2_UTF8_ONE_BYTE_LIMIT) {
525         encoded.push_back(static_cast<char>(value));
526     } else if (value < LCQI_GPT2_UTF8_TWO_BYTE_LIMIT) {
527         encoded.push_back(static_cast<char>(
528             LCQI_GPT2_UTF8_TWO_BYTE_HEAD | (value >> 6U)));
529         encoded.push_back(static_cast<char>(
530             LCQI_GPT2_UTF8_CONTINUATION_BITS |
531             (value & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
532     } else if (value < LCQI_GPT2_UTF8_THREE_BYTE_LIMIT) {
533         encoded.push_back(static_cast<char>(
534             LCQI_GPT2_UTF8_THREE_BYTE_HEAD | (value >> 12U)));
535         encoded.push_back(static_cast<char>(
536             LCQI_GPT2_UTF8_CONTINUATION_BITS |
537             ((value >> 6U) & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
538         encoded.push_back(static_cast<char>(
539             LCQI_GPT2_UTF8_CONTINUATION_BITS |

```

```

540         (value & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
541     } else {
542         encoded.push_back(static_cast<char>(
543             LCQI_GPT2_UTF8_FOUR_BYTE_HEAD | (value >> 18U)));
544         encoded.push_back(static_cast<char>(
545             LCQI_GPT2_UTF8_CONTINUATION_BITS |
546             ((value >> 12U) & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
547         encoded.push_back(static_cast<char>(
548             LCQI_GPT2_UTF8_CONTINUATION_BITS |
549             ((value >> 6U) & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
550         encoded.push_back(static_cast<char>(
551             LCQI_GPT2_UTF8_CONTINUATION_BITS |
552             (value & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
553     }
554     return encoded;
555 }
556
557 void require_positive(std::int32_t value, const char* name) {
558     if (value <= 0) {
559         throw std::runtime_error(std::string(name) + " must be positive");
560     }
561 }
562
563 void validate_config(const Gpt2Config& config) {
564     require_positive(config.hidden_size, "hidden_size");
565     require_positive(config.head_count, "head_count");
566     require_positive(config.layer_count, "layer_count");
567     require_positive(config.max_positions, "max_positions");
568     require_positive(config.vocab_size, "vocab_size");
569     require_positive(config.intermediate_size, "intermediate_size");
570     if (config.hidden_size % config.head_count != 0) {
571         throw std::runtime_error("hidden_size must be divisible by head_count")↵
572     };
573     if (config.layer_norm_epsilon <= 0.0F) {
574         throw std::runtime_error("layer_norm_epsilon must be positive");
575     }
576 }
577
578 std::int32_t head_dim(const Gpt2Config& config) {
579     return config.hidden_size / config.head_count;
580 }
581
582 std::size_t matrix_size(std::int32_t rows, std::int32_t columns) {
583     return checked_size(rows, "rows") * checked_size(columns, "columns");
584 }
585
586 Gpt2ProfileClock::time_point profile_start(const Gpt2HotspotProfile* profile) {
587     if (profile == nullptr) {
588         return Gpt2ProfileClock::time_point{};
589     }
590     return Gpt2ProfileClock::now();

```

```

591 }
592
593 void add_profile_ms(Gpt2HotspotProfile* profile,
594                   double Gpt2HotspotProfile::*field,
595                   Gpt2ProfileClock::time_point start) {
596     if (profile == nullptr) {
597         return;
598     }
599     (*profile).*field +=
600         std::chrono::duration<double, std::milli>(Gpt2ProfileClock::now() - ↵
↵ start).count());
601 }
602
603 void validate_vector_size(std::size_t actual, std::int32_t expected, const char↵
↵ * name) {
604     if (actual != checked_size(expected, name)) {
605         throw std::runtime_error(std::string(name) + " size mismatch");
606     }
607 }
608
609 void validate_linear(const Gpt2LinearF32& linear, const char* name) {
610     require_positive(linear.input_size, "linear input_size");
611     require_positive(linear.output_size, "linear output_size");
612     if (linear.weights.size() != matrix_size(linear.output_size, linear.↵
↵ input_size)) {
613         throw std::runtime_error(std::string(name) + " weight size mismatch");
614     }
615     if (!linear.bias.empty() &&
616         linear.bias.size() != checked_size(linear.output_size, "linear ↵
↵ output_size")) {
617         throw std::runtime_error(std::string(name) + " bias size mismatch");
618     }
619 }
620
621 void validate_model(const Gpt2ReferenceModel& model) {
622     validate_config(model.config);
623     if (model.layers.size() != checked_size(model.config.layer_count, "↵
↵ layer_count")) {
624         throw std::runtime_error("GPT-2 layer count mismatch");
625     }
626     if (model.token_embedding.size() !=
627         matrix_size(model.config.vocab_size, model.config.hidden_size)) {
628         throw std::runtime_error("GPT-2 token embedding size mismatch");
629     }
630     if (model.position_embedding.size() !=
631         matrix_size(model.config.max_positions, model.config.hidden_size)) {
632         throw std::runtime_error("GPT-2 position embedding size mismatch");
633     }
634     validate_vector_size(model.final_ln_weight.size(), model.config.hidden_size↵
↵ , "ln_f weight");
635     validate_vector_size(model.final_ln_bias.size(), model.config.hidden_size, ↵
↵ "ln_f bias");

```

```

636     if (model.tie_lm_head_to_embedding) {
637         if (!model.lm_head_weight.empty()) {
638             throw std::runtime_error("tied GPT-2 lm_head should not carry ↵
↵ separate weights");
639         }
640     } else if (model.lm_head_weight.size() !=
641         matrix_size(model.config.vocab_size, model.config.hidden_size)) ↵
↵ {
642         throw std::runtime_error("GPT-2 lm_head size mismatch");
643     }
644     for (const Gpt2LayerWeightsF32& layer : model.layers) {
645         validate_vector_size(layer.ln_1_weight.size(), model.config.hidden_size ↵
↵ , "ln_1 weight");
646         validate_vector_size(layer.ln_1_bias.size(), model.config.hidden_size, ↵
↵ "ln_1 bias");
647         validate_linear(layer.c_attn, "c_attn");
648         validate_linear(layer.c_proj, "c_proj");
649         validate_vector_size(layer.ln_2_weight.size(), model.config.hidden_size ↵
↵ , "ln_2 weight");
650         validate_vector_size(layer.ln_2_bias.size(), model.config.hidden_size, ↵
↵ "ln_2 bias");
651         validate_linear(layer.c_fc, "c_fc");
652         validate_linear(layer.mlp_c_proj, "mlp_c_proj");
653     }
654 }
655
656 std::span<const float> row_span(std::span<const float> values,
657     std::int32_t row,
658     std::int32_t row_width) {
659     return values.subspan(checked_size(row, "row") * checked_size(row_width, "↵
↵ row_width"),
660         checked_size(row_width, "row_width"));
661 }
662
663 void add_inplace(std::span<float> target, std::span<const float> source) {
664     if (target.size() != source.size()) {
665         throw std::runtime_error("GPT-2 residual add size mismatch");
666     }
667     for (std::size_t index = 0; index < target.size(); ++index) {
668         target[index] += source[index];
669     }
670 }
671
672 void layer_norm(std::span<const float> input,
673     std::span<const float> weight,
674     std::span<const float> bias,
675     float epsilon,
676     std::span<float> output) {
677     if (input.empty() || input.size() != weight.size() || input.size() != bias.↵
↵ size() ||
678         input.size() != output.size()) {
679         throw std::runtime_error("GPT-2 LayerNorm size mismatch");

```

```

680     }
681     float mean = 0.0F;
682     for (const float value : input) {
683         mean += value;
684     }
685     mean /= static_cast<float>(input.size());
686
687     float variance = 0.0F;
688     for (const float value : input) {
689         const float centered = value - mean;
690         variance += centered * centered;
691     }
692     variance /= static_cast<float>(input.size());
693     const float scale = 1.0F / std::sqrt(variance + epsilon);
694
695     for (std::size_t index = 0; index < input.size(); ++index) {
696         output[index] = (input[index] - mean) * scale * weight[index] + bias[↵
↵ index];
697     }
698 }
699
700 void linear_f32(const Gpt2LinearF32& linear,
701                std::span<const float> input,
702                std::span<float> output) {
703     validate_linear(linear, "GPT-2 linear");
704     if (input.size() != checked_size(linear.input_size, "linear input") ||
705         output.size() != checked_size(linear.output_size, "linear output")) {
706         throw std::runtime_error("GPT-2 linear input/output size mismatch");
707     }
708     const std::size_t input_size = checked_size(linear.input_size, "linear ↵
↵ input");
709     const std::size_t output_size = checked_size(linear.output_size, "linear ↵
↵ output");
710     for (std::size_t out = 0; out < output_size; ++out) {
711         float sum = linear.bias.empty() ? 0.0F : linear.bias[out];
712         const std::size_t row_base = out * input_size;
713         for (std::size_t in = 0; in < input_size; ++in) {
714             sum += linear.weights[row_base + in] * input[in];
715         }
716         output[out] = sum;
717     }
718 }
719
720 void linear_f32_unchecked(const Gpt2LinearF32& linear,
721                           std::span<const float> input,
722                           std::span<float> output) {
723     const std::size_t input_size = static_cast<std::size_t>(linear.input_size);
724     const std::size_t output_size = static_cast<std::size_t>(linear.output_size↵
↵ );
725     const float* weights = linear.weights.data();
726     const float* bias = linear.bias.empty() ? nullptr : linear.bias.data();
727     const float* input_data = input.data();

```

```

728 linear_f32_rows_unchecked(weights, input_data, bias, input_size, 0, ↵
↵ output_size, output.data());
729 }
730
731 [[nodiscard]] bool should_parallelize_rows(const Gpt2ExecutionOptions& options,
732                                           std::size_t rows,
733                                           std::size_t columns,
734                                           const detail::Gpt2ParallelWorkerPool↵
↵ * worker_pool) {
735     if (worker_pool == nullptr || worker_pool->worker_count() <= 1) {
736         return false;
737     }
738     const std::int32_t min_rows =
739         options.parallel_min_rows > 0 ? options.parallel_min_rows
740         : LCQI_GPT2_DEFAULT_PARALLEL_MIN_ROWS;
741     if (rows < checked_size(min_rows, "parallel_min_rows")) {
742         return false;
743     }
744     if (options.worker_count > 1) {
745         return true;
746     }
747     return columns >= checked_size(LCQI_GPT2_PARALLEL_MIN_COLUMNS, "↵
↵ parallel_min_columns") ||
748         rows * columns >= static_cast<std::size_t>(↵
↵ LCQI_GPT2_PARALLEL_MIN_OPS);
749 }
750
751 [[nodiscard]] std::size_t parallel_row_grain(std::size_t rows,
752                                              const detail::↵
↵ Gpt2ParallelWorkerPool& worker_pool) {
753     const std::size_t workers = checked_size(worker_pool.worker_count(), "↵
↵ worker_count");
754     const std::size_t target_tasks = workers * ↵
↵ LCQI_GPT2_PARALLEL_ROW_BLOCK_MULTIPLIER;
755     return std::max<std::size_t>(1, (rows + target_tasks - 1) / target_tasks);
756 }
757
758 [[nodiscard]] bool model_has_parallel_work(const Gpt2ReferenceModel& model,
759                                             const Gpt2ExecutionOptions& options)↵
↵ {
760     const std::size_t hidden_size = checked_size(model.config.hidden_size, "↵
↵ hidden_size");
761     const std::size_t vocab_size = checked_size(model.config.vocab_size, "↵
↵ vocab_size");
762     const std::size_t intermediate_size =
763         checked_size(model.config.intermediate_size, "intermediate_size");
764     const std::size_t qkv_rows =
765         checked_size(model.config.hidden_size * LCQI_GPT2_ATTENTION_PROJECTIONS↵
↵ ,
766                     "qkv output rows");
767     const std::int32_t min_rows =
768         options.parallel_min_rows > 0 ? options.parallel_min_rows

```



```

769                                     : LCQI_GPT2_DEFAULT_PARALLEL_MIN_ROWS;
770     const auto large_enough = [min_rows](std::size_t rows, std::size_t columns)↵
↵ {
771         if (rows < checked_size(min_rows, "parallel_min_rows")) {
772             return false;
773         }
774         return columns >= checked_size(LCQI_GPT2_PARALLEL_MIN_COLUMNS,
775                                         "parallel_min_columns") ||
776                rows * columns >= static_cast<std::size_t>(↵
↵ LCQI_GPT2_PARALLEL_MIN_OPS);
777     };
778     if (options.worker_count > 1) {
779         return vocab_size >= checked_size(min_rows, "parallel_min_rows") ||
780                qkv_rows >= checked_size(min_rows, "parallel_min_rows") ||
781                intermediate_size >= checked_size(min_rows, "parallel_min_rows")↵
↵ ||
782                hidden_size >= checked_size(min_rows, "parallel_min_rows");
783     }
784     if (options.worker_count == 1) {
785         return false;
786     }
787     return large_enough(vocab_size, hidden_size) ||
788            large_enough(qkv_rows, hidden_size) ||
789            large_enough(hidden_size, hidden_size) ||
790            large_enough(intermediate_size, hidden_size) ||
791            large_enough(hidden_size, intermediate_size);
792 }
793
794 [[nodiscard]] std::unique_ptr<detail::Gpt2ParallelWorkerPool> make_worker_pool(
795     const Gpt2ReferenceModel& model,
796     const Gpt2ExecutionOptions& options) {
797     if (!model_has_parallel_work(model, options)) {
798         return nullptr;
799     }
800     return std::make_unique<detail::Gpt2ParallelWorkerPool>(options.↵
↵ worker_count);
801 }
802
803 void linear_f32_unchecked_parallel(const Gpt2LinearF32& linear,
804                                     std::span<const float> input,
805                                     std::span<float> output,
806                                     const Gpt2ExecutionOptions& options,
807                                     detail::Gpt2ParallelWorkerPool* worker_pool)↵
↵ {
808     const std::size_t input_size = static_cast<std::size_t>(linear.input_size);
809     const std::size_t output_size = static_cast<std::size_t>(linear.output_size)↵
↵ );
810     if (!should_parallelize_rows(options, output_size, input_size, worker_pool)↵
↵ ) {
811         linear_f32_unchecked(linear, input, output);
812         return;
813     }

```

```

814
815     const float* weights = linear.weights.data();
816     const float* bias = linear.bias.empty() ? nullptr : linear.bias.data();
817     const float* input_data = input.data();
818     float* output_data = output.data();
819     const auto rows_fn =
820         [input_size, weights, bias, input_data, output_data](std::size_t begin,
821             std::size_t end) {
822         linear_f32_rows_unchecked(weights,
823             input_data,
824             bias,
825             input_size,
826             begin,
827             end,
828             output_data);
829     };
830     worker_pool->parallel_for_rows(0, output_size, parallel_row_grain(↵
831     ↵ output_size, *worker_pool), rows_fn);
832 }
833
834 float gelu_new(float value) {
835     const float cubic = value * value * value;
836     return 0.5F * value *
837         (1.0F + std::tanh(LCQI_GPT2_GELU_SQRT_2_OVER_PI *
838             (value + LCQI_GPT2_GELU_CUBIC_COEFF * cubic)));
839 }
840
841 std::int32_t argmax(std::span<const float> values) {
842     if (values.empty()) {
843         throw std::runtime_error("cannot take argmax of empty logits");
844     }
845     std::int32_t best_index = 0;
846     float best_value = values.front();
847     for (std::int32_t index = 1; index < static_cast<std::int32_t>(values.size(↵
848     ↵ ()); ++index) {
849         const float value = values[checked_size(index, "argmax index")];
850         if (value > best_value) {
851             best_value = value;
852             best_index = index;
853         }
854     }
855     return best_index;
856 }
857
858 void project_qkv(const Gpt2Config& config,
859     const Gpt2LayerWeightsF32& layer,
860     std::span<const float> hidden,
861     std::span<float> q,
862     std::span<float> k,
863     std::span<float> v) {
864     std::vector<float> packed(matrix_size(LCQI_GPT2_ATTENTION_PROJECTIONS,
865         config.hidden_size),

```

```

864         0.0F);
865     linear_f32(layer.c_attn, hidden, packed);
866     const std::size_t width = checked_size(config.hidden_size, "hidden_size");
867     std::copy_n(packed.begin(), config.hidden_size, q.begin());
868     std::copy_n(packed.begin() + static_cast<std::ptrdiff_t>(width),
869                 config.hidden_size,
870                 k.begin());
871     std::copy_n(packed.begin() + static_cast<std::ptrdiff_t>(width * 2),
872                 config.hidden_size,
873                 v.begin());
874 }
875
876 void project_qkv_reuse_workspace(const Gpt2Config& config,
877                                 const Gpt2LayerWeightsF32& layer,
878                                 std::span<const float> hidden,
879                                 std::span<float> packed,
880                                 std::span<float> q,
881                                 std::span<float> k,
882                                 std::span<float> v,
883                                 const Gpt2ExecutionOptions& options,
884                                 detail::Gpt2ParallelWorkerPool* worker_pool) {
885     linear_f32_unchecked_parallel(layer.c_attn, hidden, packed, options, ←
    ↪ worker_pool);
886     const std::size_t width = checked_size(config.hidden_size, "hidden_size");
887     std::copy_n(packed.begin(), config.hidden_size, q.begin());
888     std::copy_n(packed.begin() + static_cast<std::ptrdiff_t>(width),
889                 config.hidden_size,
890                 k.begin());
891     std::copy_n(packed.begin() + static_cast<std::ptrdiff_t>(width * 2),
892                 config.hidden_size,
893                 v.begin());
894 }
895
896 void attention_full_sequence(const Gpt2Config& config,
897                             std::span<const float> q_by_pos,
898                             std::span<const float> k_by_pos,
899                             std::span<const float> v_by_pos,
900                             std::int32_t sequence_length,
901                             std::span<float> output_by_pos) {
902     const std::int32_t local_head_dim = head_dim(config);
903     const float score_scale = 1.0F / std::sqrt(static_cast<float>(←
    ↪ local_head_dim));
904     std::fill(output_by_pos.begin(), output_by_pos.end(), 0.0F);
905     std::vector<float> scores(checked_size(sequence_length, "sequence_length"),
906                             LCQI_GPT2_SOFTMAX_NEGATIVE_INFINITY);
907
908     for (std::int32_t position = 0; position < sequence_length; ++position) {
909         for (std::int32_t head = 0; head < config.head_count; ++head) {
910             float max_score = LCQI_GPT2_SOFTMAX_NEGATIVE_INFINITY;
911             for (std::int32_t past = 0; past <= position; ++past) {
912                 float dot = 0.0F;
913                 for (std::int32_t dim = 0; dim < local_head_dim; ++dim) {

```

```

914         const std::size_t q_index =
915             matrix_size(position, config.hidden_size) +
916             matrix_size(head, local_head_dim) + checked_size(dim, "↵
↵ head dim");
917         const std::size_t k_index =
918             matrix_size(past, config.hidden_size) +
919             matrix_size(head, local_head_dim) + checked_size(dim, "↵
↵ head dim");
920         dot += q_by_pos[q_index] * k_by_pos[k_index];
921     }
922     const float score = dot * score_scale;
923     scores[checked_size(past, "past")] = score;
924     max_score = std::max(max_score, score);
925 }
926
927 float denominator = 0.0F;
928 for (std::int32_t past = 0; past <= position; ++past) {
929     const std::size_t index = checked_size(past, "past");
930     scores[index] = std::exp(scores[index] - max_score);
931     denominator += scores[index];
932 }
933 if (denominator <= 0.0F) {
934     throw std::runtime_error("GPT-2 attention denominator is ↵
↵ invalid");
935 }
936 for (std::int32_t past = 0; past <= position; ++past) {
937     const float probability = scores[checked_size(past, "past")] / ↵
↵ denominator;
938     for (std::int32_t dim = 0; dim < local_head_dim; ++dim) {
939         const std::size_t output_index =
940             matrix_size(position, config.hidden_size) +
941             matrix_size(head, local_head_dim) + checked_size(dim, "↵
↵ head dim");
942         const std::size_t value_index =
943             matrix_size(past, config.hidden_size) +
944             matrix_size(head, local_head_dim) + checked_size(dim, "↵
↵ head dim");
945         output_by_pos[output_index] += probability * v_by_pos[↵
↵ value_index];
946     }
947 }
948 }
949 }
950 }
951
952 void forward_gpt2_layer(const Gpt2Config& config,
953     const Gpt2LayerWeightsF32& layer,
954     std::int32_t sequence_length,
955     std::vector<float>& hidden_by_pos) {
956     std::vector<float> normed(checked_size(config.hidden_size, "hidden_size"), ↵
↵ 0.0F);
957     std::vector<float> q_by_pos(matrix_size(sequence_length, config.hidden_size, ↵

```

```

↪ ), 0.0F);
958 std::vector<float> k_by_pos(q_by_pos.size(), 0.0F);
959 std::vector<float> v_by_pos(q_by_pos.size(), 0.0F);
960 std::vector<float> attention_by_pos(q_by_pos.size(), 0.0F);
961 std::vector<float> projected(checkered_size(config.hidden_size, "hidden_size"↪
↪ ), 0.0F);
962 std::vector<float> mlp_fc(checkered_size(config.intermediate_size, "↪
↪ intermediate_size"), 0.0F);
963 std::vector<float> mlp_out(checkered_size(config.hidden_size, "hidden_size"),↪
↪ 0.0F);
964
965 for (std::int32_t position = 0; position < sequence_length; ++position) {
966     const std::span<float> hidden = std::span<float>(
967         hidden_by_pos.data() + matrix_size(position, config.hidden_size),
968         checked_size(config.hidden_size, "hidden_size"));
969     layer_norm(hidden,
970         layer.ln_1_weight,
971         layer.ln_1_bias,
972         config.layer_norm_epsilon,
973         normed);
974     std::span<float> q = std::span<float>(
975         q_by_pos.data() + matrix_size(position, config.hidden_size),
976         checked_size(config.hidden_size, "hidden_size"));
977     std::span<float> k = std::span<float>(
978         k_by_pos.data() + matrix_size(position, config.hidden_size),
979         checked_size(config.hidden_size, "hidden_size"));
980     std::span<float> v = std::span<float>(
981         v_by_pos.data() + matrix_size(position, config.hidden_size),
982         checked_size(config.hidden_size, "hidden_size"));
983     project_qkv(config, layer, normed, q, k, v);
984 }
985
986 attention_full_sequence(config, q_by_pos, k_by_pos, v_by_pos, ↪
↪ sequence_length, attention_by_pos);
987
988 for (std::int32_t position = 0; position < sequence_length; ++position) {
989     const std::span<float> hidden = std::span<float>(
990         hidden_by_pos.data() + matrix_size(position, config.hidden_size),
991         checked_size(config.hidden_size, "hidden_size"));
992     const std::span<const float> attention = std::span<const float>(
993         attention_by_pos.data() + matrix_size(position, config.hidden_size)↪
↪ ,
994         checked_size(config.hidden_size, "hidden_size"));
995     linear_f32(layer.c_proj, attention, projected);
996     add_inplace(hidden, projected);
997
998     layer_norm(hidden,
999         layer.ln_2_weight,
1000         layer.ln_2_bias,
1001         config.layer_norm_epsilon,
1002         normed);
1003     linear_f32(layer.c_fc, normed, mlp_fc);

```

```

1004     for (float& value : mlp_fc) {
1005         value = gelu_new(value);
1006     }
1007     linear_f32(layer.mlp_c_proj, mlp_fc, mlp_out);
1008     add_inplace(hidden, mlp_out);
1009 }
1010 }
1011
1012 void forward_gpt2_layer_cached(const Gpt2Config& config,
1013                               const Gpt2LayerWeightsF32& layer,
1014                               std::int32_t layer_id,
1015                               std::int32_t model_position,
1016                               Gpt2KvCache& cache,
1017                               Gpt2ForwardWorkspace& workspace,
1018                               std::span<float> hidden,
1019                               Gpt2HotspotProfile* hotspot_profile,
1020                               const Gpt2ExecutionOptions& options,
1021                               detail::Gpt2ParallelWorkerPool* worker_pool) {
1022     if (hotspot_profile != nullptr) {
1023         ++hotspot_profile->layer_steps;
1024     }
1025     std::span<float> normed = workspace.normed();
1026     std::span<float> query = workspace.query();
1027     std::span<float> key = workspace.key();
1028     std::span<float> value = workspace.value();
1029     std::span<float> attention = workspace.attention();
1030     std::span<float> projected = workspace.projected();
1031     std::span<float> qkv_packed = workspace.qkv_packed();
1032     std::span<float> mlp_fc = workspace.mlp_fc();
1033     std::span<float> mlp_out = workspace.mlp_out();
1034     std::span<float> scores = workspace.scores_prefix(model_position + 1);
1035
1036     Gpt2ProfileClock::time_point start =
1037         profile_start(hotspot_profile);
1038     layer_norm(hidden,
1039               layer.ln_1_weight,
1040               layer.ln_1_bias,
1041               config.layer_norm_epsilon,
1042               normed);
1043     add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::layer_norm_ms, start);
1044
1045     start = profile_start(hotspot_profile);
1046     project_qkv_reuse_workspace(config,
1047                                layer,
1048                                normed,
1049                                qkv_packed,
1050                                query,
1051                                key,
1052                                value,
1053                                options,
1054                                worker_pool);
1055     add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::qkv_projection_ms, ←

```

```

↪ start);
1056
1057 start = profile_start(hotspot_profile);
1058 cache.append(layer_id, model_position, key, value);
1059 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::kv_append_ms, start);
1060
1061 start = profile_start(hotspot_profile);
1062 detail::attend_cached_position(cache,
1063                                layer_id,
1064                                model_position,
1065                                query,
1066                                scores,
1067                                attention);
1068 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::attention_ms, start);
1069
1070 start = profile_start(hotspot_profile);
1071 linear_f32_unchecked_parallel(layer.c_proj, attention, projected, options, ↪
↪ worker_pool);
1072 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::↪
↪ attention_projection_ms, start);
1073
1074 start = profile_start(hotspot_profile);
1075 add_inplace(hidden, projected);
1076 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::residual_add_ms, start↪
↪ );
1077
1078 start = profile_start(hotspot_profile);
1079 layer_norm(hidden,
1080            layer.ln_2_weight,
1081            layer.ln_2_bias,
1082            config.layer_norm_epsilon,
1083            normed);
1084 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::layer_norm_ms, start);
1085
1086 start = profile_start(hotspot_profile);
1087 linear_f32_unchecked_parallel(layer.c_fc, normed, mlp_fc, options, ↪
↪ worker_pool);
1088 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::mlp_fc_ms, start);
1089
1090 start = profile_start(hotspot_profile);
1091 for (float& value_ref : mlp_fc) {
1092     value_ref = gelu_new(value_ref);
1093 }
1094 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::gelu_ms, start);
1095
1096 start = profile_start(hotspot_profile);
1097 linear_f32_unchecked_parallel(layer.mlp_c_proj, mlp_fc, mlp_out, options, ↪
↪ worker_pool);
1098 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::mlp_projection_ms, ↪
↪ start);
1099
1100 start = profile_start(hotspot_profile);

```



```

1101     add_inplace(hidden, mlp_out);
1102     add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::residual_add_ms, start↵
↵ );
1103 }
1104
1105 void compute_logits(const Gpt2ReferenceModel& model,
1106                   std::span<const float> hidden,
1107                   std::span<float> logits,
1108                   const Gpt2ExecutionOptions& options,
1109                   detail::Gpt2ParallelWorkerPool* worker_pool) {
1110     if (logits.size() != checked_size(model.config.vocab_size, "vocab_size")) {
1111         throw std::runtime_error("GPT-2 logits size mismatch");
1112     }
1113     const std::span<const float> weight =
1114         model.tie_lm_head_to_embedding ? std::span<const float>(model.↵
↵ token_embedding)
1115                                         : std::span<const float>(model.↵
↵ lm_head_weight);
1116     const std::size_t hidden_size = checked_size(model.config.hidden_size, "↵
↵ hidden_size");
1117     const std::size_t vocab_size = checked_size(model.config.vocab_size, "↵
↵ vocab_size");
1118     const float* weight_data = weight.data();
1119     const float* hidden_data = hidden.data();
1120     const auto rows_fn =
1121         [hidden_size, weight_data, hidden_data, logits_data = logits.data()](
1122             std::size_t begin,
1123             std::size_t end) {
1124             linear_f32_rows_unchecked(weight_data,
1125                                       hidden_data,
1126                                       nullptr,
1127                                       hidden_size,
1128                                       begin,
1129                                       end,
1130                                       logits_data);
1131         };
1132     if (should_parallelize_rows(options, vocab_size, hidden_size, worker_pool))↵
↵ {
1133         worker_pool->parallel_for_rows(0,
1134                                       vocab_size,
1135                                       parallel_row_grain(vocab_size, *↵
↵ worker_pool),
1136                                       rows_fn);
1137     } else {
1138         rows_fn(0, vocab_size);
1139     }
1140 }
1141
1142 void compute_logits(const Gpt2ReferenceModel& model,
1143                   std::span<const float> hidden,
1144                   std::span<float> logits) {
1145     Gpt2ExecutionOptions options;

```

```

1146     options.worker_count = 1;
1147     compute_logits(model, hidden, logits, options, nullptr);
1148 }
1149
1150 struct Gpt2TokenScore {
1151     std::int32_t token = 0;
1152     float value = -std::numeric_limits<float>::infinity();
1153 };
1154
1155 enum class Gpt2CachedStepMode {
1156     advance_only,
1157     predict_token,
1158     logits,
1159 };
1160
1161 [[nodiscard]] Gpt2TokenScore compute_predicted_token_range(const float* ↵
    ↵ weight_data,
1162                                                         const float* ↵
    ↵ hidden_data,
1163                                                         std::size_t ↵
    ↵ hidden_size,
1164                                                         std::size_t begin,
1165                                                         std::size_t end) {
1166     Gpt2TokenScore best;
1167     const F32RowMax row_max =
1168         max_dot_f32_rows_unchecked(weight_data, hidden_data, hidden_size, begin↵
    ↵ , end);
1169     best.token = static_cast<std::int32_t>(row_max.row);
1170     best.value = row_max.value;
1171     return best;
1172 }
1173
1174 std::int32_t compute_predicted_token(const Gpt2ReferenceModel& model,
1175                                     std::span<const float> hidden,
1176                                     const Gpt2ExecutionOptions& options,
1177                                     detail::Gpt2ParallelWorkerPool* ↵
    ↵ worker_pool) {
1178     const std::span<const float> weight =
1179         model.tie_lm_head_to_embedding ? std::span<const float>(model.↵
    ↵ token_embedding)
1180                                     : std::span<const float>(model.↵
    ↵ lm_head_weight);
1181     const std::size_t hidden_size = checked_size(model.config.hidden_size, "↵
    ↵ hidden_size");
1182     const std::size_t vocab_size = checked_size(model.config.vocab_size, "↵
    ↵ vocab_size");
1183     const float* weight_data = weight.data();
1184     const float* hidden_data = hidden.data();
1185     if (!should_parallelize_rows(options, vocab_size, hidden_size, worker_pool)↵
    ↵ ) {
1186         return compute_predicted_token_range(weight_data, hidden_data, ↵
    ↵ hidden_size, 0, vocab_size)

```

```

1187         .token;
1188     }
1189
1190     const std::size_t worker_count = checked_size(worker_pool->worker_count(), ↵
    ↵ "worker_count");
1191     const std::size_t chunk_count = std::min(worker_count, vocab_size);
1192     std::vector<Gpt2TokenScore> partials(chunk_count);
1193     const auto rows_fn =
1194         [chunk_count, hidden_size, vocab_size, weight_data, hidden_data, &↵
    ↵ partials](
1195         std::size_t begin,
1196         std::size_t end) {
1197         for (std::size_t chunk = begin; chunk < end; ++chunk) {
1198             const std::size_t token_begin = vocab_size * chunk / ↵
    ↵ chunk_count;
1199             const std::size_t token_end = vocab_size * (chunk + 1) / ↵
    ↵ chunk_count;
1200             partials[chunk] = compute_predicted_token_range(
1201                 weight_data,
1202                 hidden_data,
1203                 hidden_size,
1204                 token_begin,
1205                 token_end);
1206         }
1207     };
1208     worker_pool->parallel_for_rows(0, chunk_count, 1, rows_fn);
1209
1210     Gpt2TokenScore best = partials.front();
1211     for (std::size_t index = 1; index < partials.size(); ++index) {
1212         if (partials[index].value > best.value) {
1213             best = partials[index];
1214         }
1215     }
1216     return best.token;
1217 }
1218
1219 Gpt2ForwardResult run_gpt2_cached_step_unchecked(const Gpt2ReferenceModel& ↵
    ↵ model,
1220
    Gpt2KvCache& cache,
1221    Gpt2ForwardWorkspace& ↵
    ↵ workspace,
1222
    std::int32_t token_id,
1223    Gpt2CachedStepMode mode,
1224    Gpt2HotspotProfile* ↵
    ↵ hotspot_profile,
1225
    const Gpt2ExecutionOptions& ↵
    ↵ options,
1226
    detail::Gpt2ParallelWorkerPool↵
    ↵ * worker_pool) {
1227     const Gpt2ProfileClock::time_point step_start = profile_start(↵
    ↵ hotspot_profile);
1228     if (hotspot_profile != nullptr) {

```

```

1229     ++hotspot_profile->decoder_steps;
1230 }
1231 if (token_id < 0 || token_id >= model.config.vocab_size) {
1232     throw std::runtime_error("GPT-2 token id out of range");
1233 }
1234 const std::int32_t model_position = cache.filled_tokens();
1235 if (model_position >= model.config.max_positions) {
1236     throw std::runtime_error("GPT-2 cached forward would exceed ↵
↵ max_positions");
1237 }
1238
1239 std::span<float> hidden = workspace.hidden();
1240 const std::span<const float> token =
1241     row_span(model.token_embedding, token_id, model.config.hidden_size);
1242 const std::span<const float> position_embedding =
1243     row_span(model.position_embedding, model_position, model.config.↵
↵ hidden_size);
1244 Gpt2ProfileClock::time_point start = profile_start(hotspot_profile);
1245 for (std::int32_t dim = 0; dim < model.config.hidden_size; ++dim) {
1246     hidden[checked_size(dim, "dim")] =
1247         token[checked_size(dim, "dim")] + position_embedding[checked_size(↵
↵ dim, "dim")];
1248 }
1249 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::embedding_ms, start);
1250
1251 for (std::int32_t layer_id = 0;
1252     layer_id < static_cast<std::int32_t>(model.layers.size());
1253     ++layer_id) {
1254     forward_gpt2_layer_cached(model.config,
1255                             model.layers[checked_size(layer_id, "layer_id↵
↵ ")]),
1256                             layer_id,
1257                             model_position,
1258                             cache,
1259                             workspace,
1260                             hidden,
1261                             hotspot_profile,
1262                             options,
1263                             worker_pool);
1264 }
1265
1266 Gpt2ForwardResult result;
1267 if (mode == Gpt2CachedStepMode::advance_only) {
1268     add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::total_step_ms, ↵
↵ step_start);
1269     return result;
1270 }
1271
1272 std::span<float> normed = workspace.normed();
1273 start = profile_start(hotspot_profile);
1274 layer_norm(hidden,
1275             model.final_ln_weight,

```

```

1276         model.final_ln_bias,
1277         model.config.layer_norm_epsilon,
1278         normed);
1279 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::final_norm_ms, start);
1280
1281 if (mode == Gpt2CachedStepMode::logits) {
1282     std::span<float> logits = workspace.logits();
1283     start = profile_start(hotspot_profile);
1284     compute_logits(model, normed, logits, options, worker_pool);
1285     add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::lm_head_ms, start)↵
↵ ;
1286     start = profile_start(hotspot_profile);
1287     result.logits.assign(logits.begin(), logits.end());
1288     result.predicted_token = argmax(result.logits);
1289     add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::logits_result_ms, ↵
↵ start);
1290 } else {
1291     start = profile_start(hotspot_profile);
1292     result.predicted_token = compute_predicted_token(model, normed, options↵
↵ , worker_pool);
1293     add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::lm_head_ms, start)↵
↵ ;
1294 }
1295 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::total_step_ms, ↵
↵ step_start);
1296 return result;
1297 }
1298
1299 std::vector<std::string> gpt2_bytes_to_unicode() {
1300     std::vector<std::int32_t> bytes;
1301     for (std::int32_t value = LCQI_GPT2_ASCII_EXCLAMATION;
1302          value <= LCQI_GPT2_ASCII_TILDE;
1303          ++value) {
1304         bytes.push_back(value);
1305     }
1306     for (std::int32_t value = LCQI_GPT2_LATIN_INVERTED_EXCLAMATION;
1307          value <= LCQI_GPT2_LATIN_NOT_SIGN;
1308          ++value) {
1309         bytes.push_back(value);
1310     }
1311     for (std::int32_t value = LCQI_GPT2_LATIN_REGISTERED;
1312          value <= LCQI_GPT2_LATIN_Y_DIAERESIS;
1313          ++value) {
1314         bytes.push_back(value);
1315     }
1316
1317     std::unordered_set<std::int32_t> byte_set(bytes.begin(), bytes.end());
1318     std::vector<std::int32_t> chars = bytes;
1319     std::int32_t private_offset = 0;
1320     for (std::int32_t byte = 0; byte < LCQI_GPT2_BYTE_COUNT; ++byte) {
1321         if (!byte_set.contains(byte)) {
1322             bytes.push_back(byte);

```

```

1323         chars.push_back(LCQI_GPT2_UNICODE_PRIVATE_BASE + private_offset);
1324         ++private_offset;
1325     }
1326 }
1327
1328 std::vector<std::string> mapping(LCQI_GPT2_BYTE_COUNT);
1329 for (std::size_t index = 0; index < bytes.size(); ++index) {
1330     mapping[checked_size(bytes[index], "byte unicode index")] =
1331         encode_codepoint(static_cast<std::uint32_t>(chars[index]));
1332 }
1333 return mapping;
1334 }
1335
1336 std::unordered_map<std::string, unsigned char> gpt2_unicode_to_bytes(
1337     const std::vector<std::string>& byte_encoder) {
1338     std::unordered_map<std::string, unsigned char> result;
1339     for (std::size_t byte = 0; byte < byte_encoder.size(); ++byte) {
1340         result.emplace(byte_encoder[byte], static_cast<unsigned char>(byte));
1341     }
1342     return result;
1343 }
1344
1345 std::string bytes_to_bpe_alphabet(std::string_view text) {
1346     static const std::vector<std::string> BYTE_ENCODER = gpt2_bytes_to_unicode(
1347         ↪ );
1348     std::string result;
1349     for (const unsigned char byte : text) {
1350         result += BYTE_ENCODER[byte];
1351     }
1352     return result;
1353 }
1354
1355 std::vector<std::string> utf8_symbols(std::string_view text) {
1356     std::vector<std::string> symbols;
1357     std::size_t offset = 0;
1358     while (offset < text.size()) {
1359         const unsigned char lead = static_cast<unsigned char>(text[offset]);
1360         std::size_t width = 1;
1361         if ((lead & 0xE0U) == LCQI_GPT2_UTF8_TWO_BYTE_HEAD) {
1362             width = 2;
1363         } else if ((lead & 0xF0U) == LCQI_GPT2_UTF8_THREE_BYTE_HEAD) {
1364             width = 3;
1365         } else if ((lead & 0xF8U) == LCQI_GPT2_UTF8_FOUR_BYTE_HEAD) {
1366             width = 4;
1367         }
1368         if (offset + width > text.size()) {
1369             throw std::runtime_error("invalid UTF-8/BPE symbol boundary");
1370         }
1371         symbols.emplace_back(text.substr(offset, width));
1372         offset += width;
1373     }
1374     return symbols;

```

```

1374 }
1375
1376 std::string pair_key(std::string_view left, std::string_view right) {
1377     std::string key(left);
1378     key.push_back('\n');
1379     key += right;
1380     return key;
1381 }
1382
1383 std::vector<std::string> bpe_apply(const Gpt2Tokenizer& tokenizer, std::↵
    ↵ string_view token) {
1384     std::vector<std::string> word = utf8_symbols(token);
1385     if (word.size() <= 1) {
1386         return word;
1387     }
1388
1389     while (true) {
1390         std::optional<std::int32_t> best_rank;
1391         std::string best_left;
1392         std::string best_right;
1393         for (std::size_t index = 0; index + 1 < word.size(); ++index) {
1394             const auto rank = tokenizer.bpe_ranks.find(pair_key(word[index], ↵
    ↵ word[index + 1]));
1395             if (rank != tokenizer.bpe_ranks.end() &&
1396                 (!best_rank.has_value() || rank->second < *best_rank)) {
1397                 best_rank = rank->second;
1398                 best_left = word[index];
1399                 best_right = word[index + 1];
1400             }
1401         }
1402         if (!best_rank.has_value()) {
1403             return word;
1404         }
1405
1406         std::vector<std::string> merged;
1407         std::size_t index = 0;
1408         while (index < word.size()) {
1409             if (index + 1 < word.size() && word[index] == best_left &&
1410                 word[index + 1] == best_right) {
1411                 merged.push_back(best_left + best_right);
1412                 index += LCQI_GPT2_PAIR_TOKEN_COUNT;
1413             } else {
1414                 merged.push_back(word[index]);
1415                 ++index;
1416             }
1417         }
1418         word = std::move(merged);
1419         if (word.size() == 1) {
1420             return word;
1421         }
1422     }
1423 }

```



```

1424
1425 bool is_ascii_letter(unsigned char value) {
1426     return (value >= 'A' && value <= 'Z') || (value >= 'a' && value <= 'z');
1427 }
1428
1429 bool is_ascii_digit(unsigned char value) {
1430     return value >= '0' && value <= '9';
1431 }
1432
1433 bool is_ascii_space(unsigned char value) {
1434     return value == ' ' || value == '\t' || value == '\n' ||
1435            value == '\r' || value == '\f' || value == '\v';
1436 }
1437
1438 bool is_gpt2_punctuation(unsigned char value) {
1439     return !is_ascii_letter(value) && !is_ascii_digit(value) &&
1440            !is_ascii_space(value);
1441 }
1442
1443 std::size_t consume_utf8_codepoint(std::string_view text, std::size_t offset) {
1444     const unsigned char lead = static_cast<unsigned char>(text[offset]);
1445     std::size_t width = 1;
1446     if ((lead & 0xE0U) == LCQI_GPT2_UTF8_TWO_BYTE_HEAD) {
1447         width = 2;
1448     } else if ((lead & 0xF0U) == LCQI_GPT2_UTF8_THREE_BYTE_HEAD) {
1449         width = 3;
1450     } else if ((lead & 0xF8U) == LCQI_GPT2_UTF8_FOUR_BYTE_HEAD) {
1451         width = 4;
1452     }
1453     if (offset + width > text.size()) {
1454         throw std::runtime_error("invalid UTF-8 text for GPT-2 tokenizer");
1455     }
1456     return offset + width;
1457 }
1458
1459 bool starts_with_contraction(std::string_view text,
1460                             std::size_t offset,
1461                             std::string_view contraction) {
1462     if (offset + contraction.size() > text.size()) {
1463         return false;
1464     }
1465     for (std::size_t index = 0; index < contraction.size(); ++index) {
1466         char left = text[offset + index];
1467         const char right = contraction[index];
1468         if (left >= 'A' && left <= 'Z') {
1469             left = static_cast<char>(left - 'A' + 'a');
1470         }
1471         if (left != right) {
1472             return false;
1473         }
1474     }
1475     return true;

```

```

1476 }
1477
1478 std::vector<std::string> gpt2_tokenize(std::string_view text) {
1479     std::vector<std::string> pieces;
1480     std::size_t offset = 0;
1481     const std::array<std::string_view, 7> contractions{
1482         "'s", "'t", "'re", "'ve", "'m", "'ll", "'d",
1483     };
1484     while (offset < text.size()) {
1485         bool emitted_contraction = false;
1486         for (const std::string_view contraction : contractions) {
1487             if (starts_with_contraction(text, offset, contraction)) {
1488                 pieces.emplace_back(text.substr(offset, contraction.size()));
1489                 offset += contraction.size();
1490                 emitted_contraction = true;
1491                 break;
1492             }
1493         }
1494         if (emitted_contraction) {
1495             continue;
1496         }
1497
1498         const std::size_t begin = offset;
1499         bool has_prefix_space = false;
1500         if (text[offset] == ' ' && offset + 1 < text.size() &&
1501             !is_ascii_space(static_cast<unsigned char>(text[offset + 1]))) {
1502             has_prefix_space = true;
1503             ++offset;
1504         }
1505
1506         if (offset >= text.size()) {
1507             pieces.emplace_back(text.substr(begin, offset - begin));
1508             continue;
1509         }
1510
1511         const unsigned char ch = static_cast<unsigned char>(text[offset]);
1512         if (is_ascii_letter(ch)) {
1513             while (offset < text.size() &&
1514                 is_ascii_letter(static_cast<unsigned char>(text[offset]))) {
1515                 ++offset;
1516             }
1517         } else if (is_ascii_digit(ch)) {
1518             while (offset < text.size() &&
1519                 is_ascii_digit(static_cast<unsigned char>(text[offset]))) {
1520                 ++offset;
1521             }
1522         } else if (is_gpt2_punctuation(ch) && ch != LCQI_GPT2_ASCII_APOSTROPHE) ←
1523             ↪ {
1524                 while (offset < text.size() &&
1525                     is_gpt2_punctuation(static_cast<unsigned char>(text[offset])) ←
1526                     ↪ ) &&
1527                     static_cast<unsigned char>(text[offset]) != ←

```

```

    ↪ LCQI_GPT2_ASCII_APOSTROPHE) {
1526         offset = consume_utf8_codepoint(text, offset);
1527     }
1528     } else if (is_ascii_space(ch)) {
1529         while (offset < text.size() &&
1530             is_ascii_space(static_cast<unsigned char>(text[offset]))) {
1531             ++offset;
1532         }
1533     } else {
1534         offset = consume_utf8_codepoint(text, offset);
1535     }
1536
1537     if (has_prefix_space || offset > begin) {
1538         pieces.emplace_back(text.substr(begin, offset - begin));
1539     } else {
1540         ++offset;
1541     }
1542 }
1543 return pieces;
1544 }
1545
1546 std::int32_t parse_i32_field(JsonCursor& cursor) {
1547     const std::int64_t value = cursor.parse_i64();
1548     if (value < std::numeric_limits<std::int32_t>::min() ||
1549         value > std::numeric_limits<std::int32_t>::max()) {
1550         throw std::runtime_error("JSON int32 field out of range");
1551     }
1552     return static_cast<std::int32_t>(value);
1553 }
1554
1555 std::unordered_map<std::string, std::int32_t> parse_vocab_json(std::string_view ↪
    ↪ text) {
1556     JsonCursor cursor(text);
1557     std::unordered_map<std::string, std::int32_t> vocab;
1558     cursor.expect('{');
1559     if (cursor.consume('}')) {
1560         return vocab;
1561     }
1562     while (true) {
1563         const std::string key = cursor.parse_string();
1564         cursor.expect(':');
1565         const std::int32_t id = parse_i32_field(cursor);
1566         vocab.emplace(key, id);
1567         if (cursor.consume('}')) {
1568             break;
1569         }
1570         cursor.expect(',');
1571     }
1572     if (!cursor.eof_after_ws()) {
1573         throw std::runtime_error("trailing data after vocab JSON");
1574     }
1575     return vocab;

```

```

1576 }
1577
1578 std::vector<std::string> build_id_to_token(
1579     const std::unordered_map<std::string, std::int32_t>& vocab) {
1580     std::int32_t max_id = -1;
1581     for (const auto& [token, id] : vocab) {
1582         if (id < 0) {
1583             throw std::runtime_error("GPT-2 vocab id cannot be negative");
1584         }
1585         max_id = std::max(max_id, id);
1586     }
1587     std::vector<std::string> result(check_size(max_id + 1, "max vocab id"));
1588     for (const auto& [token, id] : vocab) {
1589         std::string& slot = result[checked_size(id, "vocab id")];
1590         if (!slot.empty()) {
1591             throw std::runtime_error("duplicate GPT-2 vocab id");
1592         }
1593         slot = token;
1594     }
1595     return result;
1596 }
1597
1598 std::unordered_map<std::string, std::int32_t> parse_merges(std::string_view ↵
    ↵ text) {
1599     std::unordered_map<std::string, std::int32_t> ranks;
1600     std::istringstream input{std::string(text)};
1601     std::string line;
1602     std::int32_t rank = 0;
1603     while (std::getline(input, line)) {
1604         if (line.empty() || line[0] == '#') {
1605             continue;
1606         }
1607         std::istringstream line_stream(line);
1608         std::string left;
1609         std::string right;
1610         if (!(line_stream >> left >> right)) {
1611             throw std::runtime_error("invalid GPT-2 merges line");
1612         }
1613         ranks.emplace(pair_key(left, right), rank);
1614         ++rank;
1615     }
1616     return ranks;
1617 }
1618
1619 void transpose_hf_conv1d(std::vector<float>& values,
1620                         std::int32_t input_size,
1621                         std::int32_t output_size) {
1622     if (values.size() != matrix_size(input_size, output_size)) {
1623         throw std::runtime_error("GPT-2 Conv1D tensor size mismatch");
1624     }
1625     std::vector<float> transposed(matrix_size(output_size, input_size), 0.0F);
1626     for (std::int32_t input = 0; input < input_size; ++input) {

```

```

1627     for (std::int32_t output = 0; output < output_size; ++output) {
1628         transposed[matrix_size(output, input_size) + checked_size(input, "↵
↵ input")] =
1629             values[matrix_size(input, output_size) + checked_size(output, "↵
↵ output")];
1630     }
1631 }
1632 values = std::move(transposed);
1633 }
1634
1635 std::vector<float> require_tensor(const SafeTensorFile& file,
1636                                 std::string_view name,
1637                                 std::span<const std::int64_t> expected_shape)↵
↵ {
1638     const SafeTensorEntry* entry = file.manifest.find_tensor(name);
1639     if (entry == nullptr) {
1640         throw std::runtime_error("missing GPT-2 tensor: " + std::string(name));
1641     }
1642     if (entry->shape.size() != expected_shape.size()) {
1643         throw std::runtime_error("GPT-2 tensor rank mismatch: " + std::string(↵
↵ name));
1644     }
1645     for (std::size_t index = 0; index < expected_shape.size(); ++index) {
1646         if (entry->shape[index] != expected_shape[index]) {
1647             throw std::runtime_error("GPT-2 tensor shape mismatch: " + std::↵
↵ string(name));
1648         }
1649     }
1650     return read_safetensor_f32_tensor(file, name);
1651 }
1652
1653 std::vector<float> require_first_tensor(const SafeTensorFile& file,
1654                                         std::span<const std::string_view> names↵
↵ ,
1655                                         std::span<const std::int64_t> ↵
↵ expected_shape) {
1656     for (const std::string_view name : names) {
1657         if (file.manifest.find_tensor(name) != nullptr) {
1658             return require_tensor(file, name, expected_shape);
1659         }
1660     }
1661     throw std::runtime_error("missing GPT-2 tensor: " + std::string(names.front↵
↵ ()));
1662 }
1663
1664 Gpt2LinearF32 make_linear(std::int32_t input_size,
1665                           std::int32_t output_size,
1666                           std::vector<float> weights,
1667                           std::vector<float> bias = {}) {
1668     Gpt2LinearF32 linear;
1669     linear.input_size = input_size;
1670     linear.output_size = output_size;

```

```

1671     linear.weights = std::move(weights);
1672     linear.bias = std::move(bias);
1673     validate_linear(linear, "make GPT-2 linear");
1674     return linear;
1675 }
1676
1677 std::vector<float> zeros(std::int32_t count) {
1678     return std::vector<float>(checked_size(count, "zero count"), 0.0F);
1679 }
1680
1681 std::vector<float> ones(std::int32_t count) {
1682     return std::vector<float>(checked_size(count, "one count"), 1.0F);
1683 }
1684
1685 std::string tensor_name(std::int32_t layer, std::string_view suffix) {
1686     return "transformer.h." + std::to_string(layer) + "." + std::string(suffix) ↵
    ↵ ;
1687 }
1688
1689 std::string bare_tensor_name(std::int32_t layer, std::string_view suffix) {
1690     return "h." + std::to_string(layer) + "." + std::string(suffix);
1691 }
1692
1693 std::array<std::string_view, 2> embedding_names(std::string_view bare) {
1694     if (bare == "wte.weight") {
1695         return {"transformer.wte.weight", "wte.weight"};
1696     }
1697     if (bare == "wpe.weight") {
1698         return {"transformer.wpe.weight", "wpe.weight"};
1699     }
1700     if (bare == "ln_f.weight") {
1701         return {"transformer.ln_f.weight", "ln_f.weight"};
1702     }
1703     if (bare == "ln_f.bias") {
1704         return {"transformer.ln_f.bias", "ln_f.bias"};
1705     }
1706     throw std::runtime_error("unsupported GPT-2 embedding tensor alias");
1707 }
1708
1709 std::array<std::string, 2> layer_names(std::int32_t layer, std::string_view ↵
    ↵ suffix) {
1710     return {tensor_name(layer, suffix), bare_tensor_name(layer, suffix)};
1711 }
1712
1713 std::vector<float> require_layer_tensor(const SafeTensorFile& file,
1714                                         std::int32_t layer,
1715                                         std::string_view suffix,
1716                                         std::span<const std::int64_t> ↵
    ↵ expected_shape) {
1717     const std::array<std::string, 2> names = layer_names(layer, suffix);
1718     const std::array<std::string_view, 2> views{names[0], names[1]};
1719     return require_first_tensor(file, views, expected_shape);

```

```

1720 }
1721
1722 std::filesystem::path find_gpt2_weight_file(const std::filesystem::path& ↵
    ↵ model_directory) {
1723     const std::array<const char*, 2> candidates{
1724         "model.safetensors",
1725         "pytorch_model.safetensors",
1726     };
1727     for (const char* candidate : candidates) {
1728         const std::filesystem::path path = model_directory / candidate;
1729         if (std::filesystem::exists(path)) {
1730             return path;
1731         }
1732     }
1733     throw std::runtime_error("cannot find model.safetensors in GPT-2 directory" ↵
    ↵ );
1734 }
1735
1736 } // namespace
1737
1738 Gpt2Tokenizer load_gpt2_tokenizer(const std::filesystem::path& vocab_json,
1739                                   const std::filesystem::path& merges_txt,
1740                                   std::int32_t bos_token_id,
1741                                   std::int32_t eos_token_id) {
1742     Gpt2Tokenizer tokenizer;
1743     tokenizer.bos_token_id = bos_token_id;
1744     tokenizer.eos_token_id = eos_token_id;
1745     tokenizer.vocab = parse_vocab_json(read_text_file(vocab_json));
1746     tokenizer.id_to_token = build_id_to_token(tokenizer.vocab);
1747     tokenizer.bpe_ranks = parse_merges(read_text_file(merges_txt));
1748     if (tokenizer.vocab.empty() || tokenizer.bpe_ranks.empty()) {
1749         throw std::runtime_error("GPT-2 tokenizer assets are empty");
1750     }
1751     return tokenizer;
1752 }
1753
1754 std::vector<std::int32_t> gpt2_encode(const Gpt2Tokenizer& tokenizer,
1755                                       std::string_view text) {
1756     std::vector<std::int32_t> ids;
1757     for (const std::string& piece : gpt2_pretokenize(text)) {
1758         const std::string byte_piece = bytes_to_bpe_alphabet(piece);
1759         const std::vector<std::string> bpe_tokens = bpe_apply(tokenizer, ↵
    ↵ byte_piece);
1760         for (const std::string& token : bpe_tokens) {
1761             const auto found = tokenizer.vocab.find(token);
1762             if (found == tokenizer.vocab.end()) {
1763                 throw std::runtime_error("GPT-2 BPE token is missing from vocab" ↵
    ↵ );
1764             }
1765             ids.push_back(found->second);
1766         }
1767     }

```



```

1768     return ids;
1769 }
1770
1771 std::string gpt2_decode(const Gpt2Tokenizer& tokenizer,
1772                        std::span<const std::int32_t> token_ids) {
1773     static const std::vector<std::string> BYTE_ENCODER = gpt2_bytes_to_unicode(
1774         ↪ );
1775     static const std::unordered_map<std::string, unsigned char> BYTE_DECODER =
1776         ↪ gpt2_unicode_to_bytes(BYTE_ENCODER);
1777
1778     std::string token_text;
1779     for (const std::int32_t token_id : token_ids) {
1780         if (token_id < 0 || token_id >= static_cast<std::int32_t>(tokenizer.
1781             ↪ id_to_token.size())) {
1782             throw std::runtime_error("GPT-2 token id out of range");
1783         }
1784         token_text += tokenizer.id_to_token[checked_size(token_id, "token id")
1785             ↪ ];
1786     }
1787
1788     std::string result;
1789     const std::vector<std::string> symbols = utf8_symbols(token_text);
1790     for (const std::string& symbol : symbols) {
1791         const auto found = BYTE_DECODER.find(symbol);
1792         if (found == BYTE_DECODER.end()) {
1793             throw std::runtime_error("GPT-2 token cannot be decoded to byte");
1794         }
1795         result.push_back(static_cast<char>(found->second));
1796     }
1797     return result;
1798 }
1799
1800 Gpt2Config load_gpt2_config(const std::filesystem::path& config_json) {
1801     const std::string config_text = read_text_file(config_json);
1802     JsonCursor cursor(config_text);
1803     Gpt2Config config;
1804     config.bos_token_id = LCQI_GPT2_DEFAULT_BOS_EOS;
1805     config.eos_token_id = LCQI_GPT2_DEFAULT_BOS_EOS;
1806
1807     cursor.expect('{');
1808     if (cursor.consume('}')) {
1809         throw std::runtime_error("GPT-2 config is empty");
1810     }
1811     while (true) {
1812         const std::string key = cursor.parse_string();
1813         cursor.expect(':');
1814         if (key == "n_embd") {
1815             config.hidden_size = parse_i32_field(cursor);
1816         } else if (key == "n_head") {
1817             config.head_count = parse_i32_field(cursor);
1818         } else if (key == "n_layer") {
1819             config.layer_count = parse_i32_field(cursor);

```

```

1817     } else if (key == "n_positions" || key == "n_ctx") {
1818         const std::int32_t value = parse_i32_field(cursor);
1819         config.max_positions = std::max(config.max_positions, value);
1820     } else if (key == "vocab_size") {
1821         config.vocab_size = parse_i32_field(cursor);
1822     } else if (key == "n_inner") {
1823         config.intermediate_size = parse_i32_field(cursor);
1824     } else if (key == "layer_norm_epsilon") {
1825         config.layer_norm_epsilon = static_cast<float>(cursor.parse_number()
↪ ());
1826     } else if (key == "activation_function") {
1827         config.activation_function = cursor.parse_string();
1828     } else if (key == "bos_token_id") {
1829         config.bos_token_id = parse_i32_field(cursor);
1830     } else if (key == "eos_token_id") {
1831         config.eos_token_id = parse_i32_field(cursor);
1832     } else {
1833         cursor.skip_value();
1834     }
1835     if (cursor.consume('}')) {
1836         break;
1837     }
1838     cursor.expect(',');
1839 }
1840 if (!cursor.eof_after_ws()) {
1841     throw std::runtime_error("trailing data after GPT-2 config JSON");
1842 }
1843 if (config.intermediate_size == 0 && config.hidden_size > 0) {
1844     config.intermediate_size = config.hidden_size * 4;
1845 }
1846 validate_config(config);
1847 return config;
1848 }
1849
1850 Gpt2ReferenceModel load_gpt2_reference_model(const std::filesystem::path& ↪
↪ config_json,
1851                                             const std::filesystem::path& ↪
↪ safetensors_path) {
1852     Gpt2ReferenceModel model;
1853     model.config = load_gpt2_config(config_json);
1854     const SafeTensorFile file = load_safetensors_file(safetensors_path);
1855     const Gpt2Config& config = model.config;
1856     const std::array<std::int64_t, 2> wte_shape{config.vocab_size, config.↪
↪ hidden_size};
1857     const std::array<std::int64_t, 2> wpe_shape{config.max_positions, config.↪
↪ hidden_size};
1858     const std::array<std::int64_t, 1> hidden_shape{config.hidden_size};
1859
1860     const std::array<std::string_view, 2> wte_names = embedding_names("wte.↪
↪ weight");
1861     const std::array<std::string_view, 2> wpe_names = embedding_names("wpe.↪
↪ weight");

```

```

1862 model.token_embedding = require_first_tensor(file, wte_names, wte_shape);
1863 model.position_embedding = require_first_tensor(file, wpe_names, wpe_shape);
1864 model.layers.resize(check_size(config.layer_count, "layer_count"));
1865
1866 for (std::int32_t layer_id = 0; layer_id < config.layer_count; ++layer_id) {
1867     Gpt2LayerWeightsF32& layer = model.layers[check_size(layer_id, "
1868     layer_id")];
1869     layer.ln_1_weight = require_layer_tensor(file, layer_id, "ln_1.weight",
1870     hidden_shape);
1871     layer.ln_1_bias = require_layer_tensor(file, layer_id, "ln_1.bias",
1872     hidden_shape);
1873     layer.ln_2_weight = require_layer_tensor(file, layer_id, "ln_2.weight",
1874     hidden_shape);
1875     layer.ln_2_bias = require_layer_tensor(file, layer_id, "ln_2.bias",
1876     hidden_shape);
1877
1878     const std::array<std::int64_t, 2> c_attn_shape{
1879         config.hidden_size,
1880         config.hidden_size * LCQI_GPT2_ATTENTION_PROJECTIONS};
1881     std::vector<float> c_attn_weight =
1882         require_layer_tensor(file, layer_id, "attn.c_attn.weight",
1883         c_attn_shape);
1884     transpose_hf_conv1d(c_attn_weight,
1885         config.hidden_size,
1886         config.hidden_size *
1887         LCQI_GPT2_ATTENTION_PROJECTIONS);
1888     const std::array<std::int64_t, 1> c_attn_bias_shape{
1889         config.hidden_size * LCQI_GPT2_ATTENTION_PROJECTIONS};
1890     layer.c_attn = make_linear(config.hidden_size,
1891         config.hidden_size *
1892         LCQI_GPT2_ATTENTION_PROJECTIONS,
1893         std::move(c_attn_weight),
1894         require_layer_tensor(file,
1895             layer_id,
1896             "attn.c_attn.bias",
1897             c_attn_bias_shape));
1898
1899     const std::array<std::int64_t, 2> c_proj_shape{config.hidden_size,
1900     config.hidden_size};
1901     std::vector<float> c_proj_weight =
1902         require_layer_tensor(file, layer_id, "attn.c_proj.weight",
1903         c_proj_shape);
1904     transpose_hf_conv1d(c_proj_weight, config.hidden_size, config.
1905     hidden_size);
1906     layer.c_proj = make_linear(config.hidden_size,
1907         config.hidden_size,
1908         std::move(c_proj_weight),
1909         require_layer_tensor(file,
1910             layer_id,
1911             "attn.c_proj.bias",

```

```

1901                                     hidden_shape));
1902
1903     const std::array<std::int64_t, 2> c_fc_shape{
1904         config.hidden_size,
1905         config.intermediate_size};
1906     std::vector<float> c_fc_weight =
1907         require_layer_tensor(file, layer_id, "mlp.c_fc.weight", c_fc_shape)↵
1908 ↵ ;
1909     transpose_hf_conv1d(c_fc_weight, config.hidden_size, config.↵
1910 ↵ intermediate_size);
1911     const std::array<std::int64_t, 1> intermediate_shape{config.↵
1912 ↵ intermediate_size};
1913     layer.c_fc = make_linear(config.hidden_size,
1914                             config.intermediate_size,
1915                             std::move(c_fc_weight),
1916                             require_layer_tensor(file,
1917                                                     layer_id,
1918                                                     "mlp.c_fc.bias",
1919                                                     intermediate_shape));
1920
1921     const std::array<std::int64_t, 2> mlp_proj_shape{
1922         config.intermediate_size,
1923         config.hidden_size};
1924     std::vector<float> mlp_proj_weight =
1925         require_layer_tensor(file, layer_id, "mlp.c_proj.weight", ↵
1926 ↵ mlp_proj_shape);
1927     transpose_hf_conv1d(mlp_proj_weight, config.intermediate_size, config.↵
1928 ↵ hidden_size);
1929     layer.mlp_c_proj = make_linear(config.intermediate_size,
1930                                   config.hidden_size,
1931                                   std::move(mlp_proj_weight),
1932                                   require_layer_tensor(file,
1933                                                         layer_id,
1934                                                         "mlp.c_proj.bias",
1935                                                         hidden_shape));
1936 }
1937
1938 const std::array<std::string_view, 2> ln_weight_names = embedding_names("↵
1939 ↵ ln_f.weight");
1940 const std::array<std::string_view, 2> ln_bias_names = embedding_names("ln_f↵
1941 ↵ .bias");
1942 model.final_ln_weight = require_first_tensor(file, ln_weight_names, ↵
1943 ↵ hidden_shape);
1944 model.final_ln_bias = require_first_tensor(file, ln_bias_names, ↵
1945 ↵ hidden_shape);
1946 if (file.manifest.find_tensor("lm_head.weight") != nullptr) {
1947     model.tie_lm_head_to_embedding = false;
1948     model.lm_head_weight = require_tensor(file, "lm_head.weight", wte_shape↵
1949 ↵ );
1950 } else {
1951     model.tie_lm_head_to_embedding = true;
1952 }

```

```

1943     validate_model(model);
1944     return model;
1945 }
1946
1947 Gpt2ReferenceModel load_gpt2_from_directory(const std::filesystem::path& ↵
    ↵ model_directory) {
1948     return load_gpt2_reference_model(model_directory / "config.json",
1949                                     find_gpt2_weight_file(model_directory));
1950 }
1951
1952 Gpt2ForwardWorkspace::Gpt2ForwardWorkspace(const Gpt2Config& config) : config_(↵
    ↵ config) {
1953     this->reset_for_config(config);
1954 }
1955
1956 void Gpt2ForwardWorkspace::reset_for_config(const Gpt2Config& config) {
1957     validate_config(config);
1958     this->config_ = config;
1959     const std::size_t hidden_size = checked_size(this->config_.hidden_size, "↵
    ↵ hidden_size");
1960     const std::size_t intermediate_size =
1961         checked_size(this->config_.intermediate_size, "intermediate_size");
1962     this->hidden_.assign(hidden_size, 0.0F);
1963     this->normed_.assign(hidden_size, 0.0F);
1964     this->query_.assign(hidden_size, 0.0F);
1965     this->key_.assign(hidden_size, 0.0F);
1966     this->value_.assign(hidden_size, 0.0F);
1967     this->attention_.assign(hidden_size, 0.0F);
1968     this->projected_.assign(hidden_size, 0.0F);
1969     this->qkv_packed_.assign(
1970         matrix_size(LCQI_GPT2_ATTENTION_PROJECTIONS, this->config_.hidden_size)↵
    ↵ ,
1971         0.0F);
1972     this->mlp_fc_.assign(intermediate_size, 0.0F);
1973     this->mlp_out_.assign(hidden_size, 0.0F);
1974     this->logits_.assign(checked_size(this->config_.vocab_size, "vocab_size"), ↵
    ↵ 0.0F);
1975     this->scores_.assign(checked_size(this->config_.max_positions, "↵
    ↵ max_positions"), 0.0F);
1976 }
1977
1978 std::span<float> Gpt2ForwardWorkspace::hidden() noexcept {
1979     return this->hidden_;
1980 }
1981
1982 std::span<float> Gpt2ForwardWorkspace::normed() noexcept {
1983     return this->normed_;
1984 }
1985
1986 std::span<float> Gpt2ForwardWorkspace::query() noexcept {
1987     return this->query_;
1988 }

```

```

1989
1990 std::span<float> Gpt2ForwardWorkspace::key() noexcept {
1991     return this->key_;
1992 }
1993
1994 std::span<float> Gpt2ForwardWorkspace::value() noexcept {
1995     return this->value_;
1996 }
1997
1998 std::span<float> Gpt2ForwardWorkspace::attention() noexcept {
1999     return this->attention_;
2000 }
2001
2002 std::span<float> Gpt2ForwardWorkspace::projected() noexcept {
2003     return this->projected_;
2004 }
2005
2006 std::span<float> Gpt2ForwardWorkspace::qkv_packed() noexcept {
2007     return this->qkv_packed_;
2008 }
2009
2010 std::span<float> Gpt2ForwardWorkspace::mlp_fc() noexcept {
2011     return this->mlp_fc_;
2012 }
2013
2014 std::span<float> Gpt2ForwardWorkspace::mlp_out() noexcept {
2015     return this->mlp_out_;
2016 }
2017
2018 std::span<float> Gpt2ForwardWorkspace::logits() noexcept {
2019     return this->logits_;
2020 }
2021
2022 std::span<float> Gpt2ForwardWorkspace::scores_prefix(std::int32_t count) {
2023     if (count < 0 || count > this->config_.max_positions) {
2024         throw std::runtime_error("GPT-2 attention scores count out of range");
2025     }
2026     return std::span<float>(this->scores_.data(), checked_size(count, "scores ←
↪ count"));
2027 }
2028
2029 Gpt2KvCache::Gpt2KvCache(const Gpt2Config& config) : config_(config) {
2030     validate_config(this->config_);
2031     const std::size_t element_count =
2032         checked_size(this->config_.layer_count, "layer_count") *
2033         checked_size(this->config_.max_positions, "max_positions") *
2034         checked_size(this->config_.head_count, "head_count") *
2035         checked_size(head_dim(this->config_), "head_dim");
2036     this->keys_.assign(element_count, 0.0F);
2037     this->values_.assign(element_count, 0.0F);
2038     this->written_.assign(
2039         checked_size(this->config_.layer_count, "layer_count") *

```

```

2040         checked_size(this->config_.max_positions, "max_positions"),
2041         0);
2042     }
2043
2044     void Gpt2KvCache::append(std::int32_t layer_id,
2045                             std::int32_t model_position,
2046                             std::span<const float> key,
2047                             std::span<const float> value) {
2048         const std::size_t hidden_size = checked_size(this->config_.hidden_size, "←
        hidden_size");
2049         if (key.size() != hidden_size || value.size() != hidden_size) {
2050             throw std::runtime_error("GPT-2 KV key/value size mismatch");
2051         }
2052         this->validate_address(layer_id, model_position, 0);
2053         const std::int32_t local_head_dim = head_dim(this->config_);
2054         for (std::int32_t head = 0; head < this->config_.head_count; ++head) {
2055             const std::size_t source =
2056                 checked_size(head, "head") * checked_size(local_head_dim, "head_dim←
        ");
2057             const std::size_t target = this->base_offset(layer_id, model_position, ←
        head);
2058             std::copy_n(key.begin() + static_cast<std::ptrdiff_t>(source),
2059                         local_head_dim,
2060                         this->keys_.begin() + static_cast<std::ptrdiff_t>(target));
2061             std::copy_n(value.begin() + static_cast<std::ptrdiff_t>(source),
2062                         local_head_dim,
2063                         this->values_.begin() + static_cast<std::ptrdiff_t>(target)←
        );
2064         }
2065         const std::size_t written_index =
2066             checked_size(layer_id, "layer_id") *
2067             checked_size(this->config_.max_positions, "max_positions") +
2068             checked_size(model_position, "model_position");
2069         this->written_[written_index] = 1;
2070         this->filled_tokens_ = std::max(this->filled_tokens_, model_position + 1);
2071     }
2072
2073     std::span<const float> Gpt2KvCache::key(std::int32_t layer_id,
2074                                             std::int32_t model_position,
2075                                             std::int32_t head) const {
2076         this->validate_address(layer_id, model_position, head);
2077         this->validate_written(layer_id, model_position);
2078         return this->key_unchecked(layer_id, model_position, head);
2079     }
2080
2081     std::span<const float> Gpt2KvCache::value(std::int32_t layer_id,
2082                                              std::int32_t model_position,
2083                                              std::int32_t head) const {
2084         this->validate_address(layer_id, model_position, head);
2085         this->validate_written(layer_id, model_position);
2086         return this->value_unchecked(layer_id, model_position, head);
2087     }

```



```

2088
2089 void detail::attend_cached_position(const Gpt2KvCache& cache,
2090                                     std::int32_t layer_id,
2091                                     std::int32_t model_position,
2092                                     std::span<const float> query,
2093                                     std::span<float> scores,
2094                                     std::span<float> output) {
2095     cache.validate_address(layer_id, model_position, 0);
2096     cache.validate_written(layer_id, model_position);
2097     const std::int32_t local_head_dim = head_dim(cache.config_);
2098     const std::size_t hidden_size = checked_size(cache.config_.hidden_size, "↵
↵ hidden_size");
2099     const std::size_t local_head_dim_size = checked_size(local_head_dim, "↵
↵ head_dim");
2100     const std::size_t model_position_size =
2101         checked_size(model_position, "model_position");
2102     if (query.size() != hidden_size || output.size() != hidden_size ||
2103         scores.size() < model_position_size + 1) {
2104         throw std::runtime_error("GPT-2 cached attention workspace size ↵
↵ mismatch");
2105     }
2106
2107     const float score_scale = 1.0F / std::sqrt(static_cast<float>(↵
↵ local_head_dim));
2108     std::fill(output.begin(), output.end(), 0.0F);
2109     for (std::int32_t head = 0; head < cache.config_.head_count; ++head) {
2110         const std::size_t head_base =
2111             checked_size(head, "head") * local_head_dim_size;
2112         float max_score = LCQI_GPT2_SOFTMAX_NEGATIVE_INFINITY;
2113         for (std::size_t past = 0; past <= model_position_size; ++past) {
2114             const std::int32_t past_i32 = static_cast<std::int32_t>(past);
2115             const std::span<const float> key =
2116                 cache.key_unchecked(layer_id, past_i32, head);
2117             float dot = 0.0F;
2118             for (std::size_t dim = 0; dim < local_head_dim_size; ++dim) {
2119                 dot += query[head_base + dim] * key[dim];
2120             }
2121             const float score = dot * score_scale;
2122             scores[past] = score;
2123             max_score = std::max(max_score, score);
2124         }
2125
2126         float denominator = 0.0F;
2127         for (std::size_t past = 0; past <= model_position_size; ++past) {
2128             scores[past] = std::exp(scores[past] - max_score);
2129             denominator += scores[past];
2130         }
2131         if (denominator <= 0.0F) {
2132             throw std::runtime_error("GPT-2 cached attention denominator is ↵
↵ invalid");
2133         }
2134

```

```

2135     for (std::size_t past = 0; past <= model_position_size; ++past) {
2136         const std::int32_t past_i32 = static_cast<std::int32_t>(past);
2137         const float probability = scores[past] / denominator;
2138         const std::span<const float> value =
2139             cache.value_unchecked(layer_id, past_i32, head);
2140         for (std::size_t dim = 0; dim < local_head_dim_size; ++dim) {
2141             output[head_base + dim] += probability * value[dim];
2142         }
2143     }
2144 }
2145 }
2146
2147 std::span<const float> Gpt2KvCache::key_unchecked(std::int32_t layer_id,
2148                                                  std::int32_t model_position,
2149                                                  std::int32_t head) const {
2150     ↪ noexcept {
2151         const std::size_t offset = this->base_offset_unchecked(layer_id, ↪
2152             ↪ model_position, head);
2153         return std::span<const float>(
2154             this->keys_.data() + offset,
2155             static_cast<std::size_t>(head_dim(this->config_)));
2156     }
2157
2158 std::span<const float> Gpt2KvCache::value_unchecked(std::int32_t layer_id,
2159                                                  std::int32_t model_position,
2160                                                  std::int32_t head) const {
2161     ↪ ,
2162     ↪ noexcept {
2163         const std::size_t offset = this->base_offset_unchecked(layer_id, ↪
2164             ↪ model_position, head);
2165         return std::span<const float>(
2166             this->values_.data() + offset,
2167             static_cast<std::size_t>(head_dim(this->config_)));
2168     }
2169
2170 std::int32_t Gpt2KvCache::filled_tokens() const noexcept {
2171     return this->filled_tokens_;
2172 }
2173
2174 std::size_t Gpt2KvCache::byte_size() const noexcept {
2175     return (this->keys_.size() + this->values_.size()) * sizeof(float) +
2176         this->written_.size() * sizeof(std::uint8_t);
2177 }
2178
2179 std::size_t Gpt2KvCache::base_offset(std::int32_t layer_id,
2180                                     std::int32_t model_position,
2181                                     std::int32_t head) const {
2182     return (((checked_size(layer_id, "layer_id") *
2183         checked_size(this->config_.max_positions, "max_positions")) +
2184         checked_size(model_position, "model_position")) *
2185         checked_size(this->config_.head_count, "head_count") +
2186         checked_size(head, "head")) *

```

```

2182         checked_size(head_dim(this->config_), "head_dim");
2183     }
2184
2185     std::size_t Gpt2KvCache::base_offset_unchecked(std::int32_t layer_id,
2186                                                    std::int32_t model_position,
2187                                                    std::int32_t head) const ↵
        ↵ noexcept {
2188         return (((static_cast<std::size_t>(layer_id) *
2189                 static_cast<std::size_t>(this->config_.max_positions)) +
2190                static_cast<std::size_t>(model_position)) *
2191                static_cast<std::size_t>(this->config_.head_count) +
2192                static_cast<std::size_t>(head)) *
2193                static_cast<std::size_t>(head_dim(this->config_));
2194     }
2195
2196     void Gpt2KvCache::validate_address(std::int32_t layer_id,
2197                                       std::int32_t model_position,
2198                                       std::int32_t head) const {
2199         if (layer_id < 0 || layer_id >= this->config_.layer_count) {
2200             throw std::runtime_error("GPT-2 KV layer_id out of range");
2201         }
2202         if (model_position < 0 || model_position >= this->config_.max_positions) {
2203             throw std::runtime_error("GPT-2 KV model_position out of range");
2204         }
2205         if (head < 0 || head >= this->config_.head_count) {
2206             throw std::runtime_error("GPT-2 KV head out of range");
2207         }
2208     }
2209
2210     void Gpt2KvCache::validate_written(std::int32_t layer_id,
2211                                       std::int32_t model_position) const {
2212         const std::size_t written_index =
2213             checked_size(layer_id, "layer_id") *
2214             checked_size(this->config_.max_positions, "max_positions") +
2215             checked_size(model_position, "model_position");
2216         if (written_index >= this->written_.size() || this->written_[written_index]↵
        ↵ == 0) {
2217             throw std::runtime_error("GPT-2 KV slot has not been written");
2218         }
2219     }
2220
2221     Gpt2CachedGreedyDecoder::Gpt2CachedGreedyDecoder(const Gpt2ReferenceModel& ↵
        ↵ model)
2222         : Gpt2CachedGreedyDecoder(model, nullptr) {}
2223
2224     Gpt2CachedGreedyDecoder::Gpt2CachedGreedyDecoder(
2225         const Gpt2ReferenceModel& model,
2226         Gpt2HotspotProfile* hotspot_profile)
2227         : Gpt2CachedGreedyDecoder(model, hotspot_profile, Gpt2ExecutionOptions{}) ↵
        ↵ {}
2228
2229     Gpt2CachedGreedyDecoder::Gpt2CachedGreedyDecoder(

```

```

2230     const Gpt2ReferenceModel& model,
2231     Gpt2HotspotProfile* hotspot_profile,
2232     const Gpt2ExecutionOptions& execution_options)
2233     : model_(&model),
2234       hotspot_profile_(hotspot_profile),
2235       execution_options_(execution_options),
2236       cache_(model.config),
2237       workspace_(model.config),
2238       worker_pool_(make_worker_pool(model, execution_options)) {
2239     validate_model(*this->model_);
2240 }
2241
2242 Gpt2CachedGreedyDecoder::~Gpt2CachedGreedyDecoder() = default;
2243
2244 Gpt2CachedGreedyDecoder::Gpt2CachedGreedyDecoder(Gpt2CachedGreedyDecoder&&) ↵
    ↵ noexcept =
2245     default;
2246
2247 Gpt2CachedGreedyDecoder& Gpt2CachedGreedyDecoder::operator=(
2248     Gpt2CachedGreedyDecoder&&) noexcept = default;
2249
2250 void Gpt2CachedGreedyDecoder::advance_without_prediction(std::int32_t token_id) ↵
    ↵ {
2251     static_cast<void>(run_gpt2_cached_step_unchecked(*this->model_,
2252                                                       this->cache_,
2253                                                       this->workspace_,
2254                                                       token_id,
2255                                                       Gpt2CachedStepMode::↵
    ↵ advance_only,
2256                                                       this->hotspot_profile_,
2257                                                       this->execution_options_,
2258                                                       this->worker_pool_.get()));
2259 }
2260
2261 std::int32_t Gpt2CachedGreedyDecoder::step(std::int32_t token_id) {
2262     return run_gpt2_cached_step_unchecked(*this->model_,
2263                                           this->cache_,
2264                                           this->workspace_,
2265                                           token_id,
2266                                           Gpt2CachedStepMode::predict_token,
2267                                           this->hotspot_profile_,
2268                                           this->execution_options_,
2269                                           this->worker_pool_.get())
2270         .predicted_token;
2271 }
2272
2273 Gpt2ForwardResult Gpt2CachedGreedyDecoder::step_with_logits(std::int32_t ↵
    ↵ token_id) {
2274     return run_gpt2_cached_step_unchecked(*this->model_,
2275                                           this->cache_,
2276                                           this->workspace_,
2277                                           token_id,

```

```

2278         Gpt2CachedStepMode::logits,
2279         this->hotspot_profile_,
2280         this->execution_options_,
2281         this->worker_pool_.get());
2282     }
2283
2284     const Gpt2KvCache& Gpt2CachedGreedyDecoder::cache() const noexcept {
2285         return this->cache_;
2286     }
2287
2288     std::int32_t Gpt2CachedGreedyDecoder::filled_tokens() const noexcept {
2289         return this->cache_.filled_tokens();
2290     }
2291
2292     std::size_t Gpt2CachedGreedyDecoder::kv_cache_bytes() const noexcept {
2293         return this->cache_.byte_size();
2294     }
2295
2296     std::int32_t Gpt2CachedGreedyDecoder::worker_count() const noexcept {
2297         return this->worker_pool_ == nullptr ? 1 : this->worker_pool_->worker_count();
2298     }
2299
2300     Gpt2ForwardResult run_gpt2_forward(const Gpt2ReferenceModel& model,
2301                                       std::span<const std::int32_t> token_ids) {
2302         validate_model(model);
2303         if (token_ids.empty()) {
2304             throw std::runtime_error("GPT-2 forward needs at least one token");
2305         }
2306         if (token_ids.size() > checked_size(model.config.max_positions, "max_positions")) {
2307             throw std::runtime_error("GPT-2 token_ids exceed max_positions");
2308         }
2309
2310         const std::int32_t sequence_length = static_cast<std::int32_t>(token_ids.size());
2311         std::vector<float> hidden_by_pos(matrix_size(sequence_length, model.config.hidden_size), 0.0F);
2312         for (std::int32_t position = 0; position < sequence_length; ++position) {
2313             const std::int32_t token_id = token_ids[checked_size(position, "position")];
2314             if (token_id < 0 || token_id >= model.config.vocab_size) {
2315                 throw std::runtime_error("GPT-2 token id out of range");
2316             }
2317             const std::span<const float> token =
2318                 row_span(model.token_embedding, token_id, model.config.hidden_size);
2319             const std::span<const float> position_embedding =
2320                 row_span(model.position_embedding, position, model.config.hidden_size);
2321             std::span<float> hidden = std::span<float>(
2322                 hidden_by_pos.data() + matrix_size(position, model.config.hidden_size),

```

```

    ↪ hidden_size),
2323         checked_size(model.config.hidden_size, "hidden_size"));
2324     for (std::int32_t dim = 0; dim < model.config.hidden_size; ++dim) {
2325         hidden[checked_size(dim, "dim")] =
2326             token[checked_size(dim, "dim")] + position_embedding[↪
    ↪ checked_size(dim, "dim")]);
2327     }
2328 }
2329
2330 for (const Gpt2LayerWeightsF32& layer : model.layers) {
2331     forward_gpt2_layer(model.config, layer, sequence_length, hidden_by_pos)↪
    ↪ ;
2332 }
2333
2334 std::vector<float> normed(checked_size(model.config.hidden_size, "↪
    ↪ hidden_size"), 0.0F);
2335 const std::span<const float> final_hidden = std::span<const float>(
2336     hidden_by_pos.data() + matrix_size(sequence_length - 1, model.config.↪
    ↪ hidden_size),
2337     checked_size(model.config.hidden_size, "hidden_size"));
2338 layer_norm(final_hidden,
2339     model.final_ln_weight,
2340     model.final_ln_bias,
2341     model.config.layer_norm_epsilon,
2342     normed);
2343
2344 Gpt2ForwardResult result;
2345 result.logits.assign(checked_size(model.config.vocab_size, "vocab_size"), ↪
    ↪ 0.0F);
2346 compute_logits(model, normed, result.logits);
2347 result.predicted_token = argmax(result.logits);
2348 return result;
2349 }
2350
2351 Gpt2ForwardResult run_gpt2_forward_cached(const Gpt2ReferenceModel& model,
2352     Gpt2KvCache& cache,
2353     std::int32_t token_id) {
2354     validate_model(model);
2355     Gpt2ForwardWorkspace workspace(model.config);
2356     Gpt2ExecutionOptions options;
2357     options.worker_count = 1;
2358     return run_gpt2_cached_step_unchecked(
2359         model,
2360         cache,
2361         workspace,
2362         token_id,
2363         Gpt2CachedStepMode::logits,
2364         nullptr,
2365         options,
2366         nullptr);
2367 }
2368

```

```

2369 std::vector<std::int32_t> gpt2_generate_greedy(const Gpt2ReferenceModel& model,
2370                                              std::span<const std::int32_t> ←
    prompt_token_ids,
2371                                              std::int32_t max_new_tokens) {
2372     if (max_new_tokens < 0) {
2373         throw std::runtime_error("max_new_tokens cannot be negative");
2374     }
2375     std::vector<std::int32_t> tokens(prompt_token_ids.begin(), prompt_token_ids ←
    .end());
2376     for (std::int32_t step = 0; step < max_new_tokens; ++step) {
2377         if (tokens.size() >= checked_size(model.config.max_positions, " ←
    max_positions")) {
2378             throw std::runtime_error("GPT-2 generation would exceed ←
    max_positions");
2379         }
2380         const Gpt2ForwardResult result = run_gpt2_forward(model, tokens);
2381         tokens.push_back(result.predicted_token);
2382         if (model.config.eos_token_id >= 0 && result.predicted_token == model. ←
    config.eos_token_id) {
2383             break;
2384         }
2385     }
2386     return tokens;
2387 }
2388
2389 std::vector<std::int32_t> gpt2_generate_greedy_cached(
2390     const Gpt2ReferenceModel& model,
2391     std::span<const std::int32_t> prompt_token_ids,
2392     std::int32_t max_new_tokens) {
2393     if (prompt_token_ids.empty()) {
2394         throw std::runtime_error("GPT-2 generation needs at least one prompt ←
    token");
2395     }
2396     if (max_new_tokens < 0) {
2397         throw std::runtime_error("max_new_tokens cannot be negative");
2398     }
2399     if (prompt_token_ids.size() > checked_size(model.config.max_positions, " ←
    max_positions")) {
2400         throw std::runtime_error("GPT-2 prompt exceeds max_positions");
2401     }
2402
2403     Gpt2CachedGreedyDecoder decoder(model);
2404     std::vector<std::int32_t> tokens(prompt_token_ids.begin(), prompt_token_ids ←
    .end());
2405     if (max_new_tokens == 0) {
2406         return tokens;
2407     }
2408     for (std::size_t index = 0; index + 1 < prompt_token_ids.size(); ++index) {
2409         decoder.advance_without_prediction(prompt_token_ids[index]);
2410     }
2411     std::int32_t predicted_token = decoder.step(prompt_token_ids.back());
2412     for (std::int32_t step = 0; step < max_new_tokens; ++step) {

```



```

2413         if (tokens.size() >= checked_size(model.config.max_positions, "↵
↵ max_positions")) {
2414             throw std::runtime_error("GPT-2 generation would exceed ↵
↵ max_positions");
2415         }
2416         tokens.push_back(predicted_token);
2417         if (model.config.eos_token_id >= 0 && predicted_token == model.config.↵
↵ eos_token_id) {
2418             break;
2419         }
2420         if (step + 1 < max_new_tokens) {
2421             predicted_token = decoder.step(predicted_token);
2422         }
2423     }
2424     return tokens;
2425 }
2426
2427 Gpt2ReferenceModel make_tiny_gpt2_reference_model() {
2428     Gpt2ReferenceModel model;
2429     model.config.hidden_size = 4;
2430     model.config.head_count = 2;
2431     model.config.layer_count = 1;
2432     model.config.max_positions = 8;
2433     model.config.vocab_size = 6;
2434     model.config.intermediate_size = 8;
2435     model.config.bos_token_id = 0;
2436     model.config.eos_token_id = 5;
2437     model.config.layer_norm_epsilon = 1.0e-5F;
2438     model.config.activation_function = "gelu_new";
2439
2440     model.token_embedding = {
2441         0.20F, -0.10F, 0.00F, 0.10F,
2442         -0.10F, 0.30F, 0.20F, -0.20F,
2443         0.05F, 0.10F, -0.30F, 0.25F,
2444         0.40F, -0.20F, 0.15F, 0.05F,
2445         -0.30F, 0.05F, 0.35F, -0.15F,
2446         0.10F, 0.20F, -0.10F, 0.30F,
2447     };
2448     model.position_embedding = {
2449         0.01F, 0.02F, 0.03F, 0.04F,
2450         -0.02F, 0.01F, 0.00F, 0.03F,
2451         0.03F, -0.01F, 0.02F, 0.00F,
2452         0.00F, 0.02F, -0.02F, 0.01F,
2453         0.02F, 0.00F, 0.01F, -0.01F,
2454         -0.01F, 0.03F, 0.00F, 0.02F,
2455         0.01F, -0.02F, 0.03F, 0.00F,
2456         0.00F, 0.01F, -0.01F, 0.02F,
2457     };
2458
2459     Gpt2LayerWeightsF32 layer;
2460     layer.ln_1_weight = ones(4);
2461     layer.ln_1_bias = zeros(4);

```

```

2462     layer.ln_2_weight = {0.9F, 1.1F, 1.0F, 0.95F};
2463     layer.ln_2_bias = {0.01F, -0.02F, 0.00F, 0.03F};
2464     layer.c_attn = make_linear(
2465         4,
2466         12,
2467         {
2468             0.10F, -0.20F, 0.05F, 0.15F,
2469             -0.05F, 0.12F, 0.20F, -0.10F,
2470             0.18F, 0.02F, -0.08F, 0.11F,
2471             -0.12F, 0.06F, 0.14F, 0.04F,
2472             0.07F, 0.16F, -0.09F, 0.03F,
2473             -0.15F, 0.05F, 0.13F, 0.10F,
2474             0.04F, -0.11F, 0.19F, -0.02F,
2475             0.09F, 0.08F, -0.06F, 0.17F,
2476             0.20F, -0.04F, 0.01F, 0.06F,
2477             -0.08F, 0.15F, 0.07F, -0.03F,
2478             0.05F, 0.10F, -0.14F, 0.12F,
2479             0.11F, -0.07F, 0.16F, 0.02F,
2480         },
2481         {
2482             0.01F, -0.02F, 0.00F, 0.03F,
2483             -0.01F, 0.02F, 0.01F, -0.02F,
2484             0.00F, 0.01F, -0.01F, 0.02F,
2485         });
2486     layer.c_proj = make_linear(
2487         4,
2488         4,
2489         {
2490             0.10F, 0.05F, -0.08F, 0.12F,
2491             -0.04F, 0.11F, 0.09F, -0.02F,
2492             0.07F, -0.10F, 0.13F, 0.05F,
2493             0.02F, 0.14F, -0.06F, 0.08F,
2494         },
2495         {0.01F, 0.00F, -0.01F, 0.02F});
2496     layer.c_fc = make_linear(
2497         4,
2498         8,
2499         {
2500             0.20F, -0.10F, 0.05F, 0.03F,
2501             -0.05F, 0.14F, 0.12F, -0.08F,
2502             0.09F, 0.07F, -0.11F, 0.16F,
2503             -0.12F, 0.05F, 0.18F, 0.02F,
2504             0.04F, -0.09F, 0.13F, 0.10F,
2505             0.11F, 0.03F, -0.07F, 0.15F,
2506             -0.02F, 0.17F, 0.06F, -0.04F,
2507             0.08F, -0.06F, 0.10F, 0.12F,
2508         },
2509         {0.01F, -0.02F, 0.00F, 0.03F, -0.01F, 0.02F, 0.01F, 0.00F});
2510     layer.mlp_c_proj = make_linear(
2511         8,
2512         4,
2513         {

```

```

2514         0.10F, -0.08F, 0.06F, 0.12F, -0.04F, 0.07F, 0.03F, 0.11F,
2515         -0.05F, 0.13F, 0.02F, -0.09F, 0.15F, 0.04F, -0.03F, 0.08F,
2516         0.07F, 0.01F, -0.12F, 0.10F, 0.06F, -0.05F, 0.14F, 0.02F,
2517         0.03F, 0.09F, 0.11F, -0.06F, 0.05F, 0.12F, -0.08F, 0.04F,
2518     },
2519     {0.00F, 0.01F, -0.01F, 0.02F});
2520     model.layers.push_back(std::move(layer));
2521     model.final_ln_weight = {1.0F, 0.95F, 1.05F, 1.0F};
2522     model.final_ln_bias = {0.0F, 0.01F, -0.01F, 0.02F};
2523     model.tie_lm_head_to_embedding = true;
2524     validate_model(model);
2525     return model;
2526 }
2527
2528 } // namespace lcqi

```

`gpt2_cli.cpp` 负责把模型目录、prompt、`max_new_tokens` 和 benchmark 选项接到 GPT-2 reference path。无参数时运行 tiny GPT-2 fixture；传入 HuggingFace 模型目录时加载真实资产。

**Listing 10.14** `src/gpt2_cli.cpp`

```

1  #include <lcqi/gpt2_reference.hpp>
2
3  #include <chrono>
4  #include <cstdlib>
5  #include <exception>
6  #include <filesystem>
7  #include <iostream>
8  #include <span>
9  #include <stdexcept>
10 #include <string>
11 #include <string_view>
12 #include <vector>
13
14 namespace {
15
16 constexpr int LCQI_GPT2_TINY_ARGC = 1;
17 constexpr int LCQI_GPT2_MIN_REAL_ARGC = 3;
18 constexpr std::int32_t LCQI_GPT2_DEFAULT_MAX_NEW_TOKENS = 8;
19 constexpr std::int32_t LCQI_GPT2_TINY_PROMPT_0 = 1;
20 constexpr std::int32_t LCQI_GPT2_TINY_PROMPT_1 = 2;
21 constexpr std::int32_t LCQI_GPT2_TINY_PROMPT_2 = 3;
22
23 enum class Gpt2EngineMode {
24     full_prefix,
25     cached_kv,
26 };
27
28 struct Gpt2CliOptions {
29     Gpt2EngineMode engine = Gpt2EngineMode::cached_kv;
30     bool benchmark = false;
31     bool profile_hotspots = false;
32     std::filesystem::path model_dir;

```

```

33     std::string prompt;
34     std::int32_t max_new_tokens = LCQI_GPT2_DEFAULT_MAX_NEW_TOKENS;
35     std::int32_t worker_count = 0;
36 };
37
38 struct Gpt2BenchmarkTimings {
39     double load_ms = 0.0;
40     double tokenize_ms = 0.0;
41     double prefill_ms = 0.0;
42     double decode_ms = 0.0;
43     double generate_ms = 0.0;
44     double total_ms = 0.0;
45     std::size_t prefill_steps = 0;
46     std::size_t decode_steps = 0;
47     std::size_t kv_cache_bytes = 0;
48     std::int32_t worker_count = 1;
49 };
50
51 using Clock = std::chrono::steady_clock;
52
53 void print_ids(std::span<const std::int32_t> ids) {
54     for (std::size_t index = 0; index < ids.size(); ++index) {
55         if (index != 0) {
56             std::cout << ' ';
57         }
58         std::cout << ids[index];
59     }
60 }
61
62 [[nodiscard]] double elapsed_ms(Clock::time_point start, Clock::time_point end)↵
    ↪ {
63     return std::chrono::duration<double, std::milli>(end - start).count();
64 }
65
66 [[nodiscard]] std::size_t checked_size(std::int64_t value, const char* name) {
67     if (value < 0) {
68         throw std::runtime_error(std::string(name) + " cannot be negative");
69     }
70     return static_cast<std::size_t>(value);
71 }
72
73 [[nodiscard]] const char* engine_name(Gpt2EngineMode engine) {
74     switch (engine) {
75         case Gpt2EngineMode::full_prefix:
76             return "full_prefix";
77         case Gpt2EngineMode::cached_kv:
78             return "cached_kv";
79     }
80     return "unknown";
81 }
82
83 [[nodiscard]] Gpt2EngineMode parse_engine(std::string_view value) {

```

```

84     if (value == "full" || value == "full_prefix") {
85         return Gpt2EngineMode::full_prefix;
86     }
87     if (value == "cached" || value == "cached_kv") {
88         return Gpt2EngineMode::cached_kv;
89     }
90     throw std::runtime_error("unknown --engine value, expected cached or full");
91 }
92
93 [[nodiscard]] std::int32_t parse_non_negative_i32(const std::string& text,
94                                                  const char* name) {
95     const std::int32_t value = static_cast<std::int32_t>(std::stoi(text));
96     if (value < 0) {
97         throw std::runtime_error(std::string(name) + " cannot be negative");
98     }
99     return value;
100 }
101
102 [[nodiscard]] Gpt2CliOptions parse_options(int argc, char** argv) {
103     Gpt2CliOptions options;
104     std::vector<std::string> positional;
105     for (int index = 1; index < argc; ++index) {
106         const std::string arg = argv[index];
107         if (arg == "--benchmark") {
108             options.benchmark = true;
109         } else if (arg == "--profile-hotspots") {
110             options.profile_hotspots = true;
111             options.benchmark = true;
112         } else if (arg == "--engine") {
113             if (index + 1 >= argc) {
114                 throw std::runtime_error("--engine requires a value");
115             }
116             ++index;
117             options.engine = parse_engine(argv[index]);
118         } else if (arg.rfind("--engine=", 0) == 0) {
119             options.engine = parse_engine(std::string_view(arg).substr(std::string("--engine=").size()));
120         } else if (arg == "--threads") {
121             if (index + 1 >= argc) {
122                 throw std::runtime_error("--threads requires a value");
123             }
124             ++index;
125             options.worker_count = parse_non_negative_i32(argv[index], "--threads");
126         } else if (arg.rfind("--threads=", 0) == 0) {
127             options.worker_count =
128                 parse_non_negative_i32(arg.substr(std::string("--threads=").size()), "--threads");
129         } else if (arg == "--full") {
130             options.engine = Gpt2EngineMode::full_prefix;
131         } else if (arg == "--cached") {

```

```

132     options.engine = Gpt2EngineMode::cached_kv;
133 } else if (!arg.empty() && arg.front() == '-') {
134     throw std::runtime_error("unknown option: " + arg);
135 } else {
136     positional.push_back(arg);
137 }
138 }
139
140 if (positional.size() < 2 || positional.size() > 3) {
141     throw std::runtime_error(
142         "usage: lcqi_gpt2 [--engine cached|full] [--threads N] [--benchmark↵
↵ ] "
143         "model_dir prompt [max_new_tokens]");
144 }
145 options.model_dir = positional[0];
146 options.prompt = positional[1];
147 if (positional.size() == 3) {
148     options.max_new_tokens = parse_non_negative_i32(positional[2], "↵
↵ max_new_tokens");
149 }
150 return options;
151 }
152
153 [[nodiscard]] std::vector<std::int32_t> generate_full_prefix(
154     const lcqi::Gpt2ReferenceModel& model,
155     std::span<const std::int32_t> prompt_ids,
156     std::int32_t max_new_tokens,
157     Gpt2BenchmarkTimings& timings) {
158     const Clock::time_point generate_start = Clock::now();
159     std::vector<std::int32_t> tokens(prompt_ids.begin(), prompt_ids.end());
160     for (std::int32_t step = 0; step < max_new_tokens; ++step) {
161         if (tokens.size() >= checked_size(model.config.max_positions, "↵
↵ max_positions")) {
162             throw std::runtime_error("GPT-2 generation would exceed ↵
↵ max_positions");
163         }
164         const lcqi::Gpt2ForwardResult result = lcqi::run_gpt2_forward(model, ↵
↵ tokens);
165         ++timings.decode_steps;
166         tokens.push_back(result.predicted_token);
167         if (model.config.eos_token_id >= 0 && result.predicted_token == model.↵
↵ config.eos_token_id) {
168             break;
169         }
170     }
171     const Clock::time_point generate_end = Clock::now();
172     timings.generate_ms = elapsed_ms(generate_start, generate_end);
173     timings.decode_ms = timings.generate_ms;
174     return tokens;
175 }
176
177 [[nodiscard]] std::vector<std::int32_t> generate_cached(

```

```

178     const lcqi::Gpt2ReferenceModel& model,
179     std::span<const std::int32_t> prompt_ids,
180     std::int32_t max_new_tokens,
181     Gpt2BenchmarkTimings& timings,
182     lcqi::Gpt2HotspotProfile* hotspot_profile,
183     std::int32_t worker_count) {
184     if (prompt_ids.empty()) {
185         throw std::runtime_error("GPT-2 generation needs at least one prompt ↵
↵ token");
186     }
187     lcqi::Gpt2ExecutionOptions execution_options;
188     execution_options.worker_count = worker_count;
189     lcqi::Gpt2CachedGreedyDecoder decoder(model, hotspot_profile, ↵
↵ execution_options);
190     timings.kv_cache_bytes = decoder.kv_cache_bytes();
191     timings.worker_count = decoder.worker_count();
192     std::vector<std::int32_t> tokens(prompt_ids.begin(), prompt_ids.end());
193     if (max_new_tokens == 0) {
194         return tokens;
195     }
196
197     const Clock::time_point generate_start = Clock::now();
198     const Clock::time_point prefill_start = Clock::now();
199     for (std::size_t index = 0; index + 1 < prompt_ids.size(); ++index) {
200         decoder.advance_without_prediction(prompt_ids[index]);
201         ++timings.prefill_steps;
202     }
203     std::int32_t predicted_token = decoder.step(prompt_ids.back());
204     ++timings.prefill_steps;
205     const Clock::time_point prefill_end = Clock::now();
206     timings.prefill_ms = elapsed_ms(prefill_start, prefill_end);
207
208     const Clock::time_point decode_start = Clock::now();
209     for (std::int32_t step = 0; step < max_new_tokens; ++step) {
210         if (tokens.size() >= checked_size(model.config.max_positions, "↵
↵ max_positions")) {
211             throw std::runtime_error("GPT-2 generation would exceed ↵
↵ max_positions");
212         }
213         tokens.push_back(predicted_token);
214         if (model.config.eos_token_id >= 0 && predicted_token == model.config.↵
↵ eos_token_id) {
215             break;
216         }
217         if (step + 1 < max_new_tokens) {
218             predicted_token = decoder.step(predicted_token);
219             ++timings.decode_steps;
220         }
221     }
222     const Clock::time_point decode_end = Clock::now();
223     const Clock::time_point generate_end = Clock::now();
224     timings.decode_ms = elapsed_ms(decode_start, decode_end);

```



```

225     timings.generate_ms = elapsed_ms(generate_start, generate_end);
226     return tokens;
227 }
228
229 void print_benchmark(const Gpt2BenchmarkTimings& timings,
230                     std::size_t prompt_tokens,
231                     std::size_t generated_tokens) {
232     const double new_tokens = static_cast<double>(generated_tokens);
233     const double generated_tokens_per_second =
234         timings.generate_ms > 0.0 ? new_tokens * 1000.0 / timings.generate_ms : ↵
        ↵ 0.0;
235     const double decode_tokens_per_second =
236         timings.decode_ms > 0.0
237         ? static_cast<double>(timings.decode_steps) * 1000.0 / timings.↵
        ↵ decode_ms
        : 0.0;
238
239     std::cout << "benchmark_load_ms " << timings.load_ms << "\n";
240     std::cout << "benchmark_tokenize_ms " << timings.tokenize_ms << "\n";
241     std::cout << "benchmark_prefill_ms " << timings.prefill_ms << "\n";
242     std::cout << "benchmark_decode_ms " << timings.decode_ms << "\n";
243     std::cout << "benchmark_generate_ms " << timings.generate_ms << "\n";
244     std::cout << "benchmark_total_ms " << timings.total_ms << "\n";
245     std::cout << "benchmark_prompt_tokens " << prompt_tokens << "\n";
246     std::cout << "benchmark_generated_tokens " << generated_tokens << "\n";
247     std::cout << "benchmark_prefill_steps " << timings.prefill_steps << "\n";
248     std::cout << "benchmark_decode_steps " << timings.decode_steps << "\n";
249     std::cout << "benchmark_worker_count " << timings.worker_count << "\n";
250     std::cout << "benchmark_generate_tokens_per_second "
251         << generated_tokens_per_second << "\n";
252     std::cout << "benchmark_decode_tokens_per_second "
253         << decode_tokens_per_second << "\n";
254     std::cout << "benchmark_kv_cache_bytes " << timings.kv_cache_bytes << "\n";
255 }
256
257 void print_hotspot_profile(const lcqi::Gpt2HotspotProfile& profile) {
258     const double total = profile.total_step_ms > 0.0 ? profile.total_step_ms : ↵
        ↵ 1.0;
259     const auto print_ms = [total](std::string_view name, double value) {
260         std::cout << "hotspot_" << name << "_ms " << value << "\n";
261         std::cout << "hotspot_" << name << "_pct " << (value * 100.0 / total) ↵
        ↵ << "\n";
262     };
263     std::cout << "hotspot_decoder_steps " << profile.decoder_steps << "\n";
264     std::cout << "hotspot_layer_steps " << profile.layer_steps << "\n";
265     print_ms("total_step", profile.total_step_ms);
266     print_ms("embedding", profile.embedding_ms);
267     print_ms("layer_norm", profile.layer_norm_ms);
268     print_ms("final_norm", profile.final_norm_ms);
269     print_ms("qkv_projection", profile.qkv_projection_ms);
270     print_ms("kv_append", profile.kv_append_ms);
271     print_ms("attention", profile.attention_ms);
272

```

```

273     print_ms("attention_projection", profile.attention_projection_ms);
274     print_ms("residual_add", profile.residual_add_ms);
275     print_ms("mlp_fc", profile.mlp_fc_ms);
276     print_ms("gelu", profile.gelu_ms);
277     print_ms("mlp_projection", profile.mlp_projection_ms);
278     print_ms("lm_head", profile.lm_head_ms);
279     print_ms("logits_result", profile.logits_result_ms);
280 }
281
282 int run_tiny_mode() {
283     const lcqi::Gpt2ReferenceModel model = lcqi::make_tiny_gpt2_reference_model(
284         ↵ );
285     const std::vector<std::int32_t> prompt{
286         LCQI_GPT2_TINY_PROMPT_0,
287         LCQI_GPT2_TINY_PROMPT_1,
288         LCQI_GPT2_TINY_PROMPT_2,
289     };
290     const lcqi::Gpt2ForwardResult result = lcqi::run_gpt2_forward(model, prompt(
291         ↵ );
292     const std::vector<std::int32_t> generated =
293         lcqi::gpt2_generate_greedy(model, prompt, 2);
294     std::cout << "mode tiny\n";
295     std::cout << "prompt_ids ";
296     print_ids(prompt);
297     std::cout << "\n";
298     std::cout << "predicted_token " << result.predicted_token << "\n";
299     std::cout << "logits";
300     for (const float value : result.logits) {
301         std::cout << ' ' << value;
302     }
303     std::cout << "\n";
304     std::cout << "generated_ids ";
305     print_ids(generated);
306     std::cout << "\n";
307     return EXIT_SUCCESS;
308 }
309
310 int run_real_model_mode(int argc, char** argv) {
311     if (argc < LCQI_GPT2_MIN_REAL_ARGC) {
312         throw std::runtime_error(
313             ↵ "usage: lcqi_gpt2 [--engine cached|full] [--threads N] [--benchmark(
314             ↵ "model_dir prompt [max_new_tokens]");
315     }
316     const Clock::time_point total_start = Clock::now();
317     const Gpt2CliOptions options = parse_options(argc, argv);
318
319     Gpt2BenchmarkTimings timings;
320     const Clock::time_point load_start = Clock::now();
321     const lcqi::Gpt2ReferenceModel model = lcqi::load_gpt2_from_directory(
322         ↵ options.model_dir);
323     const lcqi::Gpt2Tokenizer tokenizer = lcqi::load_gpt2_tokenizer(

```

```

321     options.model_dir / "vocab.json",
322     options.model_dir / "merges.txt",
323     model.config.bos_token_id,
324     model.config.eos_token_id);
325     timings.load_ms = elapsed_ms(load_start, Clock::now());
326
327     const Clock::time_point tokenize_start = Clock::now();
328     const std::vector<std::int32_t> prompt_ids = lcqi::gpt2_encode(tokenizer, ←
    ↪ options.prompt);
329     timings.tokenize_ms = elapsed_ms(tokenize_start, Clock::now());
330
331     std::vector<std::int32_t> generated;
332     lcqi::Gpt2HotspotProfile hotspot_profile;
333     if (options.engine == Gpt2EngineMode::full_prefix) {
334         if (options.profile_hotspots) {
335             throw std::runtime_error("--profile-hotspots currently supports ←
    ↪ cached engine only");
336         }
337         generated = generate_full_prefix(model, prompt_ids, options.←
    ↪ max_new_tokens, timings);
338     } else {
339         generated = generate_cached(model,
340                                     prompt_ids,
341                                     options.max_new_tokens,
342                                     timings,
343                                     options.profile_hotspots ? &hotspot_profile←
    ↪ : nullptr,
344                                     options.worker_count);
345     }
346     timings.total_ms = elapsed_ms(total_start, Clock::now());
347
348     std::cout << "mode gpt2_directory\n";
349     std::cout << "engine " << engine_name(options.engine) << "\n";
350     std::cout << "model_dir " << options.model_dir.string() << "\n";
351     std::cout << "prompt_ids ";
352     print_ids(prompt_ids);
353     std::cout << "\n";
354     std::cout << "generated_ids ";
355     print_ids(generated);
356     std::cout << "\n";
357     std::cout << "generated_text " << lcqi::gpt2_decode(tokenizer, generated) ←
    ↪ << "\n";
358     if (options.benchmark) {
359         const std::size_t generated_tokens = generated.size() - prompt_ids.size←
    ↪ ();
360         print_benchmark(timings, prompt_ids.size(), generated_tokens);
361     }
362     if (options.profile_hotspots) {
363         print_hotspot_profile(hotspot_profile);
364     }
365     return EXIT_SUCCESS;
366 }

```

```
367
368 } // namespace
369
370 int main(int argc, char** argv) {
371     try {
372         if (argc == LCQI_GPT2_TINY_ARGC) {
373             return run_tiny_mode();
374         }
375         return run_real_model_mode(argc, argv);
376     } catch (const std::exception& error) {
377         std::cerr << "error: " << error.what() << "\n";
378         return EXIT_FAILURE;
379     }
380 }
```

## 第 11 章实践：运行时内存计划、liveness 与 Workspace

本章对应原书中的第三册“推理运行时与内存规划”。不要把它当作概念复述，本章要把 activation arena、workspace、liveness、state edge、debug dump、buffer reuse 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `layout`，核心命令或测试入口是 `lcqi_trace`。

**Listing 11.1** 第 11 章实践：运行时内存计划、liveness 与 Workspace 的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_trace / layout
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：比较普通 activation 和 KV cache 的生命周期。读者要先手写预期结果，再运行项目观察输出。动作是：说明为什么 KV cache 不能被 activation planner 当作临时缓冲复用。如果输出里没有 `checksum` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：内存复用只在生命周期不重叠时成立；debug/oracle/materialization 会延长生命周期这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

### 四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交 liveness 表：每个 checkpoint 的 `defined_at`、`last_used_at`、是否 state、是否允许复用。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

## 第 12 章实践：IR、Pass、Lowering 与 Executable Plan

本章对应原书中的第三册“AI 编译器细化：IR、pass、lowering 与 executable plan”。不要把它当作概念复述，本章要把 typed graph、pass、fusion、layout propagation、backend capability、fallback reason 放到同一个可运行项目里检查。贯穿代码仍然是 [books/cpu-volume-3/labs/linux\\_cpu\\_inference](#)；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `run_inference`，核心命令或测试入口是 `lcqi_cli`。

**Listing 12.1** 第 12 章实践：IR、Pass、Lowering 与 Executable Plan 的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_cli / run_inference
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：把 tiny MLP 写成三节点 IR：linear、relu、linear。读者要先手写预期结果，再运行项目观察输出。动作是：说明哪些 pass 可以融合，哪些必须保留为 oracle checkpoint。如果输出里没有 `backend` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：IR 的价值不是画图，而是让 shape、layout、量化、后端选择和 fallback 诊断有稳定承载物这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 [books/cpu-volume-3/labs/linux\\_cpu\\_inference/README.md](#)、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

### 四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交 executable plan 草图：节点、输入输出 tensor、layout、kernel、fallback reason 和 profiler event。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

第零部分

## 后端优化、工业专项与验收



## 第 13 章实践：CPU 后端、微内核与性能证据

本章对应原书中的第三册“CPU 后端优化细讲”和“CPU decode 实战”。不要把它当作概念复述，本章要把 scalar baseline、packed weight、AVX2、objdump、perf boundary、shape sweep 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `avx2`、`GGUF`、`Q4_K` 和 `Q8_K`，核心命令或测试入口是 `lcqi_bench` 与 `lcqi_gguf_q4_bench`。

Listing 13.1 第 13 章实践：CPU 后端、微内核与性能证据的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_bench / lcqi_gguf_q4_bench / avx2 / Q4_K
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：对同一 kernel 保存 benchmark CSV 和反汇编摘要。读者要先手写预期结果，再运行项目观察输出。动作是：解释速度差来自地址流、SIMD lane、packing、load 或 reduction 哪一项。如果输出里没有 `average_us` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

第二个特例必须对齐真实模型：`lcqi_gguf_q4_bench` 读取 llama.cpp smoke 使用的同一个 `SmolLM2-135M-Instruct-Q4_K_M.gguf`，从 GGUF tensor 表中选出真实 `Q4_K` 矩阵，比较三条路径：预先解成 f32 权重再 matvec、`Q4_K` 权重直接乘 f32 input、`Q4_K` 权重乘临时 `Q8_K` input。报告必须同时写 tensor 名称、shape、原始量化字节、解码后 f32 字节、重复轮次、最大误差和 checksum。

从这个特例推出的规律是：CPU 后端优化必须同时有 oracle、反汇编、shape sweep 和降级边界；只有耗时表不够这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA、GGUF 模型缓存或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。真实模型量化切片用下面命令生成报告：

Listing 13.2 SmolLM2 Q4\_K 单算子证据入口

```
python3 books/cpu-volume-3/tools/run_smolm2_q4_k_benchmark.py --no-download --<
  ↪ repeat 20
```

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系



统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

#### 四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交 CPU kernel evidence：命令、机器、编译器、输入 shape、backend、checksum、反汇编摘要和未验证 PMU 边界。若使用真实 GGUF 模型，还要给出模型 repo、文件名、SHA-256、tensor 名、量化格式、同模型 llama.cpp 参考和“为什么不能把单 tensor us 直接等价于端到端 tokens/s”。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

## 第 14 章实践：GPU 后端边界、copy 与同步成本

本章对应原书中的第三册“GPU 硬件基础”和“GPU 后端优化细讲”。不要把它当作概念复述，本章要把 device buffer、stream、event、kernel launch、copy、sync、fallback 放到同一个可运行项目里检查。贯穿代码仍然是 [books/cpu-volume-3/labs/linux\\_cpu\\_inference](#)；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `StepTrace`，核心命令或测试入口是 `lcqi_trace`。

**Listing 14.1** 第 14 章实践：GPU 后端边界、copy 与同步成本的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<↩
↩ DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_trace / StepTrace
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：当前项目没有 GPU kernel，也要写清 adapter 合同。读者要先手写预期结果，再运行项目观察输出。动作是：把 CPU trace 字段扩展成 GPU StepTrace 需要哪些字段。如果输出里没有 `copy_time_ns` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：GPU 快不快不能只看 kernel；copy、sync、sampler、fallback 和 batch wait 必须进入端到端 trace 这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 [books/cpu-volume-3/labs/linux\\_cpu\\_inference/README.md](#)、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

### 四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交 GPU adapter 设计：buffer owner、stream、event、H2D/D2H 字节、sync 点、fallback 和 oracle 对照。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

## 第 15 章实践：工业 LLM 专项能力对照

本章对应原书中的第三册“工业 LLM 专项：投机解码、MoE、LoRA、多 GPU 与源码对照”。不要把它当作概念复述，本章要把 speculative decoding、MoE routing、LoRA adapter、multi-GPU、source reading 放到同一个可运行项目里检查。贯穿代码仍然是 [books/cpu-volume-3/labs/linux\\_cpu\\_inference](#)；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `bandwidth_limit`，核心命令或测试入口是 `lcqi_ledger`。

Listing 15.1 第 15 章实践：工业 LLM 专项能力对照的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  <- DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_ledger / bandwidth_limit
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：选择一个专项，不写泛泛介绍。读者要先手写预期结果，再运行项目观察输出。动作是：把它接到现有 trace、memory、backend 或 serving 字段上。如果输出里没有 `tokens/s` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：工业专项不是功能清单；必须说明它购买什么能力、增加什么状态、破坏哪些假设这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 [books/cpu-volume-3/labs/linux\\_cpu\\_inference/README.md](#)、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

### 四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交专项设计卡：动机、状态新增、trace 字段、回归测试、性能账本和不能验证的部分。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

## 第 16 章实践：工程验收、回归门禁与最终项目

本章对应原书中的第三册“工程验收与回归体系”。不要把它当作概念复述，本章要把 reference、unit test、trace golden、benchmark gate、SLO gate、coverage、CI 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

### 一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `lcqi_tests`，核心命令或测试入口是 `ctest`。

Listing 16.1 第 16 章实践：工程验收、回归门禁与最终项目的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
↳ DCMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: ctest / lcqi_tests
```

### 二、从特例推到规律：先抓一个能手算的切片

本章先看特例：把所有前面报告合成最终验收包。读者要先手写预期结果，再运行项目观察输出。动作是：运行 `ctest` 并说明每个 test 保护哪条合同。如果输出里没有 `100% tests passed` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：实践卷最终目标是长期维护：结果正确、trace 稳定、性能可解释、服务指标可回归、失败边界不伪装这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

### 三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

### 四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交最终项目：README、命令记录、报告索引、测试结果、benchmark 表、trace 摘要、风险清单和下一阶段计划。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

## 代码走读：工具、Smoke、Benchmark 与回归测试

### 工具不是附属品

LCQI 的工程证据不只在 C++ 源码里。smoke 脚本决定真实模型怎样下载、校验和运行；benchmark 程序决定性能口径；trace、ledger 和 serving probe 决定报告字段；CTest 决定哪些行为能长期回归。读者必须把这些文件当作工程合同的一部分。

### Benchmark、Trace、Ledger 与 Serving Probe

`bench.cpp` 输出 int8 kernel benchmark。它服务性能观察，但必须和 max diff 对照一起看。

Listing 16.2 src/bench.cpp

```

1 #include <lcqi/int8_kernels.hpp>
2
3 #include <algorithm>
4 #include <cstdlib>
5 #include <iostream>
6 #include <stdexcept>
7 #include <string>
8 #include <vector>
9
10 namespace {
11
12 constexpr const char* LCQI_TARGET_ARCH_X86_64 = "x86_64";
13 constexpr const char* LCQI_TARGET_ARCH_I386 = "i386";
14 constexpr const char* LCQI_TARGET_ARCH_UNKNOWN = "unknown";
15 constexpr std::int32_t LCQI_DEFAULT_REPEAT = 200;
16 constexpr int LCQI_REPEAT_ARG_INDEX = 1;
17 constexpr std::int32_t LCQI_LARGE_CASE_REPEAT_DIVISOR = 4;
18 constexpr std::int32_t LCQI_MIN_LARGE_CASE_REPEAT = 1;
19 constexpr std::int32_t LCQI_DECODE_SHAPE_SIZE = 4096;
20
21 const char* target_arch() noexcept {
22     #if defined(__x86_64__) || defined(_M_X64)
23         return LCQI_TARGET_ARCH_X86_64;
24     #elif defined(__i386__) || defined(_M_IX86)
25         return LCQI_TARGET_ARCH_I386;
26     #else
27         return LCQI_TARGET_ARCH_UNKNOWN;
28     #endif
29 }
30
31 std::int32_t parse_repeat(int argc, char** argv) {
32     if (argc <= LCQI_REPEAT_ARG_INDEX) {
33         return LCQI_DEFAULT_REPEAT;
34     }
35     const std::int32_t repeat = static_cast<std::int32_t>(std::stoi(argv[↵
↵ LCQI_REPEAT_ARG_INDEX]));
36     if (repeat <= 0) {

```

```

37     throw std::runtime_error("repeat must be positive");
38 }
39 return repeat;
40 }
41
42 std::vector<lcqi::KernelBenchmarkCase> make_cases(std::int32_t repeat) {
43     return {
44         {128, 128, 16, repeat},
45         {256, 256, 16, repeat},
46         {512, 512, 16, repeat},
47         {513, 257, 16, repeat},
48         {1024, 4096, 16, std::max(LCQI_MIN_LARGE_CASE_REPEAT,
49                                     repeat / LCQI_LARGE_CASE_REPEAT_DIVISOR)},
50         {LCQI_DECODE_SHAPE_SIZE, LCQI_DECODE_SHAPE_SIZE, 16,
51          std::max(LCQI_MIN_LARGE_CASE_REPEAT, repeat / ↵
52 ↵ LCQI_LARGE_CASE_REPEAT_DIVISOR)}},
53     };
54 }
55 } // namespace
56
57 int main(int argc, char** argv) {
58     try {
59         const std::int32_t repeat = parse_repeat(argc, argv);
60         std::cout
61             << "target_arch,input_size,output_size,output_block,repeat,backend,↵
62 ↵ available,"
63             << "average_us,max_abs_diff,checksum\n";
64         for (const lcqi::KernelBenchmarkCase& benchmark_case : make_cases(↵
65 ↵ repeat)) {
66             const lcqi::KernelBenchmarkResult result =
67                 lcqi::benchmark_linear_i8_case(benchmark_case);
68             for (const lcqi::KernelBackendBenchmark& backend : result.backends)↵
69 ↵ {
70                 std::cout << target_arch() << ','
71                     << result.benchmark_case.input_size << ','
72                     << result.benchmark_case.output_size << ','
73                     << result.benchmark_case.output_block_size << ','
74                     << result.benchmark_case.repeat << ','
75                     << backend.name << ','
76                     << (backend.available ? 1 : 0) << ','
77                     << backend.average_us << ','
78                     << backend.max_abs_diff << ','
79                     << backend.checksum << '\n';
80             }
81         }
82         return EXIT_SUCCESS;
83     } catch (const std::exception& error) {
84         std::cerr << "error: " << error.what() << '\n';
85         return EXIT_FAILURE;
86     }
87 }

```



`trace.cpp` 输出 reference decoder checkpoint。它用于定位错误边界，不应该被简化成只打印最终 token。

Listing 16.3 src/trace.cpp

```

1  #include <lcqi/reference_decoder.hpp>
2  #include <lcqi/tokenizer.hpp>
3
4  #include <algorithm>
5  #include <array>
6  #include <cstdlib>
7  #include <exception>
8  #include <filesystem>
9  #include <iostream>
10 #include <span>
11 #include <sstream>
12 #include <string>
13 #include <string_view>
14
15 namespace {
16
17 constexpr std::array<std::int32_t, 3> LCQI_TRACE_TOKEN_IDS{1, 2, 3};
18 constexpr std::string_view LCQI_TRACE_PROMPT = "tok1 tok2 tok3";
19 constexpr std::int32_t LCQI_TRACE_METADATA_LAYER = -1;
20 constexpr std::int32_t LCQI_TRACE_BOS_ID = 0;
21 constexpr std::int32_t LCQI_TRACE_EOS_ID = 4;
22 constexpr std::size_t LCQI_TRACE_VALUE_LIMIT = 8;
23
24 std::filesystem::path tokenizer_fixture_path() {
25     return std::filesystem::path(__FILE__).parent_path().parent_path() /
26         "models" / "tiny_tokenizer.txt";
27 }
28
29 std::vector<std::int32_t> decoder_tokens_from_prompt(const lcqi::TokenizerModel&
    ↪ & tokenizer,
30                                                     std::string_view prompt) {
31     std::vector<std::int32_t> full_tokens = lcqi::encode_prompt(tokenizer,
    ↪ prompt);
32     if (full_tokens.size() != LCQI_TRACE_TOKEN_IDS.size() + 2 ||
33         full_tokens.front() != LCQI_TRACE_BOS_ID ||
34         full_tokens.back() != LCQI_TRACE_EOS_ID) {
35         throw std::runtime_error("unexpected tokenizer fixture ids");
36     }
37
38     std::vector<std::int32_t> decoder_tokens(
39         full_tokens.begin() + 1,
40         full_tokens.end() - 1);
41     if (!std::equal(decoder_tokens.begin(),
42                    decoder_tokens.end(),
43                    LCQI_TRACE_TOKEN_IDS.begin(),
44                    LCQI_TRACE_TOKEN_IDS.end())) {
45         throw std::runtime_error("unexpected decoder token ids");
46     }

```



```

47     return decoder_tokens;
48 }
49
50 std::string join_ints(std::span<const std::int32_t> values) {
51     std::ostringstream stream;
52     for (std::size_t index = 0; index < values.size(); ++index) {
53         if (index != 0) {
54             stream << 'x';
55         }
56         stream << values[index];
57     }
58     return stream.str();
59 }
60
61 std::string join_int_values(std::span<const std::int32_t> values) {
62     std::ostringstream stream;
63     for (std::size_t index = 0; index < values.size(); ++index) {
64         if (index != 0) {
65             stream << ';';
66         }
67         stream << values[index];
68     }
69     return stream.str();
70 }
71
72 std::int32_t checksum_ints(std::span<const std::int32_t> values) {
73     std::int32_t result = 0;
74     for (const std::int32_t value : values) {
75         result += value;
76     }
77     return result;
78 }
79
80 std::int32_t max_abs_ints(std::span<const std::int32_t> values) {
81     std::int32_t result = 0;
82     for (const std::int32_t value : values) {
83         result = std::max(result, std::abs(value));
84     }
85     return result;
86 }
87
88 std::string join_floats(std::span<const float> values) {
89     std::ostringstream stream;
90     const std::size_t count = std::min(values.size(), LCQI_TRACE_VALUE_LIMIT);
91     for (std::size_t index = 0; index < count; ++index) {
92         if (index != 0) {
93             stream << ';';
94         }
95         stream << values[index];
96     }
97     if (values.size() > LCQI_TRACE_VALUE_LIMIT) {
98         stream << "...";

```

```

99     }
100     return stream.str();
101 }
102
103 void print_trace_csv(const lcqi::ReferenceDecodeTraceResult& trace,
104                    const lcqi::TokenizerModel& tokenizer,
105                    std::span<const std::int32_t> full_tokens) {
106     std::cout << "name,token_position,layer_id,shape,stride,dtype,layout,↵
↵ checksum,max_abs,values\n";
107     std::cout << "prompt,0," << LCQI_TRACE_METADATA_LAYER
108         << ",scalar,scalar,utf8,text,0,0," << LCQI_TRACE_PROMPT << '\n';
109     std::cout << "tokenizer,0," << LCQI_TRACE_METADATA_LAYER
110         << ', ' << full_tokens.size() << ",1,i32,contiguous,"
111         << checksum_ints(full_tokens) << ', '
112         << max_abs_ints(full_tokens) << ', '
113         << join_int_values(full_tokens) << '\n';
114     std::cout << "tokenizer_contract,0," << LCQI_TRACE_METADATA_LAYER
115         << ",scalar,scalar,u64,fnv1a,"
116         << tokenizer_contract_hash(tokenizer) << ",0,"
117         << tokenizer.chat_template_hash << '\n';
118     for (const lcqi::ReferenceTensorCheckpoint& checkpoint : trace.checkpoints)↵
↵     {
119         std::cout << checkpoint.name << ', '
120             << checkpoint.token_position << ', '
121             << checkpoint.layer_id << ', '
122             << join_ints(checkpoint.shape) << ', '
123             << join_ints(checkpoint.stride) << ', '
124             << checkpoint.dtype << ', '
125             << checkpoint.layout << ', '
126             << checkpoint.checksum << ', '
127             << checkpoint.max_abs << ', '
128             << join_floats(checkpoint.values) << '\n';
129     }
130     std::cout << "sampler_argmax,"
131         << LCQI_TRACE_TOKEN_IDS.size() - 1 << ', '
132         << LCQI_TRACE_METADATA_LAYER
133         << ', ' << trace.result.logits.size() << ",1,f32,argmax,"
134         << "0,0,token=" << trace.result.predicted_token << '\n';
135     std::cout << "generated_token,"
136         << LCQI_TRACE_TOKEN_IDS.size() << ', '
137         << LCQI_TRACE_METADATA_LAYER
138         << ",scalar,scalar,i32,generated,"
139         << trace.result.predicted_token << ', '
140         << trace.result.predicted_token << ', '
141         << trace.result.predicted_token << '\n';
142 }
143
144 } // namespace
145
146 int main() {
147     try {
148         const lcqi::TokenizerModel tokenizer =

```

```

149         lcqi::load_tokenizer(tokenizer_fixture_path());
150         const std::vector<std::int32_t> full_tokens =
151             lcqi::encode_prompt(tokenizer, LCQI_TRACE_PROMPT);
152         const std::vector<std::int32_t> decoder_tokens =
153             decoder_tokens_from_prompt(tokenizer, LCQI_TRACE_PROMPT);
154         const lcqi::ReferenceDecoderModel model = lcqi::↵
↵ make_tiny_reference_decoder_model();
155         const lcqi::ReferenceDecodeTraceResult trace =
156             lcqi::run_reference_decode_with_trace(model, decoder_tokens);
157         print_trace_csv(trace, tokenizer, full_tokens);
158         return EXIT_SUCCESS;
159     } catch (const std::exception& error) {
160         std::cerr << "error: " << error.what() << '\n';
161         return EXIT_FAILURE;
162     }
163 }

```

**ledger.cpp** 输出 7B 规模账本。它是估算，不是实测吞吐；读者要区分 FLOPs、KV 容量、权重/KV 字节和 bandwidth-only tokens/s 上限。

**Listing 16.4** src/ledger.cpp

```

1  #include <cstdlib>
2  #include <cstdint>
3  #include <exception>
4  #include <iomanip>
5  #include <iostream>
6  #include <string_view>
7
8  namespace {
9
10 constexpr double LCQI_BYTES_PER_GB = 1000.0 * 1000.0 * 1000.0;
11 constexpr double LCQI_BYTES_PER_MIB = 1024.0 * 1024.0;
12 constexpr double LCQI_FLOPS_PER_MAC = 2.0;
13 constexpr std::int32_t LCQI_SEVENB_LAYER_COUNT = 32;
14 constexpr std::int32_t LCQI_SEVENB_HIDDEN_SIZE = 4096;
15 constexpr std::int32_t LCQI_SEVENB_QUERY_HEADS = 32;
16 constexpr std::int32_t LCQI_SEVENB_KV_HEADS = 8;
17 constexpr std::int32_t LCQI_SEVENB_HEAD_DIM = 128;
18 constexpr std::int32_t LCQI_SEVENB_INTERMEDIATE_SIZE = 11008;
19 constexpr std::int32_t LCQI_SEVENB_CONTEXT_LENGTH = 4096;
20
21 struct DTypeLedger {
22     std::string_view name;
23     double weight_gb_per_token = 0.0;
24     double kv_element_bytes = 0.0;
25 };
26
27 constexpr DTypeLedger LCQI_DTYPE_LEDGERS[] = {
28     {"fp16", 14.0, 2.0},
29     {"int8", 7.0, 2.0},
30     {"int4", 3.7, 2.0},
31 };

```

```

32
33 constexpr double LCQI_BANDWIDTH_GB_PER_S[] = {
34     50.0,
35     100.0,
36     200.0,
37     600.0,
38     1000.0,
39     1600.0,
40 };
41
42 double linear_macs_per_layer() noexcept {
43     const double hidden = LCQI_SEVENB_HIDDEN_SIZE;
44     const double kv_width = LCQI_SEVENB_KV_HEADS * LCQI_SEVENB_HEAD_DIM;
45     const double qkv = hidden * (hidden + 2.0 * kv_width);
46     const double output_projection = hidden * hidden;
47     const double mlp = 3.0 * hidden * LCQI_SEVENB_INTERMEDIATE_SIZE;
48     return qkv + output_projection + mlp;
49 }
50
51 double attention_macs_per_layer() noexcept {
52     return 2.0 * LCQI_SEVENB_QUERY_HEADS * LCQI_SEVENB_CONTEXT_LENGTH *
53           LCQI_SEVENB_HEAD_DIM;
54 }
55
56 double kv_bytes_per_token_all_layers(double element_bytes) noexcept {
57     return static_cast<double>(LCQI_SEVENB_LAYER_COUNT) * 2.0 *
58           LCQI_SEVENB_KV_HEADS * LCQI_SEVENB_HEAD_DIM * element_bytes;
59 }
60
61 double kv_read_gb_per_decode_token(double element_bytes) noexcept {
62     const double bytes = static_cast<double>(LCQI_SEVENB_CONTEXT_LENGTH) *
63           kv_bytes_per_token_all_layers(element_bytes);
64     return bytes / LCQI_BYTES_PER_GB;
65 }
66
67 double kv_capacity_mib(double element_bytes, std::int32_t context_length) ↵
68     ↵ noexcept {
69     const double bytes = static_cast<double>(context_length) *
70           kv_bytes_per_token_all_layers(element_bytes);
71     return bytes / LCQI_BYTES_PER_MIB;
72 }
73
74 void print_model_rows() {
75     const double linear_macs = linear_macs_per_layer();
76     const double attention_macs = attention_macs_per_layer();
77     const double total_flops =
78         (linear_macs + attention_macs) * LCQI_SEVENB_LAYER_COUNT * ↵
79         ↵ LCQI_FLOPS_PER_MAC;
80     std::cout << "section,item,value,unit,notes\n";
81     std::cout << "model,layers," << LCQI_SEVENB_LAYER_COUNT << ",count,7B ↵
82         ↵ fixture\n";
83     std::cout << "model,hidden," << LCQI_SEVENB_HIDDEN_SIZE << ",elements,H\n";

```

```

81     std::cout << "model,query_heads," << LCQI_SEVENB_QUERY_HEADS << ",count,GQA" <<
    << query_heads << "\n";
82     std::cout << "model,kv_heads," << LCQI_SEVENB_KV_HEADS << ",count,GQA KV" <<
    << kv_heads << "\n";
83     std::cout << "model,head_dim," << LCQI_SEVENB_HEAD_DIM << ",elements,per" <<
    << head_dim << "\n";
84     std::cout << "compute,linear_macs_per_layer," << linear_macs << ",macs," <<
    << decode_token << "\n";
85     std::cout << "compute,attention_macs_per_layer," << attention_macs
86         << ",macs,context=4096\n";
87     std::cout << "compute,total_decode_flops_per_token," << total_flops
88         << ",flops,linear plus attention\n";
89 }
90
91 void print_kv_rows() {
92     for (const std::int32_t context : {1024, 2048, 4096, 8192}) {
93         std::cout << "kv,fp16_capacity_context_" << context << ', '
94             << kv_capacity_mib(2.0, context)
95             << ",MiB,single session\n";
96     }
97 }
98
99 void print_dtype_rows() {
100     for (const DTypeLedger& dtype : LCQI_DTYPE_LEDGERS) {
101         const double kv_read_gb = kv_read_gb_per_decode_token(dtype.
    << kv_element_bytes);
102         const double total_gb = dtype.weight_gb_per_token + kv_read_gb;
103         std::cout << "decode_bytes," << dtype.name << "_weight_gb,"
104             << dtype.weight_gb_per_token << ",GB/token,weight stream\n";
105         std::cout << "decode_bytes," << dtype.name << "_kv_read_gb,"
106             << kv_read_gb << ",GB/token,context=4096\n";
107         std::cout << "decode_bytes," << dtype.name << "_total_gb,"
108             << total_gb << ",GB/token,weight plus KV\n";
109         for (const double bandwidth : LCQI_BANDWIDTH_GB_PER_S) {
110             std::cout << "bandwidth_limit," << dtype.name << "_at_"
111                 << static_cast<std::int32_t>(bandwidth) << "_gbps,"
112                 << bandwidth / total_gb
113                 << ",tokens/s,bandwidth-only upper bound\n";
114         }
115     }
116 }
117
118 } // namespace
119
120 int main() {
121     try {
122         std::cout << std::fixed << std::setprecision(3);
123         print_model_rows();
124         print_kv_rows();
125         print_dtype_rows();
126         return EXIT_SUCCESS;
127     } catch (const std::exception& error) {

```

```

128         std::cerr << "error: " << error.what() << '\n';
129         return EXIT_FAILURE;
130     }
131 }

```

`serving_probe.cpp` 固定短请求、RAG 请求、取消请求和超长请求。它把 TTFT、TPOT、拒绝和取消回收变成可观察字段。

Listing 16.5 src/serving\_probe.cpp

```

1  #include <algorithm>
2  #include <cstdlib>
3  #include <cstdint>
4  #include <exception>
5  #include <cmath>
6  #include <iomanip>
7  #include <iostream>
8  #include <string_view>
9  #include <vector>
10
11 namespace {
12
13 constexpr double LCQI_BYTES_PER_MIB = 1024.0 * 1024.0;
14 constexpr std::int32_t LCQI_PROBE_LAYER_COUNT = 32;
15 constexpr std::int32_t LCQI_PROBE_KV_HEADS = 8;
16 constexpr std::int32_t LCQI_PROBE_HEAD_DIM = 128;
17 constexpr double LCQI_PROBE_KV_ELEMENT_BYTES = 2.0;
18 constexpr double LCQI_PROBE_KV_BUDGET_MIB = 512.0;
19 constexpr double LCQI_PROBE_WORKSPACE_MIB = 64.0;
20 constexpr double LCQI_PROBE_ARENA_MIB = 32.0;
21 constexpr double LCQI_PROBE_TRACE_MIB = 4.0;
22 constexpr double LCQI_PROBE_QUEUE_WAIT_PER_REQUEST_MS = 8.0;
23 constexpr double LCQI_PROBE_PREFILL_MS_PER_TOKEN = 0.08;
24 constexpr double LCQI_PROBE_DECODE_STEP_MS = 14.0;
25 constexpr double LCQI_PROBE_CANCEL_RECLAIM_MS = 2.0;
26 constexpr std::size_t LCQI_PERCENTILE_MIN_INDEX = 1;
27
28 struct ProbeRequest {
29     std::string_view request_id;
30     std::int32_t prompt_tokens = 0;
31     std::int32_t max_new_tokens = 0;
32     bool cancel_after_prefill = false;
33 };
34
35 struct ProbeResult {
36     std::string_view request_id;
37     std::int32_t accepted = 0;
38     std::string_view rejection_reason;
39     std::int32_t prompt_tokens = 0;
40     std::int32_t reserved_tokens = 0;
41     double kv_reserved_mib = 0.0;
42     double queue_wait_ms = 0.0;
43     double prefill_ms = 0.0;

```

```

44     double ttft_ms = 0.0;
45     double decode_step_p95_ms = 0.0;
46     double tpot_ms = 0.0;
47     std::int32_t completion_tokens = 0;
48     std::string_view finish_reason;
49     double cancel_reclaim_ms = 0.0;
50 };
51
52 const std::vector<ProbeRequest> LCQI_PROBE_REQUESTS{
53     {"req_short", 32, 4, false},
54     {"req_rag", 512, 8, false},
55     {"req_cancel", 128, 16, true},
56     {"req_too_long", 4096, 512, false},
57 };
58
59 double kv_mib_for_tokens(std::int32_t tokens) noexcept {
60     const double bytes = static_cast<double>(tokens) * LCQI_PROBE_LAYER_COUNT *
        ↪ 2.0 *
61         LCQI_PROBE_KV_HEADS * LCQI_PROBE_HEAD_DIM *
62         LCQI_PROBE_KV_ELEMENT_BYTES;
63     return bytes / LCQI_BYTES_PER_MIB;
64 }
65
66 double percentile(std::vector<double> values, double fraction) {
67     if (values.empty()) {
68         return 0.0;
69     }
70     std::sort(values.begin(), values.end());
71     const std::size_t max_index = values.size() - LCQI_PERCENTILE_MIN_INDEX;
72     const std::size_t index =
73         std::min(max_index,
74             static_cast<std::size_t>(
75                 std::ceil(fraction * static_cast<double>(max_index))));
76     return values[index];
77 }
78
79 std::vector<ProbeResult> run_probe() {
80     std::vector<ProbeResult> results;
81     double reserved_kv_mib = 0.0;
82     std::int32_t accepted_before = 0;
83     for (const ProbeRequest& request : LCQI_PROBE_REQUESTS) {
84         ProbeResult result;
85         result.request_id = request.request_id;
86         result.prompt_tokens = request.prompt_tokens;
87         result.reserved_tokens = request.prompt_tokens + request.max_new_tokens
        ↪ ;
88         result.kv_reserved_mib = kv_mib_for_tokens(result.reserved_tokens);
89         const double static_session_mib =
90             result.kv_reserved_mib + LCQI_PROBE_WORKSPACE_MIB +
91             LCQI_PROBE_ARENA_MIB + LCQI_PROBE_TRACE_MIB;
92         if (reserved_kv_mib + result.kv_reserved_mib > LCQI_PROBE_KV_BUDGET_MIB
        ↪ ) {

```



```

93         result.accepted = 0;
94         result.rejection_reason = "memory_budget_exceeded";
95         result.finish_reason = "rejected";
96         results.push_back(result);
97         continue;
98     }
99
100     reserved_kv_mib += result.kv_reserved_mib;
101     result.accepted = 1;
102     result.rejection_reason = "none";
103     result.queue_wait_ms = static_cast<double>(accepted_before) *
104                             LCQI_PROBE_QUEUE_WAIT_PER_REQUEST_MS;
105     result.prefill_ms = static_cast<double>(request.prompt_tokens) *
106                             LCQI_PROBE_PREFILL_MS_PER_TOKEN;
107     result.ttft_ms = result.queue_wait_ms + result.prefill_ms + ↵
↵ LCQI_PROBE_DECODE_STEP_MS;
108     result.decode_step_p95_ms = LCQI_PROBE_DECODE_STEP_MS +
109                                 static_session_mib / 512.0;
110     result.tpot_ms = result.decode_step_p95_ms;
111     if (request.cancel_after_prefill) {
112         result.completion_tokens = 0;
113         result.finish_reason = "cancelled";
114         result.cancel_reclaim_ms = LCQI_PROBE_CANCEL_RECLAIM_MS;
115         reserved_kv_mib -= result.kv_reserved_mib;
116     } else {
117         result.completion_tokens = request.max_new_tokens;
118         result.finish_reason = "max_tokens";
119     }
120     ++accepted_before;
121     results.push_back(result);
122 }
123 return results;
124 }
125
126 void print_request_rows(const std::vector<ProbeResult>& results) {
127     std::cout << "row_type,request_id,accepted,rejection_reason,prompt_tokens,↵
↵ reserved_tokens,"
128                 "kv_reserved_mib,queue_wait_ms,prefill_ms,ttft_ms,↵
↵ decode_step_p95_ms,"
129                 "tpot_ms,completion_tokens,finish_reason,cancel_reclaim_ms,↵
↵ metric_name,"
130                 "metric_value\n";
131     for (const ProbeResult& result : results) {
132         std::cout << "request," << result.request_id << ',' <<
133                     << result.accepted << ',' <<
134                     << result.rejection_reason << ',' <<
135                     << result.prompt_tokens << ',' <<
136                     << result.reserved_tokens << ',' <<
137                     << result.kv_reserved_mib << ',' <<
138                     << result.queue_wait_ms << ',' <<
139                     << result.prefill_ms << ',' <<
140                     << result.ttft_ms << ',' <<

```

```

141         << result.decode_step_p95_ms << ', '
142         << result.tpot_ms << ', '
143         << result.completion_tokens << ', '
144         << result.finish_reason << ', '
145         << result.cancel_reclaim_ms << ",,\n";
146     }
147 }
148
149 void print_summary_row(const std::vector<ProbeResult>& results) {
150     std::vector<double> ttft_values;
151     std::vector<double> queue_values;
152     double accepted = 0.0;
153     double rejected = 0.0;
154     double cancelled = 0.0;
155     double reserved_peak_mib = 0.0;
156     double running_reserved_mib = 0.0;
157     for (const ProbeResult& result : results) {
158         if (result.accepted == 0) {
159             rejected += 1.0;
160             continue;
161         }
162         accepted += 1.0;
163         ttft_values.push_back(result.ttft_ms);
164         queue_values.push_back(result.queue_wait_ms);
165         running_reserved_mib += result.kv_reserved_mib;
166         reserved_peak_mib = std::max(reserved_peak_mib, running_reserved_mib);
167         if (result.finish_reason == "cancelled") {
168             cancelled += 1.0;
169             running_reserved_mib -= result.kv_reserved_mib;
170         }
171     }
172     const double total = accepted + rejected;
173     const double rejection_rate = total == 0.0 ? 0.0 : rejected / total;
174     std::cout << "summary,all,0,summary,0,0,0,0,0,0,0,0,summary,0,"
175               << "accepted_count," << accepted << '\n';
176     std::cout << "summary,all,0,summary,0,0,0,0,0,0,0,0,summary,"
177               << LCQI_PROBE_CANCEL_RECLAIM_MS << "cancel_count," << cancelled <<
178               << '\n';
179     std::cout << "summary,all,0,summary,0,0,0,0,0,0,0,0,summary,0,"
180               << "rejected_count," << rejected << '\n';
181     std::cout << "summary,all,0,summary,0,0,0,0,0,0,0,0,summary,0,"
182               << "rejection_rate," << rejection_rate << '\n';
183     std::cout << "summary,all,0,summary,0,0,0,0,0,0,0,0,summary,0,"
184               << "kv_reserved_peak_mib," << reserved_peak_mib << '\n';
185     std::cout << "summary,all,0,summary,0,0,0,0,0,0,0,0,summary,0,"
186               << "queue_wait_p95_ms," << percentile(queue_values, 0.95) << '\n' <<
187               << ;
188     std::cout << "summary,all,0,summary,0,0,0,0,0,0,0,0,summary,0,"
189               << "ttft_p95_ms," << percentile(ttft_values, 0.95) << '\n';
190 } // namespace

```

```

191
192 int main() {
193     try {
194         std::cout << std::fixed << std::setprecision(3);
195         const std::vector<ProbeResult> results = run_probe();
196         print_request_rows(results);
197         print_summary_row(results);
198         return EXIT_SUCCESS;
199     } catch (const std::exception& error) {
200         std::cerr << "error: " << error.what() << '\n';
201         return EXIT_FAILURE;
202     }
203 }

```

`onednn_bench.cpp` 是可选外部库对标入口。它只在环境安装 oneDNN 时构建，不能把不可用平台写成已验证性能。

**Listing 16.6** `src/onednn_bench.cpp`

```

1 #include <dnnl.hpp>
2
3 #include <chrono>
4 #include <cstdint>
5 #include <cstdlib>
6 #include <iostream>
7 #include <numeric>
8 #include <stdexcept>
9 #include <string>
10 #include <unordered_map>
11 #include <vector>
12
13 namespace {
14
15 constexpr std::int64_t LCQI_ONEDNN_BATCH = 1;
16 constexpr std::int64_t LCQI_ONEDNN_K = 4096;
17 constexpr std::int64_t LCQI_ONEDNN_O = 4096;
18 constexpr std::int32_t LCQI_ONEDNN_DEFAULT_REPEAT = 20;
19 constexpr std::int32_t LCQI_ONEDNN_WARMUP = 3;
20 constexpr int LCQI_REPEAT_ARG_INDEX = 1;
21 constexpr double LCQI_SECONDS_TO_MICROSECONDS = 1000000.0;
22 constexpr float LCQI_INPUT_SCALE = 0.125F;
23 constexpr std::int32_t LCQI_INPUT_MODULUS = 17;
24 constexpr std::int32_t LCQI_INPUT_CENTER = 8;
25 constexpr std::int32_t LCQI_WEIGHT_MODULUS = 23;
26 constexpr std::int32_t LCQI_WEIGHT_CENTER = 11;
27 constexpr float LCQI_WEIGHT_SCALE = 0.03125F;
28
29 std::int32_t parse_repeat(int argc, char** argv) {
30     if (argc <= LCQI_REPEAT_ARG_INDEX) {
31         return LCQI_ONEDNN_DEFAULT_REPEAT;
32     }
33     const std::int32_t repeat = static_cast<std::int32_t>(std::stoi(argv[↵
↵ LCQI_REPEAT_ARG_INDEX]));

```

```

34     if (repeat <= 0) {
35         throw std::runtime_error("repeat must be positive");
36     }
37     return repeat;
38 }
39
40 std::vector<float> make_input() {
41     std::vector<float> input(static_cast<std::size_t>(LCQI_ONEDNN_K));
42     for (std::int64_t index = 0; index < LCQI_ONEDNN_K; ++index) {
43         input[static_cast<std::size_t>(index)] =
44             static_cast<float>((index % LCQI_INPUT_MODULUS) - LCQI_INPUT_CENTER) *
45             LCQI_INPUT_SCALE;
46     }
47     return input;
48 }
49
50 std::vector<float> make_weights_kn() {
51     std::vector<float> weights(static_cast<std::size_t>(LCQI_ONEDNN_K * LCQI_ONEDNN_O));
52     for (std::int64_t k = 0; k < LCQI_ONEDNN_K; ++k) {
53         for (std::int64_t out = 0; out < LCQI_ONEDNN_O; ++out) {
54             const std::int64_t row_major_out_k = out * LCQI_ONEDNN_K + k;
55             const float value =
56                 static_cast<float>((row_major_out_k % LCQI_WEIGHT_MODULUS) -
57                                     LCQI_WEIGHT_CENTER) *
58                 LCQI_WEIGHT_SCALE;
59             weights[static_cast<std::size_t>(k * LCQI_ONEDNN_O + out)] = value;
60         }
61     }
62     return weights;
63 }
64
65 float checksum(const std::vector<float>& values) {
66     return std::accumulate(values.begin(), values.end(), 0.0F);
67 }
68
69 } // namespace
70
71 int main(int argc, char** argv) {
72     try {
73         const std::int32_t repeat = parse_repeat(argc, argv);
74         std::vector<float> input = make_input();
75         std::vector<float> weights = make_weights_kn();
76         std::vector<float> output(static_cast<std::size_t>(LCQI_ONEDNN_BATCH * LCQI_ONEDNN_O),
77                                0.0F);
78
79         dnnl::engine engine(dnnl::engine::kind::cpu, 0);
80         dnnl::stream stream(engine);
81
82         const dnnl::memory::dims source_dims = {LCQI_ONEDNN_BATCH, LCQI_ONEDNN_K, LCQI_ONEDNN_O};

```

```

↪ LCQI_ONEDNN_K};
83     const dnnl::memory::dims weight_dims = {LCQI_ONEDNN_K, LCQI_ONEDNN_0};
84     const dnnl::memory::dims output_dims = {LCQI_ONEDNN_BATCH, ↪
↪ LCQI_ONEDNN_0};

85
86     const dnnl::memory::desc source_desc(
87         source_dims, dnnl::memory::data_type::f32, dnnl::memory::format_tag↪
↪ ::ab);
88     const dnnl::memory::desc weight_desc(
89         weight_dims, dnnl::memory::data_type::f32, dnnl::memory::format_tag↪
↪ ::ab);
90     const dnnl::memory::desc output_desc(
91         output_dims, dnnl::memory::data_type::f32, dnnl::memory::format_tag↪
↪ ::ab);
92
93     dnnl::memory source_memory(source_desc, engine, input.data());
94     dnnl::memory weight_memory(weight_desc, engine, weights.data());
95     dnnl::memory output_memory(output_desc, engine, output.data());
96
97     dnnl::matmul::primitive_desc matmul_desc(engine, source_desc, ↪
↪ weight_desc, output_desc);
98     dnnl::matmul matmul(matmul_desc);
99     const std::unordered_map<int, dnnl::memory> arguments = {
100         {DNNL_ARG_SRC, source_memory},
101         {DNNL_ARG_WEIGHTS, weight_memory},
102         {DNNL_ARG_DST, output_memory},
103     };
104
105     for (std::int32_t index = 0; index < LCQI_ONEDNN_WARMUP; ++index) {
106         matmul.execute(stream, arguments);
107         stream.wait();
108     }
109
110     const auto begin = std::chrono::steady_clock::now();
111     for (std::int32_t index = 0; index < repeat; ++index) {
112         matmul.execute(stream, arguments);
113         stream.wait();
114     }
115     const auto end = std::chrono::steady_clock::now();
116     const std::chrono::duration<double> elapsed = end - begin;
117     const double average_us =
118         elapsed.count() * LCQI_SECONDS_TO_MICROSECONDS / static_cast<double>↪
↪ >(repeat);
119
120     std::cout << "library,input_size,output_size,dtype,repeat,average_us,↪
↪ checksum\n";
121     std::cout << "onednn," << LCQI_ONEDNN_K << ',' << LCQI_ONEDNN_0
122         << ",f32," << repeat << ',' << average_us << ','
123         << checksum(output) << '\n';
124     return EXIT_SUCCESS;
125 } catch (const std::exception& error) {
126     std::cerr << "error: " << error.what() << '\n';

```

```

127         return EXIT_FAILURE;
128     }
129 }

```

`gguf_q4_bench.cpp` 是真实 GGUF 量化 tensor 的单算子 benchmark。它读取同一个 SmolLM2 Q4\_K\_M GGUF, 选择真实 Q4\_K 矩阵, 分别测 f32 解码基线、Q4\_Kxf32 和 Q4\_KxQ8\_K。它输出的 `speedup` 只适用于这个单 tensor matvec, 不能写成完整模型 tokens/s。

**Listing 16.7** `src/gguf_q4_bench.cpp`

```

1  #include <lcqi/gguf.hpp>
2  #include <lcqi/q4_k.hpp>
3
4  #include <algorithm>
5  #include <chrono>
6  #include <cmath>
7  #include <cstdlib>
8  #include <filesystem>
9  #include <iomanip>
10 #include <iostream>
11 #include <span>
12 #include <stdexcept>
13 #include <string>
14 #include <string_view>
15 #include <vector>
16
17 namespace {
18
19 constexpr int LCQI_ARG_MODEL = 1;
20 constexpr int LCQI_ARG_TENSOR = 2;
21 constexpr int LCQI_ARG_REPEAT = 3;
22 constexpr int LCQI_DEFAULT_REPEAT = 20;
23 constexpr std::int64_t LCQI_MAX_AUTO_ROWS = 4096;
24 constexpr double LCQI_SECONDS_TO_MICROSECONDS = 1000000.0;
25 constexpr float LCQI_INPUT_PATTERN_SCALE = 0.015625F;
26 constexpr int LCQI_INPUT_PATTERN_MODULUS = 31;
27 constexpr int LCQI_INPUT_PATTERN_CENTER = 15;
28
29 struct SelectedTensor {
30     const lcqi::GgufTensorInfo* tensor = nullptr;
31     std::int64_t rows = 0;
32     std::int64_t columns = 0;
33 };
34
35 std::int32_t parse_repeat(int argc, char** argv) {
36     if (argc <= LCQI_ARG_REPEAT) {
37         return LCQI_DEFAULT_REPEAT;
38     }
39     const std::int32_t repeat = static_cast<std::int32_t>(std::stoi(argv[↵
↵ LCQI_ARG_REPEAT]));
40     if (repeat <= 0) {
41         throw std::runtime_error("repeat must be positive");
42     }

```

```

43     return repeat;
44 }
45
46 bool is_q4_k_matrix(const lcqi::GgufTensorInfo& tensor) {
47     return tensor.type == lcqi::GgmlType::q4_k && tensor.shape.size() == 2 &&
48         tensor.shape[0] > 0 && tensor.shape[1] > 0 &&
49         tensor.shape[0] % lcqi::LCQI_QK_K_BLOCK_VALUES == 0;
50 }
51
52 bool preferred_tensor_name(std::string_view name) {
53     return name.find("ffn_up.weight") != std::string_view::npos ||
54         name.find("ffn_gate.weight") != std::string_view::npos ||
55         name.find("ffn_down.weight") != std::string_view::npos;
56 }
57
58 SelectedTensor select_tensor(const lcqi::GgufManifest& manifest, std::↵
    ↵ string_view requested) {
59     if (!requested.empty() && requested != "auto") {
60         const lcqi::GgufTensorInfo* tensor = manifest.find_tensor(requested);
61         if (tensor == nullptr) {
62             throw std::runtime_error("requested GGUF tensor not found");
63         }
64         if (!is_q4_k_matrix(*tensor)) {
65             throw std::runtime_error("requested GGUF tensor is not a rank-2 ↵
    ↵ Q4_K matrix");
66         }
67         return {tensor, tensor->shape[1], tensor->shape[0]};
68     }
69
70     const lcqi::GgufTensorInfo* best = nullptr;
71     for (const lcqi::GgufTensorInfo& tensor : manifest.tensors) {
72         if (!is_q4_k_matrix(tensor) || tensor.shape[1] > LCQI_MAX_AUTO_ROWS) {
73             continue;
74         }
75         if (best == nullptr) {
76             best = &tensor;
77             continue;
78         }
79         const bool tensor_preferred = preferred_tensor_name(tensor.name);
80         const bool best_preferred = preferred_tensor_name(best->name);
81         if ((tensor_preferred && !best_preferred) ||
82             (tensor_preferred == best_preferred &&
83              tensor.element_count() > best->element_count())) {
84             best = &tensor;
85         }
86     }
87     if (best == nullptr) {
88         throw std::runtime_error("no suitable rank-2 Q4_K tensor found in GGUF ↵
    ↵ model");
89     }
90     return {best, best->shape[1], best->shape[0]};
91 }

```



```

92
93 std::vector<float> make_input(std::int64_t columns) {
94     std::vector<float> input(static_cast<std::size_t>(columns), 0.0F);
95     for (std::int64_t index = 0; index < columns; ++index) {
96         input[static_cast<std::size_t>(index)] =
97             static_cast<float>((index % LCQI_INPUT_PATTERN_MODULUS) -
98                             LCQI_INPUT_PATTERN_CENTER) *
99             LCQI_INPUT_PATTERN_SCALE;
100     }
101     return input;
102 }
103
104 float max_abs_diff(std::span<const float> lhs, std::span<const float> rhs) {
105     if (lhs.size() != rhs.size()) {
106         throw std::runtime_error("max_abs_diff size mismatch");
107     }
108     float diff = 0.0F;
109     for (std::size_t index = 0; index < lhs.size(); ++index) {
110         diff = std::max(diff, std::fabs(lhs[index] - rhs[index]));
111     }
112     return diff;
113 }
114
115 float checksum(std::span<const float> values) {
116     float sum = 0.0F;
117     for (const float value : values) {
118         sum += value;
119     }
120     return sum;
121 }
122
123 void matvec_f32(std::span<const float> weights,
124                std::int64_t rows,
125                std::int64_t columns,
126                std::span<const float> input,
127                std::span<float> output) {
128     if (weights.size() != static_cast<std::size_t>(rows * columns) ||
129         input.size() != static_cast<std::size_t>(columns) ||
130         output.size() != static_cast<std::size_t>(rows)) {
131         throw std::runtime_error("F32 matvec shape mismatch");
132     }
133     for (std::int64_t row = 0; row < rows; ++row) {
134         const float* row_weights = weights.data() + row * columns;
135         float sum = 0.0F;
136         for (std::int64_t column = 0; column < columns; ++column) {
137             sum += row_weights[column] * input[static_cast<std::size_t>(column)];
138         }
139         output[static_cast<std::size_t>(row)] = sum;
140     }
141 }
142

```

```

143 template <typename Callable>
144 double measure_average_us(std::int32_t repeat, Callable&& callable) {
145     const auto begin = std::chrono::steady_clock::now();
146     for (std::int32_t index = 0; index < repeat; ++index) {
147         callable();
148     }
149     const auto end = std::chrono::steady_clock::now();
150     const std::chrono::duration<double> elapsed = end - begin;
151     return elapsed.count() * LCQI_SECONDS_TO_MICROSECONDS / static_cast<double>
    ↪ >(repeat);
152 }
153
154 void print_usage(const char* program) {
155     std::cerr << "usage: " << program << " <model.gguf> [tensor-name|auto] [
    ↪ repeat]\n";
156 }
157
158 } // namespace
159
160 int main(int argc, char** argv) {
161     try {
162         if (argc <= LCQI_ARG_MODEL) {
163             print_usage(argv[0]);
164             return EXIT_FAILURE;
165         }
166         const std::filesystem::path model_path = argv[LCQI_ARG_MODEL];
167         const std::string tensor_request =
168             argc > LCQI_ARG_TENSOR ? argv[LCQI_ARG_TENSOR] : "auto";
169         const std::int32_t repeat = parse_repeat(argc, argv);
170
171         const lcqi::GgufManifest manifest = lcqi::load_gguf_manifest(model_path
    ↪ );
172         const SelectedTensor selected = select_tensor(manifest, tensor_request)
    ↪ ;
173         const std::vector<std::uint8_t> tensor_bytes =
174             lcqi::read_gguf_tensor_bytes(model_path, *selected.tensor);
175         const std::vector<float> input = make_input(selected.columns);
176         const std::vector<lcqi::Q8KBlock> q8_input = lcqi::quantize_q8_k_input(
    ↪ input);
177         std::vector<float> q4_f32_output(static_cast<std::size_t>(selected.rows
    ↪ ), 0.0F);
178         std::vector<float> q4_q8_output(q4_f32_output.size(), 0.0F);
179         std::vector<float> f32_output(q4_f32_output.size(), 0.0F);
180
181         lcqi::matvec_q4_k_f32(tensor_bytes,
182                               selected.rows,
183                               selected.columns,
184                               input,
185                               q4_f32_output);
186         lcqi::matvec_q4_k_q8(tensor_bytes,
187                               selected.rows,
188                               selected.columns,

```

```

189             q8_input,
190             q4_q8_output);
191
192     const std::vector<float> f32_weights =
193         lcqi::dequantize_q4_k(tensor_bytes, selected.rows * selected.↵
↵ columns);
194     matvec_f32(f32_weights, selected.rows, selected.columns, input, ↵
↵ f32_output);
195
196     const double f32_us = measure_average_us(
197         repeat,
198         [&]() { matvec_f32(f32_weights, selected.rows, selected.columns, ↵
↵ input, f32_output); });
199     const double q4_f32_us = measure_average_us(
200         repeat,
201         [&]() {
202             lcqi::matvec_q4_k_f32(tensor_bytes,
203                                     selected.rows,
204                                     selected.columns,
205                                     input,
206                                     q4_f32_output);
207         });
208     const double q8_quantize_us = measure_average_us(
209         repeat,
210         [&]() {
211             const std::vector<lcqi::Q8KBlock> temporary =
212                 lcqi::quantize_q8_k_input(input);
213             static_cast<void>(temporary);
214         });
215     const double q4_q8_us = measure_average_us(
216         repeat,
217         [&]() {
218             lcqi::matvec_q4_k_q8(tensor_bytes,
219                                     selected.rows,
220                                     selected.columns,
221                                     q8_input,
222                                     q4_q8_output);
223         });
224
225     const double q4_q8_total_us = q8_quantize_us + q4_q8_us;
226     std::cout << std::fixed << std::setprecision(3);
227     std::cout << "lcqi_smol1m2_q4_k_benchmark\n";
228     std::cout << "model_path=" << model_path.string() << '\n';
229     std::cout << "gguf_version=" << manifest.version << '\n';
230     std::cout << "gguf_alignment=" << manifest.alignment << '\n';
231     std::cout << "gguf_tensors=" << manifest.tensors.size() << '\n';
232     std::cout << "tensor_name=" << selected.tensor->name << '\n';
233     std::cout << "tensor_type=" << lcqi::ggml_type_name(selected.tensor->↵
↵ type) << '\n';
234     std::cout << "tensor_shape_ne0_ne1=" << selected.columns << 'x' << ↵
↵ selected.rows << '\n';
235     std::cout << "tensor_bytes=" << selected.tensor->byte_size << '\n';

```

```

236     std::cout << "f32_dequantized_bytes=" << f32_weights.size() * sizeof(↵
↵ float) << '\n';
237     std::cout << "repeat=" << repeat << '\n';
238     std::cout << "f32_matvec_average_us=" << f32_us << '\n';
239     std::cout << "q4_f32_input_average_us=" << q4_f32_us << '\n';
240     std::cout << "q8_quantize_input_average_us=" << q8_quantize_us << '\n';
241     std::cout << "q4_q8_backend=" << lcqi::q4_k_q8_active_backend() << '\n'↵
↵ ;
242     std::cout << "q4_q8_matvec_average_us=" << q4_q8_us << '\n';
243     std::cout << "q4_q8_total_average_us=" << q4_q8_total_us << '\n';
244     std::cout << "speedup_q4_q8_total_vs_f32="
245         << (q4_q8_total_us > 0.0 ? f32_us / q4_q8_total_us : 0.0) << ↵
↵ '\n';
246     std::cout << "speedup_q4_f32_input_vs_f32="
247         << (q4_f32_us > 0.0 ? f32_us / q4_f32_us : 0.0) << '\n';
248     std::cout << "q4_f32_vs_f32_max_abs_diff="
249         << max_abs_diff(q4_f32_output, f32_output) << '\n';
250     std::cout << "q4_q8_vs_q4_f32_max_abs_diff="
251         << max_abs_diff(q4_q8_output, q4_f32_output) << '\n';
252     std::cout << "q4_q8_checksum=" << checksum(q4_q8_output) << '\n';
253     return EXIT_SUCCESS;
254 } catch (const std::exception& error) {
255     std::cerr << "error: " << error.what() << '\n';
256     return EXIT_FAILURE;
257 }
258 }

```

`ggml_tensors.cpp` 是真实 GGUF 混合量化进入 LCQI 的格式闸门。它按 GGML block 布局解码 `Q8_0`、`Q5_0`、`Q6_K` 和已有 `Q4_K`，也读取 `F32` 标量张量。测试用手造 block 固定每个格式的位布局，真实 SmolLM2 smoke 再验证同一个解码层能读完 272 个 tensor。

Listing 16.8 `src/ggml_tensors.cpp`

```

1  #include <lcqi/ggml_tensors.hpp>
2
3  #include <lcqi/q4_k.hpp>
4
5  #include <cmath>
6  #include <cstring>
7  #include <limits>
8  #include <stdexcept>
9  #include <string>
10
11 namespace lcqi {
12 namespace {
13
14 constexpr std::uint32_t LCQI_GGML_F32_EXPONENT_MASK = 0x7F800000U;
15 constexpr std::uint32_t LCQI_GGML_F16_SIGN_MASK = 0x8000U;
16 constexpr std::uint32_t LCQI_GGML_F16_EXPONENT_MASK = 0x7C00U;
17 constexpr std::uint32_t LCQI_GGML_F16_MANTISSA_MASK = 0x03FFU;
18 constexpr std::uint32_t LCQI_GGML_F16_EXPONENT_BIAS = 15U;
19 constexpr std::uint32_t LCQI_GGML_F32_EXPONENT_BIAS = 127U;
20 constexpr std::uint32_t LCQI_GGML_F32_MANTISSA_BITS = 23U;

```

```

21 constexpr std::uint32_t LCQI_GGML_F16_MANTISSA_BITS = 10U;
22 constexpr std::uint32_t LCQI_GGML_F16_TO_F32_SIGN_SHIFT = 16U;
23 constexpr std::uint32_t LCQI_GGML_BYTE_BITS = 8U;
24 constexpr std::uint8_t LCQI_GGML_LOW_NIBBLE_MASK = 0x0FU;
25 constexpr std::uint8_t LCQI_GGML_Q5_HIGH_BIT_MASK = 0x10U;
26 constexpr std::uint32_t LCQI_GGML_Q5_HIGH_BIT_SHIFT = 4U;
27 constexpr std::int32_t LCQI_GGML_Q5_SECOND_HALF_BIT_OFFSET = 16;
28 constexpr std::uint8_t LCQI_GGML_Q6_HIGH_TWO_BITS_MASK = 0x03U;
29 constexpr std::int32_t LCQI_GGML_Q5_ZERO_POINT = 16;
30 constexpr std::int32_t LCQI_GGML_Q6_ZERO_POINT = 32;
31 constexpr std::int32_t LCQI_GGML_Q6_HALF_BLOCK_VALUES = 128;
32 constexpr std::int32_t LCQI_GGML_Q6_QUARTER_BLOCK_VALUES = 64;
33 constexpr std::int32_t LCQI_GGML_Q6_SUBBLOCK_VALUES = 32;
34 constexpr std::int32_t LCQI_GGML_Q6_SCALE_STRIDE = 8;
35 constexpr std::int32_t LCQI_GGML_F32_BYTES = 4;
36
37 [[nodiscard]] std::size_t checked_size(std::int64_t value, const char* name) {
38     if (value < 0 ||
39         static_cast<std::uint64_t>(value) >
40         static_cast<std::uint64_t>(std::numeric_limits<std::size_t>::max())) ←
    ↪ ) {
41         throw std::runtime_error(std::string(name) + " is outside size_t range" ←
    ↪ );
42     }
43     return static_cast<std::size_t>(value);
44 }
45
46 void require_multiple(std::int64_t value, std::int64_t divisor, const char* ←
    ↪ name) {
47     if (value <= 0 || value % divisor != 0) {
48         throw std::runtime_error(std::string(name) + " must be a positive ←
    ↪ multiple of " +
49             std::to_string(divisor));
50     }
51 }
52
53 [[nodiscard]] std::uint16_t read_le_u16(const std::uint8_t* bytes) {
54     return static_cast<std::uint16_t>(
55         static_cast<std::uint16_t>(bytes[0]) |
56         (static_cast<std::uint16_t>(bytes[1]) << LCQI_GGML_BYTE_BITS));
57 }
58
59 [[nodiscard]] std::uint32_t read_le_u32(const std::uint8_t* bytes) {
60     std::uint32_t value = 0;
61     for (std::uint32_t index = 0; index < LCQI_GGML_F32_BYTES; ++index) {
62         value |= static_cast<std::uint32_t>(bytes[index]) <<
63             (LCQI_GGML_BYTE_BITS * index);
64     }
65     return value;
66 }
67
68 [[nodiscard]] float bits_to_float(std::uint32_t bits) {

```

```

69     float value = 0.0F;
70     std::memcpy(&value, &bits, sizeof(value));
71     return value;
72 }
73
74 void validate_quantized_bytes(std::span<const std::uint8_t> bytes,
75                             std::int64_t element_count,
76                             std::int64_t block_values,
77                             std::int64_t block_bytes,
78                             const char* name) {
79     require_multiple(element_count, block_values, name);
80     const std::int64_t expected_bytes = (element_count / block_values) * ↵
    ↵ block_bytes;
81     if (bytes.size() != checked_size(expected_bytes, name)) {
82         throw std::runtime_error(std::string(name) + " byte count does not ↵
    ↵ match element count");
83     }
84 }
85
86 void dequantize_f32_to(std::span<const std::uint8_t> bytes, std::span<float> ↵
    ↵ output) {
87     const std::size_t expected_bytes = output.size() * LCQI_GGML_F32_BYTES;
88     if (bytes.size() != expected_bytes) {
89         throw std::runtime_error("F32 byte count does not match element count")↵
    ↵ ;
90     }
91     for (std::size_t index = 0; index < output.size(); ++index) {
92         output[index] = bits_to_float(
93             read_le_u32(bytes.data() + index * LCQI_GGML_F32_BYTES));
94     }
95 }
96
97 void dequantize_q8_0_to(std::span<const std::uint8_t> bytes, std::span<float> ↵
    ↵ output) {
98     validate_quantized_bytes(bytes,
99                             static_cast<std::int64_t>(output.size()),
100                             LCQI_Q8_0_BLOCK_VALUES,
101                             LCQI_Q8_0_BLOCK_BYTES,
102                             "Q8_0");
103     const std::int64_t block_count =
104         static_cast<std::int64_t>(output.size()) / LCQI_Q8_0_BLOCK_VALUES;
105     for (std::int64_t block_index = 0; block_index < block_count; ++block_index↵
    ↵ ) {
106         const std::uint8_t* block =
107             bytes.data() + checked_size(block_index * LCQI_Q8_0_BLOCK_BYTES, "↵
    ↵ Q8_0 offset");
108         const float d = ggml_f16_to_f32(read_le_u16(block));
109         const std::uint8_t* qs = block + sizeof(std::uint16_t);
110         float* target =
111             output.data() + checked_size(block_index * LCQI_Q8_0_BLOCK_VALUES, ↵
    ↵ "Q8_0 target");
112         for (std::int32_t index = 0; index < LCQI_Q8_0_BLOCK_VALUES; ++index) {

```

```

113         target[index] = static_cast<float>(static_cast<std::int8_t>(qs[↵
↵ index])) * d;
114     }
115 }
116 }
117
118 void dequantize_q5_0_to(std::span<const std::uint8_t> bytes, std::span<float> ↵
↵ output) {
119     validate_quantized_bytes(bytes,
120                             static_cast<std::int64_t>(output.size()),
121                             LCQI_Q5_0_BLOCK_VALUES,
122                             LCQI_Q5_0_BLOCK_BYTES,
123                             "Q5_0");
124     const std::int64_t block_count =
125         static_cast<std::int64_t>(output.size()) / LCQI_Q5_0_BLOCK_VALUES;
126     for (std::int64_t block_index = 0; block_index < block_count; ++block_index↵
↵ ) {
127         const std::uint8_t* block =
128             bytes.data() + checked_size(block_index * LCQI_Q5_0_BLOCK_BYTES, "↵
↵ Q5_0 offset");
129         const float d = ggml_f16_to_f32(read_le_u16(block));
130         const std::uint32_t qh = read_le_u32(block + sizeof(std::uint16_t));
131         const std::uint8_t* qs = block + sizeof(std::uint16_t) + sizeof(std::↵
↵ uint32_t);
132         float* target =
133             output.data() + checked_size(block_index * LCQI_Q5_0_BLOCK_VALUES, ↵
↵ "Q5_0 target");
134         for (std::int32_t index = 0; index < LCQI_Q5_0_BLOCK_VALUES / 2; ++↵
↵ index) {
135             const std::uint8_t high_0 =
136                 static_cast<std::uint8_t>(((qh >> index) & 1U)
137                                             << LCQI_GGML_Q5_HIGH_BIT_SHIFT);
138             const std::uint8_t high_1 =
139                 static_cast<std::uint8_t>(((qh >> (index + ↵
↵ LCQI_GGML_Q5_SECOND_HALF_BIT_OFFSET)) &
140                                             1U)
141                                             << LCQI_GGML_Q5_HIGH_BIT_SHIFT);
142             const std::int32_t low =
143                 static_cast<std::int32_t>((qs[index] & ↵
↵ LCQI_GGML_LOW_NIBBLE_MASK) | high_0) -
144                 LCQI_GGML_Q5_ZERO_POINT;
145             const std::int32_t high =
146                 static_cast<std::int32_t>((qs[index] >> 4U) | high_1) -
147                 LCQI_GGML_Q5_ZERO_POINT;
148             target[index] = static_cast<float>(low) * d;
149             target[index + LCQI_Q5_0_BLOCK_VALUES / 2] = static_cast<float>(↵
↵ high) * d;
150         }
151     }
152 }
153
154 void dequantize_q6_k_to(std::span<const std::uint8_t> bytes, std::span<float> ↵

```



[illegible]

```

↪ )
195                                     << 4U)) -
196         LCQI_GGML_Q6_ZERO_POINT;
197         const std::int32_t q4 =
198             static_cast<std::int32_t>((ql[lane + ↪
↪ LCQI_GGML_Q6_SUBBLOCK_VALUES] >> 4U) |
199                                     (((qh[lane] >> 6U) &
200                                     LCQI_GGML_Q6_HIGH_TWO_BITS_MASK↪
↪ )
201                                     << 4U)) -
202         LCQI_GGML_Q6_ZERO_POINT;
203
204         y[lane] = d * static_cast<float>(scales[scale_index + 0]) *
205                     static_cast<float>(q1);
206         y[lane + LCQI_GGML_Q6_SUBBLOCK_VALUES] =
207             d * static_cast<float>(scales[scale_index + 2]) *
208                 static_cast<float>(q2);
209         y[lane + LCQI_GGML_Q6_QUARTER_BLOCK_VALUES] =
210             d * static_cast<float>(scales[scale_index + 4]) *
211                 static_cast<float>(q3);
212         y[lane + LCQI_GGML_Q6_QUARTER_BLOCK_VALUES +
213             LCQI_GGML_Q6_SUBBLOCK_VALUES] =
214             d * static_cast<float>(scales[scale_index + 6]) *
215                 static_cast<float>(q4);
216     }
217     y += LCQI_GGML_Q6_HALF_BLOCK_VALUES;
218     ql += LCQI_GGML_Q6_QUARTER_BLOCK_VALUES;
219     qh += LCQI_GGML_Q6_SUBBLOCK_VALUES;
220     scales += LCQI_GGML_Q6_SCALE_STRIDE;
221 }
222 }
223 }
224
225 } // namespace
226
227 float ggml_f16_to_f32(std::uint16_t bits) {
228     const std::uint32_t sign =
229         (static_cast<std::uint32_t>(bits) & LCQI_GGML_F16_SIGN_MASK)
230         << LCQI_GGML_F16_TO_F32_SIGN_SHIFT;
231     const std::uint32_t exponent =
232         static_cast<std::uint32_t>(bits) & LCQI_GGML_F16_EXPONENT_MASK;
233     const std::uint32_t mantissa =
234         static_cast<std::uint32_t>(bits) & LCQI_GGML_F16_MANTISSA_MASK;
235
236     if (exponent == 0U) {
237         if (mantissa == 0U) {
238             return bits_to_float(sign);
239         }
240         float value = static_cast<float>(mantissa) /
241                     static_cast<float>(1U << LCQI_GGML_F16_MANTISSA_BITS);
242         value = std::ldexp(value, 1 - static_cast<int>(↪
↪ LCQI_GGML_F16_EXPONENT_BIAS));

```

```

243     return sign == 0U ? value : -value;
244 }
245
246 if (exponent == LCQI_GGML_F16_EXPONENT_MASK) {
247     const std::uint32_t f32_mantissa =
248         mantissa << (LCQI_GGML_F32_MANTISSA_BITS -
↵ LCQI_GGML_F16_MANTISSA_BITS);
249     return bits_to_float(sign | LCQI_GGML_F32_EXPONENT_MASK | f32_mantissa)↵
↵ ;
250 }
251
252 const std::uint32_t f32_exponent =
253     ((exponent >> LCQI_GGML_F16_MANTISSA_BITS) -
↵ LCQI_GGML_F16_EXPONENT_BIAS +
254     LCQI_GGML_F32_EXPONENT_BIAS)
255     << LCQI_GGML_F32_MANTISSA_BITS;
256 const std::uint32_t f32_mantissa =
257     mantissa << (LCQI_GGML_F32_MANTISSA_BITS - LCQI_GGML_F16_MANTISSA_BITS)↵
↵ ;
258 return bits_to_float(sign | f32_exponent | f32_mantissa);
259 }
260
261 std::vector<float> dequantize_ggml_tensor(GgmlType type,
262                                         std::span<const std::uint8_t> bytes,
263                                         std::int64_t element_count) {
264     if (element_count < 0) {
265         throw std::runtime_error("GGML element_count cannot be negative");
266     }
267     std::vector<float> output(checked_size(element_count, "GGML element_count")↵
↵ , 0.0F);
268     dequantize_ggml_tensor_to(type, bytes, output);
269     return output;
270 }
271
272 void dequantize_ggml_tensor_to(GgmlType type,
273                                std::span<const std::uint8_t> bytes,
274                                std::span<float> output) {
275     switch (type) {
276     case GgmlType::f32:
277         dequantize_f32_to(bytes, output);
278         return;
279     case GgmlType::q8_0:
280         dequantize_q8_0_to(bytes, output);
281         return;
282     case GgmlType::q5_0:
283         dequantize_q5_0_to(bytes, output);
284         return;
285     case GgmlType::q6_k:
286         dequantize_q6_k_to(bytes, output);
287         return;
288     case GgmlType::q4_k:
289         dequantize_q4_k_to(bytes, output);

```

```

290         return;
291     default:
292         throw std::runtime_error(std::string("LCQI cannot dequantize GGML ↵
↵ type ") +
293                                   ggml_type_name(type));
294     }
295 }
296
297 } // namespace lcqi

```

`ggml_matvec.cpp` 是 `Q5_0/Q6_K/Q8_0` direct matvec 的实验实现。它有两条 oracle：一条是 `matvec_ggml_quantized_f32`，直接从 GGUF block bytes 和 f32 input 做点积，用来对照“先解成 f32 权重再算”；另一条是 `matvec_ggml_quantized_q8_0`，先把 input 临时量化成 `Q8_0InputBlock`，再做低比特权重点积。`Q5_0` 需要用 input block 的 `sum` 减掉 zero-point 贡献，`Q6_K` 需要按 256 元素 block 的 scale lane 解包。测试会同时比较 exact path 和 `Q8_0` 近似 path，避免把 block 布局写错后直接进入真实模型。

Listing 16.9 `src/ggml_matvec.cpp`

```

1  #include <lcqi/ggml_matvec.hpp>
2
3  #include <lcqi/ggml_tensors.hpp>
4
5  #include <algorithm>
6  #include <cmath>
7  #include <limits>
8  #include <stdexcept>
9  #include <string>
10
11 namespace lcqi {
12 namespace {
13
14 constexpr std::uint32_t LCQI_GGML_MATVEC_BYTE_BITS = 8U;
15 constexpr std::uint8_t LCQI_GGML_MATVEC_LOW_NIBBLE_MASK = 0x0FU;
16 constexpr std::uint8_t LCQI_GGML_MATVEC_Q6_HIGH_TWO_BITS_MASK = 0x03U;
17 constexpr std::int32_t LCQI_GGML_MATVEC_Q5_ZERO_POINT = 16;
18 constexpr std::int32_t LCQI_GGML_MATVEC_Q6_ZERO_POINT = 32;
19 constexpr std::int32_t LCQI_GGML_MATVEC_Q6_HALF_BLOCK_VALUES = 128;
20 constexpr std::int32_t LCQI_GGML_MATVEC_Q6_QUARTER_BLOCK_VALUES = 64;
21 constexpr std::int32_t LCQI_GGML_MATVEC_Q6_SUBBLOCK_VALUES = 32;
22 constexpr std::int32_t LCQI_GGML_MATVEC_Q6_SCALE_STRIDE = 8;
23 constexpr std::int32_t LCQI_GGML_MATVEC_Q8_QUANT_MAX = 127;
24
25 [[nodiscard]] std::size_t checked_size(std::int64_t value, const char* name) {
26     if (value < 0 ||
27         static_cast<std::uint64_t>(value) >
28         static_cast<std::uint64_t>(std::numeric_limits<std::size_t>::max())) ↵
↵ ) {
29         throw std::runtime_error(std::string(name) + " is outside size_t range" ↵
↵ );
30     }
31     return static_cast<std::size_t>(value);

```

```

32 }
33
34 void require_positive(std::int64_t value, const char* name) {
35     if (value <= 0) {
36         throw std::runtime_error(std::string(name) + " must be positive");
37     }
38 }
39
40 void require_multiple(std::int64_t value, std::int64_t divisor, const char* ↵
    ↵ name) {
41     if (value <= 0 || value % divisor != 0) {
42         throw std::runtime_error(std::string(name) + " must be a positive ↵
    ↵ multiple of " +
43                                 std::to_string(divisor));
44     }
45 }
46
47 [[nodiscard]] std::uint16_t read_le_u16(const std::uint8_t* bytes) {
48     return static_cast<std::uint16_t>(
49         static_cast<std::uint16_t>(bytes[0]) |
50         (static_cast<std::uint16_t>(bytes[1]) << LCQI_GGML_MATVEC_BYTE_BITS));
51 }
52
53 [[nodiscard]] std::uint32_t read_le_u32(const std::uint8_t* bytes) {
54     std::uint32_t value = 0;
55     for (std::uint32_t index = 0; index < sizeof(std::uint32_t); ++index) {
56         value |= static_cast<std::uint32_t>(bytes[index]) <<
57                 (LCQI_GGML_MATVEC_BYTE_BITS * index);
58     }
59     return value;
60 }
61
62 [[nodiscard]] float dot_q8_0_block_f32(const std::uint8_t* block, const float* ↵
    ↵ input) {
63     const float d = ggml_f16_to_f32(read_le_u16(block));
64     const auto* qs = reinterpret_cast<const std::int8_t*>(block + sizeof(std::↵
    ↵ uint16_t));
65     float sum = 0.0F;
66     for (std::int32_t index = 0; index < LCQI_Q8_0_BLOCK_VALUES; ++index) {
67         sum += static_cast<float>(qs[index]) * input[index];
68     }
69     return d * sum;
70 }
71
72 [[nodiscard]] float dot_q8_0_block_q8_0(const std::uint8_t* block,
73                                           const Q8_0InputBlock& input) {
74     const float d = ggml_f16_to_f32(read_le_u16(block));
75     const auto* qs = reinterpret_cast<const std::int8_t*>(block + sizeof(std::↵
    ↵ uint16_t));
76     std::int32_t sum = 0;
77     for (std::int32_t index = 0; index < LCQI_Q8_0_BLOCK_VALUES; ++index) {
78         sum += static_cast<std::int32_t>(qs[index]) *

```

```

79         static_cast<std::int32_t>(input.qs[index]);
80     }
81     return d * input.d * static_cast<float>(sum);
82 }
83
84 [[nodiscard]] float dot_q5_0_block_f32(const std::uint8_t* block, const float* ↵
↵ input) {
85     const float d = ggml_f16_to_f32(read_le_u16(block));
86     const std::uint32_t qh = read_le_u32(block + sizeof(std::uint16_t));
87     const std::uint8_t* qs = block + sizeof(std::uint16_t) + sizeof(std::↵
↵ uint32_t);
88     float sum = 0.0F;
89     for (std::int32_t index = 0; index < LCQI_Q5_0_BLOCK_VALUES / 2; ++index) {
90         const std::uint8_t high_0 =
91             static_cast<std::uint8_t>(((qh >> index) & 1U) << 4U);
92         const std::uint8_t high_1 =
93             static_cast<std::uint8_t>(((qh >> (index + LCQI_Q5_0_BLOCK_VALUES / ↵
↵ 2)) & 1U)
94                                     << 4U);
95         const std::int32_t first =
96             static_cast<std::int32_t>((qs[index] & ↵
↵ LCQI_GGML_MATVEC_LOW_NIBBLE_MASK) |
97                                     high_0) -
98             LCQI_GGML_MATVEC_Q5_ZERO_POINT;
99         const std::int32_t second =
100             static_cast<std::int32_t>((qs[index] >> 4U) | high_1) -
101             LCQI_GGML_MATVEC_Q5_ZERO_POINT;
102         sum += static_cast<float>(first) * input[index];
103         sum += static_cast<float>(second) * input[index + ↵
↵ LCQI_Q5_0_BLOCK_VALUES / 2];
104     }
105     return d * sum;
106 }
107
108 [[nodiscard]] float dot_q5_0_block_q8_0(const std::uint8_t* block,
109                                         const Q8_0InputBlock& input) {
110     const float d = ggml_f16_to_f32(read_le_u16(block));
111     const std::uint32_t qh = read_le_u32(block + sizeof(std::uint16_t));
112     const std::uint8_t* qs = block + sizeof(std::uint16_t) + sizeof(std::↵
↵ uint32_t);
113     std::int32_t weighted_sum = 0;
114     for (std::int32_t index = 0; index < LCQI_Q5_0_BLOCK_VALUES / 2; ++index) {
115         const std::uint8_t high_0 =
116             static_cast<std::uint8_t>(((qh >> index) & 1U) << 4U);
117         const std::uint8_t high_1 =
118             static_cast<std::uint8_t>(((qh >> (index + LCQI_Q5_0_BLOCK_VALUES / ↵
↵ 2)) & 1U)
119                                     << 4U);
120         const std::int32_t first =
121             static_cast<std::int32_t>((qs[index] & ↵
↵ LCQI_GGML_MATVEC_LOW_NIBBLE_MASK) |
122                                     high_0);

```

```

123     const std::int32_t second =
124         static_cast<std::int32_t>((qs[index] >> 4U) | high_1);
125     weighted_sum += first * static_cast<std::int32_t>(input.qs[index]);
126     weighted_sum += second *
127         static_cast<std::int32_t>(input.qs[index + ↵
↵ LCQI_Q5_0_BLOCK_VALUES / 2]);
128 }
129 weighted_sum -= LCQI_GGML_MATVEC_Q5_ZERO_POINT * static_cast<std::int32_t>(↵
↵ input.sum);
130 return d * input.d * static_cast<float>(weighted_sum);
131 }
132
133 [[nodiscard]] float dot_q6_k_block_f32(const std::uint8_t* block, const float* ↵
↵ input) {
134     const std::uint8_t* ql = block;
135     const std::uint8_t* qh = ql + LCQI_Q6_K_BLOCK_VALUES / 2;
136     const auto* scales =
137         reinterpret_cast<const std::int8_t*>(qh + LCQI_Q6_K_BLOCK_VALUES / 4);
138     const float d = ggml_f16_to_f32(
139         read_le_u16(block + LCQI_Q6_K_BLOCK_BYTES - sizeof(std::uint16_t)));
140     float sum = 0.0F;
141
142     for (std::int32_t half = 0; half < LCQI_Q6_K_BLOCK_VALUES;
143         half += LCQI_GGML_MATVEC_Q6_HALF_BLOCK_VALUES) {
144         const float* x = input + half;
145         for (std::int32_t lane = 0; lane < LCQI_GGML_MATVEC_Q6_SUBBLOCK_VALUES; ↵
↵ ++lane) {
146             const std::int32_t scale_index =
147                 lane / (LCQI_GGML_MATVEC_Q6_SUBBLOCK_VALUES / 2);
148             const std::int32_t q1 =
149                 static_cast<std::int32_t>((ql[lane] & ↵
↵ LCQI_GGML_MATVEC_LOW_NIBBLE_MASK) |
150                 (((qh[lane] >> 0U) &
151                 ↵
↵ LCQI_GGML_MATVEC_Q6_HIGH_TWO_BITS_MASK)
152                 << 4U)) -
153                 LCQI_GGML_MATVEC_Q6_ZERO_POINT;
154             const std::int32_t q2 =
155                 static_cast<std::int32_t>((ql[lane + ↵
↵ LCQI_GGML_MATVEC_Q6_SUBBLOCK_VALUES] &
156                 LCQI_GGML_MATVEC_LOW_NIBBLE_MASK) |
157                 (((qh[lane] >> 2U) &
158                 ↵
↵ LCQI_GGML_MATVEC_Q6_HIGH_TWO_BITS_MASK)
159                 << 4U)) -
160                 LCQI_GGML_MATVEC_Q6_ZERO_POINT;
161             const std::int32_t q3 =
162                 static_cast<std::int32_t>((ql[lane] >> 4U) |
163                 (((qh[lane] >> 4U) &
164                 ↵
↵ LCQI_GGML_MATVEC_Q6_HIGH_TWO_BITS_MASK)
165                 << 4U)) -

```



```

166         LCQI_GGML_MATVEC_Q6_ZERO_POINT;
167         const std::int32_t q4 =
168             static_cast<std::int32_t>((ql[lane + ↵
↵ LCQI_GGML_MATVEC_Q6_SUBBLOCK_VALUES] >>
169                 4U) |
170                 (((qh[lane] >> 6U) &
171                 ↵
↵ LCQI_GGML_MATVEC_Q6_HIGH_TWO_BITS_MASK)
172                 << 4U)) -
173         LCQI_GGML_MATVEC_Q6_ZERO_POINT;
174
175         sum += static_cast<float>(scales[scale_index + 0]) *
176             static_cast<float>(q1) * x[lane];
177         sum += static_cast<float>(scales[scale_index + 2]) *
178             static_cast<float>(q2) * x[lane + ↵
↵ LCQI_GGML_MATVEC_Q6_SUBBLOCK_VALUES];
179         sum += static_cast<float>(scales[scale_index + 4]) *
180             static_cast<float>(q3) * x[lane + ↵
↵ LCQI_GGML_MATVEC_Q6_QUARTER_BLOCK_VALUES];
181         sum += static_cast<float>(scales[scale_index + 6]) *
182             static_cast<float>(q4) *
183             x[lane + LCQI_GGML_MATVEC_Q6_QUARTER_BLOCK_VALUES +
184                 LCQI_GGML_MATVEC_Q6_SUBBLOCK_VALUES];
185     }
186     ql += LCQI_GGML_MATVEC_Q6_QUARTER_BLOCK_VALUES;
187     qh += LCQI_GGML_MATVEC_Q6_SUBBLOCK_VALUES;
188     scales += LCQI_GGML_MATVEC_Q6_SCALE_STRIDE;
189 }
190 return d * sum;
191 }
192
193 [[nodiscard]] float dot_q6_k_block_q8_0(const std::uint8_t* block,
194                                         const Q8_0InputBlock* input) {
195     const std::uint8_t* ql = block;
196     const std::uint8_t* qh = ql + LCQI_Q6_K_BLOCK_VALUES / 2;
197     const auto* scales =
198         reinterpret_cast<const std::int8_t*>(qh + LCQI_Q6_K_BLOCK_VALUES / 4);
199     const float d = ggml_f16_to_f32(
200         read_le_u16(block + LCQI_Q6_K_BLOCK_BYTES - sizeof(std::uint16_t)));
201     float sum = 0.0F;
202
203     for (std::int32_t half = 0; half < LCQI_Q6_K_BLOCK_VALUES;
204         half += LCQI_GGML_MATVEC_Q6_HALF_BLOCK_VALUES) {
205         const std::int32_t input_half_block = half / ↵
↵ LCQI_Q8_0_INPUT_BLOCK_VALUES;
206         for (std::int32_t lane = 0; lane < LCQI_GGML_MATVEC_Q6_SUBBLOCK_VALUES; ↵
↵ ++lane) {
207             const std::int32_t scale_index =
208                 lane / (LCQI_GGML_MATVEC_Q6_SUBBLOCK_VALUES / 2);
209             const std::int32_t q1 =
210                 static_cast<std::int32_t>((ql[lane] & ↵
↵ LCQI_GGML_MATVEC_LOW_NIBBLE_MASK) |

```

```

211         (((qh[lane] >> 0U) &
212             ↵
213             ↵ LCQI_GGML_MATVEC_Q6_HIGH_TWO_BITS_MASK)
214                 << 4U)) -
215             LCQI_GGML_MATVEC_Q6_ZERO_POINT;
216     const std::int32_t q2 =
217         static_cast<std::int32_t>((ql[lane + ↵
218             ↵ LCQI_GGML_MATVEC_Q6_SUBBLOCK_VALUES] &
219                 LCQI_GGML_MATVEC_LOW_NIBBLE_MASK) |
220                 (((qh[lane] >> 2U) &
221                     ↵
222                     ↵ LCQI_GGML_MATVEC_Q6_HIGH_TWO_BITS_MASK)
223                         << 4U)) -
224                 LCQI_GGML_MATVEC_Q6_ZERO_POINT;
225     const std::int32_t q3 =
226         static_cast<std::int32_t>((ql[lane] >> 4U) |
227                 (((qh[lane] >> 4U) &
228                     ↵
229                     ↵ LCQI_GGML_MATVEC_Q6_HIGH_TWO_BITS_MASK)
230                         << 4U)) -
231                 LCQI_GGML_MATVEC_Q6_ZERO_POINT;
232     const std::int32_t q4 =
233         static_cast<std::int32_t>((ql[lane + ↵
234             ↵ LCQI_GGML_MATVEC_Q6_SUBBLOCK_VALUES] >>
235                 4U) |
236                 (((qh[lane] >> 6U) &
237                     ↵
238                     ↵ LCQI_GGML_MATVEC_Q6_HIGH_TWO_BITS_MASK)
239                         << 4U)) -
240                 LCQI_GGML_MATVEC_Q6_ZERO_POINT;
241
242     const Q8_0InputBlock& x1 = input[input_half_block + 0];
243     const Q8_0InputBlock& x2 = input[input_half_block + 1];
244     const Q8_0InputBlock& x3 = input[input_half_block + 2];
245     const Q8_0InputBlock& x4 = input[input_half_block + 3];
246     sum += x1.d * static_cast<float>(scales[scale_index + 0]) *
247             static_cast<float>(q1) * static_cast<float>(x1.qs[lane]);
248     sum += x2.d * static_cast<float>(scales[scale_index + 2]) *
249             static_cast<float>(q2) * static_cast<float>(x2.qs[lane]);
250     sum += x3.d * static_cast<float>(scales[scale_index + 4]) *
251             static_cast<float>(q3) * static_cast<float>(x3.qs[lane]);
252     sum += x4.d * static_cast<float>(scales[scale_index + 6]) *
253             static_cast<float>(q4) * static_cast<float>(x4.qs[lane]);
254 }
255 ql += LCQI_GGML_MATVEC_Q6_QUARTER_BLOCK_VALUES;
256 qh += LCQI_GGML_MATVEC_Q6_SUBBLOCK_VALUES;
257 scales += LCQI_GGML_MATVEC_Q6_SCALE_STRIDE;
258 }
259 return d * sum;
260 }
261
262 [[nodiscard]] float dot_quantized_block_f32(GgmlType type,

```

```

257         const std::uint8_t* block,
258         const float* input) {
259     switch (type) {
260     case GgmlType::q8_0:
261         return dot_q8_0_block_f32(block, input);
262     case GgmlType::q5_0:
263         return dot_q5_0_block_f32(block, input);
264     case GgmlType::q6_k:
265         return dot_q6_k_block_f32(block, input);
266     default:
267         throw std::runtime_error(std::string("LCQI has no direct F32 matvec ←
↪ for ") +
268                                 ggml_type_name(type));
269     }
270 }
271
272 [[nodiscard]] float dot_quantized_block_q8_0(GgmlType type,
273         const std::uint8_t* block,
274         const Q8_0InputBlock* input) {
275     switch (type) {
276     case GgmlType::q8_0:
277         return dot_q8_0_block_q8_0(block, input[0]);
278     case GgmlType::q5_0:
279         return dot_q5_0_block_q8_0(block, input[0]);
280     case GgmlType::q6_k:
281         return dot_q6_k_block_q8_0(block, input);
282     default:
283         throw std::runtime_error(std::string("LCQI has no direct Q8_0 ←
↪ matvec for ") +
284                                 ggml_type_name(type));
285     }
286 }
287
288 void validate_q8_0_blocks(std::span<const Q8_0InputBlock> input, std::int64_t ←
↪ column_count) {
289     require_multiple(column_count, LCQI_Q8_0_INPUT_BLOCK_VALUES, "Q8_0 input ←
↪ column count");
290     if (input.size() !=
291         checked_size(column_count / LCQI_Q8_0_INPUT_BLOCK_VALUES, "Q8_0 input ←
↪ block count")) {
292         throw std::runtime_error("Q8_0 input block count mismatch");
293     }
294 }
295
296 } // namespace
297
298 bool ggml_type_has_f32_direct_matvec(GgmlType type) noexcept {
299     return type == GgmlType::q8_0 || type == GgmlType::q5_0 || type == GgmlType ←
↪ ::q6_k;
300 }
301
302 bool ggml_type_has_q8_0_direct_matvec(GgmlType type) noexcept {

```

```

303     return type == GgmlType::q8_0 || type == GgmlType::q5_0 || type == GgmlType::q4_0 ||
        ↪ ::q6_k;
304 }
305
306 #if defined(LCQI_ENABLE_AVX2)
307 bool ggml_q8_0_avx2_available() noexcept;
308
309 void matvec_ggml_quantized_q8_0_avx2_unchecked(GgmlType type,
310                                                 std::span<const std::uint8_t> ↪
        ↪ rows,
311                                                 std::int64_t row_count,
312                                                 std::int64_t column_count,
313                                                 std::span<const Q8_0InputBlock> ↪
        ↪ input,
314                                                 std::span<float> output);
315 #else
316 bool ggml_q8_0_avx2_available() noexcept {
317     return false;
318 }
319
320 void matvec_ggml_quantized_q8_0_avx2_unchecked(GgmlType,
321                                                 std::span<const std::uint8_t>,
322                                                 std::int64_t,
323                                                 std::int64_t,
324                                                 std::span<const Q8_0InputBlock>,
325                                                 std::span<float>) {}
326 #endif
327
328 std::vector<Q8_0InputBlock> quantize_q8_0_input(std::span<const float> input) {
329     require_multiple(static_cast<std::int64_t>(input.size()),
330                     LCQI_Q8_0_INPUT_BLOCK_VALUES,
331                     "Q8_0 input size");
332     std::vector<Q8_0InputBlock> output(
333         checked_size(static_cast<std::int64_t>(input.size()) / ↪
        ↪ LCQI_Q8_0_INPUT_BLOCK_VALUES,
334                     "Q8_0 input block count"));
335     quantize_q8_0_input_to(input, output);
336     return output;
337 }
338
339 void quantize_q8_0_input_to(std::span<const float> input,
340                             std::span<Q8_0InputBlock> output) {
341     require_multiple(static_cast<std::int64_t>(input.size()),
342                     LCQI_Q8_0_INPUT_BLOCK_VALUES,
343                     "Q8_0 input size");
344     if (output.size() != checked_size(static_cast<std::int64_t>(input.size()) /
345                                     LCQI_Q8_0_INPUT_BLOCK_VALUES,
346                                     "Q8_0 input block count")) {
347         throw std::runtime_error("Q8_0 output block count mismatch");
348     }
349     for (std::size_t block_index = 0; block_index < output.size(); ++↪
        ↪ block_index) {

```



```

396     }
397     require_multiple(column_count, layout.block_size, "GGML matvec column count");
398     if (input.size() != checked_size(column_count, "GGML matvec column count")) {
399         throw std::runtime_error("GGML matvec input size mismatch");
400     }
401     if (output.size() != checked_size(row_count, "GGML matvec row count")) {
402         throw std::runtime_error("GGML matvec output size mismatch");
403     }
404     const std::int64_t row_bytes =
405         (column_count / layout.block_size) * static_cast<std::int64_t>(layout.type_size);
406     if (rows.size() != checked_size(row_bytes * row_count, "GGML matvec matrix bytes")) {
407         throw std::runtime_error("GGML matvec matrix byte size mismatch");
408     }
409
410     for (std::int64_t row = 0; row < row_count; ++row) {
411         const std::uint8_t* row_bytes_begin =
412             rows.data() + checked_size(row * row_bytes, "GGML matvec row offset");
413         float sum = 0.0F;
414         for (std::int64_t block = 0; block < column_count / layout.block_size; ++block) {
415             sum += dot_quantized_block_f32(
416                 type,
417                 row_bytes_begin +
418                     checked_size(block * static_cast<std::int64_t>(layout.type_size),
419                                 "GGML matvec block offset"),
420                 input.data() + checked_size(block * layout.block_size,
421                                             "GGML matvec input block offset"));
422         }
423         output[checked_size(row, "GGML matvec row")] = sum;
424     }
425 }
426
427 void matvec_ggml_quantized_q8_0(GgmlType type,
428                                 std::span<const std::uint8_t> rows,
429                                 std::int64_t row_count,
430                                 std::int64_t column_count,
431                                 std::span<const Q8_0InputBlock> input,
432                                 std::span<float> output) {
433     require_positive(row_count, "GGML Q8_0 matvec row count");
434     const GgmlTypeLayout layout = ggml_type_layout(type);
435     if (!ggml_type_has_q8_0_direct_matvec(type)) {
436         throw std::runtime_error(std::string("LCQI has no direct Q8_0 matvec ") +
437                                 ggml_type_name(type));
438     }
439     require_multiple(column_count, layout.block_size, "GGML Q8_0 matvec column

```

```

    ↪ count");
440 validate_q8_0_blocks(input, column_count);
441 if (output.size() != checked_size(row_count, "GGML Q8_0 matvec row count")) ↪
    ↪ {
442     throw std::runtime_error("GGML Q8_0 matvec output size mismatch");
443 }
444 const std::int64_t row_bytes =
445     (column_count / layout.block_size) * static_cast<std::int64_t>(layout.↪
    ↪ type_size);
446 if (rows.size() != checked_size(row_bytes * row_count, "GGML Q8_0 matvec ↪
    ↪ matrix bytes")) {
447     throw std::runtime_error("GGML Q8_0 matvec matrix byte size mismatch");
448 }
449 #if defined(LCQI_ENABLE_AVX2)
450 if ((type == GgmlType::q8_0 || type == GgmlType::q5_0) && ↪
    ↪ ggml_q8_0_avx2_available()) {
451     matvec_ggml_quantized_q8_0_avx2_unchecked(type,
452                                                 rows,
453                                                 row_count,
454                                                 column_count,
455                                                 input,
456                                                 output);
457     return;
458 }
459 #endif
460 const std::int64_t input_blocks_per_weight_block =
461     layout.block_size / LCQI_Q8_0_INPUT_BLOCK_VALUES;
462
463 for (std::int64_t row = 0; row < row_count; ++row) {
464     const std::uint8_t* row_bytes_begin =
465         rows.data() + checked_size(row * row_bytes, "GGML Q8_0 matvec row ↪
    ↪ offset");
466     float sum = 0.0F;
467     for (std::int64_t block = 0; block < column_count / layout.block_size; ↪
    ↪ ++block) {
468         sum += dot_quantized_block_q8_0(
469             type,
470             row_bytes_begin +
471                 checked_size(block * static_cast<std::int64_t>(layout.↪
    ↪ type_size),
472                             "GGML Q8_0 matvec block offset"),
473             input.data() +
474                 checked_size(block * input_blocks_per_weight_block,
475                             "GGML Q8_0 matvec input block offset"));
476     }
477     output[checked_size(row, "GGML Q8_0 matvec row")] = sum;
478 }
479 }
480
481 } // namespace lcqi

```

`ggml_matvec_avx2.cpp` 是一次负结果驱动的优化尝试。它给 `Q8_0xQ8_0` 与 `Q5_0xQ8_0`



的 32 元素块点积加了 AVX2 `maddubs` 归约；但 `caw` 端到端报告显示，实验路径仍然比默认 `f32 fallback` 慢，因为主要成本已经转移到逐 block 解包、逐 row 调度和 `Q6_K` 标量路径。源码卷保留它，是为了说明内核优化必须接受端到端证据约束：覆盖率提高不等于性能提高。

**Listing 16.10** src/ggml\_matvec\_avx2.cpp

[illegible]

```

48     const __m256i lhs_values = _mm256_loadu_si256(reinterpret_cast<const
↳ __m256i*>(lhs));
49     const __m256i rhs_values = _mm256_loadu_si256(reinterpret_cast<const
↳ __m256i*>(rhs));
50     const __m256i absolute_lhs = _mm256_sign_epi8(lhs_values, lhs_values);
51     const __m256i signed_rhs = _mm256_sign_epi8(rhs_values, lhs_values);
52     const __m256i products_i16 = _mm256_maddubs_epi16(absolute_lhs, signed_rhs)↳
↳ ;
53     const __m256i ones = _mm256_set1_epi16(1);
54     const __m256i partial_i32 = _mm256_madd_epi16(products_i16, ones);
55     return horizontal_sum_i32(partial_i32);
56 }
57
58 [[nodiscard]] std::int32_t dot_32_u8_i8_avx2(const std::uint8_t* lhs,
59                                               const std::int8_t* rhs) noexcept {
60     const __m256i lhs_values = _mm256_loadu_si256(reinterpret_cast<const
↳ __m256i*>(lhs));
61     const __m256i rhs_values = _mm256_loadu_si256(reinterpret_cast<const
↳ __m256i*>(rhs));
62     const __m256i products_i16 = _mm256_maddubs_epi16(lhs_values, rhs_values);
63     const __m256i ones = _mm256_set1_epi16(1);
64     const __m256i partial_i32 = _mm256_madd_epi16(products_i16, ones);
65     return horizontal_sum_i32(partial_i32);
66 }
67
68 [[nodiscard]] float dot_q8_0_block_q8_0_avx2(const std::uint8_t* block,
69                                               const Q8_0InputBlock& input) ↳
↳ noexcept {
70     const float d = ggml_f16_to_f32(read_le_u16(block));
71     const auto* qs = reinterpret_cast<const std::int8_t*>(block + sizeof(std::↳
↳ uint16_t));
72     return d * input.d * static_cast<float>(dot_32_i8_i8_avx2(qs, input.qs.data↳
↳ ()));
73 }
74
75 [[nodiscard]] float dot_q5_0_block_q8_0_avx2(const std::uint8_t* block,
76                                               const Q8_0InputBlock& input) ↳
↳ noexcept {
77     const float d = ggml_f16_to_f32(read_le_u16(block));
78     const std::uint32_t qh = read_le_u32(block + sizeof(std::uint16_t));
79     const std::uint8_t* qs = block + sizeof(std::uint16_t) + sizeof(std::↳
↳ uint32_t);
80
81     alignas(32) std::array<std::uint8_t, LCQI_Q8_0_INPUT_BLOCK_VALUES> values↳
↳ {};
82     for (std::int32_t index = 0; index < LCQI_Q8_0_INPUT_BLOCK_VALUES / 2; ++↳
↳ index) {
83         const std::uint8_t high_0 =
84             static_cast<std::uint8_t>(((qh >> index) & 1U) << 4U);
85         const std::uint8_t high_1 = static_cast<std::uint8_t>(
86             ((qh >> (index + LCQI_Q8_0_INPUT_BLOCK_VALUES / 2)) & 1U) << 4U);
87         values[static_cast<std::size_t>(index)] =

```

```

88         static_cast<std::uint8_t>((qs[index] & ←
    ↪ LCQI_GGML_AVX2_LOW_NIBBLE_MASK) | high_0);
89         values[static_cast<std::size_t>(index + LCQI_Q8_0_INPUT_BLOCK_VALUES / ←
    ↪ 2)] =
90         static_cast<std::uint8_t>((qs[index] >> 4U) | high_1);
91     }
92
93     const std::int32_t weighted_sum =
94         dot_32_u8_i8_avx2(values.data(), input.qs.data()) -
95         LCQI_GGML_AVX2_Q5_ZERO_POINT * static_cast<std::int32_t>(input.sum);
96     return d * input.d * static_cast<float>(weighted_sum);
97 }
98
99 void matvec_q8_0_q8_0_avx2_unchecked(std::span<const std::uint8_t> rows,
100                                     std::int64_t row_count,
101                                     std::int64_t column_count,
102                                     std::span<const Q8_0InputBlock> input,
103                                     std::span<float> output) {
104     const std::int64_t row_blocks = column_count / LCQI_Q8_0_INPUT_BLOCK_VALUES; ←
    ↪ ;
105     const std::int64_t row_bytes = row_blocks * LCQI_GGML_AVX2_Q8_0_BLOCK_BYTES; ←
    ↪ ;
106     for (std::int64_t row = 0; row < row_count; ++row) {
107         float sum = 0.0F;
108         const std::uint8_t* row_data = rows.data() + row * row_bytes;
109         for (std::int64_t block = 0; block < row_blocks; ++block) {
110             sum += dot_q8_0_block_q8_0_avx2(
111                 row_data + block * LCQI_GGML_AVX2_Q8_0_BLOCK_BYTES,
112                 input[static_cast<std::size_t>(block)]);
113         }
114         output[static_cast<std::size_t>(row)] = sum;
115     }
116 }
117
118 void matvec_q5_0_q8_0_avx2_unchecked(std::span<const std::uint8_t> rows,
119                                     std::int64_t row_count,
120                                     std::int64_t column_count,
121                                     std::span<const Q8_0InputBlock> input,
122                                     std::span<float> output) {
123     const std::int64_t row_blocks = column_count / LCQI_Q8_0_INPUT_BLOCK_VALUES; ←
    ↪ ;
124     const std::int64_t row_bytes = row_blocks * LCQI_GGML_AVX2_Q5_0_BLOCK_BYTES; ←
    ↪ ;
125     for (std::int64_t row = 0; row < row_count; ++row) {
126         float sum = 0.0F;
127         const std::uint8_t* row_data = rows.data() + row * row_bytes;
128         for (std::int64_t block = 0; block < row_blocks; ++block) {
129             sum += dot_q5_0_block_q8_0_avx2(
130                 row_data + block * LCQI_GGML_AVX2_Q5_0_BLOCK_BYTES,
131                 input[static_cast<std::size_t>(block)]);
132         }
133         output[static_cast<std::size_t>(row)] = sum;

```

```

134     }
135 }
136
137 } // namespace
138
139 bool ggml_q8_0_avx2_available() noexcept {
140     #if defined(__GNUC__) || defined(__clang__)
141         __builtin_cpu_init();
142         return __builtin_cpu_supports("avx2");
143     #else
144         return true;
145     #endif
146 }
147
148 void matvec_ggml_quantized_q8_0_avx2_unchecked(GgmlType type,
149                                                 std::span<const std::uint8_t> ↵
150                                                 ↵ rows,
151                                                 std::int64_t row_count,
152                                                 std::int64_t column_count,
153                                                 std::span<const Q8_0InputBlock> ↵
154                                                 ↵ input,
155                                                 std::span<float> output) {
156     if (type == GgmlType::q8_0) {
157         matvec_q8_0_q8_0_avx2_unchecked(rows, row_count, column_count, input, ↵
158         ↵ output);
159         return;
160     }
161     if (type == GgmlType::q5_0) {
162         matvec_q5_0_q8_0_avx2_unchecked(rows, row_count, column_count, input, ↵
163         ↵ output);
164     }
165 }
166
167 } // namespace lcqi
168
169 #endif

```

`llama_gguf.cpp` 是自研 SmolLM2/GGUF decode path。它读取 GGUF metadata 得到 hidden size、层数、GQA 头数、RoPE base、RMSNorm epsilon 和 tokenizer id；再读取 `token_embd.weight`、每层 attention/FFN 权重和 `output_norm.weight`。最新实现把权重表示分成三类：默认启用的 shape 兼容 `Q4_K` linear tensor 保留 GGUF 原始 block bytes，前向时通过 `Q8_K` scratch 调用 `matvec_q4_k_q8`；`LCQI_LLAMA_GGML_DIRECT=1` 打开的 `Q5_0/Q6_K/Q8_0` 实验 tensor 通过 `Q8_0` scratch 调用 GGML direct matvec；其余 tensor materialize 成 f32 fallback。读这份源码时还要看 `LlamaParallelWorkerPool`：默认 `worker_count=0` 会自动选 worker，大行数 f32 fallback、tied `lm_head` 和 `Q4_K` row-range 都按输出行切分；`--threads1` 保留串行 reference。三条主线分别是：loader 如何记录 `q4_k_direct_tensors`、`ggml_direct_tensors` 与 fallback 字节，`linear_gguf` 如何在三类路径之间分派并接入 worker pool，`LlamaGgufHotspotReport` 如何把 direct/fallback call、worker count 和 per-op hotspot 输出到报告。

Listing 16.11 src/llama\_gguf.cpp

```

1 #include <lcqi/llama_gguf.hpp>

```

```

2
3 #include <lcqi/f32_kernels.hpp>
4 #include <lcqi/ggml_matvec.hpp>
5 #include <lcqi/ggml_tensors.hpp>
6 #include <lcqi/gguf.hpp>
7 #include <lcqi/q4_k.hpp>
8
9 #include <algorithm>
10 #include <array>
11 #include <atomic>
12 #include <chrono>
13 #include <cmath>
14 #include <cstdlib>
15 #include <limits>
16 #include <memory>
17 #include <sstream>
18 #include <stdexcept>
19 #include <string>
20 #include <thread>
21 #include <unordered_map>
22 #include <utility>
23
24 namespace lcqi {
25 namespace {
26
27 constexpr std::int32_t LCQI_LLAMA_ROPE_PAIR_WIDTH = 2;
28 constexpr std::int32_t LCQI_LLAMA_DEFAULT_BOS_TOKEN = 1;
29 constexpr std::int32_t LCQI_LLAMA_DEFAULT_EOS_TOKEN = 2;
30 constexpr std::int32_t LCQI_LLAMA_DEFAULT_UNK_TOKEN = 0;
31 constexpr const char* LCQI_LLAMA_ARCHITECTURE_KEY = "general.architecture";
32 constexpr const char* LCQI_LLAMA_NAME_KEY = "general.name";
33 constexpr const char* LCQI_LLAMA_EMBEDDING_LENGTH_KEY = "llama.embedding_length"↵
↵ ";
34 constexpr const char* LCQI_LLAMA_BLOCK_COUNT_KEY = "llama.block_count";
35 constexpr const char* LCQI_LLAMA_FEED_FORWARD_LENGTH_KEY = "llama.↵
↵ feed_forward_length";
36 constexpr const char* LCQI_LLAMA_HEAD_COUNT_KEY = "llama.attention.head_count";
37 constexpr const char* LCQI_LLAMA_KV_HEAD_COUNT_KEY = "llama.attention.↵
↵ head_count_kv";
38 constexpr const char* LCQI_LLAMA_ROPE_DIMENSION_KEY = "llama.rope.↵
↵ dimension_count";
39 constexpr const char* LCQI_LLAMA_ROPE_BASE_KEY = "llama.rope.freq_base";
40 constexpr const char* LCQI_LLAMA_CONTEXT_LENGTH_KEY = "llama.context_length";
41 constexpr const char* LCQI_LLAMA_VOCAB_SIZE_KEY = "llama.vocab_size";
42 constexpr const char* LCQI_LLAMA_RMS_EPSILON_KEY = "llama.attention.↵
↵ layer_norm_rms_epsilon";
43 constexpr const char* LCQI_LLAMA_TOKENIZER_TOKENS_KEY = "tokenizer.ggml.tokens"↵
↵ ;
44 constexpr const char* LCQI_LLAMA_TOKENIZER_MERGES_KEY = "tokenizer.ggml.merges"↵
↵ ;
45 constexpr const char* LCQI_LLAMA_BOS_TOKEN_KEY = "tokenizer.ggml.bos_token_id";
46 constexpr const char* LCQI_LLAMA_EOS_TOKEN_KEY = "tokenizer.ggml.eos_token_id";

```

```

47 constexpr const char* LCQI_LLAMA_UNKNOWN_TOKEN_KEY = "tokenizer.ggml.↵
    ↵ unknown_token_id";
48 constexpr const char* LCQI_LLAMA_Q4_DIRECT_ENV = "LCQI_LLAMA_Q4_DIRECT";
49 constexpr const char* LCQI_LLAMA_GGML_DIRECT_ENV = "LCQI_LLAMA_GGML_DIRECT";
50 constexpr const char* LCQI_LLAMA_FALSE_ENV = "0";
51 constexpr const char* LCQI_LLAMA_TRUE_ENV = "1";
52 constexpr std::int32_t LCQI_LLAMA_MAX_AUTO_WORKERS = 8;
53 constexpr std::int32_t LCQI_LLAMA_DEFAULT_PARALLEL_MIN_ROWS = 512;
54 constexpr std::int32_t LCQI_LLAMA_PARALLEL_MIN_COLUMNS = 512;
55 constexpr std::int64_t LCQI_LLAMA_PARALLEL_MIN_OPS = 1'000'000;
56 constexpr std::size_t LCQI_LLAMA_PARALLEL_ROW_BLOCK_MULTIPLIER = 4;
57 constexpr std::size_t LCQI_LLAMA_WORKER_SPIN_PAUSES_BEFORE_YIELD = 4096;
58
59 using Clock = std::chrono::steady_clock;
60
61 enum class LlamaGgufLinearOp : std::uint8_t {
62     wq,
63     wk,
64     wv,
65     wo,
66     w_gate,
67     w_up,
68     w_down,
69     lm_head,
70 };
71
72 void pause_worker_spin(std::size_t& pause_count) noexcept {
73     #if defined(__GNUC__) && (defined(__x86_64__) || defined(__i386__))
74         __builtin_ia32_pause();
75     #endif
76     ++pause_count;
77     if (pause_count >= LCQI_LLAMA_WORKER_SPIN_PAUSES_BEFORE_YIELD) {
78         pause_count = 0;
79         std::this_thread::yield();
80     }
81 }
82
83 class LlamaParallelWorkerPool {
84 public:
85     explicit LlamaParallelWorkerPool(std::int32_t requested_workers)
86         : worker_count_(LlamaParallelWorkerPool::normalize_worker_count(↵
    ↵ requested_workers)) {
87         if (this->worker_count_ <= 1) {
88             return;
89         }
90         const std::size_t background_count =
91             static_cast<std::size_t>(this->worker_count_ - 1);
92         this->threads_.reserve(background_count);
93         for (std::size_t index = 0; index < background_count; ++index) {
94             this->threads_.emplace_back([this, index]() { this->worker_loop(↵
    ↵ index); });
95         }

```

```

96     }
97
98     ~LlamaParallelWorkerPool() {
99         this->stopping_.store(true, std::memory_order_release);
100        this->generation_.fetch_add(1, std::memory_order_release);
101        for (std::thread& thread : this->threads_) {
102            if (thread.joinable()) {
103                thread.join();
104            }
105        }
106    }
107
108    LlamaParallelWorkerPool(const LlamaParallelWorkerPool&) = delete;
109    LlamaParallelWorkerPool& operator=(const LlamaParallelWorkerPool&) = delete;↵
110    ↵ ;
111
112    [[nodiscard]] std::int32_t worker_count() const noexcept {
113        return this->worker_count_;
114    }
115
116    template <typename Function>
117    void parallel_for_rows(std::size_t begin,
118                          std::size_t end,
119                          std::size_t grain,
120                          Function&& function) {
121        if (end <= begin) {
122            return;
123        }
124        const std::size_t row_count = end - begin;
125        if (this->worker_count_ <= 1 || row_count <= grain) {
126            function(begin, end, 0);
127            return;
128        }
129
130        const std::size_t task_count =
131            std::min<std::size_t>(static_cast<std::size_t>(this->worker_count_)↵
132            ↵ ,
133                                (row_count + grain - 1) / grain);
134
135        if (task_count <= 1) {
136            function(begin, end, 0);
137            return;
138        }
139
140        auto task_context = ParallelTaskContext<Function>{function};
141        this->task_begin_.store(begin, std::memory_order_release);
142        this->task_end_.store(end, std::memory_order_release);
143        this->task_count_.store(task_count, std::memory_order_release);
144        this->task_context_.store(&task_context, std::memory_order_release);
145        this->task_invoke_.store(&ParallelTaskContext<Function>::invoke,
146                                std::memory_order_release);
147        this->active_workers_.store(task_count - 1, std::memory_order_release);
148        this->generation_.fetch_add(1, std::memory_order_release);

```



```

146
147     const std::size_t main_worker = task_count - 1;
148     const auto [main_begin, main_end] =
149         this->chunk_bounds(begin, end, task_count, main_worker);
150     function(main_begin, main_end, main_worker);
151
152     std::size_t pause_count = 0;
153     while (this->active_workers_.load(std::memory_order_acquire) != 0) {
154         pause_worker_spin(pause_count);
155     }
156     this->task_context_.store(nullptr, std::memory_order_release);
157     this->task_invoke_.store(nullptr, std::memory_order_release);
158 }
159
160 private:
161     template <typename Function>
162     struct ParallelTaskContext {
163         Function& function;
164
165         static void invoke(void* context,
166                             std::size_t begin,
167                             std::size_t end,
168                             std::size_t worker_index) {
169             auto* typed_context = static_cast<ParallelTaskContext*>(context);
170             typed_context->function(begin, end, worker_index);
171         }
172     };
173
174     std::int32_t worker_count_ = 1;
175     std::vector<std::thread> threads_;
176     std::atomic<bool> stopping_ = false;
177     std::atomic<std::uint64_t> generation_ = 0;
178     std::atomic<std::size_t> task_begin_ = 0;
179     std::atomic<std::size_t> task_end_ = 0;
180     std::atomic<std::size_t> task_count_ = 0;
181     std::atomic<std::size_t> active_workers_ = 0;
182     std::atomic<void*> task_context_ = nullptr;
183     std::atomic<void (*) (void*, std::size_t, std::size_t, std::size_t)> ↵
    ↵ task_invoke_ = nullptr;
184
185     [[nodiscard]] static std::int32_t normalize_worker_count(std::int32_t ↵
    ↵ requested_workers) {
186         if (requested_workers > 0) {
187             return requested_workers;
188         }
189         const unsigned int hardware_workers = std::thread::hardware_concurrency; ↵
    ↵ ();
190         if (hardware_workers == 0) {
191             return 1;
192         }
193         const std::int32_t capped =
194             std::min<std::int32_t>(static_cast<std::int32_t>(hardware_workers),

```

```

195         LCQI_LLAMA_MAX_AUTO_WORKERS);
196     return std::max(capped, 1);
197 }
198
199 [[nodiscard]] std::pair<std::size_t, std::size_t> chunk_bounds(
200     std::size_t begin,
201     std::size_t end,
202     std::size_t task_count,
203     std::size_t worker_index) const noexcept {
204     const std::size_t row_count = end - begin;
205     const std::size_t chunk_begin = begin + row_count * worker_index / ↵
↵ task_count;
206     const std::size_t chunk_end = begin + row_count * (worker_index + 1) / ↵
↵ task_count;
207     return {chunk_begin, chunk_end};
208 }
209
210 void worker_loop(std::size_t worker_index) {
211     std::uint64_t observed_generation = 0;
212     std::size_t pause_count = 0;
213     while (true) {
214         const std::uint64_t current_generation =
215             this->generation_.load(std::memory_order_acquire);
216         if (current_generation == observed_generation) {
217             if (this->stopping_.load(std::memory_order_acquire)) {
218                 return;
219             }
220             pause_worker_spin(pause_count);
221             continue;
222         }
223
224         observed_generation = current_generation;
225         pause_count = 0;
226         if (this->stopping_.load(std::memory_order_acquire)) {
227             return;
228         }
229
230         const std::size_t task_count = this->task_count_.load(std::↵
↵ memory_order_acquire);
231         if (this->generation_.load(std::memory_order_acquire) != ↵
↵ current_generation ||
232             task_count == 0) {
233             continue;
234         }
235         const std::size_t background_task_count = task_count - 1;
236         if (worker_index >= background_task_count) {
237             continue;
238         }
239         void* task_context = this->task_context_.load(std::↵
↵ memory_order_acquire);
240         void (*task_invoke)(void*, std::size_t, std::size_t, std::size_t) =
241             this->task_invoke_.load(std::memory_order_acquire);

```

```

242         if (task_context == nullptr || task_invoke == nullptr) {
243             continue;
244         }
245         const std::size_t task_begin = this->task_begin_.load(std::memory_order_acquire);
246         const std::size_t task_end = this->task_end_.load(std::memory_order_acquire);
247         if (this->generation_.load(std::memory_order_acquire) != current_generation) {
248             continue;
249         }
250         const auto [begin, end] =
251             this->chunk_bounds(task_begin, task_end, task_count, worker_index);
252         task_invoke(task_context, begin, end, worker_index);
253         this->active_workers_.fetch_sub(1, std::memory_order_acq_rel);
254     }
255 }
256 };
257
258 [[nodiscard]] double elapsed_ms(Clock::time_point begin, Clock::time_point end) {
259     return std::chrono::duration<double, std::milli>(end - begin).count();
260 }
261
262 [[nodiscard]] std::size_t checked_size(std::int64_t value, const char* name) {
263     if (value < 0 ||
264         static_cast<std::uint64_t>(value) >
265         static_cast<std::uint64_t>(std::numeric_limits<std::size_t>::max())) {
266         throw std::runtime_error(std::string(name) + " is outside size_t range");
267     }
268     return static_cast<std::size_t>(value);
269 }
270
271 void require_positive(std::int32_t value, const char* name) {
272     if (value <= 0) {
273         throw std::runtime_error(std::string(name) + " must be positive");
274     }
275 }
276
277 [[nodiscard]] const GgufMetadataEntry& require_metadata(const GgufManifest& manifest,
278                                                         std::string_view key) {
279     const GgufMetadataEntry* entry = manifest.find_metadata(key);
280     if (entry == nullptr) {
281         throw std::runtime_error("missing GGUF metadata key: " + std::string(key));
282     }
283     return *entry;
284 }

```

```

285
286 [[nodiscard]] std::string metadata_string(const GgufManifest& manifest, std::string_view key) {
287     return require_metadata(manifest, key).value_preview;
288 }
289
290 [[nodiscard]] std::int32_t metadata_i32(const GgufManifest& manifest, std::string_view key) {
291     const std::int64_t value = std::stoll(require_metadata(manifest, key).value_preview);
292     if (value < std::numeric_limits<std::int32_t>::min() ||
293         value > std::numeric_limits<std::int32_t>::max()) {
294         throw std::runtime_error("GGUF metadata int32 out of range: " + std::string(key));
295     }
296     return static_cast<std::int32_t>(value);
297 }
298
299 [[nodiscard]] std::int32_t optional_metadata_i32(const GgufManifest& manifest,
300                                                  std::string_view key,
301                                                  std::int32_t fallback) {
302     const GgufMetadataEntry* entry = manifest.find_metadata(key);
303     if (entry == nullptr) {
304         return fallback;
305     }
306     const std::int64_t value = std::stoll(entry->value_preview);
307     if (value < std::numeric_limits<std::int32_t>::min() ||
308         value > std::numeric_limits<std::int32_t>::max()) {
309         throw std::runtime_error("GGUF metadata int32 out of range: " + std::string(key));
310     }
311     return static_cast<std::int32_t>(value);
312 }
313
314 [[nodiscard]] float metadata_f32(const GgufManifest& manifest, std::string_view key) {
315     return std::stof(require_metadata(manifest, key).value_preview);
316 }
317
318 void validate_shape(const GgufTensorInfo& tensor,
319                    std::span<const std::int64_t> expected_shape) {
320     if (tensor.shape.size() != expected_shape.size()) {
321         throw std::runtime_error("GGUF tensor rank mismatch: " + tensor.name);
322     }
323     for (std::size_t index = 0; index < expected_shape.size(); ++index) {
324         if (tensor.shape[index] != expected_shape[index]) {
325             throw std::runtime_error("GGUF tensor shape mismatch: " + tensor.name);
326         }
327     }
328 }
329

```

```

330 [[nodiscard]] bool can_use_q4_k_direct(const GgufTensorInfo& tensor) {
331     return tensor.type == GgmlType::q4_k && tensor.shape.size() == 2 &&
332         tensor.shape[0] > 0 && tensor.shape[1] > 0 &&
333         tensor.shape[0] % LCQI_QK_K_BLOCK_VALUES == 0;
334 }
335
336 [[nodiscard]] bool can_use_ggml_direct(const GgufTensorInfo& tensor) {
337     if (tensor.shape.size() != 2 || tensor.shape[0] <= 0 || tensor.shape[1] <= 0 ||
338         !ggml_type_has_q8_0_direct_matvec(tensor.type)) {
339         return false;
340     }
341     const GgmlTypeLayout layout = ggml_type_layout(tensor.type);
342     return tensor.shape[0] % layout.block_size == 0 &&
343         tensor.shape[0] % LCQI_Q8_0_INPUT_BLOCK_VALUES == 0;
344 }
345
346 [[nodiscard]] bool q4_k_direct_enabled() noexcept {
347     const char* enabled = std::getenv(LCQI_LLAMA_Q4_DIRECT_ENV);
348     return enabled == nullptr || std::string_view(enabled) != 0 ||
349         LCQI_LLAMA_FALSE_ENV;
350 }
351
352 [[nodiscard]] bool ggml_direct_enabled() noexcept {
353     const char* enabled = std::getenv(LCQI_LLAMA_GGML_DIRECT_ENV);
354     return enabled != nullptr && std::string_view(enabled) != 0 ||
355         LCQI_LLAMA_TRUE_ENV;
356 }
357
358 void account_f32_tensor_load(const GgufTensorInfo& tensor,
359                             std::span<const float> values,
360                             LlamaGgufLoadReport& report) {
361     report.quantized_weight_bytes += tensor.byte_size;
362     const std::uint64_t f32_bytes =
363         static_cast<std::uint64_t>(values.size()) * static_cast<std::uint64_t>(
364             sizeof(float));
365     report.f32_weight_bytes += f32_bytes;
366     report.fallback_dequantized_weight_bytes += f32_bytes;
367     ++report.tensors_loaded;
368     ++report.f32_fallback_tensors;
369 }
370
371 [[nodiscard]] std::vector<float> read_tensor_f32(const std::filesystem::path& gguf_path,
372                                                  const GgufManifest& manifest,
373                                                  std::string_view name,
374                                                  std::span<const std::int64_t> expected_shape,
375                                                  LlamaGgufLoadReport& report) {
376     const GgufTensorInfo* tensor = manifest.find_tensor(name);
377     if (tensor == nullptr) {
378         throw std::runtime_error("missing LLaMA GGUF tensor: " + std::string(

```

```

    ↵ name));
376 }
377 validate_shape(*tensor, expected_shape);
378 const std::vector<std::uint8_t> bytes = read_gguf_tensor_bytes(gguf_path, *↵
    ↵ tensor);
379 std::vector<float> values =
380     dequantize_ggml_tensor(tensor->type,
381                             bytes,
382                             static_cast<std::int64_t>(tensor->element_count↵
    ↵ ());
383 account_f32_tensor_load(*tensor, values, report);
384 return values;
385 }
386
387 void account_quantized_direct_tensor_load(const GgufTensorInfo& tensor,
388                                           LlamaGgufLoadReport& report) {
389     report.quantized_weight_bytes += tensor.byte_size;
390     report.direct_quantized_weight_bytes += tensor.byte_size;
391     ++report.tensors_loaded;
392     if (tensor.type == GgmlType::q4_k) {
393         ++report.q4_k_direct_tensors;
394     } else {
395         ++report.ggml_direct_tensors;
396     }
397 }
398
399 [[nodiscard]] LlamaGgufLinearWeights make_linear_from_gguf(
400     const std::filesystem::path& gguf_path,
401     const GgufManifest& manifest,
402     std::string_view name,
403     std::int32_t input_size,
404     std::int32_t output_size,
405     LlamaGgufLoadReport& report) {
406     const std::array<std::int64_t, 2> shape{input_size, output_size};
407     const GgufTensorInfo* tensor = manifest.find_tensor(name);
408     if (tensor == nullptr) {
409         throw std::runtime_error("missing LLaMA GGUF tensor: " + std::string(↵
    ↵ name));
410     }
411     validate_shape(*tensor, shape);
412
413     LlamaGgufLinearWeights linear;
414     linear.input_size = input_size;
415     linear.output_size = output_size;
416     linear.source_type = tensor->type;
417     const std::vector<std::uint8_t> bytes = read_gguf_tensor_bytes(gguf_path, *↵
    ↵ tensor);
418     if (q4_k_direct_enabled() && can_use_q4_k_direct(*tensor)) {
419         linear.storage = LlamaGgufLinearStorage::q4_k;
420         linear.quantized_bytes = bytes;
421         account_quantized_direct_tensor_load(*tensor, report);
422         return linear;

```

```

423     }
424     if (ggml_direct_enabled() && can_use_ggml_direct(*tensor)) {
425         linear.storage = LlamaGgufLinearStorage::ggml_quantized;
426         linear.quantized_bytes = bytes;
427         account_quantized_direct_tensor_load(*tensor, report);
428         return linear;
429     }
430
431     linear.storage = LlamaGgufLinearStorage::f32;
432     linear.f32_weights =
433         dequantize_ggml_tensor(tensor->type,
434                                bytes,
435                                static_cast<std::int64_t>(tensor->element_count ←
↳ ())),
436     account_f32_tensor_load(*tensor, linear.f32_weights, report);
437     return linear;
438 }
439
440 [[nodiscard]] std::string layer_tensor_name(std::int32_t layer_id, std::int32_t ←
↳ string_view suffix) {
441     return "blk." + std::to_string(layer_id) + "." + std::string(suffix);
442 }
443
444 void validate_llama_config(const DecoderConfig& config, std::int32_t ←
↳ layer_count) {
445     require_positive(layer_count, "layer_count");
446     require_positive(config.hidden_size, "hidden_size");
447     require_positive(config.query_heads, "query_heads");
448     require_positive(config.kv_heads, "kv_heads");
449     require_positive(config.head_dim, "head_dim");
450     require_positive(config.intermediate_size, "intermediate_size");
451     require_positive(config.vocab_size, "vocab_size");
452     require_positive(config.max_context, "max_context");
453     if (config.query_heads % config.kv_heads != 0) {
454         throw std::runtime_error("LLaMA query_heads must be divisible by ←
↳ kv_heads");
455     }
456     if (config.hidden_size != config.query_heads * config.head_dim) {
457         throw std::runtime_error("LLaMA hidden_size must equal query_heads * ←
↳ head_dim");
458     }
459     if (config.head_dim % LCQI_LLAMA_ROPE_PAIR_WIDTH != 0) {
460         throw std::runtime_error("LLaMA head_dim must be even for RoPE");
461     }
462 }
463
464 [[nodiscard]] std::span<const float> row_span(std::span<const float> values,
465                                                std::int32_t row,
466                                                std::int32_t row_width) {
467     return values.subspan(checked_size(row, "row") * checked_size(row_width, "←
↳ row_width"),
468                           checked_size(row_width, "row_width"));

```



```

469 }
470
471 void add_inplace(std::span<float> target, std::span<const float> source) {
472     if (target.size() != source.size()) {
473         throw std::runtime_error("LLaMA residual add size mismatch");
474     }
475     for (std::size_t index = 0; index < target.size(); ++index) {
476         target[index] += source[index];
477     }
478 }
479
480 [[nodiscard]] float silu(float value) {
481     return value / (1.0F + std::exp(-value));
482 }
483
484 template <typename Func>
485 double measure_ms(Func&& func) {
486     const Clock::time_point begin = Clock::now();
487     std::forward<Func>(func)();
488     return elapsed_ms(begin, Clock::now());
489 }
490
491 [[nodiscard]] double& linear_hotspot_field(LlamaGgufHotspotReport& hotspots,
492                                           LlamaGgufLinearOp operation) {
493     switch (operation) {
494         case LlamaGgufLinearOp::wq:
495             return hotspots.wq_ms;
496         case LlamaGgufLinearOp::wk:
497             return hotspots.wk_ms;
498         case LlamaGgufLinearOp::wv:
499             return hotspots.wv_ms;
500         case LlamaGgufLinearOp::wo:
501             return hotspots.wo_ms;
502         case LlamaGgufLinearOp::w_gate:
503             return hotspots.w_gate_ms;
504         case LlamaGgufLinearOp::w_up:
505             return hotspots.w_up_ms;
506         case LlamaGgufLinearOp::w_down:
507             return hotspots.w_down_ms;
508         case LlamaGgufLinearOp::lm_head:
509             return hotspots.lm_head_ms;
510     }
511     throw std::runtime_error("unknown LLaMA linear hotspot operation");
512 }
513
514 void validate_linear_shape(const LlamaGgufLinearWeights& layer,
515                           std::span<const float> input,
516                           std::span<float> output) {
517     if (input.size() != checked_size(layer.input_size, "linear input size") ||
518         output.size() != checked_size(layer.output_size, "linear output size"))↵
519     ↵ {
520         throw std::runtime_error("LLaMA GGUF linear input/output size mismatch"↵

```

```

    ↪ );
520 }
521 }
522
523 void linear_f32_fallback(const LlamaGgufLinearWeights& layer,
524                         std::span<const float> input,
525                         std::span<float> output) {
526     if (layer.f32_weights.size() !=
527         checked_size(layer.input_size, "linear input size") *
528         checked_size(layer.output_size, "linear output size")) {
529         throw std::runtime_error("LLaMA GGUF F32 linear weight size mismatch");
530     }
531     linear_f32_rows_unchecked(layer.f32_weights.data(),
532                              input.data(),
533                              nullptr,
534                              checked_size(layer.input_size, "linear input size" ↪
535     ↪ "),
536                              0,
537                              checked_size(layer.output_size, "linear output ↪
538     ↪ size"),
539                              output.data());
538 }
539
540 [[nodiscard]] bool should_parallelize_rows(const LlamaGgufExecutionOptions& ↪
541     ↪ options,
542                                         std::size_t rows,
543                                         std::size_t columns,
544                                         const LlamaParallelWorkerPool* ↪
545     ↪ worker_pool) {
546     if (worker_pool == nullptr || worker_pool->worker_count() <= 1) {
547         return false;
548     }
549     const std::int32_t min_rows =
550         options.parallel_min_rows > 0 ? options.parallel_min_rows
551         : LCQI_LLAMA_DEFAULT_PARALLEL_MIN_ROWS;
552     if (rows < checked_size(min_rows, "parallel_min_rows")) {
553         return false;
554     }
555     if (options.worker_count > 1) {
556         return true;
557     }
558     return columns >= checked_size(LCQI_LLAMA_PARALLEL_MIN_COLUMNS, "↪
559     ↪ parallel_min_columns") ||
560         rows * columns >= static_cast<std::size_t>(↪
561     ↪ LCQI_LLAMA_PARALLEL_MIN_OPS);
558 }
559
560 [[nodiscard]] std::size_t parallel_row_grain(std::size_t rows,
561                                              const LlamaParallelWorkerPool& ↪
562     ↪ worker_pool) {
563     const std::size_t workers = checked_size(worker_pool.worker_count(), "↪
564     ↪ worker_count");

```

```

563     const std::size_t target_tasks = workers * ↵
    ↵ LCQI_LLAMA_PARALLEL_ROW_BLOCK_MULTIPLIER;
564     return std::max<std::size_t>(1, (rows + target_tasks - 1) / target_tasks);
565 }
566
567 [[nodiscard]] bool model_has_parallel_work(const LlamaGgufModel& model,
568                                           const LlamaGgufExecutionOptions& ↵
    ↵ options) {
569     if (options.worker_count == 1) {
570         return false;
571     }
572     const DecoderConfig& config = model.decoder.config;
573     const std::size_t hidden_size = checked_size(config.hidden_size, "↵
    ↵ hidden_size");
574     const std::size_t kv_width = checked_size(config.kv_heads * config.head_dim↵
    ↵ , "kv_width");
575     const std::size_t intermediate_size =
576         checked_size(config.intermediate_size, "intermediate_size");
577     const std::size_t vocab_size = checked_size(config.vocab_size, "vocab_size"↵
    ↵ );
578     const std::int32_t min_rows =
579         options.parallel_min_rows > 0 ? options.parallel_min_rows
580                                         : LCQI_LLAMA_DEFAULT_PARALLEL_MIN_ROWS;
581     const auto large_enough = [min_rows](std::size_t rows, std::size_t columns)↵
    ↵ {
582         if (rows < checked_size(min_rows, "parallel_min_rows")) {
583             return false;
584         }
585         return columns >= checked_size(LCQI_LLAMA_PARALLEL_MIN_COLUMNS,
586                                         "parallel_min_columns") ||
587            rows * columns >= static_cast<std::size_t>(↵
    ↵ LCQI_LLAMA_PARALLEL_MIN_OPS);
588     };
589     if (options.worker_count > 1) {
590         return vocab_size >= checked_size(min_rows, "parallel_min_rows") ||
591            hidden_size >= checked_size(min_rows, "parallel_min_rows") ||
592            kv_width >= checked_size(min_rows, "parallel_min_rows") ||
593            intermediate_size >= checked_size(min_rows, "parallel_min_rows")↵
    ↵ ;
594     }
595     return large_enough(vocab_size, hidden_size) ||
596        large_enough(hidden_size, hidden_size) ||
597        large_enough(kv_width, hidden_size) ||
598        large_enough(intermediate_size, hidden_size) ||
599        large_enough(hidden_size, intermediate_size);
600 }
601
602 [[nodiscard]] std::unique_ptr<LlamaParallelWorkerPool> make_worker_pool(
603     const LlamaGgufModel& model,
604     const LlamaGgufExecutionOptions& options) {
605     if (!model_has_parallel_work(model, options)) {
606         return nullptr;

```



```

↪ LlamaGgufExecutionOptions& options,
655                                     LlamaParallelWorkerPool* ↪
↪ worker_pool) {
656     if (!should_parallelize_rows(options, row_count, input_size, worker_pool)) ↪
↪ {
657         return max_dot_f32_rows_unchecked(weights, input, input_size, 0, ↪
↪ row_count);
658     }
659
660     const std::size_t worker_count = checked_size(worker_pool->worker_count(), ↪
↪ "worker_count");
661     std::vector<F32RowMax> local_best(worker_count);
662     std::vector<std::uint8_t> has_result(worker_count, 0U);
663     const auto rows_fn = [weights, input, input_size, &local_best, &has_result↪
↪ ](
664         std::size_t begin,
665         std::size_t end,
666         std::size_t worker_index) {
667         if (begin >= end) {
668             return;
669         }
670         local_best[worker_index] =
671             max_dot_f32_rows_unchecked(weights, input, input_size, begin, end);
672         has_result[worker_index] = 1U;
673     };
674     worker_pool->parallel_for_rows(0,
675                                   row_count,
676                                   parallel_row_grain(row_count, *worker_pool),
677                                   rows_fn);
678
679     F32RowMax best;
680     best.row = 0;
681     best.value = -std::numeric_limits<float>::infinity();
682     bool found = false;
683     for (std::size_t index = 0; index < local_best.size(); ++index) {
684         if (has_result[index] == 0U) {
685             continue;
686         }
687         const F32RowMax& candidate = local_best[index];
688         if (!found || candidate.value > best.value) {
689             best = candidate;
690             found = true;
691         }
692     }
693     if (!found) {
694         return max_dot_f32_rows_unchecked(weights, input, input_size, 0, ↪
↪ row_count);
695     }
696     return best;
697 }
698
699 void linear_q4_k_direct(const LlamaGgufLinearWeights& layer,

```

```

700         std::span<const Q8KBlock> q8_input,
701         std::span<float> output) {
702     if (q8_input.size() !=
703         checked_size(layer.input_size / LCQI_QK_K_BLOCK_VALUES, "Q8_K input ↵
↵ block count")) {
704         throw std::runtime_error("LLaMA GGUF Q4_K scratch block count mismatch"↵
↵ );
705     }
706     matvec_q4_k_q8(layer.quantized_bytes,
707                    layer.output_size,
708                    layer.input_size,
709                    q8_input,
710                    output);
711 }
712
713 void linear_q4_k_direct_parallel(const LlamaGgufLinearWeights& layer,
714                                std::span<const Q8KBlock> q8_input,
715                                std::span<float> output,
716                                const LlamaGgufExecutionOptions& options,
717                                LlamaParallelWorkerPool* worker_pool) {
718     if (q8_input.size() !=
719         checked_size(layer.input_size / LCQI_QK_K_BLOCK_VALUES, "Q8_K input ↵
↵ block count")) {
720         throw std::runtime_error("LLaMA GGUF Q4_K scratch block count mismatch"↵
↵ );
721     }
722     const std::size_t input_size = checked_size(layer.input_size, "linear input↵
↵ size");
723     const std::size_t output_size = checked_size(layer.output_size, "linear ↵
↵ output size");
724     if (!should_parallelize_rows(options, output_size, input_size, worker_pool)↵
↵ ) {
725         linear_q4_k_direct(layer, q8_input, output);
726         return;
727     }
728
729     const auto rows_fn = [&layer, q8_input, output](
730         std::size_t begin,
731         std::size_t end,
732         std::size_t worker_index) {
733         (void)worker_index;
734         matvec_q4_k_q8_rows_unchecked(layer.quantized_bytes,
735                                       layer.output_size,
736                                       layer.input_size,
737                                       q8_input,
738                                       output,
739                                       static_cast<std::int64_t>(begin),
740                                       static_cast<std::int64_t>(end));
741     };
742     worker_pool->parallel_for_rows(0,
743                                   output_size,
744                                   parallel_row_grain(output_size, *worker_pool↵

```

```

↪ ),
745         rows_fn);
746 }
747
748 void linear_ggml_quantized_direct(const LlamaGgufLinearWeights& layer,
749                                   std::span<const Q8_0InputBlock> q8_0_input,
750                                   std::span<float> output) {
751     if (q8_0_input.size() != checked_size(layer.input_size / ↪
↪ LCQI_Q8_0_INPUT_BLOCK_VALUES,
752                                             "Q8_0 input block count")) {
753         throw std::runtime_error("LLaMA GGUF Q8_0 scratch block count mismatch"↪
↪ );
754     }
755     matvec_ggml_quantized_q8_0(layer.source_type,
756                                layer.quantized_bytes,
757                                layer.output_size,
758                                layer.input_size,
759                                q8_0_input,
760                                output);
761 }
762
763 void linear_gguf(const LlamaGgufLinearWeights& layer,
764                  std::span<const float> input,
765                  std::span<const Q8KBlock> q8_input,
766                  std::span<const Q8_0InputBlock> q8_0_input,
767                  std::span<float> output,
768                  LlamaGgufLinearOp operation,
769                  LlamaGgufHotspotReport& hotspots,
770                  const LlamaGgufExecutionOptions& options,
771                  LlamaParallelWorkerPool* worker_pool) {
772     validate_linear_shape(layer, input, output);
773     double elapsed = 0.0;
774     switch (layer.storage) {
775         case LlamaGgufLinearStorage::q4_k:
776             elapsed = measure_ms([&]() {
777                 linear_q4_k_direct_parallel(layer, q8_input, output, options, ↪
↪ worker_pool);
778             });
779             hotspots.q4_k_direct_ms += elapsed;
780             ++hotspots.q4_k_direct_calls;
781             break;
782         case LlamaGgufLinearStorage::ggml_quantized:
783             elapsed = measure_ms([&]() {
784                 linear_ggml_quantized_direct(layer, q8_0_input, output);
785             });
786             hotspots.ggml_direct_ms += elapsed;
787             ++hotspots.ggml_direct_calls;
788             break;
789         case LlamaGgufLinearStorage::f32:
790             elapsed = measure_ms([&]() {
791                 linear_f32_fallback_parallel(layer, input, output, options, ↪
↪ worker_pool);

```



```

792         });
793         hotspots.f32_fallback_ms += elapsed;
794         ++hotspots.f32_fallback_calls;
795         break;
796     }
797     linear_hotspot_field(hotspots, operation) += elapsed;
798 }
799
800 [[nodiscard]] bool uses_q4_k_direct(const LlamaGgufLinearWeights& layer) ↵
    ↵ noexcept {
801     return layer.storage == LlamaGgufLinearStorage::q4_k;
802 }
803
804 [[nodiscard]] bool uses_ggml_direct(const LlamaGgufLinearWeights& layer) ↵
    ↵ noexcept {
805     return layer.storage == LlamaGgufLinearStorage::ggml_quantized;
806 }
807
808 [[nodiscard]] std::size_t q8_scratch_blocks(std::int32_t input_size) {
809     if (input_size <= 0 || input_size % LCQI_QK_K_BLOCK_VALUES != 0) {
810         return 0;
811     }
812     return checked_size(input_size / LCQI_QK_K_BLOCK_VALUES, "Q8_K scratch ↵
    ↵ blocks");
813 }
814
815 void prepare_q8_if_needed(std::span<const float> input,
816                          std::span<Q8KBlock> scratch,
817                          bool needed) {
818     if (!needed) {
819         return;
820     }
821     quantize_q8_k_input_to(input, scratch);
822 }
823
824 [[nodiscard]] std::size_t q8_0_scratch_blocks(std::int32_t input_size) {
825     if (input_size <= 0 || input_size % LCQI_Q8_0_INPUT_BLOCK_VALUES != 0) {
826         return 0;
827     }
828     return checked_size(input_size / LCQI_Q8_0_INPUT_BLOCK_VALUES, "Q8_0 ↵
    ↵ scratch blocks");
829 }
830
831 void prepare_q8_0_if_needed(std::span<const float> input,
832                             std::span<Q8_0InputBlock> scratch,
833                             bool needed) {
834     if (!needed) {
835         return;
836     }
837     quantize_q8_0_input_to(input, scratch);
838 }
839

```

```

840 struct LlamaWorkspace {
841     std::vector<float> hidden;
842     std::vector<float> normed;
843     std::vector<float> query;
844     std::vector<float> key;
845     std::vector<float> value;
846     std::vector<float> attention;
847     std::vector<float> projected;
848     std::vector<float> gate;
849     std::vector<float> up;
850     std::vector<float> mlp_hidden;
851     std::vector<float> mlp_out;
852     std::vector<float> final_norm;
853     std::vector<Q8KBlock> normed_q8;
854     std::vector<Q8KBlock> attention_q8;
855     std::vector<Q8KBlock> mlp_hidden_q8;
856     std::vector<Q8_0InputBlock> normed_q8_0;
857     std::vector<Q8_0InputBlock> attention_q8_0;
858     std::vector<Q8_0InputBlock> mlp_hidden_q8_0;
859
860     explicit LlamaWorkspace(const DecoderConfig& config)
861         : hidden(check_size(config.hidden_size, "hidden_size"), 0.0F),
862           normed(hidden.size(), 0.0F),
863           query(hidden.size(), 0.0F),
864           key(check_size(config.kv_heads * config.head_dim, "kv_width"), 0.0F,
865             ↪ ),
866           value(key.size(), 0.0F),
867           attention(hidden.size(), 0.0F),
868           projected(hidden.size(), 0.0F),
869           gate(check_size(config.intermediate_size, "intermediate_size"), 0.0F,
870             ↪ F),
871           up(gate.size(), 0.0F),
872           mlp_hidden(gate.size(), 0.0F),
873           mlp_out(hidden.size(), 0.0F),
874           final_norm(hidden.size(), 0.0F),
875           normed_q8(q8_scratch_blocks(config.hidden_size)),
876           attention_q8(normed_q8.size()),
877           mlp_hidden_q8(q8_scratch_blocks(config.intermediate_size)),
878           normed_q8_0(q8_0_scratch_blocks(config.hidden_size)),
879           attention_q8_0(normed_q8_0.size()),
880           mlp_hidden_q8_0(q8_0_scratch_blocks(config.intermediate_size)) {}
881 };
882
883 void forward_llama_layer(const DecoderConfig& config,
884     const LlamaGgufDecoderLayerWeights& layer,
885     std::int32_t layer_id,
886     std::int32_t model_position,
887     ReferenceKVCache& cache,
888     LlamaWorkspace& workspace,
889     LlamaGgufHotspotReport& hotspots,
890     const LlamaGgufExecutionOptions& options,
891     LlamaParallelWorkerPool* worker_pool) {

```

```

890     hotspots.rms_norm_ms += measure_ms([&]() {
891         rms_norm(workspace.hidden,
892                 layer.rms_attention_weight,
893                 config.rms_epsilon,
894                 workspace.normed);
895     });
896     prepare_q8_if_needed(workspace.normed,
897                         workspace.normed_q8,
898                         uses_q4_k_direct(layer.wq) || uses_q4_k_direct(layer.↵
↵ wk) ||
899                             uses_q4_k_direct(layer.wv));
900     prepare_q8_0_if_needed(workspace.normed,
901                           workspace.normed_q8_0,
902                           uses_ggml_direct(layer.wq) || uses_ggml_direct(layer↵
↵ .wk) ||
903                             uses_ggml_direct(layer.wv));
904     linear_gguf(layer.wq,
905                workspace.normed,
906                workspace.normed_q8,
907                workspace.normed_q8_0,
908                workspace.query,
909                LlamaGgufLinearOp::wq,
910                hotspots,
911                options,
912                worker_pool);
913     linear_gguf(layer.wk,
914                workspace.normed,
915                workspace.normed_q8,
916                workspace.normed_q8_0,
917                workspace.key,
918                LlamaGgufLinearOp::wk,
919                hotspots,
920                options,
921                worker_pool);
922     linear_gguf(layer.wv,
923                workspace.normed,
924                workspace.normed_q8,
925                workspace.normed_q8_0,
926                workspace.value,
927                LlamaGgufLinearOp::wv,
928                hotspots,
929                options,
930                worker_pool);
931     hotspots.rope_ms += measure_ms([&]() {
932         apply_rope(workspace.query,
933                   config.query_heads,
934                   config.head_dim,
935                   model_position,
936                   config.rope_base);
937         apply_rope(workspace.key,
938                   config.kv_heads,
939                   config.head_dim,

```

```

940         model_position,
941         config.rope_base);
942     });
943     cache.append(layer_id, model_position, workspace.key, workspace.value);
944     hotspots.attention_ms += measure_ms([&]() {
945         attention_decode(config,
946             cache,
947             layer_id,
948             model_position,
949             workspace.query,
950             workspace.attention);
951     });
952     prepare_q8_if_needed(workspace.attention,
953         workspace.attention_q8,
954         uses_q4_k_direct(layer.wo));
955     prepare_q8_0_if_needed(workspace.attention,
956         workspace.attention_q8_0,
957         uses_ggml_direct(layer.wo));
958     linear_gguf(layer.wo,
959         workspace.attention,
960         workspace.attention_q8,
961         workspace.attention_q8_0,
962         workspace.projected,
963         LlamaGgufLinearOp::wo,
964         hotspots,
965         options,
966         worker_pool);
967     add_inplace(workspace.hidden, workspace.projected);
968
969     hotspots.rms_norm_ms += measure_ms([&]() {
970         rms_norm(workspace.hidden, layer.rms_mlp_weight, config.rms_epsilon, ↵
↵ workspace.normed);
971     });
972     prepare_q8_if_needed(workspace.normed,
973         workspace.normed_q8,
974         uses_q4_k_direct(layer.w_gate) || uses_q4_k_direct(↵
↵ layer.w_up));
975     prepare_q8_0_if_needed(workspace.normed,
976         workspace.normed_q8_0,
977         uses_ggml_direct(layer.w_gate) || uses_ggml_direct(↵
↵ layer.w_up));
978     linear_gguf(layer.w_gate,
979         workspace.normed,
980         workspace.normed_q8,
981         workspace.normed_q8_0,
982         workspace.gate,
983         LlamaGgufLinearOp::w_gate,
984         hotspots,
985         options,
986         worker_pool);
987     linear_gguf(layer.w_up,
988         workspace.normed,

```



```

1035     });
1036     return static_cast<std::int32_t>(best.row);
1037 }
1038
1039 std::vector<float> logits_checked_size(config.vocab_size, "vocab_size"), ←
    ↪ 0.0F);
1040 const std::vector<Q8KBlock> final_hidden_q8 =
1041     uses_q4_k_direct(model.decoder.lm_head)
1042     ? quantize_q8_k_input(final_hidden)
1043     : std::vector<Q8KBlock>{};
1044 const std::vector<Q8_0InputBlock> final_hidden_q8_0 =
1045     uses_ggml_direct(model.decoder.lm_head)
1046     ? quantize_q8_0_input(final_hidden)
1047     : std::vector<Q8_0InputBlock>{};
1048 linear_gguf(model.decoder.lm_head,
1049             final_hidden,
1050             final_hidden_q8,
1051             final_hidden_q8_0,
1052             logits,
1053             LlamaGgufLinearOp::lm_head,
1054             hotspots,
1055             options,
1056             worker_pool);
1057 best.row = 0;
1058 best.value = logits.front();
1059 for (std::size_t row = 1; row < logits.size(); ++row) {
1060     if (logits[row] > best.value) {
1061         best.row = row;
1062         best.value = logits[row];
1063     }
1064 }
1065 return static_cast<std::int32_t>(best.row);
1066 }
1067
1068 [[nodiscard]] std::int32_t step_llama_decoder(const LlamaGgufModel& model,
1069                                             const DecoderConfig& ←
    ↪ active_config,
1070                                             ReferenceKVCache& cache,
1071                                             LlamaWorkspace& workspace,
1072                                             std::int32_t token_id,
1073                                             bool predict,
1074                                             LlamaGgufHotspotReport& hotspots,
1075                                             const LlamaGgufExecutionOptions& ←
    ↪ options,
1076                                             LlamaParallelWorkerPool* ←
    ↪ worker_pool) {
1077     const DecoderConfig& original_config = model.decoder.config;
1078     if (token_id < 0 || token_id >= original_config.vocab_size) {
1079         throw std::runtime_error("LLaMA token id out of vocabulary");
1080     }
1081     const std::int32_t model_position = cache.filled_tokens();
1082     if (model_position >= active_config.max_context) {

```

```

1083     throw std::runtime_error("LLaMA generation exceeded active context");
1084 }
1085
1086 const std::span<const float> embedding =
1087     row_span(model.decoder.token_embedding, token_id, original_config.↵
↵ hidden_size);
1088 std::copy(embedding.begin(), embedding.end(), workspace.hidden.begin());
1089 for (std::int32_t layer_id = 0;
1090      layer_id < static_cast<std::int32_t>(model.decoder.layers.size());
1091      ++layer_id) {
1092     forward_llama_layer(active_config,
1093                        model.decoder.layers[checked_size(layer_id, "↵
↵ layer_id")],
1094                        layer_id,
1095                        model_position,
1096                        cache,
1097                        workspace,
1098                        hotspots,
1099                        options,
1100                        worker_pool);
1101 }
1102 if (!predict) {
1103     return -1;
1104 }
1105 hotspots.rms_norm_ms += measure_ms([&]() {
1106     rms_norm(workspace.hidden,
1107             model.decoder.final_norm_weight,
1108             active_config.rms_epsilon,
1109             workspace.final_norm);
1110 });
1111 return predict_next_token(model, workspace.final_norm, hotspots, options, ↵
↵ worker_pool);
1112 }
1113
1114 [[nodiscard]] std::size_t kv_cache_bytes(const DecoderConfig& config, std::↵
↵ int32_t layer_count) {
1115     return checked_size(layer_count, "layer_count") *
1116            checked_size(config.max_context, "max_context") *
1117            checked_size(config.kv_heads, "kv_heads") *
1118            checked_size(config.head_dim, "head_dim") * sizeof(float) * 2U;
1119 }
1120
1121 [[nodiscard]] std::unordered_map<std::string, std::int32_t> ↵
↵ build_vocab_from_tokens(
1122     std::span<const std::string> tokens) {
1123     std::unordered_map<std::string, std::int32_t> vocab;
1124     for (std::size_t index = 0; index < tokens.size(); ++index) {
1125         vocab.emplace(tokens[index], static_cast<std::int32_t>(index));
1126     }
1127     return vocab;
1128 }
1129

```



```

1130 [[nodiscard]] std::unordered_map<std::string, std::int32_t> build_merge_ranks(
1131     std::span<const std::string> merges) {
1132     std::unordered_map<std::string, std::int32_t> ranks;
1133     for (std::size_t index = 0; index < merges.size(); ++index) {
1134         std::istringstream line_stream(merges[index]);
1135         std::string left;
1136         std::string right;
1137         if (!(line_stream >> left >> right)) {
1138             throw std::runtime_error("invalid GGUF tokenizer merge: " + merges[↵
↵ index]);
1139         }
1140         ranks.emplace(left + "\n" + right, static_cast<std::int32_t>(index));
1141     }
1142     return ranks;
1143 }
1144
1145 [[nodiscard]] bool starts_with(std::string_view text,
1146                                std::size_t offset,
1147                                std::string_view needle) {
1148     return offset + needle.size() <= text.size() &&
1149         text.substr(offset, needle.size()) == needle;
1150 }
1151
1152 [[nodiscard]] std::vector<std::pair<std::string, std::int32_t>> special_tokens(
1153     const Gpt2Tokenizer& tokenizer) {
1154     std::vector<std::pair<std::string, std::int32_t>> result;
1155     for (const auto& [token, id] : tokenizer.vocab) {
1156         if (token.size() >= 4 && token.front() == '<' && token.find('|') != std::↵
↵ ::string::npos) {
1157             result.emplace_back(token, id);
1158         }
1159     }
1160     std::sort(result.begin(),
1161               result.end(),
1162               [](const auto& lhs, const auto& rhs) {
1163                   return lhs.first.size() > rhs.first.size();
1164               });
1165     return result;
1166 }
1167
1168 [[nodiscard]] std::vector<std::int32_t> encode_with_special_tokens(
1169     const Gpt2Tokenizer& tokenizer,
1170     std::string_view text) {
1171     const std::vector<std::pair<std::string, std::int32_t>> specials = ↵
↵ special_tokens(tokenizer);
1172     std::vector<std::int32_t> ids;
1173     std::size_t offset = 0;
1174     while (offset < text.size()) {
1175         bool matched_special = false;
1176         for (const auto& [token, id] : specials) {
1177             if (starts_with(text, offset, token)) {
1178                 ids.push_back(id);

```

```

1179         offset += token.size();
1180         matched_special = true;
1181         break;
1182     }
1183 }
1184 if (matched_special) {
1185     continue;
1186 }
1187
1188 std::size_t next_special = text.size();
1189 for (const auto& [token, id] : specials) {
1190     (void)id;
1191     const std::size_t found = text.find(token, offset);
1192     if (found != std::string_view::npos) {
1193         next_special = std::min(next_special, found);
1194     }
1195 }
1196 const std::vector<std::int32_t> chunk =
1197     gpt2_encode(tokenizer, text.substr(offset, next_special - offset));
1198 ids.insert(ids.end(), chunk.begin(), chunk.end());
1199 offset = next_special;
1200 }
1201 return ids;
1202 }
1203
1204 } // namespace
1205
1206 LlamaGgufLoadedModel load_llama_gguf_reference_model(const std::filesystem::↵
    ↵ path& gguf_path) {
1207     LlamaGgufLoadedModel loaded;
1208     const Clock::time_point manifest_begin = Clock::now();
1209     const GgufManifest manifest = load_gguf_manifest(gguf_path);
1210     loaded.report.manifest_ms = elapsed_ms(manifest_begin, Clock::now());
1211
1212     const Clock::time_point weights_begin = Clock::now();
1213     loaded.model.architecture = metadata_string(manifest, ↵
    ↵ LCQI_LLAMA_ARCHITECTURE_KEY);
1214     loaded.model.name = metadata_string(manifest, LCQI_LLAMA_NAME_KEY);
1215     DecoderConfig& config = loaded.model.decoder.config;
1216     config.hidden_size = metadata_i32(manifest, LCQI_LLAMA_EMBEDDING_LENGTH_KEY↵
    ↵ );
1217     const std::int32_t layer_count = metadata_i32(manifest, ↵
    ↵ LCQI_LLAMA_BLOCK_COUNT_KEY);
1218     config.intermediate_size = metadata_i32(manifest, ↵
    ↵ LCQI_LLAMA_FEED_FORWARD_LENGTH_KEY);
1219     config.query_heads = metadata_i32(manifest, LCQI_LLAMA_HEAD_COUNT_KEY);
1220     config.kv_heads = metadata_i32(manifest, LCQI_LLAMA_KV_HEAD_COUNT_KEY);
1221     config.head_dim = config.hidden_size / config.query_heads;
1222     const std::int32_t rope_dimension = metadata_i32(manifest, ↵
    ↵ LCQI_LLAMA_ROPE_DIMENSION_KEY);
1223     if (rope_dimension != config.head_dim) {
1224         throw std::runtime_error("LLaMA rope dimension does not match head_dim"↵

```

```

↪ );
1225 }
1226 config.rope_base = metadata_f32(manifest, LCQI_LLAMA_ROPE_BASE_KEY);
1227 config.max_context = metadata_i32(manifest, LCQI_LLAMA_CONTEXT_LENGTH_KEY);
1228 config.vocab_size = metadata_i32(manifest, LCQI_LLAMA_VOCAB_SIZE_KEY);
1229 config.rms_epsilon = metadata_f32(manifest, LCQI_LLAMA_RMS_EPSILON_KEY);
1230 loaded.model.bos_token_id =
1231     optional_metadata_i32(manifest, LCQI_LLAMA_BOS_TOKEN_KEY, ↪
↪ LCQI_LLAMA_DEFAULT_BOS_TOKEN);
1232 loaded.model.eos_token_id =
1233     optional_metadata_i32(manifest, LCQI_LLAMA_EOS_TOKEN_KEY, ↪
↪ LCQI_LLAMA_DEFAULT_EOS_TOKEN);
1234 loaded.model.unknown_token_id = optional_metadata_i32(
1235     manifest,
1236     LCQI_LLAMA_UNKNOWN_TOKEN_KEY,
1237     LCQI_LLAMA_DEFAULT_UNK_TOKEN);
1238 validate_llama_config(config, layer_count);
1239
1240 const std::array<std::int64_t, 2> embedding_shape{config.hidden_size, ↪
↪ config.vocab_size};
1241 loaded.model.decoder.token_embedding = read_tensor_f32(gguf_path,
1242                                                         manifest,
1243                                                         "token_embd.weight",
1244                                                         embedding_shape,
1245                                                         loaded.report);
1246 const std::array<std::int64_t, 1> hidden_shape{config.hidden_size};
1247 loaded.model.decoder.final_norm_weight = read_tensor_f32(gguf_path,
1248                                                         manifest,
1249                                                         "output_norm.↪
↪ weight",
1250                                                         hidden_shape,
1251                                                         loaded.report);
1252 loaded.model.decoder.layers.resize(check_size(layer_count, "layer_count")↪
↪ );
1253 const std::int32_t kv_width = config.kv_heads * config.head_dim;
1254 for (std::int32_t layer_id = 0; layer_id < layer_count; ++layer_id) {
1255     LlamaGgufDecoderLayerWeights& layer =
1256         loaded.model.decoder.layers[check_size(layer_id, "layer_id")];
1257     layer.rms_attention_weight = read_tensor_f32(
1258         gguf_path,
1259         manifest,
1260         layer_tensor_name(layer_id, "attn_norm.weight"),
1261         hidden_shape,
1262         loaded.report);
1263     layer.rms_mlp_weight = read_tensor_f32(gguf_path,
1264                                           manifest,
1265                                           layer_tensor_name(layer_id, "↪
↪ ffn_norm.weight"),
1266                                           hidden_shape,
1267                                           loaded.report);
1268     layer.wq = make_linear_from_gguf(gguf_path,
1269                                     manifest,

```

```

1270         layer_tensor_name(layer_id, "attn_q.↵
↵ weight"),
1271         config.hidden_size,
1272         config.hidden_size,
1273         loaded.report());
1274     layer.wk = make_linear_from_gguf(gguf_path,
1275         manifest,
1276         layer_tensor_name(layer_id, "attn_k.↵
↵ weight"),
1277         config.hidden_size,
1278         kv_width,
1279         loaded.report());
1280     layer.wv = make_linear_from_gguf(gguf_path,
1281         manifest,
1282         layer_tensor_name(layer_id, "attn_v.↵
↵ weight"),
1283         config.hidden_size,
1284         kv_width,
1285         loaded.report());
1286     layer.wo = make_linear_from_gguf(gguf_path,
1287         manifest,
1288         layer_tensor_name(layer_id, "↵
↵ attn_output.weight"),
1289         config.hidden_size,
1290         config.hidden_size,
1291         loaded.report());
1292     layer.w_gate = make_linear_from_gguf(gguf_path,
1293         manifest,
1294         layer_tensor_name(layer_id, "↵
↵ ffn_gate.weight"),
1295         config.hidden_size,
1296         config.intermediate_size,
1297         loaded.report());
1298     layer.w_up = make_linear_from_gguf(gguf_path,
1299         manifest,
1300         layer_tensor_name(layer_id, "ffn_up.↵
↵ weight"),
1301         config.hidden_size,
1302         config.intermediate_size,
1303         loaded.report());
1304     layer.w_down = make_linear_from_gguf(gguf_path,
1305         manifest,
1306         layer_tensor_name(layer_id, "↵
↵ ffn_down.weight"),
1307         config.intermediate_size,
1308         config.hidden_size,
1309         loaded.report());
1310 }
1311
1312 if (manifest.find_tensor("output.weight") != nullptr) {
1313     loaded.model.lm_head_tied_to_embedding = false;
1314     loaded.model.decoder.lm_head = make_linear_from_gguf(gguf_path,

```

```

1315 manifest,
1316 "output.weight",
1317 config.hidden_size↵
↵ ,
1318 config.vocab_size,
1319 loaded.report);
1320 } else {
1321     loaded.model.lm_head_tied_to_embedding = true;
1322     loaded.model.decoder.lm_head.input_size = config.hidden_size;
1323     loaded.model.decoder.lm_head.output_size = config.vocab_size;
1324 }
1325 loaded.report.weights_ms = elapsed_ms(weights_begin, Clock::now());
1326 return loaded;
1327 }
1328
1329 Gpt2Tokenizer load_gpt2_tokenizer_from_gguf(const std::filesystem::path& ↵
↵ gguf_path) {
1330     const GgufManifest manifest = load_gguf_manifest(gguf_path);
1331     const GgufMetadataEntry& tokens_entry = require_metadata(manifest,
1332                                     ↵
↵ LCQI_LLAMA_TOKENIZER_TOKENS_KEY);
1333     const GgufMetadataEntry& merges_entry = require_metadata(manifest,
1334                                     ↵
↵ LCQI_LLAMA_TOKENIZER_MERGES_KEY);
1335     if (tokens_entry.string_values.empty() || merges_entry.string_values.empty()↵
↵ ()) {
1336         throw std::runtime_error("GGUF tokenizer tokens or merges are empty");
1337     }
1338     Gpt2Tokenizer tokenizer;
1339     tokenizer.bos_token_id =
1340         optional_metadata_i32(manifest, LCQI_LLAMA_BOS_TOKEN_KEY, ↵
↵ LCQI_LLAMA_DEFAULT_BOS_TOKEN);
1341     tokenizer.eos_token_id =
1342         optional_metadata_i32(manifest, LCQI_LLAMA_EOS_TOKEN_KEY, ↵
↵ LCQI_LLAMA_DEFAULT_EOS_TOKEN);
1343     tokenizer.id_to_token = tokens_entry.string_values;
1344     tokenizer.vocab = build_vocab_from_tokens(tokenizer.id_to_token);
1345     tokenizer.bpe_ranks = build_merge_ranks(merges_entry.string_values);
1346     return tokenizer;
1347 }
1348
1349 std::vector<std::int32_t> llama_gguf_chat_prompt_ids(const Gpt2Tokenizer& ↵
↵ tokenizer,
1350                                     std::string_view ↵
↵ user_prompt,
1351                                     std::int32_t bos_token_id)↵
↵ {
1352     std::string prompt;
1353     prompt += "<|im_start|>system\n";
1354     prompt += "You are a helpful AI assistant named SmolLM, trained by Hugging ↵
↵ Face";
1355     prompt += "<|im_end|>\n";

```

```

1356     prompt += "<|im_start|>user\n";
1357     prompt += user_prompt;
1358     prompt += "<|im_end|>\n";
1359     prompt += "<|im_start|>assistant\n";
1360     std::vector<std::int32_t> ids = encode_with_special_tokens(tokenizer, ↵
↵ prompt);
1361     if (ids.empty() && bos_token_id >= 0) {
1362         ids.push_back(bos_token_id);
1363     }
1364     return ids;
1365 }
1366
1367 LlamaGgufGenerationResult llama_gguf_generate_greedy(const LlamaGgufModel& ↵
↵ model,
1368                                                         std::span<const std::↵
↵ int32_t> prompt_ids,
1369                                                         std::int32_t ↵
↵ max_new_tokens) {
1370     return llama_gguf_generate_greedy(model,
1371                                       prompt_ids,
1372                                       max_new_tokens,
1373                                       LlamaGgufExecutionOptions{});
1374 }
1375
1376 LlamaGgufGenerationResult llama_gguf_generate_greedy(
1377     const LlamaGgufModel& model,
1378     std::span<const std::int32_t> prompt_ids,
1379     std::int32_t max_new_tokens,
1380     const LlamaGgufExecutionOptions& options) {
1381     if (prompt_ids.empty()) {
1382         throw std::runtime_error("LLaMA generation requires at least one prompt↵
↵ token");
1383     }
1384     if (max_new_tokens < 0) {
1385         throw std::runtime_error("max_new_tokens cannot be negative");
1386     }
1387     const std::int32_t layer_count = static_cast<std::int32_t>(model.decoder.↵
↵ layers.size());
1388     validate_llama_config(model.decoder.config, layer_count);
1389     const std::int32_t active_context =
1390         static_cast<std::int32_t>(prompt_ids.size()) + std::max(max_new_tokens, ↵
↵ 1);
1391     if (active_context > model.decoder.config.max_context) {
1392         throw std::runtime_error("LLaMA prompt plus generation exceeds model ↵
↵ context");
1393     }
1394
1395     DecoderConfig active_config = model.decoder.config;
1396     active_config.max_context = active_context;
1397     ReferenceKVCache cache(active_config, layer_count);
1398     LlamaWorkspace workspace(active_config);
1399     std::unique_ptr<LlamaParallelWorkerPool> worker_pool = make_worker_pool(↵

```

```

1400 ↪ model, options);
1401
1402 LlamaGgufGenerationResult result;
1403 result.prompt_ids.assign(prompt_ids.begin(), prompt_ids.end());
1404 result.generated_ids.assign(prompt_ids.begin(), prompt_ids.end());
1405 result.kv_cache_bytes = kv_cache_bytes(active_config, layer_count);
1406 result.worker_count = worker_pool ? worker_pool->worker_count() : 1;
1407 if (max_new_tokens == 0) {
1408     return result;
1409 }
1410
1411 const Clock::time_point prefill_begin = Clock::now();
1412 for (std::size_t index = 0; index + 1 < prompt_ids.size(); ++index) {
1413     static_cast<void>(step_llama_decoder(model,
1414         active_config,
1415         cache,
1416         workspace,
1417         prompt_ids[index],
1418         false,
1419         result.hotspots,
1420         options,
1421         worker_pool.get()));
1422 }
1423 std::int32_t predicted = step_llama_decoder(model,
1424     active_config,
1425     cache,
1426     workspace,
1427     prompt_ids.back(),
1428     true,
1429     result.hotspots,
1430     options,
1431     worker_pool.get());
1432
1433 result.predicted_first_token = predicted;
1434 result.prefill_steps = prompt_ids.size();
1435 result.prefill_ms = elapsed_ms(prefill_begin, Clock::now());
1436
1437 const Clock::time_point decode_begin = Clock::now();
1438 for (std::int32_t step = 0; step < max_new_tokens; ++step) {
1439     result.generated_ids.push_back(predicted);
1440     if (predicted == model.eos_token_id) {
1441         break;
1442     }
1443     if (step + 1 < max_new_tokens) {
1444         predicted = step_llama_decoder(model,
1445             active_config,
1446             cache,
1447             workspace,
1448             predicted,
1449             true,
1450             result.hotspots,
1451             options,
1452             worker_pool.get());

```

```

1451         ++result.decode_steps;
1452     }
1453 }
1454 result.decode_ms = elapsed_ms(decode_begin, Clock::now());
1455 result.total_ms = result.prefill_ms + result.decode_ms;
1456 return result;
1457 }
1458
1459 } // namespace lcqi

```

`llama_cli.cpp` 是 `lcqi_llama_gguf` 的命令行入口。它支持 `--ids` 直接给 token id，也支持 `--prompt` 从 GGUF tokenizer arrays 构造 chat prompt；`--threads0` 自动选择 worker，`--threads1` 强制串行，`--threadsN` 固定 worker 数。`--benchmark` 输出 manifest、权重加载、prefill、decode、KV cache、worker count、量化权重字节、direct Q4\_K 字节、实验 `ggml_direct` 字节、fallback f32 字节、direct/fallback tensor 数、per-op hotspot 和 direct/fallback call 数。环境变量 `LCQI_LLAMA_Q4_DIRECT=0` 会关闭真实 decoder 里的 Q4 direct path，用于同二进制 A/B；`LCQI_LLAMA_GGML_DIRECT=1` 会打开 `Q5_0/Q6_K/Q8_0` 实验 direct path，用来证明覆盖率和速度必须分开验收。报告中 `decode_tokens_per_second` 只统计真正喂入新 token 后再次预测的 decode step，不能把首 token prefill 的预测误算成绩 decode。

Listing 16.12 `src/llama_cli.cpp`

```

1 #include <lcqi/llama_gguf.hpp>
2
3 #include <chrono>
4 #include <cstdlib>
5 #include <exception>
6 #include <filesystem>
7 #include <iostream>
8 #include <sstream>
9 #include <stdexcept>
10 #include <string>
11 #include <string_view>
12 #include <vector>
13
14 namespace {
15
16 constexpr std::int32_t LCQI_LLAMA_DEFAULT_MAX_NEW_TOKENS = 1;
17 constexpr std::string_view LCQI_LLAMA_IDS_PREFIX = "--ids=";
18 constexpr std::string_view LCQI_LLAMA_PROMPT_PREFIX = "--prompt=";
19 constexpr std::string_view LCQI_LLAMA_MAX_NEW_PREFIX = "--max-new=";
20 constexpr std::string_view LCQI_LLAMA_THREADS_PREFIX = "--threads=";
21 constexpr const char* LCQI_LLAMA_Q4_DIRECT_ENV = "LCQI_LLAMA_Q4_DIRECT";
22 constexpr const char* LCQI_LLAMA_GGML_DIRECT_ENV = "LCQI_LLAMA_GGML_DIRECT";
23 constexpr std::string_view LCQI_LLAMA_FALSE_ENV = "0";
24 constexpr std::string_view LCQI_LLAMA_TRUE_ENV = "1";
25
26 using Clock = std::chrono::steady_clock;
27
28 struct LlamaCliOptions {
29     std::filesystem::path gguf_path;
30     std::string prompt;
31     std::vector<std::int32_t> ids;

```



```

32     std::int32_t max_new_tokens = LCQI_LLAMA_DEFAULT_MAX_NEW_TOKENS;
33     std::int32_t threads = 0;
34     bool use_ids = false;
35     bool benchmark = false;
36     bool decode_text = false;
37 };
38
39 [[nodiscard]] double elapsed_ms(Clock::time_point begin, Clock::time_point end)↵
    ↵ {
40     return std::chrono::duration<double, std::milli>(end - begin).count();
41 }
42
43 void print_ids(std::string_view label, const std::vector<std::int32_t>& ids) {
44     std::cout << label;
45     for (const std::int32_t id : ids) {
46         std::cout << ' ' << id;
47     }
48     std::cout << "\n";
49 }
50
51 [[nodiscard]] std::int32_t parse_non_negative_i32(const std::string& text,
52                                                    const char* name) {
53     const int value = std::stoi(text);
54     if (value < 0) {
55         throw std::runtime_error(std::string(name) + " cannot be negative");
56     }
57     return static_cast<std::int32_t>(value);
58 }
59
60 [[nodiscard]] std::vector<std::int32_t> parse_ids(std::string_view text) {
61     std::vector<std::int32_t> ids;
62     std::string normalized(text);
63     for (char& ch : normalized) {
64         if (ch == ',') {
65             ch = ' ';
66         }
67     }
68     std::istringstream input(normalized);
69     std::int32_t id = 0;
70     while (input >> id) {
71         if (id < 0) {
72             throw std::runtime_error("token ids cannot be negative");
73         }
74         ids.push_back(id);
75     }
76     if (ids.empty()) {
77         throw std::runtime_error("--ids requires at least one token id");
78     }
79     return ids;
80 }
81
82 [[nodiscard]] bool consume_prefix(std::string_view arg,

```

```

83         std::string_view prefix,
84         std::string_view& value) {
85     if (arg.substr(0, prefix.size()) != prefix) {
86         return false;
87     }
88     value = arg.substr(prefix.size());
89     return true;
90 }
91
92 [[nodiscard]] LlamaCliOptions parse_options(int argc, char** argv) {
93     if (argc < 2) {
94         throw std::runtime_error(
95             "usage: lcqi_llama_gguf model.gguf [--prompt TEXT|--ids 1,2,3] "
96             "[--max-new N] [--threads N] [--benchmark] [--decode-text]");
97     }
98     LlamaCliOptions options;
99     options.gguf_path = argv[1];
100     for (int index = 2; index < argc; ++index) {
101         const std::string arg = argv[index];
102         std::string_view value;
103         if (arg == "--benchmark") {
104             options.benchmark = true;
105         } else if (arg == "--decode-text") {
106             options.decode_text = true;
107         } else if (arg == "--ids") {
108             if (index + 1 >= argc) {
109                 throw std::runtime_error("--ids requires a value");
110             }
111             ++index;
112             options.ids = parse_ids(argv[index]);
113             options.use_ids = true;
114         } else if (consume_prefix(arg, LCQI_LLAMA_IDS_PREFIX, value)) {
115             options.ids = parse_ids(value);
116             options.use_ids = true;
117         } else if (arg == "--prompt") {
118             if (index + 1 >= argc) {
119                 throw std::runtime_error("--prompt requires a value");
120             }
121             ++index;
122             options.prompt = argv[index];
123             options.use_ids = false;
124         } else if (consume_prefix(arg, LCQI_LLAMA_PROMPT_PREFIX, value)) {
125             options.prompt = std::string(value);
126             options.use_ids = false;
127         } else if (arg == "--max-new") {
128             if (index + 1 >= argc) {
129                 throw std::runtime_error("--max-new requires a value");
130             }
131             ++index;
132             options.max_new_tokens = parse_non_negative_i32(argv[index], "--max-↵
↵ -new");
133         } else if (consume_prefix(arg, LCQI_LLAMA_MAX_NEW_PREFIX, value)) {

```

```

134         options.max_new_tokens =
135             parse_non_negative_i32(std::string(value), "--max-new");
136     } else if (arg == "--threads") {
137         if (index + 1 >= argc) {
138             throw std::runtime_error("--threads requires a value");
139         }
140         ++index;
141         options.threads = parse_non_negative_i32(argv[index], "--threads");
142     } else if (consume_prefix(arg, LCQI_LLAMA_THREADS_PREFIX, value)) {
143         options.threads = parse_non_negative_i32(std::string(value), "--↵
↵ threads");
144     } else {
145         throw std::runtime_error("unknown option: " + arg);
146     }
147 }
148 if (!options.use_ids && options.prompt.empty()) {
149     options.prompt = "Hello";
150 }
151 return options;
152 }
153
154 [[nodiscard]] std::vector<std::int32_t> make_prompt_ids(const LlamaCliOptions& ↵
↵ options,
155                                                         lcqi::Gpt2Tokenizer* ↵
↵ tokenizer,
156                                                         std::int32_t ↵
↵ bos_token_id) {
157     if (options.use_ids) {
158         return options.ids;
159     }
160     return lcqi::llama_gguf_chat_prompt_ids(*tokenizer, options.prompt, ↵
↵ bos_token_id);
161 }
162
163 void print_benchmark(const lcqi::LlamaGgufLoadedModel& loaded,
164                     const lcqi::LlamaGgufGenerationResult& result,
165                     std::size_t prompt_size,
166                     std::size_t generated_new_tokens) {
167     const double first_token_tokens_per_second =
168         result.prefill_ms > 0.0 ? 1000.0 / result.prefill_ms : 0.0;
169     const double decode_tokens_per_second =
170         result.decode_ms > 0.0
171         ? static_cast<double>(result.decode_steps) * 1000.0 / result.↵
↵ decode_ms
172         : 0.0;
173     std::cout << "benchmark_manifest_ms " << loaded.report.manifest_ms << "\n";
174     std::cout << "benchmark_weight_load_ms " << loaded.report.weights_ms << "\n↵
↵ ";
175     std::cout << "benchmark_load_ms " << result.load_ms << "\n";
176     std::cout << "benchmark_prefill_ms " << result.prefill_ms << "\n";
177     std::cout << "benchmark_decode_ms " << result.decode_ms << "\n";
178     std::cout << "benchmark_total_ms " << result.total_ms << "\n";

```

```

179 std::cout << "benchmark_prompt_tokens " << prompt_size << "\n";
180 std::cout << "benchmark_generated_tokens " << generated_new_tokens << "\n";
181 std::cout << "benchmark_prefill_steps " << result.prefill_steps << "\n";
182 std::cout << "benchmark_decode_steps " << result.decode_steps << "\n";
183 std::cout << "benchmark_worker_count " << result.worker_count << "\n";
184 std::cout << "benchmark_first_token_tokens_per_second "
185         << first_token_tokens_per_second << "\n";
186 std::cout << "benchmark_decode_tokens_per_second " << ↵
↵ decode_tokens_per_second << "\n";
187 std::cout << "benchmark_kv_cache_bytes " << result.kv_cache_bytes << "\n";
188 std::cout << "benchmark_quantized_weight_bytes "
189         << loaded.report.quantized_weight_bytes << "\n";
190 std::cout << "benchmark_f32_weight_bytes " << loaded.report.↵
↵ f32_weight_bytes << "\n";
191 std::cout << "benchmark_direct_quantized_weight_bytes "
192         << loaded.report.direct_quantized_weight_bytes << "\n";
193 std::cout << "benchmark_fallback_dequantized_weight_bytes "
194         << loaded.report.fallback_dequantized_weight_bytes << "\n";
195 std::cout << "benchmark_tensors_loaded " << loaded.report.tensors_loaded <<↵
↵ "\n";
196 std::cout << "benchmark_q4_k_direct_tensors "
197         << loaded.report.q4_k_direct_tensors << "\n";
198 std::cout << "benchmark_ggml_direct_tensors "
199         << loaded.report.ggml_direct_tensors << "\n";
200 std::cout << "benchmark_f32_fallback_tensors "
201         << loaded.report.f32_fallback_tensors << "\n";
202 std::cout << "benchmark_hotspot_rms_norm_ms " << result.hotspots.↵
↵ rms_norm_ms << "\n";
203 std::cout << "benchmark_hotspot_attention_ms " << result.hotspots.↵
↵ attention_ms << "\n";
204 std::cout << "benchmark_hotspot_rope_ms " << result.hotspots.rope_ms << "\n↵
↵ ";
205 std::cout << "benchmark_hotspot_wq_ms " << result.hotspots.wq_ms << "\n";
206 std::cout << "benchmark_hotspot_wk_ms " << result.hotspots.wk_ms << "\n";
207 std::cout << "benchmark_hotspot_wv_ms " << result.hotspots.wv_ms << "\n";
208 std::cout << "benchmark_hotspot_wo_ms " << result.hotspots.wo_ms << "\n";
209 std::cout << "benchmark_hotspot_w_gate_ms " << result.hotspots.w_gate_ms <<↵
↵ "\n";
210 std::cout << "benchmark_hotspot_w_up_ms " << result.hotspots.w_up_ms << "\n↵
↵ ";
211 std::cout << "benchmark_hotspot_w_down_ms " << result.hotspots.w_down_ms <<↵
↵ "\n";
212 std::cout << "benchmark_hotspot_lm_head_ms " << result.hotspots.lm_head_ms ↵
↵ << "\n";
213 std::cout << "benchmark_hotspot_q4_k_direct_ms "
214         << result.hotspots.q4_k_direct_ms << "\n";
215 std::cout << "benchmark_hotspot_ggml_direct_ms "
216         << result.hotspots.ggml_direct_ms << "\n";
217 std::cout << "benchmark_hotspot_f32_fallback_ms "
218         << result.hotspots.f32_fallback_ms << "\n";
219 std::cout << "benchmark_hotspot_q4_k_direct_calls "
220         << result.hotspots.q4_k_direct_calls << "\n";

```

[illegible]

```

271     result.load_ms = load_ms;
272     result.total_ms = elapsed_ms(total_begin, Clock::now());
273
274     std::cout << "mode llama_gguf_reference\n";
275     std::cout << "weight_execution " << weight_execution_mode() << "\n";
276     std::cout << "model_path " << options.gguf_path.string() << "\n";
277     std::cout << "architecture " << loaded.model.architecture << "\n";
278     std::cout << "model_name " << loaded.model.name << "\n";
279     std::cout << "layers " << loaded.model.decoder.layers.size() << "\n";
280     std::cout << "hidden_size " << loaded.model.decoder.config.hidden_size <<
    ↪ << "\n";
281     std::cout << "query_heads " << loaded.model.decoder.config.query_heads <<
    ↪ << "\n";
282     std::cout << "kv_heads " << loaded.model.decoder.config.kv_heads << "\n"
    ↪ ";
283     std::cout << "head_dim " << loaded.model.decoder.config.head_dim << "\n"
    ↪ ";
284     std::cout << "vocab_size " << loaded.model.decoder.config.vocab_size <<
    ↪ "\n";
285     std::cout << "tie_lm_head_to_embedding "
286               << (loaded.model.lm_head_tied_to_embedding ? 1 : 0) << "\n";
287     print_ids("prompt_ids", result.prompt_ids);
288     print_ids("generated_ids", result.generated_ids);
289     std::cout << "predicted_first_token " << result.predicted_first_token <<
    ↪ << "\n";
290     if (options.decode_text) {
291         std::cout << "generated_text "
292               << lcqi::gpt2_decode(tokenizer, result.generated_ids) <<
    ↪ "\n";
293     }
294     if (options.benchmark) {
295         const std::size_t generated_new_tokens =
296             result.generated_ids.size() >= result.prompt_ids.size()
297             ? result.generated_ids.size() - result.prompt_ids.size()
298             : 0;
299         print_benchmark(loaded, result, result.prompt_ids.size(),
    ↪ generated_new_tokens);
300     }
301     return EXIT_SUCCESS;
302 } catch (const std::exception& error) {
303     std::cerr << "error: " << error.what() << "\n";
304     return EXIT_FAILURE;
305 }
306 }

```

**q4\_k\_avx2.cpp** 是本轮 Q4\_K 优化入口。它不是完整 llama.cpp 后端，只把 Q4\_KxQ8\_K 的 32 元素子块点积换成 AVX2 **maddubs** 归约，并提供 row-range unchecked 入口，供 LLaMA GGUF decoder 按输出行并行切分。读这份源码时要同时看 fallback：环境变量 **LCQI\_Q4K\_BACKEND=scalar** 可以强制走标量路径，报告脚本用它生成同机同模型的 A/B 证据；而端到端报告会验证 row-range 和 worker pool 是否真正改善 prefill/decode。

**Listing 16.13** src/q4\_k\_avx2.cpp

```

1 #include <lcqi/q4_k.hpp>
2
3 #if defined(LCQI_ENABLE_AVX2)
4
5 #include <immintrin.h>
6
7 #include <array>
8 #include <cmath>
9 #include <cstdint>
10 #include <cstdint>
11 #include <cstdlib>
12 #include <cstring>
13
14 namespace lcqi {
15 namespace {
16
17 constexpr std::uint32_t LCQI_Q4K_F32_EXPONENT_MASK = 0x7F800000U;
18 constexpr std::uint32_t LCQI_Q4K_F16_SIGN_MASK = 0x8000U;
19 constexpr std::uint32_t LCQI_Q4K_F16_EXPONENT_MASK = 0x7C00U;
20 constexpr std::uint32_t LCQI_Q4K_F16_MANTISSA_MASK = 0x03FFU;
21 constexpr std::uint32_t LCQI_Q4K_F16_EXPONENT_BIAS = 15U;
22 constexpr std::uint32_t LCQI_Q4K_F32_EXPONENT_BIAS = 127U;
23 constexpr std::uint32_t LCQI_Q4K_F32_MANTISSA_BITS = 23U;
24 constexpr std::uint32_t LCQI_Q4K_F16_MANTISSA_BITS = 10U;
25 constexpr std::uint32_t LCQI_Q4K_F16_TO_F32_SIGN_SHIFT = 16U;
26 constexpr std::uint32_t LCQI_Q4K_BYTE_BITS = 8U;
27 constexpr std::uint8_t LCQI_Q4K_LOW_NIBBLE_MASK = 0x0FU;
28 constexpr std::uint8_t LCQI_Q4K_SIX_BIT_MASK = 63U;
29 constexpr std::int64_t LCQI_Q4K_QS_OFFSET = 16;
30 constexpr std::int64_t LCQI_Q4K_QS_BYTES_PER_PAIR = 32;
31 constexpr std::int64_t LCQI_Q4K_SUBBLOCK_PAIR_COUNT = 4;
32 constexpr const char* LCQI_Q4K_BACKEND_ENV = "LCQI_Q4K_BACKEND";
33 constexpr const char* LCQI_Q4K_FORCE_SCALAR_ENV = "LCQI_Q4K_FORCE_SCALAR";
34 constexpr const char* LCQI_Q4K_BACKEND_SCALAR = "scalar";
35 constexpr const char* LCQI_Q4K_FORCE_TRUE = "1";
36
37 std::uint16_t read_le_u16(const std::uint8_t* bytes) noexcept {
38     return static_cast<std::uint16_t>(
39         static_cast<std::uint16_t>(bytes[0]) |
40         (static_cast<std::uint16_t>(bytes[1]) << LCQI_Q4K_BYTE_BITS));
41 }
42
43 float bits_to_float(std::uint32_t bits) noexcept {
44     float value = 0.0F;
45     std::memcpy(&value, &bits, sizeof(value));
46     return value;
47 }
48
49 float f16_to_f32(std::uint16_t bits) noexcept {
50     const std::uint32_t sign =
51         (static_cast<std::uint32_t>(bits) & LCQI_Q4K_F16_SIGN_MASK)
52         << LCQI_Q4K_F16_TO_F32_SIGN_SHIFT;

```



```

53  const std::uint32_t exponent =
54      static_cast<std::uint32_t>(bits) & LCQI_Q4K_F16_EXPONENT_MASK;
55  const std::uint32_t mantissa =
56      static_cast<std::uint32_t>(bits) & LCQI_Q4K_F16_MANTISSA_MASK;
57
58  if (exponent == 0U) {
59      if (mantissa == 0U) {
60          return bits_to_float(sign);
61      }
62      float value = static_cast<float>(mantissa) /
63          static_cast<float>(1U << LCQI_Q4K_F16_MANTISSA_BITS);
64      value = std::ldexp(value, 1 - static_cast<int>(<
    ↪ LCQI_Q4K_F16_EXPONENT_BIAS));
65      return sign == 0U ? value : -value;
66  }
67
68  if (exponent == LCQI_Q4K_F16_EXPONENT_MASK) {
69      const std::uint32_t f32_mantissa =
70          mantissa << (LCQI_Q4K_F32_MANTISSA_BITS - <
    ↪ LCQI_Q4K_F16_MANTISSA_BITS);
71      return bits_to_float(sign | LCQI_Q4K_F32_EXPONENT_MASK | f32_mantissa);
72  }
73
74  const std::uint32_t f32_exponent =
75      ((exponent >> LCQI_Q4K_F16_MANTISSA_BITS) - LCQI_Q4K_F16_EXPONENT_BIAS <
    ↪ +
76      LCQI_Q4K_F32_EXPONENT_BIAS)
77      << LCQI_Q4K_F32_MANTISSA_BITS;
78  const std::uint32_t f32_mantissa =
79      mantissa << (LCQI_Q4K_F32_MANTISSA_BITS - LCQI_Q4K_F16_MANTISSA_BITS);
80  return bits_to_float(sign | f32_exponent | f32_mantissa);
81 }
82
83 void get_scale_min_k4(std::int32_t index,
84     const std::uint8_t* packed,
85     std::uint8_t& scale,
86     std::uint8_t& min) noexcept {
87     if (index < LCQI_Q4K_SUBBLOCK_PAIR_COUNT) {
88         scale = packed[index] & LCQI_Q4K_SIX_BIT_MASK;
89         min = packed[index + LCQI_Q4K_SUBBLOCK_PAIR_COUNT] & <
    ↪ LCQI_Q4K_SIX_BIT_MASK;
90         return;
91     }
92     scale = static_cast<std::uint8_t>((packed[index + <
    ↪ LCQI_Q4K_SUBBLOCK_PAIR_COUNT] &
93         LCQI_Q4K_LOW_NIBBLE_MASK) |
94         ((packed[index - <
    ↪ LCQI_Q4K_SUBBLOCK_PAIR_COUNT] >> 6U)
95         << 4U));
96     min = static_cast<std::uint8_t>((packed[index + <
    ↪ LCQI_Q4K_SUBBLOCK_PAIR_COUNT] >> 4U) |
97         ((packed[index] >> 6U) << 4U));

```



```

98 }
99
100 std::int32_t horizontal_sum_i32(__m256i values) noexcept {
101     const __m128i low = _mm256_castsi256_si128(values);
102     const __m128i high = _mm256_extracti128_si256(values, 1);
103     __m128i sum = _mm_add_epi32(low, high);
104     sum = _mm_add_epi32(sum, _mm_shuffle_epi32(sum, _MM_SHUFFLE(1, 0, 3, 2)));
105     sum = _mm_add_epi32(sum, _mm_shuffle_epi32(sum, _MM_SHUFFLE(2, 3, 0, 1)));
106     return _mm_cvtsi128_si32(sum);
107 }
108
109 std::int32_t dot_32_u4_i8_avx2(__m256i q4_values, const std::int8_t* q8_values) ←
    ↪ noexcept {
110     const __m256i q8 = _mm256_loadu_si256(reinterpret_cast<const __m256i*>(←
    ↪ q8_values));
111     const __m256i pair_sums_i16 = _mm256_maddubs_epi16(q4_values, q8);
112     const __m256i ones = _mm256_set1_epi16(1);
113     const __m256i partial_i32 = _mm256_madd_epi16(pair_sums_i16, ones);
114     return horizontal_sum_i32(partial_i32);
115 }
116
117 float dot_block_q4_k_q8_avx2(const std::uint8_t* block, const Q8KBlock& input) ←
    ↪ noexcept {
118     const float d = f16_to_f32(read_le_u16(block));
119     const float dmin = f16_to_f32(read_le_u16(block + 2));
120     const std::uint8_t* scales = block + 4;
121     const std::uint8_t* qs = block + LCQI_Q4K_QS_OFFSET;
122     const __m256i nibble_mask = _mm256_set1_epi8(static_cast<char>(←
    ↪ LCQI_Q4K_LOW_NIBBLE_MASK));
123     std::int32_t weighted_sum = 0;
124     std::int32_t min_sum = 0;
125
126     for (std::int32_t pair = 0; pair < LCQI_Q4K_SUBBLOCK_PAIR_COUNT; ++pair) {
127         const __m256i packed =
128             _mm256_loadu_si256(reinterpret_cast<const __m256i*>(
129                 qs + pair * LCQI_Q4K_QS_BYTES_PER_PAIR));
130         const __m256i low_nibbles = _mm256_and_si256(packed, nibble_mask);
131         const __m256i high_nibbles =
132             _mm256_and_si256(_mm256_srli_epi16(packed, 4), nibble_mask);
133
134         const std::int32_t low_subblock = pair * 2;
135         const std::int32_t high_subblock = low_subblock + 1;
136         std::uint8_t low_scale = 0;
137         std::uint8_t low_min = 0;
138         std::uint8_t high_scale = 0;
139         std::uint8_t high_min = 0;
140         get_scale_min_k4(low_subblock, scales, low_scale, low_min);
141         get_scale_min_k4(high_subblock, scales, high_scale, high_min);
142
143         const std::int32_t low_dot =
144             dot_32_u4_i8_avx2(low_nibbles,
145                 input.qs.data() + low_subblock * ←

```

```

    ↪ LCQI_Q4_K_SUBBLOCK_VALUES);
146     const std::int32_t high_dot =
147         dot_32_u4_i8_avx2(high_nibbles,
148             input.qs.data() + high_subblock * ↪
    ↪ LCQI_Q4_K_SUBBLOCK_VALUES);
149     weighted_sum += static_cast<std::int32_t>(low_scale) * low_dot;
150     weighted_sum += static_cast<std::int32_t>(high_scale) * high_dot;
151
152     min_sum += static_cast<std::int32_t>(low_min) *
153         (static_cast<std::int32_t>(input.bsums[low_subblock * 2]) +
154             static_cast<std::int32_t>(input.bsums[low_subblock * 2 + ↪
    ↪ 1]));
155     min_sum += static_cast<std::int32_t>(high_min) *
156         (static_cast<std::int32_t>(input.bsums[high_subblock * 2]) +
157             static_cast<std::int32_t>(input.bsums[high_subblock * 2 + ↪
    ↪ 1]));
158 }
159
160 return (d * input.d) * static_cast<float>(weighted_sum) -
161     (dmin * input.d) * static_cast<float>(min_sum);
162 }
163
164 bool scalar_backend_forced() noexcept {
165     const char* backend = std::getenv(LCQI_Q4K_BACKEND_ENV);
166     if (backend != nullptr && std::strcmp(backend, LCQI_Q4K_BACKEND_SCALAR) == ↪
    ↪ 0) {
167         return true;
168     }
169     const char* force_scalar = std::getenv(LCQI_Q4K_FORCE_SCALAR_ENV);
170     return force_scalar != nullptr && std::strcmp(force_scalar, ↪
    ↪ LCQI_Q4K_FORCE_TRUE) == 0;
171 }
172
173 } // namespace
174
175 bool q4_k_q8_avx2_available() noexcept {
176     if (scalar_backend_forced()) {
177         return false;
178     }
179     #if defined(__GNUC__) || defined(__clang__)
180     __builtin_cpu_init();
181     return __builtin_cpu_supports("avx2");
182     #else
183     return true;
184     #endif
185 }
186
187 const char* q4_k_q8_active_backend() noexcept {
188     return q4_k_q8_avx2_available() ? "avx2_maddubs" : "scalar";
189 }
190
191 void matvec_q4_k_q8_avx2_rows_unchecked(std::span<const std::uint8_t> q4_rows,

```

```

192         std::int64_t row_count,
193         std::int64_t column_count,
194         std::span<const Q8KBlock> input,
195         std::span<float> output,
196         std::int64_t row_begin,
197         std::int64_t row_end) noexcept;
198
199 void matvec_q4_k_q8_avx2_unchecked(std::span<const std::uint8_t> q4_rows,
200                                     std::int64_t row_count,
201                                     std::int64_t column_count,
202                                     std::span<const Q8KBlock> input,
203                                     std::span<float> output) {
204     matvec_q4_k_q8_avx2_rows_unchecked(q4_rows,
205                                         row_count,
206                                         column_count,
207                                         input,
208                                         output,
209                                         0,
210                                         row_count);
211 }
212
213 void matvec_q4_k_q8_avx2_rows_unchecked(std::span<const std::uint8_t> q4_rows,
214                                         std::int64_t row_count,
215                                         std::int64_t column_count,
216                                         std::span<const Q8KBlock> input,
217                                         std::span<float> output,
218                                         std::int64_t row_begin,
219                                         std::int64_t row_end) noexcept {
220     (void)row_count;
221     const std::int64_t row_blocks = column_count / LCQI_QK_K_BLOCK_VALUES;
222     const std::int64_t row_bytes = row_blocks * LCQI_Q4_K_BLOCK_BYTES;
223     for (std::int64_t row = row_begin; row < row_end; ++row) {
224         float sum = 0.0F;
225         const std::uint8_t* row_data = q4_rows.data() + row * row_bytes;
226         for (std::int64_t block = 0; block < row_blocks; ++block) {
227             sum += dot_block_q4_k_q8_avx2(
228                 row_data + block * LCQI_Q4_K_BLOCK_BYTES,
229                 input[static_cast<std::size_t>(block)]);
230         }
231         output[static_cast<std::size_t>(row)] = sum;
232     }
233 }
234
235 } // namespace lcqi
236
237 #endif

```

## 外部模型 Smoke 与对比脚本

`run_smollm2_small_smoke.py` 下载并校验 SmolLM2-135M-Instruct 的 GGUF 量化文件，构建固定 commit 的 `llama-simple-chat`，在 CPU 上运行一次真实开源 small 模型生成，并输出可复查报告。它是外部真实模型 smoke，不替代 LCQI tiny golden trace。

Listing 16.14 tools/run\_smollm2\_small\_smoke.py

```

1  #!/usr/bin/env python3
2  from __future__ import annotations
3
4  import argparse
5  import datetime as dt
6  import hashlib
7  import os
8  import re
9  import shlex
10 import shutil
11 import subprocess
12 import sys
13 from dataclasses import dataclass
14 from pathlib import Path
15
16
17 MODEL_SOURCE_ID = "HuggingFaceTB/SmolLM2-135M-Instruct"
18 MODEL_GGUF_REPO_ID = "bartowski/SmolLM2-135M-Instruct-GGUF"
19 MODEL_FILE_NAME = "SmolLM2-135M-Instruct-Q4_K_M.gguf"
20 MODEL_URL = (
21     "https://huggingface.co/bartowski/SmolLM2-135M-Instruct-GGUF"
22     f"/resolve/main/{MODEL_FILE_NAME}"
23 )
24 MODEL_EXPECTED_BYTES = 105454432
25 MODEL_EXPECTED_SHA256 = (
26     "2e8040ceae7815abe0dcb3540b9995eaa1fa0d2ca9e797d0a635ae4433c68c2d"
27 )
28 LLAMA_CPP_REPO_URL = "https://github.com/ggml-org/llama.cpp.git"
29 LLAMA_CPP_COMMIT = "b3fed31b99f9bd37725833674252bccb429bb183"
30 LLAMA_TARGET = "llama-simple-chat"
31 DEFAULT_PROMPT = "What is 2+3? Answer in one short sentence."
32 DEFAULT_CONTEXT_TOKENS = 512
33 DEFAULT_TIMEOUT_SECONDS = 180
34 DEFAULT_BUILD_JOBS = 2
35
36
37 @dataclass(frozen=True)
38 class CommandResult:
39     args: list[str]
40     returncode: int
41     stdout: str
42     stderr: str
43
44
45 def script_path() -> Path:
46     return Path(__file__).resolve()
47
48
49 def volume_dir() -> Path:
50     return script_path().parents[1]
51

```

```

52
53 def repo_dir() -> Path:
54     return script_path().parents[3]
55
56
57 def default_cache_dir() -> Path:
58     explicit_cache = os.environ.get("LCQI_SMOLLM2_CACHE_DIR")
59     if explicit_cache:
60         return Path(explicit_cache).expanduser()
61     xdg_cache = os.environ.get("XDG_CACHE_HOME")
62     cache_root = Path(xdg_cache).expanduser() if xdg_cache else Path.home() / "↵
↵ .cache"
63     return cache_root / "lcqi-smolllm2-small-smoke"
64
65
66 def default_report_path() -> Path:
67     return volume_dir() / "results" / "lcqi-smolllm2-small-model-smoke.txt"
68
69
70 def command_line(args: list[str]) -> str:
71     return " ".join(shlex.quote(arg) for arg in args)
72
73
74 def require_tool(name: str) -> str:
75     path = shutil.which(name)
76     if path is None:
77         raise RuntimeError(f"required tool not found in PATH: {name}")
78     return path
79
80
81 def run_command(
82     args: list[str],
83     *,
84     cwd: Path | None = None,
85     input_text: str | None = None,
86     timeout_seconds: int | None = None,
87 ) -> CommandResult:
88     try:
89         completed = subprocess.run(
90             args,
91             cwd=str(cwd) if cwd else None,
92             input=input_text,
93             text=True,
94             capture_output=True,
95             timeout=timeout_seconds,
96             check=False,
97         )
98     except subprocess.TimeoutExpired as exc:
99         stdout = exc.stdout if isinstance(exc.stdout, str) else ""
100         stderr = exc.stderr if isinstance(exc.stderr, str) else ""
101         return CommandResult(args=args, returncode=124, stdout=stdout, stderr=↵
↵ stderr)

```

```

102
103     return CommandResult(
104         args=args,
105         returncode=completed.returncode,
106         stdout=completed.stdout,
107         stderr=completed.stderr,
108     )
109
110
111 def ensure_success(result: CommandResult, action: str) -> None:
112     if result.returncode == 0:
113         return
114     raise RuntimeError(
115         f"{action} failed with exit code {result.returncode}\n"
116         f"command: {command_line(result.args)}\n"
117         f"stdout tail:\n{tail(result.stdout)}\n"
118         f"stderr tail:\n{tail(result.stderr)}"
119     )
120
121
122 def tail(text: str, max_chars: int = 4000) -> str:
123     if len(text) <= max_chars:
124         return text
125     return text[-max_chars:]
126
127
128 def sha256_file(path: Path) -> str:
129     digest = hashlib.sha256()
130     with path.open("rb") as handle:
131         for block in iter(lambda: handle.read(1024 * 1024), b''):
132             digest.update(block)
133     return digest.hexdigest()
134
135
136 def verify_model_file(path: Path) -> tuple[bool, str, int]:
137     if not path.exists():
138         return False, "", 0
139     actual_bytes = path.stat().st_size
140     actual_sha = sha256_file(path)
141     return (
142         actual_bytes == MODEL_EXPECTED_BYTES
143         and actual_sha == MODEL_EXPECTED_SHA256,
144         actual_sha,
145         actual_bytes,
146     )
147
148
149 def download_model(model_path: Path, *, force_download: bool, no_download: bool ↵
    ↵ ) -> None:
150     model_path.parent.mkdir(parents=True, exist_ok=True)
151
152     if force_download and model_path.exists():

```

```

153     model_path.unlink()
154
155     is_valid, actual_sha, actual_bytes = verify_model_file(model_path)
156     if is_valid:
157         print(f"[lcqi-smoke] model cache hit: {model_path}")
158         return
159
160     if model_path.exists():
161         print(
162             "[lcqi-smoke] cached model hash mismatch, redownloading: "
163             f"bytes={actual_bytes}, sha256={actual_sha}"
164         )
165         model_path.unlink()
166
167     if no_download:
168         raise RuntimeError(
169             f"model file is missing or invalid and --no-download was set: {↵
↵ model_path}"
170         )
171
172     require_tool("curl")
173     part_path = model_path.with_suffix(model_path.suffix + ".part")
174     if part_path.exists():
175         part_path.unlink()
176
177     print(f"[lcqi-smoke] downloading small GGUF model: {MODEL_GGUF_REPO_ID}/{↵
↵ MODEL_FILE_NAME}")
178     result = run_command(
179         [
180             "curl",
181             "-L",
182             "--fail",
183             "--retry",
184             "3",
185             "-o",
186             str(part_path),
187             MODEL_URL,
188         ]
189     )
190     ensure_success(result, "download SmolLM2 GGUF")
191
192     downloaded_ok, downloaded_sha, downloaded_bytes = verify_model_file(↵
↵ part_path)
193     if not downloaded_ok:
194         raise RuntimeError(
195             "downloaded model did not match expected identity: "
196             f"bytes={downloaded_bytes}, sha256={downloaded_sha}"
197         )
198     part_path.replace(model_path)
199     print(f"[lcqi-smoke] model verified: sha256={MODEL_EXPECTED_SHA256}")
200
201

```

```

202 def ensure_llama_simple_chat(cache_dir: Path, *, jobs: int, skip_build: bool) ←
    ↪ -> Path:
203     cached_binary = cache_dir / "llama.cpp" / "build" / "bin" / LLAMA_TARGET
204     source_dir = cache_dir / "llama.cpp"
205     build_dir = source_dir / "build"
206     if cached_binary.exists() and os.access(cached_binary, os.X_OK) and ↪
    ↪ is_pinned_llama_checkout(source_dir):
207         print(f"[lcqi-smoke] llama.cpp binary cache hit: {cached_binary}")
208         return cached_binary
209
210     if skip_build:
211         raise RuntimeError(f"llama.cpp binary is missing and --skip-build was ↪
    ↪ set: {cached_binary}")
212
213     require_tool("git")
214     require_tool("cmake")
215
216     cache_dir.mkdir(parents=True, exist_ok=True)
217
218     if not (source_dir / ".git").exists():
219         print(f"[lcqi-smoke] cloning llama.cpp into cache: {source_dir}")
220         result = run_command(
221             [
222                 "git",
223                 "clone",
224                 "--filter=blob:none",
225                 "--no-checkout",
226                 LLAMA_CPP_REPO_URL,
227                 str(source_dir),
228             ]
229         )
230         ensure_success(result, "clone llama.cpp")
231
232     print(f"[lcqi-smoke] checking out pinned llama.cpp commit: {↪
    ↪ LLAMA_CPP_COMMIT}")
233     fetch = run_command(
234         ["git", "-C", str(source_dir), "fetch", "--depth", "1", "origin", ↪
    ↪ LLAMA_CPP_COMMIT]
235     )
236     ensure_success(fetch, "fetch pinned llama.cpp commit")
237     checkout = run_command(["git", "-C", str(source_dir), "checkout", "--detach↪
    ↪ ", LLAMA_CPP_COMMIT])
238     ensure_success(checkout, "checkout pinned llama.cpp commit")
239
240     print(f"[lcqi-smoke] configuring llama.cpp build: {build_dir}")
241     configure = run_command(
242         [
243             "cmake",
244             "-S",
245             str(source_dir),
246             "-B",
247             str(build_dir),

```



```

248         "-DCMAKE_BUILD_TYPE=Release",
249         "-DLLAMA_BUILD_TESTS=OFF",
250         "-DLLAMA_BUILD_EXAMPLES=ON",
251         "-DLLAMA_BUILD_SERVER=OFF",
252         "-DGGML_NATIVE=OFF",
253     ]
254 )
255 ensure_success(configure, "configure llama.cpp")
256
257 print(f"[lcqi-smoke] building {LLAMA_TARGET} with -j{jobs}")
258 build = run_command(
259     ["cmake", "--build", str(build_dir), "--target", LLAMA_TARGET, "-j", ←
↪ str(jobs)]
260 )
261 ensure_success(build, f"build {LLAMA_TARGET}")
262
263 if not cached_binary.exists():
264     raise RuntimeError(f"expected binary was not produced: {cached_binary}" ←
↪ )
265 return cached_binary
266
267
268 def is_pinned_llama_checkout(source_dir: Path) -> bool:
269     if not (source_dir / ".git").exists():
270         return False
271     result = run_command(["git", "-C", str(source_dir), "rev-parse", "HEAD"])
272     if result.returncode != 0:
273         return False
274     return result.stdout.strip() == LLAMA_CPP_COMMIT
275
276
277 def strip_ansi(text: str) -> str:
278     return re.sub(r"\x1b\[([0-?]*[ -/]*[@-~])", "", text)
279
280
281 def extract_response(stdout: str) -> str:
282     cleaned = strip_ansi(stdout).replace("\r", "")
283     chunks = [chunk.strip() for chunk in re.split(r"(:^|\n)> ?", cleaned) if ←
↪ chunk.strip()]
284     if not chunks:
285         return ""
286     return chunks[0]
287
288
289 def run_model_smoke(
290     llama_binary: Path,
291     model_path: Path,
292     *,
293     prompt: str,
294     context_tokens: int,
295     timeout_seconds: int,
296 ) -> tuple[CommandResult, str]:

```

```

297     command = [
298         str(llama_binary),
299         "-m",
300         str(model_path),
301         "-ngl",
302         "0",
303         "-c",
304         str(context_tokens),
305     ]
306     print("[lcqi-smoke] running real small model generation")
307     result = run_command(
308         command,
309         input_text=prompt.rstrip("\n") + "\n\n",
310         timeout_seconds=timeout_seconds,
311     )
312     ensure_success(result, "run small model smoke")
313
314     response = extract_response(result.stdout)
315     if not response:
316         raise RuntimeError(
317             "small model smoke exited successfully but produced no assistant ↵
↵ text\n"
318             f"stdout:\n{result.stdout}\n"
319             f"stderr:\n{result.stderr}"
320         )
321     return result, response
322
323
324 def current_repo_commit() -> str:
325     result = run_command(["git", "-C", str(repo_dir()), "rev-parse", "--short", ↵
↵ "HEAD"])
326     if result.returncode != 0:
327         return "unknown"
328     return result.stdout.strip()
329
330
331 def current_uname() -> str:
332     result = run_command(["uname", "-a"])
333     if result.returncode != 0:
334         return "unknown"
335     return result.stdout.strip()
336
337
338 def write_report(
339     report_path: Path,
340     *,
341     cache_dir: Path,
342     llama_binary: Path,
343     model_path: Path,
344     prompt: str,
345     result: CommandResult,
346     response: str,

```

```

347 ) -> None:
348     report_path.parent.mkdir(parents=True, exist_ok=True)
349     clean_stdout = strip_ansi(result.stdout).strip()
350     report = "\n".join(
351         [
352             "LCQI SmolLM2 Small Model Smoke",
353             "",
354             f"timestamp_utc={dt.datetime.now(dt.UTC).isoformat()}",
355             f"repo_commit={current_repo_commit()}",
356             f"host={current_uname()}",
357             f"python={sys.version.split()[0]}",
358             f"cache_dir={cache_dir}",
359             "",
360             f"model_source_id={MODEL_SOURCE_ID}",
361             f"model_gguf_repo_id={MODEL_GGUF_REPO_ID}",
362             f"model_file={MODEL_FILE_NAME}",
363             f"model_url={MODEL_URL}",
364             f"model_expected_bytes={MODEL_EXPECTED_BYTES}",
365             f"model_expected_sha256={MODEL_EXPECTED_SHA256}",
366             f"model_path={model_path}",
367             "",
368             f"llama_cpp_repo={LLAMA_CPP_REPO_URL}",
369             f"llama_cpp_commit={LLAMA_CPP_COMMIT}",
370             f"llama_target={LLAMA_TARGET}",
371             f"llama_binary={llama_binary}",
372             "",
373             f"prompt={prompt}",
374             f"command={command_line(result.args)}",
375             f"exit_code={result.returncode}",
376             "",
377             "validation=PASS: model hash matched, llama-simple-chat exited 0, "
378             "assistant response was non-empty",
379             "",
380             "assistant_response:",
381             response,
382             "",
383             "stdout_clean_tail:",
384             tail(clean_stdout, 2000),
385             "",
386             "stderr_tail:",
387             tail(result.stderr.strip(), 2000),
388             "",
389         ]
390     )
391     report_path.write_text(report, encoding="utf-8")
392
393
394 def parse_args() -> argparse.Namespace:
395     parser = argparse.ArgumentParser(
396         description=(
397             "Run a real local small open-source model smoke for CPU Volume 3. "
398             "The script caches llama.cpp and the GGUF model outside the Git ←

```

```

↪ repository."
399     )
400 )
401 parser.add_argument("--cache-dir", type=Path, default=default_cache_dir())
402 parser.add_argument("--model-path", type=Path, default=None)
403 parser.add_argument("--llama-bin", type=Path, default=None)
404 parser.add_argument("--report", type=Path, default=default_report_path())
405 parser.add_argument("--prompt", default=DEFAULT_PROMPT)
406 parser.add_argument("--ctx", type=int, default=DEFAULT_CONTEXT_TOKENS)
407 parser.add_argument("--timeout", type=int, default=DEFAULT_TIMEOUT_SECONDS)
408 parser.add_argument("--jobs", type=int, default=DEFAULT_BUILD_JOBS)
409 parser.add_argument("--force-download", action="store_true")
410 parser.add_argument("--no-download", action="store_true")
411 parser.add_argument("--skip-build", action="store_true")
412 return parser.parse_args()
413
414
415 def main() -> int:
416     args = parse_args()
417     cache_dir = args.cache_dir.expanduser().resolve()
418     model_path = (
419         args.model_path.expanduser().resolve()
420         if args.model_path
421         else cache_dir / "models" / MODEL_FILE_NAME
422     )
423     report_path = args.report.expanduser().resolve()
424
425     try:
426         download_model(
427             model_path,
428             force_download=args.force_download,
429             no_download=args.no_download,
430         )
431         if args.llama_bin:
432             llama_binary = args.llama_bin.expanduser().resolve()
433             if not llama_binary.exists() or not os.access(llama_binary, os.X_OK↪
↪ ):
434                 raise RuntimeError(f"--llama-bin is not executable: {↪
↪ llama_binary}")
435             else:
436                 llama_binary = ensure_llama_simple_chat(
437                     cache_dir,
438                     jobs=args.jobs,
439                     skip_build=args.skip_build,
440                 )
441
442             result, response = run_model_smoke(
443                 llama_binary,
444                 model_path,
445                 prompt=args.prompt,
446                 context_tokens=args.ctx,
447                 timeout_seconds=args.timeout,

```

```

448         )
449         write_report(
450             report_path,
451             cache_dir=cache_dir,
452             llama_binary=llama_binary,
453             model_path=model_path,
454             prompt=args.prompt,
455             result=result,
456             response=response,
457         )
458     except RuntimeError as exc:
459         print(f"[lcqi-smoke] ERROR: {exc}", file=sys.stderr)
460         return 1
461
462     print(f"report={report_path}")
463     print(f"model_sha256={MODEL_EXPECTED_SHA256}")
464     print(f"assistant_response={response}")
465     return 0
466
467
468 if __name__ == "__main__":
469     raise SystemExit(main())

```

`run_smollm2_q4_k_benchmark.py` 复用同一个 SmolLM2 GGUF 缓存，构建 LCQI 的 `lcqi_gguf_q4_bench`，并可附带 `llama.cpp llama-bench` 端到端参考。它会先用 `LCQI_Q4K_BACKEND=scalar` 跑标量路径，再跑 `auto` 后端，因此报告里能直接看到 `avx2_maddubs` 相对 `scalar` 的提升。报告要同时说明模型 SHA-256、tensor 名、shape、量化字节、f32 解码字节、误差、checksum 和成熟引擎参考，避免把不同层级数字混在一起。

**Listing 16.15** `tools/run_smollm2_q4_k_benchmark.py`

```

1  #!/usr/bin/env python3
2  from __future__ import annotations
3
4  import argparse
5  import datetime as dt
6  import os
7  import re
8  import shlex
9  import subprocess
10 import sys
11 from dataclasses import dataclass
12 from pathlib import Path
13
14 import run_smollm2_small_smoke
15
16
17 DEFAULT_BUILD_JOBS = 2
18 DEFAULT_REPEAT = 20
19 DEFAULT_TIMEOUT_SECONDS = 300
20 BENCH_TARGET = "lcqi_gguf_q4_bench"
21
22

```

```

23 @dataclass(frozen=True)
24 class CommandResult:
25     name: str
26     args: list[str]
27     returncode: int
28     stdout: str
29     stderr: str
30     metrics: dict[str, str]
31
32
33 def volume_dir() -> Path:
34     return Path(__file__).resolve().parents[1]
35
36
37 def repo_dir() -> Path:
38     return Path(__file__).resolve().parents[3]
39
40
41 def default_build_dir() -> Path:
42     return volume_dir() / "build" / "lcqi-release"
43
44
45 def default_report_path() -> Path:
46     return volume_dir() / "results" / "lcqi-smollm2-q4-k-benchmark.txt"
47
48
49 def command_line(args: list[str]) -> str:
50     return " ".join(shlex.quote(arg) for arg in args)
51
52
53 def run_command(
54     name: str,
55     args: list[str],
56     *,
57     timeout_seconds: int,
58     cwd: Path | None = None,
59     env: dict[str, str] | None = None,
60 ) -> CommandResult:
61     command_env = os.environ.copy()
62     if env:
63         command_env.update(env)
64     try:
65         completed = subprocess.run(
66             args,
67             cwd=str(cwd) if cwd else str(repo_dir()),
68             env=command_env,
69             text=True,
70             capture_output=True,
71             timeout=timeout_seconds,
72             check=False,
73         )
74     except subprocess.TimeoutExpired as exc:

```

```

75         stdout = exc.stdout if isinstance(exc.stdout, str) else ""
76         stderr = exc.stderr if isinstance(exc.stderr, str) else ""
77         return CommandResult(name, args, 124, stdout, stderr, {})
78     return CommandResult(
79         name=name,
80         args=args,
81         returncode=completed.returncode,
82         stdout=completed.stdout,
83         stderr=completed.stderr,
84         metrics=parse_key_value_metrics(completed.stdout),
85     )
86
87
88 def ensure_success(result: CommandResult, action: str) -> None:
89     if result.returncode == 0:
90         return
91     raise RuntimeError(
92         f"{action} failed with exit code {result.returncode}\n"
93         f"command: {command_line(result.args)}\n"
94         f"stdout tail:\n{tail(result.stdout)}\n"
95         f"stderr tail:\n{tail(result.stderr)}"
96     )
97
98
99 def tail(text: str, max_chars: int = 4000) -> str:
100     return text if len(text) <= max_chars else text[-max_chars:]
101
102
103 def parse_key_value_metrics(stdout: str) -> dict[str, str]:
104     metrics: dict[str, str] = {}
105     for line in stdout.splitlines():
106         key, sep, value = line.partition("=")
107         if sep:
108             metrics[key.strip()] = value.strip()
109     return metrics
110
111
112 def parse_llama_bench_metrics(stdout: str) -> dict[str, str]:
113     metrics: dict[str, str] = {}
114     for line in stdout.splitlines():
115         stripped = line.strip()
116         if not stripped.startswith("|"):
117             continue
118         columns = [column.strip() for column in stripped.strip("|").split("|")]
119         if len(columns) != 7 or columns[0] == "model" or columns[0].startswith(↵
↵ "-"):
120             continue
121         metrics["llama_model"] = columns[0]
122         metrics["llama_size"] = columns[1]
123         metrics["llama_params"] = columns[2]
124         metrics["llama_backend"] = columns[3]
125         metrics["llama_threads"] = columns[4]

```

```

126     tokens_per_second = re.sub(r"\s+.*$", "", columns[6])
127     if columns[5].startswith("pp"):
128         metrics["llama_prefill_test"] = columns[5]
129         metrics["llama_prefill_tps"] = tokens_per_second
130     if columns[5].startswith("tg"):
131         metrics["llama_decode_test"] = columns[5]
132         metrics["llama_decode_tps"] = tokens_per_second
133     return metrics
134
135
136 def build_lcqi(build_dir: Path, jobs: int, timeout_seconds: int) -> Path:
137     configure = run_command(
138         "cmake_configure",
139         [
140             "cmake",
141             "-S",
142             str(volume_dir()),
143             "-B",
144             str(build_dir),
145             "-DCMAKE_BUILD_TYPE=Release",
146         ],
147         timeout_seconds=timeout_seconds,
148     )
149     ensure_success(configure, "configure LCQI")
150     build = run_command(
151         "cmake_build_lcqi_gguf_q4_bench",
152         [
153             "cmake",
154             "--build",
155             str(build_dir),
156             "--target",
157             BENCH_TARGET,
158             "-j",
159             str(jobs),
160         ],
161         timeout_seconds=timeout_seconds,
162     )
163     ensure_success(build, "build LCQI Q4_K benchmark")
164     binary = build_dir / "labs" / "linux_cpu_inference" / BENCH_TARGET
165     if not binary.exists() or not os.access(binary, os.X_OK):
166         raise RuntimeError(f"LCQI Q4_K benchmark binary not found: {binary}")
167     return binary
168
169
170 def llama_bench_binary(cache_dir: Path) -> Path:
171     return cache_dir / "llama.cpp" / "build" / "bin" / "llama-bench"
172
173
174 def model_path(cache_dir: Path) -> Path:
175     return cache_dir / "models" / run_smollm2_small_smoke.MODEL_FILE_NAME
176
177

```



```

178 def run_lcqi_benchmark(
179     binary: Path,
180     model: Path,
181     *,
182     name: str,
183     tensor: str,
184     repeat: int,
185     timeout_seconds: int,
186     env: dict[str, str] | None = None,
187 ) -> CommandResult:
188     return run_command(
189         name,
190         [str(binary), str(model), tensor, str(repeat)],
191         timeout_seconds=timeout_seconds,
192         env=env,
193     )
194
195
196 def run_llama_bench(cache_dir: Path, timeout_seconds: int) -> CommandResult | ←
    ↪ None:
197     binary = llama_bench_binary(cache_dir)
198     model = model_path(cache_dir)
199     if not binary.exists() or not os.access(binary, os.X_OK) or not model.↪
    ↪ exists():
200         return None
201     result = run_command(
202         "llama_cpp_smollm2_q4_k_m",
203         [
204             str(binary),
205             "-m",
206             str(model),
207             "-ngl",
208             "0",
209             "-p",
210             "5",
211             "-n",
212             "4",
213             "-r",
214             "1",
215         ],
216         timeout_seconds=timeout_seconds,
217     )
218     return CommandResult(
219         result.name,
220         result.args,
221         result.returncode,
222         result.stdout,
223         result.stderr,
224         parse_llama_bench_metrics(result.stdout),
225     )
226
227

```

```

228 def current_commit() -> str:
229     result = subprocess.run(
230         ["git", "-C", str(repo_dir()), "rev-parse", "--short", "HEAD"],
231         text=True,
232         capture_output=True,
233         check=False,
234     )
235     return result.stdout.strip() if result.returncode == 0 else "unknown"
236
237
238 def working_tree_dirty() -> str:
239     result = subprocess.run(
240         ["git", "-C", str(repo_dir()), "status", "--porcelain"],
241         text=True,
242         capture_output=True,
243         check=False,
244     )
245     if result.returncode != 0:
246         return "unknown"
247     return "true" if result.stdout.strip() else "false"
248
249
250 def write_report(path: Path, runs: list[CommandResult], *, model: Path) -> None ←
    ↪ :
251     path.parent.mkdir(parents=True, exist_ok=True)
252     model_matches, actual_sha256, actual_bytes = run_smolllm2_small_smoke. ←
    ↪ verify_model_file(model)
253     lines = [
254         "LCQI SmolLM2 Q4_K Benchmark",
255         "",
256         f"timestamp_utc={dt.datetime.now(dt.UTC).isoformat()}",
257         f"repo_commit={current_commit()}",
258         f"working_tree_dirty={working_tree_dirty()}",
259         f"model={model}",
260         f"model_source={run_smolllm2_small_smoke.MODEL_GGUF_REPO_ID}",
261         f"model_file={run_smolllm2_small_smoke.MODEL_FILE_NAME}",
262         f"model_expected_bytes={run_smolllm2_small_smoke.MODEL_EXPECTED_BYTES}",
263         f"model_actual_bytes={actual_bytes}",
264         f"model_expected_sha256={run_smolllm2_small_smoke.MODEL_EXPECTED_SHA256} ←
    ↪ ",
265         f"model_actual_sha256={actual_sha256}",
266         f"model_identity_match={str(model_matches).lower()}",
267         f"llama_cpp_commit={run_smolllm2_small_smoke.LLAMA_CPP_COMMIT}",
268         "",
269         "scope=LCQI reads the same SmolLM2 Q4_K_M GGUF used by llama.cpp and ←
    ↪ runs a "
270         "single real Q4_K tensor matvec slice. llama.cpp numbers are end-to-end ←
    ↪ mature "
271         "engine context, not same-function microkernel parity.",
272         "",
273     ]
274     for run in runs:

```

```

275     lines.extend(
276         [
277             f"[{run.name}]",
278             f"returncode={run.returncode}",
279             f"command={command_line(run.args)}",
280         ]
281     )
282     for key in sorted(run.metrics):
283         lines.append(f"{key}={run.metrics[key]}")
284     lines.extend(
285         [
286             "stdout_tail:",
287             tail(run.stdout.strip(), 2400),
288             "stderr_tail:",
289             tail(run.stderr.strip(), 2400),
290             "",
291         ]
292     )
293     scalar = next((run for run in runs if run.name == "lcqi_smollm2_q4_k_scalar"
↵"), None)
294     auto = next((run for run in runs if run.name == "lcqi_smollm2_q4_k_auto"),
↵ None)
295     if scalar is not None and auto is not None:
296         scalar_matvec = metric_as_float(scalar.metrics, "
↵ q4_q8_matvec_average_us")
297         auto_matvec = metric_as_float(auto.metrics, "q4_q8_matvec_average_us")
298         scalar_total = metric_as_float(scalar.metrics, "q4_q8_total_average_us"
↵ )
299         auto_total = metric_as_float(auto.metrics, "q4_q8_total_average_us")
300         if scalar_matvec is not None and auto_matvec is not None and
↵ auto_matvec > 0.0:
301             lines.append("[lcqi_auto_vs_scalar]")
302             lines.append(f"q4_q8_matvec_speedup_auto_vs_scalar={scalar_matvec /
↵ auto_matvec:.3f}")
303             if scalar_total is not None and auto_total is not None and
↵ auto_total > 0.0:
304                 lines.append(f"q4_q8_total_speedup_auto_vs_scalar={scalar_total
↵ / auto_total:.3f}")
305             lines.append("")
306     while lines and lines[-1] == "":
307         lines.pop()
308     path.write_text("\n".join(lines) + "\n", encoding="utf-8")
309
310
311 def metric_as_float(metrics: dict[str, str], key: str) -> float | None:
312     value = metrics.get(key)
313     if value is None:
314         return None
315     try:
316         return float(value)
317     except ValueError:
318         return None

```

```

319
320
321 def parse_args() -> argparse.Namespace:
322     parser = argparse.ArgumentParser(
323         description="Benchmark LCQI Q4_K matvec on the same SmolLM2 GGUF used ←
↪ by llama.cpp."
324     )
325     parser.add_argument("--cache-dir", type=Path, default=↪
↪ run_smolllm2_small_smoke.default_cache_dir())
326     parser.add_argument("--build-dir", type=Path, default=default_build_dir())
327     parser.add_argument("--report", type=Path, default=default_report_path())
328     parser.add_argument("--tensor", default="auto")
329     parser.add_argument("--repeat", type=int, default=DEFAULT_REPEAT)
330     parser.add_argument("--jobs", type=int, default=DEFAULT_BUILD_JOBS)
331     parser.add_argument("--timeout-seconds", type=int, default=↪
↪ DEFAULT_TIMEOUT_SECONDS)
332     parser.add_argument("--no-download", action="store_true")
333     parser.add_argument("--skip-llama", action="store_true")
334     return parser.parse_args()
335
336
337 def main() -> int:
338     args = parse_args()
339     if args.repeat <= 0:
340         print("[lcqi-smolllm2-q4] --repeat must be positive", file=sys.stderr)
341         return 1
342     try:
343         cache_dir = args.cache_dir.expanduser().resolve()
344         model = model_path(cache_dir)
345         run_smolllm2_small_smoke.download_model(
346             model,
347             force_download=False,
348             no_download=args.no_download,
349         )
350         binary = build_lcqi(args.build_dir, args.jobs, args.timeout_seconds)
351         runs = [
352             run_lcqi_benchmark(
353                 binary,
354                 model,
355                 name="lcqi_smolllm2_q4_k_scalar",
356                 tensor=args.tensor,
357                 repeat=args.repeat,
358                 timeout_seconds=args.timeout_seconds,
359                 env={"LCQI_Q4K_BACKEND": "scalar"},
360             ),
361             run_lcqi_benchmark(
362                 binary,
363                 model,
364                 name="lcqi_smolllm2_q4_k_auto",
365                 tensor=args.tensor,
366                 repeat=args.repeat,
367                 timeout_seconds=args.timeout_seconds,

```

```

368     )
369 ]
370 if not args.skip_llama:
371     llama = run_llama_bench(cache_dir, args.timeout_seconds)
372     if llama is not None:
373         runs.append(llama)
374     write_report(args.report, runs, model=model)
375 except Exception as error:
376     print(f"[lcqi-smollm2-q4] error: {error}", file=sys.stderr)
377     return 1
378
379 for run in runs:
380     print(f"{run.name}: returncode={run.returncode}")
381     for key in sorted(run.metrics):
382         print(f"  {key}={run.metrics[key]}")
383     print(f"report={args.report}")
384     return 0 if all(run.returncode == 0 for run in runs) else 1
385
386
387 if __name__ == "__main__":
388     raise SystemExit(main())

```

`run_smollm2_same_input_compare.py` 是 SmolLM2 同输入对齐脚本。它构建 LCQI 和固定 commit 的 llama.cpp, 先用 `LCQI_LLAMA_Q4_DIRECT=0` 跑 LCQI baseline, 再用 `--threads1` 跑串行默认 Q4 direct, 再跑自动 worker 的默认 Q4 direct 路径; 可选再用 `LCQI_LLAMA_GGML_DIRECT=1` 跑 `Q5_0/Q6_K/Q8_0` 实验 direct 路径, 最后跑 `llama-simple`、`llama-tokenize` 和 `llama-bench`。报告必须同时看 `same_input_prompt_token_ids_match`、LCQI 线程 A/B、Q4 direct A/B、llama.cpp 对比和 hotspot 字段。caw 最新报告显示, 默认 8 worker Q4 direct 把 prefill 从串行 `363.759ms` 降到 `161.436ms`, decode step 从 `15.1652ms` 降到 `7.26678ms`; 关闭 Q4 direct 后 prefill 为 `175.634ms`, 默认 Q4 direct 为 `161.436ms`, 因此 Q4 direct 在并行后端里仍有 `1.087948x` prefill 收益, decode 中位数 `0.995660x` 基本持平。实验 GGML direct 的旧负结果仍然有效: direct 量化字节覆盖率从 `7.68%` 提高到 `70.85%`, 但 prefill 从 `370.847ms` 退化到 `1211.710ms`。这正是源码卷要训练的工程判断: 覆盖率、内存、局部 kernel、线程调度和端到端速度必须分开证明。

Listing 16.16 `tools/run_smollm2_same_input_compare.py`

```

1  #!/usr/bin/env python3
2  from __future__ import annotations
3
4  import argparse
5  import datetime as dt
6  import os
7  import re
8  import shlex
9  import statistics
10 import subprocess
11 import sys
12 from dataclasses import dataclass
13 from pathlib import Path
14
15 import run_smollm2_small_smoke

```

```

16
17
18 DEFAULT_PROMPT = "Hello"
19 DEFAULT_MAX_NEW_TOKENS = 2
20 DEFAULT_ROUNDS = 5
21 DEFAULT_BUILD_JOBS = 2
22 DEFAULT_TIMEOUT_SECONDS = 300
23 LCQI_TARGET = "lcqi_llama_gguf"
24 LLAMA_SIMPLE_TARGET = "llama-simple"
25 LLAMA_BENCH_TARGET = "llama-bench"
26 LLAMA_TOKENIZE_TARGET = "llama-tokenize"
27 LCQI_Q4_DIRECT_ENV = "LCQI_LLAMA_Q4_DIRECT"
28 LCQI_GGML_DIRECT_ENV = "LCQI_LLAMA_GGML_DIRECT"
29 LCQI_Q4_DIRECT_OFF = "0"
30 LCQI_GGML_DIRECT_ON = "1"
31 LCQI_SUMMARY_KEYS = [
32     "benchmark_load_ms",
33     "benchmark_prefill_ms",
34     "derived_prefill_ms_per_prompt_token",
35     "derived_prefill_prompt_tokens_per_second",
36     "benchmark_decode_ms",
37     "derived_decode_ms_per_step",
38     "benchmark_decode_tokens_per_second",
39     "benchmark_worker_count",
40     "derived_f32_weight_inflation_ratio",
41     "derived_direct_quantized_weight_byte_share",
42     "benchmark_hotspot_rms_norm_ms",
43     "benchmark_hotspot_attention_ms",
44     "benchmark_hotspot_rope_ms",
45     "benchmark_hotspot_wq_ms",
46     "benchmark_hotspot_wk_ms",
47     "benchmark_hotspot_wv_ms",
48     "benchmark_hotspot_wo_ms",
49     "benchmark_hotspot_w_gate_ms",
50     "benchmark_hotspot_w_up_ms",
51     "benchmark_hotspot_w_down_ms",
52     "benchmark_hotspot_lm_head_ms",
53     "benchmark_hotspot_q4_k_direct_ms",
54     "benchmark_hotspot_ggml_direct_ms",
55     "benchmark_hotspot_f32_fallback_ms",
56     "benchmark_hotspot_q4_k_direct_calls",
57     "benchmark_hotspot_ggml_direct_calls",
58     "benchmark_hotspot_f32_fallback_calls",
59     "benchmark_q4_k_direct_tensors",
60     "benchmark_ggml_direct_tensors",
61     "benchmark_f32_fallback_tensors",
62 ]
63
64
65 @dataclass(frozen=True)
66 class CommandResult:
67     name: str

```

```
68     args: list[str]
69     returncode: int
70     stdout: str
71     stderr: str
72     metrics: dict[str, str]
73
74
75 def volume_dir() -> Path:
76     return Path(__file__).resolve().parents[1]
77
78
79 def repo_dir() -> Path:
80     return Path(__file__).resolve().parents[3]
81
82
83 def default_build_dir() -> Path:
84     return volume_dir() / "build" / "lcqi-release"
85
86
87 def default_report_path() -> Path:
88     return volume_dir() / "results" / "lcqi-smollm2-same-input-compare-caw.txt"
89
90
91 def command_line(args: list[str]) -> str:
92     return " ".join(shlex.quote(arg) for arg in args)
93
94
95 def tail(text: str, max_chars: int = 4000) -> str:
96     return text if len(text) <= max_chars else text[-max_chars:]
97
98
99 def run_command(
100     name: str,
101     args: list[str],
102     *,
103     timeout_seconds: int,
104     cwd: Path | None = None,
105     env: dict[str, str] | None = None,
106 ) -> CommandResult:
107     command_env = os.environ.copy()
108     if env:
109         command_env.update(env)
110     try:
111         completed = subprocess.run(
112             args,
113             cwd=cwd if cwd else str(repo_dir()),
114             env=command_env,
115             text=True,
116             capture_output=True,
117             timeout=timeout_seconds,
118             check=False,
119         )
```

```

120     except subprocess.TimeoutExpired as exc:
121         stdout = exc.stdout if isinstance(exc.stdout, str) else ""
122         stderr = exc.stderr if isinstance(exc.stderr, str) else ""
123         return CommandResult(name, args, 124, stdout, stderr, {})
124     return CommandResult(
125         name=name,
126         args=args,
127         returncode=completed.returncode,
128         stdout=completed.stdout,
129         stderr=completed.stderr,
130         metrics={},
131     )
132
133
134 def ensure_success(result: CommandResult, action: str) -> None:
135     if result.returncode == 0:
136         return
137     raise RuntimeError(
138         f"{action} failed with exit code {result.returncode}\n"
139         f"command: {command_line(result.args)}\n"
140         f"stdout tail:\n{tail(result.stdout)}\n"
141         f"stderr tail:\n{tail(result.stderr)}"
142     )
143
144
145 def smollm2_chat_prompt(user_prompt: str) -> str:
146     return (
147         "<|im_start|>system\n"
148         "You are a helpful AI assistant named SmolLM, trained by Hugging Face"
149         "<|im_end|>\n"
150         "<|im_start|>user\n"
151         f"{user_prompt}"
152         "<|im_end|>\n"
153         "<|im_start|>assistant\n"
154     )
155
156
157 def escape_newlines(text: str) -> str:
158     return text.replace("\\", "\\\").replace("\n", "\\n")
159
160
161 def parse_int_list(value: str) -> list[int]:
162     ids: list[int] = []
163     for item in value.replace(",", " ").split():
164         ids.append(int(item))
165     return ids
166
167
168 def parse_lcqi_metrics(stdout: str) -> dict[str, str]:
169     metrics: dict[str, str] = {}
170     generated_ids: list[int] = []
171     prompt_ids: list[int] = []

```



```
172     for line in stdout.splitlines():
173         key, sep, value = line.partition(" ")
174         if not sep:
175             continue
176         if key in {
177             "mode",
178             "weight_execution",
179             "architecture",
180             "model_name",
181             "layers",
182             "hidden_size",
183             "query_heads",
184             "kv_heads",
185             "head_dim",
186             "vocab_size",
187             "tie_lm_head_to_embedding",
188             "predicted_first_token",
189             "benchmark_manifest_ms",
190             "benchmark_weight_load_ms",
191             "benchmark_load_ms",
192             "benchmark_prefill_ms",
193             "benchmark_decode_ms",
194             "benchmark_total_ms",
195             "benchmark_prompt_tokens",
196             "benchmark_generated_tokens",
197             "benchmark_prefill_steps",
198             "benchmark_decode_steps",
199             "benchmark_worker_count",
200             "benchmark_first_token_tokens_per_second",
201             "benchmark_decode_tokens_per_second",
202             "benchmark_kv_cache_bytes",
203             "benchmark_quantized_weight_bytes",
204             "benchmark_f32_weight_bytes",
205             "benchmark_direct_quantized_weight_bytes",
206             "benchmark_fallback_dequantized_weight_bytes",
207             "benchmark_tensors_loaded",
208             "benchmark_q4_k_direct_tensors",
209             "benchmark_ggml_direct_tensors",
210             "benchmark_f32_fallback_tensors",
211             "benchmark_hotspot_rms_norm_ms",
212             "benchmark_hotspot_attention_ms",
213             "benchmark_hotspot_rope_ms",
214             "benchmark_hotspot_wq_ms",
215             "benchmark_hotspot_wk_ms",
216             "benchmark_hotspot_wv_ms",
217             "benchmark_hotspot_wo_ms",
218             "benchmark_hotspot_w_gate_ms",
219             "benchmark_hotspot_w_up_ms",
220             "benchmark_hotspot_w_down_ms",
221             "benchmark_hotspot_lm_head_ms",
222             "benchmark_hotspot_q4_k_direct_ms",
223             "benchmark_hotspot_ggml_direct_ms",
```

```

224         "benchmark_hotspot_f32_fallback_ms",
225         "benchmark_hotspot_q4_k_direct_calls",
226         "benchmark_hotspot_ggml_direct_calls",
227         "benchmark_hotspot_f32_fallback_calls",
228     }:
229         metrics[key] = value.strip()
230     elif key == "prompt_ids":
231         prompt_ids = parse_int_list(value)
232         metrics["prompt_ids"] = " ".join(str(item) for item in prompt_ids)
233         metrics["prompt_token_count_from_ids"] = str(len(prompt_ids))
234     elif key == "generated_ids":
235         generated_ids = parse_int_list(value)
236         metrics["generated_ids"] = " ".join(str(item) for item in ←
↪ generated_ids)
237         metrics["generated_token_count_from_ids"] = str(len(generated_ids))
238     if prompt_ids and generated_ids and len(generated_ids) >= len(prompt_ids):
239         new_ids = generated_ids[len(prompt_ids):]
240         metrics["generated_new_ids"] = " ".join(str(item) for item in new_ids)
241     add_lcqi_derived_metrics(metrics)
242     return metrics
243
244
245 def add_lcqi_derived_metrics(metrics: dict[str, str]) -> None:
246     prefill_ms = metric_as_float(metrics, "benchmark_prefill_ms")
247     prompt_tokens = metric_as_float(metrics, "benchmark_prompt_tokens")
248     decode_ms = metric_as_float(metrics, "benchmark_decode_ms")
249     decode_steps = metric_as_float(metrics, "benchmark_decode_steps")
250     quantized_bytes = metric_as_float(metrics, "←
↪ benchmark_quantized_weight_bytes")
251     f32_bytes = metric_as_float(metrics, "benchmark_f32_weight_bytes")
252     direct_bytes = metric_as_float(metrics, "←
↪ benchmark_direct_quantized_weight_bytes")
253     if prefill_ms is not None and prefill_ms > 0.0 and prompt_tokens is not ←
↪ None:
254         metrics["derived_prefill_prompt_tokens_per_second"] = (
255             f"{prompt_tokens * 1000.0 / prefill_ms:.6f}"
256         )
257         metrics["derived_prefill_ms_per_prompt_token"] = (
258             f"{prefill_ms / prompt_tokens:.6f}"
259         )
260     if decode_ms is not None and decode_ms > 0.0 and decode_steps is not None:
261         metrics["derived_decode_steps_per_second"] = f"{decode_steps * 1000.0 /←
↪ decode_ms:.6f}"
262         metrics["derived_decode_ms_per_step"] = f"{decode_ms / max(decode_steps←
↪ , 1.0):.6f}"
263     if quantized_bytes is not None and quantized_bytes > 0.0 and f32_bytes is ←
↪ not None:
264         metrics["derived_f32_weight_inflation_ratio"] = f"{f32_bytes / ←
↪ quantized_bytes:.6f}"
265     if quantized_bytes is not None and quantized_bytes > 0.0 and direct_bytes ←
↪ is not None:
266         metrics["derived_direct_quantized_weight_byte_share"] = (

```

```

267         f"{direct_bytes / quantized_bytes:.6f}"
268     )
269
270
271 def parse_llama_simple_metrics(stderr: str, stdout: str, full_prompt: str) -> dict[str, str]:
272     metrics: dict[str, str] = {}
273     parse_llama_perf_metrics(stderr, metrics)
274     clean_stdout = strip_ansi(stdout).replace("\r", "")
275     metrics["stdout_prompt_prefix_match"] = str(clean_stdout.startswith(
276         full_prompt)).lower()
277     if clean_stdout.startswith(full_prompt):
278         generated = clean_stdout[len(full_prompt):].strip()
279         metrics["generated_suffix"] = escape_newlines(generated)
280     decoded_match = re.search(
281         r"main:\s+decoded\s+(?P<count>\d+)\s+tokens\s+in\s+(?P<seconds>[0-9.]+)
282         \s+s,"
283         r"\s+speed:\s+(?P<tps>[0-9.]+)\s+t/s",
284         stderr,
285     )
286     if decoded_match:
287         metrics["llama_simple_decoded_tokens"] = decoded_match.group("count")
288         metrics["llama_simple_main_seconds"] = decoded_match.group("seconds")
289         metrics["llama_simple_main_tokens_per_second"] = decoded_match.group("
290         tps")
291     return metrics
292
293
294 def parse_llama_perf_metrics(text: str, metrics: dict[str, str]) -> None:
295     load_match = re.search(
296         r"llama_(?:perf_context_print|print_timings):\s+load time\s*=\s*"
297         r"(?P<ms>[0-9.]+)\s*ms",
298         text,
299     )
300     if load_match:
301         metrics["llama_load_ms"] = load_match.group("ms")
302
303     prompt_match = re.search(
304         r"llama_(?:perf_context_print|print_timings):\s+prompt eval time\s*=\s*"
305         r"(?P<ms>[0-9.]+)\s*ms\s*/\s*(?P<count>\d+)\s+tokens\s*"
306         r"\(\s*(?P<ms_per>[0-9.]+)\s*ms per token,\s*"
307         r"(?P<tps>[0-9.]+)\s*tokens per second\s*\)",
308         text,
309     )
310     if prompt_match:
311         metrics["llama_prompt_eval_ms"] = prompt_match.group("ms")
312         metrics["llama_prompt_eval_tokens"] = prompt_match.group("count")
313         metrics["llama_prompt_eval_ms_per_token"] = prompt_match.group("ms_per")
314         metrics["llama_prompt_eval_tokens_per_second"] = prompt_match.group("
315         tps")

```

```

312
313 eval_match = re.search(
314     r"llama_(?:perf_context_print|print_timings):\s+eval time\s*=\s*"
315     r"(?P<ms>[0-9.]+)\s*ms\s*/\s*(?P<count>\d+)\s+runs\s*"
316     r"\(\s*(?P<ms_per>[0-9.]+)\s*ms per token,\s*"
317     r"(?P<tps>[0-9.]+)\s*tokens per second\s*\)",
318     text,
319 )
320 if eval_match:
321     metrics["llama_eval_ms"] = eval_match.group("ms")
322     metrics["llama_eval_runs"] = eval_match.group("count")
323     metrics["llama_eval_ms_per_run"] = eval_match.group("ms_per")
324     metrics["llama_eval_tokens_per_second"] = eval_match.group("tps")
325
326 total_match = re.search(
327     r"llama_(?:perf_context_print|print_timings):\s+total time\s*=\s*"
328     r"(?P<ms>[0-9.]+)\s*ms(?:\s*/\s*(?P<count>\d+)\s+tokens)?",
329     text,
330 )
331 if total_match:
332     metrics["llama_total_ms"] = total_match.group("ms")
333     if total_match.group("count"):
334         metrics["llama_total_tokens"] = total_match.group("count")
335
336
337 def parse_llama_bench_metrics(stdout: str) -> dict[str, str]:
338     metrics: dict[str, str] = {}
339     for line in stdout.splitlines():
340         stripped = line.strip()
341         if not stripped.startswith("|"):
342             continue
343         columns = [column.strip() for column in stripped.strip("|").split("|")]
344         if len(columns) != 7 or columns[0] == "model" or columns[0].startswith(↵
↵ "-"):
345             continue
346         metrics["llama_bench_model"] = columns[0]
347         metrics["llama_bench_size"] = columns[1]
348         metrics["llama_bench_params"] = columns[2]
349         metrics["llama_bench_backend"] = columns[3]
350         metrics["llama_bench_threads"] = columns[4]
351         test_name = columns[5]
352         tokens_per_second = re.sub(r"\s+.*$", "", columns[6])
353         if test_name.startswith("pp"):
354             metrics["llama_bench_prefill_test"] = test_name
355             metrics["llama_bench_prefill_tokens_per_second"] = ↵
↵ tokens_per_second
356         elif test_name.startswith("tg"):
357             metrics["llama_bench_decode_test"] = test_name
358             metrics["llama_bench_decode_tokens_per_second"] = tokens_per_second
359     return metrics
360
361

```

```

362 def parse_llama_tokenize_metrics(stdout: str) -> dict[str, str]:
363     metrics: dict[str, str] = {}
364     ids_match = re.search(r"\[(?P<ids>[0-9,\s-]+)\]", stdout)
365     if ids_match:
366         ids = parse_int_list(ids_match.group("ids"))
367         metrics["llama_tokenize_ids"] = " ".join(str(item) for item in ids)
368         metrics["llama_tokenize_token_count_from_ids"] = str(len(ids))
369     count_match = re.search(r"Total number of tokens:\s*(?P<count>\d+)", stdout↵
↵ )
370     if count_match:
371         metrics["llama_tokenize_show_count"] = count_match.group("count")
372     return metrics
373
374
375 def strip_ansi(text: str) -> str:
376     return re.sub(r"\x1b\[([0-?]*[ -/]*[@-~])", "", text)
377
378
379 def metric_as_float(metrics: dict[str, str], key: str) -> float | None:
380     value = metrics.get(key)
381     if value is None:
382         return None
383     try:
384         return float(value)
385     except ValueError:
386         return None
387
388
389 def metric_values(runs: list[CommandResult], key: str) -> list[float]:
390     values: list[float] = []
391     for run in runs:
392         value = metric_as_float(run.metrics, key)
393         if value is not None:
394             values.append(value)
395     return values
396
397
398 def median_metric(runs: list[CommandResult], key: str) -> float | None:
399     values = metric_values(runs, key)
400     return statistics.median(values) if values else None
401
402
403 def format_summary(name: str, runs: list[CommandResult], keys: list[str]) -> ↵
↵ list[str]:
404     lines = [f"summary:{name}", f"rounds={len(runs)}"]
405     for key in keys:
406         values = metric_values(runs, key)
407         if not values:
408             continue
409         lines.append(
410             f"{key}_median={statistics.median(values):.6f} "
411             f"{key}_min={min(values):.6f} "

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```

412         f"{key}_max={max(values):.6f}"
413     )
414     return lines
415
416
417 def build_lcqi(build_dir: Path, jobs: int, timeout_seconds: int) -> Path:
418     configure = run_command(
419         "cmake_configure_lcqi",
420         [
421             "cmake",
422             "-S",
423             str(volume_dir()),
424             "-B",
425             str(build_dir),
426             "-DCMAKE_BUILD_TYPE=Release",
427         ],
428         timeout_seconds=timeout_seconds,
429     )
430     ensure_success(configure, "configure LCQI")
431     build = run_command(
432         "cmake_build_lcqi_llama_gguf",
433         ["cmake", "--build", str(build_dir), "--target", LCQI_TARGET, "-j", str(
↵ (jobs)],
434         timeout_seconds=timeout_seconds,
435     )
436     ensure_success(build, "build LCQI llama GGUF target")
437     binary = build_dir / "labs" / "linux_cpu_inference" / LCQI_TARGET
438     if not binary.exists() or not os.access(binary, os.X_OK):
439         raise RuntimeError(f"LCQI binary not found or not executable: {binary}"↵
↵ )
440     return binary
441
442
443 def ensure_llama_targets(
444     cache_dir: Path,
445     *,
446     jobs: int,
447     skip_build: bool,
448     timeout_seconds: int,
449 ) -> tuple[Path, Path, Path]:
450     run_smollm2_small_smoke.ensure_llama_simple_chat(
451         cache_dir,
452         jobs=jobs,
453         skip_build=skip_build,
454     )
455     build_dir = cache_dir / "llama.cpp" / "build"
456     binaries = {
457         LLAMA_SIMPLE_TARGET: build_dir / "bin" / LLAMA_SIMPLE_TARGET,
458         LLAMA_BENCH_TARGET: build_dir / "bin" / LLAMA_BENCH_TARGET,
459         LLAMA_TOKENIZE_TARGET: build_dir / "bin" / LLAMA_TOKENIZE_TARGET,
460     }
461     for target, binary in binaries.items():

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462     if binary.exists() and os.access(binary, os.X_OK):
463         continue
464     if skip_build:
465         raise RuntimeError(f"{target} is missing and --skip-llama-build was ←
↪ set: {binary}")
466     build = run_command(
467         f"cmake_build_{target}",
468         ["cmake", "--build", str(build_dir), "--target", target, "-j", str(↪
↪ jobs)],
469         timeout_seconds=timeout_seconds,
470     )
471     ensure_success(build, f"build llama.cpp target {target}")
472     if not binary.exists() or not os.access(binary, os.X_OK):
473         raise RuntimeError(f"{target} was not produced: {binary}")
474     return (
475         binaries[LLAMA_SIMPLE_TARGET],
476         binaries[LLAMA_BENCH_TARGET],
477         binaries[LLAMA_TOKENIZE_TARGET],
478     )
479
480
481 def run_lcqi_round(
482     binary: Path,
483     model_path: Path,
484     *,
485     user_prompt: str,
486     max_new_tokens: int,
487     round_index: int,
488     timeout_seconds: int,
489     name_prefix: str = "lcqi_round",
490     env: dict[str, str] | None = None,
491     threads: int | None = None,
492 ) -> CommandResult:
493     command = [
494         str(binary),
495         str(model_path),
496         "--prompt",
497         user_prompt,
498         "--max-new",
499         str(max_new_tokens),
500         "--benchmark",
501         "--decode-text",
502     ]
503     if threads is not None:
504         command.extend(["--threads", str(threads)])
505     result = run_command(
506         f"{name_prefix}_{round_index}",
507         command,
508         timeout_seconds=timeout_seconds,
509         env=env,
510     )
511     return CommandResult(

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```
512     result.name,
513     result.args,
514     result.returncode,
515     result.stdout,
516     result.stderr,
517     parse_lcqi_metrics(result.stdout),
518 )
519
520
521 def run_llama_simple_round(
522     binary: Path,
523     model_path: Path,
524     *,
525     full_prompt: str,
526     max_new_tokens: int,
527     round_index: int,
528     timeout_seconds: int,
529 ) -> CommandResult:
530     command = [
531         str(binary),
532         "-m",
533         str(model_path),
534         "-ngl",
535         "0",
536         "-n",
537         str(max_new_tokens),
538         full_prompt,
539     ]
540     result = run_command(
541         f"llama_simple_round_{round_index}",
542         command,
543         timeout_seconds=timeout_seconds,
544         env={"LC_ALL": "C"},
545     )
546     return CommandResult(
547         result.name,
548         result.args,
549         result.returncode,
550         result.stdout,
551         result.stderr,
552         parse_llama_simple_metrics(result.stderr, result.stdout, full_prompt),
553     )
554
555
556 def run_llama_tokenize(
557     binary: Path,
558     model_path: Path,
559     *,
560     full_prompt: str,
561     timeout_seconds: int,
562 ) -> CommandResult:
563     command = [
```



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564     str(binary),
565     "-m",
566     str(model_path),
567     "--ids",
568     "--show-count",
569     "--log-disable",
570     "-p",
571     full_prompt,
572 ]
573 result = run_command(
574     "llama_tokenize_same_prompt",
575     command,
576     timeout_seconds=timeout_seconds,
577     env={"LC_ALL": "C"},
578 )
579 return CommandResult(
580     result.name,
581     result.args,
582     result.returncode,
583     result.stdout,
584     result.stderr,
585     parse_llama_tokenize_metrics(result.stdout),
586 )
587
588
589 def run_llama_bench(
590     binary: Path,
591     model_path: Path,
592     *,
593     prompt_tokens: int,
594     max_new_tokens: int,
595     rounds: int,
596     timeout_seconds: int,
597 ) -> CommandResult:
598     command = [
599         str(binary),
600         "-m",
601         str(model_path),
602         "-ngl",
603         "0",
604         "-p",
605         str(prompt_tokens),
606         "-n",
607         str(max_new_tokens),
608         "-r",
609         str(rounds),
610     ]
611     result = run_command(
612         "llama_bench_same_token_count",
613         command,
614         timeout_seconds=timeout_seconds,
615         env={"LC_ALL": "C"},
```

```

616     )
617     return CommandResult(
618         result.name,
619         result.args,
620         result.returncode,
621         result.stdout,
622         result.stderr,
623         parse_llama_bench_metrics(result.stdout),
624     )
625
626
627 def current_commit() -> str:
628     result = subprocess.run(
629         ["git", "-C", str(repo_dir()), "rev-parse", "--short", "HEAD"],
630         text=True,
631         capture_output=True,
632         check=False,
633     )
634     return result.stdout.strip() if result.returncode == 0 else "unknown"
635
636
637 def working_tree_dirty() -> str:
638     result = subprocess.run(
639         ["git", "-C", str(repo_dir()), "status", "--porcelain"],
640         text=True,
641         capture_output=True,
642         check=False,
643     )
644     if result.returncode != 0:
645         return "unknown"
646     return "true" if result.stdout.strip() else "false"
647
648
649 def command_output(args: list[str]) -> str:
650     completed = subprocess.run(args, text=True, capture_output=True, check=↵
↵ False)
651     if completed.returncode != 0:
652         return "unknown"
653     return completed.stdout.strip()
654
655
656 def write_report(
657     report_path: Path,
658     *,
659     cache_dir: Path,
660     model_path: Path,
661     user_prompt: str,
662     full_prompt: str,
663     max_new_tokens: int,
664     lcqi_runs: list[CommandResult],
665     lcqi_serial_runs: list[CommandResult],
666     lcqi_q4_direct_off_runs: list[CommandResult],

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667     lcqi_ggml_direct_runs: list[CommandResult],
668     llama_tokenize: CommandResult,
669     llama_simple_runs: list[CommandResult],
670     llama_bench: CommandResult,
671 ) -> None:
672     report_path.parent.mkdir(parents=True, exist_ok=True)
673     model_matches, actual_sha256, actual_bytes = run_smollm2_small_smoke.↵
↵ verify_model_file(
674         model_path
675     )
676     lines = [
677         "LCQI vs llama.cpp SmolLM2 Same Input Compare",
678         "",
679         f"timestamp_utc={dt.datetime.now(dt.UTC).isoformat()}",
680         f"repo_commit={current_commit()}",
681         f"working_tree_dirty={working_tree_dirty()}",
682         f"host={command_output(['hostname'])}",
683         f"uname={command_output(['uname', '-a'])}",
684         f"nproc={command_output(['nproc'])}",
685         f"python={sys.version.split()[0]}",
686         f"cache_dir={cache_dir}",
687         "",
688         f"model_source={run_smollm2_small_smoke.MODEL_GGUF_REPO_ID}",
689         f"model_file={run_smollm2_small_smoke.MODEL_FILE_NAME}",
690         f"model_path={model_path}",
691         f"model_expected_bytes={run_smollm2_small_smoke.MODEL_EXPECTED_BYTES}",
692         f"model_actual_bytes={actual_bytes}",
693         f"model_expected_sha256={run_smollm2_small_smoke.MODEL_EXPECTED_SHA256}↵
↵ ",
694         f"model_actual_sha256={actual_sha256}",
695         f"model_identity_match={str(model_matches).lower()}",
696         f"llama_cpp_commit={run_smollm2_small_smoke.LLAMA_CPP_COMMIT}",
697         "",
698         f"user_prompt={escape_newlines(user_prompt)}",
699         f"max_new_tokens={max_new_tokens}",
700         f"expanded_chat_prompt={escape_newlines(full_prompt)}",
701         "",
702         "scope=LCQI uses its internal SmolLM2 chat-prompt builder from the same↵
↵ user "
703         "prompt. llama-simple receives the expanded chat prompt string directly↵
↵ . "
704         "The same-input check is prompt-token-count equality plus matching ↵
↵ visible "
705         "generated suffix; llama-bench is reported separately as same-token-↵
↵ count "
706         "synthetic throughput, not same text.",
707         "",
708     ]
709
710     if lcqi_q4_direct_off_runs:
711         lines.extend(format_summary(
712             "lcqi_q4_direct_off_same_input",

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```

713         lcqi_q4_direct_off_runs,
714         LCQI_SUMMARY_KEYS,
715     ))
716     lines.append("")
717
718     if lcqi_serial_runs:
719         lines.extend(format_summary(
720             "lcqi_serial_same_input",
721             lcqi_serial_runs,
722             LCQI_SUMMARY_KEYS,
723         ))
724         lines.append("")
725
726     lines.extend(format_summary(
727         "lcqi_same_input",
728         lcqi_runs,
729         LCQI_SUMMARY_KEYS,
730     ))
731     lines.append("")
732     if lcqi_ggml_direct_runs:
733         lines.extend(format_summary(
734             "lcqi_ggml_direct_experimental_same_input",
735             lcqi_ggml_direct_runs,
736             LCQI_SUMMARY_KEYS,
737         ))
738         lines.append("")
739     lines.extend(format_summary(
740         "llama_simple_same_input",
741         llama_simple_runs,
742         [
743             "llama_load_ms",
744             "llama_prompt_eval_ms",
745             "llama_prompt_eval_ms_per_token",
746             "llama_prompt_eval_tokens_per_second",
747             "llama_eval_ms",
748             "llama_eval_ms_per_run",
749             "llama_eval_tokens_per_second",
750             "llama_total_ms",
751         ],
752     ))
753     lines.append("")
754
755     add_ratio_summary(
756         lines,
757         lcqi_runs,
758         llama_tokenize,
759         llama_simple_runs,
760         llama_bench,
761         lcqi_serial_runs,
762         lcqi_q4_direct_off_runs,
763         lcqi_ggml_direct_runs,
764     )

```

```

765
766     for run in (
767         lcqi_q4_direct_off_runs +
768         lcqi_serial_runs +
769         lcqi_runs +
770         lcqi_ggml_direct_runs +
771         [llama_tokenize] +
772         llama_simple_runs +
773         [llama_bench]
774     ):
775         lines.extend(
776             [
777                 "",
778                 f"[{run.name}]",
779                 f"returncode={run.returncode}",
780                 f"command={command_line(run.args)}",
781             ]
782         )
783         for key in sorted(run.metrics):
784             lines.append(f"{key}={run.metrics[key]}")
785         lines.extend(
786             [
787                 "stdout_tail:",
788                 tail(strip_ansi(run.stdout).strip(), 3000),
789                 "stderr_tail:",
790                 tail(strip_ansi(run.stderr).strip(), 3000),
791             ]
792         )
793
794     while lines and lines[-1] == "":
795         lines.pop()
796     report_path.write_text("\n".join(lines) + "\n", encoding="utf-8")
797
798
799 def add_ratio_summary(
800     lines: list[str],
801     lcqi_runs: list[CommandResult],
802     llama_tokenize: CommandResult,
803     llama_simple_runs: list[CommandResult],
804     llama_bench: CommandResult,
805     lcqi_serial_runs: list[CommandResult],
806     lcqi_q4_direct_off_runs: list[CommandResult],
807     lcqi_ggml_direct_runs: list[CommandResult],
808 ) -> None:
809     lcqi_prompt_tokens = median_metric(lcqi_runs, "benchmark_prompt_tokens")
810     lcqi_prompt_ids = lcqi_runs[0].metrics.get("prompt_ids") if lcqi_runs else ←
811     ↪ None
812     llama_tokenize_ids = llama_tokenize.metrics.get("llama_tokenize_ids")
813     llama_tokenize_count = metric_as_float(llama_tokenize.metrics, "↵
814     ↪ llama_tokenize_show_count")
815     llama_prompt_tokens = median_metric(llama_simple_runs, "↵
816     ↪ llama_prompt_eval_tokens")

```

```

814 lcqi_prefill = median_metric(lcqi_runs, "benchmark_prefill_ms")
815 llama_prompt = median_metric(llama_simple_runs, "llama_prompt_eval_ms")
816 lcqi_decode_step = median_metric(lcqi_runs, "derived_decode_ms_per_step")
817 llama_eval_step = median_metric(llama_simple_runs, "llama_eval_ms_per_run")
818 lcqi_weight_ratio = median_metric(lcqi_runs, "←
↪ derived_f32_weight_inflation_ratio")
819 lcqi_direct_share = median_metric(lcqi_runs, "←
↪ derived_direct_quantized_weight_byte_share")
820 bench_pp_tps = metric_as_float(llama_bench.metrics, "←
↪ llama_bench_prefill_tokens_per_second")
821 bench_tg_tps = metric_as_float(llama_bench.metrics, "←
↪ llama_bench_decode_tokens_per_second")

822
823 lines.extend(["interpretation", f"prompt_tokens_lcqi_median={←
↪ lcqi_prompt_tokens}"])
824 lines.append(f"prompt_tokens_llama_tokenize={llama_tokenize_count}")
825 lines.append(f"prompt_tokens_llama_simple_median={llama_prompt_tokens}")
826 if lcqi_prompt_tokens == llama_prompt_tokens == llama_tokenize_count:
827     lines.append("same_input_prompt_token_count_match=true")
828 else:
829     lines.append("same_input_prompt_token_count_match=false")
830 lines.append(
831     "same_input_prompt_token_ids_match="
832     f"{str(lcqi_prompt_ids == llama_tokenize_ids).lower()}")
833 )
834 if lcqi_prefill is not None and llama_prompt is not None and llama_prompt >←
↪ 0.0:
835     lines.append(f"prefill_ms_ratio_lcqi_over_llama_simple={lcqi_prefill / ←
↪ llama_prompt:.6f}")
836 if lcqi_decode_step is not None and llama_eval_step is not None and ←
↪ llama_eval_step > 0.0:
837     lines.append(f"decode_step_ms_ratio_lcqi_over_llama_simple={←
↪ lcqi_decode_step / llama_eval_step:.6f}")
838 if lcqi_weight_ratio is not None:
839     lines.append(f"lcqi_f32_dequantized_weight_inflation_ratio={←
↪ lcqi_weight_ratio:.6f}")
840 if lcqi_direct_share is not None:
841     lines.append(f"lcqi_direct_quantized_weight_byte_share={←
↪ lcqi_direct_share:.6f}")
842 add_lcqi_threaded_ratios(lines, lcqi_serial_runs, lcqi_runs)
843 add_lcqi_ab_ratios(lines, lcqi_q4_direct_off_runs, lcqi_runs)
844 add_lcqi_ggml_experimental_ratios(lines, lcqi_runs, lcqi_ggml_direct_runs)
845 if bench_pp_tps is not None:
846     lines.append(f"llama_bench_same_token_count_prefill_tps={bench_pp_tps←
↪ :.6f}")
847 if bench_tg_tps is not None:
848     lines.append(f"llama_bench_same_token_count_decode_tps={bench_tg_tps:.6←
↪ f}")
849 lines.append(
850     "root_cause=LCQI keeps shape-compatible Q4_K matrices in the default ←
↪ direct "
851     "quantized path. The experimental Q5_0/Q6_K/Q8_0 direct path increases ←

```

```

↪ "
852     "quantized-byte coverage, but its current scalar block kernels are ↪
↪ slower "
853     "than the dequantized f32 AVX2 fallback, so it is not enabled by ↪
↪ default. "
854     "The default LCQI run now uses a persistent row worker pool for large ↪
↪ f32 "
855     "fallback matrices; --threads 1 keeps the serial reference. llama.cpp "
856     "still batches prompt evaluation, uses ggml graph scheduling, tuned low↪
↪ -bit "
857     "CPU kernels, prefetch/thread scheduling, and more compact KV/cache ↪
↪ execution."
858 )
859
860
861 def add_lcqi_threaded_ratios(
862     lines: list[str],
863     lcqi_serial_runs: list[CommandResult],
864     lcqi_runs: list[CommandResult],
865 ) -> None:
866     if not lcqi_serial_runs:
867         return
868     serial_prefill = median_metric(lcqi_serial_runs, "benchmark_prefill_ms")
869     threaded_prefill = median_metric(lcqi_runs, "benchmark_prefill_ms")
870     serial_decode = median_metric(lcqi_serial_runs, "derived_decode_ms_per_step↪
↪ ")
871     threaded_decode = median_metric(lcqi_runs, "derived_decode_ms_per_step")
872     serial_f32 = median_metric(lcqi_serial_runs, "↪
↪ benchmark_hotspot_f32_fallback_ms")
873     threaded_f32 = median_metric(lcqi_runs, "benchmark_hotspot_f32_fallback_ms"↪
↪ )
874     worker_count = median_metric(lcqi_runs, "benchmark_worker_count")
875     if worker_count is not None:
876         lines.append(f"lcqi_threaded_worker_count_median={worker_count:.0f}")
877     if serial_prefill is not None and threaded_prefill is not None and ↪
↪ threaded_prefill > 0.0:
878         lines.append(
879             "lcqi_threaded_prefill_speedup_serial_over_threaded="
880             f"{serial_prefill / threaded_prefill:.6f}"
881         )
882     if serial_decode is not None and threaded_decode is not None and ↪
↪ threaded_decode > 0.0:
883         lines.append(
884             "lcqi_threaded_decode_step_speedup_serial_over_threaded="
885             f"{serial_decode / threaded_decode:.6f}"
886         )
887     if serial_f32 is not None and threaded_f32 is not None and threaded_f32 > ↪
↪ 0.0:
888         lines.append(
889             "lcqi_threaded_f32_hotspot_speedup_serial_over_threaded="
890             f"{serial_f32 / threaded_f32:.6f}"
891         )

```

```

892
893
894 def add_lcqi_ab_ratios(
895     lines: list[str],
896     lcqi_q4_direct_off_runs: list[CommandResult],
897     lcqi_runs: list[CommandResult],
898 ) -> None:
899     if not lcqi_q4_direct_off_runs:
900         return
901     off_prefill = median_metric(lcqi_q4_direct_off_runs, "benchmark_prefill_ms" ←
    ↪ )
902     on_prefill = median_metric(lcqi_runs, "benchmark_prefill_ms")
903     off_decode = median_metric(lcqi_q4_direct_off_runs, " ←
    ↪ derived_decode_ms_per_step")
904     on_decode = median_metric(lcqi_runs, "derived_decode_ms_per_step")
905     off_down = median_metric(lcqi_q4_direct_off_runs, " ←
    ↪ benchmark_hotspot_w_down_ms")
906     on_down = median_metric(lcqi_runs, "benchmark_hotspot_w_down_ms")
907     off_f32_bytes = median_metric(lcqi_q4_direct_off_runs, " ←
    ↪ benchmark_f32_weight_bytes")
908     on_f32_bytes = median_metric(lcqi_runs, "benchmark_f32_weight_bytes")
909     if off_prefill is not None and on_prefill is not None and on_prefill > 0.0:
910         lines.append(f"lcqi_q4_direct_prefill_speedup_off_over_on={off_prefill ←
    ↪ / on_prefill:.6f}")
911     if off_decode is not None and on_decode is not None and on_decode > 0.0:
912         lines.append(f"lcqi_q4_direct_decode_step_speedup_off_over_on={ ←
    ↪ off_decode / on_decode:.6f}")
913     if off_down is not None and on_down is not None and on_down > 0.0:
914         lines.append(f"lcqi_q4_direct_w_down_speedup_off_over_on={off_down / ←
    ↪ on_down:.6f}")
915     if off_f32_bytes is not None and on_f32_bytes is not None:
916         lines.append(f"lcqi_q4_direct_f32_weight_bytes_saved={off_f32_bytes - ←
    ↪ on_f32_bytes:.0f}")
917
918
919 def add_lcqi_ggml_experimental_ratios(
920     lines: list[str],
921     lcqi_runs: list[CommandResult],
922     lcqi_ggml_direct_runs: list[CommandResult],
923 ) -> None:
924     if not lcqi_ggml_direct_runs:
925         return
926     default_prefill = median_metric(lcqi_runs, "benchmark_prefill_ms")
927     ggml_prefill = median_metric(lcqi_ggml_direct_runs, "benchmark_prefill_ms")
928     default_decode = median_metric(lcqi_runs, "derived_decode_ms_per_step")
929     ggml_decode = median_metric(lcqi_ggml_direct_runs, " ←
    ↪ derived_decode_ms_per_step")
930     default_direct_share = median_metric(lcqi_runs, " ←
    ↪ derived_direct_quantized_weight_byte_share")
931     ggml_direct_share = median_metric(lcqi_ggml_direct_runs, " ←
    ↪ derived_direct_quantized_weight_byte_share")
932     ggml_hotspot = median_metric(lcqi_ggml_direct_runs, " ←

```



```

    ↪ benchmark_hotspot_ggml_direct_ms")
933 if default_prefill is not None and ggml_prefill is not None and ↪
    ↪ ggml_prefill > 0.0:
934     lines.append(
935         "lcqi_ggml_experimental_prefill_speedup_default_over_experimental="
936         f"{default_prefill / ggml_prefill:.6f}"
937     )
938 if default_decode is not None and ggml_decode is not None and ggml_decode >↪
    ↪ 0.0:
939     lines.append(
940         "↪
    ↪ lcqi_ggml_experimental_decode_step_speedup_default_over_experimental="
941         f"{default_decode / ggml_decode:.6f}"
942     )
943 if default_direct_share is not None and ggml_direct_share is not None:
944     lines.append(
945         "lcqi_ggml_experimental_direct_share_delta="
946         f"{ggml_direct_share - default_direct_share:.6f}"
947     )
948 if ggml_hotspot is not None:
949     lines.append(f"lcqi_ggml_experimental_hotspot_ms={ggml_hotspot:.6f}")
950
951
952 def parse_args() -> argparse.Namespace:
953     parser = argparse.ArgumentParser(
954         description="Run same-input SmolLM2 comparison between LCQI and llama.↪
    ↪ cpp."
955     )
956     parser.add_argument("--cache-dir", type=Path, default=↪
    ↪ run_smolllm2_small_smoke.default_cache_dir())
957     parser.add_argument("--model-path", type=Path, default=None)
958     parser.add_argument("--build-dir", type=Path, default=default_build_dir())
959     parser.add_argument("--report", type=Path, default=default_report_path())
960     parser.add_argument("--prompt", default=DEFAULT_PROMPT)
961     parser.add_argument("--max-new", type=int, default=DEFAULT_MAX_NEW_TOKENS)
962     parser.add_argument("--rounds", type=int, default=DEFAULT_ROUNDS)
963     parser.add_argument("--jobs", type=int, default=DEFAULT_BUILD_JOBS)
964     parser.add_argument("--timeout-seconds", type=int, default=↪
    ↪ DEFAULT_TIMEOUT_SECONDS)
965     parser.add_argument("--force-download", action="store_true")
966     parser.add_argument("--no-download", action="store_true")
967     parser.add_argument("--skip-llama-build", action="store_true")
968     parser.add_argument("--include-lcqi-q4-direct-off", action=argparse.↪
    ↪ BooleanOptionalAction, default=True)
969     parser.add_argument("--include-lcqi-ggml-direct", action=argparse.↪
    ↪ BooleanOptionalAction, default=True)
970     return parser.parse_args()
971
972
973 def main() -> int:
974     args = parse_args()
975     if args.rounds <= 0:

```

```

976     print("[lcqi-same-input] --rounds must be positive", file=sys.stderr)
977     return 1
978 if args.max_new < 0:
979     print("[lcqi-same-input] --max-new cannot be negative", file=sys.stderr)
980     ↪ )
981     return 1
982
983 cache_dir = args.cache_dir.expanduser().resolve()
984 model_path = (
985     args.model_path.expanduser().resolve()
986     if args.model_path
987     else cache_dir / "models" / run_smollm2_small_smoke.MODEL_FILE_NAME
988 )
989 build_dir = args.build_dir.expanduser().resolve()
990 report_path = args.report.expanduser().resolve()
991 full_prompt = smollm2_chat_prompt(args.prompt)
992
993 try:
994     run_smollm2_small_smoke.download_model(
995         model_path,
996         force_download=args.force_download,
997         no_download=args.no_download,
998     )
999     lcqi_binary = build_lcqi(build_dir, args.jobs, args.timeout_seconds)
1000     llama_simple_binary, llama_bench_binary, llama_tokenize_binary = ↪
1001     ↪ ensure_llama_targets(
1002         cache_dir,
1003         jobs=args.jobs,
1004         skip_build=args.skip_llama_build,
1005         timeout_seconds=args.timeout_seconds,
1006     )
1007
1008     lcqi_q4_direct_off_runs: list[CommandResult] = []
1009     lcqi_serial_runs: list[CommandResult] = []
1010     lcqi_runs: list[CommandResult] = []
1011     lcqi_ggml_direct_runs: list[CommandResult] = []
1012     llama_simple_runs: list[CommandResult] = []
1013     for round_index in range(1, args.rounds + 1):
1014         if args.include_lcqi_q4_direct_off:
1015             print(f"[lcqi-same-input] LCQI Q4 direct off round {round_index} ↪
1016             ↪ }/{args.rounds}")
1017             lcqi_q4_direct_off_runs.append(
1018                 run_lcqi_round(
1019                     lcqi_binary,
1020                     model_path,
1021                     user_prompt=args.prompt,
1022                     max_new_tokens=args.max_new,
1023                     round_index=round_index,
1024                     timeout_seconds=args.timeout_seconds,
1025                     name_prefix="lcqi_q4_direct_off_round",
1026                     env={LCQI_Q4_DIRECT_ENV: LCQI_Q4_DIRECT_OFF},
1027                 )

```

```

1025         )
1026         ensure_success(
1027             lcqi_q4_direct_off_runs[-1],
1028             "run LCQI Q4 direct off same-input round",
1029         )
1030         print(f"[lcqi-same-input] LCQI serial round {round_index}/{args.↵
↵ rounds}")
1031         lcqi_serial_runs.append(
1032             run_lcqi_round(
1033                 lcqi_binary,
1034                 model_path,
1035                 user_prompt=args.prompt,
1036                 max_new_tokens=args.max_new,
1037                 round_index=round_index,
1038                 timeout_seconds=args.timeout_seconds,
1039                 name_prefix="lcqi_serial_round",
1040                 threads=1,
1041             )
1042         )
1043         ensure_success(lcqi_serial_runs[-1], "run LCQI serial same-input ↵
↵ round")
1044         print(f"[lcqi-same-input] LCQI round {round_index}/{args.rounds}")
1045         lcqi_runs.append(
1046             run_lcqi_round(
1047                 lcqi_binary,
1048                 model_path,
1049                 user_prompt=args.prompt,
1050                 max_new_tokens=args.max_new,
1051                 round_index=round_index,
1052                 timeout_seconds=args.timeout_seconds,
1053             )
1054         )
1055         ensure_success(lcqi_runs[-1], "run LCQI same-input round")
1056         if args.include_lcqi_ggml_direct:
1057             print(f"[lcqi-same-input] LCQI GGML direct experimental round {↵
↵ round_index}/{args.rounds}")
1058             lcqi_ggml_direct_runs.append(
1059                 run_lcqi_round(
1060                     lcqi_binary,
1061                     model_path,
1062                     user_prompt=args.prompt,
1063                     max_new_tokens=args.max_new,
1064                     round_index=round_index,
1065                     timeout_seconds=args.timeout_seconds,
1066                     name_prefix="lcqi_ggml_direct_experimental_round",
1067                     env={LCQI_GGML_DIRECT_ENV: LCQI_GGML_DIRECT_ON},
1068                 )
1069             )
1070             ensure_success(
1071                 lcqi_ggml_direct_runs[-1],
1072                 "run LCQI GGML direct experimental same-input round",
1073             )

```

```

1074         print(f"[lcqi-same-input] llama-simple round {round_index}/{args.↵
↵ rounds}")
1075         llama_simple_runs.append(
1076             run_llama_simple_round(
1077                 llama_simple_binary,
1078                 model_path,
1079                 full_prompt=full_prompt,
1080                 max_new_tokens=args.max_new,
1081                 round_index=round_index,
1082                 timeout_seconds=args.timeout_seconds,
1083             )
1084         )
1085         ensure_success(llama_simple_runs[-1], "run llama-simple same-input ↵
↵ round")
1086
1087         prompt_tokens = int(lcqi_runs[0].metrics["benchmark_prompt_tokens"])
1088         print("[lcqi-same-input] llama-tokenize prompt identity check")
1089         llama_tokenize = run_llama_tokenize(
1090             llama_tokenize_binary,
1091             model_path,
1092             full_prompt=full_prompt,
1093             timeout_seconds=args.timeout_seconds,
1094         )
1095         ensure_success(llama_tokenize, "run llama-tokenize same-input check")
1096         print("[lcqi-same-input] llama-bench same-token-count reference")
1097         llama_bench = run_llama_bench(
1098             llama_bench_binary,
1099             model_path,
1100             prompt_tokens=prompt_tokens,
1101             max_new_tokens=args.max_new,
1102             rounds=args.rounds,
1103             timeout_seconds=args.timeout_seconds,
1104         )
1105         ensure_success(llama_bench, "run llama-bench same-token-count reference↵
↵ ")
1106
1107         write_report(
1108             report_path,
1109             cache_dir=cache_dir,
1110             model_path=model_path,
1111             user_prompt=args.prompt,
1112             full_prompt=full_prompt,
1113             max_new_tokens=args.max_new,
1114             lcqi_runs=lcqi_runs,
1115             lcqi_serial_runs=lcqi_serial_runs,
1116             lcqi_q4_direct_off_runs=lcqi_q4_direct_off_runs,
1117             lcqi_ggml_direct_runs=lcqi_ggml_direct_runs,
1118             llama_tokenize=llama_tokenize,
1119             llama_simple_runs=llama_simple_runs,
1120             llama_bench=llama_bench,
1121         )
1122     except Exception as error:

```

```

1123     print(f"[lcqi-same-input] ERROR: {error}", file=sys.stderr)
1124     return 1
1125
1126     print(f"report={report_path}")
1127     for line in format_summary(
1128         "lcqi_same_input",
1129         lcqi_runs,
1130         ["benchmark_prefill_ms", "derived_prefill_prompt_tokens_per_second", "↵
↵ derived_decode_ms_per_step"],
1131     ):
1132         print(line)
1133     for line in format_summary(
1134         "llama_simple_same_input",
1135         llama_simple_runs,
1136         ["llama_prompt_eval_ms", "llama_prompt_eval_tokens_per_second", "↵
↵ llama_eval_ms_per_run"],
1137     ):
1138         print(line)
1139     return 0
1140
1141
1142 if __name__ == "__main__":
1143     raise SystemExit(main())

```

`run_gpt2_smoke.py` 下载标准 GPT-2 的 `config.json`、`vocab.json`、`merges.txt` 和 `model.safetensors`，再调用 LCQI 自研 `lcqi_gpt2` reference path 生成 token。它证明自研 `safetensors`、BPE、GPT-2 forward、KV cache decode 和 greedy generation 已连成闭环。

**Listing 16.17** `tools/run_gpt2_smoke.py`

```

1  #!/usr/bin/env python3
2  from __future__ import annotations
3
4  import argparse
5  import datetime as dt
6  import hashlib
7  import os
8  import shlex
9  import subprocess
10 import sys
11 import urllib.request
12 from dataclasses import dataclass
13 from pathlib import Path
14
15
16 MODEL_REPO_ID = "openai-community/gpt2"
17 MODEL_REVISION = "main"
18 MODEL_FILE_IDENTITIES = {
19     "config.json": (
20         665,
21         "0daed7749b4f02b8f76240d5444551d7b08712dab4d0adb8239c56ba823bb7b4",
22     ),
23     "vocab.json": (

```

```

24     1042301,
25     "196139668be63f3b5d6574427317ae82f612a97c5d1cdaf36ed2256dbf636783",
26 ),
27 "merges.txt": (
28     456318,
29     "1ce1664773c50f3e0cc8842619a93edc4624525b728b188a9e0be33b7726adc5",
30 ),
31 "model.safetensors": (
32     548105171,
33     "248dfc3911869ec493c76e65bf2fcf7f615828b0254c12b473182f0f81d3a707",
34 ),
35 }
36 DEFAULT_PROMPT = "Hello, my name is"
37 DEFAULT_MAX_NEW_TOKENS = 1
38 DEFAULT_TIMEOUT_SECONDS = 300
39 DEFAULT_BUILD_JOBS = 2
40
41
42 @dataclass(frozen=True)
43 class CommandResult:
44     args: list[str]
45     returncode: int
46     stdout: str
47     stderr: str
48
49
50 def script_path() -> Path:
51     return Path(__file__).resolve()
52
53
54 def volume_dir() -> Path:
55     return script_path().parents[1]
56
57
58 def repo_dir() -> Path:
59     return script_path().parents[3]
60
61
62 def default_cache_dir() -> Path:
63     explicit_cache = os.environ.get("LCQI_GPT2_CACHE_DIR")
64     if explicit_cache:
65         return Path(explicit_cache).expanduser()
66     xdg_cache = os.environ.get("XDG_CACHE_HOME")
67     cache_root = Path(xdg_cache).expanduser() if xdg_cache else Path.home() / "←
↪ .cache"
68     return cache_root / "lcqi-gpt2-smoke" / MODEL_REPO_ID.replace("/", "--")
69
70
71 def default_build_dir() -> Path:
72     return volume_dir() / "build" / "lcqi-release"
73
74

```

```

75 def default_report_path() -> Path:
76     return volume_dir() / "results" / "lcqi-gpt2-smoke.txt"
77
78
79 def command_line(args: list[str]) -> str:
80     return " ".join(shlex.quote(arg) for arg in args)
81
82
83 def run_command(
84     args: list[str],
85     *,
86     cwd: Path | None = None,
87     timeout_seconds: int | None = None,
88 ) -> CommandResult:
89     try:
90         completed = subprocess.run(
91             args,
92             cwd=str(cwd) if cwd else None,
93             text=True,
94             capture_output=True,
95             timeout=timeout_seconds,
96             check=False,
97         )
98     except subprocess.TimeoutExpired as exc:
99         stdout = exc.stdout if isinstance(exc.stdout, str) else ""
100         stderr = exc.stderr if isinstance(exc.stderr, str) else ""
101         return CommandResult(args=args, returncode=124, stdout=stdout, stderr=↵
↵ stderr)
102     return CommandResult(
103         args=args,
104         returncode=completed.returncode,
105         stdout=completed.stdout,
106         stderr=completed.stderr,
107     )
108
109
110 def ensure_success(result: CommandResult, action: str) -> None:
111     if result.returncode == 0:
112         return
113     raise RuntimeError(
114         f"{action} failed with exit code {result.returncode}\n"
115         f"command: {command_line(result.args)}\n"
116         f"stdout tail:\n{tail(result.stdout)}\n"
117         f"stderr tail:\n{tail(result.stderr)}"
118     )
119
120
121 def tail(text: str, max_chars: int = 4000) -> str:
122     if len(text) <= max_chars:
123         return text
124     return text[-max_chars:]
125

```

```

126
127 def sha256_file(path: Path) -> str:
128     digest = hashlib.sha256()
129     with path.open("rb") as handle:
130         for block in iter(lambda: handle.read(1024 * 1024), b''):
131             digest.update(block)
132     return digest.hexdigest()
133
134
135 def model_url(file_name: str) -> str:
136     return (
137         f"https://huggingface.co/{MODEL_REPO_ID}/resolve/"
138         f"{MODEL_REVISION}/{file_name}"
139     )
140
141
142 def download_file(url: str, target: Path, timeout_seconds: int) -> None:
143     part_path = target.with_suffix(target.suffix + ".part")
144     if part_path.exists():
145         part_path.unlink()
146     request = urllib.request.Request(url, headers={"User-Agent": "lcqi-gpt2-↵
↵ smoke"})
147     with urllib.request.urlopen(request, timeout=timeout_seconds) as response:
148         with part_path.open("wb") as output:
149             while True:
150                 block = response.read(1024 * 1024)
151                 if not block:
152                     break
153                 output.write(block)
154     part_path.replace(target)
155
156
157 def file_identity_matches(path: Path, expected_bytes: int, expected_sha256: str↵
↵ ) -> bool:
158     return (
159         path.exists()
160         and path.stat().st_size == expected_bytes
161         and sha256_file(path) == expected_sha256
162     )
163
164
165 def ensure_model_files(cache_dir: Path, *, no_download: bool, timeout_seconds: ↵
↵ int) -> None:
166     cache_dir.mkdir(parents=True, exist_ok=True)
167     missing = [
168         name
169         for name, (expected_bytes, expected_sha256) in MODEL_FILE_IDENTITYIES.↵
↵ items()
170         if not file_identity_matches(cache_dir / name, expected_bytes, ↵
↵ expected_sha256)
171     ]
172     if not missing:

```



```

173     print(f"[lcqi-gpt2] model cache hit: {cache_dir}")
174     return
175 if no_download:
176     raise RuntimeError(
177         f"missing GPT-2 model files and --no-download was set: {missing}"
178     )
179 for file_name in missing:
180     target = cache_dir / file_name
181     if target.exists():
182         target.unlink()
183     print(f"[lcqi-gpt2] downloading {MODEL_REPO_ID}/{file_name}")
184     download_file(model_url(file_name), target, timeout_seconds)
185     expected_bytes, expected_sha256 = MODEL_FILE_IDENTITIES[file_name]
186     if not file_identity_matches(target, expected_bytes, expected_sha256):
187         raise RuntimeError(
188             f"downloaded GPT-2 file identity mismatch: {file_name}"
189         )
190
191
192 def build_lcqi_gpt2(build_dir: Path, jobs: int) -> Path:
193     configure = run_command(
194         [
195             "cmake",
196             "-S",
197             str(volume_dir()),
198             "-B",
199             str(build_dir),
200             "-DCMAKE_BUILD_TYPE=Release",
201         ],
202         timeout_seconds=DEFAULT_TIMEOUT_SECONDS,
203     )
204     ensure_success(configure, "configure LCQI")
205     build = run_command(
206         [
207             "cmake",
208             "--build",
209             str(build_dir),
210             "--target",
211             "lcqi_gpt2",
212             "-j",
213             str(jobs),
214         ],
215         timeout_seconds=DEFAULT_TIMEOUT_SECONDS,
216     )
217     ensure_success(build, "build lcqi_gpt2")
218     binary = build_dir / "labs" / "linux_cpu_inference" / "lcqi_gpt2"
219     if not binary.exists():
220         raise RuntimeError(f"lcqi_gpt2 binary not found: {binary}")
221     return binary
222
223
224 def write_report(

```

```

225     report_path: Path,
226     *,
227     cache_dir: Path,
228     binary: Path,
229     prompt: str,
230     max_new_tokens: int,
231     engine: str,
232     result: CommandResult,
233 ) -> None:
234     report_path.parent.mkdir(parents=True, exist_ok=True)
235     lines = [
236         f"timestamp_utc={dt.datetime.now(dt.UTC).isoformat()}",
237         f"model_repo_id={MODEL_REPO_ID}",
238         f"model_revision={MODEL_REVISION}",
239         f"model_cache_dir={cache_dir}",
240     ]
241     for file_name, (expected_bytes, expected_sha256) in MODEL_FILE_IDENTITIES.items():
242         path = cache_dir / file_name
243         lines.append(f"file.{file_name}.bytes={path.stat().st_size}")
244         lines.append(f"file.{file_name}.sha256={sha256_file(path)}")
245         lines.append(f"file.{file_name}.expected_bytes={expected_bytes}")
246         lines.append(f"file.{file_name}.expected_sha256={expected_sha256}")
247     lines.extend(
248         [
249             f"binary={binary}",
250             f"prompt={prompt}",
251             f"max_new_tokens={max_new_tokens}",
252             f"engine={engine}",
253             f"command={command_line(result.args)}",
254             f"exit_code={result.returncode}",
255             "stdout_begin",
256             result.stdout.rstrip(),
257             "stdout_end",
258             "stderr_begin",
259             tail(result.stderr).rstrip(),
260             "stderr_end",
261         ]
262     )
263     passed = result.returncode == 0 and "generated_text " in result.stdout
264     lines.append(f"validation={'PASS' if passed else 'FAIL'}")
265     report_path.write_text("\n".join(lines) + "\n", encoding="utf-8")
266
267
268 def parse_args() -> argparse.Namespace:
269     parser = argparse.ArgumentParser(
270         description="Run LCQI self-owned GPT-2 smoke on a HuggingFace
271         ← safetensors directory."
272     )
273     parser.add_argument("--cache-dir", type=Path, default=default_cache_dir())
274     parser.add_argument("--build-dir", type=Path, default=default_build_dir())
275     parser.add_argument("--report", type=Path, default=default_report_path())

```

```

275     parser.add_argument("--prompt", default=DEFAULT_PROMPT)
276     parser.add_argument("--max-new-tokens", type=int, default=↵
↵     DEFAULT_MAX_NEW_TOKENS)
277     parser.add_argument("--engine", choices=("cached", "full"), default="cached↵
↵     ")
278     parser.add_argument("--benchmark", action="store_true")
279     parser.add_argument("--timeout-seconds", type=int, default=↵
↵     DEFAULT_TIMEOUT_SECONDS)
280     parser.add_argument("--jobs", type=int, default=DEFAULT_BUILD_JOBS)
281     parser.add_argument("--no-download", action="store_true")
282     return parser.parse_args()
283
284
285 def main() -> int:
286     args = parse_args()
287     try:
288         if args.max_new_tokens < 0:
289             raise RuntimeError("--max-new-tokens cannot be negative")
290         ensure_model_files(
291             args.cache_dir,
292             no_download=args.no_download,
293             timeout_seconds=args.timeout_seconds,
294         )
295         binary = build_lcqi_gpt2(args.build_dir, args.jobs)
296         command = [
297             str(binary),
298             "--engine",
299             args.engine,
300             str(args.cache_dir),
301             args.prompt,
302             str(args.max_new_tokens),
303         ]
304         if args.benchmark:
305             command.insert(1, "--benchmark")
306         result = run_command(command, cwd=repo_dir(), timeout_seconds=args.↵
↵         timeout_seconds)
307         write_report(
308             args.report,
309             cache_dir=args.cache_dir,
310             binary=binary,
311             prompt=args.prompt,
312             max_new_tokens=args.max_new_tokens,
313             engine=args.engine,
314             result=result,
315         )
316         print(f"[lcqi-gpt2] report: {args.report}")
317         print(tail(result.stdout, max_chars=2000))
318         return 0 if result.returncode == 0 else result.returncode
319     except Exception as error:
320         print(f"[lcqi-gpt2] error: {error}", file=sys.stderr)
321         return 1
322

```

```

323
324 if __name__ == "__main__":
325     raise SystemExit(main())

```

`run_gpt2_benchmark_compare.py` 用同一个 GPT-2 F32 safetensors 模型分别运行 LCQI full-prefix 与 cached-KV 路径，并在本机存在 llama.cpp/SmolLM2 缓存时附带成熟引擎参考。它的报告说明算法差距、kernel 差距和模型口径差异。

**Listing 16.18** tools/run\_gpt2\_benchmark\_compare.py

```

1  #!/usr/bin/env python3
2  from __future__ import annotations
3
4  import argparse
5  import datetime as dt
6  import os
7  import re
8  import shlex
9  import subprocess
10 import sys
11 from dataclasses import dataclass
12 from pathlib import Path
13
14 import run_gpt2_smoke
15
16
17 DEFAULT_PROMPT = "Hello, my name is"
18 DEFAULT_MAX_NEW_TOKENS = 4
19 DEFAULT_BUILD_JOBS = 2
20 DEFAULT_TIMEOUT_SECONDS = 300
21 DEFAULT_SMOLLM2_CACHE_DIR = Path.home() / ".cache" / "lcqi-smolllm2-small-smoke"
22 SMOLLM2_GGUF = "SmolLM2-135M-Instruct-Q4_K_M.gguf"
23
24
25 @dataclass(frozen=True)
26 class EngineRun:
27     name: str
28     command: list[str]
29     returncode: int
30     stdout: str
31     stderr: str
32     metrics: dict[str, str]
33
34
35 def volume_dir() -> Path:
36     return Path(__file__).resolve().parents[1]
37
38
39 def repo_dir() -> Path:
40     return Path(__file__).resolve().parents[3]
41
42
43 def default_report_path() -> Path:

```

```

44     return volume_dir() / "results" / "lcqi-gpt2-benchmark-compare.txt"
45
46
47 def default_build_dir() -> Path:
48     return volume_dir() / "build" / "lcqi-release"
49
50
51 def command_line(args: list[str]) -> str:
52     return " ".join(shlex.quote(arg) for arg in args)
53
54
55 def tail(text: str, max_chars: int = 4000) -> str:
56     return text if len(text) <= max_chars else text[-max_chars:]
57
58
59 def run_command(args: list[str], *, timeout_seconds: int) -> tuple[int, str, ←
    ↪ str]:
60     try:
61         completed = subprocess.run(
62             args,
63             cwd=repo_dir(),
64             text=True,
65             capture_output=True,
66             timeout=timeout_seconds,
67             check=False,
68         )
69     except subprocess.TimeoutExpired as exc:
70         stdout = exc.stdout if isinstance(exc.stdout, str) else ""
71         stderr = exc.stderr if isinstance(exc.stderr, str) else ""
72         return 124, stdout, stderr
73     return completed.returncode, completed.stdout, completed.stderr
74
75
76 def parse_lcqi_metrics(stdout: str) -> dict[str, str]:
77     metrics: dict[str, str] = {}
78     for line in stdout.splitlines():
79         if not line:
80             continue
81         key, _, value = line.partition(" ")
82         if key in {
83             "engine",
84             "generated_text",
85             "benchmark_load_ms",
86             "benchmark_tokenize_ms",
87             "benchmark_prefill_ms",
88             "benchmark_decode_ms",
89             "benchmark_generate_ms",
90             "benchmark_total_ms",
91             "benchmark_prompt_tokens",
92             "benchmark_generated_tokens",
93             "benchmark_prefill_steps",
94             "benchmark_decode_steps",

```

```

95         "benchmark_generate_tokens_per_second",
96         "benchmark_decode_tokens_per_second",
97         "benchmark_kv_cache_bytes",
98     }:
99         metrics[key] = value
100     return metrics
101
102
103 def run_lcqi(
104     binary: Path,
105     model_dir: Path,
106     *,
107     engine: str,
108     prompt: str,
109     max_new_tokens: int,
110     timeout_seconds: int,
111 ) -> EngineRun:
112     command = [
113         str(binary),
114         "--benchmark",
115         "--engine",
116         engine,
117         str(model_dir),
118         prompt,
119         str(max_new_tokens),
120     ]
121     returncode, stdout, stderr = run_command(command, timeout_seconds=↵
↵ timeout_seconds)
122     return EngineRun(
123         name=f"lcqi_{engine}",
124         command=command,
125         returncode=returncode,
126         stdout=stdout,
127         stderr=stderr,
128         metrics=parse_lcqi_metrics(stdout),
129     )
130
131
132 def llama_bench_binary(cache_dir: Path) -> Path:
133     return cache_dir / "llama.cpp" / "build" / "bin" / "llama-bench"
134
135
136 def smollm2_model_path(cache_dir: Path) -> Path:
137     return cache_dir / "models" / SMOLLM2_GGUF
138
139
140 def parse_llama_bench_metrics(stdout: str) -> dict[str, str]:
141     metrics: dict[str, str] = {}
142     lines = [line.strip() for line in stdout.splitlines() if line.strip()]
143     for line in lines:
144         if not line.startswith("|"):
145             continue

```

```

146     columns = [column.strip() for column in line.strip("|").split("|")]
147     if len(columns) != 7 or columns[0] == "model" or columns[0].startswith(↵
↵ "-"):
148         continue
149     metrics["llama_model"] = columns[0]
150     metrics["llama_size"] = columns[1]
151     metrics["llama_params"] = columns[2]
152     metrics["llama_backend"] = columns[3]
153     metrics["llama_threads"] = columns[4]
154     test_name = columns[5]
155     tokens_per_second = re.sub(r"\s+.*$", "", columns[6])
156     if test_name.startswith("pp"):
157         metrics["llama_prefill_test"] = test_name
158         metrics["llama_prefill_tps"] = tokens_per_second
159     elif test_name.startswith("tg"):
160         metrics["llama_decode_test"] = test_name
161         metrics["llama_decode_tps"] = tokens_per_second
162     return metrics
163
164
165 def run_llama_bench(cache_dir: Path, *, timeout_seconds: int) -> EngineRun | ↵
↵ None:
166     binary = llama_bench_binary(cache_dir)
167     model = smollm2_model_path(cache_dir)
168     if not binary.exists() or not os.access(binary, os.X_OK) or not model.↵
↵ exists():
169         return None
170     command = [
171         str(binary),
172         "-m",
173         str(model),
174         "-ngl",
175         "0",
176         "-p",
177         "5",
178         "-n",
179         "4",
180         "-r",
181         "1",
182     ]
183     returncode, stdout, stderr = run_command(command, timeout_seconds=↵
↵ timeout_seconds)
184     return EngineRun(
185         name="llama_cpp_smollm2_q4_k_m",
186         command=command,
187         returncode=returncode,
188         stdout=stdout,
189         stderr=stderr,
190         metrics=parse_llama_bench_metrics(stdout),
191     )
192
193

```

```

194 def write_report(path: Path, runs: list[EngineRun], *, prompt: str, ↵
    ↵ max_new_tokens: int) -> None:
195     path.parent.mkdir(parents=True, exist_ok=True)
196     lines = [
197         "LCQI GPT-2 Benchmark Compare",
198         "",
199         f"timestamp_utc={dt.datetime.now(dt.UTC).isoformat()}",
200         f"repo_commit={current_commit()}",
201         f"working_tree_dirty={working_tree_dirty()}",
202         f"python={sys.version.split()[0]}",
203         f"prompt={prompt}",
204         f"max_new_tokens={max_new_tokens}",
205         "",
206         "scope=LCQI full_prefix and cached_kv run the same GPT-2 F32 ↵
    ↵ safetensors model. "
207         "llama.cpp uses SmolLM2-135M-Instruct Q4_K_M as an external mature-↵
    ↵ engine reference, "
208         "so its numbers are engineering context, not same-model quality ranking↵
    ↵ .",
209         "",
210     ]
211     for run in runs:
212         lines.extend(
213             [
214                 f"[{run.name}]",
215                 f"returncode={run.returncode}",
216                 f"command={command_line(run.command)}",
217             ]
218         )
219         for key in sorted(run.metrics):
220             lines.append(f"{key}={run.metrics[key]}")
221         lines.extend(
222             [
223                 "stdout_tail:",
224                 tail(run.stdout.strip(), 2000),
225                 "stderr_tail:",
226                 tail(run.stderr.strip(), 2000),
227                 "",
228             ]
229         )
230     while lines and lines[-1] == "":
231         lines.pop()
232     path.write_text("\n".join(lines) + "\n", encoding="utf-8")
233
234
235 def current_commit() -> str:
236     result = subprocess.run(
237         ["git", "-C", str(repo_dir()), "rev-parse", "--short", "HEAD"],
238         text=True,
239         capture_output=True,
240         check=False,
241     )

```



```

242     return result.stdout.strip() if result.returncode == 0 else "unknown"
243
244
245 def working_tree_dirty() -> str:
246     result = subprocess.run(
247         ["git", "-C", str(repo_dir()), "status", "--porcelain"],
248         text=True,
249         capture_output=True,
250         check=False,
251     )
252     if result.returncode != 0:
253         return "unknown"
254     return "true" if result.stdout.strip() else "false"
255
256
257 def parse_args() -> argparse.Namespace:
258     parser = argparse.ArgumentParser(
259         description="Compare LCQI GPT-2 full-prefix and KV-cache paths, with ↵
↵ optional llama.cpp context."
260     )
261     parser.add_argument("--cache-dir", type=Path, default=run_gpt2_smoke.↵
↵ default_cache_dir())
262     parser.add_argument("--build-dir", type=Path, default=default_build_dir())
263     parser.add_argument("--report", type=Path, default=default_report_path())
264     parser.add_argument("--prompt", default=DEFAULT_PROMPT)
265     parser.add_argument("--max-new-tokens", type=int, default=↵
↵ DEFAULT_MAX_NEW_TOKENS)
266     parser.add_argument("--jobs", type=int, default=DEFAULT_BUILD_JOBS)
267     parser.add_argument("--timeout-seconds", type=int, default=↵
↵ DEFAULT_TIMEOUT_SECONDS)
268     parser.add_argument("--no-download", action="store_true")
269     parser.add_argument("--skip-llama", action="store_true")
270     parser.add_argument("--smollm2-cache-dir", type=Path, default=↵
↵ DEFAULT_SMOLLM2_CACHE_DIR)
271     return parser.parse_args()
272
273
274 def main() -> int:
275     args = parse_args()
276     if args.max_new_tokens < 0:
277         print("[lcqi-compare] --max-new-tokens cannot be negative", file=sys.↵
↵ stderr)
278         return 1
279     try:
280         run_gpt2_smoke.ensure_model_files(
281             args.cache_dir,
282             no_download=args.no_download,
283             timeout_seconds=args.timeout_seconds,
284         )
285         binary = run_gpt2_smoke.build_lcqi_gpt2(args.build_dir, args.jobs)
286         runs = [
287             run_lcqi(

```

```

288         binary,
289         args.cache_dir,
290         engine="full",
291         prompt=args.prompt,
292         max_new_tokens=args.max_new_tokens,
293         timeout_seconds=args.timeout_seconds,
294     ),
295     run_lcqi(
296         binary,
297         args.cache_dir,
298         engine="cached",
299         prompt=args.prompt,
300         max_new_tokens=args.max_new_tokens,
301         timeout_seconds=args.timeout_seconds,
302     ),
303 ]
304 if not args.skip_llama:
305     llama_run = run_llama_bench(
306         args.smollm2_cache_dir.expanduser().resolve(),
307         timeout_seconds=args.timeout_seconds,
308     )
309     if llama_run is not None:
310         runs.append(llama_run)
311 write_report(args.report, runs, prompt=args.prompt, max_new_tokens=args.
↪ .max_new_tokens)
312 except Exception as error:
313     print(f"[lcqi-compare] error: {error}", file=sys.stderr)
314     return 1
315
316 for run in runs:
317     print(f"{run.name}: returncode={run.returncode}")
318     for key in sorted(run.metrics):
319         print(f"  {key}={run.metrics[key]}")
320 print(f"report={args.report}")
321 return 0 if all(run.returncode == 0 for run in runs) else 1
322
323
324 if __name__ == "__main__":
325     raise SystemExit(main())

```

`run_gpt2_optimization_ab.py` 专门比较 LCQI 自身改动前后。它从 baseline commit 建临时 worktree，baseline 和当前工作区都通过实践卷 CMake 入口构建 Release，再多轮运行同一个 GPT-2 F32 模型。报告中的 `median/min/max` 用来约束机器波动，`generated_text` 用来确认优化前后输出没有漂移，`benchmark_worker_count` 用来确认 cached decoder 到底是串行、自动线程数还是固定线程数。

**Listing 16.19** tools/run\_gpt2\_optimization\_ab.py

```

1 #!/usr/bin/env python3
2 from __future__ import annotations
3
4 import argparse
5 import datetime as dt

```

```
6 import shutil
7 import statistics
8 import subprocess
9 import sys
10 import tempfile
11 from dataclasses import dataclass
12 from pathlib import Path
13
14
15 DEFAULT_BASELINE_REF = "4febf3f"
16 DEFAULT_PROMPT = "Hello, my name is"
17 DEFAULT_MAX_NEW_TOKENS = 4
18 DEFAULT_ROUNDS = 3
19 DEFAULT_BUILD_JOBS = 2
20 DEFAULT_TIMEOUT_SECONDS = 300
21
22 METRIC_KEYS = [
23     "benchmark_prefill_ms",
24     "benchmark_decode_ms",
25     "benchmark_generate_ms",
26     "benchmark_worker_count",
27     "benchmark_generate_tokens_per_second",
28     "benchmark_decode_tokens_per_second",
29 ]
30
31
32 @dataclass(frozen=True)
33 class RunSample:
34     variant: str
35     engine: str
36     round_id: int
37     metrics: dict[str, float]
38     generated_text: str
39
40
41 def script_path() -> Path:
42     return Path(__file__).resolve()
43
44
45 def repo_dir() -> Path:
46     return script_path().parents[3]
47
48
49 def volume_dir() -> Path:
50     return script_path().parents[1]
51
52
53 def practice_dir(root: Path) -> Path:
54     return root / "books" / "cpu-volume-3-practice"
55
56
57 def default_report_path() -> Path:
```

```

58     return volume_dir() / "results" / "lcqi-gpt2-optimization-ab.txt"
59
60
61 def default_model_dir() -> Path:
62     return Path.home() / ".cache" / "lcqi-gpt2-smoke" / "openai-community--gpt2"
63     ↪ "
64
65 def run_command(
66     args: list[str],
67     *,
68     cwd: Path,
69     timeout_seconds: int,
70 ) -> subprocess.CompletedProcess[str]:
71     return subprocess.run(
72         args,
73         cwd=cwd,
74         text=True,
75         capture_output=True,
76         timeout=timeout_seconds,
77         check=False,
78     )
79
80
81 def ensure_success(
82     completed: subprocess.CompletedProcess[str],
83     action: str,
84 ) -> None:
85     if completed.returncode == 0:
86         return
87     command = (
88         " ".join(completed.args)
89         if isinstance(completed.args, list)
90         else str(completed.args)
91     )
92     raise RuntimeError(
93         f"{action} failed with exit code {completed.returncode}\n"
94         f"command: {command}\n"
95         f"stdout tail:\n{completed.stdout[-2000:]} \n"
96         f"stderr tail:\n{completed.stderr[-2000:]} \n"
97     )
98
99
100 def git_text(args: list[str]) -> str:
101     completed = run_command(["git", *args], cwd=repo_dir(), timeout_seconds=60)
102     ensure_success(completed, "git command")
103     return completed.stdout.strip()
104
105
106 def working_tree_dirty() -> bool:
107     completed = run_command(
108         ["git", "status", "--porcelain"],

```

```

109     cwd=repo_dir(),
110     timeout_seconds=60,
111 )
112 ensure_success(completed, "git status")
113 return bool(completed.stdout.strip())
114
115
116 def build_binary(root: Path, build_dir: Path, jobs: int, timeout_seconds: int) ←
    ↪ -> Path:
117     configure = run_command(
118         [
119             "cmake",
120             "-S",
121             str(practice_dir(root)),
122             "-B",
123             str(build_dir),
124             "-DCMAKE_BUILD_TYPE=Release",
125         ],
126         cwd=root,
127         timeout_seconds=timeout_seconds,
128     )
129     ensure_success(configure, f"configure {root}")
130     build = run_command(
131         [
132             "cmake",
133             "--build",
134             str(build_dir),
135             "--target",
136             "lcqi_gpt2",
137             "-j",
138             str(jobs),
139         ],
140         cwd=root,
141         timeout_seconds=timeout_seconds,
142     )
143     ensure_success(build, f"build {root}")
144     binary = build_dir / "linux_cpu_inference" / "lcqi_gpt2"
145     if not binary.exists():
146         raise RuntimeError(f"lcqi_gpt2 binary not found: {binary}")
147     return binary
148
149
150 def parse_metrics(stdout: str) -> tuple[dict[str, float], str]:
151     metrics: dict[str, float] = {}
152     generated_text = ""
153     for line in stdout.splitlines():
154         key, _, value = line.partition(" ")
155         if key in METRIC_KEYS:
156             metrics[key] = float(value)
157         elif key == "generated_text":
158             generated_text = value
159     metrics.setdefault("benchmark_worker_count", 1.0)

```

```

160     missing = sorted(set(METRIC_KEYS) - metrics.keys())
161     if missing:
162         raise RuntimeError(f"missing benchmark metrics: {missing}")
163     return metrics, generated_text
164
165
166 def run_sample(
167     binary: Path,
168     *,
169     variant: str,
170     engine: str,
171     round_id: int,
172     model_dir: Path,
173     prompt: str,
174     max_new_tokens: int,
175     threads: int,
176     timeout_seconds: int,
177 ) -> RunSample:
178     command = [
179         str(binary),
180         "--benchmark",
181         "--engine",
182         engine,
183     ]
184     if variant == "current":
185         command.extend(["--threads", str(threads)])
186     command.extend([
187         str(model_dir),
188         prompt,
189         str(max_new_tokens),
190     ])
191     completed = run_command(
192         command,
193         cwd=repo_dir(),
194         timeout_seconds=timeout_seconds,
195     )
196     ensure_success(completed, f"run {variant} {engine}")
197     metrics, generated_text = parse_metrics(completed.stdout)
198     return RunSample(
199         variant=variant,
200         engine=engine,
201         round_id=round_id,
202         metrics=metrics,
203         generated_text=generated_text,
204     )
205
206
207 def summarize(samples: list[RunSample]) -> list[str]:
208     lines: list[str] = []
209     variants = sorted({sample.variant for sample in samples})
210     engines = sorted({sample.engine for sample in samples})
211     for variant in variants:

```

```

212     for engine in engines:
213         selected = [
214             sample for sample in samples
215             if sample.variant == variant and sample.engine == engine
216         ]
217         if not selected:
218             continue
219         lines.append(f"[{variant} {engine}]")
220         for key in METRIC_KEYS:
221             values = [sample.metrics[key] for sample in selected]
222             lines.append(
223                 f"{key}=median:{statistics.median(values):.6g},"
224                 f"min:{min(values):.6g},max:{max(values):.6g}"
225             )
226         texts = sorted({sample.generated_text for sample in selected})
227         lines.append(f"generated_text={texts[0] if len(texts) == 1 else ↵
↵ texts}")
228         lines.append("")
229     return lines
230
231
232 def write_report(
233     path: Path,
234     samples: list[RunSample],
235     *,
236     baseline_ref: str,
237     baseline_commit: str,
238     current_commit: str,
239     prompt: str,
240     max_new_tokens: int,
241     rounds: int,
242     current_threads: int,
243 ) -> None:
244     path.parent.mkdir(parents=True, exist_ok=True)
245     lines = [
246         "LCQI GPT-2 Optimization A/B",
247         "",
248         f"timestamp_utc={dt.datetime.now(dt.UTC).isoformat()}",
249         f"baseline_ref={baseline_ref}",
250         f"baseline_commit={baseline_commit}",
251         f"current_commit={current_commit}",
252         f"current_working_tree_dirty={str(working_tree_dirty()).lower()}",
253         f"prompt={prompt}",
254         f"max_new_tokens={max_new_tokens}",
255         f"rounds={rounds}",
256         f"baseline_threads=1",
257         f"current_threads={current_threads}",
258         "",
259         "scope=Both variants are built through books/cpu-volume-3-practice with↵
↵ "
260         "CMAKE_BUILD_TYPE=Release and run on the same GPT-2 F32 safetensors ↵
↵ model. "

```

```

261     "The llama.cpp comparison remains in lcqi-gpt2-benchmark-compare.txt; ←
↪ this "
262     "report isolates LCQI before/after code changes.",
263     "",
264     "[samples]",
265 ]
266 for sample in samples:
267     metric_text = ",".join(
268         f"{key}:{sample.metrics[key]:.6g}" for key in METRIC_KEYS
269     )
270     lines.append(
271         f"round={sample.round_id},variant={sample.variant},engine={sample.↪
↪ engine},"
272         f"{metric_text},generated_text={sample.generated_text}"
273     )
274     lines.append("")
275     lines.extend(summarize(samples))
276     while lines and lines[-1] == "":
277         lines.pop()
278     path.write_text("\n".join(lines) + "\n", encoding="utf-8")
279
280
281 def parse_args() -> argparse.Namespace:
282     parser = argparse.ArgumentParser(description="Run LCQI GPT-2 before/after A↪
↪ /B benchmark.")
283     parser.add_argument("--baseline-ref", default=DEFAULT_BASELINE_REF)
284     parser.add_argument("--model-dir", type=Path, default=default_model_dir())
285     parser.add_argument("--report", type=Path, default=default_report_path())
286     parser.add_argument("--prompt", default=DEFAULT_PROMPT)
287     parser.add_argument("--max-new-tokens", type=int, default=↪
↪ DEFAULT_MAX_NEW_TOKENS)
288     parser.add_argument("--rounds", type=int, default=DEFAULT_ROUNDS)
289     parser.add_argument("--threads", type=int, default=0)
290     parser.add_argument("--jobs", type=int, default=DEFAULT_BUILD_JOBS)
291     parser.add_argument("--timeout-seconds", type=int, default=↪
↪ DEFAULT_TIMEOUT_SECONDS)
292     return parser.parse_args()
293
294
295 def main() -> int:
296     args = parse_args()
297     if args.rounds <= 0:
298         print("[lcqi-ab] --rounds must be positive", file=sys.stderr)
299         return 1
300     if args.max_new_tokens < 0:
301         print("[lcqi-ab] --max-new-tokens cannot be negative", file=sys.stderr)
302         return 1
303     if args.threads < 0:
304         print("[lcqi-ab] --threads cannot be negative", file=sys.stderr)
305         return 1
306     if not args.model_dir.exists():
307         print(f"[lcqi-ab] model directory not found: {args.model_dir}", file=↪

```



```

↪ sys.stderr)
308     return 1
309
310     baseline_commit = git_text(["rev-parse", "--short", args.baseline_ref])
311     current_commit = git_text(["rev-parse", "--short", "HEAD"])
312     with tempfile.TemporaryDirectory(prefix="lcqi-gpt2-ab-") as tmp:
313         tmp_path = Path(tmp)
314         baseline_root = tmp_path / "baseline"
315         current_build = tmp_path / "current-build"
316         baseline_build = tmp_path / "baseline-build"
317         ensure_success(
318             run_command(
319                 ↪ .baseline_ref],
320                 ["git", "worktree", "add", "--detach", str(baseline_root), args↪
321                 cwd=repo_dir(),
322                 timeout_seconds=args.timeout_seconds,
323             ),
324             "create baseline worktree",
325         )
326         try:
327             baseline_binary = build_binary(
328                 baseline_root,
329                 baseline_build,
330                 args.jobs,
331                 args.timeout_seconds,
332             )
333             current_binary = build_binary(
334                 repo_dir(),
335                 current_build,
336                 args.jobs,
337                 args.timeout_seconds,
338             )
339
340             samples: list[RunSample] = []
341             for round_id in range(1, args.rounds + 1):
342                 for variant, binary in [
343                     ("baseline", baseline_binary),
344                     ("current", current_binary),
345                 ]:
346                     for engine in ["full", "cached"]:
347                         sample = run_sample(
348                             binary,
349                             variant=variant,
350                             engine=engine,
351                             round_id=round_id,
352                             model_dir=args.model_dir,
353                             prompt=args.prompt,
354                             max_new_tokens=args.max_new_tokens,
355                             threads=1 if variant == "baseline" else args.↪
356                             ↪ threads,
357                             timeout_seconds=args.timeout_seconds,
358                         )

```

```

357         samples.append(sample)
358         print(
359             f"round={round_id} variant={variant} engine={engine}↵
↵ } "
360             f"generate_ms={sample.metrics['↵
↵ benchmark_generate_ms']:.3f} "
361             f"workers={sample.metrics['benchmark_worker_count↵
↵ ']:.0f} "
362             f"gen_tps={sample.metrics['↵
↵ benchmark_generate_tokens_per_second']:.5f}"
363         )
364         write_report(
365             args.report,
366             samples,
367             baseline_ref=args.baseline_ref,
368             baseline_commit=baseline_commit,
369             current_commit=current_commit,
370             prompt=args.prompt,
371             max_new_tokens=args.max_new_tokens,
372             rounds=args.rounds,
373             current_threads=args.threads,
374         )
375     finally:
376         subprocess.run(
377             ["git", "worktree", "remove", "--force", str(baseline_root)],
378             cwd=repo_dir(),
379             text=True,
380             capture_output=True,
381             check=False,
382         )
383     print(f"report={args.report}")
384     return 0
385
386
387 if __name__ == "__main__":
388     raise SystemExit(main())

```

`run_gpt2_hotspot_profile.py` 专门收集 cached decode 内部热点。它构建 Release `lcqi_gpt2`，用 `--profile-hotspots` 输出 `hotspot_lm_head_pct`、`hotspot_mlp_fc_pct`、`hotspot_mlp_projection_pct`、`hotspot_qkv_projection_pct`、`hotspot_attention_projection_pct`、`hotspot_attention_pct` 和 `benchmark_worker_count`，再按 token 数和轮次计算中位数。这个脚本回答“优化应该打哪里”，不是回答“某次提交快了多少”。本轮线程优化就是靠它发现第一次 auto 阈值漏掉了 attention output projection，随后把行数和列数阈值一起纳入并行判定。

**Listing 16.20** `tools/run_gpt2_hotspot_profile.py`

```

1 #!/usr/bin/env python3
2 from __future__ import annotations
3
4 import argparse
5 import datetime as dt
6 import statistics

```

```

7 import subprocess
8 from dataclasses import dataclass
9 from pathlib import Path
10
11
12 DEFAULT_PROMPT = "Hello, my name is"
13 DEFAULT_TOKEN_COUNTS = "4,16"
14 DEFAULT_ROUNDS = 3
15 DEFAULT_BUILD_JOBS = 2
16 DEFAULT_TIMEOUT_SECONDS = 900
17
18 METRIC_PREFIXES = ("benchmark_", "hotspot_")
19
20
21 @dataclass(frozen=True)
22 class HotspotSample:
23     max_new_tokens: int
24     round_id: int
25     metrics: dict[str, float]
26     generated_text: str
27
28
29 def script_path() -> Path:
30     return Path(__file__).resolve()
31
32
33 def repo_dir() -> Path:
34     return script_path().parents[3]
35
36
37 def volume_dir() -> Path:
38     return script_path().parents[1]
39
40
41 def practice_dir() -> Path:
42     return repo_dir() / "books" / "cpu-volume-3-practice"
43
44
45 def default_build_dir() -> Path:
46     return volume_dir() / "build" / "lcqi-hotspot-profile"
47
48
49 def default_model_dir() -> Path:
50     return Path.home() / ".cache" / "lcqi-gpt2-smoke" / "openai-community--gpt2↵
    ↵ "
51
52
53 def default_report_path() -> Path:
54     return volume_dir() / "results" / "lcqi-gpt2-hotspot-profile.txt"
55
56
57 def run_command(

```

```

58     args: list[str],
59     *,
60     cwd: Path,
61     timeout_seconds: int,
62 ) -> subprocess.CompletedProcess[str]:
63     return subprocess.run(
64         args,
65         cwd=cwd,
66         text=True,
67         capture_output=True,
68         timeout=timeout_seconds,
69         check=False,
70     )
71
72
73 def ensure_success(completed: subprocess.CompletedProcess[str], action: str) -> None:
74     ↪ None:
75     if completed.returncode == 0:
76         return
77     command = (
78         " ".join(completed.args)
79         if isinstance(completed.args, list)
80         else str(completed.args)
81     )
82     raise RuntimeError(
83         f"{action} failed with exit code {completed.returncode}\n"
84         f"command: {command}\n"
85         f"stdout tail:\n{completed.stdout[-2000:]]}\n"
86         f"stderr tail:\n{completed.stderr[-2000:]]}"
87     )
88
89 def build_binary(build_dir: Path, jobs: int, timeout_seconds: int) -> Path:
90     configure = run_command(
91         [
92             "cmake",
93             "-S",
94             str(practice_dir()),
95             "-B",
96             str(build_dir),
97             "-DCMAKE_BUILD_TYPE=Release",
98         ],
99         cwd=repo_dir(),
100         timeout_seconds=timeout_seconds,
101     )
102     ensure_success(configure, "configure hotspot profile build")
103     build = run_command(
104         [
105             "cmake",
106             "--build",
107             str(build_dir),
108             "--target",

```

```

109         "lcqi_gpt2",
110         "-j",
111         str(jobs),
112     ],
113     cwd=repo_dir(),
114     timeout_seconds=timeout_seconds,
115 )
116 ensure_success(build, "build lcqi_gpt2")
117 binary = build_dir / "linux_cpu_inference" / "lcqi_gpt2"
118 if not binary.exists():
119     raise RuntimeError(f"lcqi_gpt2 binary not found: {binary}")
120 return binary
121
122
123 def parse_numeric_list(value: str) -> list[int]:
124     result: list[int] = []
125     for part in value.split(","):
126         stripped = part.strip()
127         if not stripped:
128             continue
129         parsed = int(stripped)
130         if parsed <= 0:
131             raise RuntimeError("--token-counts entries must be positive")
132         result.append(parsed)
133     if not result:
134         raise RuntimeError("--token-counts must contain at least one positive ←
135     ↪ integer")
136     return result
137
138 def parse_stdout(stdout: str) -> tuple[dict[str, float], str]:
139     metrics: dict[str, float] = {}
140     generated_text = ""
141     for line in stdout.splitlines():
142         key, _, value = line.partition(" ")
143         if key.startswith(METRIC_PREFIXES):
144             metrics[key] = float(value)
145         elif key == "generated_text":
146             generated_text = value
147     required = {
148         "benchmark_generate_ms",
149         "benchmark_worker_count",
150         "hotspot_total_step_ms",
151         "hotspot_lm_head_pct",
152         "hotspot_mlp_fc_pct",
153         "hotspot_mlp_projection_pct",
154         "hotspot_qkv_projection_pct",
155         "hotspot_attention_pct",
156     }
157     missing = sorted(required - metrics.keys())
158     if missing:
159         raise RuntimeError(f"missing hotspot metrics: {missing}")

```

```

160     if not generated_text:
161         raise RuntimeError("missing generated_text")
162     return metrics, generated_text
163
164
165 def run_sample(
166     binary: Path,
167     *,
168     model_dir: Path,
169     prompt: str,
170     max_new_tokens: int,
171     round_id: int,
172     threads: int,
173     timeout_seconds: int,
174 ) -> HotspotSample:
175     completed = run_command(
176         [
177             str(binary),
178             "--profile-hotspots",
179             "--engine",
180             "cached",
181             "--threads",
182             str(threads),
183             str(model_dir),
184             prompt,
185             str(max_new_tokens),
186         ],
187         cwd=repo_dir(),
188         timeout_seconds=timeout_seconds,
189     )
190     ensure_success(completed, f"run hotspot profile tokens={max_new_tokens}")
191     metrics, generated_text = parse_stdout(completed.stdout)
192     return HotspotSample(
193         max_new_tokens=max_new_tokens,
194         round_id=round_id,
195         metrics=metrics,
196         generated_text=generated_text,
197     )
198
199
200 def summarize(samples: list[HotspotSample]) -> list[str]:
201     lines: list[str] = []
202     metric_keys = [
203         "benchmark_load_ms",
204         "benchmark_generate_ms",
205         "benchmark_generate_tokens_per_second",
206         "benchmark_decode_tokens_per_second",
207         "hotspot_total_step_ms",
208         "benchmark_worker_count",
209         "hotspot_lm_head_pct",
210         "hotspot_mlp_fc_pct",
211         "hotspot_mlp_projection_pct",

```

```

212     "hotspot_qkv_projection_pct",
213     "hotspot_attention_projection_pct",
214     "hotspot_attention_pct",
215 ]
216 for token_count in sorted({sample.max_new_tokens for sample in samples}):
217     selected = [sample for sample in samples if sample.max_new_tokens == ←
↪ token_count]
218     lines.append(f"[summary max_new_tokens={token_count}]")
219     for key in metric_keys:
220         values = [sample.metrics[key] for sample in selected if key in ←
↪ sample.metrics]
221         if not values:
222             continue
223         lines.append(
224             f"{key}=median:{statistics.median(values):.6g},"
225             f"min:{min(values):.6g},max:{max(values):.6g}"
226         )
227         texts = sorted({sample.generated_text for sample in selected})
228         lines.append(f"generated_text={texts[0] if len(texts) == 1 else texts}"↪
↪ )
229         lines.append("")
230     return lines
231
232
233 def write_report(
234     path: Path,
235     samples: list[HotspotSample],
236     *,
237     model_dir: Path,
238     prompt: str,
239     token_counts: list[int],
240     rounds: int,
241     threads: int,
242 ) -> None:
243     path.parent.mkdir(parents=True, exist_ok=True)
244     lines = [
245         "LCQI GPT-2 Hotspot Profile",
246         "",
247         f"timestamp_utc={dt.datetime.now(dt.UTC).isoformat()}",
248         f"model_dir={model_dir}",
249         f"prompt={prompt}",
250         f"token_counts='{','.join(str(value) for value in token_counts)}'",
251         f"rounds={rounds}",
252         f"requested_threads={threads}",
253         "engine=cached",
254         "build=CMAKE_BUILD_TYPE=Release through books/cpu-volume-3-practice",
255         "",
256         "[samples]",
257     ]
258     for sample in samples:
259         metric_text = ",".join(
260             f"{key}:{sample.metrics[key]:.6g}"

```

```

261         for key in sorted(sample.metrics)
262         if key.startswith(METRIC_PREFIXES)
263     )
264     lines.append(
265         f"round={sample.round_id},max_new_tokens={sample.max_new_tokens},"
266         f"{metric_text},generated_text={sample.generated_text}"
267     )
268     lines.append("")
269     lines.extend(summarize(samples))
270     while lines and lines[-1] == "":
271         lines.pop()
272     path.write_text("\n".join(lines) + "\n", encoding="utf-8")
273
274
275 def parse_args() -> argparse.Namespace:
276     parser = argparse.ArgumentParser(description="Run LCQI GPT-2 cached hotspot↵
↵ profile.")
277     parser.add_argument("--model-dir", type=Path, default=default_model_dir())
278     parser.add_argument("--build-dir", type=Path, default=default_build_dir())
279     parser.add_argument("--report", type=Path, default=default_report_path())
280     parser.add_argument("--prompt", default=DEFAULT_PROMPT)
281     parser.add_argument("--token-counts", default=DEFAULT_TOKEN_COUNTS)
282     parser.add_argument("--rounds", type=int, default=DEFAULT_ROUNDS)
283     parser.add_argument("--threads", type=int, default=0)
284     parser.add_argument("--jobs", type=int, default=DEFAULT_BUILD_JOBS)
285     parser.add_argument("--timeout-seconds", type=int, default=↵
↵ DEFAULT_TIMEOUT_SECONDS)
286     return parser.parse_args()
287
288
289 def main() -> int:
290     args = parse_args()
291     if args.rounds <= 0:
292         print("[lcqi-hotspot] --rounds must be positive")
293         return 1
294     if args.threads < 0:
295         print("[lcqi-hotspot] --threads cannot be negative")
296         return 1
297     if not args.model_dir.exists():
298         print(f"[lcqi-hotspot] model directory not found: {args.model_dir}")
299         return 1
300     token_counts = parse_numeric_list(args.token_counts)
301     binary = build_binary(args.build_dir, args.jobs, args.timeout_seconds)
302     samples: list[HotspotSample] = []
303     for round_id in range(1, args.rounds + 1):
304         for token_count in token_counts:
305             sample = run_sample(
306                 binary,
307                 model_dir=args.model_dir,
308                 prompt=args.prompt,
309                 max_new_tokens=token_count,
310                 round_id=round_id,

```



```

311         threads=args.threads,
312         timeout_seconds=args.timeout_seconds,
313     )
314     samples.append(sample)
315     print(
316         f"round={round_id} max_new_tokens={token_count} "
317         f"generate_ms={sample.metrics['benchmark_generate_ms']:.3f} "
318         f"workers={sample.metrics['benchmark_worker_count']:.0f} "
319         f"lm_head_pct={sample.metrics['hotspot_lm_head_pct']:.3f} "
320         f"mlp_pct={sample.metrics['hotspot_mlp_fc_pct']} + sample.↵
↵ metrics['hotspot_mlp_projection_pct']:.3f}"
321     )
322     write_report(
323         args.report,
324         samples,
325         model_dir=args.model_dir,
326         prompt=args.prompt,
327         token_counts=token_counts,
328         rounds=args.rounds,
329         threads=args.threads,
330     )
331     print(f"report={args.report}")
332     return 0
333
334
335 if __name__ == "__main__":
336     raise SystemExit(main())

```

`run_lcqi_benchmark.sh` 是本地 benchmark 收集脚本。它适合在固定机器上反复运行，把同一组 shape 和 backend 的结果落到报告文件。

**Listing 16.21** tools/run\_lcqi\_benchmark.sh

```

1  #!/usr/bin/env bash
2  set -euo pipefail
3
4  SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
5  VOLUME_DIR="$(cd "${SCRIPT_DIR}/.." && pwd)"
6  REPO_DIR="$(cd "${VOLUME_DIR}/../.." && pwd)"
7  BUILD_DIR="${VOLUME_DIR}/build/lcqi-release"
8  RESULT_DIR="${VOLUME_DIR}/results"
9  REPEAT="${1:-200}"
10 STAMP="$(date -u +%Y%m%dT%H%M%S)"
11 CSV_PATH="${RESULT_DIR}/lcqi-bench-${STAMP}.csv"
12 META_PATH="${RESULT_DIR}/lcqi-bench-${STAMP}.meta.txt"
13 PERF_PATH="${RESULT_DIR}/lcqi-bench-${STAMP}.perf.txt"
14
15 mkdir -p "${RESULT_DIR}"
16
17 cmake -S "${VOLUME_DIR}" -B "${BUILD_DIR}" -DCMAKE_BUILD_TYPE=Release
18 cmake --build "${BUILD_DIR}"
19 ctest --test-dir "${BUILD_DIR}" --output-on-failure
20

```

```

21 {
22     echo "timestamp_utc=${STAMP}"
23     echo "repeat=${REPEAT}"
24     echo "commit=$(git -C "${REPO_DIR}" rev-parse --short HEAD)"
25     echo "uname=$(uname -a)"
26     echo "compiler=$((${CXX:-c++} --version | sed -n '1p')"
27     echo "cmake=$(cmake --version | sed -n '1p')"
28     echo "bench=${BUILD_DIR}/labs/linux_cpu_inference/lcqi_bench"
29 } > "${META_PATH}"
30
31 "${BUILD_DIR}/labs/linux_cpu_inference/lcqi_bench" "${REPEAT}" > "${CSV_PATH}"
32
33 if command -v perf >/dev/null 2>&1; then
34     perf stat \
35         -e cycles,instructions,cache-misses,branches,branch-misses \
36         "${BUILD_DIR}/labs/linux_cpu_inference/lcqi_bench" "${REPEAT}" \
37         > /dev/null 2> "${PERF_PATH}" || true
38 else
39     echo "perf_unavailable=1" > "${PERF_PATH}"
40 fi
41
42 echo "csv=${CSV_PATH}"
43 echo "meta=${META_PATH}"
44 echo "perf=${PERF_PATH}"

```

## 完整测试

`lcqi_tests.cpp` 把 safetensors、tokenizer、tiny MLP、int8 kernel、reference decoder、GPT-2 tiny path、trace、ledger 和 serving probe 串成验收网。GPT-2 优化不是只看速度：测试会把 full-prefix logits、旧 cached logits、优化 decoder logits、强制两 worker 的并行 decoder logits 和 logits-free greedy token 放在同一个 tiny prompt 下对齐，确认 predicted token、logits、KV filled token 数和 generated ids 不漂移。新增优化或 shape 时，必须先扩展这里的 golden 和边界样例。

Listing 16.22 tests/lcqi\_tests.cpp

```

1 #include <lcqi/gpt2_reference.hpp>
2 #include <lcqi/f32_kernels.hpp>
3 #include <lcqi/ggml_matvec.hpp>
4 #include <lcqi/ggml_tensors.hpp>
5 #include <lcqi/gguf.hpp>
6 #include <lcqi/inference.hpp>
7 #include <lcqi/int8_kernels.hpp>
8 #include <lcqi/llama_gguf.hpp>
9 #include <lcqi/model.hpp>
10 #include <lcqi/q4_k.hpp>
11 #include <lcqi/reference_decoder.hpp>
12 #include <lcqi/safetensors.hpp>
13 #include <lcqi/tokenizer.hpp>
14
15 #include <array>
16 #include <cmath>
17 #include <cstring>

```

```

18 #include <cstdlib>
19 #include <filesystem>
20 #include <fstream>
21 #include <iostream>
22 #include <span>
23 #include <sstream>
24 #include <stdexcept>
25 #include <string>
26 #include <utility>
27 #include <vector>
28
29 namespace {
30
31 constexpr float LCQI_TEST_EPSILON = 1.0e-5F;
32 constexpr std::int32_t LCQI_TEST_INPUT_SIZE = 4;
33 constexpr std::int32_t LCQI_TEST_HIDDEN_SIZE = 3;
34 constexpr std::int32_t LCQI_TEST_OUTPUT_SIZE = 2;
35 constexpr std::int32_t LCQI_TEST_PREDICTED_CLASS = 1;
36 constexpr float LCQI_TEST_LOGIT_0 = -0.48F;
37 constexpr float LCQI_TEST_LOGIT_1 = 1.06F;
38 constexpr float LCQI_TEST_HIDDEN_0 = 0.0F;
39 constexpr float LCQI_TEST_HIDDEN_1 = 0.4F;
40 constexpr float LCQI_TEST_HIDDEN_2 = 1.4F;
41 constexpr float LCQI_RMS_EXPECTED_0 = 0.848528F;
42 constexpr float LCQI_RMS_EXPECTED_1 = 0.565685F;
43 constexpr float LCQI_ATTENTION_EXPECTED_0 = 2.0F;
44 constexpr float LCQI_ATTENTION_EXPECTED_1 = 3.0F;
45 constexpr float LCQI_KERNEL_MAX_DIFF = 1.0e-5F;
46 constexpr float LCQI_F32_KERNEL_MAX_DIFF = 1.0e-4F;
47 constexpr std::int32_t LCQI_SIMD_TEST_BLOCK = 8;
48 constexpr std::size_t LCQI_F32_DOT_TEST_SIZE = 257;
49 constexpr std::size_t LCQI_F32_ROW_TEST_INPUT_SIZE = 65;
50 constexpr std::size_t LCQI_F32_ROW_TEST_OUTPUT_SIZE = 17;
51 constexpr std::int32_t LCQI_REFERENCE_DECODER_PREDICTED_TOKEN = 0;
52 constexpr std::size_t LCQI_REFERENCE_DECODER_VOCAB_SIZE = 5;
53 constexpr std::array<float, LCQI_REFERENCE_DECODER_VOCAB_SIZE> ↵
    ↵ LCQI_REFERENCE_DECODER_LOGITS{
54     0.352516979F,
55     -0.126884326F,
56     0.0797837451F,
57     -0.29797405F,
58     0.127030417F,
59 };
60 constexpr std::int32_t LCQI_REFERENCE_TRACE_TOKEN_COUNT = 3;
61 constexpr std::int32_t LCQI_REFERENCE_TRACE_HIDDEN_SIZE = 4;
62 constexpr std::int32_t LCQI_REFERENCE_TRACE_FIRST_TOKEN = 0;
63 constexpr std::uint64_t LCQI_TOKENIZER_HASH_EXPECTED = 13452902845388333734ULL;
64 constexpr std::int32_t LCQI_SAFETENSORS_PREFIX_BYTES = 8;
65 constexpr std::int32_t LCQI_TEST_BYTE_BITS = 8;
66 constexpr std::uint64_t LCQI_SAFETENSORS_VALID_PAYLOAD_BYTES = 48;
67 constexpr std::uint64_t LCQI_SAFETENSORS_EMBED_BEGIN = 0;
68 constexpr std::uint64_t LCQI_SAFETENSORS_EMBED_END = 24;

```

```

69 constexpr std::uint64_t LCQI_SAFETENSORS_QPROJ_BEGIN = 24;
70 constexpr std::uint64_t LCQI_SAFETENSORS_QPROJ_END = 48;
71 constexpr unsigned int LCQI_BYTE_MASK = 0xFFU;
72 constexpr std::int32_t LCQI_GPT2_PREDICTED_TOKEN = 3;
73 constexpr std::size_t LCQI_GPT2_VOCAB_SIZE = 6;
74 constexpr std::array<float, LCQI_GPT2_VOCAB_SIZE> LCQI_GPT2_LOGITS{
75     0.407358F,
76     -0.504049F,
77     -0.107834F,
78     0.840337F,
79     -0.450123F,
80     -0.156763F,
81 };
82 constexpr std::int32_t LCQI_GPT2_GENERATED_LENGTH = 5;
83 constexpr std::int32_t LCQI_GPT2_TOKENIZER_HELLO_ID = 7;
84 constexpr std::uint16_t LCQI_TEST_F16_ONE = 0x3C00U;
85 constexpr std::uint16_t LCQI_TEST_F16_HALF = 0x3800U;
86 constexpr std::uint16_t LCQI_TEST_F16_NEGATIVE_TWO = 0xC000U;
87 constexpr std::uint16_t LCQI_TEST_BF16_ONE_POINT_FIVE = 0x3FC0U;
88 constexpr std::uint16_t LCQI_TEST_BF16_NEGATIVE_HALF = 0xBF00U;
89 constexpr std::uint32_t LCQI_GGUF_TEST_VERSION = 3;
90 constexpr std::uint32_t LCQI_GGUF_TEST_ALIGNMENT = 32;
91 constexpr std::int64_t LCQI_GGUF_TEST_TENSOR_COUNT = 1;
92 constexpr std::int64_t LCQI_GGUF_TEST_METADATA_COUNT = 3;
93 constexpr std::int32_t LCQI_GGUF_TYPE_UINT32 = 4;
94 constexpr std::int32_t LCQI_GGUF_TYPE_FLOAT32 = 6;
95 constexpr std::int32_t LCQI_GGUF_TYPE_STRING = 8;
96 constexpr std::int32_t LCQI_GGUF_TYPE_ARRAY = 9;
97 constexpr std::int32_t LCQI_GGUF_TENSOR_TYPE_Q4_K = 12;
98 constexpr std::uint64_t LCQI_GGUF_TEST_TENSOR_OFFSET = 0;
99 constexpr float LCQI_Q4K_TEST_BLOCK_SCALE = 0.5F;
100 constexpr float LCQI_Q4K_TEST_BLOCK_MIN = 0.25F;
101 constexpr float LCQI_Q4K_Q8_DIFF_LIMIT = 0.08F;
102 constexpr float LCQI_GGML_MATVEC_MAX_DIFF = 1.0e-4F;
103 constexpr float LCQI_GGML_Q8_MATVEC_MAX_DIFF = 0.2F;
104 constexpr std::int32_t LCQI_GGML_TEST_Q5_HALF_VALUES = 16;
105 constexpr std::int32_t LCQI_GGML_TEST_Q5_ZERO_POINT = 16;
106 constexpr std::int32_t LCQI_GGML_TEST_Q6_LANE = 5;
107 constexpr std::int32_t LCQI_LLAMA_TEST_HIDDEN = 4;
108 constexpr std::int32_t LCQI_LLAMA_TEST_HEADS = 2;
109 constexpr std::int32_t LCQI_LLAMA_TEST_KV_HEADS = 1;
110 constexpr std::int32_t LCQI_LLAMA_TEST_HEAD_DIM = 2;
111 constexpr std::int32_t LCQI_LLAMA_TEST_INTERMEDIATE = 4;
112 constexpr std::int32_t LCQI_LLAMA_TEST_VOCAB = 4;
113 constexpr std::int32_t LCQI_LLAMA_TEST_CONTEXT = 8;
114 constexpr std::int32_t LCQI_LLAMA_TEST_LAYER_COUNT = 1;
115 constexpr std::int32_t LCQI_LLAMA_TEST_EXPECTED_TOKEN = 2;
116 constexpr float LCQI_LLAMA_TEST_SMALL_ATTENTION_VALUE = 0.1F;
117 constexpr float LCQI_LLAMA_TEST_SMALL_OUTPUT_SCALE = 0.1F;
118 constexpr std::int32_t LCQI_LLAMA_Q4_TEST_HIDDEN = 256;
119 constexpr std::int32_t LCQI_LLAMA_Q4_TEST_HEADS = 1;
120 constexpr std::int32_t LCQI_LLAMA_Q4_TEST_KV_HEADS = 1;

```

```

121 constexpr std::int32_t LCQI_LLAMA_Q4_TEST_HEAD_DIM = 256;
122 constexpr std::int32_t LCQI_LLAMA_Q4_TEST_INTERMEDIATE = 256;
123 constexpr std::int32_t LCQI_LLAMA_Q4_TEST_VOCAB = 2;
124 constexpr std::int32_t LCQI_LLAMA_Q4_TEST_CONTEXT = 4;
125 constexpr std::int32_t LCQI_LLAMA_Q4_TEST_LAYER_COUNT = 1;
126 constexpr const char* LCQI_TEST_LLAMA_GGML_DIRECT_ENV = "LCQI_LLAMA_GGML_DIRECT"↵
    ↵ ";
127 constexpr const char* LCQI_TEST_TRUE_ENV = "1";
128
129 struct KernelShapeCase {
130     std::int32_t input_size = 0;
131     std::int32_t output_size = 0;
132     std::int32_t output_block_size = 0;
133 };
134
135 struct RawTensorFixture {
136     std::string name;
137     std::string dtype;
138     std::vector<std::int64_t> shape;
139     std::vector<std::uint8_t> bytes;
140 };
141
142 struct FloatTensorFixture {
143     std::string name;
144     std::vector<std::int64_t> shape;
145     std::vector<float> values;
146 };
147
148 struct GgufTensorFixture {
149     std::string name;
150     std::vector<std::int64_t> shape;
151     lcqi::GgmlType type = lcqi::GgmlType::f32;
152     std::vector<std::uint8_t> bytes;
153     std::uint64_t relative_offset = 0;
154 };
155
156 struct ScopedEnvVar {
157     std::string name;
158     std::string old_value;
159     bool had_old_value = false;
160
161     ScopedEnvVar(const char* env_name, const char* value) : name(env_name) {
162         const char* old = std::getenv(env_name);
163         if (old != nullptr) {
164             this->old_value = old;
165             this->had_old_value = true;
166         }
167         if (setenv(env_name, value, 1) != 0) {
168             throw std::runtime_error("failed to set test environment variable")↵
                ↵ ;
169         }
170     }

```







```

272         std::uint64_t payload_bytes) {
273     std::ofstream output(path, std::ios::binary);
274     if (!output) {
275         throw std::runtime_error("cannot create safetensors test fixture");
276     }
277     write_le_u64(output, static_cast<std::uint64_t>(header.size()));
278     output.write(header.data(), static_cast<std::streamsize>(header.size()));
279     for (std::uint64_t i = 0; i < payload_bytes; ++i) {
280         const char byte = static_cast<char>(i & 0xFFU);
281         output.write(&byte, 1);
282     }
283 }
284
285 std::string json_shape(std::span<const std::int64_t> shape) {
286     std::ostringstream stream;
287     stream << '[';
288     for (std::size_t index = 0; index < shape.size(); ++index) {
289         if (index != 0) {
290             stream << ',';
291         }
292         stream << shape[index];
293     }
294     stream << ']';
295     return stream.str();
296 }
297
298 void write_safetensors_raw_fixture(const std::filesystem::path& path,
299                                   std::span<const RawTensorFixture> tensors) {
300     std::ostringstream header;
301     header << "{\"__metadata__\":{\"format\":\"lcqi-test\"}";
302     std::uint64_t offset = 0;
303     for (const RawTensorFixture& tensor : tensors) {
304         header << ",\"\" << tensor.name << "\":{\"dtype\":\"\" << tensor.dtype
305             << "\",\"shape\":\"\" << json_shape(tensor.shape)
306             << "\",\"data_offsets\":[\" << offset << ', '
307             << offset + tensor.bytes.size() << "]}\"";
308         offset += tensor.bytes.size();
309     }
310     header << '}'';
311
312     std::ofstream output(path, std::ios::binary);
313     if (!output) {
314         throw std::runtime_error("cannot create safetensors raw test fixture");
315     }
316     const std::string header_text = header.str();
317     write_le_u64(output, static_cast<std::uint64_t>(header_text.size()));
318     output.write(header_text.data(), static_cast<std::streamsize>(header_text.
↵ size()));
319     for (const RawTensorFixture& tensor : tensors) {
320         output.write(reinterpret_cast<const char*>(tensor.bytes.data()),
321                     static_cast<std::streamsize>(tensor.bytes.size()));
322     }

```



```

323 }
324
325 std::vector<std::uint8_t> f32_payload(std::span<const float> values) {
326     std::vector<std::uint8_t> bytes;
327     bytes.reserve(values.size() * 4);
328     for (const float value : values) {
329         append_le_f32(bytes, value);
330     }
331     return bytes;
332 }
333
334 std::vector<float> transpose_linear_to_hf_conv1d(const lcqi::Gpt2LinearF32& ←
    ↪ linear) {
335     std::vector<float> transposed(
336         static_cast<std::size_t>(linear.input_size) *
337         static_cast<std::size_t>(linear.output_size),
338         0.0F);
339     for (std::int32_t out = 0; out < linear.output_size; ++out) {
340         for (std::int32_t in = 0; in < linear.input_size; ++in) {
341             transposed[static_cast<std::size_t>(in) *
342                 static_cast<std::size_t>(linear.output_size) +
343                 static_cast<std::size_t>(out)] =
344                 linear.weights[static_cast<std::size_t>(out) *
345                     static_cast<std::size_t>(linear.input_size) ←
    ↪ +
346                     static_cast<std::size_t>(in)];
347         }
348     }
349     return transposed;
350 }
351
352 void add_f32_tensor(std::vector<RawTensorFixture>& tensors,
353                     std::string name,
354                     std::vector<std::int64_t> shape,
355                     std::span<const float> values) {
356     tensors.push_back(RawTensorFixture{
357         std::move(name),
358         "F32",
359         std::move(shape),
360         f32_payload(values),
361     });
362 }
363
364 void write_text_file(const std::filesystem::path& path, const std::string& text ←
    ↪ ) {
365     std::ofstream output(path);
366     if (!output) {
367         throw std::runtime_error("cannot create text fixture");
368     }
369     output << text;
370 }
371

```

```

372 std::vector<std::uint8_t> make_q4_k_test_block() {
373     std::vector<std::uint8_t> block(
374         static_cast<std::size_t>(lcqi::LCQI_Q4_K_BLOCK_BYTES),
375         0);
376     block[0] = 0x00U;
377     block[1] = 0x38U;
378     block[2] = 0x00U;
379     block[3] = 0x34U;
380
381     std::array<std::uint8_t, static_cast<std::size_t>(lcqi::LCQI_Q4_K_SUBBLOCKS↵
↵ )> scales{
382         1, 2, 3, 4, 5, 6, 7, 8,
383     };
384     std::array<std::uint8_t, static_cast<std::size_t>(lcqi::LCQI_Q4_K_SUBBLOCKS↵
↵ )> mins{
385         1, 1, 2, 2, 3, 3, 4, 4,
386     };
387     for (std::size_t index = 0; index < 4; ++index) {
388         block[4 + index] = static_cast<std::uint8_t>(
389             (scales[index] & 0x3FU) | ((scales[index + 4] >> 4U) << 6U));
390         block[8 + index] = static_cast<std::uint8_t>(
391             (mins[index] & 0x3FU) | ((mins[index + 4] >> 4U) << 6U));
392         block[12 + index] = static_cast<std::uint8_t>(
393             (scales[index + 4] & 0x0FU) | ((mins[index + 4] & 0x0FU) << 4U));
394     }
395
396     for (std::int32_t pair = 0; pair < 4; ++pair) {
397         const std::int32_t low_subblock = pair * 2;
398         const std::int32_t high_subblock = low_subblock + 1;
399         for (std::int32_t index = 0; index < lcqi::LCQI_Q4_K_SUBBLOCK_VALUES; ↵
↵ ++index) {
400             const std::uint8_t low = static_cast<std::uint8_t>(
401                 (low_subblock + index) & 0x0F);
402             const std::uint8_t high = static_cast<std::uint8_t>(
403                 (high_subblock + index + 1) & 0x0F);
404             block[static_cast<std::size_t>(16 + pair * 32 + index)] =
405                 static_cast<std::uint8_t>(low | (high << 4U));
406         }
407     }
408     return block;
409 }
410
411 std::vector<float> expected_q4_k_test_values() {
412     const std::array<float, static_cast<std::size_t>(lcqi::LCQI_Q4_K_SUBBLOCKS)↵
↵ > scales{
413         1, 2, 3, 4, 5, 6, 7, 8,
414     };
415     const std::array<float, static_cast<std::size_t>(lcqi::LCQI_Q4_K_SUBBLOCKS)↵
↵ > mins{
416         1, 1, 2, 2, 3, 3, 4, 4,
417     };
418     std::vector<float> values(

```

```

419     static_cast<std::size_t>(lcqi::LCQI_QK_K_BLOCK_VALUES),
420     0.0F);
421     for (std::int32_t subblock = 0; subblock < lcqi::LCQI_Q4_K_SUBBLOCKS; ++subblock) {
422         for (std::int32_t index = 0; index < lcqi::LCQI_Q4_K_SUBBLOCK_VALUES; ++index) {
423             const std::int32_t quantized =
424                 (subblock % 2 == 0)
425                 ? ((subblock + index) & 0x0F)
426                 : ((subblock + index + 1) & 0x0F);
427             values[static_cast<std::size_t>(
428                 subblock * lcqi::LCQI_Q4_K_SUBBLOCK_VALUES + index)] =
429                 LCQI_Q4K_TEST_BLOCK_SCALE * scales[static_cast<std::size_t>(subblock)] *
430                 static_cast<float>(quantized) -
431                 LCQI_Q4K_TEST_BLOCK_MIN * mins[static_cast<std::size_t>(subblock)];
432         }
433     }
434     return values;
435 }
436
437 std::vector<std::uint8_t> make_q8_0_test_block() {
438     std::vector<std::uint8_t> block(
439         static_cast<std::size_t>(lcqi::LCQI_Q8_0_BLOCK_BYTES),
440         0);
441     block[0] = static_cast<std::uint8_t>(LCQI_TEST_F16_HALF & LCQI_BYTE_MASK);
442     block[1] = static_cast<std::uint8_t>((LCQI_TEST_F16_HALF >>
443         LCQI_TEST_BYTE_BITS) &
444         LCQI_BYTE_MASK);
445     for (std::int32_t index = 0; index < lcqi::LCQI_Q8_0_BLOCK_VALUES; ++index) {
446         block[static_cast<std::size_t>(2 + index)] =
447             static_cast<std::uint8_t>(static_cast<std::int8_t>(index - 16));
448     }
449     return block;
450 }
451
452 std::vector<std::uint8_t> make_q5_0_test_block() {
453     std::vector<std::uint8_t> block(
454         static_cast<std::size_t>(lcqi::LCQI_Q5_0_BLOCK_BYTES),
455         0);
456     block[0] = static_cast<std::uint8_t>(LCQI_TEST_F16_HALF & LCQI_BYTE_MASK);
457     block[1] = static_cast<std::uint8_t>((LCQI_TEST_F16_HALF >>
458         LCQI_TEST_BYTE_BITS) &
459         LCQI_BYTE_MASK);
460
461     std::uint32_t qh = 0;
462     for (std::int32_t index = 0; index < lcqi::LCQI_Q5_0_BLOCK_VALUES / 2; ++index) {
463         const std::int32_t first_quantized = index -
464             LCQI_GGML_TEST_Q5_ZERO_POINT;

```

```

462     const std::int32_t second_quantized = index;
463     const std::uint8_t first_encoded =
464         static_cast<std::uint8_t>(first_quantized + ↵
↵ LCQI_GGML_TEST_Q5_ZERO_POINT);
465     const std::uint8_t second_encoded =
466         static_cast<std::uint8_t>(second_quantized + ↵
↵ LCQI_GGML_TEST_Q5_ZERO_POINT);
467     if ((first_encoded & 0x10U) != 0U) {
468         qh |= (1U << index);
469     }
470     if ((second_encoded & 0x10U) != 0U) {
471         qh |= (1U << (index + LCQI_GGML_TEST_Q5_HALF_VALUES));
472     }
473     const std::uint8_t low = static_cast<std::uint8_t>(first_encoded & 0↵
↵ 0xFU);
474     const std::uint8_t high = static_cast<std::uint8_t>(second_encoded & 0↵
↵ 0xFU);
475     block[static_cast<std::size_t>(6 + index)] =
476         static_cast<std::uint8_t>(low | (high << 4U));
477 }
478 block[2] = static_cast<std::uint8_t>(qh & LCQI_BYTE_MASK);
479 block[3] = static_cast<std::uint8_t>((qh >> LCQI_TEST_BYTE_BITS) & ↵
↵ LCQI_BYTE_MASK);
480 block[4] = static_cast<std::uint8_t>((qh >> (LCQI_TEST_BYTE_BITS * 2)) & ↵
↵ LCQI_BYTE_MASK);
481 block[5] = static_cast<std::uint8_t>((qh >> (LCQI_TEST_BYTE_BITS * 3)) & ↵
↵ LCQI_BYTE_MASK);
482 return block;
483 }
484
485 std::vector<std::uint8_t> make_q6_k_test_block() {
486     std::vector<std::uint8_t> block(
487         static_cast<std::size_t>(lcqi::LCQI_Q6_K_BLOCK_BYTES),
488         0);
489     const std::size_t ql_offset = 0;
490     const std::size_t qh_offset = static_cast<std::size_t>(lcqi::↵
↵ LCQI_Q6_K_BLOCK_VALUES / 2);
491     const std::size_t scales_offset =
492         qh_offset + static_cast<std::size_t>(lcqi::LCQI_Q6_K_BLOCK_VALUES / 4);
493     for (std::int32_t index = 0; index < 16; ++index) {
494         block[scales_offset + static_cast<std::size_t>(index)] = 1;
495     }
496     block[static_cast<std::size_t>(lcqi::LCQI_Q6_K_BLOCK_BYTES - 2)] =
497         static_cast<std::uint8_t>(LCQI_TEST_F16_ONE & LCQI_BYTE_MASK);
498     block[static_cast<std::size_t>(lcqi::LCQI_Q6_K_BLOCK_BYTES - 1)] =
499         static_cast<std::uint8_t>((LCQI_TEST_F16_ONE >> LCQI_TEST_BYTE_BITS) &
500             LCQI_BYTE_MASK);
501
502     const auto store_q6 = [&block, ql_offset, qh_offset](std::int32_t position,
503                                                         std::uint8_t encoded) {
504         const std::int32_t half = position / 128;
505         const std::int32_t local = position % 128;

```

```

506     const std::size_t ql_base = ql_offset + static_cast<std::size_t>(half *
↪ 64);
507     const std::size_t qh_base = qh_offset + static_cast<std::size_t>(half *
↪ 32);
508     const std::uint8_t low = encoded & 0x0FU;
509     const std::uint8_t high = static_cast<std::uint8_t>((encoded >> 4U) & 0
↪ x03U);
510     if (local < 32) {
511         block[ql_base + static_cast<std::size_t>(local)] =
512             static_cast<std::uint8_t>(
513                 (block[ql_base + static_cast<std::size_t>(local)] & 0xF0U)
↪ | low);
514         block[qh_base + static_cast<std::size_t>(local)] =
515             static_cast<std::uint8_t>(
516                 (block[qh_base + static_cast<std::size_t>(local)] & 0xFCU)
↪ | high);
517     } else if (local < 64) {
518         const std::size_t lane = static_cast<std::size_t>(local - 32);
519         block[ql_base + static_cast<std::size_t>(32) + lane] =
520             static_cast<std::uint8_t>(
521                 (block[ql_base + static_cast<std::size_t>(32) + lane] & 0
↪ xF0U) | low);
522         block[qh_base + lane] =
523             static_cast<std::uint8_t>((block[qh_base + lane] & 0xF3U) | (
↪ high << 2U));
524     } else if (local < 96) {
525         const std::size_t lane = static_cast<std::size_t>(local - 64);
526         block[ql_base + lane] =
527             static_cast<std::uint8_t>(
528                 (block[ql_base + lane] & 0x0FU) | (low << 4U));
529         block[qh_base + lane] =
530             static_cast<std::uint8_t>((block[qh_base + lane] & 0xCFU) | (
↪ high << 4U));
531     } else {
532         const std::size_t lane = static_cast<std::size_t>(local - 96);
533         block[ql_base + static_cast<std::size_t>(32) + lane] =
534             static_cast<std::uint8_t>(
535                 (block[ql_base + static_cast<std::size_t>(32) + lane] & 0
↪ x0FU) |
536                 (low << 4U));
537         block[qh_base + lane] =
538             static_cast<std::uint8_t>((block[qh_base + lane] & 0x3FU) | (
↪ high << 6U));
539     }
540 };
541
542 store_q6(LCQI_GGML_TEST_Q6_LANE, 35);
543 store_q6(32 + LCQI_GGML_TEST_Q6_LANE, 31);
544 store_q6(64 + LCQI_GGML_TEST_Q6_LANE, 63);
545 store_q6(96 + LCQI_GGML_TEST_Q6_LANE, 0);
546 store_q6(128 + LCQI_GGML_TEST_Q6_LANE, 36);
547 return block;

```

```

548 }
549
550 void write_gguf_q4_k_fixture(const std::filesystem::path& path) {
551     std::vector<std::uint8_t> bytes;
552     bytes.push_back('G');
553     bytes.push_back('G');
554     bytes.push_back('U');
555     bytes.push_back('F');
556     append_le_u32(bytes, LCQI_GGUF_TEST_VERSION);
557     append_le_i64(bytes, LCQI_GGUF_TEST_TENSOR_COUNT);
558     append_le_i64(bytes, LCQI_GGUF_TEST_METADATA_COUNT);
559
560     append_gguf_string(bytes, "general.alignment");
561     append_le_i32(bytes, LCQI_GGUF_TYPE_UINT32);
562     append_le_u32(bytes, LCQI_GGUF_TEST_ALIGNMENT);
563     append_gguf_string(bytes, "general.architecture");
564     append_le_i32(bytes, LCQI_GGUF_TYPE_STRING);
565     append_gguf_string(bytes, "lcqi-test");
566     append_gguf_string(bytes, "tokenizer.ggml.tokens");
567     append_le_i32(bytes, LCQI_GGUF_TYPE_ARRAY);
568     append_le_i32(bytes, LCQI_GGUF_TYPE_STRING);
569     append_le_u64(bytes, 2);
570     append_gguf_string(bytes, "<bos>");
571     append_gguf_string(bytes, "hello");
572
573     append_gguf_string(bytes, "blk.0.ffn_up.weight");
574     append_le_u32(bytes, 2);
575     append_le_i64(bytes, lcqi::LCQI_QK_K_BLOCK_VALUES);
576     append_le_i64(bytes, 1);
577     append_le_i32(bytes, LCQI_GGUF_TENSOR_TYPE_Q4_K);
578     append_le_u64(bytes, LCQI_GGUF_TEST_TENSOR_OFFSET);
579     while (bytes.size() % LCQI_GGUF_TEST_ALIGNMENT != 0) {
580         bytes.push_back(0);
581     }
582     const std::vector<std::uint8_t> block = make_q4_k_test_block();
583     bytes.insert(bytes.end(), block.begin(), block.end());
584
585     std::ofstream output(path, std::ios::binary);
586     if (!output) {
587         throw std::runtime_error("cannot create GGUF fixture");
588     }
589     output.write(reinterpret_cast<const char*>(bytes.data()),
590                 static_cast<std::streamsize>(bytes.size()));
591 }
592
593 void append_gguf_uint32_metadata(std::vector<std::uint8_t>& bytes,
594                                 const std::string& key,
595                                 std::uint32_t value) {
596     append_gguf_string(bytes, key);
597     append_le_i32(bytes, LCQI_GGUF_TYPE_UINT32);
598     append_le_u32(bytes, value);
599 }

```

```

600
601 void append_gguf_float32_metadata(std::vector<std::uint8_t>& bytes,
602                                   const std::string& key,
603                                   float value) {
604     append_gguf_string(bytes, key);
605     append_le_i32(bytes, LCQI_GGUF_TYPE_FLOAT32);
606     append_le_f32(bytes, value);
607 }
608
609 void append_gguf_string_metadata(std::vector<std::uint8_t>& bytes,
610                                  const std::string& key,
611                                  const std::string& value) {
612     append_gguf_string(bytes, key);
613     append_le_i32(bytes, LCQI_GGUF_TYPE_STRING);
614     append_gguf_string(bytes, value);
615 }
616
617 void write_gguf_fixture(const std::filesystem::path& path,
618                        std::span<GgufTensorFixture> tensors,
619                        const std::vector<std::uint8_t>& metadata_bytes) {
620     std::uint64_t relative_offset = 0;
621     for (GgufTensorFixture& tensor : tensors) {
622         tensor.relative_offset = relative_offset;
623         relative_offset += tensor.bytes.size();
624         while (relative_offset % LCQI_GGUF_TEST_ALIGNMENT != 0) {
625             ++relative_offset;
626         }
627     }
628
629     std::vector<std::uint8_t> bytes;
630     bytes.push_back('G');
631     bytes.push_back('G');
632     bytes.push_back('U');
633     bytes.push_back('F');
634     append_le_u32(bytes, LCQI_GGUF_TEST_VERSION);
635     append_le_u64(bytes, static_cast<std::uint64_t>(tensors.size()));
636     append_le_u64(bytes, 14);
637     bytes.insert(bytes.end(), metadata_bytes.begin(), metadata_bytes.end());
638
639     for (const GgufTensorFixture& tensor : tensors) {
640         append_gguf_string(bytes, tensor.name);
641         append_le_u32(bytes, static_cast<std::uint32_t>(tensor.shape.size()));
642         for (const std::int64_t extent : tensor.shape) {
643             append_le_i64(bytes, extent);
644         }
645         append_le_i32(bytes, static_cast<std::int32_t>(tensor.type));
646         append_le_u64(bytes, tensor.relative_offset);
647     }
648     while (bytes.size() % LCQI_GGUF_TEST_ALIGNMENT != 0) {
649         bytes.push_back(0);
650     }
651     std::uint64_t written_relative = 0;

```



```

652     for (const GgufTensorFixture& tensor : tensors) {
653         while (written_relative < tensor.relative_offset) {
654             bytes.push_back(0);
655             ++written_relative;
656         }
657         bytes.insert(bytes.end(), tensor.bytes.begin(), tensor.bytes.end());
658         written_relative += tensor.bytes.size();
659     }
660
661     std::ofstream output(path, std::ios::binary);
662     if (!output) {
663         throw std::runtime_error("cannot create GGUF fixture");
664     }
665     output.write(reinterpret_cast<const char*>(bytes.data()),
666                 static_cast<std::streamsize>(bytes.size()));
667 }
668
669 void add_gguf_f32_tensor(std::vector<GgufTensorFixture>& tensors,
670                          std::string name,
671                          std::vector<std::int64_t> shape,
672                          std::span<const float> values) {
673     tensors.push_back(GgufTensorFixture{
674         std::move(name),
675         std::move(shape),
676         lcqi::GgmlType::f32,
677         f32_payload(values),
678         0,
679     });
680 }
681
682 void add_gguf_q4_k_tensor(std::vector<GgufTensorFixture>& tensors,
683                           std::string name,
684                           std::vector<std::int64_t> shape,
685                           const std::vector<std::uint8_t>& bytes) {
686     tensors.push_back(GgufTensorFixture{
687         std::move(name),
688         std::move(shape),
689         lcqi::GgmlType::q4_k,
690         bytes,
691         0,
692     });
693 }
694
695 void add_gguf_quantized_tensor(std::vector<GgufTensorFixture>& tensors,
696                                std::string name,
697                                std::vector<std::int64_t> shape,
698                                lcqi::GgmlType type,
699                                const std::vector<std::uint8_t>& bytes) {
700     tensors.push_back(GgufTensorFixture{
701         std::move(name),
702         std::move(shape),
703         type,

```



```

704     bytes,
705     0,
706 });
707 }
708
709 std::vector<std::uint8_t> repeated_ggml_block(const std::vector<std::uint8_t>& ↵
    ↵ block,
710
711         std::int32_t input_size,
712         std::int32_t output_size,
713         std::int64_t block_values) {
714     const std::size_t block_count =
715         static_cast<std::size_t>((input_size / block_values) * output_size);
716     std::vector<std::uint8_t> bytes;
717     bytes.reserve(block.size() * block_count);
718     for (std::size_t index = 0; index < block_count; ++index) {
719         bytes.insert(bytes.end(), block.begin(), block.end());
720     }
721     return bytes;
722 }
723
724 void write_tiny_llama_gguf_fixture(const std::filesystem::path& path) {
725     std::vector<std::uint8_t> metadata;
726     append_gguf_uint32_metadata(metadata, "general.alignment", ↵
    ↵ LCQI_GGUF_TEST_ALIGNMENT);
727     append_gguf_string_metadata(metadata, "general.architecture", "llama");
728     append_gguf_string_metadata(metadata, "general.name", "lcqi tiny llama");
729     append_gguf_uint32_metadata(metadata, "llama.embedding_length", ↵
    ↵ LCQI_LLAMA_TEST_HIDDEN);
730     append_gguf_uint32_metadata(metadata, "llama.block_count", ↵
    ↵ LCQI_LLAMA_TEST_LAYER_COUNT);
731     append_gguf_uint32_metadata(metadata, "llama.feed_forward_length", ↵
    ↵ LCQI_LLAMA_TEST_INTERMEDIATE);
732     append_gguf_uint32_metadata(metadata, "llama.attention.head_count", ↵
    ↵ LCQI_LLAMA_TEST_HEADS);
733     append_gguf_uint32_metadata(metadata, "llama.attention.head_count_kv", ↵
    ↵ LCQI_LLAMA_TEST_KV_HEADS);
734     append_gguf_uint32_metadata(metadata, "llama.rope.dimension_count", ↵
    ↵ LCQI_LLAMA_TEST_HEAD_DIM);
735     append_gguf_float32_metadata(metadata, "llama.rope.freq_base", 10000.0F);
736     append_gguf_uint32_metadata(metadata, "llama.context_length", ↵
    ↵ LCQI_LLAMA_TEST_CONTEXT);
737     append_gguf_uint32_metadata(metadata, "llama.vocab_size", ↵
    ↵ LCQI_LLAMA_TEST_VOCAB);
738     append_gguf_float32_metadata(metadata, "llama.attention.↵
    ↵ layer_norm_rms_epsilon", 1.0e-5F);
739     append_gguf_uint32_metadata(metadata, "tokenizer.ggml.eos_token_id", 3);
740
741     std::vector<GgufTensorFixture> tensors;
742     const std::vector<float> embedding{
743         1.0F, 0.0F, 0.0F, 0.0F,
744         0.0F, 1.0F, 0.0F, 0.0F,
745         0.0F, 2.0F, 0.0F, 0.0F,

```

```

745     0.0F, 0.0F, 1.0F, 0.0F,
746 };
747 add_gguf_f32_tensor(tensors,
748     "token_embd.weight",
749     {LCQI_LLAMA_TEST_HIDDEN, LCQI_LLAMA_TEST_VOCAB},
750     embedding);
751 add_gguf_f32_tensor(tensors,
752     "output_norm.weight",
753     {LCQI_LLAMA_TEST_HIDDEN},
754     std::vector<float>(LCQI_LLAMA_TEST_HIDDEN, 1.0F));
755 add_gguf_f32_tensor(tensors,
756     "blk.0.attn_norm.weight",
757     {LCQI_LLAMA_TEST_HIDDEN},
758     std::vector<float>(LCQI_LLAMA_TEST_HIDDEN, 1.0F));
759 add_gguf_f32_tensor(tensors,
760     "blk.0.ffn_norm.weight",
761     {LCQI_LLAMA_TEST_HIDDEN},
762     std::vector<float>(LCQI_LLAMA_TEST_HIDDEN, 1.0F));
763 const std::vector<float> hidden_to_hidden(
764     static_cast<std::size_t>(LCQI_LLAMA_TEST_HIDDEN * ←
↪ LCQI_LLAMA_TEST_HIDDEN),
765     0.0F);
766 std::vector<float> hidden_to_kv(
767     static_cast<std::size_t>(LCQI_LLAMA_TEST_HIDDEN * ←
↪ LCQI_LLAMA_TEST_HEAD_DIM),
768     0.0F);
769 hidden_to_kv[1] = LCQI_LLAMA_TEST_SMALL_ATTENTION_VALUE;
770 std::vector<float> attention_output = hidden_to_hidden;
771 attention_output[0] = LCQI_LLAMA_TEST_SMALL_OUTPUT_SCALE;
772 attention_output[10] = LCQI_LLAMA_TEST_SMALL_OUTPUT_SCALE;
773 const std::vector<float> hidden_to_intermediate(
774     static_cast<std::size_t>(LCQI_LLAMA_TEST_HIDDEN * ←
↪ LCQI_LLAMA_TEST_INTERMEDIATE),
775     0.0F);
776 const std::vector<float> intermediate_to_hidden(
777     static_cast<std::size_t>(LCQI_LLAMA_TEST_INTERMEDIATE * ←
↪ LCQI_LLAMA_TEST_HIDDEN),
778     0.0F);
779 add_gguf_f32_tensor(tensors,
780     "blk.0.attn_q.weight",
781     {LCQI_LLAMA_TEST_HIDDEN, LCQI_LLAMA_TEST_HIDDEN},
782     hidden_to_hidden);
783 add_gguf_f32_tensor(tensors,
784     "blk.0.attn_k.weight",
785     {LCQI_LLAMA_TEST_HIDDEN, LCQI_LLAMA_TEST_HEAD_DIM},
786     hidden_to_kv);
787 add_gguf_f32_tensor(tensors,
788     "blk.0.attn_v.weight",
789     {LCQI_LLAMA_TEST_HIDDEN, LCQI_LLAMA_TEST_HEAD_DIM},
790     hidden_to_kv);
791 add_gguf_f32_tensor(tensors,
792     "blk.0.attn_output.weight",

```

```

793         {LCQI_LLAMA_TEST_HIDDEN, LCQI_LLAMA_TEST_HIDDEN},
794         attention_output);
795     add_gguf_f32_tensor(tensors,
796         "blk.0.ffn_gate.weight",
797         {LCQI_LLAMA_TEST_HIDDEN, LCQI_LLAMA_TEST_INTERMEDIATE},
798         hidden_to_intermediate);
799     add_gguf_f32_tensor(tensors,
800         "blk.0.ffn_up.weight",
801         {LCQI_LLAMA_TEST_HIDDEN, LCQI_LLAMA_TEST_INTERMEDIATE},
802         hidden_to_intermediate);
803     add_gguf_f32_tensor(tensors,
804         "blk.0.ffn_down.weight",
805         {LCQI_LLAMA_TEST_INTERMEDIATE, LCQI_LLAMA_TEST_HIDDEN},
806         intermediate_to_hidden);
807     write_gguf_fixture(path, tensors, metadata);
808 }
809
810 void write_tiny_llama_ggml_direct_gguf_fixture(const std::filesystem::path& ↵
    ↵ path) {
811     std::vector<std::uint8_t> metadata;
812     append_gguf_uint32_metadata(metadata, "general.alignment", ↵
    ↵ LCQI_GGUF_TEST_ALIGNMENT);
813     append_gguf_string_metadata(metadata, "general.architecture", "llama");
814     append_gguf_string_metadata(metadata, "general.name", "lcqi ggml direct ↵
    ↵ tiny llama");
815     append_gguf_uint32_metadata(metadata, "llama.embedding_length", ↵
    ↵ LCQI_LLAMA_Q4_TEST_HIDDEN);
816     append_gguf_uint32_metadata(metadata, "llama.block_count", ↵
    ↵ LCQI_LLAMA_Q4_TEST_LAYER_COUNT);
817     append_gguf_uint32_metadata(metadata,
818         "llama.feed_forward_length",
819         LCQI_LLAMA_Q4_TEST_INTERMEDIATE);
820     append_gguf_uint32_metadata(metadata, "llama.attention.head_count", ↵
    ↵ LCQI_LLAMA_Q4_TEST_HEADS);
821     append_gguf_uint32_metadata(metadata,
822         "llama.attention.head_count_kv",
823         LCQI_LLAMA_Q4_TEST_KV_HEADS);
824     append_gguf_uint32_metadata(metadata,
825         "llama.rope.dimension_count",
826         LCQI_LLAMA_Q4_TEST_HEAD_DIM);
827     append_gguf_float32_metadata(metadata, "llama.rope.freq_base", 10000.0F);
828     append_gguf_uint32_metadata(metadata, "llama.context_length", ↵
    ↵ LCQI_LLAMA_Q4_TEST_CONTEXT);
829     append_gguf_uint32_metadata(metadata, "llama.vocab_size", ↵
    ↵ LCQI_LLAMA_Q4_TEST_VOCAB);
830     append_gguf_float32_metadata(metadata, "llama.attention.↵
    ↵ layer_norm_rms_epsilon", 1.0e-5F);
831     append_gguf_uint32_metadata(metadata, "tokenizer.ggml.eos_token_id", 1);
832
833     std::vector<GgufTensorFixture> tensors;
834     std::vector<float> embedding(
835         static_cast<std::size_t>(LCQI_LLAMA_Q4_TEST_HIDDEN * ↵

```

```

836     ↪ LCQI_LLAMA_Q4_TEST_VOCAB),
        0.0F);
837 embedding[0] = 1.0F;
838 embedding[static_cast<std::size_t>(LCQI_LLAMA_Q4_TEST_HIDDEN)] = 0.5F;
839 add_gguf_f32_tensor(tensors,
840                     "token_embd.weight",
841                     {LCQI_LLAMA_Q4_TEST_HIDDEN, LCQI_LLAMA_Q4_TEST_VOCAB},
842                     embedding);
843 add_gguf_f32_tensor(tensors,
844                     "output_norm.weight",
845                     {LCQI_LLAMA_Q4_TEST_HIDDEN},
846                     std::vector<float>(LCQI_LLAMA_Q4_TEST_HIDDEN, 1.0F));
847 add_gguf_f32_tensor(tensors,
848                     "blk.0.attn_norm.weight",
849                     {LCQI_LLAMA_Q4_TEST_HIDDEN},
850                     std::vector<float>(LCQI_LLAMA_Q4_TEST_HIDDEN, 1.0F));
851 add_gguf_f32_tensor(tensors,
852                     "blk.0.ffn_norm.weight",
853                     {LCQI_LLAMA_Q4_TEST_HIDDEN},
854                     std::vector<float>(LCQI_LLAMA_Q4_TEST_HIDDEN, 1.0F));
855
856 const std::vector<std::uint8_t> q5_matrix = repeated_ggml_block(
857     std::vector<std::uint8_t>(static_cast<std::size_t>(lcqi::↪
858     ↪ LCQI_Q5_0_BLOCK_BYTES), 0),
859     LCQI_LLAMA_Q4_TEST_HIDDEN,
860     LCQI_LLAMA_Q4_TEST_HIDDEN,
861     lcqi::LCQI_Q5_0_BLOCK_VALUES);
862 const std::vector<std::uint8_t> q8_matrix = repeated_ggml_block(
863     std::vector<std::uint8_t>(static_cast<std::size_t>(lcqi::↪
864     ↪ LCQI_Q8_0_BLOCK_BYTES), 0),
865     LCQI_LLAMA_Q4_TEST_HIDDEN,
866     LCQI_LLAMA_Q4_TEST_HIDDEN,
867     lcqi::LCQI_Q8_0_BLOCK_VALUES);
868 const std::vector<std::uint8_t> q6_matrix = repeated_ggml_block(
869     std::vector<std::uint8_t>(static_cast<std::size_t>(lcqi::↪
870     ↪ LCQI_Q6_K_BLOCK_BYTES), 0),
871     LCQI_LLAMA_Q4_TEST_INTERMEDIATE,
872     LCQI_LLAMA_Q4_TEST_HIDDEN,
873     lcqi::LCQI_Q6_K_BLOCK_VALUES);
874
875 add_gguf_quantized_tensor(tensors,
876                           "blk.0.attn_q.weight",
877                           {LCQI_LLAMA_Q4_TEST_HIDDEN, ↪
878     ↪ LCQI_LLAMA_Q4_TEST_HIDDEN},
879                           lcqi::GgmlType::q5_0,
880                           q5_matrix);
881 add_gguf_quantized_tensor(tensors,
882                           "blk.0.attn_k.weight",
883                           {LCQI_LLAMA_Q4_TEST_HIDDEN, ↪
884     ↪ LCQI_LLAMA_Q4_TEST_HEAD_DIM},
885                           lcqi::GgmlType::q5_0,
886                           q5_matrix);

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```

882     add_gguf_quantized_tensor(tensors,
883                               "blk.0.attn_v.weight",
884                               {LCQI_LLAMA_Q4_TEST_HIDDEN, ←
↪ LCQI_LLAMA_Q4_TEST_HEAD_DIM},
885                               lcqi::GgmlType::q8_0,
886                               q8_matrix);
887     add_gguf_quantized_tensor(tensors,
888                               "blk.0.attn_output.weight",
889                               {LCQI_LLAMA_Q4_TEST_HIDDEN, ←
↪ LCQI_LLAMA_Q4_TEST_HIDDEN},
890                               lcqi::GgmlType::q5_0,
891                               q5_matrix);
892     add_gguf_quantized_tensor(tensors,
893                               "blk.0.ffn_gate.weight",
894                               {LCQI_LLAMA_Q4_TEST_HIDDEN, ←
↪ LCQI_LLAMA_Q4_TEST_INTERMEDIATE},
895                               lcqi::GgmlType::q5_0,
896                               q5_matrix);
897     add_gguf_quantized_tensor(tensors,
898                               "blk.0.ffn_up.weight",
899                               {LCQI_LLAMA_Q4_TEST_HIDDEN, ←
↪ LCQI_LLAMA_Q4_TEST_INTERMEDIATE},
900                               lcqi::GgmlType::q8_0,
901                               q8_matrix);
902     add_gguf_quantized_tensor(tensors,
903                               "blk.0.ffn_down.weight",
904                               {LCQI_LLAMA_Q4_TEST_INTERMEDIATE, ←
↪ LCQI_LLAMA_Q4_TEST_HIDDEN},
905                               lcqi::GgmlType::q6_k,
906                               q6_matrix);
907     write_gguf_fixture(path, tensors, metadata);
908 }
909
910 void write_tiny_llama_q4_direct_gguf_fixture(const std::filesystem::path& path)↪
↪ {
911     std::vector<std::uint8_t> metadata;
912     append_gguf_uint32_metadata(metadata, "general.alignment", ←
↪ LCQI_GGUF_TEST_ALIGNMENT);
913     append_gguf_string_metadata(metadata, "general.architecture", "llama");
914     append_gguf_string_metadata(metadata, "general.name", "lcqi q4 direct tiny ↪
↪ llama");
915     append_gguf_uint32_metadata(metadata, "llama.embedding_length", ←
↪ LCQI_LLAMA_Q4_TEST_HIDDEN);
916     append_gguf_uint32_metadata(metadata, "llama.block_count", ←
↪ LCQI_LLAMA_Q4_TEST_LAYER_COUNT);
917     append_gguf_uint32_metadata(metadata, "llama.feed_forward_length", ←
↪ LCQI_LLAMA_Q4_TEST_INTERMEDIATE);
918     append_gguf_uint32_metadata(metadata, "llama.attention.head_count", ←
↪ LCQI_LLAMA_Q4_TEST_HEADS);
919     append_gguf_uint32_metadata(metadata, "llama.attention.head_count_kv", ←
↪ LCQI_LLAMA_Q4_TEST_KV_HEADS);
920     append_gguf_uint32_metadata(metadata, "llama.rope.dimension_count", ←

```

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    ↪ LCQI_LLAMA_Q4_TEST_HEAD_DIM);
921 append_gguf_float32_metadata(metadata, "llama.rope.freq_base", 10000.0F);
922 append_gguf_uint32_metadata(metadata, "llama.context_length", ↪
    ↪ LCQI_LLAMA_Q4_TEST_CONTEXT);
923 append_gguf_uint32_metadata(metadata, "llama.vocab_size", ↪
    ↪ LCQI_LLAMA_Q4_TEST_VOCAB);
924 append_gguf_float32_metadata(metadata, "llama.attention.↪
    ↪ layer_norm_rms_epsilon", 1.0e-5F);
925 append_gguf_uint32_metadata(metadata, "tokenizer.ggml.eos_token_id", 1);
926
927 std::vector<GgufTensorFixture> tensors;
928 std::vector<float> embedding(
929     static_cast<std::size_t>(LCQI_LLAMA_Q4_TEST_HIDDEN * ↪
    ↪ LCQI_LLAMA_Q4_TEST_VOCAB),
930     0.0F);
931 embedding[0] = 1.0F;
932 embedding[static_cast<std::size_t>(LCQI_LLAMA_Q4_TEST_HIDDEN)] = 0.5F;
933 add_gguf_f32_tensor(tensors,
934     "token_embd.weight",
935     {LCQI_LLAMA_Q4_TEST_HIDDEN, LCQI_LLAMA_Q4_TEST_VOCAB},
936     embedding);
937 add_gguf_f32_tensor(tensors,
938     "output_norm.weight",
939     {LCQI_LLAMA_Q4_TEST_HIDDEN},
940     std::vector<float>(LCQI_LLAMA_Q4_TEST_HIDDEN, 1.0F));
941 add_gguf_f32_tensor(tensors,
942     "blk.0.attn_norm.weight",
943     {LCQI_LLAMA_Q4_TEST_HIDDEN},
944     std::vector<float>(LCQI_LLAMA_Q4_TEST_HIDDEN, 1.0F));
945 add_gguf_f32_tensor(tensors,
946     "blk.0.ffn_norm.weight",
947     {LCQI_LLAMA_Q4_TEST_HIDDEN},
948     std::vector<float>(LCQI_LLAMA_Q4_TEST_HIDDEN, 1.0F));
949
950 const std::vector<std::uint8_t> zero_q4_matrix(
951     static_cast<std::size_t>(LCQI_LLAMA_Q4_TEST_HIDDEN) *
952     static_cast<std::size_t>(lcqi::LCQI_Q4_K_BLOCK_BYTES),
953     0);
954 add_gguf_q4_k_tensor(tensors,
955     "blk.0.attn_q.weight",
956     {LCQI_LLAMA_Q4_TEST_HIDDEN, LCQI_LLAMA_Q4_TEST_HIDDEN↪
    ↪ },
957     zero_q4_matrix);
958 add_gguf_q4_k_tensor(tensors,
959     "blk.0.attn_k.weight",
960     {LCQI_LLAMA_Q4_TEST_HIDDEN, ↪
    ↪ LCQI_LLAMA_Q4_TEST_HEAD_DIM},
961     zero_q4_matrix);
962 add_gguf_q4_k_tensor(tensors,
963     "blk.0.attn_v.weight",
964     {LCQI_LLAMA_Q4_TEST_HIDDEN, ↪
    ↪ LCQI_LLAMA_Q4_TEST_HEAD_DIM},

```

```

965         zero_q4_matrix);
966     add_gguf_q4_k_tensor(tensors,
967         "blk.0.attn_output.weight",
968         {LCQI_LLAMA_Q4_TEST_HIDDEN, LCQI_LLAMA_Q4_TEST_HIDDEN}
↪ },
969         zero_q4_matrix);
970     add_gguf_q4_k_tensor(tensors,
971         "blk.0.ffn_gate.weight",
972         {LCQI_LLAMA_Q4_TEST_HIDDEN,
↪ LCQI_LLAMA_Q4_TEST_INTERMEDIATE},
973         zero_q4_matrix);
974     add_gguf_q4_k_tensor(tensors,
975         "blk.0.ffn_up.weight",
976         {LCQI_LLAMA_Q4_TEST_HIDDEN,
↪ LCQI_LLAMA_Q4_TEST_INTERMEDIATE},
977         zero_q4_matrix);
978     add_gguf_q4_k_tensor(tensors,
979         "blk.0.ffn_down.weight",
980         {LCQI_LLAMA_Q4_TEST_INTERMEDIATE,
↪ LCQI_LLAMA_Q4_TEST_HIDDEN},
981         zero_q4_matrix);
982     write_gguf_fixture(path, tensors, metadata);
983 }
984
985 void write_tiny_gpt2_safetensors(const std::filesystem::path& path,
986     const lcqi::Gpt2ReferenceModel& model) {
987     std::vector<RawTensorFixture> tensors;
988     add_f32_tensor(tensors,
989         "transformer.wte.weight",
990         {model.config.vocab_size, model.config.hidden_size},
991         model.token_embedding);
992     add_f32_tensor(tensors,
993         "transformer.wpe.weight",
994         {model.config.max_positions, model.config.hidden_size},
995         model.position_embedding);
996     for (std::int32_t layer_id = 0; layer_id < model.config.layer_count; ++
↪ layer_id) {
997         const lcqi::Gpt2LayerWeightsF32& layer =
998             model.layers[static_cast<std::size_t>(layer_id)];
999         const std::string prefix = "transformer.h." + std::to_string(layer_id)
↪ + ".";
1000         add_f32_tensor(tensors, prefix + "ln_1.weight", {model.config.
↪ hidden_size}, layer.ln_1_weight);
1001         add_f32_tensor(tensors, prefix + "ln_1.bias", {model.config.hidden_size
↪ }, layer.ln_1_bias);
1002         const std::vector<float> c_attn_hf = transpose_linear_to_hf_conv1d(
↪ layer.c_attn);
1003         add_f32_tensor(tensors,
1004             prefix + "attn.c_attn.weight",
1005             {model.config.hidden_size,
1006                 model.config.hidden_size * 3},
1007             c_attn_hf);

```



```

1008     add_f32_tensor(tensors,
1009                     prefix + "attn.c_attn.bias",
1010                     {model.config.hidden_size * 3},
1011                     layer.c_attn.bias);
1012     const std::vector<float> c_proj_hf = transpose_linear_to_hf_conv1d(↵
↵ layer.c_proj);
1013     add_f32_tensor(tensors,
1014                     prefix + "attn.c_proj.weight",
1015                     {model.config.hidden_size, model.config.hidden_size},
1016                     c_proj_hf);
1017     add_f32_tensor(tensors,
1018                     prefix + "attn.c_proj.bias",
1019                     {model.config.hidden_size},
1020                     layer.c_proj.bias);
1021     add_f32_tensor(tensors, prefix + "ln_2.weight", {model.config.↵
↵ hidden_size}, layer.ln_2_weight);
1022     add_f32_tensor(tensors, prefix + "ln_2.bias", {model.config.hidden_size↵
↵ }, layer.ln_2_bias);
1023     const std::vector<float> c_fc_hf = transpose_linear_to_hf_conv1d(layer.↵
↵ c_fc);
1024     add_f32_tensor(tensors,
1025                     prefix + "mlp.c_fc.weight",
1026                     {model.config.hidden_size, model.config.↵
↵ intermediate_size},
1027                     c_fc_hf);
1028     add_f32_tensor(tensors,
1029                     prefix + "mlp.c_fc.bias",
1030                     {model.config.intermediate_size},
1031                     layer.c_fc.bias);
1032     const std::vector<float> mlp_proj_hf =
1033         transpose_linear_to_hf_conv1d(layer.mlp_c_proj);
1034     add_f32_tensor(tensors,
1035                     prefix + "mlp.c_proj.weight",
1036                     {model.config.intermediate_size, model.config.↵
↵ hidden_size},
1037                     mlp_proj_hf);
1038     add_f32_tensor(tensors,
1039                     prefix + "mlp.c_proj.bias",
1040                     {model.config.hidden_size},
1041                     layer.mlp_c_proj.bias);
1042 }
1043 add_f32_tensor(tensors,
1044                 "transformer.ln_f.weight",
1045                 {model.config.hidden_size},
1046                 model.final_ln_weight);
1047 add_f32_tensor(tensors,
1048                 "transformer.ln_f.bias",
1049                 {model.config.hidden_size},
1050                 model.final_ln_bias);
1051 write_safetensors_raw_fixture(path, tensors);
1052 }
1053

```



```

1054 void test_tiny_mlp() {
1055     const lcqi::TinyMlpModel model = lcqi::load_model(test_model_path());
1056     require(model.input_size == LCQI_TEST_INPUT_SIZE, "input size mismatch");
1057     require(model.hidden_size == LCQI_TEST_HIDDEN_SIZE, "hidden size mismatch")↵
    ↵ ;
1058     require(model.output_size == LCQI_TEST_OUTPUT_SIZE, "output size mismatch")↵
    ↵ ;
1059
1060     const std::vector<float> input{1.0F, 2.0F, -1.0F, 0.5F};
1061     const lcqi::InferenceResult result = lcqi::run_inference(model, input);
1062
1063     require(result.predicted_class == LCQI_TEST_PREDICTED_CLASS, "unexpected ↵
    ↵ predicted class");
1064     require(result.logits.size() == static_cast<std::size_t>(↵
    ↵ LCQI_TEST_OUTPUT_SIZE),
1065             "unexpected logits size");
1066     require_close(result.logits[0], LCQI_TEST_LOGIT_0, "logit 0 mismatch");
1067     require_close(result.logits[1], LCQI_TEST_LOGIT_1, "logit 1 mismatch");
1068
1069     std::vector<float> hidden(static_cast<std::size_t>(LCQI_TEST_HIDDEN_SIZE), ↵
    ↵ 0.0F);
1070     lcqi::linear_i8(model.hidden, input, hidden);
1071     lcqi::relu_inplace(hidden);
1072     require_close(hidden[0], LCQI_TEST_HIDDEN_0, "hidden 0 mismatch");
1073     require_close(hidden[1], LCQI_TEST_HIDDEN_1, "hidden 1 mismatch");
1074     require_close(hidden[2], LCQI_TEST_HIDDEN_2, "hidden 2 mismatch");
1075 }
1076
1077 void test_tokenizer_contract() {
1078     const lcqi::TokenizerModel tokenizer = lcqi::load_tokenizer(↵
    ↵ test_tokenizer_path());
1079     require(tokenizer.bos_id == 0, "tokenizer BOS id mismatch");
1080     require(tokenizer.eos_id == 4, "tokenizer EOS id mismatch");
1081     require(tokenizer.unk_id == -1, "tokenizer UNK id mismatch");
1082     require(tokenizer.chat_template_hash == "tiny-chat-template-v1",
1083             "tokenizer template hash mismatch");
1084
1085     const std::vector<std::int32_t> token_ids =
1086         lcqi::encode_prompt(tokenizer, "tok1 tok2 tok3");
1087     const std::vector<std::int32_t> expected{0, 1, 2, 3, 4};
1088     require(token_ids == expected, "tokenizer prompt ids mismatch");
1089     require(lcqi::tokenizer_contract_hash(tokenizer) == ↵
    ↵ LCQI_TOKENIZER_HASH_EXPECTED,
1090             "tokenizer contract hash mismatch");
1091 }
1092
1093 void test_tokenizer_rejects_unknown_without_unk() {
1094     const lcqi::TokenizerModel tokenizer = lcqi::load_tokenizer(↵
    ↵ test_tokenizer_path());
1095     bool threw = false;
1096     try {
1097         static_cast<void>(lcqi::encode_prompt(tokenizer, "tok1 missing_token"))↵

```

```

    ↪ ;
1098 } catch (const std::runtime_error&) {
1099     threw = true;
1100 }
1101 require(threw, "tokenizer should reject unknown token when unk id is absent"↪
    ↪ ");
1102 }
1103
1104 void test_safetensors_manifest_contract() {
1105     const std::filesystem::path path =
1106         temp_file_path("lcqi_safetensors_manifest_contract.safetensors");
1107     const std::string header =
1108         R"({"__metadata__":{"format":"lcqi-fixture"},"model.embed_tokens.weight"↪
    ↪ "":{"dtype":"F32","shape":[2,3],"data_offsets":[0,24]},"model.layers.0."↪
    ↪ attn.q_proj.weight":{"dtype":"F16","shape":[4,3],"data_offsets"↪
    ↪ :[24,48]}})";
1109     write_safetensors_fixture(path, header, ↪
    ↪ LCQI_SAFETENSORS_VALID_PAYLOAD_BYTES);
1110
1111     const lcqi::SafeTensorManifest manifest = lcqi::load_safetensors_manifest(↪
    ↪ path);
1112     std::filesystem::remove(path);
1113
1114     require(manifest.header_size == static_cast<std::uint64_t>(header.size()),
1115         "safetensors header size mismatch");
1116     require(manifest.data_start_offset ==
1117         static_cast<std::uint64_t>(LCQI_SAFETENSORS_PREFIX_BYTES) +
1118         static_cast<std::uint64_t>(header.size()),
1119         "safetensors data start mismatch");
1120     require(manifest.metadata.size() == 1, "safetensors metadata count mismatch"↪
    ↪ );
1121     require(manifest.metadata[0].first == "format" &&
1122         manifest.metadata[0].second == "lcqi-fixture",
1123         "safetensors metadata mismatch");
1124     require(manifest.tensors.size() == 2, "safetensors tensor count mismatch");
1125
1126     const lcqi::SafeTensorEntry* embedding =
1127         manifest.find_tensor("model.embed_tokens.weight");
1128     require(embedding != nullptr, "safetensors embedding tensor missing");
1129     require(embedding->dtype == "F32", "safetensors embedding dtype mismatch");
1130     require(embedding->shape == std::vector<std::int64_t>({2, 3}),
1131         "safetensors embedding shape mismatch");
1132     require(embedding->data_begin == LCQI_SAFETENSORS_EMBED_BEGIN &&
1133         embedding->data_end == LCQI_SAFETENSORS_EMBED_END,
1134         "safetensors embedding offset mismatch");
1135     require(embedding->byte_size() ==
1136         LCQI_SAFETENSORS_EMBED_END - LCQI_SAFETENSORS_EMBED_BEGIN,
1137         "safetensors embedding byte size mismatch");
1138
1139     const lcqi::SafeTensorEntry* q_proj =
1140         manifest.find_tensor("model.layers.0.attn.q_proj.weight");
1141     require(q_proj != nullptr, "safetensors q projection tensor missing");

```

```

1142     require(q_proj->dtype == "F16", "safetensors q projection dtype mismatch");
1143     require(q_proj->data_begin == LCQI_SAFETENSORS_QPROJ_BEGIN &&
1144           q_proj->data_end == LCQI_SAFETENSORS_QPROJ_END,
1145           "safetensors q projection offset mismatch");
1146 }
1147
1148 void test_safetensors_rejects_bad_offsets() {
1149     const std::filesystem::path path =
1150         temp_file_path("lcqi_safetensors_bad_offsets.safetensors");
1151     const std::string header =
1152         R"({"tensor":{"dtype":"F32","shape":[4],"data_offsets":[0,32]}})";
1153     write_safetensors_fixture(path, header, 4);
1154
1155     bool threw = false;
1156     try {
1157         static_cast<void>(lcqi::load_safetensors_manifest(path));
1158     } catch (const std::runtime_error&) {
1159         threw = true;
1160     }
1161     std::filesystem::remove(path);
1162     require(threw, "safetensors parser should reject offsets beyond payload");
1163 }
1164
1165 void test_safetensors_rejects_bad_dtype_shape_size() {
1166     const std::filesystem::path path =
1167         temp_file_path("lcqi_safetensors_bad_dtype_shape_size.safetensors");
1168     const std::string header =
1169         R"({"tensor":{"dtype":"F32","shape":[4],"data_offsets":[0,8]}})";
1170     write_safetensors_fixture(path, header, 8);
1171
1172     bool threw = false;
1173     try {
1174         static_cast<void>(lcqi::load_safetensors_manifest(path));
1175     } catch (const std::runtime_error&) {
1176         threw = true;
1177     }
1178     std::filesystem::remove(path);
1179     require(threw, "safetensors parser should reject dtype/shape byte mismatch"↵
1180     ↵ );
1181 }
1182
1183 void test_safetensors_reads_float_tensors() {
1184     const std::filesystem::path path =
1185         temp_file_path("lcqi_safetensors_float_read.safetensors");
1186     std::vector<std::uint8_t> f16_bytes;
1187     append_le_u16(f16_bytes, LCQI_TEST_F16_ONE);
1188     append_le_u16(f16_bytes, LCQI_TEST_F16_NEGATIVE_TWO);
1189     std::vector<std::uint8_t> bf16_bytes;
1190     append_le_u16(bf16_bytes, LCQI_TEST_BF16_ONE_POINT_FIVE);
1191     append_le_u16(bf16_bytes, LCQI_TEST_BF16_NEGATIVE_HALF);
1192
1193     const std::array<RawTensorFixture, 3> tensors{{

```

```

1193     {"f32", "F32", {2}, f32_payload(std::array<float, 2>{1.25F, -2.5F})},
1194     {"f16", "F16", {2}, f16_bytes},
1195     {"bf16", "BF16", {2}, bf16_bytes},
1196 };
1197 write_safetensors_raw_fixture(path, tensors);
1198
1199 const lcqi::SafeTensorFile file = lcqi::load_safetensors_file(path);
1200 std::filesystem::remove(path);
1201 const std::vector<float> f32 = lcqi::read_safetensor_f32_tensor(file, "f32"↵
↵ );
1202 const std::vector<float> f16 = lcqi::read_safetensor_f32_tensor(file, "f16"↵
↵ );
1203 const std::vector<float> bf16 = lcqi::read_safetensor_f32_tensor(file, "↵
↵ bf16");
1204
1205 require_close(f32[0], 1.25F, "safetensors F32 value 0 mismatch");
1206 require_close(f32[1], -2.5F, "safetensors F32 value 1 mismatch");
1207 require_close(f16[0], 1.0F, "safetensors F16 value 0 mismatch");
1208 require_close(f16[1], -2.0F, "safetensors F16 value 1 mismatch");
1209 require_close(bf16[0], 1.5F, "safetensors BF16 value 0 mismatch");
1210 require_close(bf16[1], -0.5F, "safetensors BF16 value 1 mismatch");
1211 }
1212
1213 void test_gguf_manifest_reads_q4_k_tensor() {
1214     const std::filesystem::path path = temp_file_path("lcqi_q4_k_fixture.gguf")↵
↵ ;
1215     write_gguf_q4_k_fixture(path);
1216
1217     const lcqi::GgufManifest manifest = lcqi::load_gguf_manifest(path);
1218     const lcqi::GgufTensorInfo* tensor =
1219         manifest.find_tensor("blk.0.ffn_up.weight");
1220     require(tensor != nullptr, "GGUF Q4_K tensor missing");
1221     const std::vector<std::uint8_t> bytes =
1222         lcqi::read_gguf_tensor_bytes(path, *tensor);
1223     std::filesystem::remove(path);
1224
1225     require(manifest.version == LCQI_GGUF_TEST_VERSION, "GGUF version mismatch"↵
↵ );
1226     require(manifest.alignment == LCQI_GGUF_TEST_ALIGNMENT, "GGUF alignment ↵
↵ mismatch");
1227     require(manifest.metadata.size() == LCQI_GGUF_TEST_METADATA_COUNT,
1228         "GGUF metadata count mismatch");
1229     const lcqi::GgufMetadataEntry* tokens = manifest.find_metadata("tokenizer.↵
↵ ggml.tokens");
1230     require(tokens != nullptr, "GGUF tokenizer tokens metadata missing");
1231     require(tokens->string_values == std::vector<std::string>({"<bos>", "hello"↵
↵ })),
1232         "GGUF tokenizer string array mismatch");
1233     require(manifest.tensors.size() == LCQI_GGUF_TEST_TENSOR_COUNT,
1234         "GGUF tensor count mismatch");
1235     require(tensor->type == lcqi::GgmlType::q4_k, "GGUF tensor type mismatch");
1236     require(tensor->shape == std::vector<std::int64_t>({lcqi::↵

```

```

    ↪ LCQI_QK_K_BLOCK_VALUES, 1})),
1237     "GGUF tensor shape mismatch");
1238     require(tensor->byte_size == lcqi::LCQI_Q4_K_BLOCK_BYTES,
1239     "GGUF tensor byte size mismatch");
1240     require(bytes == make_q4_k_test_block(), "GGUF tensor payload mismatch");
1241 }
1242
1243 void test_q4_k_dequant_and_dot_paths() {
1244     const std::vector<std::uint8_t> block = make_q4_k_test_block();
1245     const std::vector<float> expected = expected_q4_k_test_values();
1246     const std::vector<float> decoded =
1247         lcqi::dequantize_q4_k(block, lcqi::LCQI_QK_K_BLOCK_VALUES);
1248
1249     require(decoded.size() == expected.size(), "Q4_K decoded size mismatch");
1250     for (std::size_t index = 0; index < expected.size(); ++index) {
1251         require_close(decoded[index], expected[index], "Q4_K decoded value ↪
    ↪ mismatch");
1252     }
1253
1254     std::vector<float> input(expected.size(), 0.0F);
1255     for (std::size_t index = 0; index < input.size(); ++index) {
1256         input[index] = static_cast<float>(static_cast<std::int32_t>(index % 11U↪
    ↪ ) - 5) *
1257             0.125F;
1258     }
1259     float expected_dot = 0.0F;
1260     for (std::size_t index = 0; index < expected.size(); ++index) {
1261         expected_dot += expected[index] * input[index];
1262     }
1263     const float q4_f32_dot = lcqi::dot_q4_k_f32(block, input);
1264     require_close(q4_f32_dot, expected_dot, "Q4_K F32 dot mismatch");
1265
1266     const std::vector<lcqi::Q8KBlock> q8_input = lcqi::quantize_q8_k_input(↪
    ↪ input);
1267     const float q4_q8_dot = lcqi::dot_q4_k_q8(block, q8_input);
1268     require(std::fabs(q4_q8_dot - q4_f32_dot) <= LCQI_Q4K_Q8_DIFF_LIMIT,
1269     "Q4_K Q8 dot drift is too large");
1270
1271     std::vector<float> matvec_output(1, 0.0F);
1272     lcqi::matvec_q4_k_q8(block, 1, lcqi::LCQI_QK_K_BLOCK_VALUES, q8_input, ↪
    ↪ matvec_output);
1273     require_close(matvec_output[0], q4_q8_dot, "Q4_K Q8 matvec mismatch");
1274 }
1275
1276 void test_ggml_mixed_dequantizers() {
1277     std::vector<std::uint8_t> f32_bytes;
1278     append_le_f32(f32_bytes, 1.25F);
1279     append_le_f32(f32_bytes, -2.5F);
1280     const std::vector<float> f32 =
1281         lcqi::dequantize_ggml_tensor(lcqi::GgmlType::f32, f32_bytes, 2);
1282     require_close(f32[0], 1.25F, "GGML F32 decoded value 0 mismatch");
1283     require_close(f32[1], -2.5F, "GGML F32 decoded value 1 mismatch");

```

```

1284
1285 const std::vector<std::uint8_t> q8 = make_q8_0_test_block();
1286 const std::vector<float> q8_values =
1287     lcqi::dequantize_ggml_tensor(lcqi::GgmlType::q8_0,
1288                                 q8,
1289                                 lcqi::LCQI_Q8_0_BLOCK_VALUES);
1290 require_close(q8_values[0], -8.0F, "Q8_0 decoded value 0 mismatch");
1291 require_close(q8_values[16], 0.0F, "Q8_0 decoded value 16 mismatch");
1292 require_close(q8_values[31], 7.5F, "Q8_0 decoded value 31 mismatch");
1293
1294 const std::vector<std::uint8_t> q5 = make_q5_0_test_block();
1295 const std::vector<float> q5_values =
1296     lcqi::dequantize_ggml_tensor(lcqi::GgmlType::q5_0,
1297                                 q5,
1298                                 lcqi::LCQI_Q5_0_BLOCK_VALUES);
1299 require_close(q5_values[0], -8.0F, "Q5_0 decoded value 0 mismatch");
1300 require_close(q5_values[1], -7.5F, "Q5_0 decoded value 1 mismatch");
1301 require_close(q5_values[15], -0.5F, "Q5_0 decoded value 15 mismatch");
1302 require_close(q5_values[16], 0.0F, "Q5_0 decoded value 16 mismatch");
1303 require_close(q5_values[31], 7.5F, "Q5_0 decoded value 31 mismatch");
1304
1305 const std::vector<std::uint8_t> q6 = make_q6_k_test_block();
1306 const std::vector<float> q6_values =
1307     lcqi::dequantize_ggml_tensor(lcqi::GgmlType::q6_k,
1308                                 q6,
1309                                 lcqi::LCQI_Q6_K_BLOCK_VALUES);
1310 require_close(q6_values[LCQI_GGML_TEST_Q6_LANE], 3.0F, "Q6_K q1 mismatch");
1311 require_close(q6_values[32 + LCQI_GGML_TEST_Q6_LANE], -1.0F, "Q6_K q2 ↵
1312 ↵ mismatch");
1313 require_close(q6_values[64 + LCQI_GGML_TEST_Q6_LANE], 31.0F, "Q6_K q3 ↵
1314 ↵ mismatch");
1315 require_close(q6_values[96 + LCQI_GGML_TEST_Q6_LANE], -32.0F, "Q6_K q4 ↵
1316 ↵ mismatch");
1317 require_close(q6_values[128 + LCQI_GGML_TEST_Q6_LANE], 4.0F, "Q6_K second ↵
1318 ↵ half mismatch");
1319
1320 bool rejected_bad_size = false;
1321 try {
1322     static_cast<void>(lcqi::dequantize_ggml_tensor(
1323         lcqi::GgmlType::q8_0,
1324         std::span<const std::uint8_t>(q8.data(), q8.size() - 1U),
1325         lcqi::LCQI_Q8_0_BLOCK_VALUES));
1326 } catch (const std::runtime_error&) {
1327     rejected_bad_size = true;
1328 }
1329 require(rejected_bad_size, "GGML dequantizer accepted a truncated Q8_0 ↵
1330 ↵ block");
1331
1332 }
1333
1334 void test_ggml_quantized_f32_matvec_paths() {
1335     const std::vector<float> input32 = [] {
1336         std::vector<float> values(static_cast<std::size_t>(lcqi::↵

```

```

    ↪ LCQI_Q8_0_BLOCK_VALUES), 0.0F);
1331     for (std::size_t index = 0; index < values.size(); ++index) {
1332         values[index] =
1333             ↪ static_cast<float>(static_cast<std::int32_t>(index % 9U) - 4) *
    ↪ 0.125F;
1334     }
1335     return values;
1336 }();
1337 const std::vector<float> input256 = [] {
1338     std::vector<float> values(static_cast<std::size_t>(lcqi::
    ↪ LCQI_Q6_K_BLOCK_VALUES), 0.0F);
1339     for (std::size_t index = 0; index < values.size(); ++index) {
1340         values[index] =
1341             ↪ static_cast<float>(static_cast<std::int32_t>(index % 13U) - 6)
    ↪ * 0.0625F;
1342     }
1343     return values;
1344 }();
1345
1346 const auto assert_matvec_matches_dequantized =
1347     [](lcqi::GgmlType type,
1348     ↪ const std::vector<std::uint8_t>& block,
1349     ↪ std::span<const float> input,
1350     ↪ const char* message) {
1351     ↪ const std::int64_t columns = static_cast<std::int64_t>(input.size())
    ↪ );
1352     const std::vector<std::uint8_t> matrix = [&] {
1353         std::vector<std::uint8_t> bytes;
1354         bytes.reserve(block.size() * 2U);
1355         bytes.insert(bytes.end(), block.begin(), block.end());
1356         bytes.insert(bytes.end(), block.begin(), block.end());
1357         return bytes;
1358     }();
1359     std::vector<float> direct(2, 0.0F);
1360     ↪ lcqi::matvec_ggml_quantized_f32(type, matrix, 2, columns, input,
    ↪ direct);
1361     const std::vector<lcqi::Q8_0InputBlock> q8_0_input =
1362         ↪ lcqi::quantize_q8_0_input(input);
1363     std::vector<float> q8_0_direct(2, 0.0F);
1364     ↪ lcqi::matvec_ggml_quantized_q8_0(type,
1365     ↪ matrix,
1366     ↪ 2,
1367     ↪ columns,
1368     ↪ q8_0_input,
1369     ↪ q8_0_direct);
1370     const std::vector<float> weights =
1371         ↪ lcqi::dequantize_ggml_tensor(type, matrix, columns * 2);
1372     std::array<float, 2> reference{0.0F, 0.0F};
1373     for (std::size_t row = 0; row < reference.size(); ++row) {
1374         for (std::size_t column = 0; column < input.size(); ++column) {
1375             ↪ reference[row] +=
1376                 ↪ weights[row * input.size() + column] * input[column];

```



```

1377     }
1378     if (std::fabs(reference[row] - direct[row]) > ↵
↵ LCQI_GGML_MATVEC_MAX_DIFF) {
1379         throw std::runtime_error(message);
1380     }
1381     if (std::fabs(reference[row] - q8_0_direct[row]) >
1382         LCQI_GGML_Q8_MATVEC_MAX_DIFF) {
1383         throw std::runtime_error(message);
1384     }
1385 }
1386 };
1387
1388 require(lcqi::ggml_type_has_f32_direct_matvec(lcqi::GgmlType::q5_0),
1389     "Q5_0 direct matvec support flag mismatch");
1390 require(lcqi::ggml_type_has_f32_direct_matvec(lcqi::GgmlType::q6_k),
1391     "Q6_K direct matvec support flag mismatch");
1392 require(lcqi::ggml_type_has_f32_direct_matvec(lcqi::GgmlType::q8_0),
1393     "Q8_0 direct matvec support flag mismatch");
1394 require(!lcqi::ggml_type_has_f32_direct_matvec(lcqi::GgmlType::f32),
1395     "F32 should not be reported as quantized direct matvec");
1396 require(lcqi::ggml_type_has_q8_0_direct_matvec(lcqi::GgmlType::q5_0),
1397     "Q5_0 Q8_0 direct matvec support flag mismatch");
1398 require(lcqi::ggml_type_has_q8_0_direct_matvec(lcqi::GgmlType::q6_k),
1399     "Q6_K Q8_0 direct matvec support flag mismatch");
1400 require(lcqi::ggml_type_has_q8_0_direct_matvec(lcqi::GgmlType::q8_0),
1401     "Q8_0 Q8_0 direct matvec support flag mismatch");
1402 require(!lcqi::ggml_type_has_q8_0_direct_matvec(lcqi::GgmlType::f32),
1403     "F32 should not be reported as Q8_0 quantized direct matvec");
1404
1405 assert_matvec_matches_dequantized(lcqi::GgmlType::q8_0,
1406     make_q8_0_test_block(),
1407     input32,
1408     "Q8_0 direct matvec mismatch");
1409 assert_matvec_matches_dequantized(lcqi::GgmlType::q5_0,
1410     make_q5_0_test_block(),
1411     input32,
1412     "Q5_0 direct matvec mismatch");
1413 assert_matvec_matches_dequantized(lcqi::GgmlType::q6_k,
1414     make_q6_k_test_block(),
1415     input256,
1416     "Q6_K direct matvec mismatch");
1417 }
1418
1419 void test_llama_gguf_loader_and_generation() {
1420     const std::filesystem::path path = temp_file_path("lcqi_tiny_llama_fixture.↵
↵ gguf");
1421     write_tiny_llama_gguf_fixture(path);
1422
1423     const lcqi::LlamaGgufLoadedModel loaded =
1424         lcqi::load_llama_gguf_reference_model(path);
1425     const std::vector<std::int32_t> prompt{1};
1426     const lcqi::LlamaGgufGenerationResult result =

```



```

1427     lcqi::llama_gguf_generate_greedy(loader.model, prompt, 1);
1428     std::filesystem::remove(path);
1429
1430     require(loader.model.architecture == "llama", "LLaMA GGUF architecture ↵
↵ mismatch");
1431     require(loader.model.decoder.config.hidden_size == LCQI_LLAMA_TEST_HIDDEN,
1432             "LLaMA GGUF hidden size mismatch");
1433     require(loader.model.decoder.layers.size() == LCQI_LLAMA_TEST_LAYER_COUNT,
1434             "LLaMA GGUF layer count mismatch");
1435     require(loader.model.lm_head_tied_to_embedding,
1436             "LLaMA GGUF tiny fixture should tie lm head to embeddings");
1437     require(result.generated_ids == std::vector<std::int32_t>({1, ↵
↵ LCQI_LLAMA_TEST_EXPECTED_TOKEN}),
1438             "LLaMA GGUF generated ids mismatch");
1439     require(result.predicted_first_token == LCQI_LLAMA_TEST_EXPECTED_TOKEN,
1440             "LLaMA GGUF predicted token mismatch");
1441     require(loader.report.tensors_loaded == 11, "LLaMA GGUF tensor load count ↵
↵ mismatch");
1442     require(loader.report.f32_weight_bytes == loader.report.↵
↵ quantized_weight_bytes,
1443             "LLaMA GGUF F32 fixture byte accounting mismatch");
1444     require(loader.report.q4_k_direct_tensors == 0,
1445             "LLaMA GGUF F32 fixture should not use direct Q4_K tensors");
1446     require(loader.report.ggml_direct_tensors == 0,
1447             "LLaMA GGUF F32 fixture should not use direct GGML tensors");
1448     require(result.hotspots.f32_fallback_calls == 7,
1449             "LLaMA GGUF F32 fixture linear fallback call count mismatch");
1450     require(result.hotspots.q4_k_direct_calls == 0,
1451             "LLaMA GGUF F32 fixture should not call Q4_K direct matvec");
1452     require(result.hotspots.ggml_direct_calls == 0,
1453             "LLaMA GGUF F32 fixture should not call GGML direct matvec");
1454 }
1455
1456 void test_llama_gguf_parallel_generation_matches_serial() {
1457     const std::filesystem::path path = temp_file_path("lcqi_tiny_llama_parallel↵
↵ .gguf");
1458     write_tiny_llama_gguf_fixture(path);
1459
1460     const lcqi::LlamaGgufLoadedModel loader =
1461         lcqi::load_llama_gguf_reference_model(path);
1462     const std::vector<std::int32_t> prompt{1};
1463     const lcqi::LlamaGgufGenerationResult serial =
1464         lcqi::llama_gguf_generate_greedy(loader.model, prompt, 1);
1465     lcqi::LlamaGgufExecutionOptions parallel_options;
1466     parallel_options.worker_count = 2;
1467     parallel_options.parallel_min_rows = 1;
1468     const lcqi::LlamaGgufGenerationResult parallel =
1469         lcqi::llama_gguf_generate_greedy(loader.model, prompt, 1, ↵
↵ parallel_options);
1470     std::filesystem::remove(path);
1471
1472     require(parallel.worker_count == parallel_options.worker_count,

```

```

1473         "parallel LLaMA worker count mismatch");
1474     require(parallel.generated_ids == serial.generated_ids,
1475             "parallel LLaMA generated ids mismatch");
1476     require(parallel.predicted_first_token == serial.predicted_first_token,
1477             "parallel LLaMA predicted token mismatch");
1478     require(parallel.hotspots.f32_fallback_calls == serial.hotspots.↵
↵ f32_fallback_calls,
1479             "parallel LLaMA fallback call count mismatch");
1480     require(parallel.hotspots.q4_k_direct_calls == serial.hotspots.↵
↵ q4_k_direct_calls,
1481             "parallel LLaMA Q4 direct call count mismatch");
1482     require(parallel.hotspots.ggml_direct_calls == serial.hotspots.↵
↵ ggml_direct_calls,
1483             "parallel LLaMA GGML direct call count mismatch");
1484 }
1485
1486 void test_llama_gguf_q4_direct_generation() {
1487     const std::filesystem::path path = temp_file_path("↵
↵ lcqi_tiny_llama_q4_direct.gguf");
1488     write_tiny_llama_q4_direct_gguf_fixture(path);
1489
1490     const lcqi::LlamaGgufLoadedModel loaded =
1491         lcqi::load_llama_gguf_reference_model(path);
1492     const std::vector<std::int32_t> prompt{0};
1493     const lcqi::LlamaGgufGenerationResult result =
1494         lcqi::llama_gguf_generate_greedy(loaded.model, prompt, 1);
1495     std::filesystem::remove(path);
1496
1497     require(loaded.model.decoder.config.hidden_size == ↵
↵ LCQI_LLAMA_Q4_TEST_HIDDEN,
1498             "LLaMA Q4_K fixture hidden size mismatch");
1499     require(loaded.report.tensors_loaded == 11, "LLaMA Q4_K fixture tensor load↵
↵ count mismatch");
1500     require(loaded.report.q4_k_direct_tensors == 7,
1501             "LLaMA Q4_K fixture direct tensor count mismatch");
1502     require(loaded.report.ggml_direct_tensors == 0,
1503             "LLaMA Q4_K fixture should not use GGML direct tensors");
1504     require(loaded.report.f32_fallback_tensors == 4,
1505             "LLaMA Q4_K fixture fallback tensor count mismatch");
1506     require(loaded.report.direct_quantized_weight_bytes >
1507             loaded.report.fallback_dequantized_weight_bytes,
1508             "LLaMA Q4_K fixture should keep direct quantized matrix bytes");
1509     require(result.hotspots.q4_k_direct_calls == 7,
1510             "LLaMA Q4_K fixture direct matvec call count mismatch");
1511     require(result.hotspots.ggml_direct_calls == 0,
1512             "LLaMA Q4_K fixture should not call GGML direct matvec");
1513     require(result.hotspots.f32_fallback_calls == 0,
1514             "LLaMA Q4_K fixture should not call fallback linear matvec");
1515     require(result.generated_ids == std::vector<std::int32_t>({0, 0}),
1516             "LLaMA Q4_K fixture generated ids mismatch");
1517 }
1518

```

```

1519 void test_llama_gguf_ggml_direct_generation() {
1520     const ScopedEnvVar enable_ggml_direct(LCQI_TEST_LLAMA_GGML_DIRECT_ENV,
1521                                             LCQI_TEST_TRUE_ENV);
1522     const std::filesystem::path path = temp_file_path("↵
↵ lcqi_tiny_llama_ggml_direct.gguf");
1523     write_tiny_llama_ggml_direct_gguf_fixture(path);
1524
1525     const lcqi::LlamaGgufLoadedModel loaded =
1526         lcqi::load_llama_gguf_reference_model(path);
1527     const std::vector<std::int32_t> prompt{0};
1528     const lcqi::LlamaGgufGenerationResult result =
1529         lcqi::llama_gguf_generate_greedy(loaded.model, prompt, 1);
1530     std::filesystem::remove(path);
1531
1532     require(loaded.model.decoder.config.hidden_size == ↵
↵ LCQI_LLAMA_Q4_TEST_HIDDEN,
1533             "LLaMA GGML fixture hidden size mismatch");
1534     require(loaded.report.tensors_loaded == 11, "LLaMA GGML fixture tensor load↵
↵ count mismatch");
1535     require(loaded.report.q4_k_direct_tensors == 0,
1536             "LLaMA GGML fixture should not use Q4_K direct tensors");
1537     require(loaded.report.ggml_direct_tensors == 7,
1538             "LLaMA GGML fixture direct tensor count mismatch");
1539     require(loaded.report.f32_fallback_tensors == 4,
1540             "LLaMA GGML fixture fallback tensor count mismatch");
1541     require(result.hotspots.ggml_direct_calls == 7,
1542             "LLaMA GGML fixture direct matvec call count mismatch");
1543     require(result.hotspots.q4_k_direct_calls == 0,
1544             "LLaMA GGML fixture should not call Q4_K direct matvec");
1545     require(result.hotspots.f32_fallback_calls == 0,
1546             "LLaMA GGML fixture should not call fallback linear matvec");
1547     require(result.generated_ids == std::vector<std::int32_t>({0, 0}),
1548             "LLaMA GGML fixture generated ids mismatch");
1549 }
1550
1551 void test_gpt2_tiny_forward_and_generation() {
1552     const lcqi::Gpt2ReferenceModel model = lcqi::make_tiny_gpt2_reference_model↵
↵ ();
1553     const std::vector<std::int32_t> tokens{1, 2, 3};
1554     const lcqi::Gpt2ForwardResult result = lcqi::run_gpt2_forward(model, tokens↵
↵ );
1555
1556     require(result.logits.size() == LCQI_GPT2_VOCAB_SIZE, "GPT-2 logits size ↵
↵ mismatch");
1557     require(result.predicted_token == LCQI_GPT2_PREDICTED_TOKEN,
1558             "GPT-2 predicted token mismatch");
1559     for (std::size_t index = 0; index < LCQI_GPT2_LOGITS.size(); ++index) {
1560         require_close(result.logits[index], LCQI_GPT2_LOGITS[index], "GPT-2 ↵
↵ logit mismatch");
1561     }
1562
1563     const std::vector<std::int32_t> generated =

```

```

1564     lcqi::gpt2_generate_greedy(model, tokens, 2);
1565     const std::vector<std::int32_t> expected{1, 2, 3, 3, 3};
1566     require(generated == expected, "GPT-2 greedy generation mismatch");
1567     require(static_cast<std::int32_t>(generated.size()) == ←
    ↪ LCQI_GPT2_GENERATED_LENGTH,
1568            "GPT-2 generated length mismatch");
1569
1570     lcqi::Gpt2KvCache cache(model.config);
1571     lcqi::Gpt2ForwardResult cached_result;
1572     for (const std::int32_t token : tokens) {
1573         cached_result = lcqi::run_gpt2_forward_cached(model, cache, token);
1574     }
1575     require(cache.filled_tokens() == static_cast<std::int32_t>(tokens.size()),
1576            "GPT-2 KV cache filled token count mismatch");
1577     require(cache.byte_size() > 0, "GPT-2 KV cache byte size should be non-zero↵
    ↪ ");
1578     require(cached_result.predicted_token == result.predicted_token,
1579            "cached GPT-2 predicted token mismatch");
1580     require(cached_result.logits.size() == result.logits.size(),
1581            "cached GPT-2 logits size mismatch");
1582     for (std::size_t index = 0; index < result.logits.size(); ++index) {
1583         require_close(cached_result.logits[index],
1584            result.logits[index],
1585            "cached GPT-2 logit mismatch");
1586     }
1587
1588     lcqi::Gpt2CachedGreedyDecoder decoder_with_logits(model);
1589     lcqi::Gpt2ForwardResult decoder_result;
1590     for (const std::int32_t token : tokens) {
1591         decoder_result = decoder_with_logits.step_with_logits(token);
1592     }
1593     require(decoder_with_logits.filled_tokens() == static_cast<std::int32_t>(↵
    ↪ tokens.size()),
1594            "GPT-2 optimized decoder filled token count mismatch");
1595     require(decoder_with_logits.kv_cache_bytes() == cache.byte_size(),
1596            "GPT-2 optimized decoder KV byte size mismatch");
1597     require(decoder_result.predicted_token == result.predicted_token,
1598            "optimized cached GPT-2 predicted token mismatch");
1599     require(decoder_result.logits.size() == result.logits.size(),
1600            "optimized cached GPT-2 logits size mismatch");
1601     for (std::size_t index = 0; index < result.logits.size(); ++index) {
1602         require_close(decoder_result.logits[index],
1603            result.logits[index],
1604            "optimized cached GPT-2 logit mismatch");
1605     }
1606
1607     lcqi::Gpt2ExecutionOptions parallel_options;
1608     parallel_options.worker_count = 2;
1609     parallel_options.parallel_min_rows = 1;
1610     lcqi::Gpt2CachedGreedyDecoder parallel_decoder(model, nullptr, ↵
    ↪ parallel_options);
1611     lcqi::Gpt2ForwardResult parallel_result;

```

```

1612     for (const std::int32_t token : tokens) {
1613         parallel_result = parallel_decoder.step_with_logits(token);
1614     }
1615     require(parallel_decoder.worker_count() == parallel_options.worker_count,
1616             "parallel GPT-2 decoder worker count mismatch");
1617     require(parallel_result.predicted_token == result.predicted_token,
1618             "parallel cached GPT-2 predicted token mismatch");
1619     require(parallel_result.logits.size() == result.logits.size(),
1620             "parallel cached GPT-2 logits size mismatch");
1621     for (std::size_t index = 0; index < result.logits.size(); ++index) {
1622         require_close(parallel_result.logits[index],
1623                       result.logits[index],
1624                       "parallel cached GPT-2 logit mismatch");
1625     }
1626
1627     lcqi::Gpt2CachedGreedyDecoder greedy_decoder(model);
1628     std::int32_t greedy_predicted = 0;
1629     for (const std::int32_t token : tokens) {
1630         greedy_predicted = greedy_decoder.step(token);
1631     }
1632     require(greedy_predicted == result.predicted_token,
1633             "optimized logits-free GPT-2 predicted token mismatch");
1634
1635     lcqi::Gpt2CachedGreedyDecoder prefill_decoder(model);
1636     for (std::size_t index = 0; index + 1 < tokens.size(); ++index) {
1637         prefill_decoder.advance_without_prediction(tokens[index]);
1638     }
1639     require(prefill_decoder.filled_tokens() ==
1640             static_cast<std::int32_t>(tokens.size() - 1U),
1641             "GPT-2 prefill-only decoder filled token count mismatch");
1642     const std::int32_t prefill_predicted = prefill_decoder.step(tokens.back());
1643     require(prefill_decoder.filled_tokens() == static_cast<std::int32_t>(tokens.size()),
1644             "GPT-2 prefill decoder final filled token count mismatch");
1645     require(prefill_predicted == result.predicted_token,
1646             "GPT-2 prefill-skipped predicted token mismatch");
1647
1648     lcqi::Gpt2HotspotProfile hotspot_profile;
1649     lcqi::Gpt2CachedGreedyDecoder profiled_decoder(model, &hotspot_profile);
1650     std::int32_t profiled_predicted = 0;
1651     for (const std::int32_t token : tokens) {
1652         profiled_predicted = profiled_decoder.step(token);
1653     }
1654     require(profiled_predicted == result.predicted_token,
1655             "profiled GPT-2 predicted token mismatch");
1656     require(hotspot_profile.decoder_steps == static_cast<std::int64_t>(tokens.size()),
1657             "profiled GPT-2 decoder step count mismatch");
1658     require(hotspot_profile.layer_steps ==
1659             static_cast<std::int64_t>(tokens.size()) * model.config.layer_count,
1660             "profiled GPT-2 layer step count mismatch");

```

```

1661     require(hotspot_profile.total_step_ms >= hotspot_profile.lm_head_ms,
1662             "profiled GPT-2 total time should include lm head time");
1663
1664     const std::vector<std::int32_t> cached_generated =
1665         lcqi::gpt2_generate_greedy_cached(model, tokens, 2);
1666     require(cached_generated == expected, "cached GPT-2 greedy generation ↵
↵ mismatch");
1667 }
1668
1669 void test_gpt2_byte_bpe_tokenizer() {
1670     const std::filesystem::path vocab_path = temp_file_path("lcqi_gpt2_vocab.↵
↵ json");
1671     const std::filesystem::path merges_path = temp_file_path("lcqi_gpt2_merges.↵
↵ txt");
1672     const std::string space_marker = "\xC4\xA0";
1673     const std::string vocab =
1674         "{\n\"Hello\":7,\n\",\":8,\n\"\" + space_marker + "world\":9,\n\"!\":10}";
1675     const std::string merges =
1676         "#version: 0.2\n"
1677         "H e\n"
1678         "He l\n"
1679         "Hel l\n"
1680         "Hell o\n" +
1681         space_marker + " w\n" +
1682         space_marker + "w o\n" +
1683         space_marker + "wo r\n" +
1684         space_marker + "wor l\n" +
1685         space_marker + "worl d\n";
1686     write_text_file(vocab_path, vocab);
1687     write_text_file(merges_path, merges);
1688
1689     const lcqi::Gpt2Tokenizer tokenizer =
1690         lcqi::load_gpt2_tokenizer(vocab_path, merges_path, -1, -1);
1691     std::filesystem::remove(vocab_path);
1692     std::filesystem::remove(merges_path);
1693
1694     const std::vector<std::int32_t> ids =
1695         lcqi::gpt2_encode(tokenizer, "Hello, world!");
1696     const std::vector<std::int32_t> expected{
1697         LCQI_GPT2_TOKENIZER_HELLO_ID,
1698         8,
1699         9,
1700         10,
1701     };
1702     require(ids == expected, "GPT-2 BPE ids mismatch");
1703     require(lcqi::gpt2_decode(tokenizer, ids) == "Hello, world!",
1704             "GPT-2 BPE decode mismatch");
1705 }
1706
1707 void test_gpt2_loads_hf_style_tiny_checkpoint() {
1708     const std::filesystem::path directory =
1709         temp_file_path("lcqi_gpt2_tiny_checkpoint_dir");

```

```

1710     std::filesystem::remove_all(directory);
1711     std::filesystem::create_directories(directory);
1712
1713     const lcqi::Gpt2ReferenceModel original = lcqi::↵
↵ make_tiny_gpt2_reference_model();
1714     write_text_file(
1715         directory / "config.json",
1716         R"({"model_type":"gpt2","n_embd":4,"n_head":2,"n_layer":1,"n_positions"↵
↵ :8,"n_ctx":8,"vocab_size":6,"n_inner":8,"layer_norm_epsilon":0.00001,"↵
↵ activation_function":"gelu_new","bos_token_id":0,"eos_token_id":5})");
1717     write_tiny_gpt2_safetensors(directory / "model.safetensors", original);
1718
1719     const lcqi::Gpt2ReferenceModel loaded = lcqi::load_gpt2_from_directory(↵
↵ directory);
1720     const std::vector<std::int32_t> tokens{1, 2, 3};
1721     const lcqi::Gpt2ForwardResult result = lcqi::run_gpt2_forward(loaded, ↵
↵ tokens);
1722     std::filesystem::remove_all(directory);
1723
1724     require(result.predicted_token == LCQI_GPT2_PREDICTED_TOKEN,
1725         "loaded GPT-2 predicted token mismatch");
1726     for (std::size_t index = 0; index < LCQI_GPT2_LOGITS.size(); ++index) {
1727         require_close(result.logits[index],
1728             LCQI_GPT2_LOGITS[index],
1729             "loaded GPT-2 logit mismatch");
1730     }
1731 }
1732
1733 void test_gpt2_loader_rejects_bad_shape() {
1734     const std::filesystem::path directory =
1735         temp_file_path("lcqi_gpt2_bad_shape_dir");
1736     std::filesystem::remove_all(directory);
1737     std::filesystem::create_directories(directory);
1738
1739     const lcqi::Gpt2ReferenceModel original = lcqi::↵
↵ make_tiny_gpt2_reference_model();
1740     write_text_file(
1741         directory / "config.json",
1742         R"({"model_type":"gpt2","n_embd":5,"n_head":1,"n_layer":1,"n_positions"↵
↵ :8,"n_ctx":8,"vocab_size":6,"n_inner":8,"layer_norm_epsilon":0.00001,"↵
↵ activation_function":"gelu_new","bos_token_id":0,"eos_token_id":5})");
1743     write_tiny_gpt2_safetensors(directory / "model.safetensors", original);
1744
1745     bool threw = false;
1746     try {
1747         static_cast<void>(lcqi::load_gpt2_from_directory(directory));
1748     } catch (const std::runtime_error&) {
1749         threw = true;
1750     }
1751     std::filesystem::remove_all(directory);
1752     require(threw, "GPT-2 loader should reject bad tensor shape");
1753 }

```



```

1754
1755 void test_reference_rms_norm() {
1756     const std::vector<float> input{3.0F, 4.0F};
1757     const std::vector<float> weight{1.0F, 0.5F};
1758     std::vector<float> output(2, 0.0F);
1759     lcqi::rms_norm(input, weight, 0.0F, output);
1760     require_close(output[0], LCQI_RMS_EXPECTED_0, "RMSNorm output 0 mismatch");
1761     require_close(output[1], LCQI_RMS_EXPECTED_1, "RMSNorm output 1 mismatch");
1762 }
1763
1764 void test_reference_rope() {
1765     std::vector<float> heads{1.0F, 0.0F, 0.0F, 1.0F};
1766     lcqi::apply_rope(heads, 1, 4, 1, 10000.0F);
1767     require_close(heads[0], std::cos(1.0F), "RoPE pair 0 cosine mismatch");
1768     require_close(heads[1], std::sin(1.0F), "RoPE pair 0 sine mismatch");
1769     require_close(heads[2], -std::sin(0.01F), "RoPE pair 1 negative sine ←
    ↪ mismatch");
1770     require_close(heads[3], std::cos(0.01F), "RoPE pair 1 cosine mismatch");
1771 }
1772
1773 void test_reference_kv_attention() {
1774     lcqi::DecoderConfig config;
1775     config.hidden_size = 4;
1776     config.query_heads = 2;
1777     config.kv_heads = 1;
1778     config.head_dim = 2;
1779     config.intermediate_size = 4;
1780     config.vocab_size = 4;
1781     config.max_context = 4;
1782
1783     lcqi::ReferenceKVCache cache(config, 1);
1784     const std::vector<float> key{1.0F, 0.0F};
1785     const std::vector<float> value{LCQI_ATTENTION_EXPECTED_0, ←
    ↪ LCQI_ATTENTION_EXPECTED_1};
1786     cache.append(0, 0, key, value);
1787     require(cache.filled_tokens() == 1, "KV filled token count mismatch");
1788     require_close(cache.key(0, 0, 0)[0], 1.0F, "KV key read mismatch");
1789
1790     const std::vector<float> query{1.0F, 0.0F, 0.0F, 1.0F};
1791     std::vector<float> output(4, 0.0F);
1792     lcqi::attention_decode(config, cache, 0, 0, query, output);
1793     require_close(output[0], LCQI_ATTENTION_EXPECTED_0, "attention head 0 dim 0←
    ↪ mismatch");
1794     require_close(output[1], LCQI_ATTENTION_EXPECTED_1, "attention head 0 dim 1←
    ↪ mismatch");
1795     require_close(output[2], LCQI_ATTENTION_EXPECTED_0, "attention head 1 dim 0←
    ↪ mismatch");
1796     require_close(output[3], LCQI_ATTENTION_EXPECTED_1, "attention head 1 dim 1←
    ↪ mismatch");
1797 }
1798
1799 void test_reference_decoder_end_to_end() {

```



```

1800     const lcqi::ReferenceDecoderModel model = lcqi::↵
↵ make_tiny_reference_decoder_model();
1801     const std::vector<std::int32_t> tokens{1, 2, 3};
1802     const lcqi::ReferenceDecodeResult result = lcqi::run_reference_decode(model↵
↵ , tokens);
1803     require(result.logits.size() == LCQI_REFERENCE_DECODER_VOCAB_SIZE,
1804             "reference decoder logits size mismatch");
1805     require(result.predicted_token == LCQI_REFERENCE_DECODER_PREDICTED_TOKEN,
1806             "reference decoder predicted token mismatch");
1807     for (std::size_t index = 0; index < LCQI_REFERENCE_DECODER_LOGITS.size(); ↵
↵ ++index) {
1808         require_close(result.logits[index],
1809                       LCQI_REFERENCE_DECODER_LOGITS[index],
1810                       "reference decoder golden logit mismatch");
1811     }
1812 }
1813
1814 void test_reference_decoder_trace_contract() {
1815     const lcqi::ReferenceDecoderModel model = lcqi::↵
↵ make_tiny_reference_decoder_model();
1816     const std::vector<std::int32_t> tokens{1, 2, 3};
1817     const lcqi::ReferenceDecodeTraceResult trace =
1818         lcqi::run_reference_decode_with_trace(model, tokens);
1819
1820     require(trace.result.predicted_token == ↵
↵ LCQI_REFERENCE_DECODER_PREDICTED_TOKEN,
1821             "reference trace predicted token mismatch");
1822     require(!trace.checkpoints.empty(), "reference trace checkpoints missing");
1823
1824     const lcqi::ReferenceTensorCheckpoint& first = trace.checkpoints.front();
1825     require(first.name == "embedding", "first checkpoint should be embedding");
1826     require(first.token_position == LCQI_REFERENCE_TRACE_FIRST_TOKEN,
1827             "first checkpoint token position mismatch");
1828     require(first.dtype == "f32", "first checkpoint dtype mismatch");
1829     require(first.layout == "row-major", "first checkpoint layout mismatch");
1830     require(first.shape.size() == 1 &&
1831             first.shape[0] == LCQI_REFERENCE_TRACE_HIDDEN_SIZE,
1832             "first checkpoint shape mismatch");
1833     require(first.stride.size() == 1 && first.stride[0] == 1,
1834             "first checkpoint stride mismatch");
1835
1836     bool saw_logits = false;
1837     bool saw_kv_slot = false;
1838     for (const lcqi::ReferenceTensorCheckpoint& checkpoint : trace.checkpoints)↵
↵ {
1839         if (checkpoint.name == "kv_cache_key_slot") {
1840             saw_kv_slot = true;
1841             require(checkpoint.shape.size() == 4, "KV checkpoint rank mismatch"↵
↵ );
1842             require(checkpoint.layout == "kv_contiguous", "KV checkpoint layout↵
↵ mismatch");
1843         }

```

```

1844     if (checkpoint.name == "logits") {
1845         saw_logits = true;
1846         require(checkpoint.token_position == ↵
↵ LCQI_REFERENCE_TRACE_TOKEN_COUNT - 1,
1847             "logits checkpoint token position mismatch");
1848         require(checkpoint.shape.size() == 1 &&
1849             checkpoint.shape[0] ==
1850             static_cast<std::int32_t>(↵
↵ LCQI_REFERENCE_DECODER_VOCAB_SIZE),
1851             "logits checkpoint shape mismatch");
1852         require(checkpoint.values.size() == LCQI_REFERENCE_DECODER_LOGITS.↵
↵ size(),
1853             "logits checkpoint value size mismatch");
1854         for (std::size_t index = 0; index < LCQI_REFERENCE_DECODER_LOGITS.↵
↵ size(); ++index) {
1855             require_close(checkpoint.values[index],
1856                 LCQI_REFERENCE_DECODER_LOGITS[index],
1857                 "trace logits checkpoint mismatch");
1858         }
1859     }
1860 }
1861 require(saw_kv_slot, "KV checkpoint missing");
1862 require(saw_logits, "logits checkpoint missing");
1863 }
1864
1865 void test_packed_i8_kernel_matches_scalar() {
1866     for (const KernelShapeCase& shape_case : LCQI_KERNEL_SHAPE_CASES) {
1867         const lcqi::QuantizedLinearLayer layer =
1868             lcqi::make_deterministic_i8_layer(shape_case.input_size,
1869                 shape_case.output_size,
1870                 0.03125F);
1871         const lcqi::PackedLinearI8 packed =
1872             lcqi::pack_linear_i8(layer, shape_case.output_block_size);
1873         const std::vector<float> input = lcqi::make_deterministic_input(layer.↵
↵ input_size);
1874         std::vector<float> scalar_output(static_cast<std::size_t>(layer.↵
↵ output_size), 0.0F);
1875         std::vector<float> packed_output(static_cast<std::size_t>(layer.↵
↵ output_size), 0.0F);
1876         lcqi::linear_i8(layer, input, scalar_output);
1877         lcqi::linear_i8_packed(packed, input, packed_output);
1878         require(lcqi::max_abs_diff(scalar_output, packed_output) <= ↵
↵ LCQI_KERNEL_MAX_DIFF,
1879             "packed int8 kernel output differs from scalar");
1880
1881         const lcqi::PackedLinearI8 simd_packed =
1882             lcqi::pack_linear_i8(layer, LCQI_SIMD_TEST_BLOCK);
1883         if (lcqi::linear_i8_packed_avx2_available()) {
1884             std::vector<float> avx2_output(static_cast<std::size_t>(layer.↵
↵ output_size), 0.0F);
1885             lcqi::linear_i8_packed_avx2(simd_packed, input, avx2_output);
1886             require(lcqi::max_abs_diff(scalar_output, avx2_output) <= ↵

```

```

    ↪ LCQI_KERNEL_MAX_DIFF,
1887         "AVX2 int8 kernel output differs from scalar");
1888     }
1889 }
1890 }
1891
1892 void test_f32_dot_kernel_matches_scalar() {
1893     std::vector<float> lhs(LCQI_F32_DOT_TEST_SIZE, 0.0F);
1894     std::vector<float> rhs(LCQI_F32_DOT_TEST_SIZE, 0.0F);
1895     for (std::size_t index = 0; index < LCQI_F32_DOT_TEST_SIZE; ++index) {
1896         lhs[index] = static_cast<float>(static_cast<std::int32_t>(index % 19U) ↪
    ↪ + 1) *
1897             0.03125F;
1898         rhs[index] = static_cast<float>(static_cast<std::int32_t>(index % 23U) ↪
    ↪ - 11) *
1899             0.0625F;
1900     }
1901
1902     const float scalar =
1903         lcqi::dot_f32_scalar_unchecked(lhs.data(), rhs.data(), lhs.size());
1904     const float dispatched = lcqi::dot_f32(lhs, rhs);
1905     require(std::fabs(scalar - dispatched) <= LCQI_F32_KERNEL_MAX_DIFF,
1906         "F32 dot kernel output differs from scalar");
1907
1908     if (lcqi::dot_f32_avx2_available()) {
1909         const float avx2 =
1910             lcqi::dot_f32_avx2_unchecked(lhs.data(), rhs.data(), lhs.size());
1911         require(std::fabs(scalar - avx2) <= LCQI_F32_KERNEL_MAX_DIFF,
1912             "AVX2 F32 dot kernel output differs from scalar");
1913     }
1914
1915     bool rejected_mismatch = false;
1916     try {
1917         (void)lcqi::dot_f32(std::span<const float>(lhs.data(), lhs.size() - 1U) ↪
    ↪ , rhs);
1918     } catch (const std::runtime_error&) {
1919         rejected_mismatch = true;
1920     }
1921     require(rejected_mismatch, "F32 dot kernel accepted mismatched sizes");
1922 }
1923
1924 void test_f32_row_kernels_match_scalar() {
1925     std::vector<float> weights(LCQI_F32_ROW_TEST_INPUT_SIZE *
1926                               LCQI_F32_ROW_TEST_OUTPUT_SIZE,
1927                               0.0F);
1928     std::vector<float> input(LCQI_F32_ROW_TEST_INPUT_SIZE, 0.0F);
1929     std::vector<float> bias(LCQI_F32_ROW_TEST_OUTPUT_SIZE, 0.0F);
1930     for (std::size_t index = 0; index < weights.size(); ++index) {
1931         weights[index] = static_cast<float>(static_cast<std::int32_t>(index % ↪
    ↪ 29U) - 14) *
1932             0.015625F;
1933     }

```

```

1934     for (std::size_t index = 0; index < input.size(); ++index) {
1935         input[index] = static_cast<float>(static_cast<std::int32_t>(index % 13U) ←
↪ ) - 6) *
1936             0.03125F;
1937     }
1938     for (std::size_t index = 0; index < bias.size(); ++index) {
1939         bias[index] = static_cast<float>(static_cast<std::int32_t>(index % 7U) ←
↪ - 3) *
1940             0.125F;
1941     }
1942
1943     std::vector<float> scalar_output(LCQI_F32_ROW_TEST_OUTPUT_SIZE, 0.0F);
1944     std::vector<float> dispatched_output(LCQI_F32_ROW_TEST_OUTPUT_SIZE, 0.0F);
1945     lcqi::linear_f32_rows_scalar_unchecked(weights.data(),
1946                                             input.data(),
1947                                             bias.data(),
1948                                             input.size(),
1949                                             0,
1950                                             LCQI_F32_ROW_TEST_OUTPUT_SIZE,
1951                                             scalar_output.data());
1952     lcqi::linear_f32_rows_unchecked(weights.data(),
1953                                     input.data(),
1954                                     bias.data(),
1955                                     input.size(),
1956                                     0,
1957                                     LCQI_F32_ROW_TEST_OUTPUT_SIZE,
1958                                     dispatched_output.data());
1959     for (std::size_t index = 0; index < scalar_output.size(); ++index) {
1960         require(std::fabs(scalar_output[index] - dispatched_output[index]) <=
1961                 LCQI_F32_KERNEL_MAX_DIFF,
1962                 "F32 linear row kernel output differs from scalar");
1963     }
1964
1965     const lcqi::F32RowMax scalar_max =
1966         lcqi::max_dot_f32_rows_scalar_unchecked(weights.data(),
1967                                                  input.data(),
1968                                                  input.size(),
1969                                                  0,
1970                                                  LCQI_F32_ROW_TEST_OUTPUT_SIZE);
1971     const lcqi::F32RowMax dispatched_max =
1972         lcqi::max_dot_f32_rows_unchecked(weights.data(),
1973                                          input.data(),
1974                                          input.size(),
1975                                          0,
1976                                          LCQI_F32_ROW_TEST_OUTPUT_SIZE);
1977     require(scalar_max.row == dispatched_max.row,
1978             "F32 row max kernel selected a different row");
1979     require(std::fabs(scalar_max.value - dispatched_max.value) <= ←
↪ LCQI_F32_KERNEL_MAX_DIFF,
1980             "F32 row max kernel value differs from scalar");
1981
1982     if (lcqi::dot_f32_avx2_available()) {

```

```

1983     std::vector<float> avx2_output(LCQI_F32_ROW_TEST_OUTPUT_SIZE, 0.0F);
1984     lcqi::linear_f32_rows_avx2_unchecked(weights.data(),
1985                                         input.data(),
1986                                         bias.data(),
1987                                         input.size(),
1988                                         0,
1989                                         LCQI_F32_ROW_TEST_OUTPUT_SIZE,
1990                                         avx2_output.data());
1991     for (std::size_t index = 0; index < scalar_output.size(); ++index) {
1992         require(std::fabs(scalar_output[index] - avx2_output[index]) <=
1993                 LCQI_F32_KERNEL_MAX_DIFF,
1994                 "AVX2 F32 linear row kernel output differs from scalar");
1995     }
1996
1997     const lcqi::F32RowMax avx2_max =
1998         lcqi::max_dot_f32_rows_avx2_unchecked(weights.data(),
1999                                               input.data(),
2000                                               input.size(),
2001                                               0,
2002                                               LCQI_F32_ROW_TEST_OUTPUT_SIZE↵
2003     ↵ );
2004     require(scalar_max.row == avx2_max.row,
2005             "AVX2 F32 row max kernel selected a different row");
2006     require(std::fabs(scalar_max.value - avx2_max.value) <= ↵
2007     ↵ LCQI_F32_KERNEL_MAX_DIFF,
2008             "AVX2 F32 row max kernel value differs from scalar");
2009 }
2010 } // namespace
2011
2012 int main() {
2013     try {
2014         test_tiny_mlp();
2015         test_tokenizer_contract();
2016         test_tokenizer_rejects_unknown_without_unk();
2017         test_safetensors_manifest_contract();
2018         test_safetensors_rejects_bad_offsets();
2019         test_safetensors_rejects_bad_dtype_shape_size();
2020         test_safetensors_reads_float_tensors();
2021         test_gguf_manifest_reads_q4_k_tensor();
2022         test_q4_k_dequant_and_dot_paths();
2023         test_ggml_mixed_dequantizers();
2024         test_ggml_quantized_f32_matvec_paths();
2025         test_llama_gguf_loader_and_generation();
2026         test_llama_gguf_parallel_generation_matches_serial();
2027         test_llama_gguf_q4_direct_generation();
2028         test_llama_gguf_ggml_direct_generation();
2029         test_gpt2_tiny_forward_and_generation();
2030         test_gpt2_byte_bpe_tokenizer();
2031         test_gpt2_loads_hf_style_tiny_checkpoint();
2032         test_gpt2_loader_rejects_bad_shape();

```

```
2033     test_reference_rms_norm();
2034     test_reference_rope();
2035     test_reference_kv_attention();
2036     test_reference_decoder_end_to_end();
2037     test_reference_decoder_trace_contract();
2038     test_packed_i8_kernel_matches_scalar();
2039     test_f32_dot_kernel_matches_scalar();
2040     test_f32_row_kernels_match_scalar();
2041
2042     std::cout << "lcqi tests passed\n";
2043     return EXIT_SUCCESS;
2044 } catch (const std::exception& error) {
2045     std::cerr << "lcqi test failure: " << error.what() << "\n";
2046     return EXIT_FAILURE;
2047 }
2048 }
```

# 实践与代码总索引

本卷已经把实践任务、代码走读路线和完整代码清单合并到同一套内容目录。目录中不再存在一个独立的“源码卷部分”：构建入口、调用链、公共合同、tiny MLP、int8 kernel、reference decoder、GPT-2、工具脚本和测试文件都插入到相应主题之后，共同解释同一份 `books/cpu-volume-3/labs/linux_cpu_inference` 源码。

## 源码阅读任务

读者不需要再在实践卷和源码卷之间来回切换，也不需要先读完 16 个实践章再去末尾翻代码。实践章节提出任务，紧邻的代码走读解释完整实现，测试和 benchmark 章节给出回归证据；三者已经合并到这一本实践与代码卷里，共同构成一条可复查的工程证据链。

## 一、实践主题到代码走读的合并位置

- 第 1 章之后 插入“代码走读：构建入口、项目边界与调用链总图”，先把 CMake、README、模型资产、target、CLI、benchmark、trace、ledger、serving probe 和测试入口放到一张可复现地图里。
- 第 4 章之后 插入“代码走读：公共合同、Tiny MLP 与 Int8 Kernel”，把模型、推理、tokenizer、safetensors、reference decoder、GPT-2 头文件，以及 tiny MLP、基础 CLI、scalar/packed/AVX2 kernel 连起来。
- 第 10 章之后 插入“代码走读：Reference Decoder、Tokenizer、Safetensors 与 GPT-2”，把真实模型加载、byte-level BPE、safetensors、GPT-2 forward、cached KV 和 greedy generation 放在模型导入主题内部讲清楚。
- 第 16 章之后 插入“代码走读：工具、Smoke、Benchmark 与回归测试”，把 benchmark、trace、ledger、serving probe、SmolLM2 smoke、GPT-2 smoke、benchmark compare 和 CTest 回归证据集中收束。

## 二、最小复现命令

```
1 cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -↵  
   ↵ DCMMAKE_BUILD_TYPE=Debug  
2 cmake --build books/cpu-volume-3/build/lcqi-debug  
3 ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure  
4 make cpu3p-pdf  
5 make cpu3p-epub
```

完成这组命令后，读者应同时检查相邻实践章节的作业结果、对应代码走读、测试断言和报告字段，确认代码、测试、报告字段和章节讲解是一条证据链。

# 术语表

| 术语                | 实践含义                                                                |
|-------------------|---------------------------------------------------------------------|
| manifest          | 模型资产清单，记录 tensor name、dtype、shape、offset、checksum 和 tokenizer/配置版本。 |
| shape verifier    | 在 kernel 运行前检查 shape、dtype、layout、量化 descriptor 和 state edge 是否匹配。  |
| reference decoder | 以清楚正确为目标的 decoder 实现，用来生成 trace 和 golden。                           |
| oracle            | 比较候选实现与 reference 的判定器，可使用精确相等或误差界。                                 |
| trace checkpoint  | 推理链路中的可复查节点，至少包含 shape、stride、dtype、layout、checksum 和摘要值。           |
| KV cache          | attention 历史 K/V 状态，跨 token 存活，不能被普通 activation planner 复用。         |
| packing           | 把权重改写成更适合热循环和 SIMD 的布局。                                             |
| backend           | 执行计划落到机器的实现，例如 scalar、AVX2、AVX-512、GPU 或库调用。                        |
| fallback          | 后端能力不满足时显式降级到可验证路径，并记录原因。                                           |
| SLO               | 服务目标，例如 TTFT、TPOT、queue wait、拒绝率和取消回收时间。                            |